

Percentage प्रतिशत

①

$$\left\{ \begin{array}{l} 1\% = \frac{1}{100} \\ 100\% = 1 \end{array} \right\}$$

Fraction Percentage
 In decimals Percentage

$$\frac{1}{2} = \frac{1}{2} \times 100\% = 50\%$$

$$\frac{1}{3} = \frac{1}{3} \times 100\% = 33\frac{1}{3}\% = 33.33\%$$

$$\frac{1}{4} = \frac{1}{4} \times 100\% = 25\%$$

$$\frac{1}{5} = \frac{1}{5} \times 100\% = 20\%$$

$$\frac{1}{6} = 16.66\% = 16\frac{2}{3}\%$$

$$\frac{1}{7} = 14.28\% = 14\frac{2}{7}\%$$

$$\frac{1}{8} = 12.5\% = 12\frac{1}{2}\%$$

$$\frac{1}{9} = 11.11\% = 11\frac{1}{9}\%$$

$$\frac{1}{10} = 10\%$$

$$\frac{1}{11} = 9.09\% = 9\frac{1}{11}\%$$

Fraction

Percentage in decimal

Percentage

$$\frac{1}{12}$$

$$= 8.33\%$$

$$= 8\frac{1}{3}\%$$

$$\frac{1}{13}$$

$$=$$

$$7.69\%$$

$$= 7\frac{9}{13}\%$$

$$\frac{1}{14}$$

$$=$$

$$7.1429\%$$

$$= 7\frac{1}{7}\%$$

$$\frac{1}{15}$$

$$=$$

$$6.66\%$$

$$=$$

$$6\frac{2}{3}\%$$

$$\frac{1}{16}$$

$$=$$

$$6.25\%$$

$$=$$

$$6\frac{1}{4}\%$$

$$\frac{1}{17}$$

$$=$$

$$5.88\%$$

$$=$$

$$5\frac{15}{17}\%$$

$$\frac{1}{18}$$

$$=$$

$$5.55\%$$

$$=$$

$$5\frac{5}{9}\%$$

$$\frac{1}{19}$$

$$=$$

$$5.26\%$$

$$=$$

$$5\frac{5}{19}\%$$

$$\frac{1}{20}$$

$$=$$

$$5\%$$

$$=$$

$$5\%$$

$$\frac{1}{21}$$

$$=$$

$$4.74\%$$

$$=$$

$$4\frac{16}{21}\%$$

$$\frac{1}{22}$$

$$=$$

$$4.5454\%$$

$$=$$

$$4\frac{6}{11}\%$$

Handwritten

Fraction

Percentage in decimal

Percentage^③

$$\frac{1}{23}$$

=

$$4.34\%$$

$$= 4\frac{8}{23}\%$$

$$\frac{1}{24}$$

=

$$4.16\%$$

$$= 4\frac{1}{6}\%$$

$$\frac{1}{25}$$

=

$$4\%$$

$$= 4\%$$

$$\frac{1}{30}$$

=

$$3.33\% = \frac{1}{3}\%$$

$$\frac{1}{40}$$

=

$$2.5\%$$

$$= 2\frac{1}{2}\%$$

$$\frac{1}{50}$$

=

$$2\%$$

$$= 2\%$$

Fraction conversion → multiply $\Rightarrow \frac{1}{100}$

eg: $15\% = 15 \times \frac{1}{100} = \frac{3}{20}$

$22.5\% = \frac{9}{40} \times \frac{1}{100} = \frac{9}{40}$

$16\frac{2}{3}\% = \frac{50}{3}\% = \frac{50}{3} \times \frac{1}{100} = \frac{1}{6}$

How to find $\rightarrow$ One fraction to other

$$* \frac{1}{4} = 25\%$$

$$\frac{3}{4} = 3 \times 25\% = 75\%$$

$$* \frac{1}{7} = 14 \frac{2}{7}\%$$

$$\frac{4}{7} = 4 \times 14 \frac{2}{7}\%$$

$$= 4 \times (14 + \frac{2}{7})\%$$

$$= 56\% + \frac{8}{7}\%$$

$$= 56\% + 1\% + \frac{1}{7}\%$$

$$\boxed{\frac{4}{7} = 57 \frac{1}{7}\%}$$

$$* \frac{1}{6} = 16 \frac{2}{3}\%$$

$$\frac{5}{6} = 80\% + 3\% + \frac{1}{3}\%$$

$$= 83 \frac{1}{3}\% \checkmark$$

Q:

$$\frac{5}{6} = 1 - \frac{1}{6}$$

$$= 100\% - 16 \frac{2}{3}\% \Rightarrow 83\% + \frac{1}{3}\%$$

$$= 83 \frac{1}{3}\%$$

$\frac{1}{8} = 12.5\%$

$\frac{3}{8} = 36\% + 1.5\% = 37.5\%$

$\frac{5}{8} = 60\% + 2.5\% = 62.5\%$

$\frac{1}{9} = 11\frac{1}{9}\% = 11.11\%$

$\frac{4}{9} = 44\frac{4}{9}\% = 44.44\%$

$\frac{7}{9} = 77\frac{7}{9}\% = 77.77\%$

* $\frac{1}{11} = 9\frac{1}{11}\% = 9.09\%$

* $\frac{3}{11} = 27\frac{3}{11}\% = 27.27\%$

* $\frac{8}{11} = 72\frac{8}{11}\% = 72.72\%$

$\left\{ \begin{array}{l} * \frac{2}{11} = 18\frac{2}{11}\% = 18.18\% \\ \frac{3}{11} = 18\frac{3}{11}\% = 18.18\% \\ \frac{4}{11} = 18\frac{4}{11}\% = 18.18\% \\ \frac{5}{11} = 18\frac{5}{11}\% = 18.18\% \\ \frac{6}{11} = 18\frac{6}{11}\% = 18.18\% \\ \frac{7}{11} = 18\frac{7}{11}\% = 18.18\% \\ \frac{8}{11} = 18\frac{8}{11}\% = 18.18\% \\ \frac{9}{11} = 18\frac{9}{11}\% = 18.18\% \\ \frac{10}{11} = 18\frac{10}{11}\% = 18.18\% \end{array} \right.$

$$\# \frac{3}{12} = ? \quad \boxed{9.375\%}$$

$$\frac{1}{16} = 6.25\%$$

$$\frac{1}{32} = 3.125\%$$

$$\frac{3}{32} = 9.375\%$$

(6)

How to Break Fraction $\rightarrow$

Eg^o

$$\frac{13}{7} = ?$$

Ist Method

$$= 1 + \frac{6}{7}$$

$$= 100\% + 85\frac{5}{7}\%$$

$$\left(\frac{1}{7} = 14\frac{2}{3}\% \Rightarrow \frac{6}{7} = 85\frac{5}{7}\%\right)$$

$$\boxed{\frac{13}{7} = 185\frac{5}{7}\%}$$

OR IInd Method

$$\frac{13}{7} = 2 - \frac{1}{7} = 200\% - 14\frac{2}{7}\%$$

$$= 185\% + \frac{5}{7}\%$$

$$= \boxed{185\frac{5}{7}\%}$$

$\checkmark$

Q# $\frac{35}{6} = ?$

$583\frac{1}{3}\%$

$$\Rightarrow 6 - \frac{1}{6} = 600\% - 16\frac{2}{3}\%$$

$$= 600\% - 16\frac{2}{3}\%$$

$$= \left(584 - \frac{2}{3}\right)\%$$

$$= \left(583 + 1 - \frac{2}{3}\right)\%$$

$$= 583\frac{1}{3}\%$$

$\frac{13}{7} = ?$

$$= 1 + \frac{6}{7} \Rightarrow 100\% + 85\frac{5}{7}\% = 185\frac{5}{7}\%$$

$\frac{29}{3} = 9 + \frac{2}{3} = 966\frac{2}{3}\%$

$\frac{71}{12} = 5 + \frac{11}{12} = 591\frac{2}{3}\%$

$\frac{37}{15} = 2 + \frac{7}{15} = 246\frac{2}{3}\%$

$\frac{41}{8} = 5 + \frac{1}{8} = 512\frac{1}{2}\%$

How to Convert Decimal to Fraction - Percentage (9)

दशमलव से भिन्न

eg ① $255.55\% = ?$

$\Rightarrow 200\% + 55.55\%$
 $\downarrow$
 $\boxed{2 + \frac{5}{9} = \frac{23}{9}}$

$\left\{ 5 \times \frac{1}{9} \rightarrow 11.11\% \times 5 \right.$
 $\left. = \frac{5}{9} = 55.55\% \right\}$

② $731.25\% = ?$

$700\% = 7$

Divide in part -

$\rightarrow 700\% + 25\% + 6.25\%$

$= 7 + \frac{1}{4} + \frac{1}{16}$

$= \frac{112 + 4 + 1}{16}$

$= \boxed{\frac{117}{16}}$

Litram Academy

③ $846.66\% = ?$

Divide in Part -

$= 800\% + 40\% + 6.66\%$

$= 8 + \frac{2}{5} + \frac{1}{15}$

$= \frac{120 + 6 + 1}{15}$

$= \boxed{\frac{127}{15}}$

④ $528.56\% = ?$

Divide in Part -

$$= 500\% + 28.56\%$$

$$= 5 + \frac{2}{7} = \left(\frac{37}{7}\right)$$

$$\left\{ 2 \times \frac{1}{7} = 14.28\% \times 2 \right\}$$

Note $\Rightarrow$ ①

$$x\% \text{ of } y = y\% \text{ of } x$$

$\downarrow$ (CHT) $\downarrow$ (CHT)

(multiply)

Proof

$$x \times \frac{1}{100} \times y = y \times \frac{1}{100} \times x$$

$$\frac{xy}{100} = \frac{yx}{100}$$

eg. ① $64.28\% \text{ of } 259$

$$(50\% + 14.28\%) \times 259$$

$$= \left(\frac{1}{2} + \frac{1}{7}\right) \times 259$$

$$= 129.5 + 37$$

$$= \boxed{166.5}$$

② $237\% \text{ of } 640$

$$(200\% + 40\% - 3\%) \times 640$$

$$= \left(2 + 0.4 - \frac{3}{100}\right) \times 640$$

$$= 1280 + 256 - 19.2$$

$$= \boxed{1516.8}$$

$$\# \quad 36\% = \frac{36}{100} = \frac{9}{25}$$

(11)
Increased or decreased value
(बढ़ी या घटी मान)

Original Value
or Initial Value
(मूल संख्या)

$$36\% \uparrow = +\frac{36}{100}$$

Initial		Final
100	:	136
25	:	34

Points — ① x is what % of y

$$\Rightarrow \boxed{\frac{x}{y} \times 100\%}$$

Base

Base value

eg. 63 is what % of 72

$$\begin{aligned} & \frac{63}{72} \times 100\% \\ & = 87.5\% \end{aligned}$$

y is what % of x
Base value

$$\Rightarrow \boxed{\frac{y}{x} \times 100\%}$$

Base

eg. 72 is what % of 63

$$\begin{aligned} & = \frac{72}{63} \times 100\% \\ & = 100\% + 14.28\% \\ & = 114.28\% \end{aligned}$$

② x is what % less than y

$$= \frac{y-x}{y} \times 100\%$$

eg. 96 is what % less than 144.

$$= \frac{144-96}{144} \times 100\%$$

$$= \frac{48}{144} \times 100\%$$

$$= 33.33\%$$

OR

$$\frac{3}{4} \div \frac{2}{3}$$

$$= \frac{1}{3} \times 100 = 33.33$$

③ y is what % more than x

$$\frac{y-x}{x} \times 100\%$$

eg. 144 is what % more than 96.

$$\frac{144-96}{96} \times 100\%$$

$$= \frac{48}{96} \times 100\% = 50\%$$

OR

$$\frac{144}{96} = \frac{3}{2}$$

$$\frac{1}{2} \times 100 = 50\%$$

* $\text{Income} = \text{Expenditure} + \text{Saving}$

आय = खर्चा + बचत

eg.

$$100 \text{ ₹} = 70 \text{ ₹} + 30 \text{ ₹}$$

$\downarrow +20\%$ $\downarrow +10\%$ $\downarrow$ $\downarrow$

$$120 \text{ ₹} \quad 77 \text{ ₹} \quad 43 \text{ ₹}$$

$\uparrow +13$ $\uparrow$

$$\frac{13 \times 100}{30} = \frac{130\%}{3} \uparrow$$

* $\text{Net Income} = \text{Total Income} - \text{Income Tax}$

(शुद्ध आय)

$$70 \text{ ₹} = 100 \text{ ₹} - 30 \text{ ₹}$$

$\downarrow$ $\downarrow$ $\downarrow +10\%$

$$60 \text{ ₹} \quad 100 \text{ ₹} \quad 40 \text{ ₹}$$

net Income
 -10 ₹ कम होगा

Conclusion —

$$\text{Income tax} \propto \frac{1}{\text{Net Income}}$$

Successive % change —

$\Rightarrow$ [क्रमशः %]

eg. 1st Method

Initial value $\leftarrow 100 \text{ ₹}$

$$100 \text{ ₹} \xrightarrow{+10\%} 110 \text{ ₹} \xrightarrow{+20\%} 132 \text{ ₹} \xrightarrow{+30\%} 171.6 \text{ ₹}$$

Final value $\uparrow$

$$\text{net \% change} = 71.6\%$$

Satish Upadhyay

10%↑, 20%↑, 30%↑

$$\left. \begin{aligned} * 110\% &= \frac{110}{100} \\ &= 1.1 \end{aligned} \right\}$$

$$100 \text{ ₹} = \left[(100 \text{ ₹} \times 1.1) \times 1.2 \right] \times 1.3$$

$$= \frac{66 \times 13}{5} = \frac{858}{5} = 171.6$$

$$\begin{aligned} 10\% \uparrow &= \frac{1}{10} \\ 20\% \uparrow &= \frac{1}{5} = \frac{2}{10} \\ 30\% \uparrow &= \frac{3}{10} \end{aligned}$$

<u>Initial value</u>	<u>final value</u>
10	11
10	12
10	13

$$1000 \div 5 = 1716$$

but change!

divide 10

change

divide 10

100 ₹ : 171.6 ₹

final

+71.6%

formula approach →

$x\%$, $y\%$. successively change.

##*
net% change = $\left(x + y + \frac{xy}{100}\right)\%$

Initial $\boxed{100}$ $\xrightarrow{x\% \uparrow}$ $(100+x)$ $\xrightarrow{y\%}$ $(100+x) \times (100+y)\%$

$$= \frac{(100+x)(100+y)}{100}$$

$$= \frac{100,00 + 100y + 100x + xy}{100}$$

Net % change $\rightarrow \times \left(\frac{x+y+\frac{xy}{100}}{100} \right) \%$

Final $\boxed{100 + y + x + \frac{xy}{100}}$

Proved (होती सिद्ध)

Decrement $\rightarrow$

Income:

1st $100 \text{ ₹} \xrightarrow[-10\%]{-10 \text{ ₹}} 90 \text{ ₹} \xrightarrow[-20\%]{-18 \text{ ₹}} 72 \text{ ₹} \xrightarrow[-30\%]{21.6 \text{ ₹}} 50.4 \text{ ₹}$

49.6 ₹ खर्च

बचत $\downarrow$

2nd $100 \times \frac{9}{10} \times \frac{4}{5} \times \frac{7}{5} = 50.4 \text{ ₹} = \frac{50.4}{100} = \frac{252}{500}$

Saving

3rd

10	:	9
5	:	4
10	:	7
500	:	252

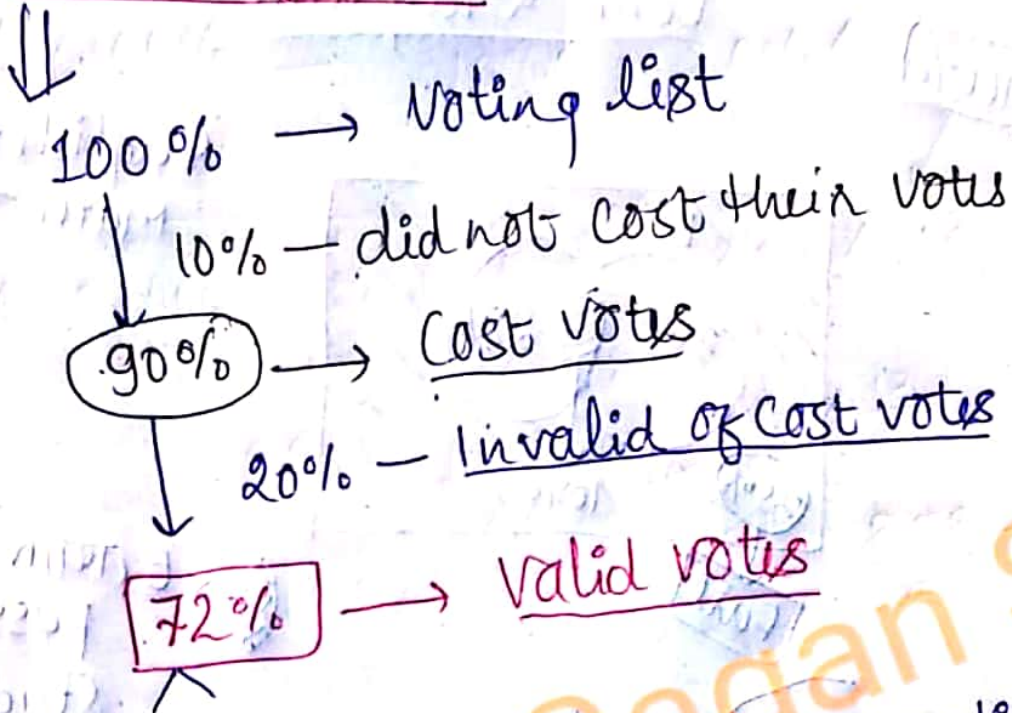
Income $\leftarrow$ $\rightarrow$ Saving.

Satnam Upadhyay

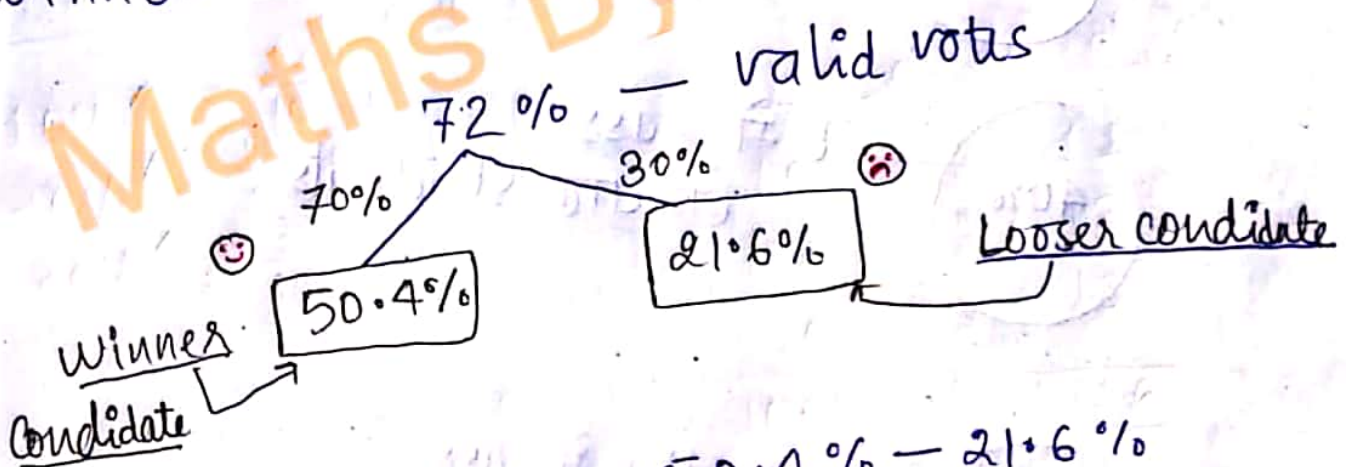
Election (चुनाव)

(16)

Learn Structure



Winner Candidate secured → 70% of valid votes



$$\begin{aligned} \% \text{ winner} &= 50.4\% - 21.6\% \\ &= 28.8\% \text{ जीता } \checkmark \end{aligned}$$



Venn Diagram.

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Student: 100%

English - E
Math - M



35% → M में Pass होने वाले Students

40% → E में Pass होने वाले Students

30% → केवल English में Pass

25% → केवल Math में Pass Student

$$\begin{array}{l} 30\% \text{ (केवल E)} + 10\% \text{ (E+M)} + 25\% \text{ (M)} = 65\% \rightarrow \text{Pass} \\ 35\% \rightarrow \text{Fail} \end{array}$$

LCM - HCF

①

multiple $\rightarrow$ (गुणज)

15 $\rightarrow$ 15, 30, 45, 60

factors (गुणनखंड)

15 $\rightarrow$ 1, 3, 5, 15

L.C.M - Least common multiple.
(लघुत्तम समापवर्त्य)

$\therefore$ eg -

5 $\rightarrow$ 5, 10, 15, 20, 25, 30, 35, 40

6 $\rightarrow$ 6, 12, 18, 24, 30, 36, 42, 48

10 $\rightarrow$ 10, 20, 30, 40, 50, ...

{ 5, 6, 10 से कच्चे वाली सबसे छोटी-से छोटी संख्या ही LCM होती है। }

$$\text{Here } \boxed{\text{LCM}(5, 6, 10) = 30}$$

Note - LCM of $[a, b, c]$ is the smallest number which is completely divisible of all $[a, b, c]$.

How to find LCM

Method - 1 - Division method (भाज विधि)

2	5, 6, 10
3	5, 3, 5
5	5, 1, 1
	1, 1, 1

$$2 \times 3 \times 5 = 30$$

Byadhyape

Method-2 - prime factorization method,

(2)

1. $24 \rightarrow 2^3 \times 3$

$30 \rightarrow 2 \times 3 \times 5$

$40 \rightarrow 2^3 \times 5$

Height power;

$\text{LCM} \rightarrow 2^3 \times 3 \times 5$
 $= 120 \checkmark$

2. $5 - 5^1$
 $6 - 2 \times 3^1$
 $10 - 2 \times 5^1$

Height power $\rightarrow \text{LCM} = 2 \times 3 \times 5$
 $= 30 \checkmark$

Method-3

1. $24 - 2 \times 3$

$30 - 3 \times 10 - 5 \times 2$

$40 - 10 \times 4$

let $\text{LCM} \rightarrow 40 - 8 \times 5$

$40 \times 3 = 120$

~~Search~~ (Search hidden number
जो 40 में ना हो)

2. 5
 $6 - 2 \times 3 \checkmark$
 10

let $\text{LCM} - 10$

Search $\rightarrow 10 \times 3 = 30 \checkmark$

HCF (Height Common Factor)

OR GCD (Greatest Common Divisor)

eg- 32 - 1, 2, 4, 8, 16, 32

40 - 1, 2, 4, 5, 8, 10, 20, 40

Common factor $\rightarrow$ (1, 2, 4, 8)

[Largest Common]
(Height Common factor)
होता है -

Note $\rightarrow$ HCF of given number is the largest number which divides all the given number's completely.

$\Rightarrow$ 32 - $(8) \times (4)$
40 - $(8) \times (5)$
HCF = 8
co-prime (होगे बचे हुए)

$\Rightarrow$ यदि दो संख्याओं का HCF - "H" है तो वे संख्याएँ -

Hx & Hy

Always co-prime होंगे $(x \neq y)$

Super
Solving

How to find HCF

1. Prime factorization method

$$72 \rightarrow 2^3 \times 3^2$$

$$76 \rightarrow 2^2 \times 19$$

$$32 \rightarrow 2^5$$

[जो सभी में common height factor हो] power

$$\text{HCF} \rightarrow 2^2 = 4 \text{ A}$$

2. Long Division method →

eg - ① 945, 693

होटे no से,

$$\begin{array}{r} 693 \overline{) 945} \quad 1 \\ \underline{693} \\ 252 \end{array} \quad \begin{array}{r} 252 \overline{) 693} \quad 2 \\ \underline{504} \\ 189 \end{array} \quad \begin{array}{r} 189 \overline{) 252} \quad 1 \\ \underline{189} \\ 63 \end{array} \quad \begin{array}{r} 63 \overline{) 189} \quad 3 \\ \underline{189} \\ 0 \end{array}$$

{final divisor ही HCF है}

② 147, 651, 546

$$\begin{array}{r} 147 \overline{) 546} \quad 3 \\ \underline{441} \\ 105 \end{array} \quad \begin{array}{r} 105 \overline{) 147} \quad 1 \\ \underline{105} \\ 42 \end{array} \quad \begin{array}{r} 42 \overline{) 105} \quad 2 \\ \underline{84} \\ 21 \end{array} \quad \begin{array}{r} 21 \overline{) 42} \quad 2 \\ \underline{42} \\ 0 \end{array}$$

21 is HCF

of 147, 546

$$\begin{array}{r} 21 \overline{) 651} \quad 31 \\ \underline{63} \\ 21 \end{array} \quad \begin{array}{r} 21 \overline{) 21} \quad 1 \\ \underline{21} \\ 0 \end{array}$$

Here, 21 is HCF of 147, 651, 546

* अगर 21 divide नही होता तो 21 का factor निकालें और divide करें नही होता तो 1 HCF होगा

③ difference method,

— 96 HCF of two number's $\geq H$
co-prime $\frac{a}{b}$.

संख्याये -

Hx Hy

difference $\Rightarrow Hx(x-y)$

$x-y=1$, $x-y > 1$

HCF is difference of two numbers

HCF will be factor of difference of two numbers

Note $\rightarrow$ [HCF cannot be $\downarrow$ than difference of two numbers $\uparrow$ greater]

eg $\rightarrow$

40, 50
 10×4 10×5

HCF $\rightarrow 10$ $\leftarrow$ co-prime [difference - 1]

Then HCF $\rightarrow [50-40 = 10]$ $\odot$

② 306, 391 $\rightarrow 17 \times 23$
 12×18

diff $\rightarrow 85$ $\times$
 $\leftarrow$ Factor

① $\rightarrow$ 17 $\rightarrow$ ⑤ $\times$

HCF = 17

③ 42, 54, 150
 Smallest diff. $\rightarrow 12 \times$
 Factor
 0 6 12
 HCF = 6

④ 323, 456, 703
 Smallest difference = 133 $\times$
 Factor
 19 $\times$ 7
 19 divide an $\rightarrow$ HCF $\rightarrow$ 19

⑤ 1785, 1995, $\frac{3381}{21}$ — divide $\rightarrow$ 21
 Smallest difference = 210 $\times$
 Factor
 HCF $\rightarrow$ 21

$\Rightarrow$ If HCF of two number = H

संख्याये — Hx Hy $\Rightarrow$ $LCM = Hxy$ *
 co-prime.

Sum = $H(x+y)$
 Diffe = $H(x-y)$

Common

product of two numbers

Group $\rightarrow$ $LCM \times HCF = \text{पहली संख्या} \times \text{दूसरी सं.}$
 $Hxy \times H = Hx \times Hy$
 $H^2xy = H^2xy$

Concept,

eg → 1

75, 40

difference = 35
 35 → factor
 7 → factor
 5

Remainder → Same

diff or factors. $\nexists$ divide ane $\nexists$

#

Note → If two numbers are divided by their difference individually, then remainder will be same in each case.

2.

96, 108, 140
 12, 32, 44

diff. $\rightarrow$ HCF = 4

remainders will be same

#

<https://t.me/yourmathbook>Unit digit (इकाई का अंक)

digit - 1, 2, 3, 4, 5, 6, 7, 8, 9, 0

①

unit digit

**

0, 1, 5, 6 → 0, 1, 5, 6 Same (multiple)

eg

$$1. (78121)^{878} = 78121 \times 78121 \dots (878 \text{ times})$$

{ unit digit = 1 }

$$2. (890)^{879} = ?$$

{ unit digit = 0 }

4, 9 ⇒ unit digitdetail

$$1. \begin{aligned} 4^1 &\rightarrow 4 \\ 4^2 &\rightarrow 6 \\ 4^3 &\rightarrow 4 \\ 4^4 &\rightarrow 6 \\ 4^5 &\rightarrow 4 \end{aligned}$$

Pattern repeated

4^{odd} → 4 unit digit
4^{even} → 6 unit digit

$$\text{eg } \left\{ \begin{aligned} (284)^{83} &\rightarrow \text{unit digit} \rightarrow 4 \\ (284)^{824} &\rightarrow \text{unit digit} \rightarrow 6 \end{aligned} \right\}$$

$$2. \begin{aligned} 9^1 &= 9 \\ 9^2 &= 1 \end{aligned} \} \rightarrow \text{pattern repeated}$$

$$\begin{aligned} 9^3 &= 9 \\ 9^4 &= 1 \\ 9^5 &= 9 \\ 9^6 &= 1 \end{aligned}$$

9^{odd} → 9 unit digit
9^{even} → 1 unit digit

$$\text{eg } \left\{ \begin{aligned} (789)^{63} &\rightarrow 9 \text{ unit digit} \\ (789)^{62} &\rightarrow 1 \text{ unit digit} \end{aligned} \right\}$$

2, 3, 7, 8 →

(2)

(2) →

$2^1 \rightarrow 2$ (unit digit 2, Remainder 1)
 $2^2 \rightarrow 4$ (unit digit 4, Remainder 2)
 $2^3 \rightarrow 8$ (unit digit 8, Remainder 3)
 $2^4 \rightarrow 6$ (unit digit 6, Remainder 0)

pattern repeated every 4 terms.

2 → Cyclicity = 4

$2^5 \rightarrow 2$

$2^6 \rightarrow 4$

$2^7 \rightarrow 8$

$2^8 \rightarrow 6$

$2^9 \rightarrow 2$

eg: $1. (132)^{457} = 2^{\overline{457}}$

$\frac{457}{4} = \text{remainder} = 1$

unit digit = 2

2. $(752)^{894} = ?$

$\frac{894}{4} = 2 \text{ remainder}$

unit digit = 4

(3)

$3^1 = 3$ (unit digit 3, Remainder 1)
 $3^2 = 9$ (unit digit 9, Remainder 2)
 $3^3 = 7$ (unit digit 7, Remainder 3)
 $3^4 = 1$ (unit digit 1, Remainder 0)

pattern repeated every 4 terms.

3 Cyclicity = 4

$3^5 = 3$

$3^6 = 9$

$3^7 = 7$

$3^8 = 1$

$3^9 = 3$

eg: $(863)^{851} \rightarrow$
 सिर्फ 2 अंक को ही Divide करके होता है
 $\frac{851}{4} = 2 \text{ remainder}$
 4th divisibility rule है
 = 3 Remainder
 unit digit → 7

⑦

$7^1 = 7$ ← Unit digit
 $7^2 = 9$ ← Remainder = ①
 $7^3 = 3$ ← Remainder = ②
 $7^4 = 1$ ← Remainder = ③
 $7^5 = 7$ ← Remainder = ④
 $7^6 = 9$
 $7^7 = 3$
 $7^8 = 1$
 $7^9 = 7$

pattern repeat every 4 terms

7 - Cyclicity = 4

③

eg. $(817)^{288}$

$\frac{288}{4} = 0$

Unit digit = 1

⑧

$8^1 = 8$ ← Unit digit
 $8^2 = 4$ ← Remainder = ①
 $8^3 = 2$ ← Remainder = ②
 $8^4 = 6$ ← Remainder = ③
 $8^5 = 8$ ← Remainder = ④
 $8^6 = 4$
 $8^7 = 2$
 $8^8 = 6$
 $8^9 = 8$

pattern repeat

8 → Cyclicity = 4

eg. $(9818)^{874}$

$\frac{874}{4} = 2 \text{ Remainder}$

Unit digit → ④

Divisibility Rule, भाज्यता के नियम

④

1- ^(भाज्य) is not divisible by any number except 1.
but 1 is an universal factor.

2 → ^(अंतिम संख्या) last digit should be even or last digit should
- be divisible by 2.

4 → last 2 digit should be divisible by 4.

8 → Last 3 digit should be divisible by 8.

16 → last 4 digit should be divisible by 16.

32 → last 5 digit should be divisible by 32.

64 → Last 6 digit should be divisible by 64.

→ If any Number cuts "a" and "b" then number
will be cut "a x b" but a & b are
co-prime.

eg- Remainder.

315 ÷ 7	=	0
315 ÷ 5	=	0
315 ÷ 7 x 5	=	0
315 ÷ 35	=	0

3, 9 → divisibility rule →

③ → Sum of digits should be divisible by 3.

⑨ → Sum of digit should be divisible by 9.

eg: Express no $\rightarrow$
<https://t.me/yoursmathtool>

$\rightarrow 1672 \rightarrow 1 \times 1000 + 6 \times 100 + 7 \times 10 + 2$

eg $\rightarrow$ चार अंकों की संख्या $\rightarrow 1000w + 100z + 10y + x$
 (wzyx) $\rightarrow \frac{(999w + 99z + 9y) + (w + z + y + x)}{9}$

$= \boxed{\frac{w + z + y + x}{9}}$

Sum of digit
 ① $\rightarrow 50769 \rightarrow \frac{27}{9} = 3$

② $- 7105104 = 18$ ✓ यह संख्या 3, 9 दोनों से भाज्य है।

③ $- 356468792 = \frac{50}{9} \times \{ \text{संख्या न तो 9 से भाज्य है और ना ही उसे -} \}$
 Remainder (5) ✓

or $3564 \cancel{6} 8 7 9 2 \rightarrow 9$ हटा दो
 $6+8 = 14 - 9 = 5$
 or $1+4 = 5$

digit sum should be 9 or 0

➡ Digital sum of a number is the remainder when it is divided 9.

⇒ ⑤ → divisibility rule

5 → Last digit should be 0 or 5.

eg. 130 ✓ 1665 ✓

Remainder → 0 ← 98 divisible by 5

②⑤ → Last two digit should be divisible by 25

eg. 6125

①②⑤ - Last three digit should be divisible by 125.

eg. 6125

⑥ → 6 → 2 × 3 → यदि कोई संख्या "2" से भाज्य है और "3" से भी तो वह संख्या 6 से भी divisible होगी।

###

{ 7, 11, 13 : divisibility rule → }

7 →

eg. 005922 = 917 ✓ [7 divisible होगी]

006489 → 483 ✓

045724 → 679 ✓

380287 → 133

006891647

Sum → 891 - 653 = 238 ✓

Capgemini Padhyani

13 →

eg-

① ⇒ 0 | 1200644

sum 655 - difference 200 = 455 ✓ 7, 13

7 ✓ 11 ✗ 13 ✓

7 × 13 = 91 — सभी divisible होंगी.

② ⇒ 0 | 2478375

sum → 387 difference →

91
7 11 ✗ 13 ✓

{ दूसरी संख्या 7, 13 से काटेगी और 7 × 13 से भी काटेगी }

eg-

00 | 026389

sum → 390 - 26 = 364

7 ✓ 11 ✗ 13 ✓

ये संख्या 7, 13 से काटेगी

97824519

sum - 824 - 616 → 208

7 ✗ 11 ✗ 13 ✓

किन्तु 13 से काटेगी दूसरी संख्या

093962

difference → 869

7 ✗ 11 ✗ 13 ✗

किन्तु 11 से काटेगी दूसरी संख्या

040676372

sum 676 - 412 = 264

7 ✗ 11 ✗ 13 ✗

11 से काटेगी वही संख्या

111 333 222

sum - 333 - 333 = 0

7 ✓ 11 ✓ 13 ✓

{ यह संख्या 7, 11, 13 सभी से काटेगी }

Sham ③

https://t.me/coursesmathbook

⇒ 92

कोई संख्या

abab में है। इसे किससे divide होगी -

- a) 101
- b) 107
- c) 109
- d) 111

$1000a + 100b + 10a + b$

$1010a + 101b$

$101(10a + b)$

(101 के multiple में होगी) तो 101 से divide भी होगी

⇒ 637637, 895895 is always divisible by

- a) 7 only
- b) 11 only
- c) 13 only

~~d) 1001~~

→ $\underline{637637}$

difference ⇒ 0

$\underline{895895}$

difference

$\left. \begin{matrix} 7 \\ 11 \\ 13 \end{matrix} \right\}$

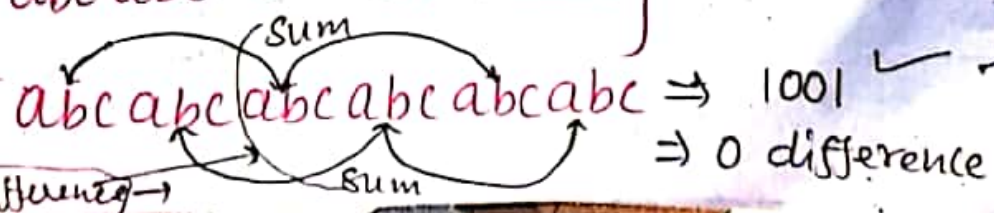
$abcabc \Rightarrow abc \times \begin{matrix} 1001 \\ \swarrow \quad \searrow \\ 7 \quad 11 \quad 13 \end{matrix}$

Any number

{ $ababab = ab \times 10101$ } ***

{ $abcabcabc = abc \times 1001001$ }

{ $abcabc = abc \times 1001$ }



Even time repeat होगी तो 7, 11, 13

eg $\overbrace{867} \overbrace{867} \overbrace{867} \dots \dots \dots 867$ time पर
 $\underbrace{\hspace{10em}}_{13}$
 remainder आयेगी?

$\Rightarrow \overbrace{867867867 \dots \dots \dots 866 \text{ times } 867}$
 $\underbrace{\hspace{10em}}_{13}$

$\Rightarrow \frac{867}{13} \rightarrow 66$
 Remainder = 9

Special 11 $\Rightarrow$

eg $\rightarrow \left\{ \begin{array}{l} \overbrace{7945938}^{\text{Sum}} \rightarrow 28 \\ \text{Sum} = 2+8 = 10 \end{array} \right.$ $\left(28 - 17 = 11 \text{ K } \right)$ हो तो संख्या 11 से भाज्य होगी।
 $\left\{ \begin{array}{l} \overbrace{166452}^{\text{Sum}} \rightarrow 24 \\ \text{Sum} = 2+4 = 6 \end{array} \right.$ $12 - 12 = 0$ यह संख्या 11 से divisible होगी

$\Rightarrow$ कोई संख्या अगर -
 $\left\{ \begin{array}{l} 88 - 11 \times 8 \\ 99 - 11 \times 9 \end{array} \right.$
 न 2 से divisible है तो 8 और 9 से भाज्य होगी
 11, 8 से भाज्य संख्या अगर 88 से divisible है तो 11 से और 9 से भाज्य संख्या

Remainder theorem

Case-I

Divisor
(भाजक)

$$\begin{array}{r}
 12 \overline{) 55} \\
 \underline{48} \\
 7
 \end{array}$$

divident भाज्य

quotient (भागफल)
[quotient]

Remainder (शेषफल)

$$\{ 55 = 12 \times 4 + 7 \}$$

#

$$\text{divident} = \text{divisor} \times \text{quotient} + \text{Remainder}$$

Case-II

eg

$$\left\{ \frac{13}{18} \rightarrow \text{[शेषफल]} \right\}$$

13

$$18 \overline{) 13} (0$$

$$\begin{array}{r}
 0 \\
 \underline{18} \\
 13
 \end{array}$$

शेषफल

Case-III

$$\left\{ \frac{11 \times 18 = 198}{18} = 0 \leftarrow \text{Remainder} \right\}$$

Case-IV

$$\left\{ \frac{18}{18} = 1 \right. \\
 \left. \text{[Remainder} = 0] \right\}$$

eg

$$\frac{40}{9} = 4$$

$$\frac{42}{9} = 6$$

$$\frac{43}{9} = 7$$

किसी संख्या d से divide करते होते तो
Remainder d से छोटा (small) है
होगा।

24. $(3^{25} + 3^{26} + 3^{27} + 3^{28})$ विभाज्य है -

- a) ~~11~~ b) ~~16~~ c) ~~25~~ d) 30

हल:-

$$3^{25} (1 + 3 + 3^2 + 3^3)$$

$$= 3^{25} \times 4 \times 2 \times 5$$

$\xrightarrow{3} \quad \times \quad \xrightarrow{10} = 30 \text{ से भी काटेगी} -$

Detail

$$\frac{130}{19} = \text{Rem} - 16$$

$$\frac{80 + 50}{19} \rightarrow 16 \leftarrow \text{Remainder}$$

$$\frac{100 + 30}{19} \rightarrow 16$$

$$\frac{70 + 42 + 133}{18} \rightarrow 11$$

Remainder

$$= \frac{29}{18} \rightarrow 11 \leftarrow \text{Remainder}$$

Actual Remainder = ? (कड़ है)

OR. $\frac{245}{18} = 11 \text{ Remainder}$

Concept
⇒

$$\frac{65}{25} = ? \text{ Remainder} = ?$$

$$25 \overline{) 65} \begin{array}{r} 2 \\ 50 \\ \hline 15 \end{array}$$

Remainder 15 ← दोषफल

OR $\frac{65}{25} \div \frac{13}{5} = 3$

* NOTE → जिससे Divide करोगे उसीसे Remainder से गुणा कराला है

Final Rem = $5 \times 3 = 15$

$$\frac{N}{7} \rightarrow \textcircled{3} \times 5$$

$$\frac{SN}{7} = \frac{15}{7} = \frac{14+1}{7} = \textcircled{1} \text{ Remainder}$$

Concept, $\rightarrow [a \neq b, n \rightarrow \forall n \text{ are natural number}]$

② $a^n + b^n$ [$n \rightarrow \text{even}$]

factors, $(a+b)$ $\times$
 $(a-b)$ $\times$

④ $a^n - b^n$ [n-even]

factors - $(a+b)$ ✓
 $(a-b)$ ✓

$k \rightarrow \text{odd}$

$$\rightarrow (64^k + 25^k)$$

$64 + 25 \rightarrow 89$ इसका एक गुणनखण्ड होगा (factor)

$$2 - (31^{20} - 1024) \rightarrow$$

is not divisible by 1 -

a) 137
c) 9

⑥ 107
④ 32

odd-

[Signature]

{ n है odd संख्या even संख्या से जोड़ा होगा।}

Concept of Negative remainder

$$\begin{array}{r} 13 \overline{)60} \text{ (4} \\ \underline{52} \\ +8 \end{array}$$

$$\begin{array}{r} 13 \overline{)60} \text{ (5} \\ \underline{65} \\ -5 \end{array}$$

$\frac{60}{13} = \text{Remainder}$

$+8 \leftarrow (-5 + 13)$

-5

eg- $\frac{130}{17}$

$+11$

-6

$\text{sum} = 17$

$\frac{84}{11}$

$+7$

-4

$\frac{61}{8}$

$+5 \leftarrow \{3 + 8\}$

-3

$\frac{(-)70}{19}$

$\xrightarrow{\text{Rem}}$

$-13 \leftarrow \{+19 - 13 = 6\}$

$+6$

$\Rightarrow$

$(+2) \times (+4) \Rightarrow +8 \text{ remainder}$

$\frac{79 \times 103}{11}$

OR

$(+2) \times (-7) = -14 + 11 = -3 \Rightarrow +8$

$\frac{79 \times 103}{11}$

Fermat's Theorem,

$$\rightarrow \frac{N^{P-1}}{P} \Rightarrow \boxed{\text{Remainder} = 1}$$

$\Rightarrow$ if P is a prime number.
(अभाज्य संख्या)

$\Rightarrow N$ & P should be co-prime.

$$\boxed{\text{HCF}(N, P) = 1}$$

Find Remainder,

eg 1. $\frac{12^{16}}{17} = 1 \text{ Remainder.}$

$12, 17 \rightarrow \text{co-prime.}$

$17 - \text{prime number}$

2. $\frac{38^{102}}{103} = 1 \text{ Remainder.}$
 $\swarrow$
 co-prime

Wilson's Theorem,

$$\boxed{\frac{(P-1)!}{P} = \text{Remainder} = -1 \text{ OR } P-1}$$

$P - \text{prime number}$

Theorem,

$$\Rightarrow \left\{ \begin{array}{l} \frac{(ax+k)^n}{a} \Rightarrow \text{Remainder} = k^n \rightarrow \frac{(ax+1)^n}{a} = 1^n = 1 \\ \frac{(ax-k)^n}{a} \Rightarrow \text{Remainder} = (-k)^n \rightarrow \frac{(ax-1)^n}{a} = (-1)^n \end{array} \right.$$

egⁿ 1. $\frac{6^{153}}{5} = \text{Remainder} = ?$

$$= \frac{(5+1)^{153}}{5} = \frac{(5+1) \times (5+1) \times \dots \times (5+1)}{5} \text{--- 153 times} = \text{--- 1}$$

Remainder = 1

2. $\frac{(77)^{67}}{39} = \frac{(2 \times 39 - 1)^{67}}{39} = \frac{(2 \times 39 - 1) \times (2 \times 39 - 1) \times \dots \times (2 \times 39 - 1)}{39} \text{--- 67 times}$

$$= (-1)^{67} = -1 \text{ or } 39$$

Remainder.

Concept,

Rem-

$$\begin{array}{l} \frac{4^1}{6} = 4 \\ \frac{4^2}{6} = 4 \\ \frac{4^3}{6} = 4 \end{array}$$

 $\frac{4^n}{6} = \text{Rem} 4$

Shyam Pradyumn

Euler's theorem, (Totient concept)*

$$\boxed{\frac{a^{\phi(N)}}{N} = \text{rem} \rightarrow 1}$$

→ Any Natural Number.
 ** $a \nmid N \rightarrow$ should be co-prime.

$\{\phi(N) = \text{Totient function of } N\}$

8 13 8 9
 $\sqrt{40}, \sqrt{65}, \sqrt{72}$

eg $N = 16$ Composite Number =

prime number - 17, 19, 23

How to find totient function →

$N = 16 = 2^4$

$\phi(16) = 16 \times (1 - \frac{1}{2}) = 8$

$N = 30 = 2^1 \times 3^1 \times 5^1$

$\phi(30) = 30 \times (1 - \frac{1}{2}) \times (1 - \frac{1}{3}) \times (1 - \frac{1}{5})$
 $= 30 \times \frac{1}{2} \times \frac{2}{3} \times \frac{4}{5} = 8$

$N = 100 = 2^2 \times 5^2$

$\phi(100) = 100 \times (1 - \frac{1}{2}) \times (1 - \frac{1}{5})$ Totient function
 $= 100 \times \frac{1}{2} \times \frac{4}{5} = 40$

eg $\frac{37^{25}}{72} \Rightarrow \frac{37^{24} \times 37}{72}$

→ remainder.
 $1 \times 37 = 37$
 $\boxed{\phi(72) = 24}$

$$\# \left\{ \phi(p) = p-1 \right\} \rightarrow \text{total of prime number} \quad (13)$$

$\uparrow$
prime no

eg $\downarrow$ prime number.
 $19 \equiv 19^1$

$$\phi(19) = 19 \times \left(1 - \frac{1}{19}\right) = (18)$$

$$\Rightarrow \frac{a^{\phi(N)}}{N} \equiv \text{Rem}(1)$$

$$\boxed{\frac{a^{p-1}}{p} \equiv \text{Rem}(1)}$$

Fermat's theorem

a & p should be
co-prime.

CHINESE REMAINDER THEOREM,

(12)

$\Rightarrow \frac{N}{abc}$ Where a, b, c are co-prime numbers

Case-I If N is divided by a, b, c respectively then they leave same remainder R in each case then remainder $\boxed{\frac{N}{abc} = R}$

Case-II If N is divided by a, b, c respectively and leaves remainder r_1, r_2, r_3 but $(a-r_1 = b-r_2 = c-r_3 = k)$ then remainder $\boxed{\frac{N}{abc} = \text{lcm of } a, b, c - k}$
 $\boxed{= abc - k}$

Case-III - If N is divided by a, b, c and leaves remainder r_1, r_2, r_3 then remainder $\frac{N}{abc} =$

$$\boxed{R = ax + r_1 = by + r_2 = cz + r_3}$$

* Number of zeroes

(19)

Q- 1. $75 \times 16 = 1200$

$2 \times 5 = 10$ ^{Zeroes}

2 Zeroes.

Explain, $3 \times 5 \times 5 \times 2 \times 2 \times 2 \times 2$

pair $\rightarrow 2$

इसलिए - 2 zeroes होंगे।

2. $2^9 \times 3^{11} \times 2^7 \times 5^7 \times 7^6 \times 8^9 \times 9^6$

Find no. of zeroes - ?

$\Rightarrow 2^9 \times 3^{11} \times 2^7 \times 5^7 \times 7^6 \times 8^9 \times 9^6$

7 pair

no. of zeroes = 7

(2 & 5)

जो सबसे कम भा है $\uparrow$

$75 \times 256 \times 51 \times 2250 \times 79 \times 88$

no. of zeroes = ?

$\Rightarrow 3 \times 2 \times 3 \times 5^2 \times 2^8 \times 51 \times 5^2 \times 9 \times 10 \times 79 \times 88$

no. of 5 = 5

no. of 2 = more than 5

no. of zeroes = जो सबसे कम भा है -

= 5 zeroes

Spair

⇒ eg- $1 \times 2 \times 3 \times 4 \times \dots \times 1000 = 1000!$ (20)
no. of zeroes?

⇒ How to find no. of 5

$$\frac{1000}{5} = \frac{200}{5} = \frac{40}{5} = \frac{8}{5} = 1$$

↑
(जितना) होना है
आम 5 से
तब तक
Divide होगा

= 249 OR 5 आया है
लेकिन 2 से 249 से
अधिक का आया है
इसलिए 5, 2 के केवल 249
pair होंगे.
no. of zeroes = 249

eg- $1 \times 3 \times 5 \times 7 \times 9 \times 11 \times \dots \times 125 \times 128$

$$\downarrow$$

$$2^7$$

कुछ ज्यादा का - 5 आ रहा है?
but, लेकिन pair केवल 7 ही होंगे
∴ no. of zeroes = 7 ही हैं

eg- $4 \times 8 \times 12 \times 16 \times \dots \times 560$ no. of zeroes = ?

$$\Rightarrow (4 \times 1) \times (4 \times 2) \times (4 \times 3) \times \dots \times 4 \times 140$$

$$= 4^{140} [1 \times 2 \times 3 \times 4 \times \dots \times 140]$$

$$\frac{140}{5} = \frac{28}{5} = \frac{5}{5} = 1$$

$$\text{no. of } 5 = 34$$

pair = 34
Zeroes = 34 Ans.

Ap. $\rightarrow$

$T_n = n^{\text{th}} \text{ term}$

$$T_n = [a + (n-1)d]$$
$$[\text{Last term} = l = a + (n-1)d]$$

$$\frac{l-a}{d} \geq n-1$$

$$n = \frac{l-a}{d} + 1$$

$$\begin{cases} l \rightarrow \text{last term} \\ a \rightarrow \text{first term} \\ d \rightarrow \text{common difference} \end{cases}$$

米

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

OR -

$$S_n = \frac{n}{2} [a + l]$$

$$\left\{ \begin{array}{l} a - \text{First term} \\ l - \text{last term} \end{array} \right\}$$

$\sum n =$ Sum of first natural no $\rightarrow n$

$$S_n = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$\sum n^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\left\{ \begin{array}{l} \sum n^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} \\ \sum n^3 = 1^3 + 2^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2 \text{ or } \rightarrow \left[\frac{n(n+1)}{2} \right]^2 \end{array} \right.$$

⇒ Sum of first n even numbers = $n(n+1)$

(29)

$$\rightarrow 2 \times 1 + 2 \times 2 + 2 \times 3 + \dots + 2 \times n$$

$$= 2 [1 + 2 + 3 + \dots + n]$$

$$= 2 \times \frac{n(n+1)}{2} = \boxed{n(n+1)}$$

⇒ Sum of first n odd numbers = n^2

$$1 + 3 + 5 + \dots + (2n-1) = \frac{n}{2} [1 + 2n-1]$$

$$= \boxed{n^2}$$

G.P. $a, ar, ar^2, ar^3, \dots$

$\left\{ \begin{array}{l} a - \text{First term} \\ r = \frac{T_{n+1}}{T_n} \end{array} \right\}$

$\left\{ \begin{array}{l} \text{Common} \\ \text{ratio} \end{array} \right\}$

$$n^{\text{th}} \text{ term} = T_n = ar^{n-1}$$

$$\text{Sum of first } n \text{ terms} = \frac{a(r^n - 1)}{r - 1}$$

$$S_{\infty} = \frac{a}{1-r} \text{ if } r < 1$$

[Sum of Infinite terms]

H.P - [harmonic progression]
(हरात्मक श्रेणी)

$$\Rightarrow \text{A.P.} \rightarrow a, a+d, a+2d, a+3d, \dots, a+(n-1)d$$

समान्ता श्रेणी

$$\text{H.P.} \rightarrow \frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \dots, \frac{1}{a+(n-1)d}$$

हरात्मक श्रेणी

$$\text{H.P.} = \frac{1}{\text{A.P.}}$$

If there are two positive number a & b

$$a \text{ \& \> } b \text{ at A.M. (समान्ता माध्य)} = \frac{a+b}{2} = 6.5$$

$$a \text{ \& \> } b \text{ at G.M. (उज्जोता माध्य)} = \sqrt{ab} = 6$$

$$a \text{ \& \> } b \text{ at H.M. (हरात्मक माध्य)} = \frac{2ab}{a+b} = 5.53$$

Result

$$\boxed{AM \geq GM \geq HM}$$

$$\frac{a+b}{2} \geq \sqrt{ab} \geq \frac{2ab}{a+b}$$

$$\Rightarrow \boxed{AM \times HM = GM^2} \Rightarrow \frac{a+b}{2} \times \frac{2ab}{a+b} = ab = (\sqrt{ab})^2$$

Setam

Classification of numbers →

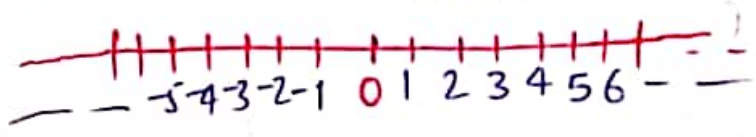
Numbers

Real Numbers
(वास्तविक संख्या)

Imaginary Numbers
(काल्पनिक संख्याएँ)

⇒ Square root of a negative no. →

eg. (Numbers represented on Numbers line)



eg. $i = \sqrt{-1}$

$\sqrt{-7}, \sqrt{-19}, \sqrt{-26}$
↓
 $\sqrt{7} \times \sqrt{-1}$
 $\sqrt{7}i$

Complex numbers ⇒
(समिश्र संख्याएँ)

Real + Imaginary
 $a + ib$

Real Numbers →

Integers -
(पूर्णांक)

Decimal
(दशमलव)

Terminating Decimal
(शांत दशमलव)
eg. 0.86, 0.346, 0.125

Non-terminating Decimal
(अशांत दशमलव)

Repeating Decimal

eg. $0.\bar{3} = .33333 \dots$
 $0.\bar{56} = .565656 \dots$

non-repeating decimal
eg. 4587362018

eg. $\begin{cases} 0.6666 \dots \\ 0.45454545 \dots \\ 0.645645645645 \dots \end{cases}$

Repeating Decimal

$$\begin{aligned} 0.1500000 & \dots = \frac{15}{100} = \frac{3}{20} \\ 0.33333 & \dots = \frac{3}{10} = \frac{3}{10} \\ 0.873873873 & \dots = \frac{873}{999} \end{aligned}$$

Also called Rational numbers $\rightarrow$ write $\left(\frac{p}{q}\right)$ Form
(परिमेय संख्याएँ)

में लिखा जा सकता है। - परिमेय संख्या को $\left(\frac{p}{q}\right)$

$p, q \in \text{Integers } q \neq 0$

eg. $\frac{1}{3}, \frac{3}{4}, \frac{873}{872}$ Rational No-

Non repeating - Nonterminating Decimal $\rightarrow$

eg. $0.286538208789 \dots$

Also called Irrational Numbers -
(अपरिमेय संख्याएँ)

जिन्हें $\frac{p}{q}$ में नहीं लिखा जा सकता।

eg. $\sqrt{3}, \sqrt{7}, \sqrt{11}, \pi \dots$

$\sqrt{3} \rightarrow 1.732 \dots$

$\pi \neq \frac{22}{7} = 3.1416 \dots$

Approximate value है।
not exact value

π का लगभग मान $\frac{22}{7}$ होता है। यह Irrational Number

इसे $\frac{p}{q}$ के रूप में नहीं लिखा जा सकता।

Real Number $\rightarrow$ {it can be negative or positive}

Rational Numbers
(परिमेय संख्याये)

Expressed in $\frac{p}{q}$ form, $q \neq 0$
{where p & q are integers}

Irrational Numbers
(अपरिमेय संख्याये)

Non Terminating & Non Repeating, cannot be expressed in $\frac{p}{q}$ form
eg. $\pm\sqrt{3}$, $\pm\pi$ etc.

^(पूर्णांक) Integers - $\frac{p}{q}$ * $\boxed{q=1}$

Negative
(ऋणात्मक)
eg. $\rightarrow -6, -5, -4, -3, -2, -1, \dots$

Non-negative
(अऋणात्मक)

eg. $0, 1, 2, 3, 4, 5, \dots$

* Whole numbers - $\{0, 1, 2, 3, 4, \dots\}$
(पूर्ण संख्याये)

Negative Integer $\{-6, -5, -4, \dots\}$

Integers $\rightarrow 0$

Positive Integer $\{+1, +2, +3, +4, \dots\}$

* Natural Number = $\{+1, +2, +3, +4, \dots\}$
(प्राकृतिक संख्याये)

Integers

Even (सम)

which can be expressed in $2n$ form
 where $n \rightarrow$ natural no's -
 $n \rightarrow \{1, 2, 3, 4, 5, \dots\}$

Odd (विषम)

$\{-6, -4, -2, 0, +2, +4, \dots\}$

which are divisible by 2.

eg. $\{-7, -5, -3, -1, +1, +3, +5, +7, \dots\}$

can represent in $(2n+1)$ form.

Point

 $\Rightarrow$

* $\left\{ \begin{array}{l} \text{Odd} \pm \text{odd} = \text{Even} \\ \text{odd} \pm \text{even} = \text{odd} \\ \text{even} \pm \text{even} = \text{even} \end{array} \right\}$

eg. 1.

$\left\{ \begin{array}{l} 3+3=6 \\ 3+2=5 \\ 2+2=4 \end{array} \right\}$

2.

$\left\{ \begin{array}{l} 3+5=8 \\ 3+8=11 \\ 2+4=6 \end{array} \right\}$

*

$\left\{ \begin{array}{l} \text{Odd} \times \text{odd} = \text{odd} \\ \text{odd} \times \text{even} = \text{even} \\ \text{even} \times \text{even} = \text{even} \end{array} \right\}$

Natural Numbers — $\{1, 2, 3, 4, 5, 6, \dots\}$
 (प्राकृतिक संख्याएँ)

Prime Numbers

अभाज्य संख्याएँ

eg. (3, 11, 17, 19 etc)

Has exactly two factors.
 1 and itself.

Composite Numbers

(भाज्य संख्याएँ)

not prime.

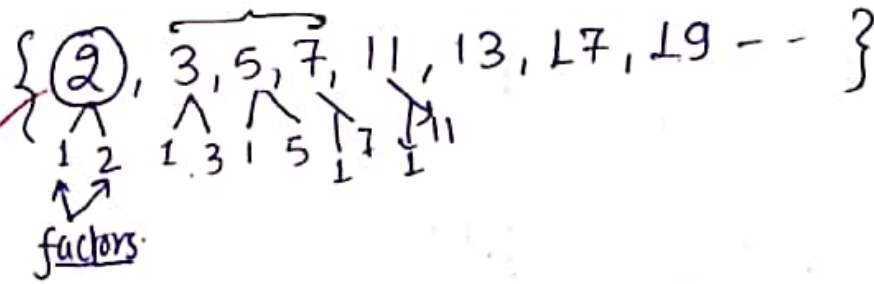
has at least 3 factors.

eg. $\frac{10}{1 \times 2 \times 5 \times 10}$

Prime Numbers (अभाज्य संख्याएँ)

5

eg.



Smallest prime number.

Only even prime number.

3, 5, 7 $\Rightarrow$ Three consecutive odd prime numbers.

[1-50] $\rightarrow$ 15 prime numbers

[50-100] $\rightarrow$ 10 prime numbers

[1-100] $\rightarrow$ 25 prime numbers

Prime numbers ^{##*} $\left\{ (6k \pm 1) \text{ form में होगा} \right\}$

① $\rightarrow$ 1 factor

[one is not prime number.]

Composite Numbers $\rightarrow$
(भाज्य संख्याएँ)

[1 $\rightarrow$ is not composite number]

(1 is neither prime nor composite number)

4, 6, 8, 10, 12, ...

9, $\sqrt{9} = 3$ $\rightarrow$ composite no -

$\left\{ 4, 6, 8, 9, 10, 12, \dots \right\}$
Smallest odd composite no - 9
Smallest composite numbers

Relatively prime or Co-prime (सहअभाज्य संख्याएँ)

→ { ऐसे दो numbers जिनका HCF = 1 है }

eg.

(5, 7)

(43, 31)

(16, 25)

~~(40, 35)~~

(9, 32)

#

Perfect numbers → (If sum of all the factors of a number is equal to that number, then this is called perfect number.)

(कुल गणना)

(excluding that number)

eg.

perfect no → ① 6 → factors 1, 2, 3, ~~6~~ ←
 $1 + 2 + 3 = 6$

② 28 → 1, 2, 4, 7, 14, ~~28~~
 $1 + 2 + 4 + 7 + 14 = 28$
 [Perfect number]

* Smallest perfect number ⇒ 6

6, 28, 496, 8128

3 digit

* 1 is not a perfect number

Face value and place value

eg.

894675

* 5 का face value = 5

7 " " " = 7

8 " " " = 8

9 " " " = 9

{ 5 की place value = $5 - 5 \times 1$
7 की " " = $70 - 7 \times 10$
6 की " " = $600 - 6 \times 100$
4 की " " = $4000 - 4 \times 1000$
9 " " = $90,000 - 9 \times 10,000$

Factors (गुणकखण्ड) :-

→ multiples (गुणक) → 12, 24, 36, 48, 60, 72 - - -

factors (गुणकखण्ड) → 1, 2, 3, 4, 6, 12 → itself

factor

universal factor

6 factor

4 even factor

2 odd factor

4 के multiple → 2 factor हैं

Number of factors →

eg. ①

$N=200 \rightarrow 1, 2, 4, 5, 8, 10, 20, 25, 40, 50, 100, 200 \dots$

12 factor



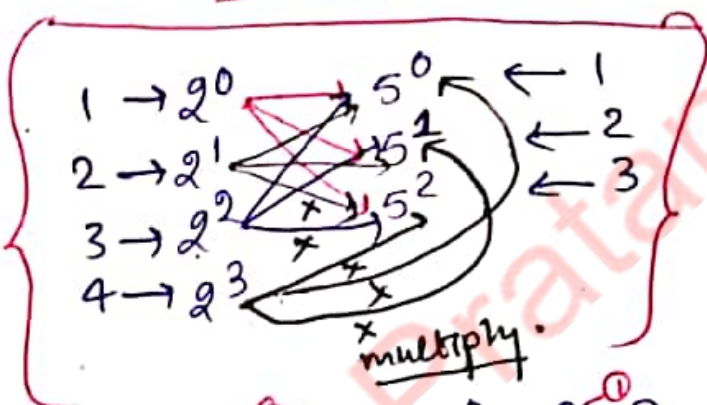
Method

prime factorization

$2^3 \times 5^2$

(prime number)

$x^0 = 1$



$2^3 \times 5^0 = 8$ ⑫	$2^2 \times 5^0 = 4$ ③	$2^0 \times 5^0 = 1$ ①	$2^1 \times 5^0 = 2$ ①
$2^3 \times 5^1 = 40$ ⑪	$2^2 \times 5^1 = 20$ ⑧	$2^0 \times 5^1 = 5$ ②	$2^1 \times 5^1 = 10$ ②
$2^3 \times 5^2 = 200$ ⑩	$2^2 \times 5^2 = 100$ ⑦	$2^0 \times 5^2 = 25$ ④	$2^1 \times 5^2 = 50$ ③

(12 factor)

⇒ Trick (Method) → $4 \times 3 = 12$

$N = 200 = 2^3 \times 5^2$

number of factor = $(3+1) \times (2+1) = 12$ factor.

eg. ②

2160

no. of factor = ?

$2160 \rightarrow 6^3 \times 2 \times 5 = 2^3 \times 3^3 \times 2 \times 5$
 $\rightarrow 2^4 \times 3^3 \times 5 \leftarrow$ prime factorization

number of factors, $(4+1) \times (3+1) \times (1+1)$
 $= 40$ factor # ↓

Sydney

$$2160 - 2^4 \times 3^3 \times 5^1$$

(9)

no. of even factors $\rightarrow (2k) =$

$$2 \times [2^3 \times 3^3 \times 5^1]$$

****** factors = $(3+1) \times (3+1) \times (1+1) = 4 \times 4 \times 2 = \boxed{32}$ factor even factors.

$$2160 - 2^4 \times 3^3 \times 5^1$$

number of odd factors $\rightarrow \frac{2^4}{2^1} 3^3 \times 5^1$

factors = $(3+1) \times (1+1) = 4 \times 2 = 8$ factor odd

odd factor OR = Total No. of factor - total no. of even factor

$$= 40 - 32$$

$$= \boxed{8} \text{ factor odd factor}$$

$$N = 2160 = 2^4 \times 3^3 \times 5^1$$

Total no. of factors which are multiple of 18 = ?
 $\swarrow \searrow$
 2×3^2

$$= 2 \times 3^2 \times [2^3 \times 3^1 \times 5^1]$$

factor = $(3+1) \times (1+1) \times (1+1) = 4 \times 2 \times 2 = 16$ factors.
 which are multiple of 18

Total no. of factors which are multiple of 60 $\rightarrow ?$

$$2 \times 3 \times 2 \times 5 = 2^2 \times 3^1 \times 5^1$$

$$= 2^2 \times 3^1 \times 5^1 [2^2 \times 3^2]$$

factors — $(2+1) \times (2+1) = 3 \times 3 = \boxed{9} \text{ factor}$

Satnam Pathway

Sum of factors →
(गुणनखंडों का योग)

eg. ① $N = 72 = 2^3 \times 3^2$

$72 = 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.$
factors

$N = 72 = 2^3 \times 3^2$

Sum of factors = $[2^0 + 2^1 + 2^2 + 2^3] \times [3^0 + 3^1 + 3^2]$
 $= [1 + 2 + 4 + 8] \times [1 + 3 + 9]$
 $= 15 \times 13$
 $= 195$

eg. 2

$N = 2160 = 2^4 \times 3^3 \times 5^1$

Sum of factors = $[2^0 + 2^1 + 2^2 + 2^3 + 2^4] \times [3^0 + 3^1 + 3^2 + 3^3] \times [5^0 + 5^1]$
 $= [1 + 2 + 4 + 8 + 16] \times [1 + 3 + 9 + 27] \times 6$
 $= 31 \times 40 \times 6$
 $= 31 \times 240$
 $= 7440 = \text{Ans.}$

Sum of even factor = $[2^1 + 2^2 + 2^3 + 2^4] \times [3^0 + 3^1 + 3^2 + 3^3] \times [5^0 + 5^1]$
 $\times 2$
 $= 30 \times 40 \times 6$
 $= 7200 \text{ Ans.}$

Sum of odd factor = $[3^0 + 3^1 + 3^2 + 3^3] \times (5^0 + 5^1)$ OR $= 7440 - 7200$
 $= 40 \times 6 = 240 \text{ Ans.}$

eg.

$$= \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8} + \frac{1}{12} + \frac{1}{24}$$

$$\frac{24 + 12 + 8 + 6 + 4 + 3 + 2 + 1}{24}$$

Perfect Square factors of a number $\rightarrow x$

3 number) Ans.

[divide the prime base power by 2
and add 1]

$$= \frac{4}{2} + 1 = 2 + 1 = \textcircled{3} \text{ Ans.}$$

OR. $3^2 \times 2 \times 5 = \frac{2}{2} + 1 = (2)$

A perfect square always has odd no. of factors.

if a number has odd no. of factor, then no- perfect square

Product of factor (गुणनखण्डों का गुणनफल)

eg.

$$100 \Rightarrow 1, 2, 4, 5, 10, 20, 25, 50, 100$$

$$60 = 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60$$

12 factors

multiply

$$\text{Product of factor} = 60^6$$

$$\text{OR} = N^{\frac{\text{no. of factors}}{2}}$$

$$= 60^{\frac{12}{2}} = 60^6$$

$$N = 100 = 1, 2, 4, 5, 10, 20, 25, 50, 100$$

(Perfect square)

9 factors

$\sqrt{N} = \sqrt{100} = 10$

$$N = 64 = 1, 2, 4, 8, 16, 32, 64$$

7 factor

$\sqrt{N} = \sqrt{64} = 8$

$$\text{Product of factor} = 100^{9/2} = 100^{4.5}$$

$$= 64^{7/2} = 64^{3.5}$$



#

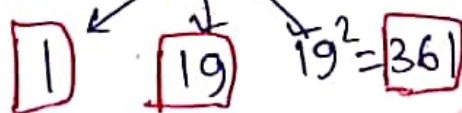
Square of a prime number has exactly 3 factor.

13

किसी अभाज्य संख्या के वर्ग के केवल 3 गुणनखण्ड होते हैं।

eg.

$$19 = 19^2 = 361$$



$$5 = 5^2 = 25$$



#

<https://t.me/yourmathbook>Unit digit (इकाई का अंक)

digit - 1, 2, 3, 4, 5, 6, 7, 8, 9, 0

①

unit digit

**

0, 1, 5, 6 → 0, 1, 5, 6 Same (multiple)

eg

$$1. (78121)^{878} = 78121 \times 78121 \dots (878 \text{ times})$$

{ unit digit = 1 }

$$2. (890)^{879} = ?$$

{ unit digit = 0 }

4, 9 ⇒ unit digitdetail

$$1. \begin{aligned} 4^1 &\rightarrow 4 \\ 4^2 &\rightarrow 6 \\ 4^3 &\rightarrow 4 \\ 4^4 &\rightarrow 6 \\ 4^5 &\rightarrow 4 \end{aligned}$$

Pattern repeated

4^{odd} → 4 unit digit
4^{even} → 6 unit digit

$$\text{eg } \left\{ \begin{aligned} (284)^{83} &\rightarrow \text{unit digit} \rightarrow 4 \\ (284)^{824} &\rightarrow \text{unit digit} \rightarrow 6 \end{aligned} \right\}$$

$$2. \begin{aligned} 9^1 &= 9 \\ 9^2 &= 1 \end{aligned} \} \rightarrow \text{pattern repeated.}$$

$$9^3 = 9$$

$$9^4 = 1$$

$$9^5 = 9$$

$$9^6 = 1$$

9^{odd} → 9 unit digit
9^{even} → 1 unit digit

$$\text{eg } \left\{ \begin{aligned} (789)^{63} &\rightarrow 9 \text{ unit digit} \\ (789)^{62} &\rightarrow 1 \text{ unit digit} \end{aligned} \right\}$$

2, 3, 7, 8 →

(2)

(2) →

$2^1 \rightarrow 2$ (unit digit 2, Remainder 1)
 $2^2 \rightarrow 4$ (unit digit 4, Remainder 2)
 $2^3 \rightarrow 8$ (unit digit 8, Remainder 3)
 $2^4 \rightarrow 6$ (unit digit 6, Remainder 0)

pattern repeated every 4 terms.

2 → Cyclicity = 4

$2^5 \rightarrow 2$

$2^6 \rightarrow 4$

$2^7 \rightarrow 8$

$2^8 \rightarrow 6$

$2^9 \rightarrow 2$

eg: $1. (132)^{457} = 2^{\overline{457}}$

$\frac{457}{4} = \text{remainder} = 1$

unit digit = 2

2. $(752)^{894} = ?$

$\frac{894}{4} = 2 \text{ remainder}$

unit digit = 4

(3)

$3^1 = 3$ (unit digit 3, Remainder 1)
 $3^2 = 9$ (unit digit 9, Remainder 2)
 $3^3 = 7$ (unit digit 7, Remainder 3)
 $3^4 = 1$ (unit digit 1, Remainder 0)

pattern repeated every 4 terms.

3 Cyclicity = 4

$3^5 = 3$

$3^6 = 9$

$3^7 = 7$

$3^8 = 1$

$3^9 = 3$

eg: $(863)^{851} \rightarrow$
 सिर्फ 2 अंक को ही Divide करके होता है
 $\frac{851}{4} = 212 \text{ remainder } 3$
 4th divisibility rule है
 = 3 Remainder
 unit digit → 7

⑦

$7^1 = 7$ ← Unit digit
 $7^2 = 9$ ← Remainder = ①
 $7^3 = 3$ ← Remainder = ②
 $7^4 = 1$ ← Remainder = ③
 $7^5 = 7$ ← Remainder = ④
 $7^6 = 9$
 $7^7 = 3$
 $7^8 = 1$
 $7^9 = 7$

pattern repeat every 4 terms

7 - Cyclicity = 4

eg. $(817)^{288}$

$\frac{288}{4} = 0$

Unit digit = 1

⑧

$8^1 = 8$ ← Unit digit
 $8^2 = 4$ ← Remainder = ①
 $8^3 = 2$ ← Remainder = ②
 $8^4 = 6$ ← Remainder = ③
 $8^5 = 8$ ← Remainder = ④
 $8^6 = 4$
 $8^7 = 2$
 $8^8 = 6$
 $8^9 = 8$

pattern repeat

8 → Cyclicity = 4

eg. $(9818)^{874}$

$\frac{874}{4} = 2 \text{ Remainder}$

Unit digit → ④

Divisibility Rule, भाज्यता के नियम

④

1- ^(भाज्य) is not divisible by any number except 1.
but 1 is an universal factor.

2 → ^(अंतिम संख्या) last digit should be even or last digit should
- be divisible by 2.

4 → last 2 digit should be divisible by 4.

8 → Last 3 digit should be divisible by 8.

16 → last 4 digit should be divisible by 16.

32 → last 5 digit should be divisible by 32.

64 → Last 6 digit should be divisible by 64.

→ If any Number cuts "a" and "b" then number
will be cut "a x b" but a & b are
co-prime.

eg- Remainder.

315 ÷ 7	=	0
315 ÷ 5	=	0
315 ÷ 7 × 5	=	0
315 ÷ 35	=	0

3, 9 → divisibility rule →

③ → Sum of digits should be divisible by 3.

⑨ → Sum of digit should be divisible by 9.

eg: Express no $\rightarrow$
<https://t.me/yoursmathtool>

$\rightarrow 1672 \rightarrow 1 \times 1000 + 6 \times 100 + 7 \times 10 + 2$

eg $\rightarrow$ चार अंकों की संख्या $\rightarrow 1000w + 100z + 10y + x$
 (wzyx) $\rightarrow \frac{(999w + 99z + 9y) + (w + z + y + x)}{9}$

$= \boxed{\frac{w + z + y + x}{9}}$

$\rightarrow$ ① $\rightarrow 50769 \xrightarrow{\text{sum of digit}} \frac{27}{9} = 3$

② $- 7105104 = 18$ ✓ यह संख्या 3, 9 दोनों से भाज्य है।

③ $- 356468792 = \frac{50}{9} \times \left\{ \begin{array}{l} \text{संख्या 1 तो 9 से भाज्य है और नहीं उसे -} \\ \text{Remainder } 5 \end{array} \right.$

or $3564 \cancel{6} \cancel{8} \cancel{7} \cancel{9} \cancel{2} \rightarrow 9$ हटा दो
 $6+8=14-9=5$
 or $1+4=5$

digit sum should be 9 or 0

$\Rightarrow$ Digital sum of a number is the remainder when it is divided 9.

<https://t.me/yourmathbook>

⇒ 5 → divisibility rule

5 → Last digit should be 0 or 5.

eg. 130 ✓ 1665 ✓

Remainder → 0 ← 98 divisible by 5

(25) → Last two digit should be divisible by 25

eg. 6125

(125) - Last three digit should be divisible by 125.

eg. 6125

(6) → 6 → 2 × 3 → यदि कोई संख्या "2" से भाज्य है और "3" से भी तो वह संख्या 6 से भी divisible होगी।

{ 7, 11, 13 : divisibility rule → }

7 → eg. 005922 = 917 ✓ [7 divisible होगी]

006489 → 483 ✓

045724 → 679 ✓

380287 → 133

006891647

Sum → 891 - 653 = 238 ✓

Signature

13 →

eg-

① ⇒ 0 | 1200644

sum 655 - difference 200 = 455 ✓ 7, 13

7 ✓ 11 ✗ 13 ✓

7 × 13 = 91 — सभी divisible होंगी.

② ⇒ 0 | 2478375

sum → 387 difference →

91
7 11 ✗ 13 ✓

{ दूसरी संख्या 7, 13 से काटेगी और 7 × 13 से भी काटेगी }

eg-

00 | 026389

sum → 390 - 26 = 364

7 ✓ 11 ✗ 13 ✓

ये संख्या 7, 13 से काटेगी

97824519

sum - 824 - 616 → 208

7 ✗ 11 ✗ 13 ✓

किन्तु 13 से काटेगी दूसरी संख्या

093962

difference → 869

7 ✗ 11 ✗ 13 ✗

किन्तु 11 से काटेगी दूसरी संख्या

040676372

sum 676 - 412 = 264

7 ✗ 11 ✗ 13 ✗

11 से काटेगी वही संख्या

111 333 222

sum - 333 - 333 = 0

7 ✓ 11 ✓ 13 ✓

{ यह संख्या 7, 11, 13 सभी से काटेगी }

Sham ③

⇒

कोई संख्या
abab में है। इसे किससे divide होगी -

- a) 101
- b) 107
- c) 109
- d) 111

$$1000a + 100b + 10a + b$$

$$1010a + 101b$$

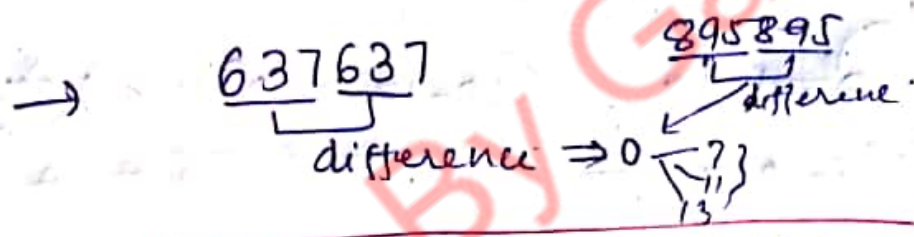
$$101(10a + b)$$

(101 के multiple में होगी)
तो 101
से divide भी होगी

⇒ 637637, 895895 is always divisible by

- a) 7 only
- b) 11 only
- c) 13 only

~~d) 1001~~



$$abcabc \Rightarrow abc \times 1001$$

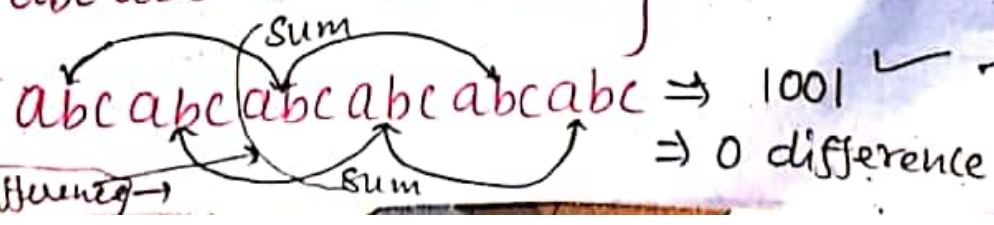
7 11 13

Any number

{ ababab = ab × 10101 } ***

{ abcabcabc = abc × 1001001 }

{ abcabc = abc × 1001 }



Even time repeat
होगा तो
7, 11, 13

eg $\overbrace{867} \overbrace{867} \overbrace{867} \dots \dots \dots 867$ time पर
 $\underbrace{\hspace{10em}}_{13}$
 remainder आयेगी?

$\Rightarrow \overbrace{867867867} \dots \dots \dots 866 \text{ times } 867$
 $\underbrace{\hspace{10em}}_{13}$

$\Rightarrow \frac{867}{13} \rightarrow 66$
 Remainder = 9

Special 11 $\Rightarrow$

eg $\rightarrow \left\{ \begin{array}{l} \overbrace{7945938}^{\text{Sum}} \rightarrow 28 \\ \text{Sum} = 217 \end{array} \right.$ $28 - 17 = 0^{th} 11 K$ तो संख्या
भाज्य होगी 11 से

$\left\{ \begin{array}{l} \overbrace{166452} \rightarrow 12 - 12 = 0 \end{array} \right.$ यह संख्या 11 से divisible होगी

$\Rightarrow$ कोई संख्या अगर - $n-2$ से divisible है तो 8089 से भा
 देंगी

$\left\{ \begin{array}{l} 88 - 11 \times 8 \\ 99 - 11 \times 9 \end{array} \right.$

$11, 8$ से आयेगी संख्या अगर 88 से divisible है तो
11 से और 9 से आयेगी संख्या

Remainder theorem

Case-I

Divisor
(भाजक)

$$\begin{array}{r}
 12 \overline{) 55} \\
 \underline{48} \\
 7
 \end{array}$$

divident भाज्य

quotient (भागफल)
[quotient]

Remainder (शेषफल)

$$\{ 55 = 12 \times 4 + 7 \}$$

#

$$\text{divident} = \text{divisor} \times \text{quotient} + \text{Remainder}$$

Case-II

eg

$$\left\{ \frac{13}{18} \rightarrow \text{[शेषफल]} \right\}$$

13

$$18 \overline{) 13} \quad (0$$

$$\begin{array}{r}
 0 \\
 \underline{18} \\
 13
 \end{array}$$

शेषफल

Case-III

$$\left\{ \frac{11 \times 18 = 198}{18} = 0 \leftarrow \text{Remainder} \right\}$$

Case-IV

$$\left\{ \frac{18}{18} = 1 \right. \\
 \left. \text{[Remainder=0]} \right\}$$

eg

$$\frac{40}{9} = 4$$

$$\frac{42}{9} = 6$$

$$\frac{43}{9} = 7$$

किसी संख्या d से divide करते होते तो
Remainder d से छोटा (small) है
होगा।

24. $(3^{25} + 3^{26} + 3^{27} + 3^{28})$ विभाज्य है -

- a) ~~11~~ b) ~~16~~ c) ~~25~~ d) 30

हल:-

$$3^{25} (1 + 3 + 3^2 + 3^3)$$

$$= 3^{25} \times 4 \times 2 \times 5$$

$\xrightarrow{3} \quad \times \quad \xrightarrow{10} = 30 \text{ से भी काटेगी} -$

Detail

$$\frac{130}{19} = \text{Rem} - 16$$

$$\textcircled{4} + \textcircled{12} \rightarrow \textcircled{16} \leftarrow \text{Remainder}$$

$$\frac{80 + 50}{19}$$

$$\textcircled{6} + \textcircled{11} \rightarrow \textcircled{16}$$

$$\frac{100 + 30}{19}$$

$$\textcircled{16} + \textcircled{6} + \textcircled{7}$$

$$\frac{70 + 42 + 133}{18}$$

Remainder

$$= ? \text{ (कड़िए) } \leftarrow \text{Actual Remainder}$$

$$= \frac{29}{18} \textcircled{11} \leftarrow \text{Remainder}$$

eg-

OR. $\frac{245}{18} = 11 \text{ Remainder}$

Concept
⇒

$$\frac{65}{25} = ? \text{ Remainder} = ?$$

$$25 \overline{) 65}$$

Remainder 50

$$\frac{50}{15} \leftarrow \text{बेसफल}$$

OR $\frac{65}{25} \frac{13}{5} = 3$

* NOTE

{ जिससे Divide करोगे उसीसे } $5 \leftarrow \text{divide}$ Final Rem = $5 \times 3 = \textcircled{15}$

$\Rightarrow \frac{N}{7} \rightarrow \overset{\text{Rem}}{\textcircled{3}}$
 $\times 5$
 $\frac{5N}{7} = \frac{15}{7} = \frac{14+1}{7} = \textcircled{1} \text{ Remainder.}$

Concept, $\rightarrow [a \& b, n \rightarrow \text{All are natural number}]$

- $\rightarrow \textcircled{1} a^n + b^n [n \rightarrow \text{odd}]$ $\textcircled{2} a^n + b^n [n \rightarrow \text{even}]$
factors $\left\{ \begin{array}{l} (a+b) \checkmark \\ (a-b) \times \end{array} \right.$ factors $\left\{ \begin{array}{l} (a+b) \times \\ (a-b) \times \end{array} \right.$

- $\textcircled{3} a^n - b^n [n \rightarrow \text{odd}]$ $\textcircled{4} a^n - b^n [n \rightarrow \text{even}]$
factors $\left\{ \begin{array}{l} (a+b) \times \\ (a-b) \checkmark \end{array} \right.$ factors $\left\{ \begin{array}{l} (a+b) \checkmark \\ (a-b) \checkmark \end{array} \right.$

eg- $\rightarrow (8^{2k} + 5^{2k})$ $k \rightarrow \text{odd}$
 $\rightarrow \left\{ \begin{array}{l} 64^k + 25^k \\ 64 + 25 \rightarrow 89 \text{ इसका एक गुणखंड होगा (factor)} \end{array} \right.$
 $2 - (31^{20} - 1024) \rightarrow \text{is not divisible by:}$
 $\rightarrow \text{odd}$
 $\textcircled{a} 137$
 $\textcircled{c} 9$
 $\textcircled{b} 107$
 $\textcircled{d} 32$

 $\{ \text{कौन odd संख्या even} \}$
 $\{ \text{असली से गलीवांगी} \}$

Concept of Negative remainder

$$\begin{array}{r} 13 \overline{) 60} (4 \\ \underline{52} \\ + 8 \end{array}$$

$$\begin{array}{r} 13 \overline{) 60} (5 \\ \underline{65} \\ - 5 \end{array}$$

$\frac{60}{13} = \text{Remainder}$

$+8 \leftarrow (-5 + 13)$

-5

eg- $\frac{130}{17}$

$+11$

-6

$\text{sum} = 17$

$\frac{84}{11}$

$+7$

-4

$\frac{61}{8}$

$+5 \leftarrow \{3 + 8\}$

-3

$\frac{(-)70}{19}$

$\xrightarrow{\text{Rem}}$

$-13 \leftarrow \{+19 - 13 = 6\}$

$+6$

$\Rightarrow$

$(+2) \times (+4) \Rightarrow +8 \text{ remainder}$

$\frac{79 \times 103}{11}$

OR

$(+2) \times (-7) = -14 + 11 = -3 \Rightarrow +8$

$\frac{79 \times 103}{11}$

Fermat's Theorem,

$$\rightarrow \frac{N^{P-1}}{P} \Rightarrow \boxed{\text{Remainder} = 1}$$

$\Rightarrow$ if P is a prime number.
(अभाज्य संख्या)

$\Rightarrow N$ & P should be co-prime.

$$\boxed{\text{HCF}(N, P) = 1}$$

Find Remainder,

eg 1. $\frac{12^{16}}{17} = 1 \text{ Remainder.}$

$12, 17 \rightarrow \text{co-prime.}$

$17 - \text{prime number}$

2. $\frac{38^{102}}{103} = 1 \text{ Remainder.}$
 $\swarrow$
 co-prime

Wilson's Theorem,

$$\boxed{\frac{(P-1)!}{P} = \text{Remainder} = -1 \text{ OR } P-1}$$

$P - \text{prime number}$

Theorem,

$$\Rightarrow \left\{ \begin{aligned} \frac{(ax+k)^n}{a} &\Rightarrow \text{Remainder} = k^n \rightarrow \frac{(ax+1)^n}{a} = 1^n = 1 \\ \frac{(ax-k)^n}{a} &= \text{Remainder} \Rightarrow (-k)^n \rightarrow \frac{(ax-1)^n}{a} = (-1)^n = -1 \end{aligned} \right.$$

egⁿ 1. $\frac{6^{153}}{5} = \text{Remainder} = ?$

$$= \frac{(5+1)^{153}}{5} = \frac{(5+1) \times (5+1) \times \dots \times (5+1)}{5} \text{ --- 153 times} = 1$$

Remainder = 1

2. $\frac{(77)^{67}}{39} = \frac{(2 \times 39 - 1)^{67}}{39} = \frac{(2 \times 39 - 1) \times (2 \times 39 - 1) \times \dots \times (2 \times 39 - 1)}{39} \text{ --- 67 times}$

$$= (-1)^{67} = -1 \text{ or } 39$$

Remainder.

Concept,

$$\begin{aligned} \frac{4^1}{6} &= 4 \\ \frac{4^2}{6} &= 4 \\ \frac{4^3}{6} &= 4 \end{aligned}$$

$$\frac{4^n}{6} = \text{Rem} 4$$

Shyam Pradyumn

Euler's theorem, (Totient concept)*

$$\boxed{\frac{a^{\phi(N)}}{N} = \text{rem} \rightarrow 1}$$

→ Any Natural Number.
 ** $a \nmid N \rightarrow$ should be co-prime.

$\{\phi(N) = \text{Totient function of } N\}$

8 13 8 9
 $\sqrt{40}, \sqrt{65}, \sqrt{72}$

eg $N = 16$ Composite Number =

prime number - 17, 19, 23

How to find totient function →

$N = 16 = 2^4$

$\phi(16) = 16 \times (1 - \frac{1}{2}) = 8$

$N = 30 = 2^1 \times 3^1 \times 5^1$

$\phi(30) = 30 \times (1 - \frac{1}{2}) \times (1 - \frac{1}{3}) \times (1 - \frac{1}{5})$
 $= 30 \times \frac{1}{2} \times \frac{2}{3} \times \frac{4}{5} = 8$

$N = 100 = 2^2 \times 5^2$

$\phi(100) = 100 \times (1 - \frac{1}{2}) \times (1 - \frac{1}{5})$ Totient function
 $= 100 \times \frac{1}{2} \times \frac{4}{5} = 40$

eg $\frac{37^{25}}{72} \Rightarrow \frac{37^{24} \times 37}{72}$

→ remainder.
 $1 \times 37 = 37$
 $\boxed{\phi(72) = 24}$

$$\# \left\{ \phi(p) = p-1 \right\} \rightarrow \text{total of prime number} \quad (13)$$

$\uparrow$
prime no

eg $\downarrow$ prime number.
 $19 \equiv 19^1$

$$\phi(19) = 19 \times \left(1 - \frac{1}{19}\right) = (18)$$

$$\Rightarrow \frac{a^{\phi(N)}}{N} \equiv \text{Rem}(1)$$

$$\boxed{\frac{a^{p-1}}{p} \equiv \text{Rem}(1)}$$

Fermat's theorem

a & p should be
co prime.

CHINESE REMAINDER THEOREM,

(12)

$\Rightarrow \frac{N}{abc}$ Where a, b, c are co-prime numbers

Case-I If N is divided by a, b, c respectively then they leave same remainder R in each case then remainder $\boxed{\frac{N}{abc} = R}$

Case-II If N is divided by a, b, c respectively and leaves remainder r_1, r_2, r_3 but $(a-r_1 = b-r_2 = c-r_3 = k)$ then remainder $\boxed{\frac{N}{abc} = \text{lcm of } a, b, c - k}$
 $\boxed{= abc - k}$

Case-III - If N is divided by a, b, c and leaves remainder r_1, r_2, r_3 then remainder $\frac{N}{abc} =$

$$\boxed{R = ax + r_1 = by + r_2 = cz + r_3}$$

* Number of zeroes

(19)

Q- 1. $75 \times 16 = 1200$

$2 \times 5 = 10$ ^{Zeroes}

2 Zeroes.

Explain, $3 \times 5 \times 5 \times 2 \times 2 \times 2 \times 2$

pair $\rightarrow 2$

इसलिए - 2 zeroes होंगे।

2. $2^9 \times 3^{11} \times 2^7 \times 5^7 \times 7^6 \times 8^9 \times 9^6$

Find no. of zeroes - ?

$\Rightarrow 2^9 \times 3^{11} \times 2^7 \times 5^7 \times 7^6 \times 8^9 \times 9^6$

7 pair

no. of zeroes = 7

(2 & 5)

जो सबसे कम भा है $\uparrow$

$75 \times 256 \times 51 \times 2250 \times 79 \times 88$

no. of zeroes = ?

$\Rightarrow 3 \times 2 \times 3 \times 5^2 \times 2^8 \times 51 \times 5^2 \times 9 \times 10 \times 79 \times 88$

no. of 5 = 5

no. of 2 = more than 5

no. of zeroes = जो सबसे कम भा है -

= 5 zeroes

Spair

→ eg-

$$1 \times 2 \times 3 \times 4 \times \dots \times 1000 = 1000!$$

(20)

no. of zeroes?

→ How to find no. of 5

$$\frac{1000}{5} = \frac{200}{5} = \frac{40}{5} = \frac{8}{5} = 1$$

जब तक
होना न हो
आम
5 से
तब
तक
Divide
होगा

= 249 OR 5 आया है
लेकिन 2 से 249 से
अधिक का आया है
इसलिए 5, 2 के केवल 249
pair होंगे.
no. of zeroes = 249

eg-

$$1 \times 2 \times 3 \times 5 \times 7 \times 9 \times 11 \times \dots \times 125 \times 128$$

$$\downarrow$$

$$2^7$$

कुछ ज्यादा का - 5 आ रहा है?

but, लेकिन pair केवल 7 ही होंगे

∴ no. of zeroes = 7 ही हैं

eg-

$$4 \times 8 \times 12 \times 16 \times \dots \times 560$$

no. of zeroes = ?

$$\Rightarrow (4 \times 1) \times (4 \times 2) \times (4 \times 3) \times \dots \times 4 \times 140$$

$$= 4^{140} [1 \times 2 \times 3 \times 4 \times \dots \times 140]$$

$$\frac{140}{5} = \frac{28}{5} = \frac{5}{5} = 1$$

no. of 5 = 34

pair = 34
Zeroes = 34 Ans.

Get Solution
Spademy

$\Rightarrow$ * Sum of first n even numbers $= n(n+1)$

(29)

$$\rightarrow 2 \times 1 + 2 \times 2 + 2 \times 3 + \dots + 2 \times n$$

$$= 2 [1 + 2 + 3 + \dots + n]$$

$$= 2 \times \frac{n(n+1)}{2} = \boxed{n(n+1)}$$

$\Rightarrow$ * Sum of first n odd numbers $= n^2$

$$1 + 3 + 5 + \dots + (2n-1) = \frac{n}{2} [1 + 2n-1]$$

$$= \boxed{n^2}$$

G.P., $a, ar, ar^2, ar^3, \dots$

$\left\{ \begin{array}{l} a - \text{First term} \\ r = \frac{T_{n+1}}{T_n} \end{array} \right\}$

$\left\{ \begin{array}{l} \text{Common} \\ \text{ratio} \end{array} \right\}$

$$n^{\text{th}} \text{ term} = T_n = ar^{n-1}$$

$$\text{Sum of first } n \text{ terms} = \frac{a(r^n - 1)}{r - 1}$$

$$S_{\infty} = \frac{a}{1-r} \text{ if } r < 1$$

[Sum of Infinite terms]

H.P - [harmonic progression]
(हरात्मक श्रेणी)

$$\Rightarrow A.P \rightarrow a, a+d, a+2d, a+3d \dots a+(n-1)d$$

समान्ता श्रेणी

$$H.P \rightarrow \frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d} \dots \frac{1}{a+(n-1)d}$$

हरात्मक श्रेणी

$$H.P = \frac{1}{A.P}$$

If there are two positive number a & b

$$a \text{ \& \; } b \text{ at A.M. (समान्ता माध्य)} = \frac{a+b}{2} = 6.5$$

$$a \text{ \& \; } b \text{ at G.M. (उज्जोता माध्य)} = \sqrt{ab} = 6$$

$$a \text{ \& \; } b \text{ at H.M. (हरात्मक माध्य)} = \frac{2ab}{a+b} = 5.53$$

Result

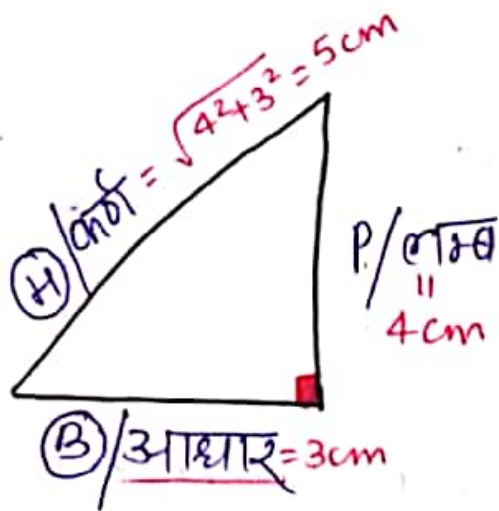
$$A.M \geq G.M \geq H.M$$

$$\frac{a+b}{2} \geq \sqrt{ab} \geq \frac{2ab}{a+b}$$

$$\Rightarrow A.M \times H.M = G.M^2 \Rightarrow \frac{a+b}{2} \times \frac{2ab}{a+b} = ab = (\sqrt{ab})^2$$

Setam

⇒ Trigonometry त्रिकोणमिति ①



$$H^2 = P^2 + B^2$$

$$\begin{cases} = 3, 4, 5 \\ = 8, 15, 17 \end{cases} \begin{matrix} \text{Triplet} \\ \text{Pa} \end{matrix}$$

Some Important, Basic Triplet

(3, 4, 5)

(5, 12, 13)

(8, 15, 17)

(7, 24, 25)

(9, 40, 41)

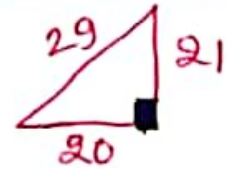
(11, 60, 61)

(12, 35, 37)

(16, 63, 65)

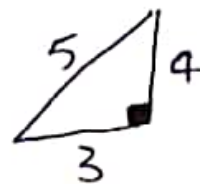
(20, 21, 29)

always Right angle Triangle



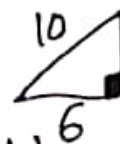
eg: 3, 4, 5

{Same no multiply} $\times 2 \times 2 \times 2$
 6 8 10



(6, 8, 10)

भी (Triplet) (होगा)



यै भी समकोण Δ होगा ?

② 8, 15, 17
 $\times 1.2 \times 1.2 \times 1.2$

[9.6, 18, 20.4] - ये भी समकोण Δ होगा।

Odd no' ka Triplet

(2)

$(3) \rightarrow 3^2 = 9$

$\begin{matrix} & (4) \\ & \swarrow \searrow \\ (3) & & (5) \end{matrix}$

$\left. \begin{matrix} (4) \\ (5) \end{matrix} \right\} - 1 \text{ difference में divide करना है}$

$(7) \rightarrow 7^2 = 49$

$\begin{matrix} & (24) \\ & \swarrow \searrow \\ (7) & & (25) \end{matrix}$

$\left. \begin{matrix} (24) \\ (25) \end{matrix} \right\} - 1 \text{ difference में divide}$

Even no' ka Triplet

$(8) \rightarrow 8^2 = \frac{64}{2} = 32$

$\begin{matrix} & (15) \\ & \swarrow \searrow \\ (8) & & (17) \end{matrix}$

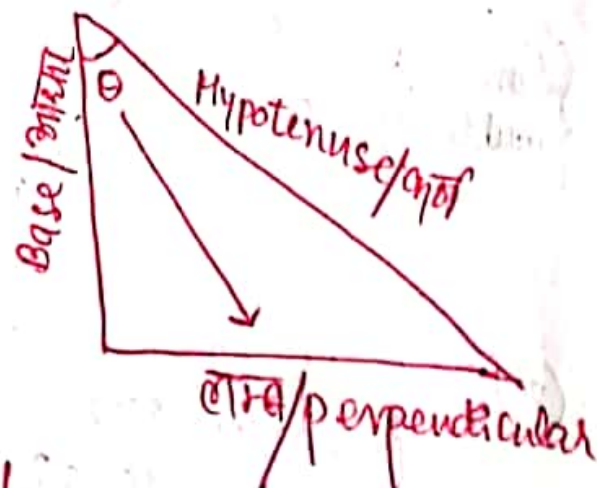
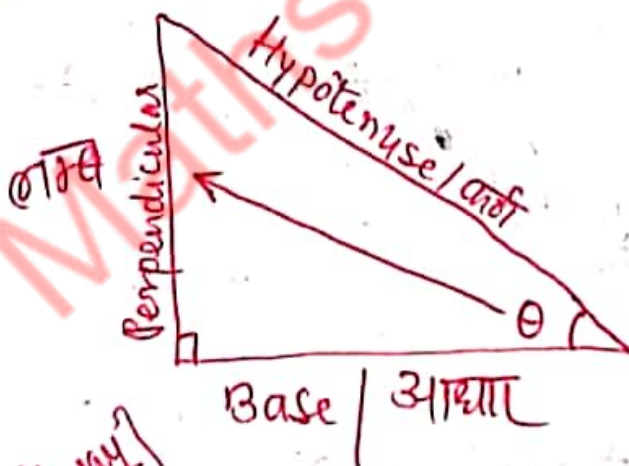
$\left. \begin{matrix} (15) \\ (17) \end{matrix} \right\} - 2 \text{ [2 के difference में divide]}$

$(12) \rightarrow 12^2 = \frac{144}{2} = 72$

$\begin{matrix} & (35) \\ & \swarrow \searrow \\ (12) & & (37) \end{matrix}$

$\left. \begin{matrix} (35) \\ (37) \end{matrix} \right\} - 2$

#



[Ayudhyay]

Trigonometry ratio

(3)

$$\sin \theta = \frac{P}{H} = \frac{\text{लम्ब}}{\text{कर्ण}}$$

$$\cos \theta = \frac{B}{H} = \frac{\text{आधार}}{\text{कर्ण}}$$

$$\tan \theta = \frac{P}{B} = \frac{\text{लम्ब}}{\text{आधार}}$$

$$\operatorname{cosec} \theta = \frac{\text{कर्ण}}{\text{लम्ब}} = \frac{H}{P}$$

$$\sec \theta = \frac{\text{कर्ण}}{\text{आधार}} = \frac{H}{B}$$

$$\cot \theta = \frac{\text{आधार}}{\text{लम्ब}} = \frac{B}{P}$$

$$\Rightarrow \sin \theta \times \operatorname{cosec} \theta = \frac{P}{H} \times \frac{H}{P} = 1$$

$$\sin \theta = \frac{1}{\operatorname{cosec} \theta}$$

OR

$$\sin \theta \times \operatorname{cosec} \theta = 1$$

$$\cos \theta \times \sec \theta = 1$$

$$\tan \theta \times \cot \theta = 1$$

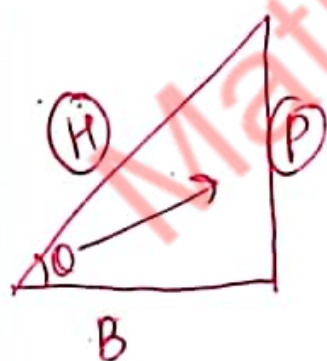
$$\cos \theta = \frac{1}{\sec \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

Basic Identities of trigonometry



$$\frac{P^2 + B^2}{H^2} = \frac{H^2}{H^2} \quad [\text{divide } H^2 \text{ both side}]$$

$$\left(\frac{P}{H}\right)^2 + \left(\frac{B}{H}\right)^2 = 1$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\cos^2 \theta = 1 - \sin^2 \theta$$

Signature

②

$$\frac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta}$$

④
[Divide $\cos^2 \theta$ Both Side]

$$\Rightarrow \left(\frac{\sin \theta}{\cos \theta} \right)^2 + 1 = \frac{1}{\cos^2 \theta}$$

$$= * \boxed{\tan^2 \theta + 1 = \sec^2 \theta}$$

$$\text{OR } * \boxed{\sec^2 \theta - 1 = \tan^2 \theta}$$

$$** \boxed{\sec^2 \theta - \tan^2 \theta = 1}$$

↓

$$(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$$

$$\boxed{(\sec \theta + \tan \theta) = \frac{1}{(\sec \theta - \tan \theta)}}$$

③

$$\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta}$$

divide $\sin^2 \theta$

$$\Rightarrow 1 + \left(\frac{\cos \theta}{\sin \theta} \right)^2 = \left(\frac{1}{\sin \theta} \right)^2$$

$$= \boxed{1 + \cot^2 \theta = \operatorname{cosec}^2 \theta}$$

$$\boxed{\operatorname{cosec}^2 \theta - \cot^2 \theta = 1}$$

$$\boxed{\cot^2 \theta = \operatorname{cosec}^2 \theta - 1}$$

$$\boxed{(\operatorname{cosec} \theta + \cot \theta) = \frac{1}{(\operatorname{cosec} \theta - \cot \theta)}}$$

Trigonometry Table

5

	0°	30°	45°	60°	90°	-
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined	
$\operatorname{cosec} \theta$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1	
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined	
$\cot \theta$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	

Concept:-

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta \times \sin \theta = (1 + \cos \theta)(1 - \cos \theta)$$

$$\frac{\sin \theta}{1 + \cos \theta} = \frac{1 - \cos \theta}{\sin \theta}$$

$$\cos^2 \theta = 1 - \sin^2 \theta$$

$$\frac{\cos \theta}{1 + \sin \theta} = \frac{1 - \sin \theta}{\cos \theta}$$

L.H.S

$$\frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}$$

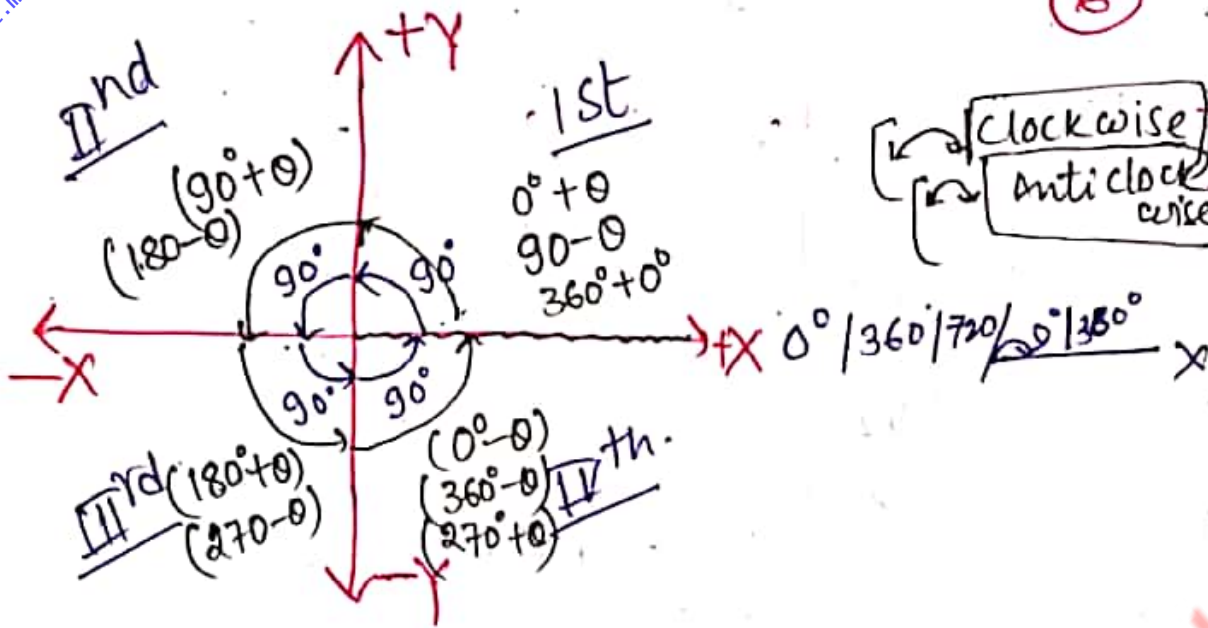
$$\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cdot \cos^2 \theta}$$

$$= \operatorname{cosec}^2 \theta \times \sec^2 \theta$$

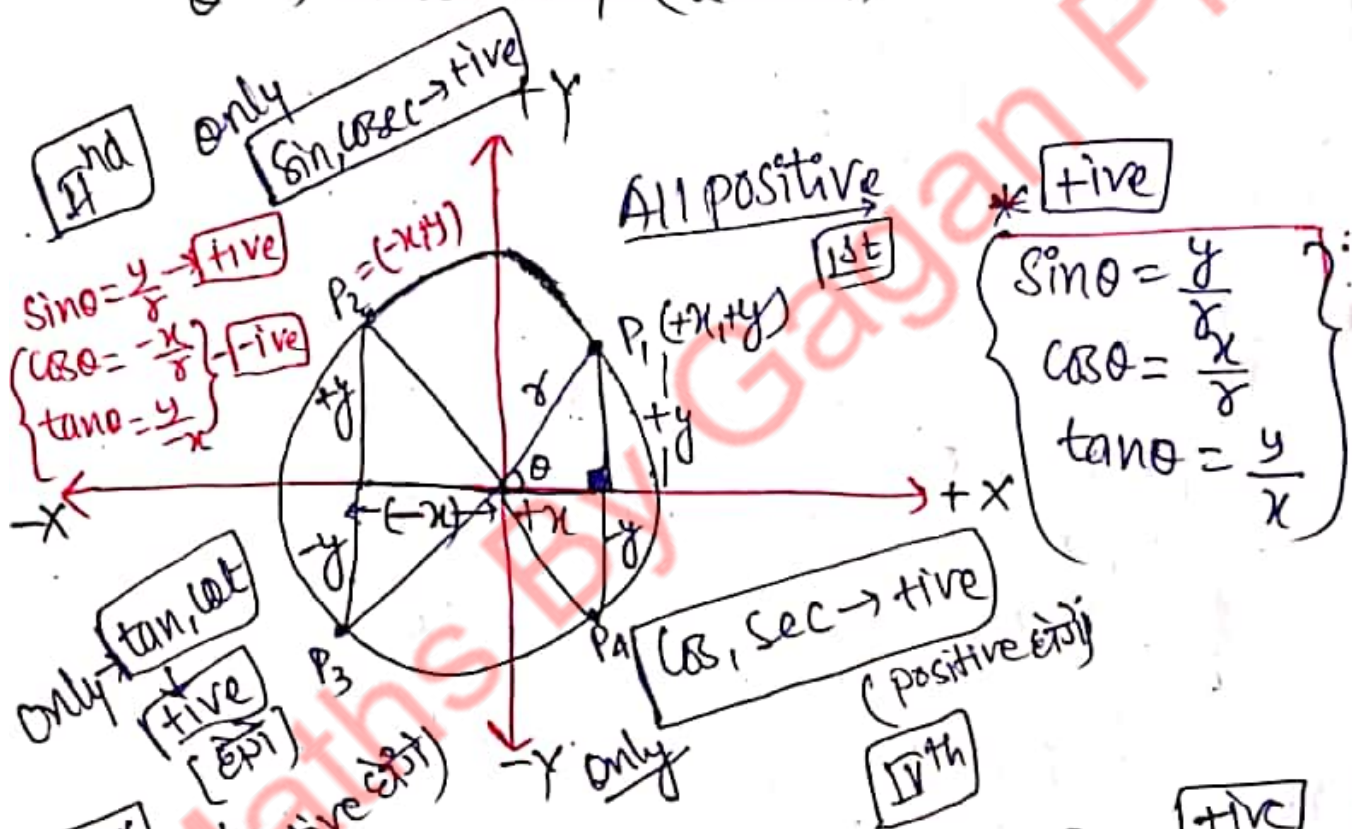
R.H.S Ans.

$$\# \operatorname{Sec}^2 \theta + \operatorname{cosec}^2 \theta = \operatorname{Sec}^2 \theta \times \operatorname{cosec}^2 \theta$$

⑥



$0 \rightarrow$ Acute Angle (न्यूनकोण)

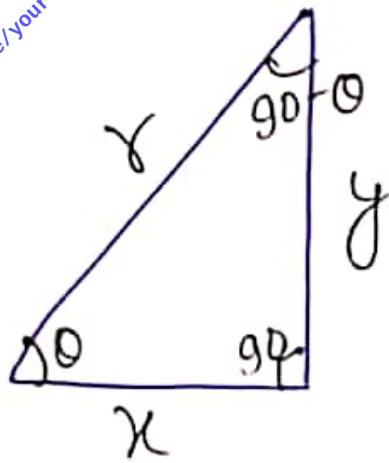


Result, $\sin, \csc \rightarrow$ +ve, All positive

$\tan, \cot \rightarrow$ +ve, $\sec, \csc \rightarrow$ +ve

Signs for other trigonometric functions:

- $\sin, \csc \rightarrow$ +ve
- $\cos, \sec \rightarrow$ -ve
- $\tan, \cot \rightarrow$ -ve



***#

(7)

$$\sin \theta = \cos(90 - \theta) = \frac{y}{r}$$

$$\cos \theta = \sin(90 - \theta) = \frac{x}{r}$$

$$\tan \theta = \cot(90 - \theta) = \frac{y}{x}$$

*

$$\sin(90 - \theta) = \cos \theta$$

$$\cos(90 - \theta) = \sin \theta$$

$$\operatorname{cosec}(90 - \theta) = \sec \theta$$

$$\sec(90 - \theta) = \operatorname{cosec} \theta$$

$$\tan(90 - \theta) = \cot \theta$$

$$\cot(90 - \theta) = \tan \theta$$

*

→ 90°, 270° → Change
0°, 180°, 360° → no change

Sin ↔ Cos
tan ↔ cot
Sec ↔ cosec

* $\sin(180 - \theta) = \sin \theta$

$$\begin{aligned} \sin(90 + (90 - \theta)) \\ \cos(90 - \theta) \\ = \sin \theta \end{aligned}$$

$$\begin{aligned} \sin(-\theta) &= -\sin \theta \\ \cos(-\theta) &= \cos \theta \\ \tan(-\theta) &= -\tan \theta \\ \cot(-\theta) &= -\cot \theta \\ \sec(-\theta) &= \sec \theta \\ \operatorname{cosec}(-\theta) &= -\operatorname{cosec} \theta \end{aligned}$$

0° ↓ → IVth co-ordinate
only - [Cos, Sec]
[+ive] होता है

Exam

IInd Co-ordinate only $\sin, \csc \rightarrow +ive \rightarrow no\ change$

(8)

$\sin(180^\circ - \theta) = \sin \theta$

$\cos(180^\circ - \theta) = -\cos \theta$

$\tan(180^\circ - \theta) = -\tan \theta$

$\cot(180^\circ - \theta) = -\cot \theta$

$\sec(180^\circ - \theta) = -\sec \theta$

$\csc(180^\circ - \theta) = \csc \theta$

IIIrd Co-ordinate only - $\tan, \cot \rightarrow +ive$ (ऐसे) $\rightarrow no\ change$

$\sin(180^\circ + \theta) = -\sin \theta$

$\cos(180^\circ + \theta) = -\cos \theta$

$\tan(180^\circ + \theta) = \tan \theta$

$\cot(180^\circ + \theta) = \cot \theta$

$\sec(180^\circ + \theta) = -\sec \theta$

$\csc(180^\circ + \theta) = -\csc \theta$

$\sin(90^\circ + \theta) = +\cos \theta$

$\cos(90^\circ + \theta) = -\sin \theta$

$\tan(90^\circ + \theta) = -\cot \theta$

IVth Co-ordinate

$\sin(270^\circ - \theta) = -\cos \theta$

$\cos(270^\circ - \theta) = -\sin \theta$

$\tan(270^\circ - \theta) = \cot \theta$

$\cot(270^\circ - \theta) = \tan \theta$

$\sec(270^\circ - \theta) = -\csc \theta$

$\csc(270^\circ - \theta) = -\sec \theta$

$\sec(270^\circ + \theta) = \csc \theta$

$\cot(270^\circ + \theta) = -\tan \theta$

$\tan(270^\circ + \theta) = -\cot \theta$

$\sin(270^\circ + \theta) = -\cos \theta$

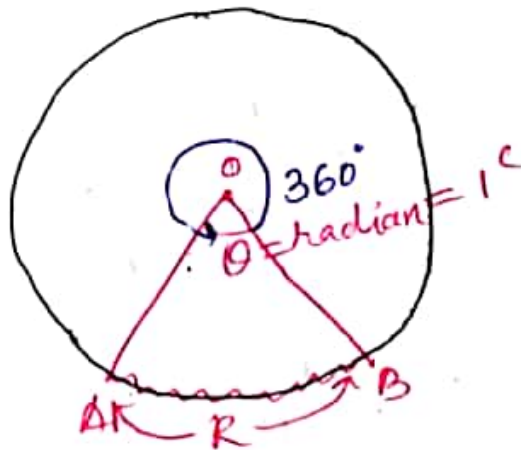
$\cos(270^\circ + \theta) = +\sin \theta$

(+ signs)

Radian →

9

AB = Radius of circle



$$R \rightarrow 1^c$$

$$2\pi R \rightarrow 2\pi^c$$

$$2\pi^c = 360^\circ$$

$$\pi^c = 180^\circ$$

$$1^c = \frac{180^\circ}{\pi}$$

If $a \sin \theta + b \cos \theta = c$ — ① $\boxed{x = ?}$
 $a \cos \theta - b \sin \theta = x$ — ②

en! eqⁿ ①² + eqⁿ ②²

$$\Rightarrow a^2 \sin^2 \theta + b^2 \cos^2 \theta + 2ab \sin \theta \cos \theta + a^2 \cos^2 \theta + b^2 \sin^2 \theta - 2ab \sin \theta \cos \theta = c^2 + x^2$$

$$\Rightarrow a^2 + b^2 = c^2 + x^2$$

$$\boxed{\sin^2 \theta + \cos^2 \theta = 1}$$

$$\Rightarrow \boxed{x = \pm \sqrt{a^2 + b^2 - c^2}}$$

Given - $a \sin \theta + b \cos \theta = c$ (10)
 $c^2 = a^2 + b^2$

Then -

$$\frac{a}{c} \sin \theta + \frac{b}{c} \cos \theta = 1$$

$$\sin \theta \times \sin \theta + \cos \theta \times \cos \theta = 1$$

Compare;

$$\left\{ \begin{array}{l} \sin \theta = \frac{a \rightarrow P}{c \rightarrow H} \\ \cos \theta = \frac{b \rightarrow B}{c \rightarrow H} \end{array} \right\}$$

$$H^2 = P^2 + B^2$$

$c^2 = a^2 + b^2$ - Condition Satisfied

Then

$$\left\{ \begin{array}{l} a \sin \theta + b \cos \theta = c \\ \uparrow \quad \quad \uparrow \quad \quad \uparrow \\ (P) \quad \quad (B) \quad \quad (H) \end{array} \right\}$$

If $a \sec \theta + b \tan \theta = c$ - (1)
 $a \tan \theta + b \sec \theta = x$ - (2) x = ?

eqⁿ (1)² - eqⁿ (2)²

$$\Rightarrow a^2 \sec^2 \theta + b^2 \tan^2 \theta + 2ab \sec \theta \tan \theta - a^2 \tan^2 \theta - b^2 \sec^2 \theta - 2ab \sec \theta \tan \theta = c^2 - x^2$$

$$\Rightarrow a^2 - b^2 = c^2 - x^2$$

$\sec^2 \theta - \tan^2 \theta = 1$

$$\Rightarrow x = \pm \sqrt{c^2 - a^2 + b^2}$$

$$1. \sin(A+B) = \sin A \cdot \cos B + \cos A \cdot \sin B \quad \text{--- (1)}$$

Put $B = -B$

$$2. \sin(A-B) = \sin A \cos B - \cos A \cdot \sin B \quad \text{--- (2)}$$

$$3. \cos(A+B) = \cos A \cdot \cos B - \sin A \cdot \sin B \quad \text{--- (3)}$$

$$\left\{ \begin{aligned} \cos(A+B) &= \sin\left\{\frac{\pi}{2} - (A+B)\right\} = \sin\left\{\left(\frac{\pi}{2} - A\right) - B\right\} \\ &= \sin\left(\frac{\pi}{2} - A\right) \cos B - \cos\left(\frac{\pi}{2} - A\right) \sin B \\ &= \cos A \cos B - \sin A \sin B \end{aligned} \right.$$

Put $B = -B$

$$4. \cos(A-B) = \cos A \cdot \cos B + \sin A \sin B \quad \text{--- (4)}$$

eqⁿ ① × eqⁿ ②

$$\begin{aligned} 5. \sin(A+B) \times \sin(A-B) &= \sin^2 A \cos^2 B - \cos^2 A \sin^2 B \\ &= \sin^2 A (1 - \sin^2 B) - (1 - \sin^2 A) \sin^2 B \\ &= \sin^2 A - \sin^2 A \sin^2 B - \sin^2 B + \sin^2 A \sin^2 B \\ &= \boxed{\sin^2 A - \sin^2 B} \end{aligned}$$

$$\text{OR} = 1 - \cos^2 A - 1 + \cos^2 B$$

$$\text{OR} = \boxed{\cos^2 B - \cos^2 A}$$

$\{eq^n ⑤ \times eq^n ④\}$

$$\begin{aligned} \rightarrow \cos(A+B) \cdot \cos(A-B) &= \cos^2 A \cdot \cos^2 B - \sin^2 A \times \sin^2 B \\ &= \cos^2 A (1 - \sin^2 B) - (1 - \cos^2 A) \times \sin^2 B \\ &= \cos^2 A - \cancel{\cos^2 A \times \sin^2 B} - \sin^2 B + \cancel{\cos^2 A \times \sin^2 B} \\ &= \boxed{\cos^2 A - \sin^2 B} \end{aligned}$$

OR.

$$\begin{aligned} &= \cancel{1 - \sin^2 A} + \cos^2 B \\ &= \boxed{\cos^2 B - \sin^2 A} \end{aligned}$$

$eq^n ① + eq^n ②$

⑦. $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$

$eq^n ① - eq^n ②$

⑧. $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$

$eq^n ⑤ + eq^n ④$

⑨. $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$

$eq^n ④ - eq^n ③$

⑩. $\boxed{2 \sin A \sin B = \cos(A-B) - \cos(A+B)}$

$eq^n ① \text{ put } B=A$

⑪. $\sin 2A = \sin A \times \cos A + \cos A \times \sin A$
 $= \boxed{2 \sin A \times \cos A}$

Pradyot

(12)

$$\sin 2A =$$

$$\frac{2 \sin A \times \cos A / \cancel{\cos^2 A}}{\cancel{\cos^2 A + \sin^2 A} / \cancel{\cos^2 A}}$$

$$= \frac{2 \tan A}{1 + \tan^2 A}$$

(13)

(13)

$$\text{eqn } 3 \quad \boxed{B = A}$$

$$\cos 2A =$$

$$\cos^2 A - \sin^2 A = (1 - \sin^2 A) - \sin^2 A = 1 - 2 \sin^2 A$$

$$= \cos^2 A - (1 - \cos^2 A) = \frac{2 \cos^2 A - 1}{1}$$

(14)

$$\cos 2A = \frac{\cos^2 A - \sin^2 A / \cancel{\cos^2 A}}{\sin^2 A + \cos^2 A / \cancel{\cos^2 A}}$$

$$= \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$= \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

(15)

$$\text{eqn. put } \boxed{B = 2A}$$

$$\begin{aligned} \sin(3A) &= \sin A \cdot \cos 2A + \cos A \cdot \sin 2A \\ &= \sin A (1 - 2 \sin^2 A) + 2 \sin A \times \cos^2 A \quad \times (1 - \sin^2 A) \\ &= \sin A - 2 \sin^3 A + 2 \sin A - 2 \sin^3 A \\ &= \boxed{3 \sin A - 4 \sin^3 A} \end{aligned}$$

Similarly.

(16)

$$\{ \cos 3A = 4 \cos^3 A - 3 \cos A \}$$

Gudhyay.

$$\frac{\sin(A+B)}{\cos(A+B)} = \frac{\frac{\sin A \cdot \cos B + \cos A \cdot \sin B}{\cos A \cdot \cos B}}{\frac{\cos A \cos B - \sin A \sin B}{\cos A \times \cos B}}$$

$$\left\{ \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} \right\}$$

(18) put $B = -B$

$$\left\{ \tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} \right\}$$

$$\begin{aligned} (19) \quad \cot(A+B) &= \frac{1 - \tan A \cdot \tan B}{\tan A + \tan B} \\ &= \left\{ \frac{\cot A \times \cot B - 1}{\cot B + \cot A} \right\} \end{aligned}$$

$$(20) \quad \cot(A-B) = \left\{ \frac{\cot A \times \cot B + 1}{\cot B - \cot A} \right\}$$

Satyam
padhyay

$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$
 $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$
 $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$
 $2 \sin A \sin B = \cos(A-B) - \cos(A+B)$

$$A+B = C$$

$$A-B = D$$

$$A = \frac{C+D}{2}, B = \frac{C-D}{2}$$

put the value - A & B →

C & D formula;

$$\sin C + \sin D = 2 \sin \frac{C+D}{2} \cdot \cos \frac{C-D}{2}$$

$$\sin C - \sin D = 2 \cos \frac{C+D}{2} \cdot \sin \frac{C-D}{2}$$

$$\cos C + \cos D = 2 \cos \frac{C+D}{2} \cdot \cos \frac{C-D}{2}$$

$$\cos C - \cos D = 2 \sin \frac{C+D}{2} \cdot \sin \frac{D-C}{2}$$

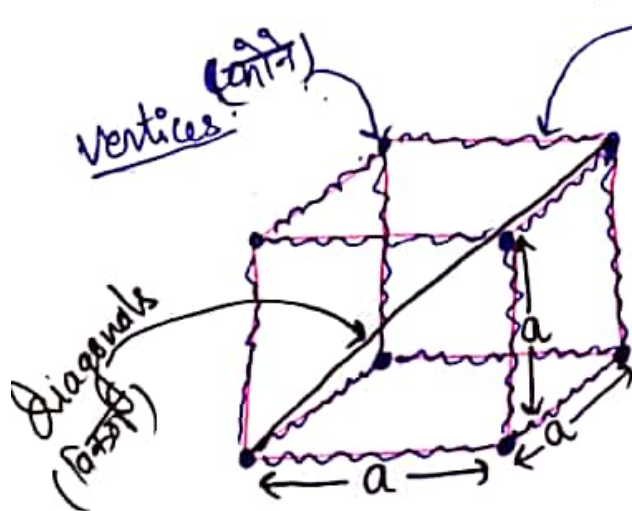


Satyam

Mensuration (क्षेत्रमिति) [3-D]

①

= Cube (घन)



- 8 vertices (कोने)
- 12 किनारे (edge)
- 6 Faces (सतह)
↓
Square (वर्ग)

⇒

Euler's formula — $V + F - E = 2$ * [For any 3-D figure]

$\left\{ \begin{array}{l} V \rightarrow \text{vertices} \\ F \rightarrow \text{Face} \\ E \rightarrow \text{Edge} \end{array} \right\}$

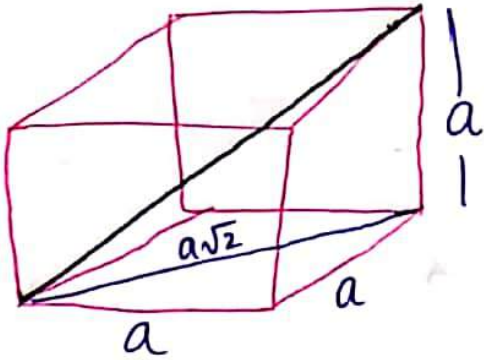
⇒ L.S.A (lateral Surface Area) = $4a^2$
(पार्श्व पृष्ठ का क्षेत्रफल)

⇒ T.S.A (Total Surface Area) = $6a^2$
(सम्पूर्ण पृष्ठ का क्षेत्रफल)

⇒ Volume (आयतन) = $a \times a \times a = a^3$

Diagonals विकर्ण = $a\sqrt{3}$

②



$$\text{diagonals} = \sqrt{(a\sqrt{2})^2 + a^2}$$

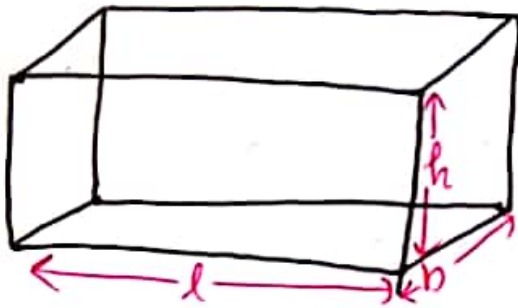
$$* \boxed{\text{diagonals} = \sqrt{3} a}$$

no. of diagonals = 4 diagonals.

#

Cuboid - (घनाभ)

③



(l, h) (b, h) (l, b)

6 - Face
12 - edge
8 → vertices

⇒

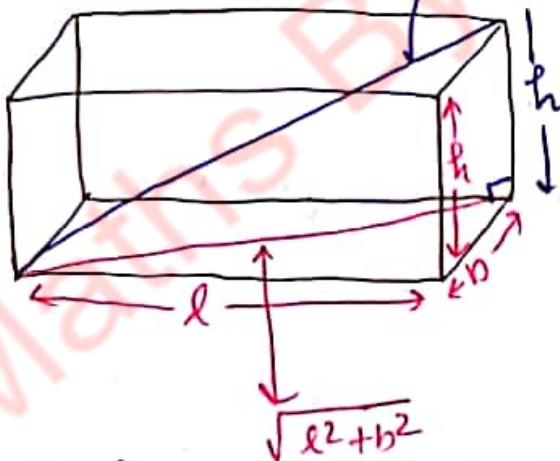
$$\text{Volume} = l \times b \times h$$

$$\Rightarrow \text{L.S.A (Lateral Surface Area)} = (lh + bh) \times 2$$

⇒

$$\text{T.S.A} = 2(lb + bh + hl)$$

⇒



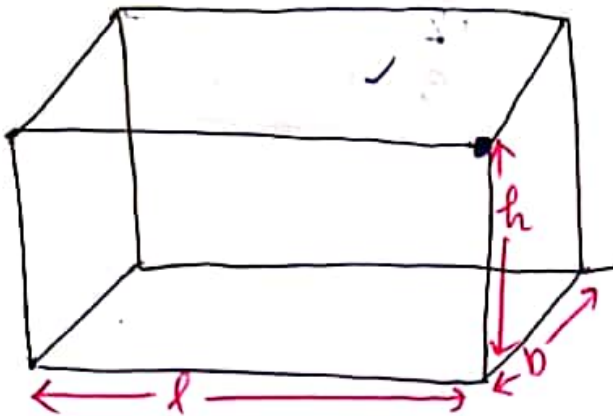
$$\text{diagonal} = \sqrt{l^2 + b^2 + h^2}$$

पाइथागोरस प्रमेय -

$$\text{diagonal} = \sqrt{l^2 + b^2 + h^2}$$

Adjacent faces:—

④



$(l, b), (b, h), (l, h) \rightarrow$ adjacent faces

eg.

$\Rightarrow$ Cuboid के तीन adjacent surface के Area x, y, z हैं
Find volume = ?

Soln—

$$x = lb \quad \text{--- ①}$$

$$y = bh \quad \text{--- ②}$$

$$z = hl \quad \text{--- ③}$$

multiply—

$$xyz = l^2 b^2 h^2$$

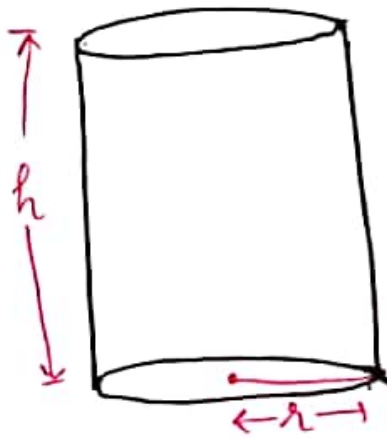
$$xyz = v^2$$

$$v = \sqrt{xyz}$$

$$v = lbh$$

Cylinder दोस बैलन

(5)



$$\text{Volume} = \pi r^2 h$$

(आयतन)

$$\text{C.S.A} = 2\pi r h$$

(Curved Surface Area)
(धातुव सतह का क्षेत्र)

Cut cylinder
(Along the height)



(rectangle) (आयत)
 $l = 2\pi r$, $b = h$

$$\text{Area} = 2\pi r h$$

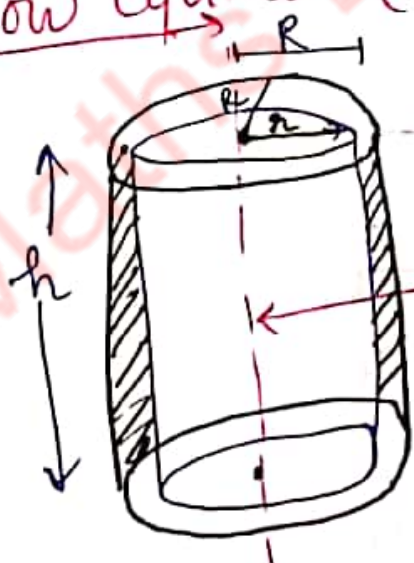
$$\text{Total Surface Area} = \text{C.S.A} + \text{Area of two bases}$$

(T.S.A)
(सम्पूर्ण सतह का क्षेत्र)

$$\text{T.S.A} = 2\pi r h + \pi r^2 \times 2$$

$$\text{T.S.A} = 2\pi r (h + r)$$

Hollow Cylinder (खोखला बेलन)



Internal radius = r
External radius = R

axis of cylinder
(अक्ष)

$$\text{Volume} = \pi R^2 h - \pi r^2 h$$

$$= \pi h (R^2 - r^2)$$

$$\text{Volume} = \pi (R + r)(R - r) \times h$$

$$C.S.A = 2\pi R h + 2\pi r h$$

$$C.S.A = 2\pi h (R + r)$$



$$T.S.A = C.S.A + \text{Area of two bases}$$

$$T.S.A = 2\pi(R+r) \times h + 2(\pi R^2 - \pi r^2)$$

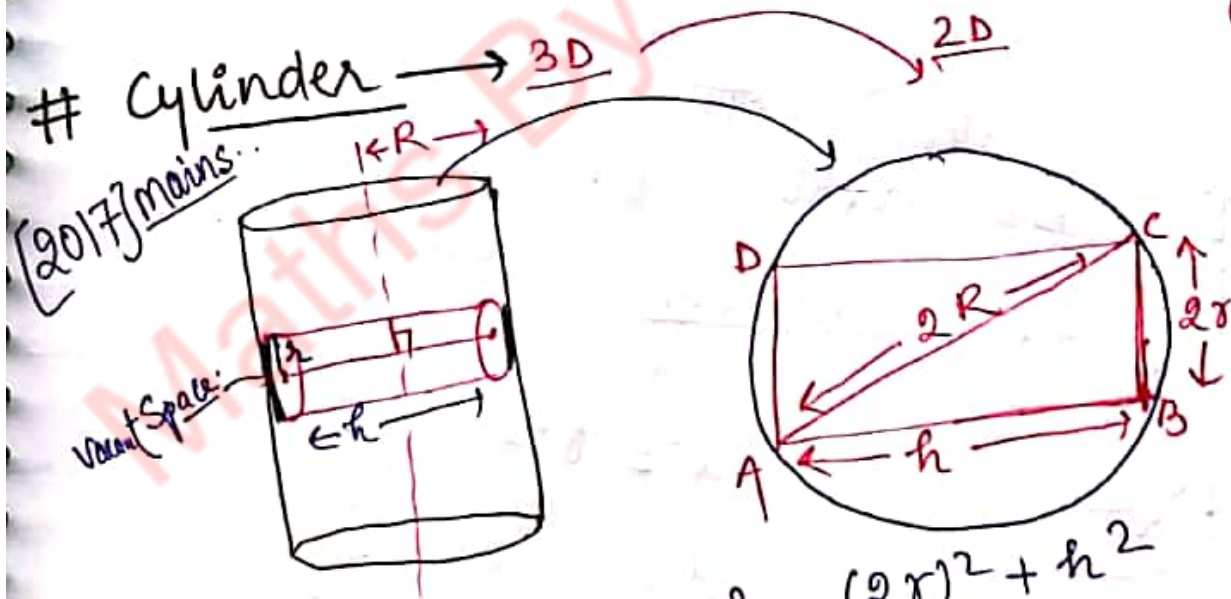
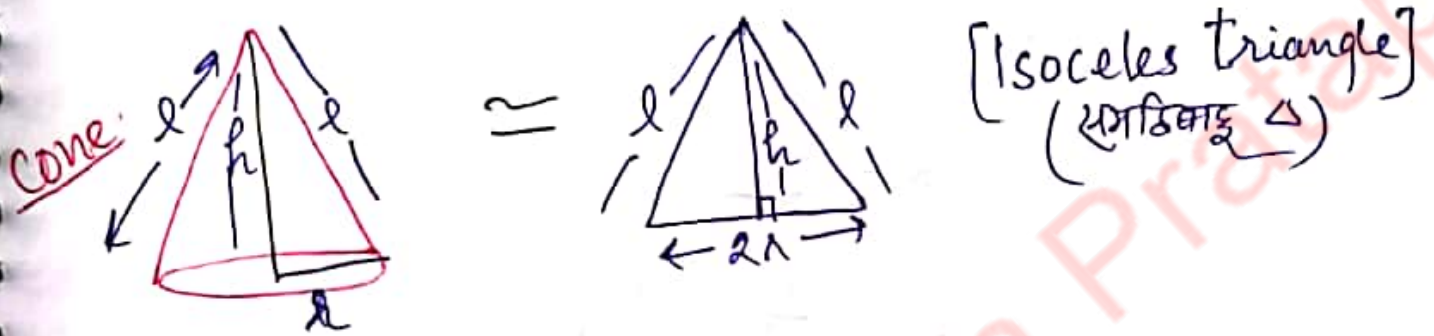
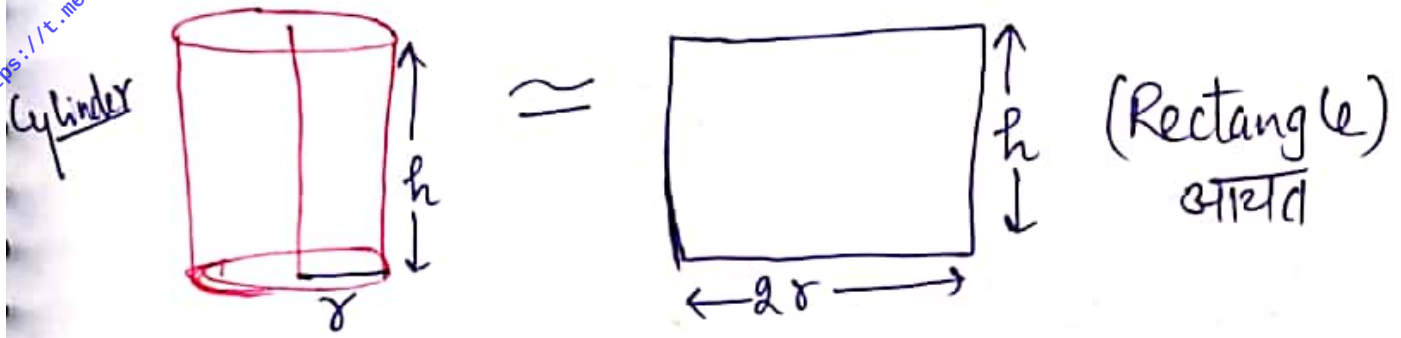
$$T.S.A = 2\pi(R+r)h + 2\pi(R^2 - r^2)$$

$$T.S.A = 2\pi(R+r)[h + R - r]$$



Cross section Area
(अनुप्रस्थ काट का क्षेत्रफल)
↓

Cross section area - 2D (7)



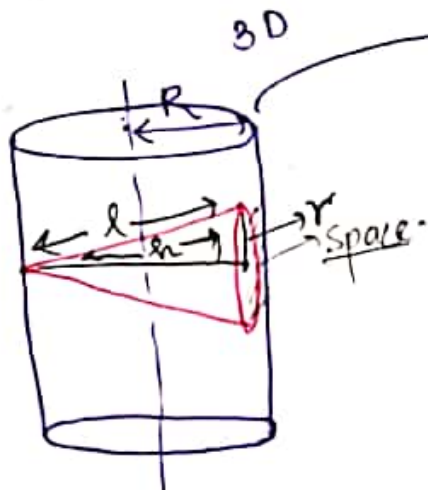
$$(2R)^2 = (2r)^2 + h^2$$

$$4R^2 = 4r^2 + h^2$$

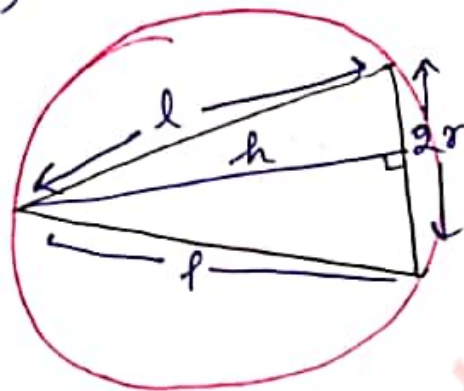
$$\text{OR } \left[\frac{R^2 - r^2}{h^2} = \frac{1}{4} \right]$$

Derivation → concept → Cone inside a cylinder

⑧



2D



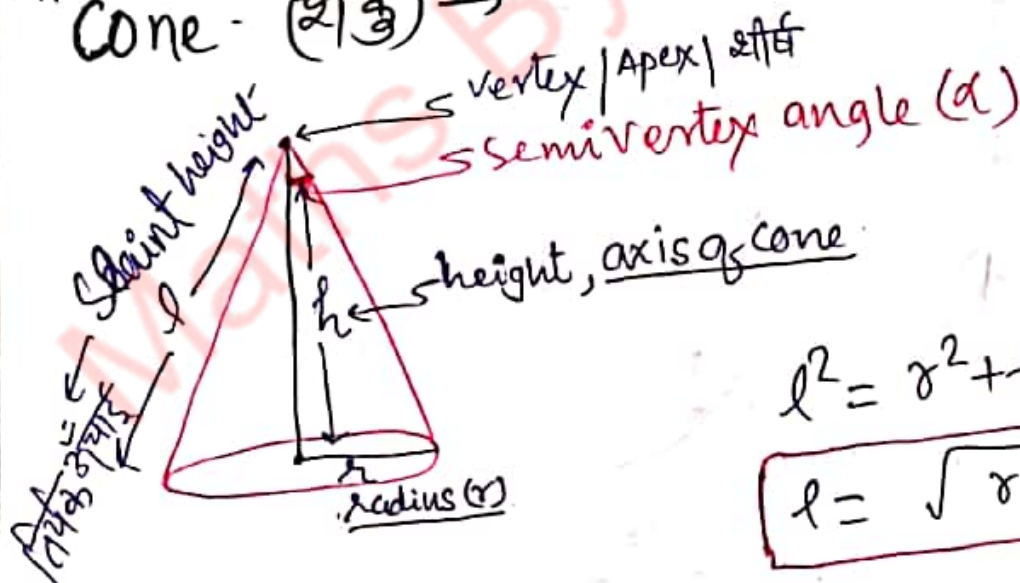
Isosceles Δ

$$R = \frac{abc}{4\Delta}$$

$$R = \frac{l \times l \times 2r}{2 \times \frac{1}{2} \times 2r \times h}$$

$$R = \frac{l^2}{2h}$$

Cone - (शंकु) →



पाइथागोरस प्रमेय से -

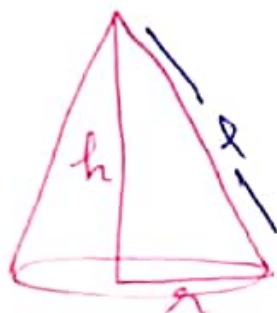
$$l^2 = r^2 + h^2$$

$$l = \sqrt{r^2 + h^2}$$

$$\text{Volume} = \frac{1}{3} \times \pi r^2 h$$

Area of base $\times$ height

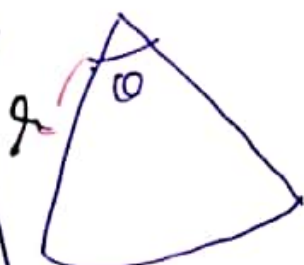
Curved Surface Area = $\pi r l$
(C.S.A) (पार्श्व पृष्ठ का क्षेत्रफल)
(वक्र)



Open cone.



$$C.S.A = \frac{1}{2} \times 2\pi r \times l = \pi r l$$



$$\text{Arc length} = 2\pi r \times \frac{\theta}{360}$$

$$\text{Sector area} = \pi r^2 \times \frac{\theta}{360}$$

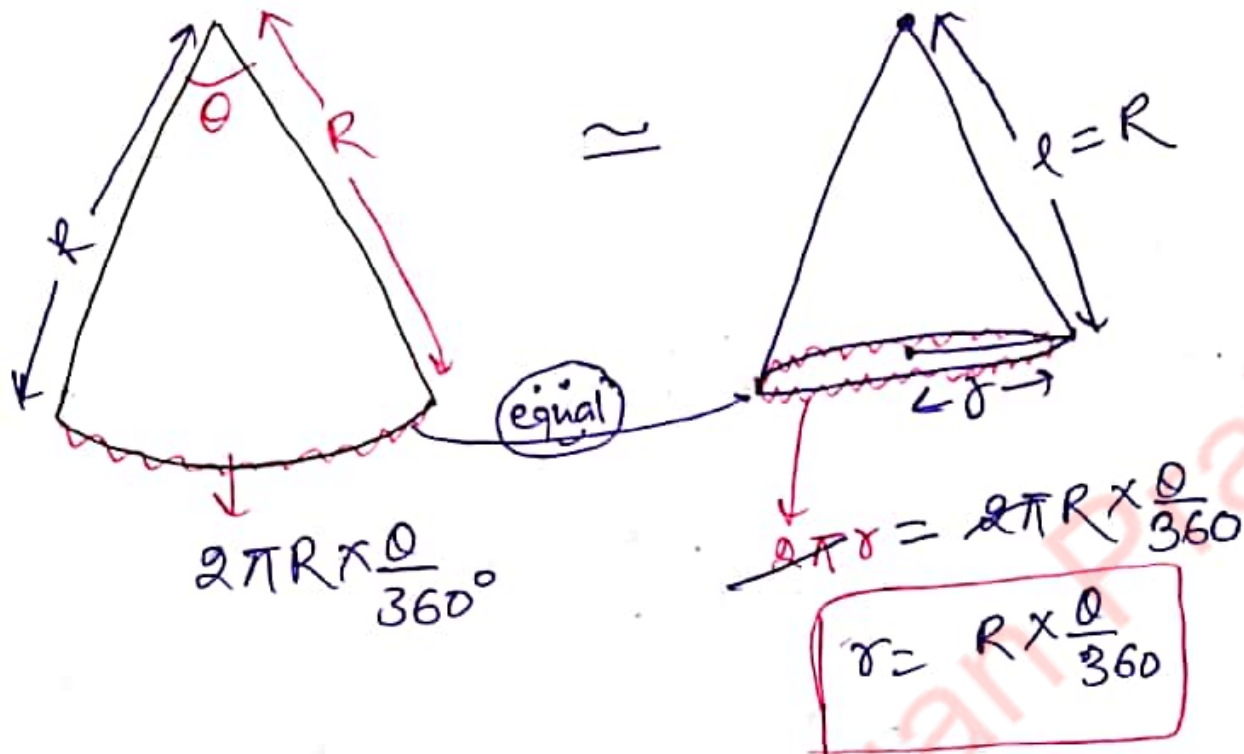
$$= \frac{1}{2} \left(2\pi r \times \frac{\theta}{360} \right) \times r$$

$$= \frac{1}{2} l \times r$$

Total Surface Area = C.S.A + Area of circular base
(सम्पूर्ण पृष्ठ का क्षेत्रफल) = $\pi r l + \pi r^2$

$T.S.A = \pi r (l + r)$

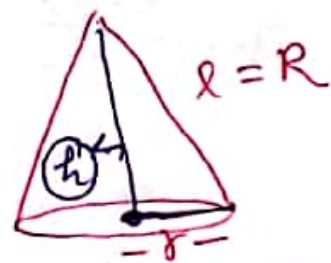
Cone formed by a sector $\rightarrow$



Case-1 Semicircle



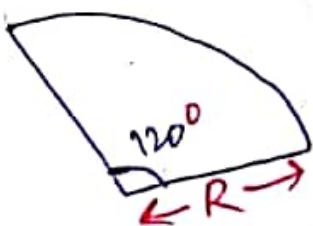
$\approx$



$$r = R \times \frac{180}{360}$$

$$r = \frac{R}{2}$$

120° sector



$\approx$



$$r = R \times \frac{120}{360}$$

$$r = \frac{R}{3}$$

By using पाइथागोरस प्रमेय
we easily find height



#

Sphere inside a cone →

(11)

Find radius of Sphere —

Using Similarity —

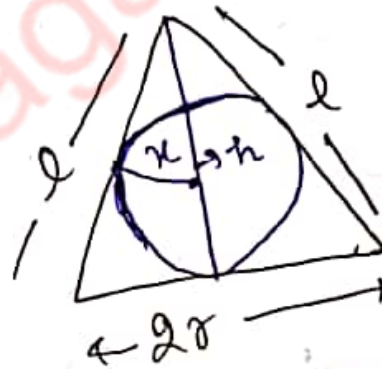
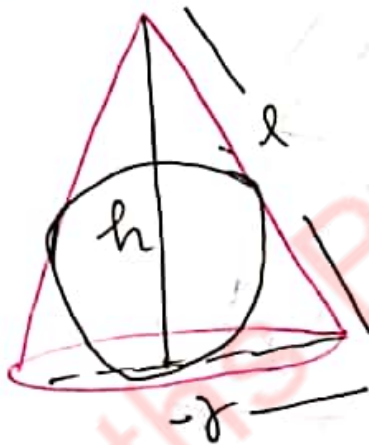
$$\frac{x}{r} = \frac{h-x}{l}$$

$$xl = hr - xr$$

$$x(l+r) = hr$$

$$x = \frac{hr}{l+r}$$

IInd Method [By using cross-section Area —]



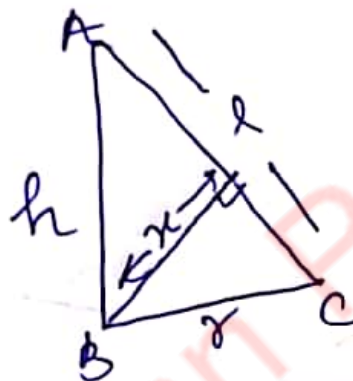
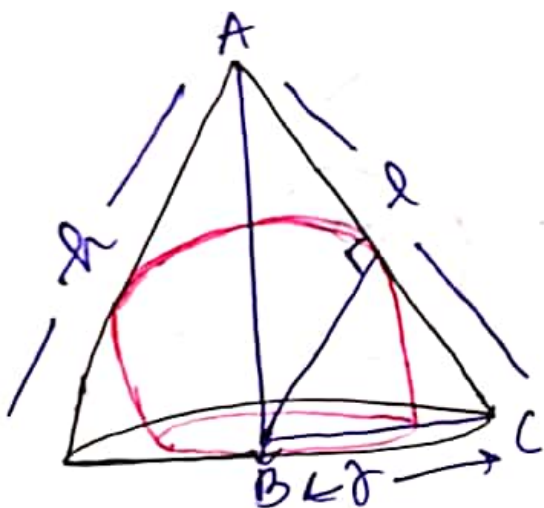
$$x = \frac{\Delta}{s} = \frac{\frac{1}{2} \times 2r \times h}{l+r}$$

$$x = \frac{rh}{l+r}$$

$$x = \frac{r \times h}{l+r}$$

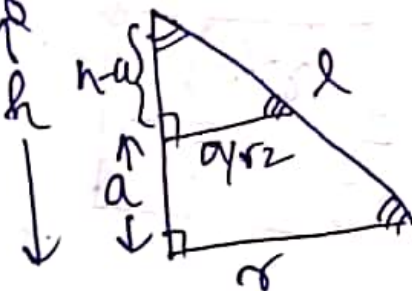
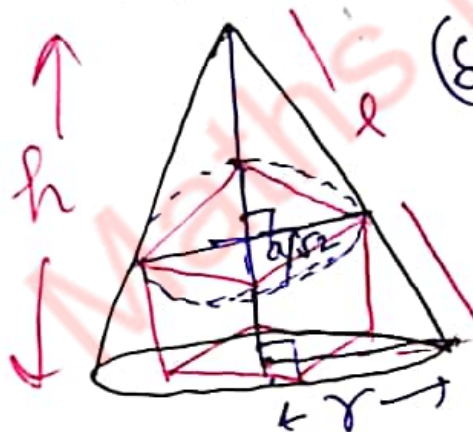
* max size
Hemisphere inside a cone →

(12)



$$x = \frac{r \times h}{l}$$

* max. size
Cube inside a cone →
 (side of cube = a)



Similarity

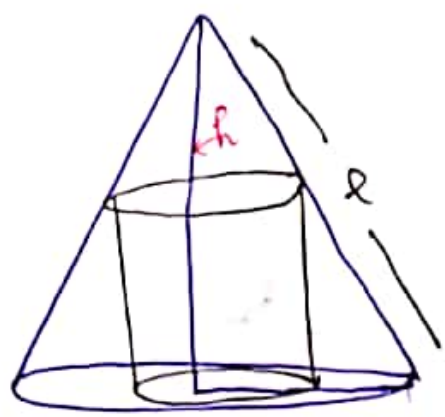
$$\frac{a}{\sqrt{2}} = \frac{h-a}{h}$$

$$\frac{a}{\sqrt{2}r} = \frac{h-a}{h}$$

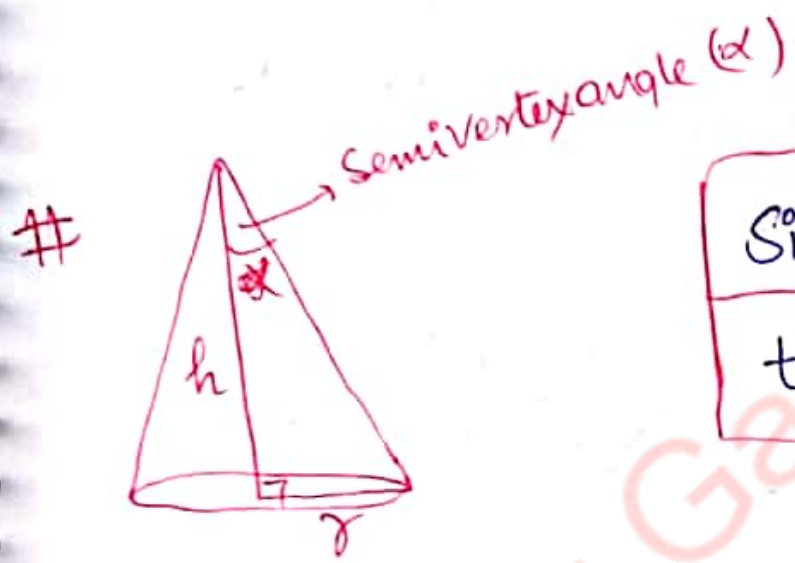
$$ah = \sqrt{2}rh - \sqrt{2}ra$$

$$a = \frac{\sqrt{2}rh}{\sqrt{2}r+h}$$

Cylinder inside a cone $\rightarrow$



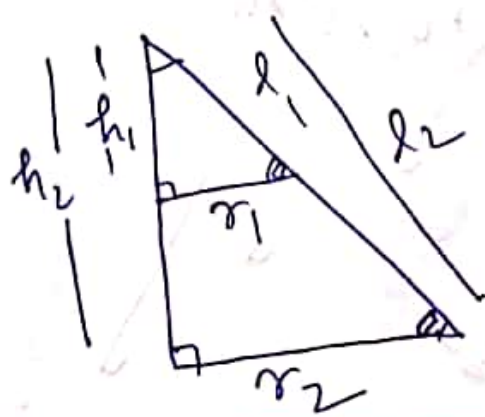
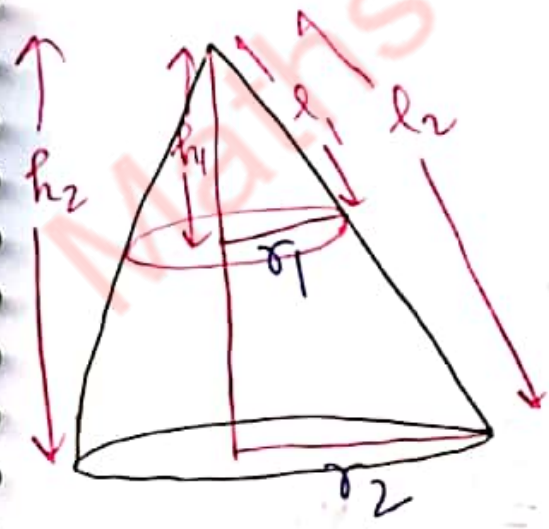
$$V_{\max} = \frac{4}{27} \pi r^2 h$$



$$\sin \alpha = \frac{r}{l}$$

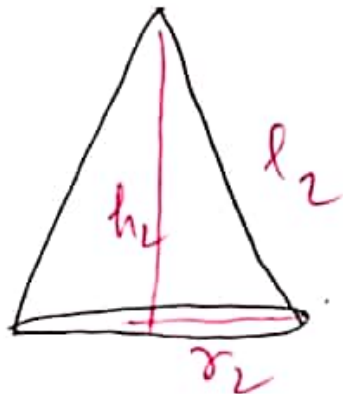
$$\tan \alpha = \frac{r}{h}$$

* Cutting of cone $\rightarrow$



$$\frac{r_1}{r_2} = \frac{h_1}{h_2} = \frac{l_1}{l_2}$$

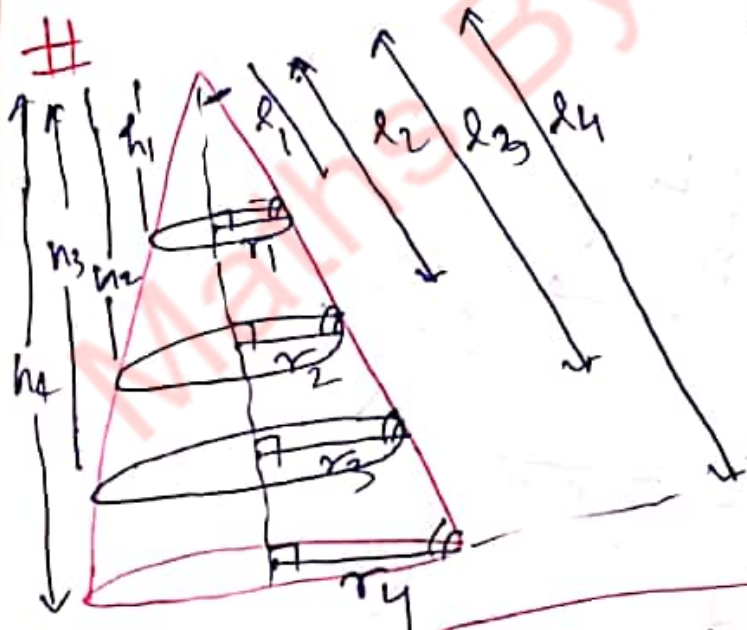
cut from top by axis



C.S.A ratio $= \pi r_1 l_1 : \pi r_2 l_2$

$$\frac{C.S.A_1}{C.S.A_2} = \frac{\pi r_1 l_1}{\pi r_2 l_2} = \frac{r_1^2}{r_2^2} = \frac{h_1^2}{h_2^2} = \frac{l_1^2}{l_2^2}$$

$$\frac{\text{Volume}_1}{\text{Volume}_2} = \frac{\frac{1}{3} \pi r_1^2 h_1}{\frac{1}{3} \pi r_2^2 h_2} = \frac{r_1^3}{r_2^3} = \frac{h_1^3}{h_2^3} = \frac{l_1^3}{l_2^3}$$

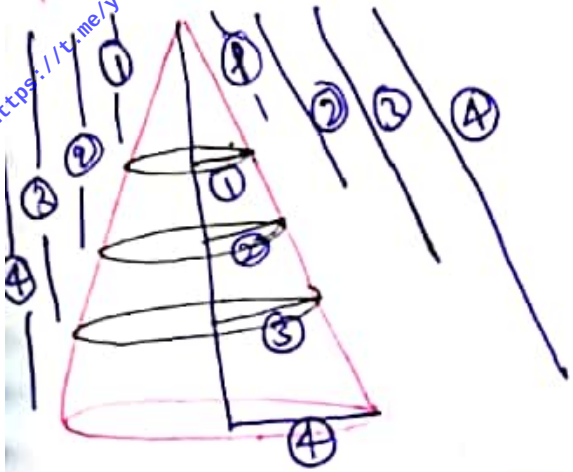


$$r_1 : r_2 : r_3 : r_4 : \dots = h_1 : h_2 : h_3 : h_4 : \dots$$

$$= l_1 : l_2 : l_3 : l_4 : \dots$$

Equal height

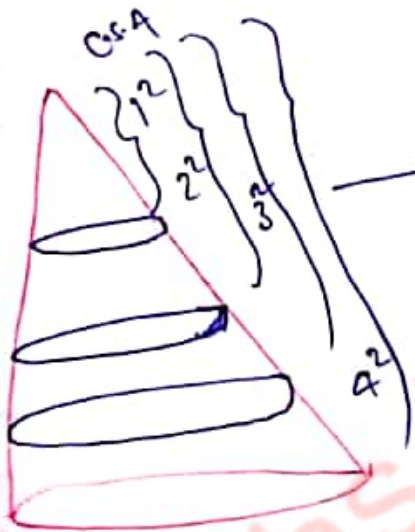
(15)



$$r_1 : r_2 : r_3 : r_4 = h_1 : h_2 : h_3 : h_4 \\ = l_1 : l_2 : l_3 : l_4 \\ = 1 : 2 : 3 : 4$$

Q*

$$C.S.A = 1^2 : 2^2 : 3^2 : 4^2 \\ \text{Volumes} = 1^3 : 2^3 : 3^3 : 4^3$$



C.S.A



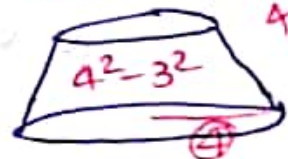
volume
13



$2^3 - 1^3$



$3^3 - 2^3$



$4^3 - 3^3$

$$C.S.A = 1^2 : 2^2 - 1^2 : 3^2 - 2^2 : 4^2 - 3^2$$

$$C.S.A = 1 : 3 : 5 : 7 : 9$$

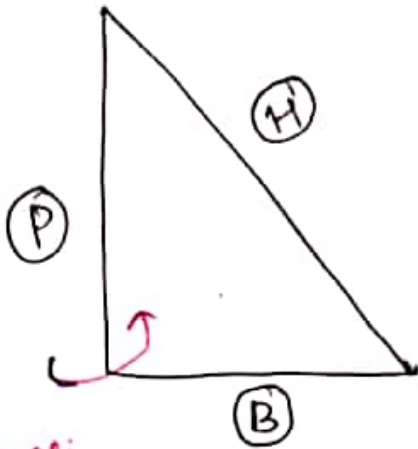
$$\text{Volume} = 1^3 : 2^3 - 1^3 : 3^3 - 2^3 : 4^3 - 3^3$$

$$\text{Volume} = 1 : 7 : 19 : 37$$

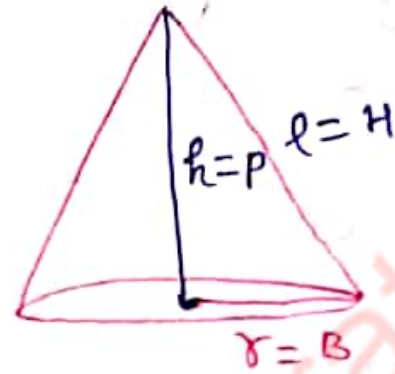
Signature

Cone formed by rotating a right angle triangle. $\rightarrow$ (16)

1st case:

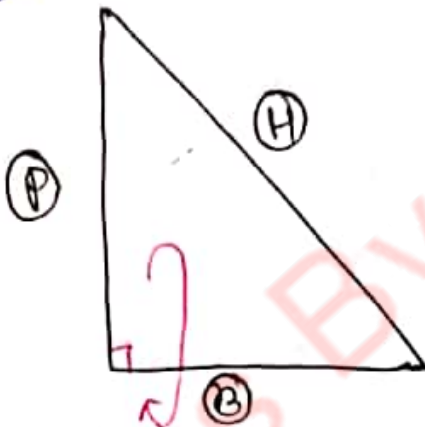


$\approx$

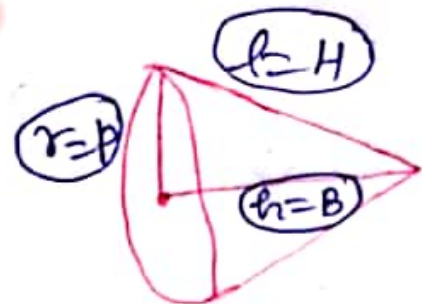


Along Base:

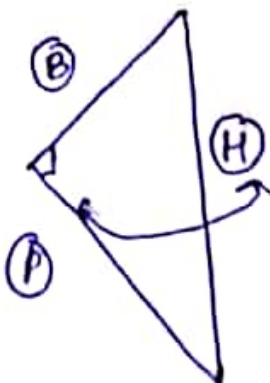
2nd case:



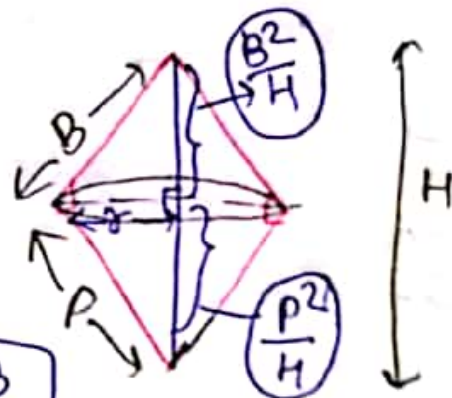
$\approx$



Double Cone



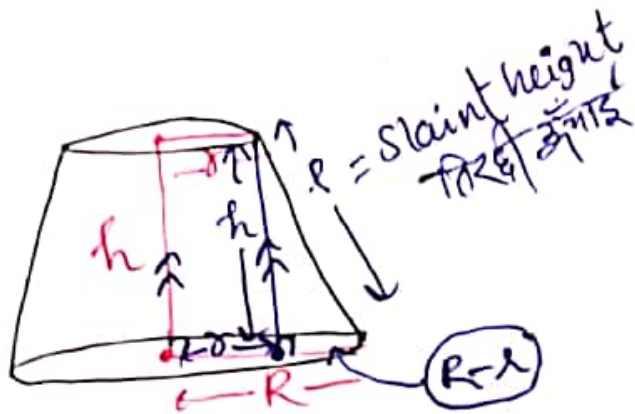
$\approx$



$$r = \frac{P \times B}{H}$$

Frustum (खंड) →

(17)



पाइथागोरस प्रमेय →

$$l^2 = h^2 + (R-r)^2$$

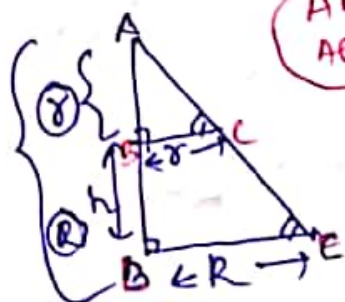
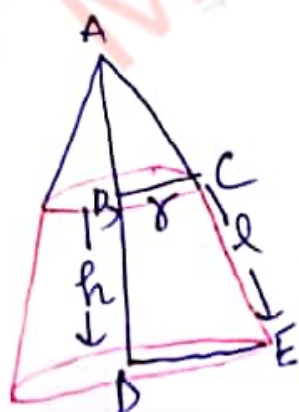
$$l = \sqrt{h^2 + (R-r)^2}$$

$$\text{Volume} = \frac{1}{3} \pi (R^2 + r^2 + Rr) h$$

$$C.S.A = \pi (R+r) l$$

$$T.S.A = \pi (R+r) l + \pi R^2 + \pi r^2$$

$$T.S.A = \pi (R+r) l + \pi (R^2 + r^2)$$



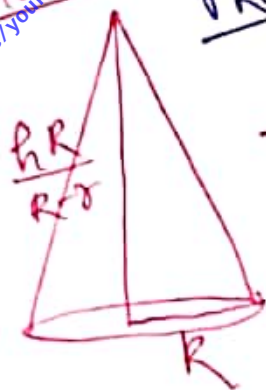
$$\begin{aligned} AB & \text{ \& } AD = r \text{ \& } R \\ AE & \text{ \& } AE = r \text{ \& } R \end{aligned}$$

$$\begin{aligned} AD - AB &= BD \\ (R-r) \text{ unit} &\rightarrow h \\ 1 \text{ unit} &\rightarrow \frac{h}{R-r} \end{aligned}$$

$$r \text{ unit} = AB = \frac{hr}{R-r}, R \text{ unit} = AD = \frac{hR}{R-r}$$

PROOF

(18)



=

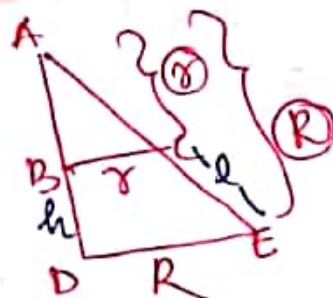
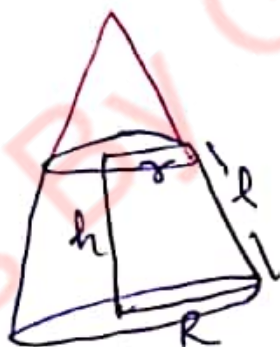


$$\text{Volume of frustum} = \frac{1}{3} \pi R^2 \times \frac{hR}{R-r} - \frac{1}{3} \pi r^2 \times \frac{hr}{R-r}$$

$$= \frac{\pi}{3} \frac{h}{(R-r)} [R^3 - r^3] = \frac{\pi}{3} \times \frac{h}{R-r} \times (R-r)(R^2 + Rr + r^2)$$

$$V = \frac{\pi}{3} h (R^2 + Rr + r^2)$$

CoS.A = ?



$$Ae : AE = r : R$$

$$(R-r) \text{ unit} = l$$

$$1 \text{ --- } \frac{l}{(R-r)}$$

$$\left\{ Ae = \frac{lr}{R-r}, AE = \frac{lR}{R-r} \right\}$$

frustum
CoS.A =

$$\pi R \times \frac{lR}{R-r} - \pi r \times \frac{lr}{R-r}$$

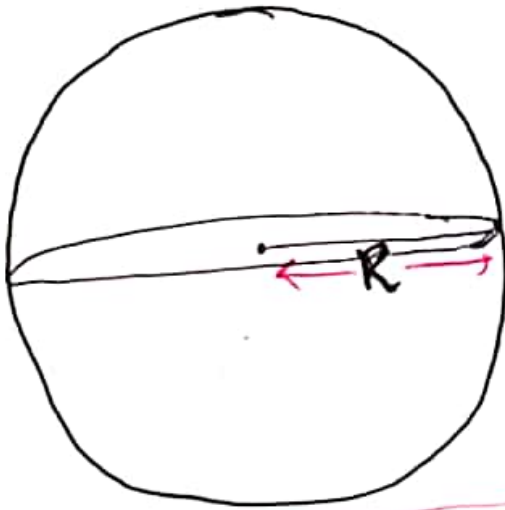
$$= \frac{\pi l}{(R-r)} (R^2 - r^2) \Rightarrow$$

$$\boxed{\pi l (R+r)}$$

इति सिद्धम् ॐ

[Sphere (गैसगोला)]

①

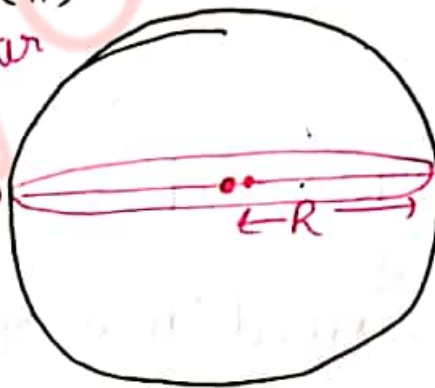


$$* T.S.A = C.S.A = 4\pi R^2$$

$$* \text{Volume} = \frac{4}{3}\pi R^3$$

* hemi-sphere (अर्धगोला) -

cut along diameter



Hemisphere



$$* C.S.A = 2\pi R^2$$

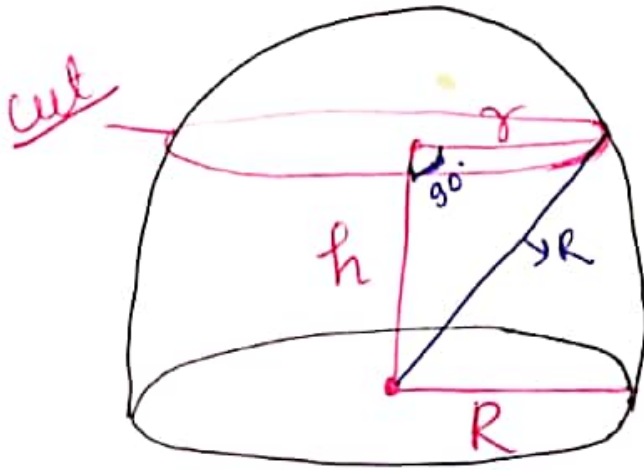
$$T.S.A = 2\pi R^2 + \pi R^2$$

$$* T.S.A = 3\pi R^2$$

$$* \text{Volume} = \frac{2}{3}\pi R^3$$

(2)

at distance of "h" meters



$$R^2 = r^2 + h^2$$

$$* \quad r = \sqrt{R^2 - h^2}$$

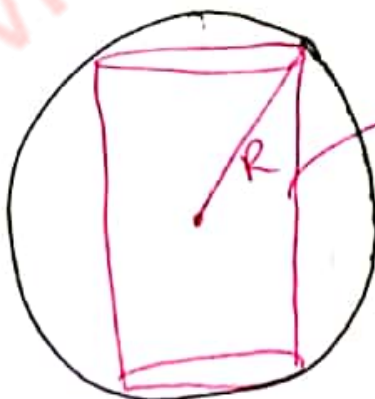
#

max. Size ^{cut} of Cube.



$$\text{Side of Cube (a)} = \sqrt{\frac{2}{3}} R$$

Cylinder inscribed in a Sphere —



Cylinder at ^{volume} $V_{\max}$

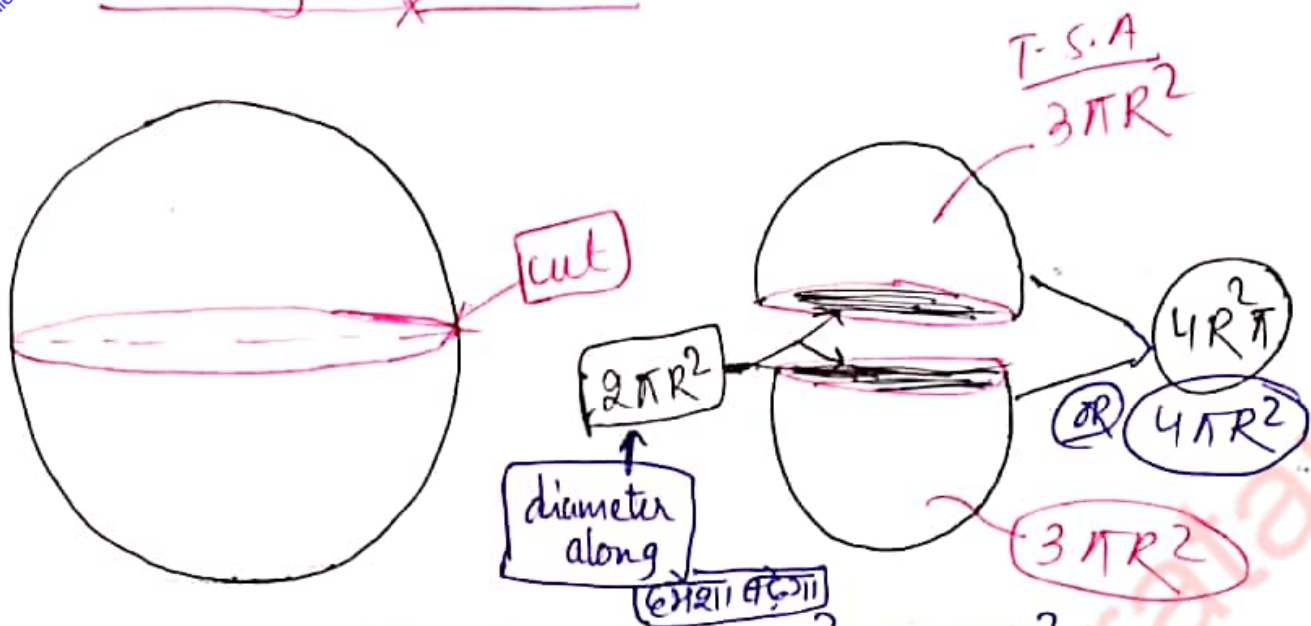
$$= \frac{1}{\sqrt{3}} \times \frac{4}{3} \pi R^3$$

Sphere at volume at $\frac{1}{\sqrt{3}}$ time

Scanned

* Cutting of Sphere →

(4)

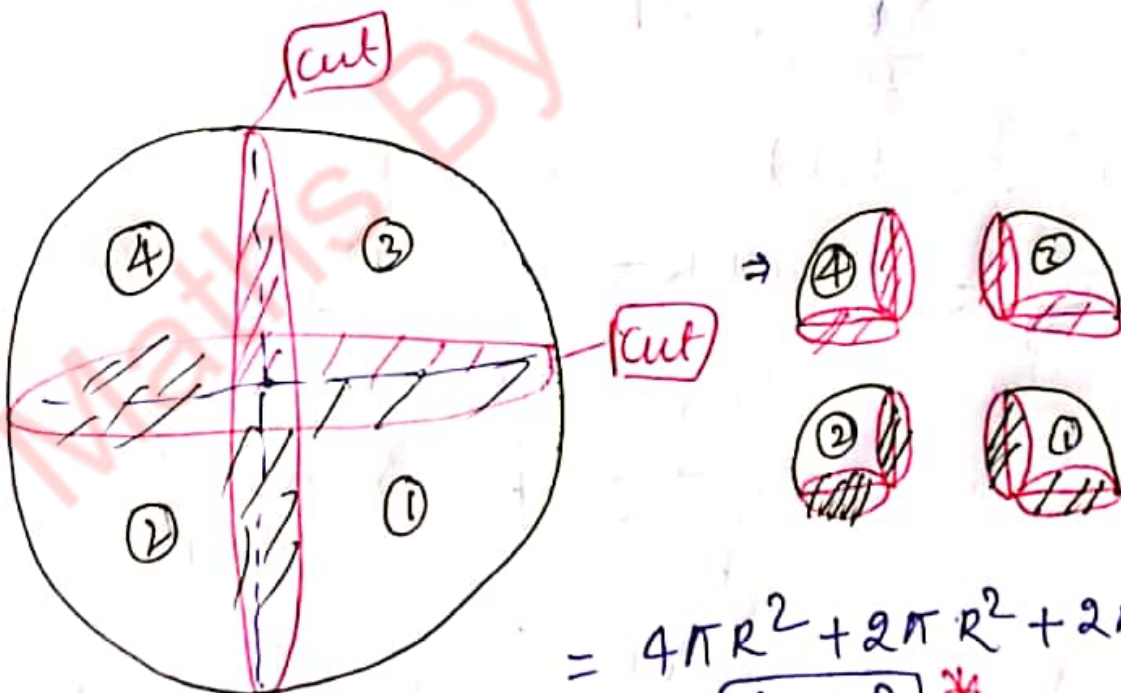


New - T.S.A = $4\pi R^2 + 2\pi R^2$

T.S.A = $6\pi R^2$

T.S.A of each part = $3\pi R^2$

#

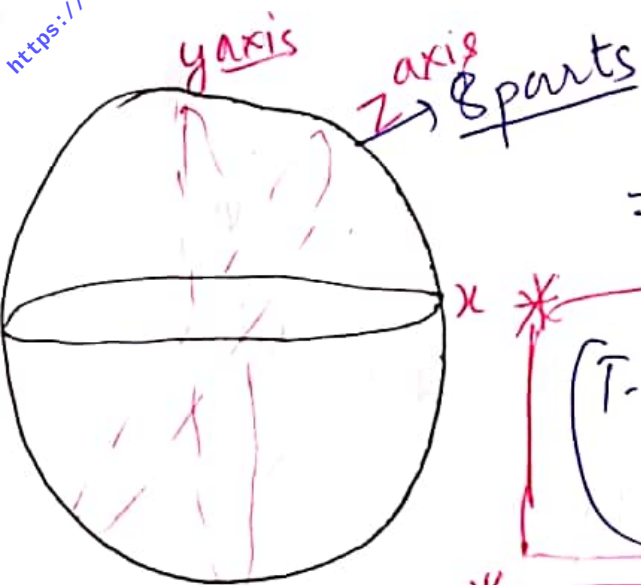


T.S.A of each part = $2\pi R^2$

= $4\pi R^2 + 2\pi R^2 + 2\pi R^2$

= $8\pi R^2$

T.S.A of all 4 part -



x, y, z axis ~~is~~ long. (5)
3 cuts - Increased Area of each $\rightarrow 2\pi R^2$
 $\Rightarrow 4\pi R^2 + 2\pi R^2 + 2\pi R^2 + 2\pi R^2$

$$\left(\begin{array}{l} \text{T.S.A of all} \\ \text{8 parts together} \end{array} \right) = 10\pi R^2$$

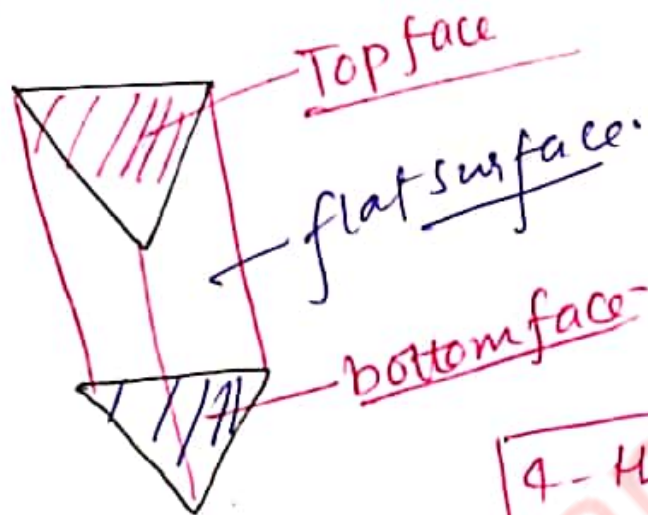
$$\begin{aligned} \text{T.S.A of each part} &= \frac{10\pi R^2}{8} \\ &= \frac{5}{4} \pi R^2 \end{aligned}$$

Prism प्रिझम →

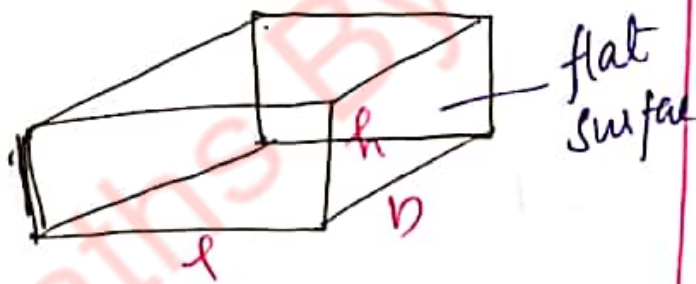
(6)

Top & bottom are same.

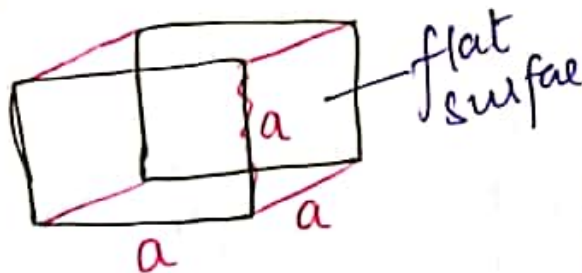
Prism with triangular base.



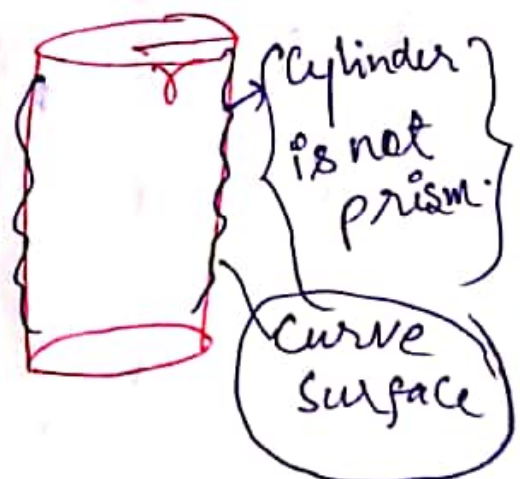
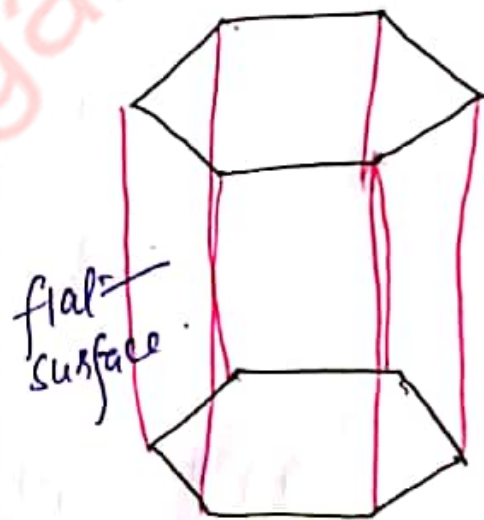
Prism 2- with base rectangle -



3- Prism with Square base.



4- Hexagon base.



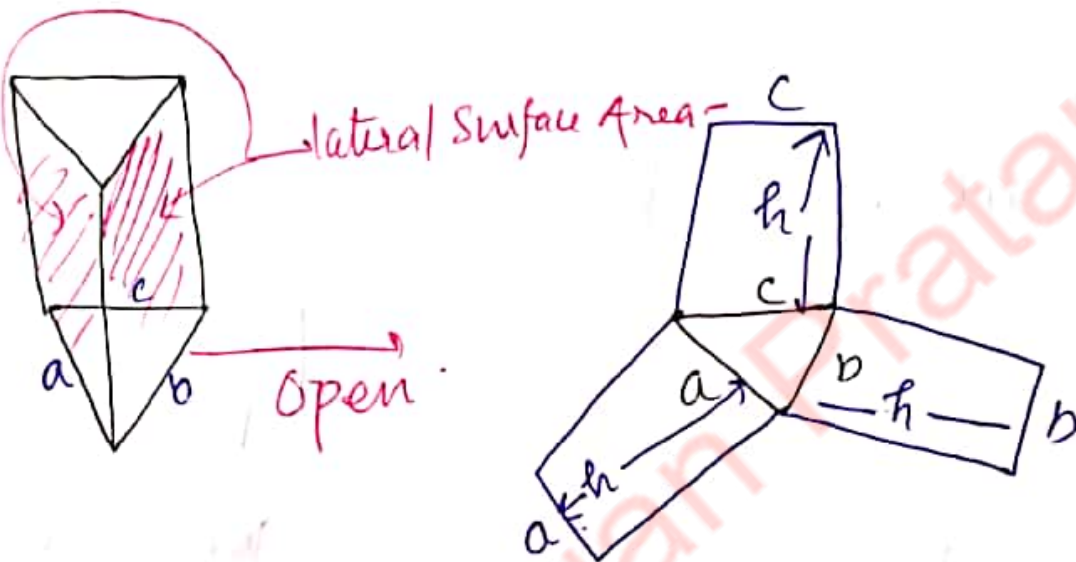
Satnam Gadhvi

formula of Prism,

(7)

① $L.S.A = \text{perimeter of base} \times \text{height}$
(पार्श्व पृष्ठ का क्षेत्र) = (आधार का परिमाप $\times$ ऊँचाई)

eg.



$$\begin{aligned} \text{Area} &= ah + bh + ch \\ &= (a + b + c) \times h \\ &\quad \downarrow \\ &\quad \text{[perimeter of triangle]} \times \text{height} \end{aligned}$$

② $T.S.A = L.S.A + \text{Area of two bases.}$

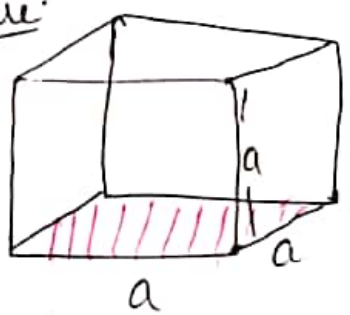
(सम्पूर्ण पृष्ठ का क्षेत्र)

③ $\text{Volume of prism} = \text{Area of base} \times \text{height}$

(आयतन = आधार का क्षेत्रफल $\times$ ऊँचाई)

eg.

Square.

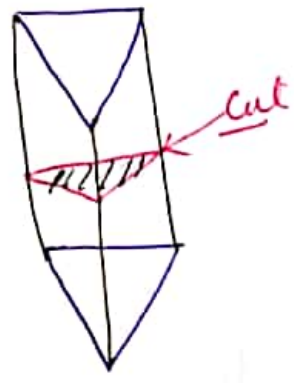


Volume of cube = a^3

$$V = a^2 \times a$$

Area of base $\rightarrow$ height

Cutting of Prism $\rightarrow$



{ 2 new surface }
equal to base area.
(create हो जाते हैं)

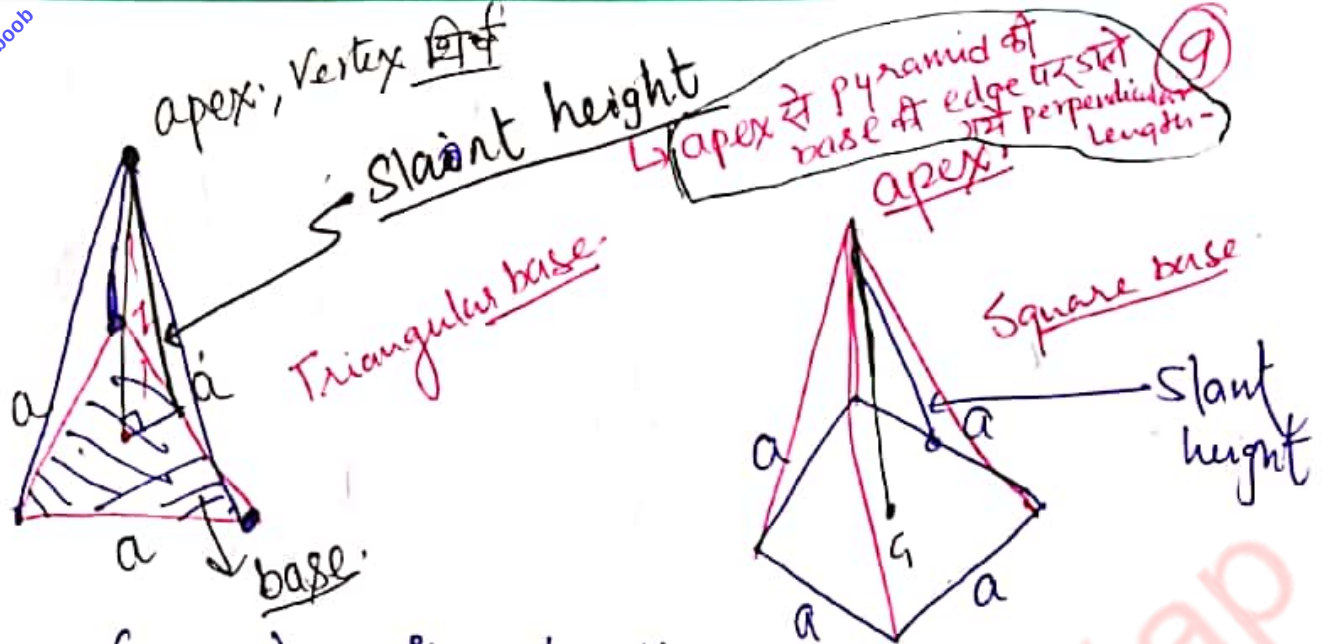
Pyramid - (पिरामिड)

$\rightarrow$ polyhedron (बहुफलक)

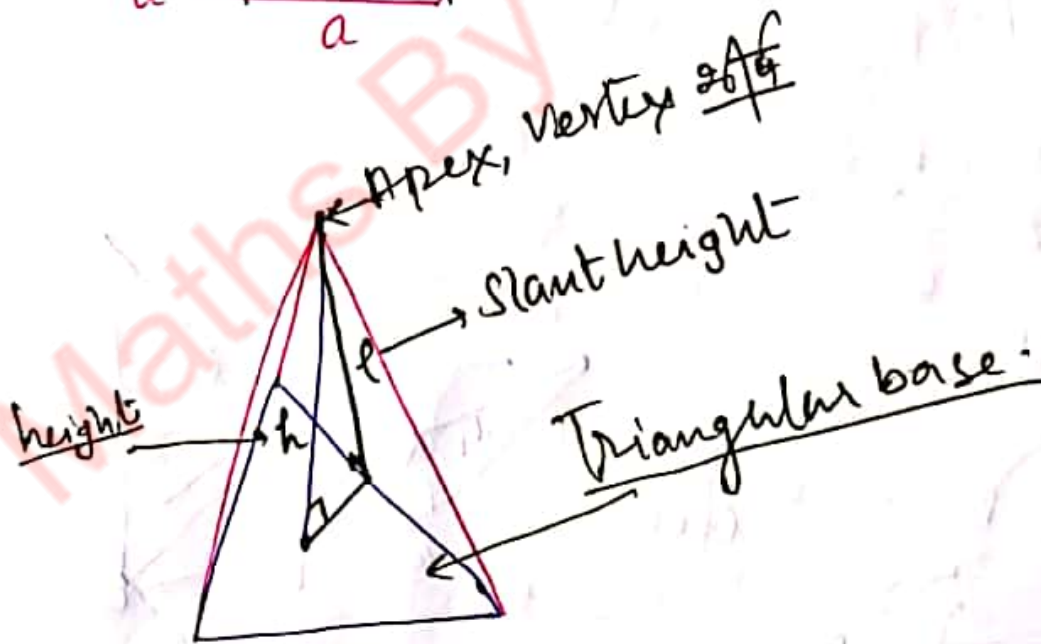
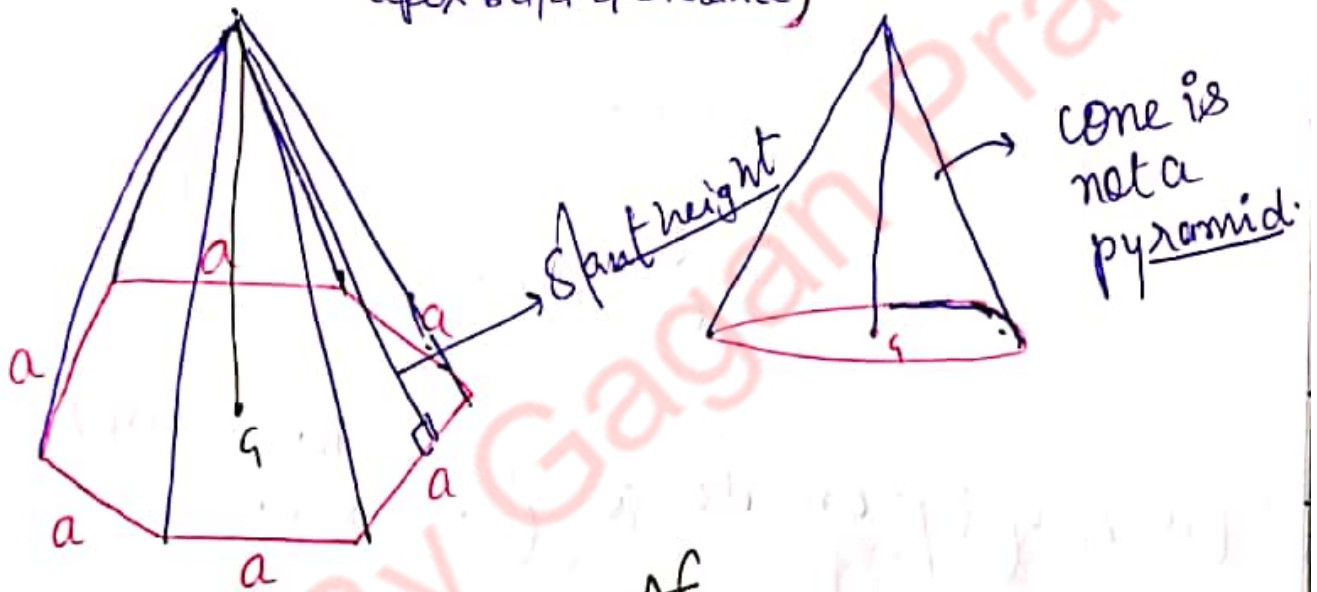
$\rightarrow$ { Pyramid is a conical solid with polygon base. }

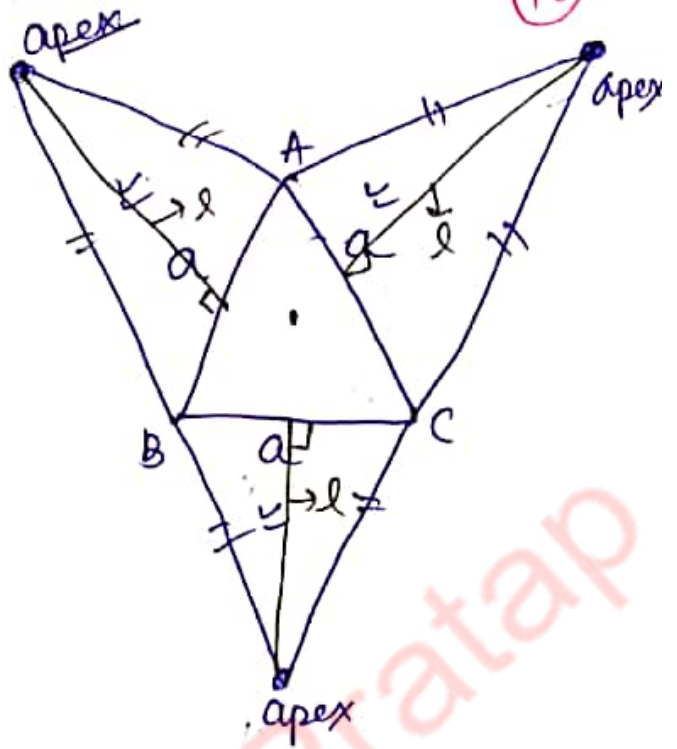
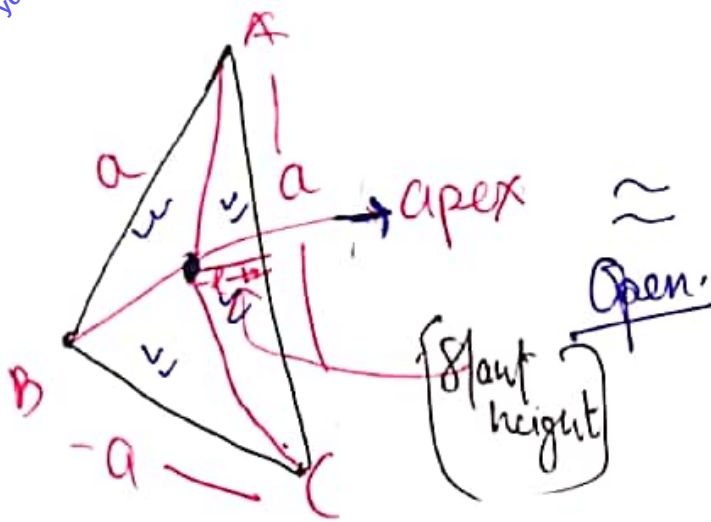
Summary

$\rightarrow$ Pyramid with Regular Triangle:



height $\rightarrow$ (base on gravity centre और apex के बीच की Distance)





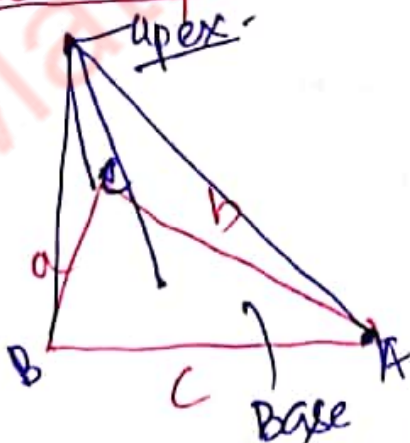
$$\begin{aligned} \text{Area} &= \frac{1}{2} a \times l \times 3 \\ &= \frac{1}{2} (3a) \times l \end{aligned}$$

↓
perimeter of base

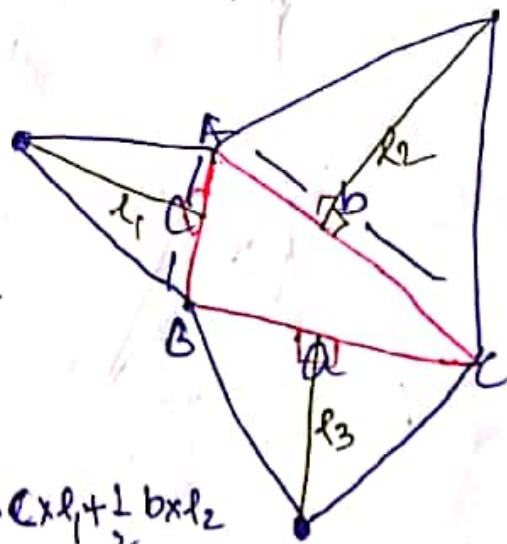
slant height

L.S.A of Pyramid = $\frac{1}{2} \times \text{perimeter of base} \times \text{slant height}$
 (आधार का परिमाप $\times$ तिरछी ऊँचाई)

Scalene triangle



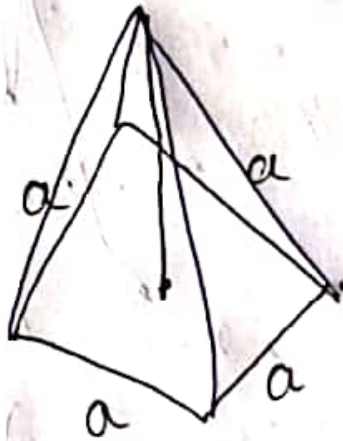
Open.



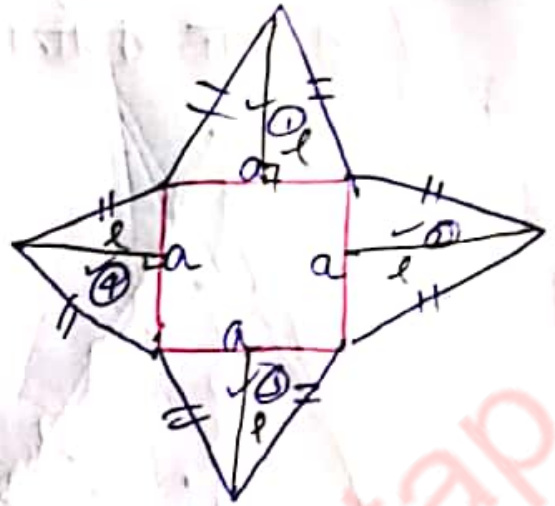
$$\begin{aligned} \text{Area} &= \frac{1}{2} \times c \times l_1 + \frac{1}{2} \times b \times l_2 + \frac{1}{2} \times a \times l_3 \\ \text{L.S.A} &= \frac{1}{2} [c l_1 + b l_2 + a l_3] \end{aligned}$$

Pyramid with Square base.

(11)



$\approx$
Open



$$\text{Area} = \frac{1}{2} a \times l \times 4$$

$$\text{L.S.A of pyramid} = 2al \text{ OR } = \frac{1}{2} [4a] \times l$$

Slant height

perimeter of base

regular figures

$$\text{L.S.A of pyramid} = \frac{1}{2} \times \text{perimeter of base} \times \text{Slant height}$$

(लार्थ वृष्ठ का क्षेत्र)

$$\text{T.S.A of pyramid} = \text{L.S.A} + \text{area of base}$$

(सम्पूर्ण वृष्ठ का क्षेत्र)

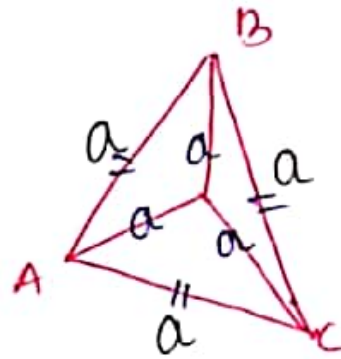
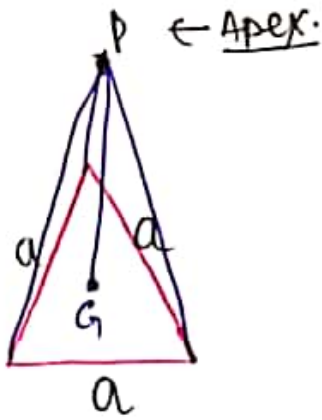
$$\Rightarrow \text{Volume of pyramid} = \frac{1}{3} \times \text{area of base} \times \text{height}$$

(आयतन)

Tetrahedron [समचतुष्फलक] —

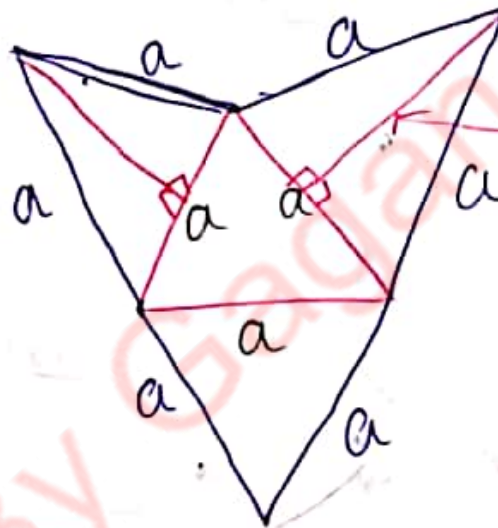
(13)

↳ [All 4 surfaces are equilateral Δ with same Side.]



[Slanted edge = a]

Open



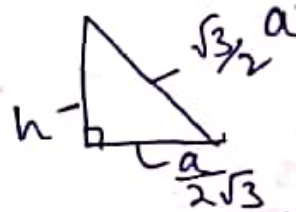
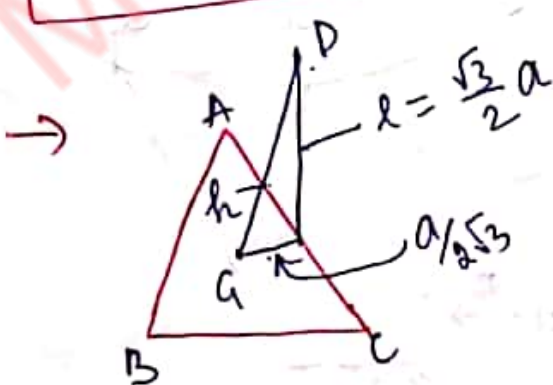
Slant height of Tetrahedron:

$$h = \frac{\sqrt{3}}{2} a$$

of Tetrahedron.

$$L.S.A = \frac{\sqrt{3}}{4} a^2 \times 3$$

$$T.S.A = \frac{\sqrt{3}}{4} \times a^2 \times 4 = \sqrt{3} a^2$$



$$h^2 + \left(\frac{a}{2\sqrt{3}}\right)^2 = \left(\frac{\sqrt{3}}{2} a\right)^2$$

$$h^2 = \frac{3}{4} a^2 - \frac{a^2}{12} = \frac{9a^2 - a^2}{12}$$

$$h^2 = \frac{8a^2}{12} = \frac{2a^2}{3}$$

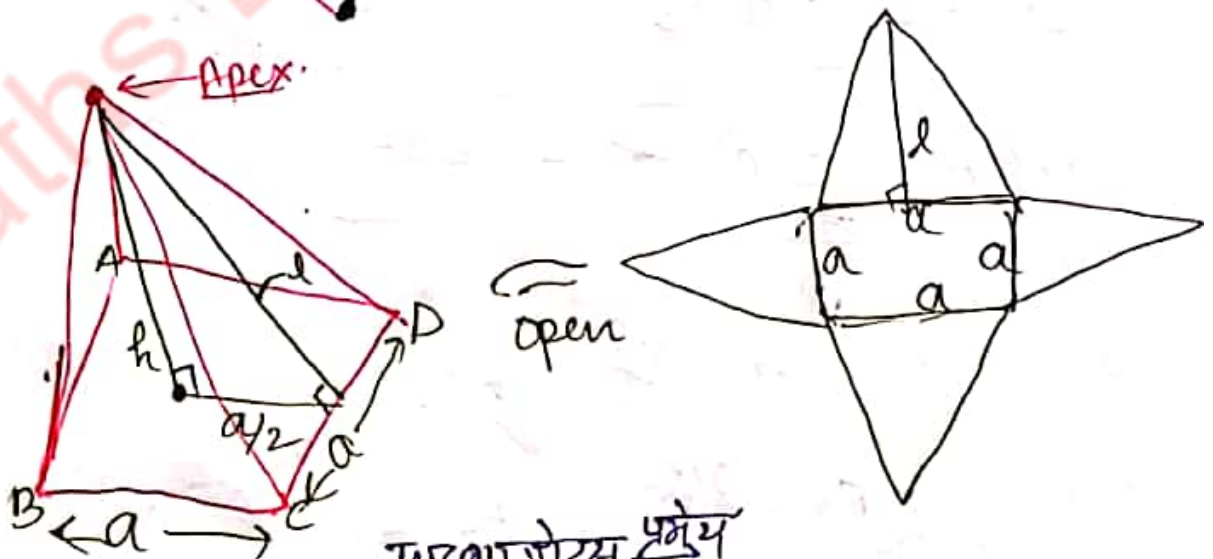
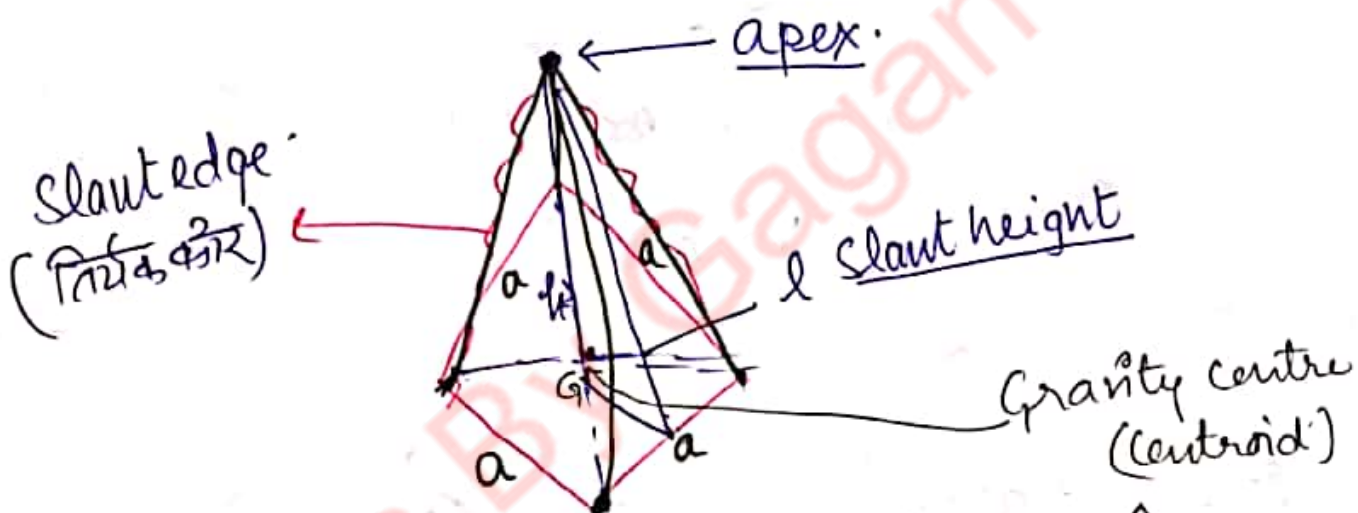
$$h = \frac{\sqrt{2}}{3} a$$

$$\text{Volume of tetrahedron} = \frac{1}{3} \times \frac{\sqrt{3}}{4} a^2 \times \frac{\sqrt{2}}{\sqrt{3}} a$$

$$= \frac{\sqrt{2} a^3}{12}$$

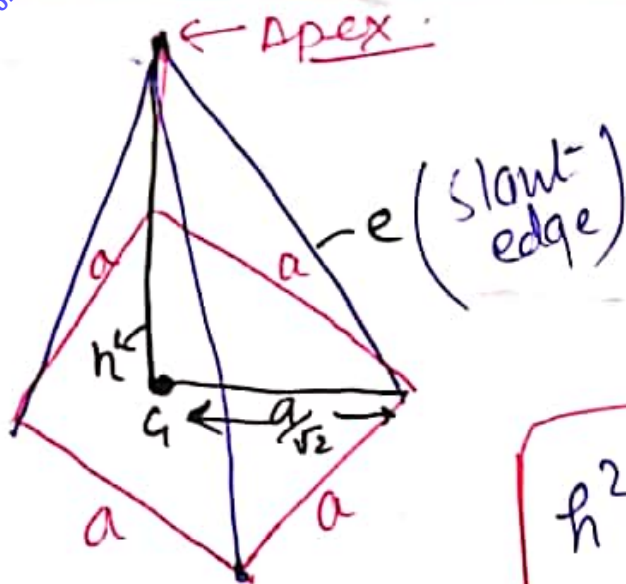
$$V = \frac{a^3}{6\sqrt{2}}$$

A pyramid with Square base —



पाइथागोरस प्रमेय

$$l^2 = h^2 + \left(\frac{a}{2}\right)^2$$

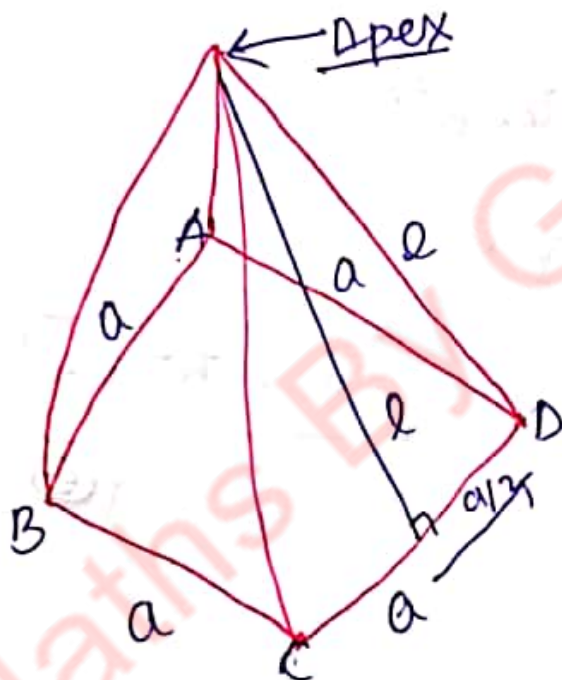


(diagonals)
 $\frac{d}{2} = \frac{a\sqrt{2}}{2}$
 $= \frac{a}{\sqrt{2}}$

(15)

$$h^2 = e^2 - \left(\frac{a}{\sqrt{2}}\right)^2$$

$\Rightarrow$ l, e, a relation Square base pyramid



$$l^2 + \left(\frac{a}{2}\right)^2 = e^2$$

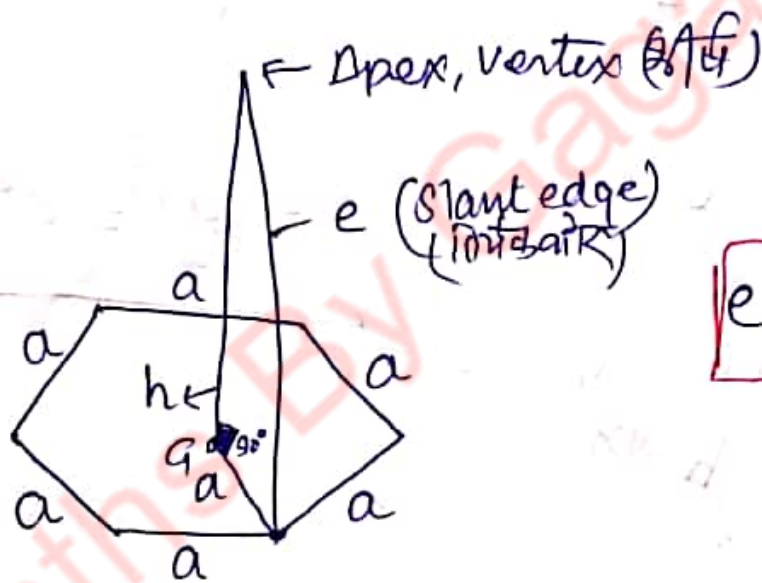
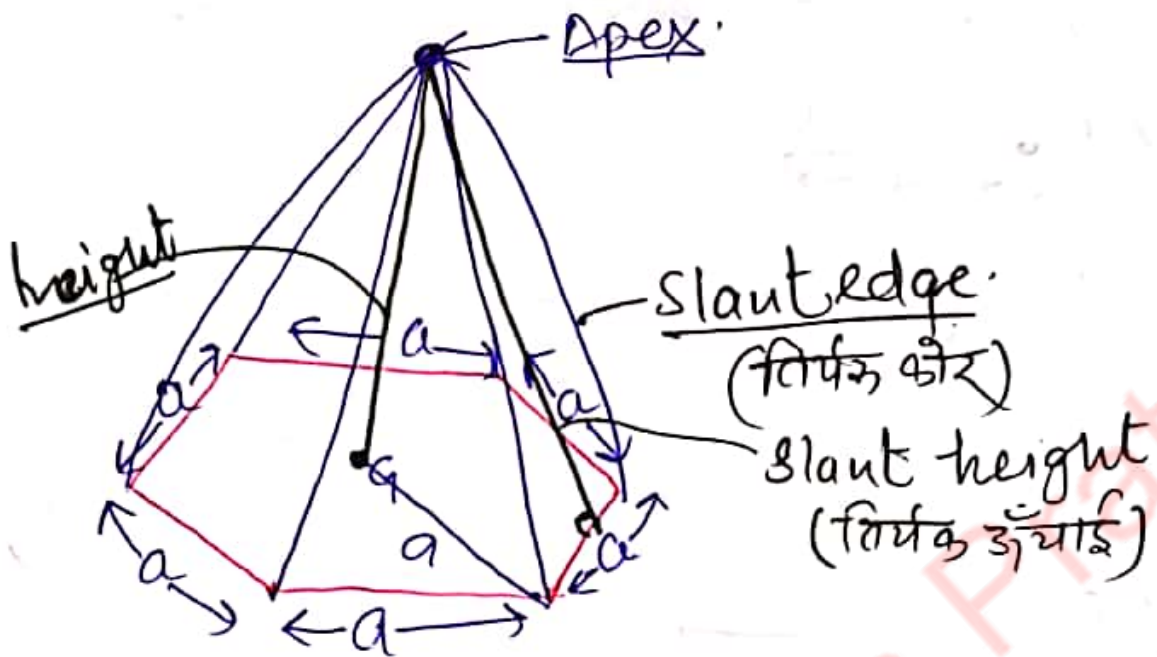
*

$$l^2 = h^2 + \left(\frac{a}{2}\right)^2$$

$$h^2 = e^2 - \left(\frac{a}{\sqrt{2}}\right)^2$$

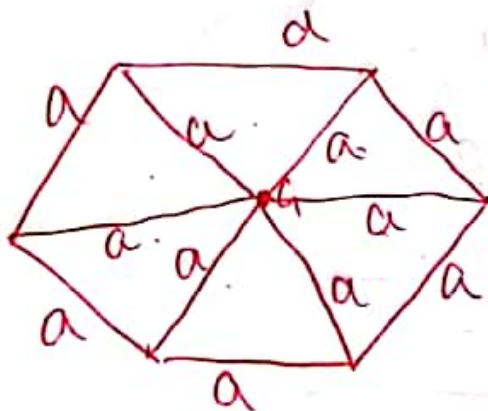
$$e^2 = l^2 + \left(\frac{a}{2}\right)^2$$

Pyramid with hexagon base — (16)



$$e = \sqrt{h^2 + a^2}$$

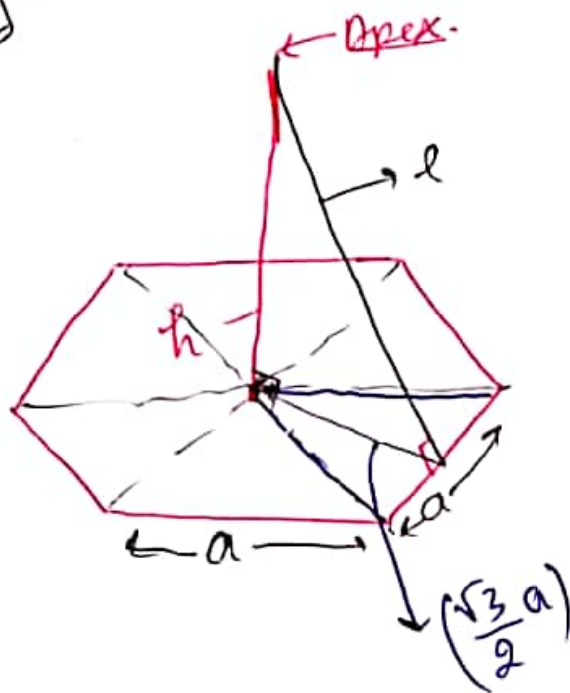
(2)



* { 6-equilateral Δ होते हैं
hexagon Centroid से मिलाने पर }

Exhame

3-D →

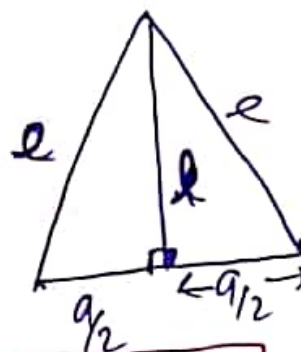
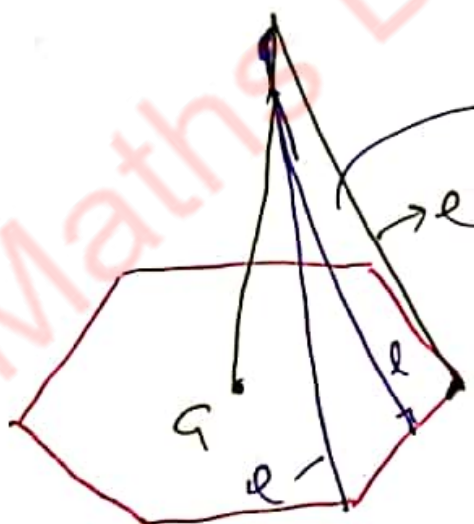


Open



equilateral Δ of height h

$$l = \sqrt{h^2 + \left(\frac{\sqrt{3}}{2}a\right)^2}$$



$$e^2 = l^2 + \left(\frac{a}{2}\right)^2$$

Hexagon:

$$T.S.A = \frac{1}{2} \times (6a) \times l + \frac{\sqrt{3}}{4} a^2 \times 6$$

$$Volume = \frac{1}{3} \times \left(6 \times \frac{\sqrt{3}}{4} a^2\right) \times h$$

End
Measurement

- * line and angle.
- * Inequality of Δ
- * Angle in Δ
- * Sine rule
- * Cosine rule
- * area related problem of Δ
- * MPG (Mass point Geometry)
- * Similarity
- * Congruency
- * Centres of Δ 's
 - Incentre
 - Circumcentre
 - Orthocentre
 - Centroid
 - Exocentre

- * Equilateral Δ
- * Right angle Δ
- * Isosceles Δ
- * acute / obtuse angle Δ
- * Square.
- * Rectangle
- * Parallelogram (11)
- * Rhombus
- * Trapezium.
- * Circle.
 - * Angles
 - * Chords
 - * Secants
 - * Tangents.

Line and Angle (रेखा और कोण)

- * Point (बिन्दु)
 - (लम्बाई) No length, (चौड़ाई) No breadth, (ऊँचाई) no height
 - Points is a zero dimensional figure.
 - Each point has unique location. (प्रत्येक बिन्दु)

(•) Point (A)

[Always Capital letter]

* Line (रेखा)

- A line is a set of points having only length.
- Line is one (1) dimensional.

eg.



denotes $\overleftrightarrow{AB}$ OR line $\overleftrightarrow{AB}$

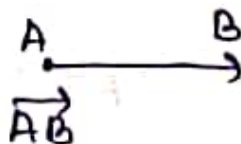


* Line Segment (रेखाखण्ड) —

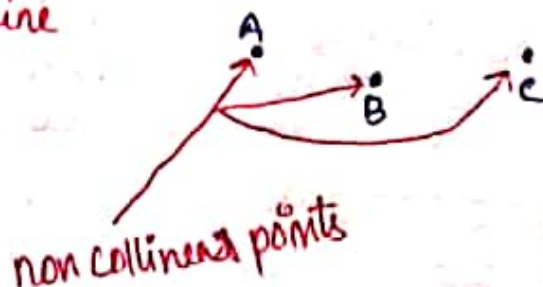
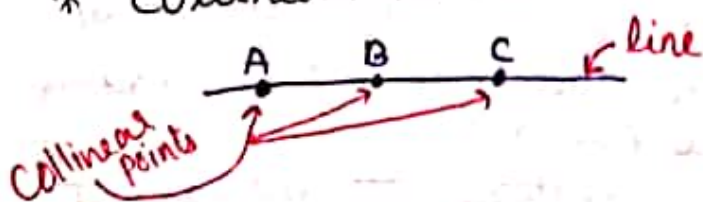


Denotes $\overline{AB}$

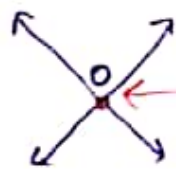
* Rays (किरण) —



* Collinear and non-collinear points (संरेखीय और असंरेखीय बिन्दु)

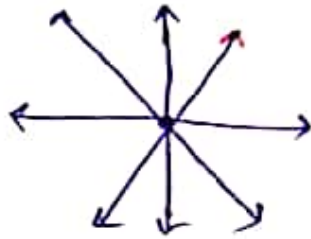


Intersecting lines (प्रतिच्छेदी रेखाएँ)

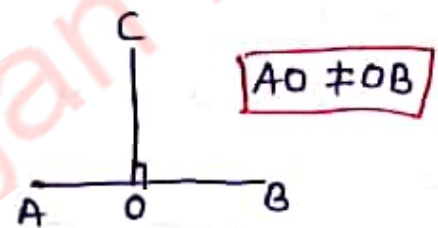


Intersecting points
प्रतिच्छेदी रेखाएँ

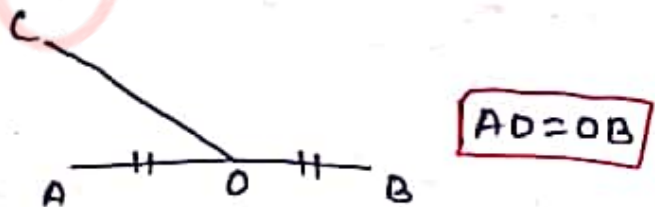
* Con-current lines: - [बिन्दुगामी रेखाएँ] → तीन या तीन से अधिक रेखाएँ एक ही बिन्दु से होकर जाती हैं।



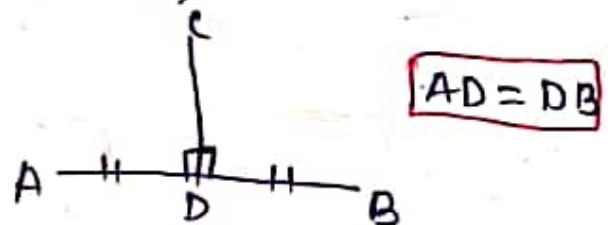
* Perpendicular line (लम्ब रेखा)



* Bisector अर्धक -



* Perpendicular Bi-sector (लम्ब अर्धक रेखा)



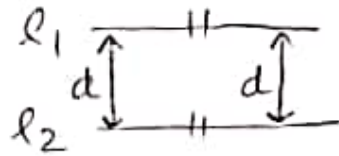
* Parallel lines :- ("समान्तर रेखाएँ")

↳ Parallel lines do not have any common point or they do not intersect each other but parallel line must be co-planar.

→ समान्तर रेखाएँ के पास न तो कोई अभ्यनिष्ठ बिन्दु होता है और न ही वे एक दूसरे को स्पर्श करते हैं (अनन्त तक) लेकिन

लेकिन समान्तर रेखाओं के बीच की दूरी सर्वत्र बराबर होती है।-

eg. <https://t.me/yourmathsnotes>



$$l_1 \parallel l_2$$

#



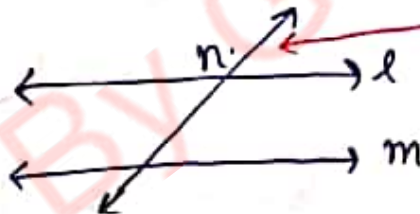
$$l_1 \parallel l_2$$

$$l_2 \parallel l_3$$

THEN $l_1 \parallel l_3$ [एक ही समतल में]

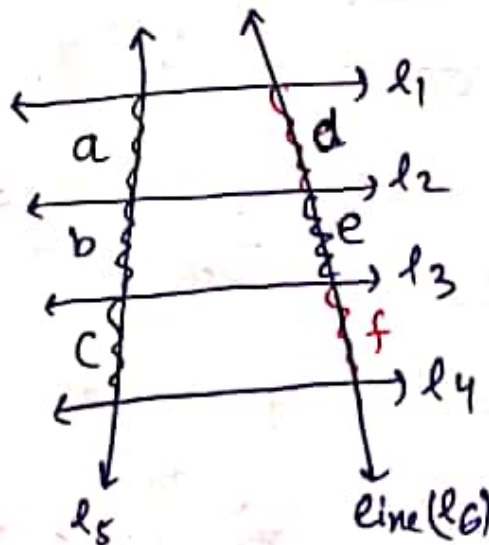
Transversal line (तिर्यकच्छेदी रेखाएं)

Transversal line



THEOREM (प्रमेय) → (I)

Corresponding ratios are always same.



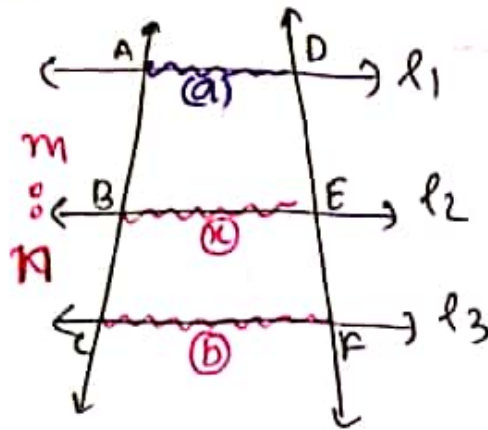
$l_1 \parallel l_2 \parallel l_3 \parallel l_4$
ratios are equal

$$\left\{ \begin{array}{l} \frac{a}{b} = \frac{d}{e} \\ \frac{b}{c} = \frac{e}{f} \end{array} \right\}$$

$$\text{OR } \left[\frac{a+b}{b+c} = \frac{d+e}{e+f} \right]$$

$$\text{OR } \frac{b}{a+b+c} = \frac{e}{d+e+f}$$

THEOREM प्रमेय - (II)



Given:

$$AD = a$$

$$CF = b$$

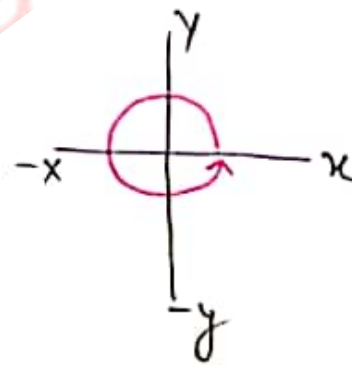
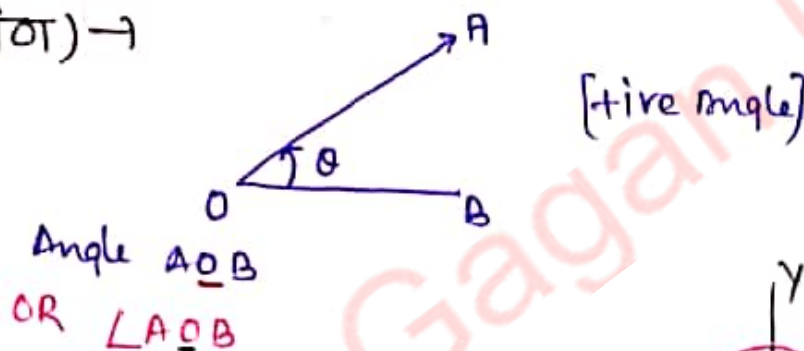
$$AB : BC = m : n$$

$$DE : EF = m : n$$

Find $x = ?$

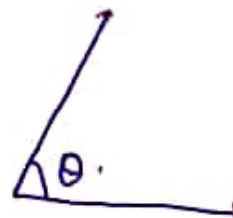
$$x = \frac{axn + bxm}{m+n}$$

Angles (कोण) $\rightarrow$



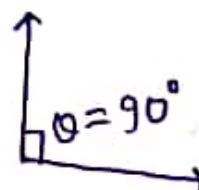
TYPES OF ANGLE (कोण के प्रकार)

1. Acute Angle (न्यून कोण)



$$0^\circ < \theta < 90^\circ$$

2. Right angle (समकोण)



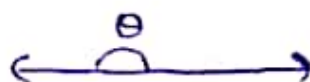
$$\theta = 90^\circ$$

3. Obtuse Angle (अधिक कोण)



$$90^\circ < \theta < 180^\circ$$

4. Straight angle (सुक्ष्म कोण)



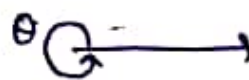
$$\theta = 180^\circ$$

5. Reflex Angle (वृद्ध कोण)



$$180^\circ < \theta < 360^\circ$$

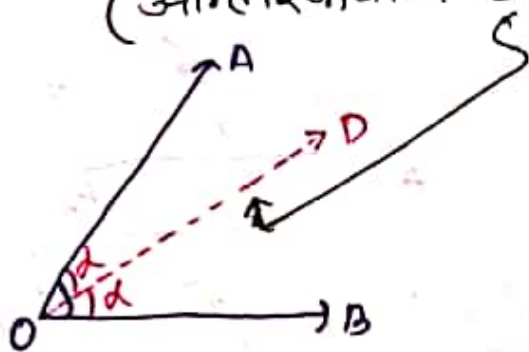
6. Complete angle (सम्पूर्ण कोण)



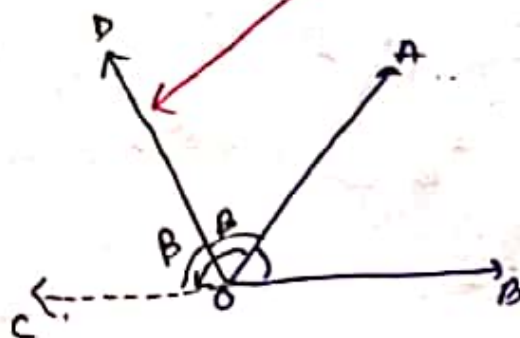
$$\theta = 360^\circ$$

Angle bisector :- (कोण समद्विभाजक)

1) Interior angle bisector (आन्तरिक कोण द्विभाजक)



2) Exterior angle bisector (बाह्य कोण समद्विभाजक)



$\angle AOC$
↓
[Exterior angle]



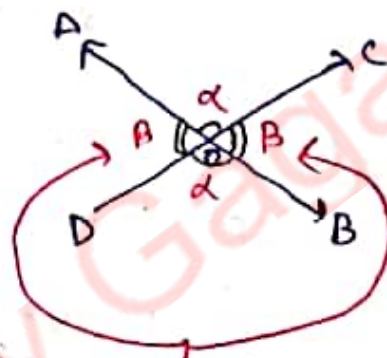
$$\alpha + \alpha + \beta + \beta = 180$$

$$\alpha + \beta = 90^\circ$$

Exterior Angle
(आन्तरिक कोण)

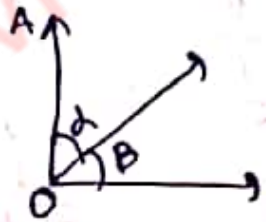
Exterior Angle
(अन्तर कोण)

Vertically opposite angle (शीर्षा विमुख कोण)



Vertically opposite angle

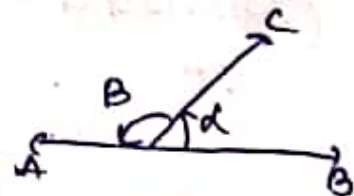
* Complementary angle
(पूरक कोण)



$$\alpha + \beta = 90^\circ$$

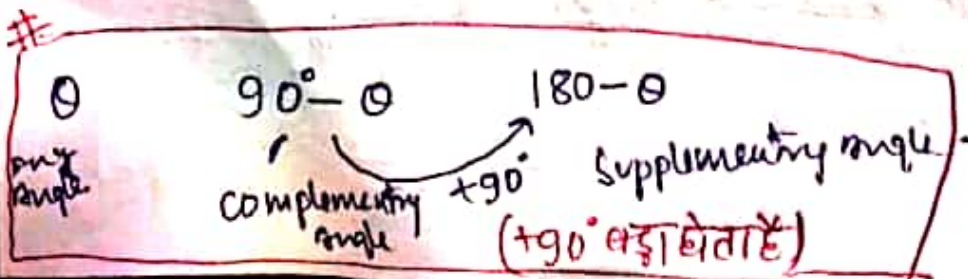
complementary to each other

* Supplementary angle
(सम्पूरक कोण)



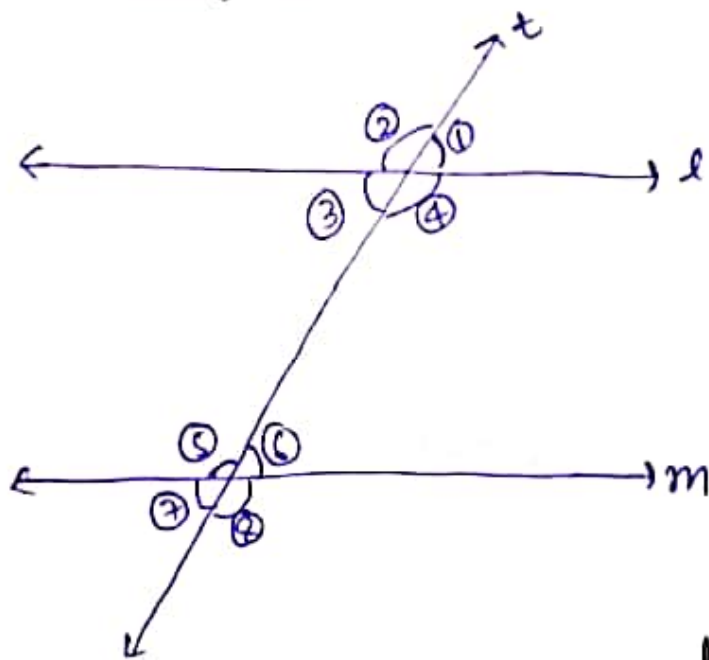
$$\alpha + \beta = 180^\circ$$

Supplementary to each other

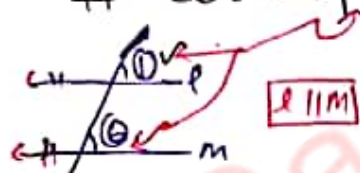


If "l" and "m" are two parallel lines and "t" is a transversal line then..

(यदि "l" और "m" दो समांतर रेखाओं और "t" तिर्यक रेखा हो तब) —



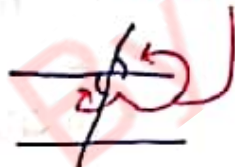
corresponding angle (संगत कोण)



$$\angle 1 = \angle 5, \angle 4 = \angle 8$$

$$\angle 2 = \angle 6, \angle 3 = \angle 7$$

Vertically opposite angle (शीर्षाभिमुख कोण)



$$\angle 1 = \angle 3, \angle 2 = \angle 4$$

$$\angle 5 = \angle 7, \angle 6 = \angle 8$$

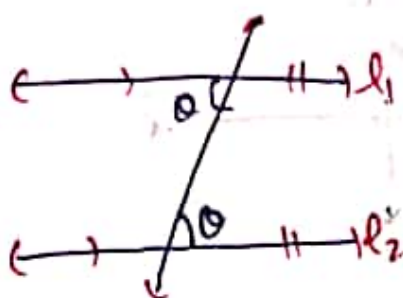
Alternate angle (संक्रान्त कोण)



$$\angle 3 = \angle 5, \angle 4 = \angle 6$$

→ Its vice versa are true →

eg.



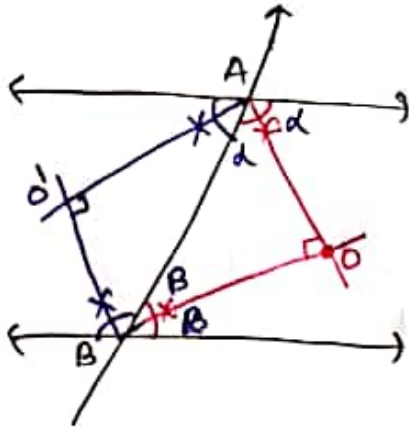
then $\boxed{l_1 \parallel l_2}$

(Alternate angle)

Interior Angle (आन्तरिक कोण)

$$[\angle 4 + \angle 6 = \angle 3 + \angle 5 = 180^\circ]$$

#



$$\angle AOB = 90^\circ$$

$$\angle A'O'B' = 90^\circ$$

PROOF $\Rightarrow \alpha + \alpha + \beta + \beta = 180$

$$\boxed{\alpha + \beta = 90^\circ}$$

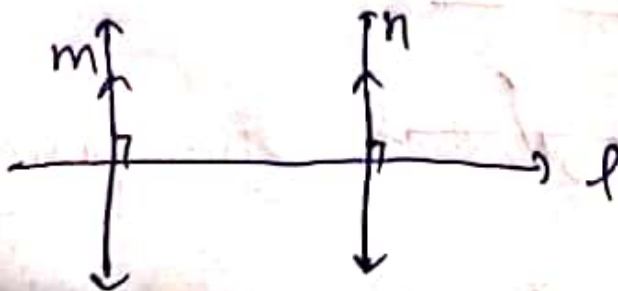
then $\angle O = 90^\circ$

Because

$\triangle AOB$ में दो Angle का सम 90° है तो तिसरा कोण 90° होगा क्योंकि $\triangle$ के तीनों भुजाओं का योग 180° होता है।

Point * ~~When two lines are perpendicular to each other~~

if $l \perp m$ and $l \perp n$ then $m \parallel n$

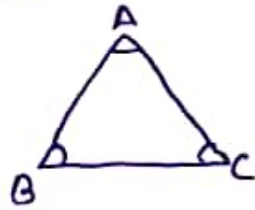


TYPES OF TRIANGLE

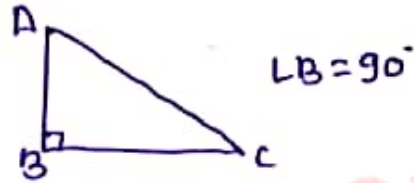
Δ - त्रिभुज

① Based upon Angles (कोणों के आधार पर)

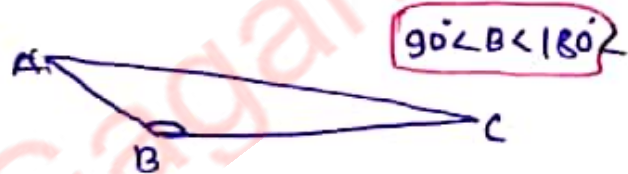
a) Acute angle triangle (न्यून कोण त्रिभुज)



b) Right angle Δ
(समकोण Δ)

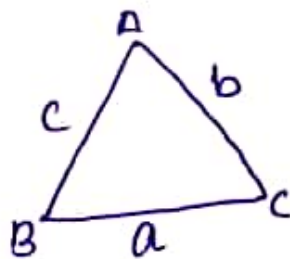


③ obtuse angle Δ
(अधिक कोण Δ)



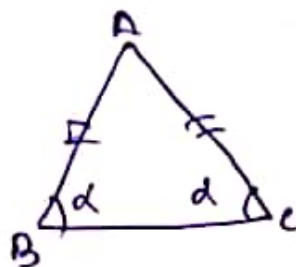
② Based upon Sides (भुजाओं के आधार पर)

① Scalene Δ
(विषमबाहु Δ)



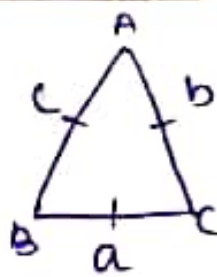
$$a \neq b \neq c \\ \angle A \neq \angle B \neq \angle C$$

② Isosceles Δ
(समबाहु Δ)

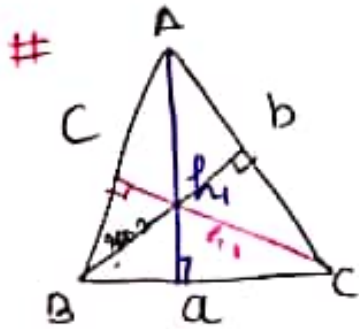


$$AB = AC \\ \angle B = \angle C$$

③ Equilateral Δ (समबाहु Δ)



$$\left[a = b = c \right. \\ \left. \angle A = \angle B = \angle C = 60^\circ \right]$$

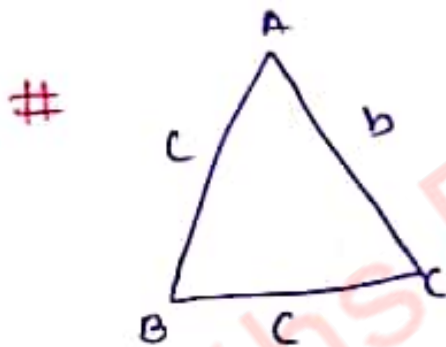


$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\left[\text{Area} = \frac{1}{2} a h_1 = \frac{1}{2} b h_2 = \frac{1}{2} c h_3 \right]$$

$$a h_1 = b h_2 = c h_3$$

$$h_1 : h_2 : h_3 = \frac{1}{a} : \frac{1}{b} : \frac{1}{c}$$



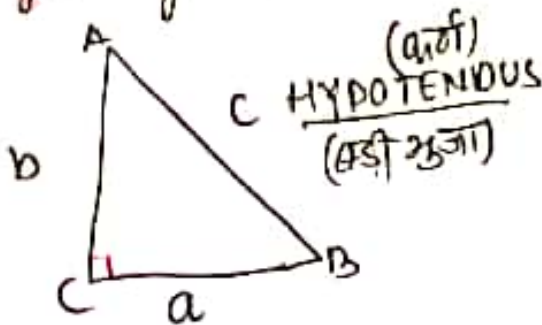
$$\text{Perimeter (परिमाप)} = a + b + c$$

$$\text{Semiperimeter (अर्धपरिमाप)} = \frac{a + b + c}{2}$$

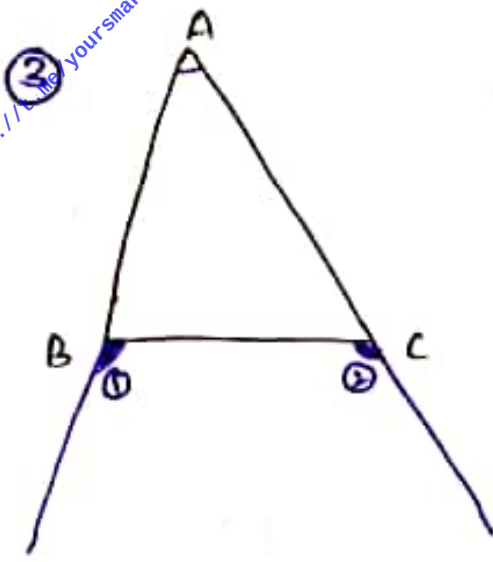
$$\text{Area of } \Delta = \sqrt{S(S-a)(S-b)(S-c)}$$

Heron's FORMULA

Right angle Δ -



$$c^2 = a^2 + b^2$$



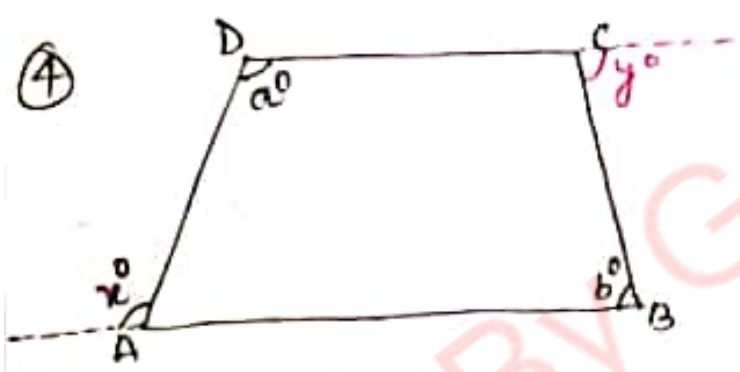
$$* \quad \textcircled{1} + \textcircled{2} = 180^\circ + A^\circ$$

PROOF -

$$\begin{aligned} \textcircled{1} + \textcircled{2} &= 180^\circ - LB + 180^\circ - LC \\ &= 360^\circ - (LB + LC) \end{aligned}$$

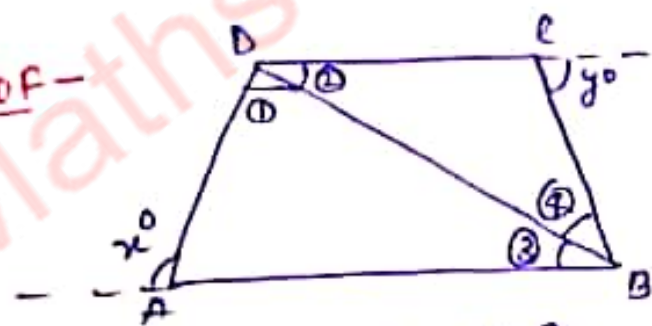
$$\therefore LB + LC = 180^\circ - LA$$

$$\textcircled{1} + \textcircled{2} = 180^\circ + LA^\circ$$



$$* \quad a^\circ + b^\circ = x^\circ + y^\circ$$

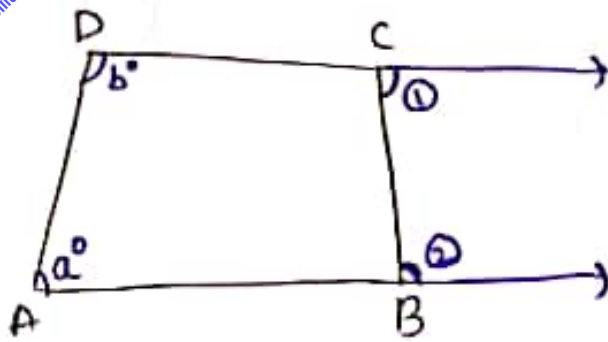
PROOF -



$$\begin{aligned} x^\circ &= \textcircled{1} + \textcircled{3} \\ y^\circ &= \textcircled{2} + \textcircled{4} \end{aligned}$$

$$* \quad x + y = a + b$$

5



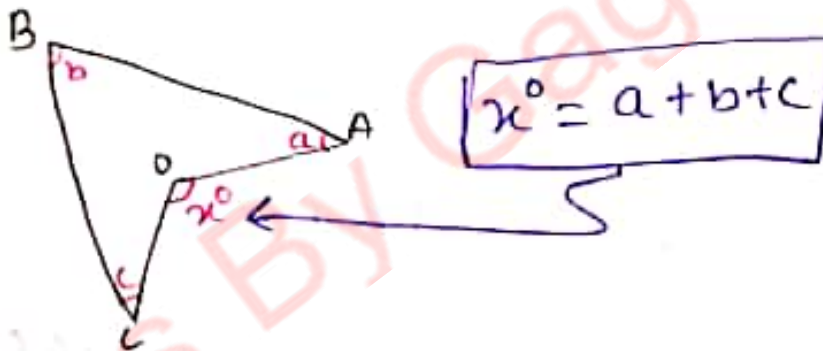
Result

$$\textcircled{1} + \textcircled{2} = a^\circ + b^\circ$$

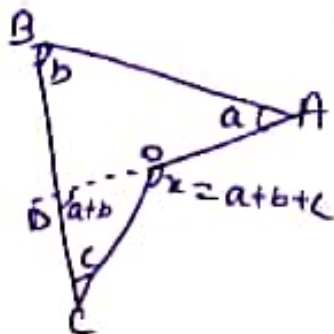
PROOF :-

$$\begin{aligned} \textcircled{1} + \textcircled{2} &= a^\circ + b^\circ \\ &\downarrow \quad \quad \downarrow \\ 180^\circ - \angle C + 180^\circ - \angle B \\ &= 360^\circ - (\angle B + \angle C) \\ &\downarrow \\ \angle A + \angle B + \angle C + \angle D - \angle B - \angle C \\ &= \boxed{a^\circ + b^\circ} \quad \# \end{aligned}$$

6

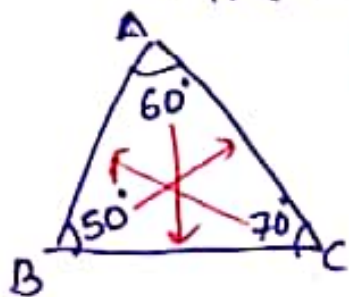


PROOF



[By sum of oppo interior angle is equal to its opposite exterior angle]

① किसी Δ में बड़े कोण के सामने की भुजा बड़ी तथा छोटे कोण के सामने की भुजा छोटी तथा बराबर कोण के सामने की भुजा बराबर होती है। -



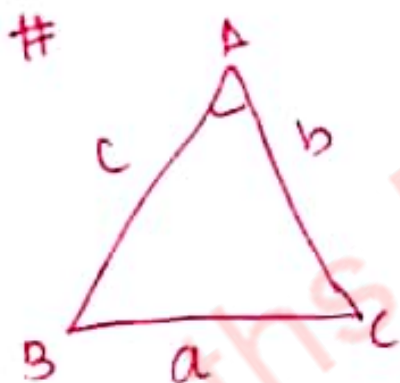
$$AB > BC > AC$$

(Sides)

② किसी Δ में बड़ी भुजा के सामने का कोण बड़ा छोटी भुजा के सामने का कोण छोटा व बराबर भुजा के सामने का कोण बराबर होता है।



$$LA > LB > LC$$



Angle $\downarrow$
 $LA = \text{largest} \rightarrow a = \text{largest}$
 $LB = \text{smallest} \Rightarrow b = \text{smallest}$

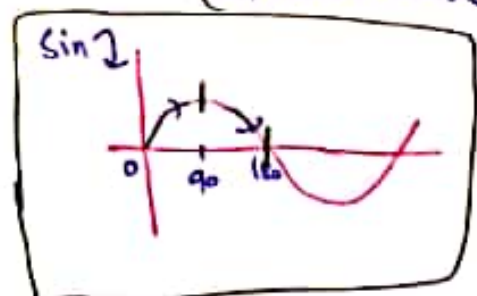
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k = (2R)$$

(Circumradius)

$$a = k \sin A$$

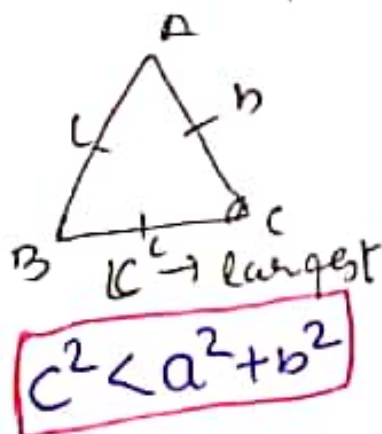
$$b = k \sin B$$

$$c = k \sin C$$

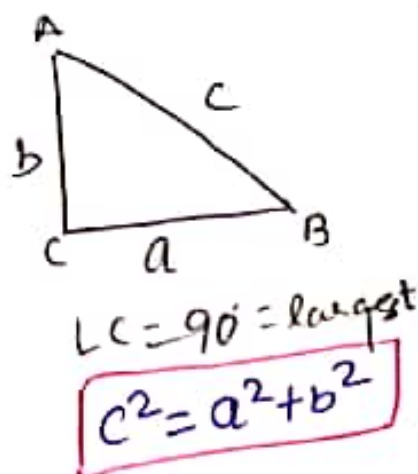


9th Δ $\rightarrow$

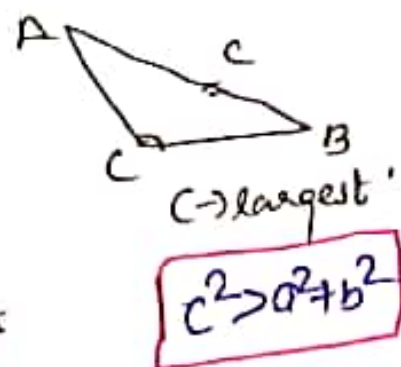
① Acute angle Δ



② Right angle Δ

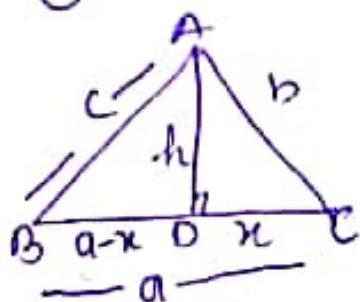


③ Obtuse angle Δ



PROOF :-

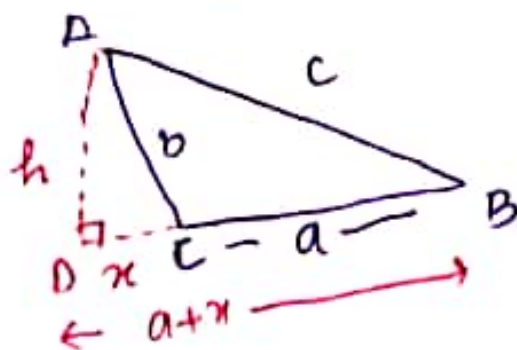
① Acute Angle (न्यून कोण) $\Delta \Rightarrow$



$$\begin{aligned} \Rightarrow c^2 &= (a-x)^2 + h^2 \\ c^2 &= a^2 + x^2 - 2ax + h^2 \\ c^2 &= a^2 + b^2 - 2ax \\ \boxed{c^2 + 2ax &= a^2 + b^2} \end{aligned}$$

$\Rightarrow c^2 \leq a^2 + b^2$
(always)

② Obtuse angle Δ
(अधिक कोण Δ)



$$\begin{aligned} c^2 &= h^2 + (a+x)^2 \\ c^2 &= h^2 + a^2 + x^2 + 2ax \\ c^2 &= a^2 + b^2 + 2ax \end{aligned}$$

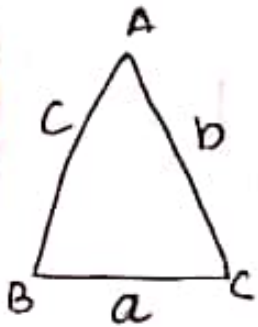
$$c^2 = a^2 + b^2 + 2ax$$

$\boxed{c^2 > a^2 + b^2}$

(add sth add कर रहे हैं)

* Sum of any two sides of a Δ is always greater than Semiperimeter of Δ .

(Δ में किसी दो भुजाओं का योग ~~हिसा~~ अर्द्धपरिमाप से सदैव बड़ा होता है)



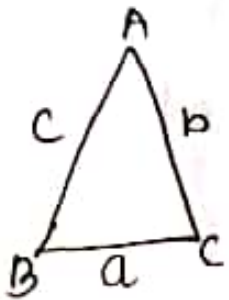
PROOF

$$\begin{aligned} a+b &> c \rightarrow a+b + \overset{\text{add}}{a+b} > c + \overset{\text{add}}{a+b} \quad \text{Perimeter} \\ b+c &> a \quad 2(a+b) > P \quad \therefore (P = a+b+c) \\ a+c &> b \quad a+b > \frac{P}{2} \quad (\because \frac{P}{2} = S) \end{aligned}$$

(Conclusion) $\rightarrow$ $\boxed{a+b > S}$ (Semiperimeter)

* Length of any side of a Δ is always less than its Semiperimeter.

(Δ की किसी भी भुजा की लम्बाई उसे अर्द्धपरिमाप से छोटी होती है)



$$\begin{aligned} a+b &> c \Rightarrow a+b+c > c+c \quad \text{Add} \\ P &> 2c \quad \therefore P = (a+b+c) \\ \frac{P}{2} &> c \quad (\frac{P}{2} = S) \end{aligned}$$

Conclusion $\rightarrow$ $\boxed{S > c}$ Semiperimeter

Similarly - $\begin{cases} S > a \\ S > b \end{cases}$

OR Δ की $\boxed{\text{largest side} < \frac{P}{2}}$

* Smallest side of a Δ is always less than or equal to one-third of Perimeter.

(Δ की छोटी भुजा $\frac{P}{3}$ से छोटी या उसके बराबर होगी)

* $\boxed{\text{Smallest side} \leq \frac{P}{3}}$

PROOF:-

$$a+b+c = P$$

$$\downarrow \downarrow \downarrow$$

$$c+c+c \leq P$$

$$c \leq \frac{P}{3}$$

$$a \geq b \geq c$$

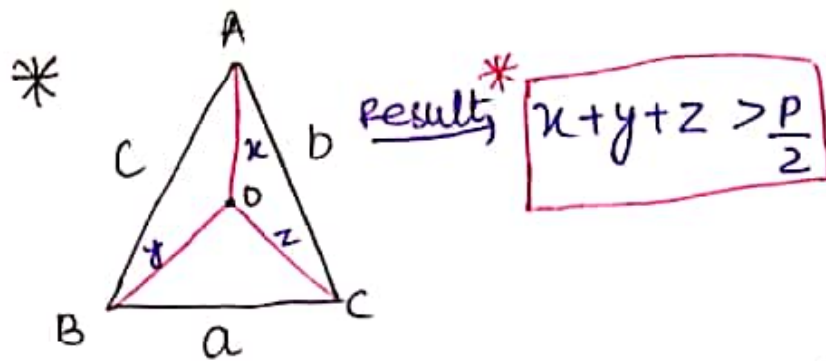
$$\downarrow \downarrow$$

$$a \leq \frac{P}{2} \quad c \leq \frac{P}{3}$$

eg. $5+4+3 = 12$

$$\downarrow \downarrow \downarrow$$

$$3 \ 3 \ 3 \leq 9$$



PROOF,

$$\begin{aligned} \text{In } \triangle OBA & \rightarrow x+y > c \quad \text{--- (1)} \\ \triangle OCB & \rightarrow y+z > a \quad \text{--- (2)} \\ \triangle OAC & \rightarrow x+z > b \quad \text{--- (3)} \end{aligned}$$

$$\text{Add} \rightarrow 2(x+y+z) > a+b+c \quad \therefore (P=a+b+c)$$

$$\#* \quad (x+y+z) > \frac{P}{2}$$

Number of possible Triangle with perimeter of Δ (Integer Side)

no. of possible Δ (Integer Side)	Perimeter	no. of possible Δ Integer side	Perimeter
X	1	2 ✓	← 7
X	2	1	← 8
(equilateral) ✓ (1)	3		
X	← 4		
1 ✓	← 5		
equilateral 1 ✓	← 6		
		Result →	

Triangle (त्रिभुज)

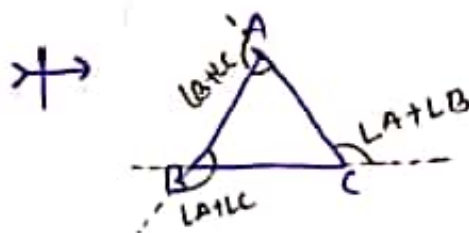


9n a Triangle:—(त्रिभुज में)

- { 3- vertices (शीर्ष)
 3- Interior angle / exterior angle. (आन्तरिक व बाहरी कोण)
 3- Sides (भुजा), Perpendicular, height etc.

$$L\alpha + L\beta + L\gamma = 180$$

$\angle A + \angle B + \angle C = 180$ * Triangle is smallest polygon.
(Δ सबसे छोटा बहुभुज है)



$\Rightarrow \Delta$ Exterior angle is equal to sum of its opposite angle.

Δ बाहरी कोण आन्तरिक दो कोणों के सम के बराबर होता है।

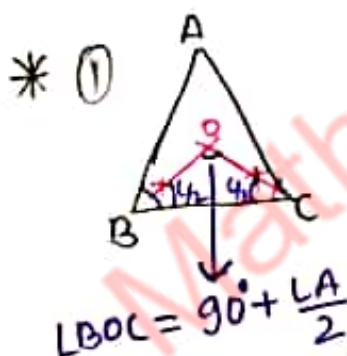
* किसी Δ के बाहरी कोण का योग 360° होता है।

Sum of $\rightarrow$ All exterior angle in a triangle is 360° .

Conclusion:- किसी बहुभुज में n भुजा हो तो -

Sum of ~~interior~~ Angle = $(n-2) \times 180^\circ$
आन्तरिक

* Sum of exterior angle = 360°



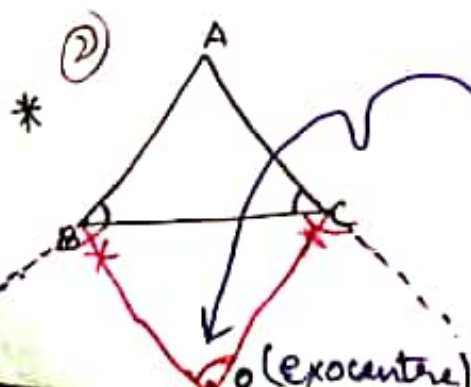
PROOF $\rightarrow \angle AOC = 180^\circ - \left(\frac{B}{2} + \frac{C}{2}\right)$

$$= 180^\circ - \left(\frac{B+C}{2}\right)$$

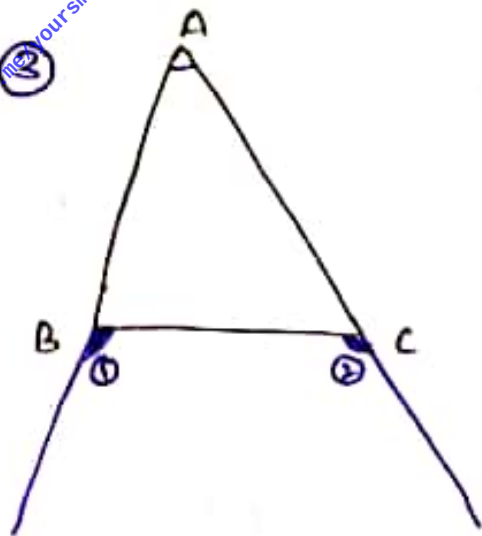
$$= 180 - 90 + \frac{LA}{2}$$

$$\angle BOC = 90 + \frac{\angle A}{2}$$

$$\therefore B+C=(180^\circ-A)$$



$$LBOC = 90 - \frac{LA}{2}$$



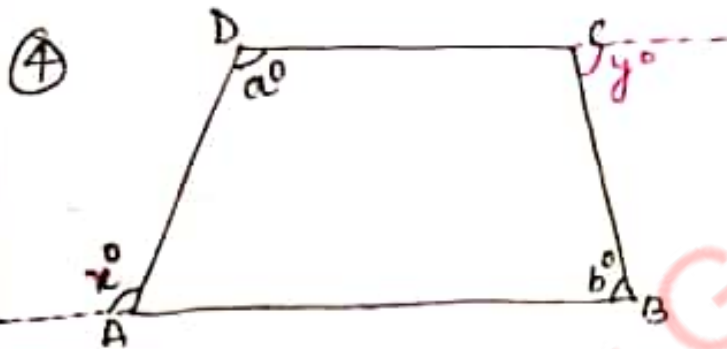
$$* \quad \textcircled{1} + \textcircled{2} = 180^\circ + A^\circ$$

PROOF -

$$\begin{aligned} \textcircled{1} + \textcircled{2} &= 180^\circ - LB + 180^\circ - LC \\ &= 360^\circ - (LB + LC) \end{aligned}$$

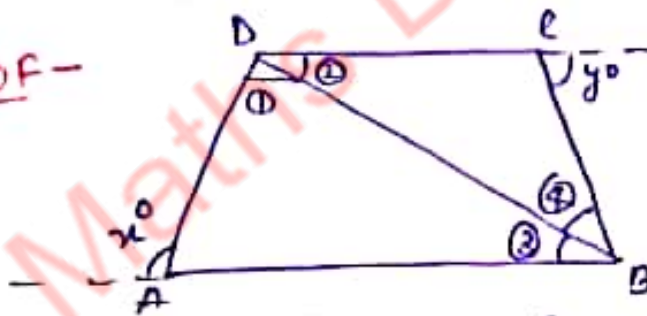
$$\therefore LB + LC = 180^\circ - LA$$

$$\textcircled{1} + \textcircled{2} = 180^\circ + LA^\circ$$



$$* \quad a^\circ + b^\circ = x^\circ + y^\circ$$

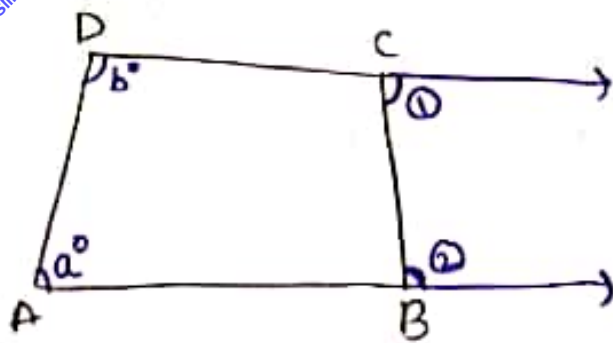
PROOF -



$$\begin{aligned} x^\circ &= \textcircled{1} + \textcircled{3} \\ y^\circ &= \textcircled{2} + \textcircled{4} \end{aligned}$$

$$* \quad x + y = a + b$$

⑤



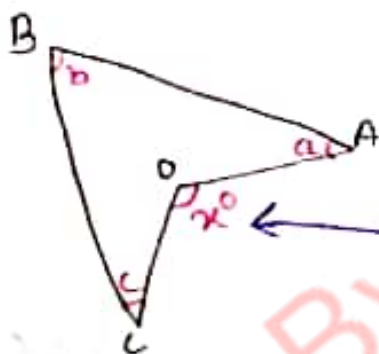
Result

$$\textcircled{1} + \textcircled{2} = a^\circ + b^\circ$$

PROOF :-

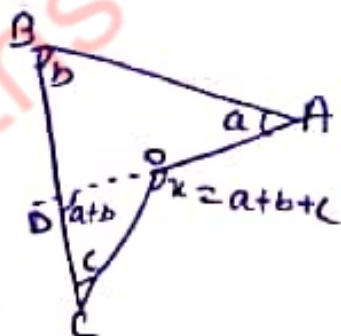
$$\begin{aligned} \textcircled{1} + \textcircled{2} &= a^\circ + b^\circ \\ &\downarrow \quad \quad \downarrow \\ 180^\circ - \angle C + 180^\circ - \angle B \\ &= 360^\circ - (\angle B + \angle C) \\ &\downarrow \\ \angle A + \angle B + \angle C + \angle D - \angle B - \angle C \\ &= \boxed{a^\circ + b^\circ} \quad \# \end{aligned}$$

⑥



$$x^\circ = a + b + c$$

PROOF



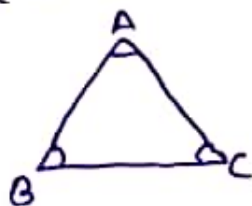
[By sum of oppo interior angle is equal to its opposite exterior angle]

TYPES OF TRIANGLE $\Rightarrow$

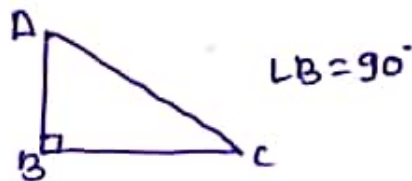
Δ - त्रिभुज

① Based upon Angles (कोणों के आधार पर)

a) Acute angle triangle (न्यून कोण त्रिभुज)



b) Right angle Δ
(समकोण Δ)

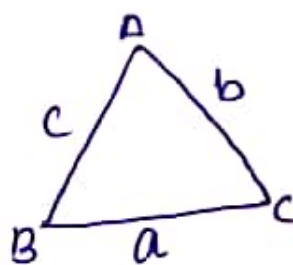


③ obtuse angle Δ
(अधिक कोण Δ)



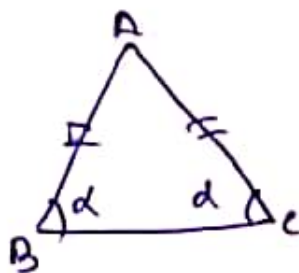
② Based upon Sides (भुजाओं के आधार पर)

① Scalene Δ
(विषमबाहु Δ)



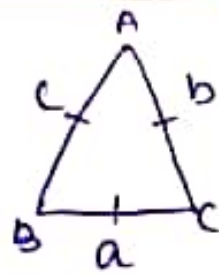
$$a \neq b \neq c \\ LA \neq LB \neq LC$$

② Isosceles Δ
(समबाहु Δ)

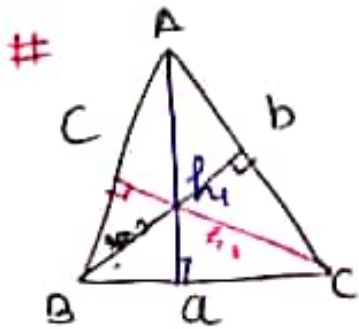


$$AB = AC \\ LB = LC$$

③ Equilateral Δ (समबाहु Δ)



$$\left[\begin{aligned} a &= b = c \\ \angle A &= \angle B = \angle C = 60^\circ \end{aligned} \right]$$

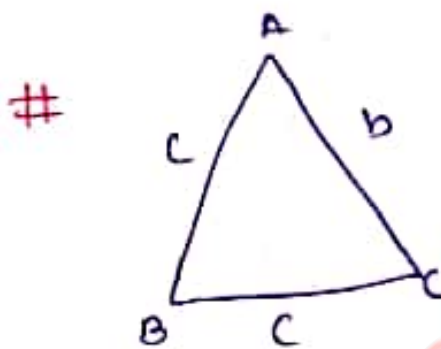


$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\left[\text{Area} = \frac{1}{2} a h_1 = \frac{1}{2} b h_2 = \frac{1}{2} c h_3 \right]$$

$$a h_1 = b h_2 = c h_3$$

$$\left[h_1 : h_2 : h_3 = \frac{1}{a} : \frac{1}{b} : \frac{1}{c} \right]$$

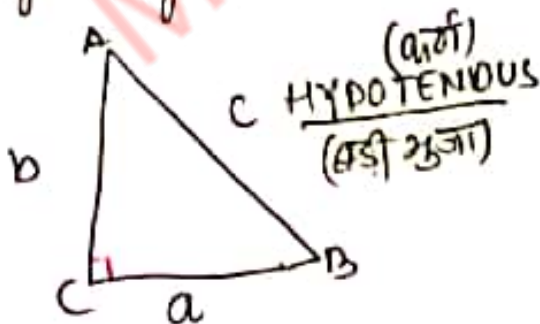


$$\text{Perimeter (परिमाप)} = a + b + c$$

$$\text{Semiperimeter (अर्धपरिमाप)} = \frac{a+b+c}{2}$$

$$\left[\begin{aligned} \text{Area of } \Delta &= \sqrt{S(S-a)(S-b)(S-c)} \\ \text{Heron's FORMULA} \end{aligned} \right]$$

Right angle Δ -



$$\left[c^2 = a^2 + b^2 \right]$$

① किसी Δ में बड़े कोण के सामने की भुजा बड़ी तथा छोटे कोण के सामने की भुजा छोटी तथा बराबर कोण के सामने की भुजा बराबर होती है। -



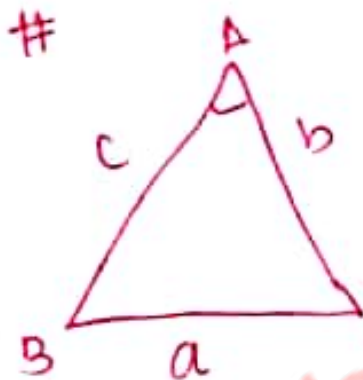
$$AB > BC > AC$$

(Sides)

② किसी Δ में बड़ी भुजा के सामने का कोण बड़ा छोटी भुजा के सामने का कोण छोटा व बराबर भुजा के सामने का कोण बराबर होता है।



$$\angle A > \angle B > \angle C$$



Angle $\downarrow$
 $\angle A = \text{largest} \rightarrow a = \text{largest}$
 $\angle B = \text{smallest} \Rightarrow b = \text{smallest}$

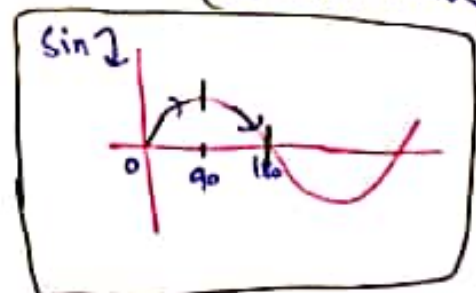
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k = (2R)$$

(Circumradius)

$$a = k \sin A$$

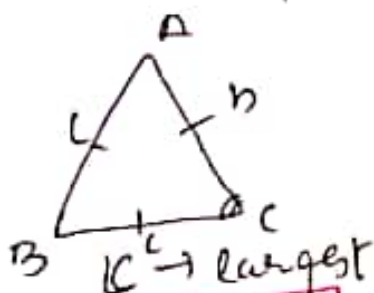
$$b = k \sin B$$

$$c = k \sin C$$



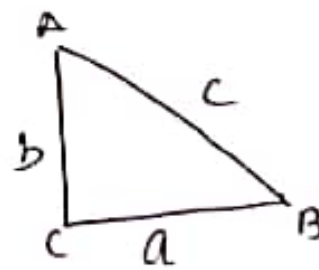
9th → Δ

① Acute angle Δ



$$c^2 < a^2 + b^2$$

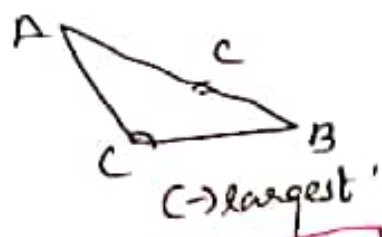
② Right angle Δ



$\angle C = 90^\circ = \text{largest}$

$$c^2 = a^2 + b^2$$

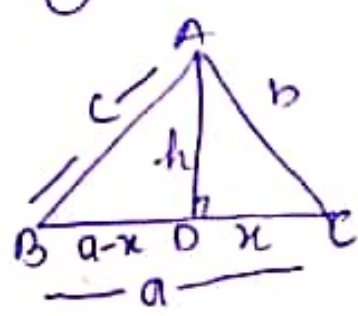
③ Obtuse angle Δ



$$c^2 > a^2 + b^2$$

PROOF :-

① Acute Angle (मनकोण) Δ ⇒

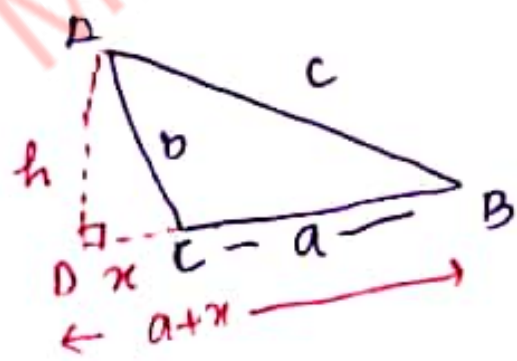


$$\begin{aligned} c^2 &= (a-x)^2 + h^2 \\ c^2 &= a^2 + x^2 - 2ax + h^2 \\ c^2 &= a^2 + b^2 - 2ax \end{aligned}$$

$$c^2 + 2ax = a^2 + b^2$$

↑
more add करने पर
⇒ $c^2 < a^2 + b^2$
(always)

② Obtuse angle Δ (अधिक कोण Δ)



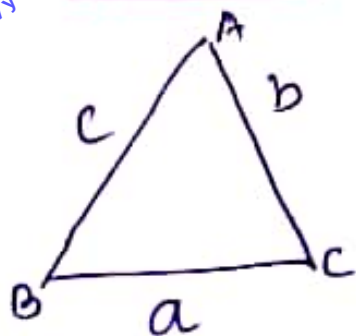
$$\begin{aligned} c^2 &= h^2 + (a+x)^2 \\ c^2 &= h^2 + a^2 + x^2 + 2ax \\ c^2 &= a^2 + b^2 + 2ax \end{aligned}$$

$$c^2 = a^2 + b^2 + 2ax$$

$$c^2 > a^2 + b^2$$

↑
(add sth add कर रहे हैं)

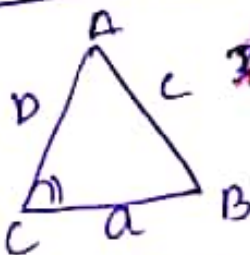
Cosine rule -



$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

By Cosine rule:-

Acute



$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

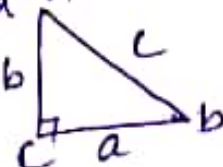
↑
+ve

$$+ve = a^2 + b^2 - c^2$$

$$c^2 + ve = a^2 + b^2$$

$$c^2 < a^2 + b^2$$

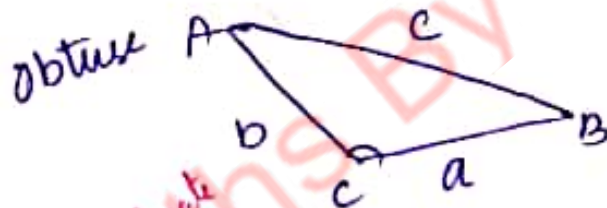
Right angle A



$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

↑
0

$$a^2 + b^2 = c^2$$



obtuse A

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

↑
-ve

$$c^2 - k = a^2 + b^2$$

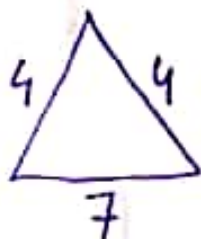
$$c^2 > a^2 + b^2$$

Properties of Δ -

① Sum of any sides of a triangle is always greater than third side.

(Δ की दो भुजाओं का योग तिसरी भुजा से सदैव बड़ा होता है)

eg:



$$4 + 4 > 7$$

✓ possible Δ



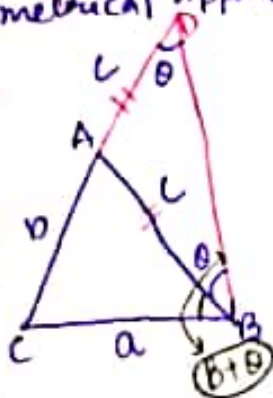
$$4 + 3 = 7$$

$$7 = 7$$

X Δ नहीं बनेगा
not possible.

Geometrical Approach -

②



ΔBCD में

$$LB + LD > LB$$

$$CD > BC$$

$$b + c > a \Rightarrow b > a - c$$

Similarly $\begin{cases} c + a > b \\ a + b > c \end{cases} \Rightarrow \begin{cases} c > b - a \\ a > b - c \end{cases}$

$b > |a - c|$ Always +ve value

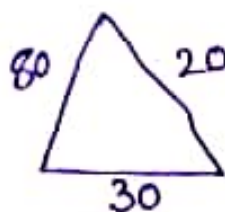
② difference of any side of a triangle is always less than the third side.

(Δ की दो भुजाओं का अन्तर तीसरी भुजा से छोटा होता है)



$$4 - 4 < 7$$

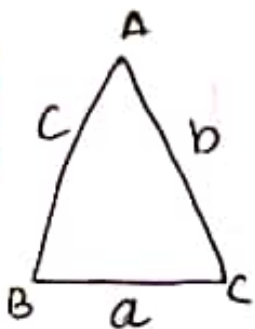
✓ possible.



X not possible.

* Sum of any two sides of a Δ is always greater than Semiperimeter of Δ .

(Δ में किसी दो भुजाओं का योग किसी भी अर्धपरिभाप से सदैव बड़ा होता है)



PROOF

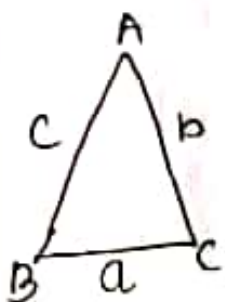
$$\begin{aligned} a+b &> c \rightarrow a+b + \overset{\text{add}}{a+b} > c + \overset{\text{add}}{a+b} \quad \text{Perimeter} \\ b+c &> a \quad 2(a+b) > P \quad \therefore (P = a+b+c) \\ a+c &> b \quad a+b > \frac{P}{2} \quad (\because \frac{P}{2} = S) \end{aligned}$$

(Conclusion) $\rightarrow \boxed{a+b > S}$

(Semiperimeter)

* Length of any side of a Δ is always less than its Semiperimeter.

(Δ का किसी भी भुजा की लम्बाई उसे अर्धपरिभाप से छोटी होती है)



$$\begin{aligned} a+b &> c \Rightarrow a+b + \overset{\text{add}}{c} > c + \overset{\text{add}}{c} \\ P &> 2c \quad \therefore P = a+b+c \\ \frac{P}{2} &> c \quad (\frac{P}{2} = S) \end{aligned}$$

Conclusion $\rightarrow \boxed{S > c}$

Semiperimeter

Similarly - $\begin{cases} S > a \\ S > b \end{cases}$

OR Δ की $\boxed{\text{largest side} < \frac{P}{2}}$

* Smallest side of a Δ is always less than or equal to one-third of Perimeter.

(Δ की छोटी भुजा $\frac{P}{3}$ से छोटी या उसके बराबर होती है)

* $\boxed{\text{Smallest side} \leq \frac{P}{3}}$

PROOF:-

$$a+b+c = P$$

Small $\downarrow \downarrow \downarrow$

$$c+c+c \leq P$$

$$c \leq \frac{P}{3}$$

$$a \geq b \geq c$$

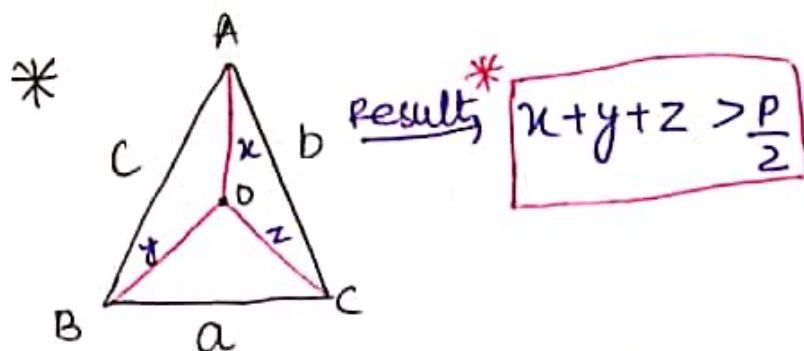
$$\downarrow \quad \downarrow$$

$$a \leq \frac{P}{2} \quad c \leq \frac{P}{3}$$

eg. $5+4+3 = 12$

Small $\downarrow \downarrow \downarrow$

$$3 \quad 3 \quad 3 \leq 9$$



PROOF,

$$\begin{aligned} \text{In } \triangle OBA & \rightarrow x+y > c \quad \text{--- (1)} \\ \triangle OCB & \rightarrow y+z > a \quad \text{--- (2)} \\ \triangle OAC & \rightarrow x+z > b \quad \text{--- (3)} \end{aligned}$$

Add $\rightarrow$

$$2(x+y+z) > a+b+c$$

$$\therefore (P = a+b+c)$$

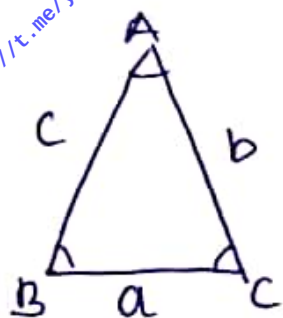
$$x+y+z > \frac{P}{2}$$

Number of possible Triangle with perimeter of Δ (Integer Side)

no. of possible Δ (Integer Side)	Perimeter	no. of possible Δ Integer side	Perimeter
X	1	2 $\checkmark$	7
X	2	1	8
(equilateral) Δ (1)	3		
X	4		
1 $\checkmark$	5		
equilateral Δ 1 $\checkmark$	6		

Result $\rightarrow$

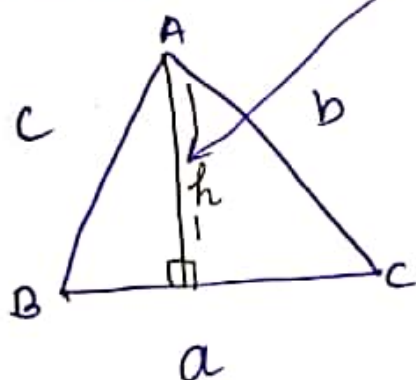
Sine rule:-



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k \text{ (Constant)}$$

$2R$
Circumradius
(परित्रिज्या)

PROOF →



(Perpendicular)

In $\triangle ABD$

$$\sin B = \frac{h}{c}$$

$$h = c \sin B$$

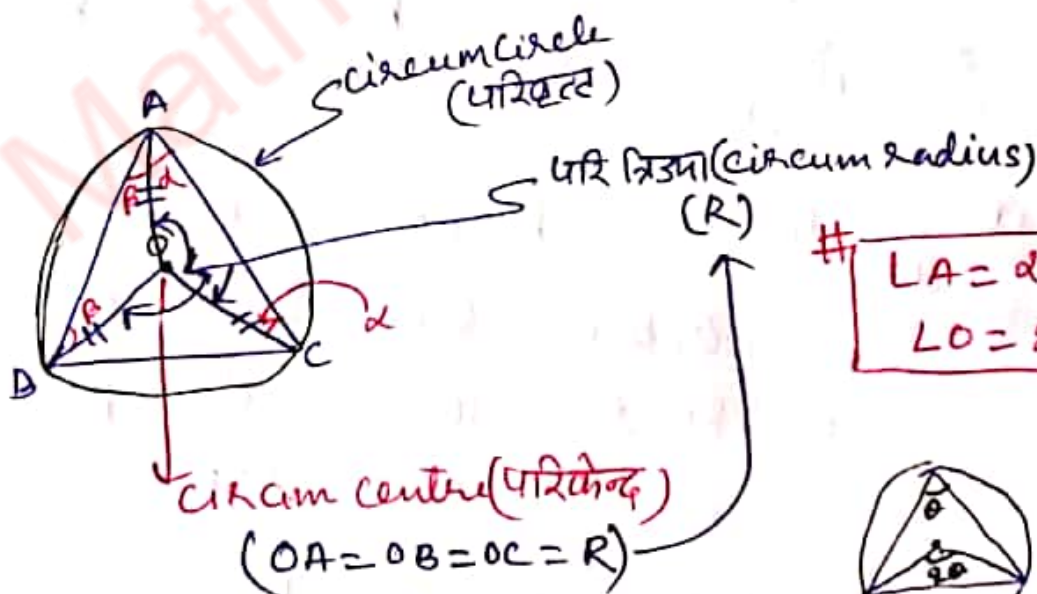
In $\triangle ADC$

$$\sin C = \frac{h}{b}$$

$$h = b \sin C$$

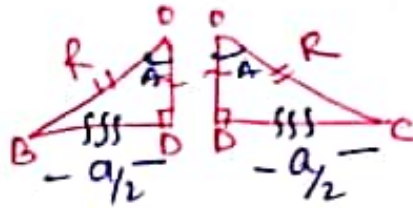
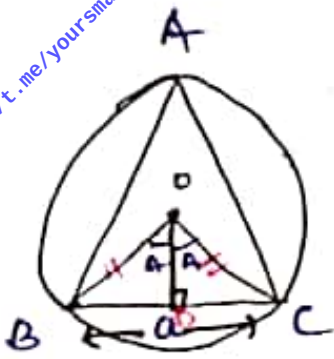
$$c \sin B = b \sin C$$

$$\frac{b}{\sin B} = \frac{c}{\sin C} = \frac{a}{\sin A} \checkmark$$



$$\begin{aligned} \angle A &= \alpha + \beta \\ \angle O &= 2(\alpha + \beta) \end{aligned}$$





We know that $\angle BOC = 2\angle A$

$$\sin A = \frac{a/2}{R}$$

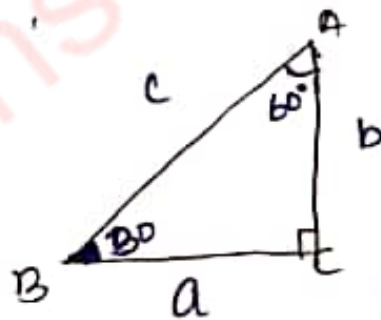
$$2R = \frac{a}{\sin A}$$

$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$$

↓
Circumradius
(परित्रिज्या)

* Standard Result:-

①-



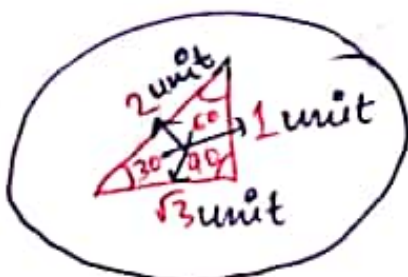
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$$

$$a = k \sin A, b = k \sin B, c = k \sin C$$

$$a : b : c = \sin A : \sin B : \sin C$$

$$a : b : c = \sin 30^\circ : \sin 60^\circ : \sin 90^\circ$$

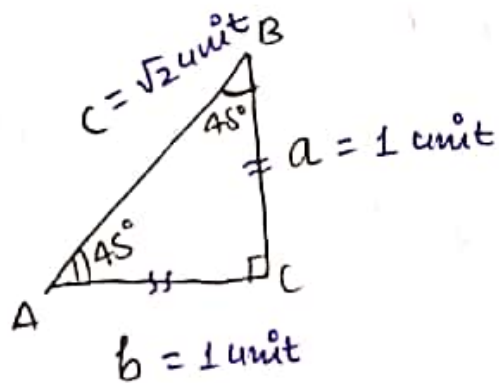
$$= \frac{1}{2} : \frac{\sqrt{3}}{2} : 1 \times 2$$



$$a : b : c = 1 : \sqrt{3} : 2$$

↓ ↓ ↓
30° 60° 90°

2-



By Sine rule:-

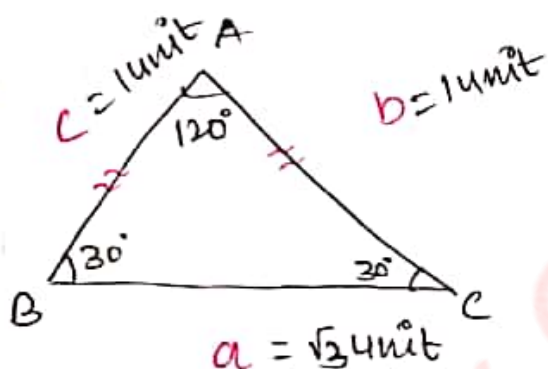
$$a:b:c = \sin 45^\circ : \sin 45^\circ : \sin 90^\circ$$

$$= \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{2}} : 1 \times \sqrt{2}$$

$$a:b:c = 1:1:\sqrt{2}$$

Arrows indicate the correspondence: $1 \rightarrow 45^\circ$, $1 \rightarrow 45^\circ$, $\sqrt{2} \rightarrow 90^\circ$.

3-



By Sine rule,

$$a:b:c = \sin 120^\circ : \sin 30^\circ : \sin 30^\circ$$

$$= \frac{\sqrt{3}}{2} \times 2 : \frac{1}{2} \times 2 : \frac{1}{2} \times 2$$

$$a:b:c = \sqrt{3}:1:1$$

Arrows indicate the correspondence: $\sqrt{3} \rightarrow 120^\circ$, $1 \rightarrow 30^\circ$, $1 \rightarrow 30^\circ$.

Imp:

[Result value]-

$$\# \sin 15^\circ = \cos 75^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$\# \sin 75^\circ = \cos 15^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$$

$$\# \tan 15^\circ = [\cot 75^\circ] = \frac{\sin 15^\circ}{\cos 15^\circ} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2-\sqrt{3}$$

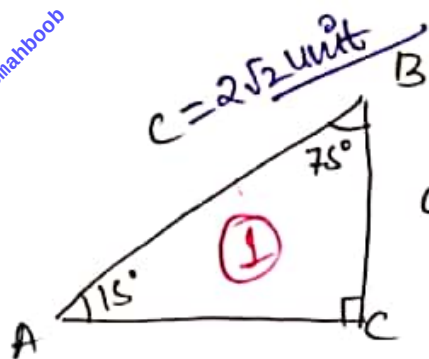
$$\# \tan 75^\circ = [\cot 15^\circ] = \frac{\sqrt{3}+1}{\sqrt{3}-1} = 2+\sqrt{3}$$

$$\# \textcircled{1} \sin 15^\circ = \sin(45^\circ - 30^\circ)$$

$$= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ$$

$$= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} = \frac{\sqrt{3}-1}{2\sqrt{2}} \checkmark$$

14-



$$b = (\sqrt{3}+1) \text{ unit}$$

$$a = (\sqrt{3}-1) \text{ unit}$$

Sine rule:-

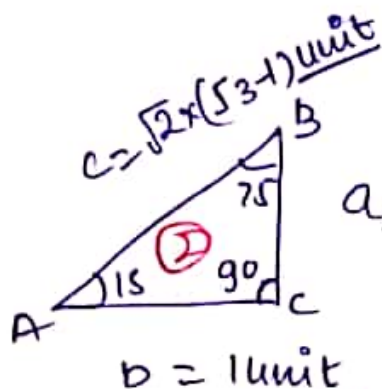
$$a : b : c = \sin 15^\circ : \sin 75^\circ : \sin 90^\circ$$

$\downarrow \quad \downarrow \quad \downarrow$
 $15 \quad 75 \quad 90$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}} : \frac{\sqrt{3}+1}{2\sqrt{2}} : 1$$

$$a : b : c = \sqrt{3}-1 : \sqrt{3}+1 : 2\sqrt{2}$$

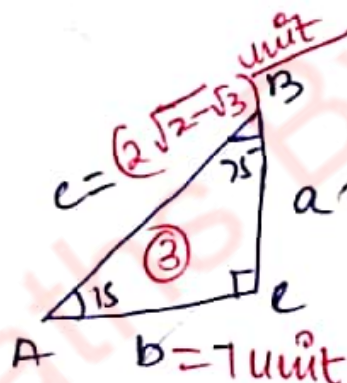
OR.



$$a = \frac{\sqrt{3}-1}{\sqrt{3}+1} \text{ unit}$$

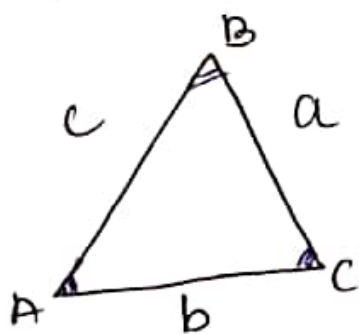
$$\tan 15^\circ = \frac{\sqrt{3}-1}{\sqrt{3}+1} \Leftarrow \frac{a}{b}$$

OR.



$$a = (2-\sqrt{3}) \text{ unit} \quad \tan 15^\circ = \frac{2-\sqrt{3}}{1} \Leftarrow \frac{a}{b}$$

Cosine rule :-



$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

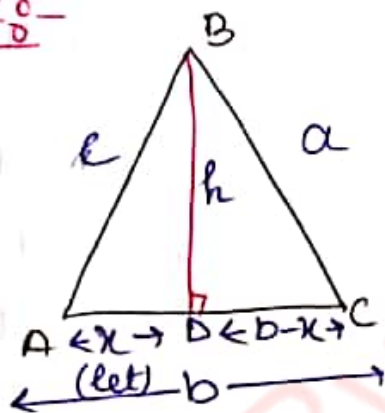
$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\left\{ \begin{array}{l} a^2 = b^2 + c^2 - 2bc \cos A \\ b^2 = a^2 + c^2 - 2ac \cos B \\ c^2 = a^2 + b^2 - 2ab \cos C \end{array} \right.$$

When one angle given ~~one~~ two side given.
Then by using cosine rule we
~~can~~ easily find Side Third.

PROOFS-



$$[h = h]$$

$$c^2 - x^2 = a^2 - (b-x)^2$$

$$(b-x)^2 - x^2 = a^2 - c^2$$

$$b^2 + \cancel{x^2} - 2bx - \cancel{x^2} = a^2 - c^2$$

$$\# \quad x = \frac{b^2 + c^2 - a^2}{2b}$$

In $\triangle ADB$

$$\Rightarrow \boxed{\cos A = \frac{x}{c}}$$

$$\frac{x}{c} = \frac{b^2 + c^2 - a^2}{2bc}$$

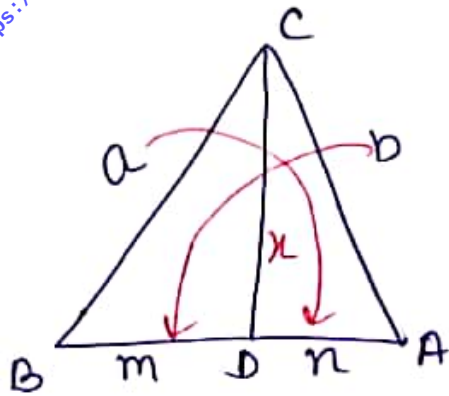
$$\# \quad \boxed{\cos A = \frac{b^2 + c^2 - a^2}{2bc}}$$

Similarly: - we find:-

$$\boxed{\cos B, \cos C}$$

(Stewart)

Stewart Theorem: $\rightarrow$

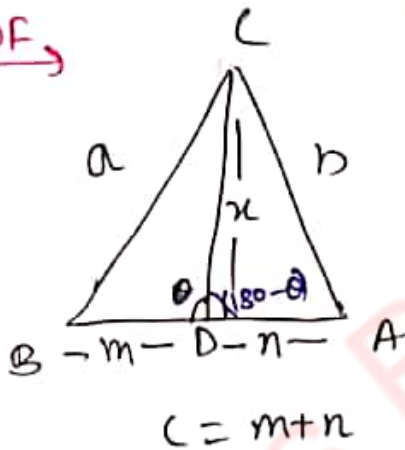


$$a^2n + b^2m = x^2c + mnc$$

Stewart Theorem

$$AB = c = m + n$$

PROOF $\rightarrow$



Using cosine rule $\rightarrow$

$$\cos(180 - \theta) = -\cos \theta$$

$$a^2 = x^2 + m^2 - 2mx \cos \theta \quad \text{--- (1)}$$

$$b^2 = x^2 + n^2 - 2nx \cos(180 - \theta)$$

$$b^2 = x^2 + n^2 + 2nx \cos \theta \quad \text{--- (2)}$$

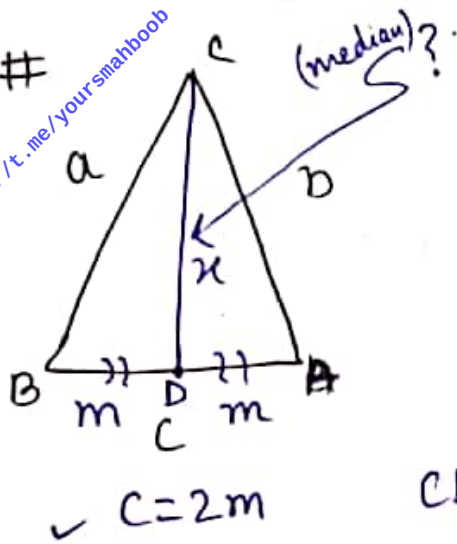
multiply (1) $\rightarrow$ n
(2) $\rightarrow$ m And Add both

$$a^2n + b^2m = x^2(n+m) + m^2n + n^2m$$

$$a^2n + b^2m = x^2(n+m) + mn(m+n)$$

$$m+n = c$$

$$a^2n + b^2m = x^2c + mnc$$



Ex

Apolloneus theorem:-

$$a^2 + b^2 = 2(x^2 + m^2)$$

$$m = \frac{c}{2}$$

OR $\left\{ a^2 + b^2 = 2\left(x^2 + \frac{c^2}{4}\right) \right\}$

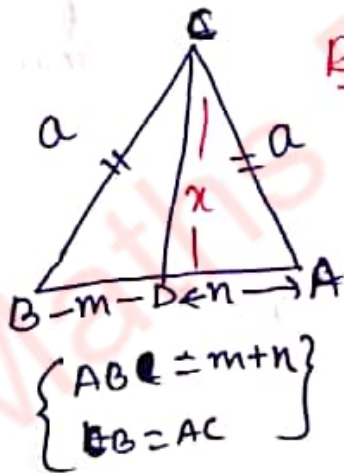
CD $\rightarrow$ median

Stewart theorem

$$a^2m + b^2m = x^2 \times 2m + m \times m \times 2m$$

$$a^2 + b^2 = 2(x^2 + m^2)$$

* Special Case in Isosceles $\Delta \rightarrow$



* Result $\rightarrow x^2 = a^2 - mn$

Stewart theorem

$$a^2n + a^2m = x^2c + mnc$$

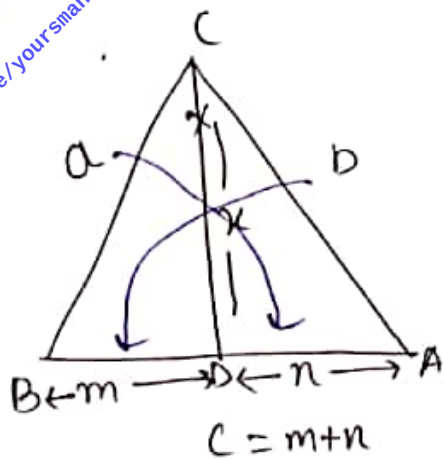
$$a^2x = x(x^2 + mn)$$

$$a^2 = x^2 + mn$$

$$x^2 = a^2 - mn$$

[length of angle bisector] $\rightarrow$ [Same result]

#



CD $\rightarrow$ angle bisector.

#

$$x^2 = axb - m \times n$$

$$\boxed{x^2 = ab - mn}$$

By using Stewart's theorem,

$$a^2n + b^2m = x^2c + mnc$$

$$\frac{a}{b} = \frac{m}{n} \left[\text{By interior angle bisector theorem} \right]$$

$$\boxed{an = bm}$$

$$\Rightarrow \underline{an} \times a + \underline{bm} \times b = c(x^2 + mn)$$

$$\Rightarrow bma + anb = c(x^2 + mn)$$

$$\Rightarrow ab(m+n) = c(x^2 + mn) \quad (\because m+n=c)$$

$$\boxed{x^2 = ab - mn}$$

SPECIAL TRIANGLE $\rightarrow$

(बड़ा कोण)

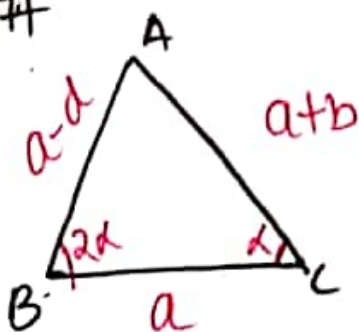
$\Rightarrow$ All side of a Δ are in A.P. largest angle is twice the smallest angle (दोय कोण)
(दोगुना)

Side are

(Always) Side ratio

$$\rightarrow \boxed{4 : 5 : 6}$$

#



(a-d)
 $\downarrow$
Smallest

(a) (a+d)
 $\downarrow$
largest

$\rightarrow$ Using sine rule $\rightarrow$

By using Sine rule:-

$$\frac{a+d}{a-d} = \frac{\sin 2\alpha}{\sin \alpha} = \frac{2\cancel{\sin \alpha} \cdot \cos \alpha}{\cancel{\sin \alpha}}$$

$$\boxed{\cos \alpha = \frac{a+d}{2(a-d)}} \quad \text{--- (1)}$$

By using Cosine rule:-

$$\cos \alpha = \frac{a^2 + (a+d)^2 - (a-d)^2}{2a(a+d)}$$

OR.

$$\cos \alpha = \frac{a+4d}{2(a+d)} \quad \text{--- (2)}$$

$$\textcircled{1} = \textcircled{2}$$

$$\frac{a+b}{2(a-d)} = \frac{a+4d}{2(a+d)}$$

$$a^2 + d^2 + 2ad = a^2 + 3ad - 4d^2$$

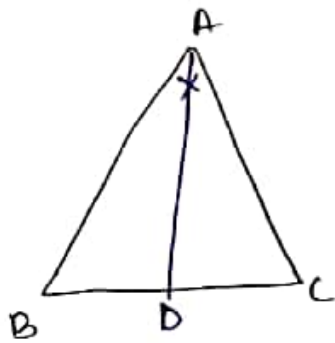
$$5d^2 = ad$$

$$\frac{a}{d} = \frac{5}{1}$$

$$(a-d) : a : (a+d) \quad \textcircled{\sim}$$

$$* \boxed{4 : 5 : 6}$$

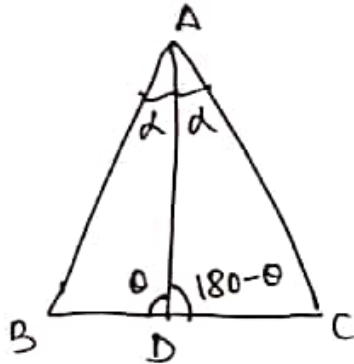
Interior angle bisector theorem:- (आन्तरिक)



AD $\rightarrow$ angle bisector of angle A

$$\boxed{\frac{AB}{AC} = \frac{BD}{DC}} \quad \#$$

PROOF



Sine rule:- ①

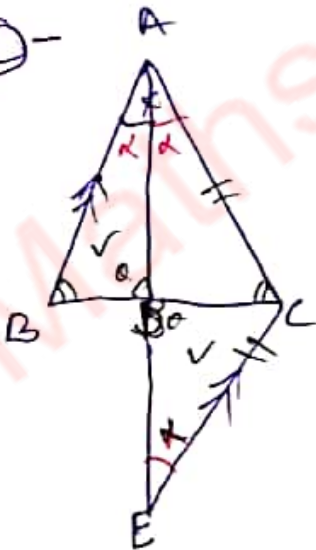
$$\frac{AB}{BD} = \frac{\sin \theta}{\sin \alpha}$$

$$\frac{AC}{DC} = \frac{\sin(180-\theta)}{\sin \alpha} = \frac{\sin \theta}{\sin \alpha}$$

$$\frac{AB}{BD} = \frac{AC}{DC}$$

OR. $\boxed{\frac{AB}{AC} = \frac{BD}{DC}}$ (Interior angle bisector theorem)

OR. ②-

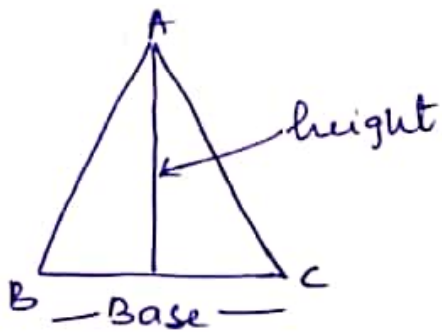


$$\frac{AB}{EC} = \frac{BD}{DC}$$

$$\# \boxed{EC = AC}$$

$$* \boxed{\frac{AB}{AC} = \frac{BD}{DC}}$$

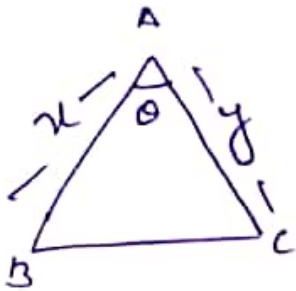
Area of Triangle!—



#①

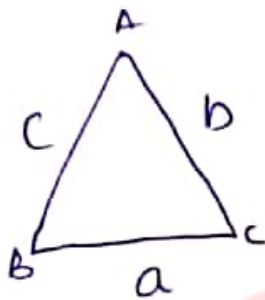
$$\Delta = \frac{1}{2} \times \text{Base} \times \text{height}$$

Δ - Area of triangle



#②

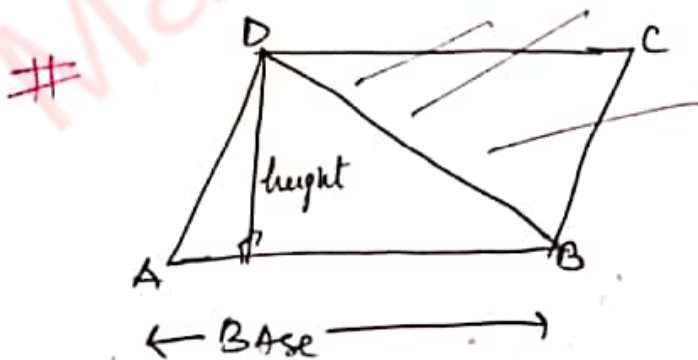
$$\Delta = \frac{1}{2} xy \sin \theta$$



#③

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

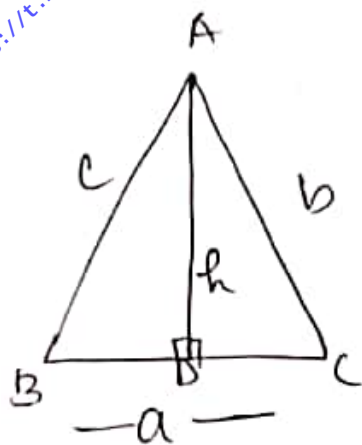
$s \rightarrow$ semiperimeter (अर्धपरिमाप)
(Heron's formula) $\left(\frac{a+b+c}{2} \right)$



Area of parallelogram Area = Base $\times$ height

Area of Δ Area = $\frac{1}{2} \times \text{Base} \times \text{height}$

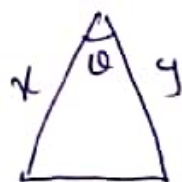
आप



$$\sin B = \frac{h}{c} \Rightarrow h = c \sin B$$

$$\sin C = \frac{h}{b} \Rightarrow h = b \sin C$$

$$\left\{ \begin{aligned} \Delta &= \frac{1}{2} \times a \times h \\ \Delta &= \frac{1}{2} ac \sin B \\ \text{Similarly } \Delta &= \frac{1}{2} ab \sin C \\ \Delta &= \frac{1}{2} bc \sin A \end{aligned} \right.$$



$$\boxed{\text{area} = \frac{1}{2} xy \sin \theta}$$

PROOF HERON'S FORMULA



$$\Delta = \frac{1}{2} bc \sin A$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} \Rightarrow \sin A = \sqrt{1 - \left(\frac{b^2 + c^2 - a^2}{2bc} \right)^2}$$

$$\Delta = \frac{1}{2} \times bc \times \sqrt{1 - \left(\frac{b^2 + c^2 - a^2}{2bc} \right)^2}$$

$$= \frac{1}{2} \times \cancel{bc} \times \frac{1}{\cancel{2bc}} \times \sqrt{(2bc)^2 - (b^2 + c^2 - a^2)^2}$$

$$= \frac{1}{4} \sqrt{(2bc + b^2 + c^2 - a^2) \times (2bc - b^2 - c^2 + a^2)}$$

$$\Delta = \frac{1}{4} \sqrt{\{(b+c)^2 - a^2\} \times \{a^2 - (b+c)^2\}}$$

$$\Delta = \frac{1}{4} \sqrt{(b+c+a)(b+c-a) \times (a+b-c)(a-b+c)}$$

$$\Delta = \sqrt{\frac{a+b+c}{2} \times \frac{b+c-a}{2} \times \frac{a+b-c}{2} \times \frac{a-b+c}{2}}$$

we know that $\left[s = \frac{a+b+c}{2} \right]$

*

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

(इति सिद्धम्)
(Proved)

☺

* # Similarity (समरूपता)

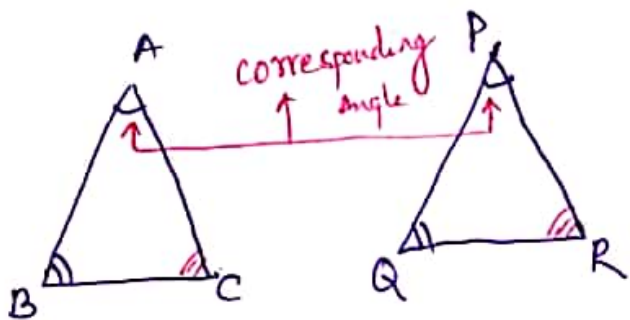
* Similarity of triangles $\rightarrow$ gbs. ~~Size~~ Size may be different or not but their shape is same.
[आकार $\rightarrow$ समान होता है।]

How to Find: —

In Triangle: —

A - Angle

(I) A - A - A Similarity! —
OR A - A Similarity! —



gbs $LA = LP$
 $LB = LQ$
then we know —
 $LC = LR$

$$\triangle ABC \sim \triangle PQR$$

[Similarity Indication]
[Identification]

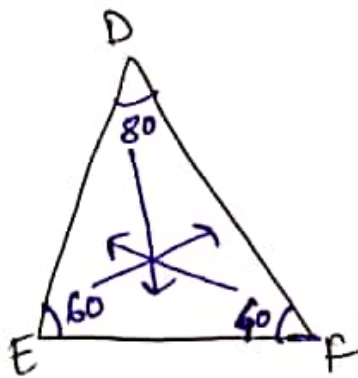
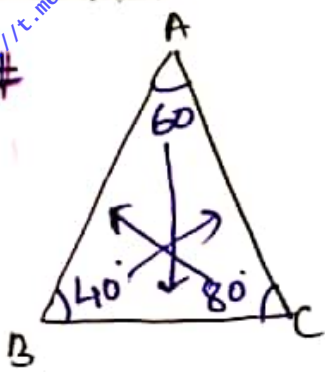
Corresponding Side: —



* Equal Angles के सामने की Side. Corresponding Side होती है।

$$\left\{ \frac{BC}{EF} = \frac{AC}{DF} = \frac{AB}{DE} \right\} = K \text{ (Scaling Factor)} \\ \text{OR (multiplying factor)}$$

Example:-

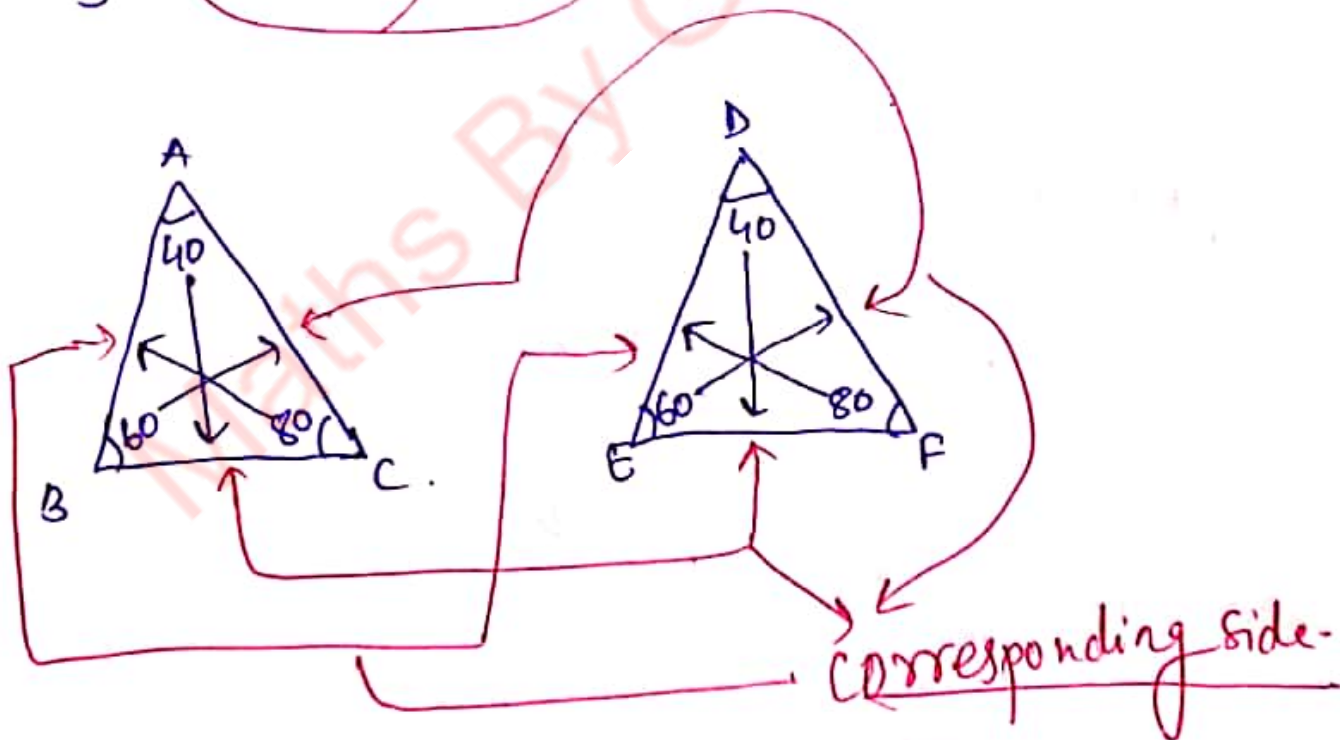
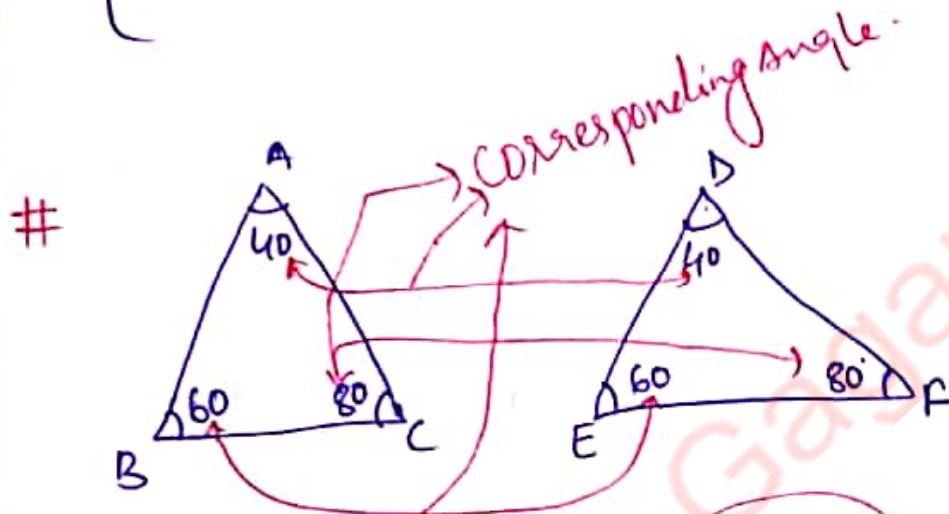


[Angle के सामने वाली भुजा संगत भुजा होगी]

$$\triangle ABC \sim \triangle FED \quad [LA=LE, LB=LF, LC=LD]$$

Its ratio is same [In Similarity]

$$\left\{ \frac{AB}{FE} = \frac{BC}{FD} = \frac{AC}{ED} = k \right\} \text{[Scaling factor]}$$

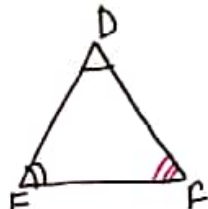
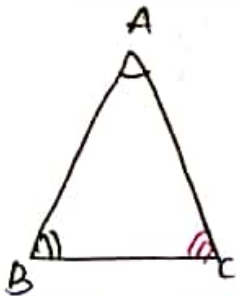


Then -

$$\triangle ABC \sim \triangle DEF$$

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = k \text{ (scaling factor)}$$

Result:-



of $\triangle ABC \sim \triangle DEF$

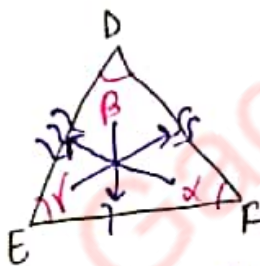
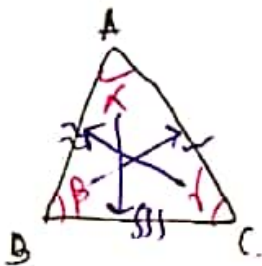
Where $\rightarrow$ $\left\{ \begin{array}{l} r \rightarrow \text{Inradius of } \triangle \\ R \rightarrow \text{Circumradius of } \triangle \\ m \rightarrow \text{medians of } \triangle \\ h \rightarrow \text{Altitude of } \triangle \\ a \rightarrow \text{Angle bisector of } \triangle \\ A \rightarrow \text{Area of } \triangle \end{array} \right.$

then:

$$\frac{BC}{EF} = \frac{AC}{DF} = \frac{AB}{DE} = \frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle DEF} = \frac{r_1}{r_2} = \frac{R_1}{R_2} = \frac{h_1}{h_2} = \frac{m_1}{m_2} = \frac{a_1}{a_2} = k \text{ (Scaling factor)}$$

$$= \left(\frac{A_1}{A_2} \right)^{\frac{1}{2}}$$

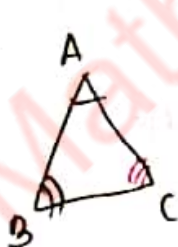
#



$$\triangle ABC \sim \triangle FDE$$

$$\frac{BC}{DE} = \frac{AC}{EF} = \frac{AB}{DF}$$

#



$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle DEF} = \left(\frac{BC}{EF} \right)^2 = \left(\frac{AC}{DF} \right)^2 = \left(\frac{AB}{DE} \right)^2 = k^2$$



$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle DEF} = \frac{\frac{1}{2} \times BC \times h_1}{\frac{1}{2} \times EF \times h_2}$$

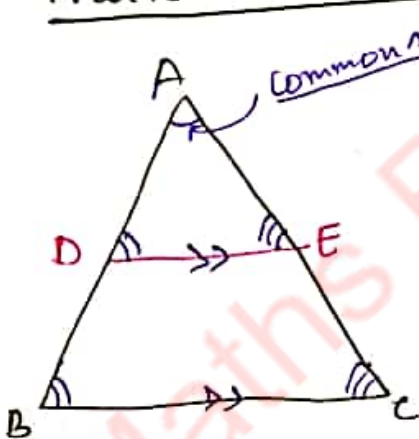
$$= \frac{BC}{EF} \times \frac{BC}{EF} = \left(\frac{BC}{EF}\right)^2$$

We know that in similar

$$\text{where } \frac{h_1}{h_2} = \frac{BC}{EF}$$

Corresponding length — $\times K$
 Area — $\times K^2$
 Volume — $\times K^3$

Thales theorem:-



$$\frac{AD}{BD} = \frac{AE}{EC}$$

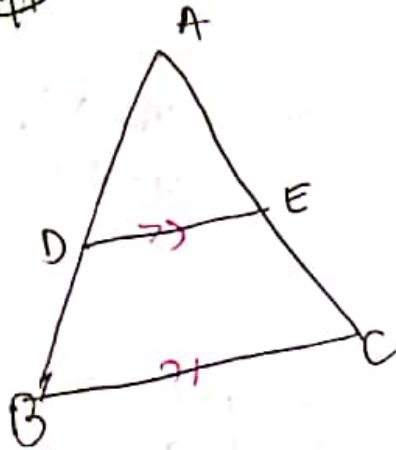
$$\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$$

By similarity

$$\frac{AB}{AD} = \frac{AC}{AE} - 1$$

$$\frac{AB - AD}{AD} = \frac{AC - AE}{AE} \Rightarrow \frac{BD}{AD} = \frac{EC}{AE} \Rightarrow \frac{AD}{BD} = \frac{AE}{EC}$$

#



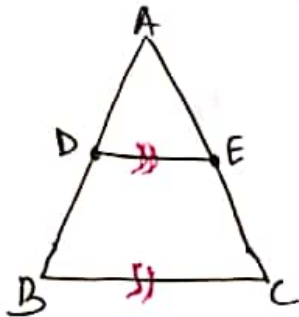
$$\left\{ \begin{array}{l} \frac{AD}{BD} = \frac{AE}{EC} \text{ (given)} \\ \text{OR } \frac{AD}{AB} = \frac{AE}{AC} \text{ (given)} \end{array} \right\}$$

Then,

$$\boxed{DE \parallel BC}$$

(Conversion is also true)

Mid point theorem :-



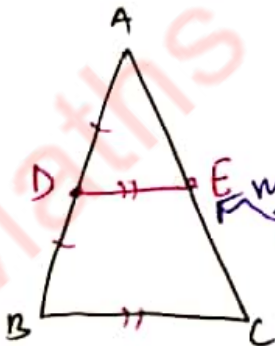
D - mid point of AB (given)
E - mid point of AC

then

$$\boxed{\begin{array}{l} DE \parallel BC \\ DE = \frac{BC}{2} \end{array}}$$

Result #

Convergence of mid point theorem: →

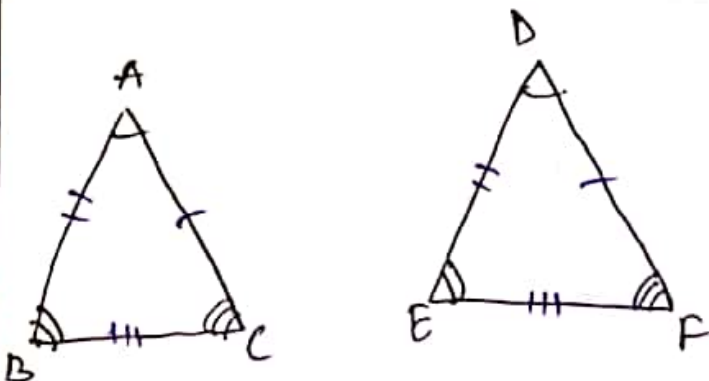


D is mid point of AB D से BC के parallel line खिंचते हैं।

∴ E will be the mid point of AC

Congruency of triangle:-

[Size and Shape $\rightarrow$ Same] then the Δ is congruent.



Given-

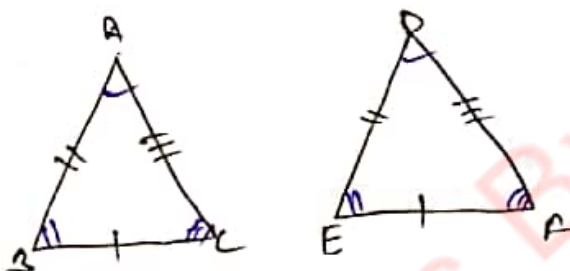
$$\begin{cases} \angle A = \angle D \\ \angle B = \angle E \\ \angle C = \angle F \end{cases}$$

Then $AB = DE$
 $BC = EF$
 $AC = DF$

$$\triangle ABC \cong \triangle DEF$$

(Congruency Identification)

① SSS - (Side Side Side)
 भुजा - भुजा - भुजा



$$\begin{aligned} AB &= DE \\ BC &= EF \\ AC &= DF \end{aligned}$$

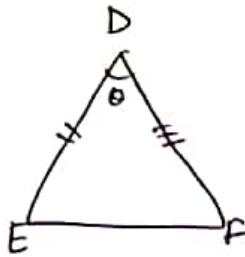
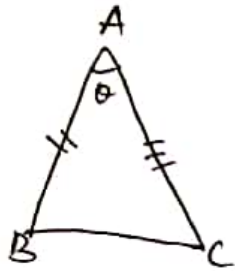
Then $\triangle ABC \cong \triangle DEF$

$$\begin{aligned} \text{Area of } \triangle ABC &= \text{Area of } \triangle DEF \\ \frac{1}{2} \times AB \times AC \sin A &= \frac{1}{2} \times DE \times DF \sin D \end{aligned}$$

$$\sin A = \sin D$$

$$\begin{cases} \angle A = \angle D \\ \angle B = \angle E \\ \angle C = \angle F \end{cases}$$

② SAS (Side-Angle-Side) Congruency.
(शुजा-कोण-शुजा)

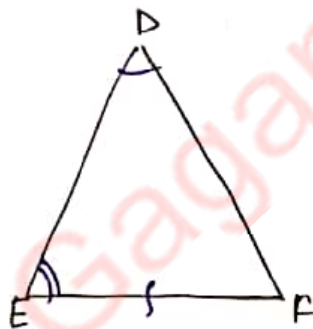
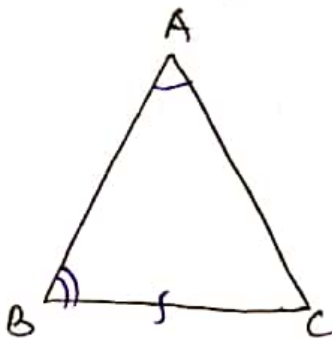


given:
 $\angle A = \angle D$
 $AB = DE$
 $AC = DF$

then $BC = EF$

$$\triangle ABC \cong \triangle DEF$$

③ AAS (Angle-Angle-Side) Congruency.
कोण-कोण-शुजा



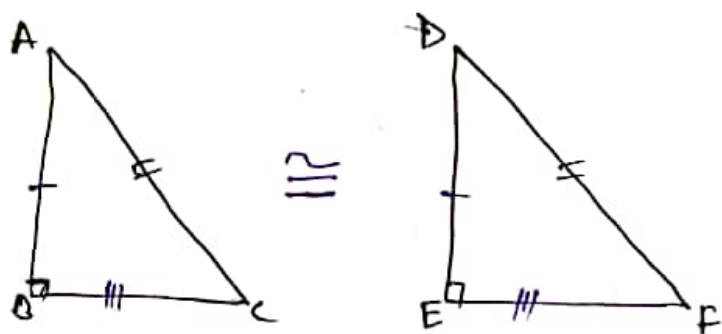
given:
 $\angle A = \angle D$
 $\angle B = \angle E$
 $BC = EF$

then

$$\begin{aligned} \angle C &= \angle F \\ AB &= DE \\ AC &= DF \end{aligned}$$

$$\triangle ABC \cong \triangle DEF$$

④ RHS = (Right angle-hypotenuse-side)
(Congruency)

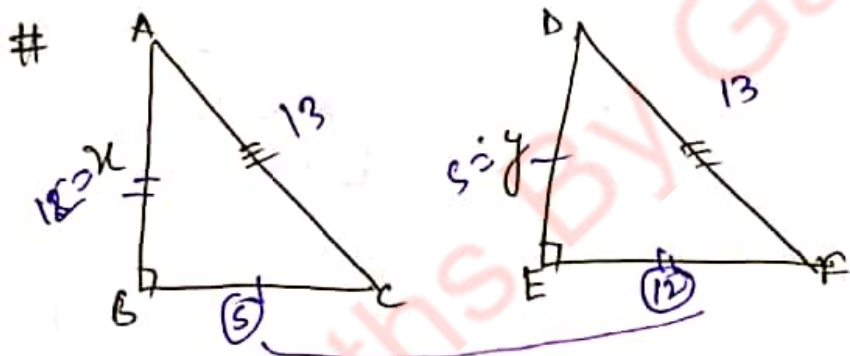


Given $\angle B = \angle E = 90^\circ$

$$AC = DF$$

$$AB = DE$$

then $\left\{ \begin{array}{l} \triangle ABC \cong \triangle DEF \\ BC = EF \end{array} \right\}$ Result



Given $\rightarrow \left\{ \begin{array}{l} BC + EF = x + y \\ BC \sim EF = x \sim y \end{array} \right\}$

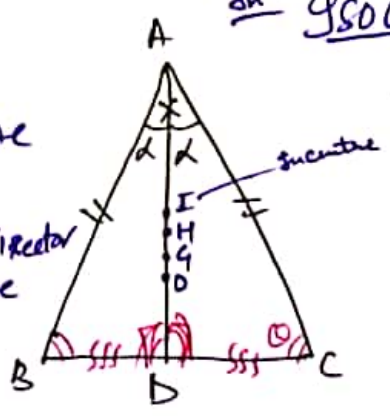
then $\triangle ABC \cong \triangle DEF$
congruent

#

Ex: Isosceles Δ

$AB = AC$

AD is angle bisector of $\angle BAC$



$\left\{ \begin{array}{l} AD - \text{Angle bisector} \\ \text{Altitude/median} \end{array} \right\}$

$AB = AC$

$\angle BAD = \angle CAD$

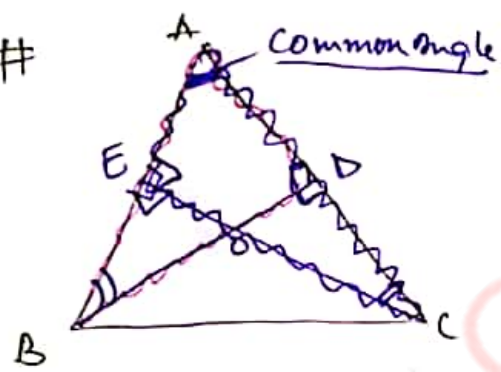
$AD = AD \text{ (Common)}$

$BD = DC$

Then AD is median.

Isosceles Δ में चारो centre एक ही लाइन में होते हैं (AD)

#



Common angle

Altitude $(BD = CE)$ दो Altitude

$\Downarrow$
then Δ is isosceles Δ

$(BD = CE = AF)$ (तीनो altitude)
equilateral Δ

$BD = CE$

$\Rightarrow AB = AC$

2nd method.

$\frac{1}{2} \times AC \times BD = \frac{1}{2} \times AB \times CE$

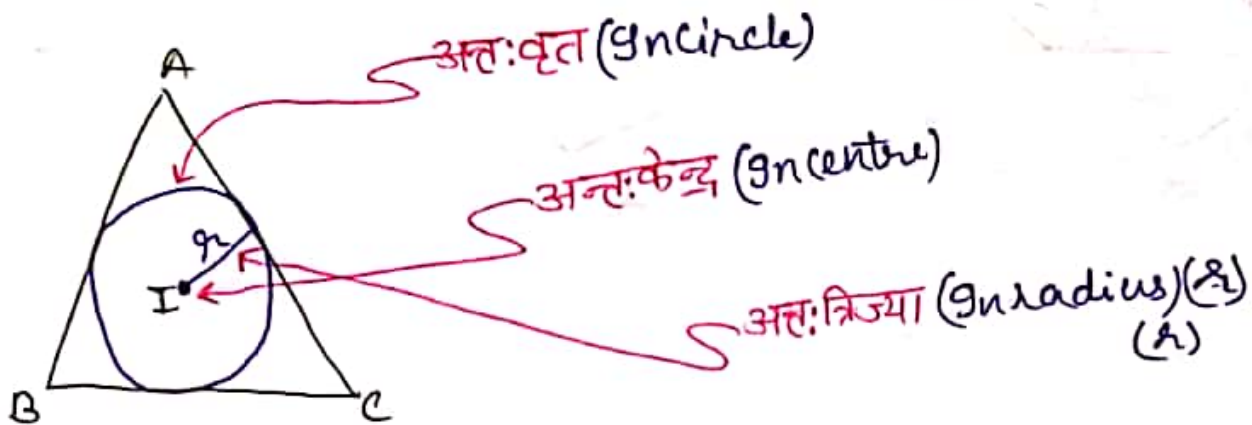
$AC = AB$



Centre of Triangle.

①

(I) - Incentre (अंतःकेन्द्र)

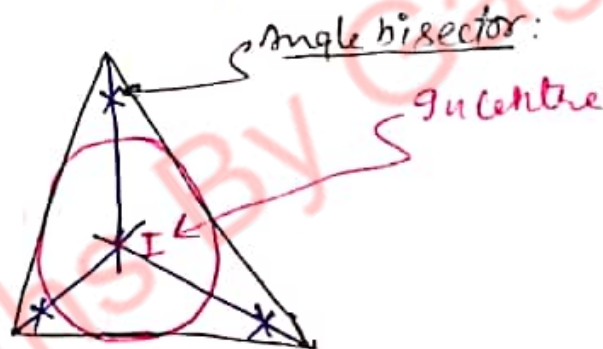


⇒ Incentre is equidistant from all the side.
(बराबर दूरी)

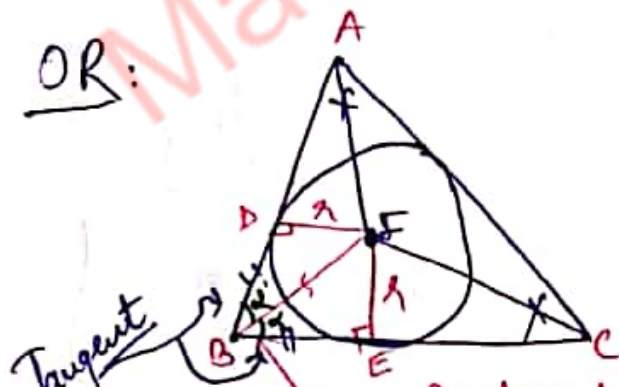
⇒ Incentre always lies inside the Δ .
(अन्दर ही होगा Incentre)
(हमेशा)

⇒ Incentre is Intersection of all the internal angle bisector of Δ triangle.

eg.



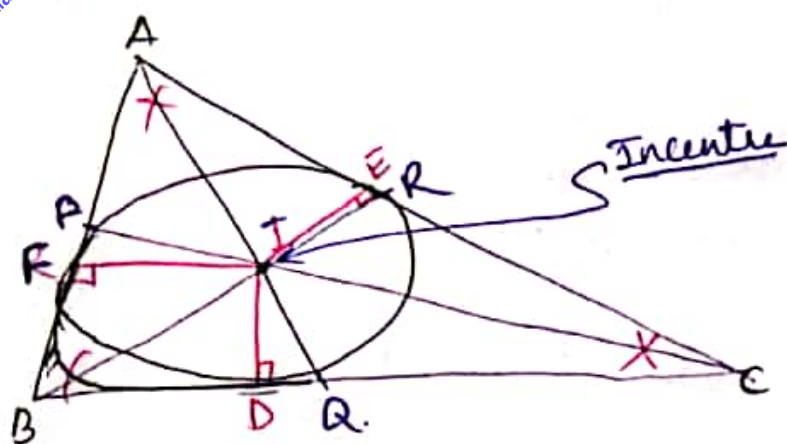
OR:



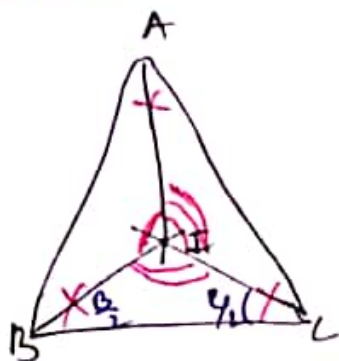
$$\triangle BID \cong \triangle CIE$$

[are congruent]

मेरी interior angle bisector होगा।



Result -



$$\begin{aligned} \angle BIC &= 90^\circ + \frac{\angle A}{2} \\ \angle AIC &= 90^\circ + \frac{\angle B}{2} \\ \angle AIB &= 90^\circ + \frac{\angle C}{2} \end{aligned}$$

PROOF

$$\begin{aligned} \angle BIC &= 180^\circ - \left(\frac{\angle B}{2} + \frac{\angle C}{2} \right) \\ &= 180^\circ - \left(\frac{\angle B + \angle C}{2} \right) \end{aligned}$$

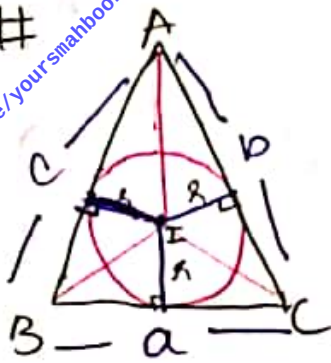
$$\because \angle B + \angle C = 180^\circ - \angle A$$

$$= 180^\circ - 90^\circ + \frac{\angle A}{2}$$

$$\angle BIC = 90^\circ + \frac{\angle A}{2}$$

Similarly,

$$\begin{cases} \angle AIC = 90^\circ + \frac{\angle B}{2} \\ \angle AIB = 90^\circ + \frac{\angle C}{2} \end{cases}$$



$$r = \frac{\Delta}{S}$$

(Ar - Area)

$$Ar \Delta ABC = Ar \Delta BIC + Ar \Delta AIC + Ar \Delta AIB$$

$$\Delta = \frac{1}{2} \times a \times r + \frac{1}{2} b \times r + \frac{1}{2} c \times r$$

$$\Delta = r \left(\frac{a+b+c}{2} \right)$$

$$\therefore \frac{a+b+c}{2} = S$$

$$\Delta = r \cdot S$$

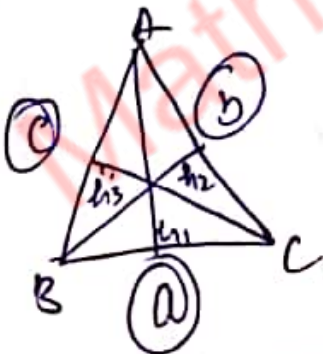
$$r = \frac{\Delta}{S} \text{ (General formula)}$$

h_1, h_2 & h_3 are the height of a Δ . the find Inradius of Δ .

$\Rightarrow$ Result $\rightarrow$

$$\frac{1}{r} = \frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3}$$

PROOF



$$\Delta = \frac{1}{2} a h_1 = \frac{1}{2} b h_2 = \frac{1}{2} c h_3$$

$$a = \frac{2\Delta}{h_1}, b = \frac{2\Delta}{h_2}, c = \frac{2\Delta}{h_3}$$

$$a+b+c = 2\Delta \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3} \right)$$

$$\frac{a+b+c}{2\Delta} = \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3} \right)$$

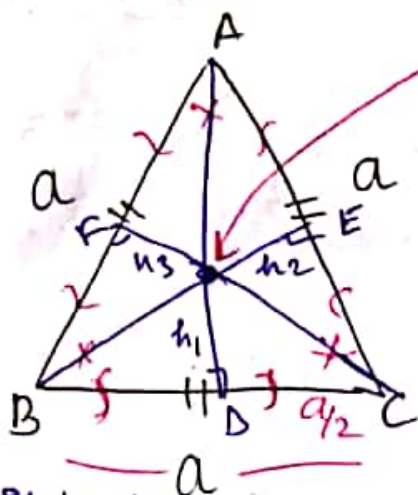
$$\frac{1}{r} = \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3} \right)$$



$$\therefore \frac{a+b+c}{2} = S$$

* Inradius of Equilateral Triangle -

(4)



I, G, H, O

equ! - Area of Δ

$$\Delta = \frac{\sqrt{3}a^2}{4}$$

$$r = \frac{\Delta}{s}$$

$$r = \frac{\frac{\sqrt{3}a^2}{4}}{\frac{3a}{2}}$$

$$r = \frac{a}{2\sqrt{3}}$$

😊

In equilateral Δ

$$h = h_1 = h_2 = h_3 = m$$

median

$$AD = \sqrt{a^2 - \frac{a^2}{4}}$$

$$AD = \frac{\sqrt{3}a}{2} = h$$

OR

$$h = \frac{\sqrt{3}a}{2} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$h = 3 \times \frac{a}{2\sqrt{3}}$$

$\therefore \frac{a}{2\sqrt{3}} = r$ Inradius. [Only equilateral Δ]

$$r = \frac{h}{3}$$

median = angle bisector

$$R = \frac{2H}{3}$$

$$AG : GD = 2 : 1$$

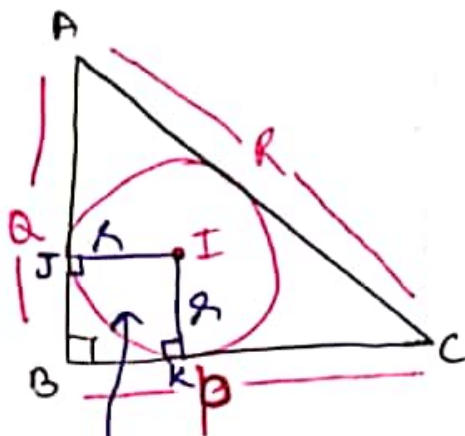
$$R : r = 2 : 1$$

R $\rightarrow$ Circumradius

r $\rightarrow$ Inradius

Radius of Right angle Triangle →

⑤



$$* \quad r = \frac{Q + P - R}{2}$$

$$\# \quad r = \frac{\text{Base} + \text{Perpendicular} - \text{Hypotenuse}}{2}$$

(Square (अनेम)) (JIKB is square)
at.

PROOF

$$r = \frac{\Delta}{S}$$

$$r = \frac{\frac{1}{2} PQ}{\frac{Q+P+R}{2}} \times \frac{P+Q-R}{P+Q-R}$$

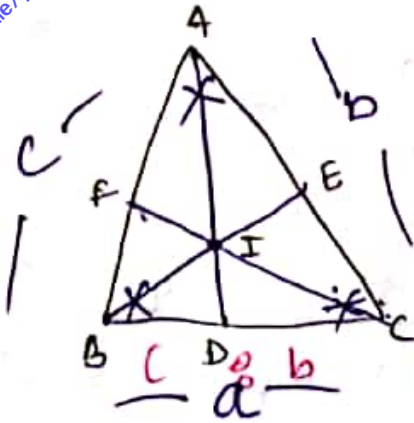
$$r = \frac{PQ (P+Q-R)}{(Q+P)^2 - R^2}$$

$$r = \frac{PQ (P+Q-R)}{Q^2 + P^2 + 2PQ - R^2}$$

$$r = \frac{P+Q-R}{2}$$



Result:-



$$\left\{ \begin{array}{l} \frac{AI}{ID} = \frac{b+c}{a} \\ \frac{BI}{IE} = \frac{a+c}{b} \\ \frac{CI}{IF} = \frac{a+b}{c} \end{array} \right.$$

I - Incentre
AD, BE & CF are angle bisector.

In $\triangle ABD$

$$BD : DC = c : b$$

By angle bisector theorem

$$\rightarrow BD = \left[a \times \frac{c}{c+b} \right]$$

PROOF -

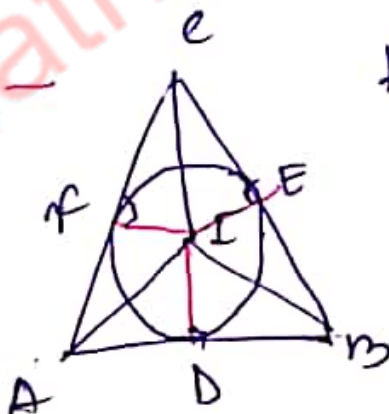
$$\frac{AI}{ID} = \frac{AB}{BD}$$

$$= \frac{c}{a \times \frac{c}{c+b}} = \frac{b+c}{a} \quad \text{Proved}$$

Similarly

$$\left\{ \begin{array}{l} \frac{BI}{IE} = \frac{a+c}{b} \\ \frac{CI}{IF} = \frac{a+b}{c} \end{array} \right.$$

Result:-



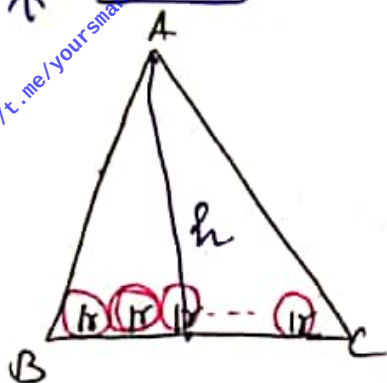
Result

$$\Delta = r^2 \left(\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} \right)$$



* Result -

⑦

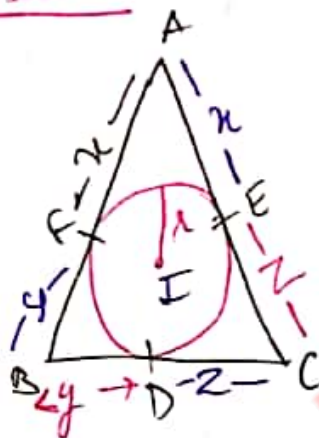


n Circle of equal radius.

$$r = \frac{\Delta}{s + (n-1)h}$$

- Δ - area of $\triangle ABC$
- s - Semiperimeter (अर्धपरिमाप)
- n - no. of circles
- * h - corresponding height (संगत अर्धवृत्त)

Result -



$$r = \sqrt{\frac{xyz}{x+y+z}}$$

Proof: Sides $\rightarrow$ $(x+y), (y+z), (z+x)$

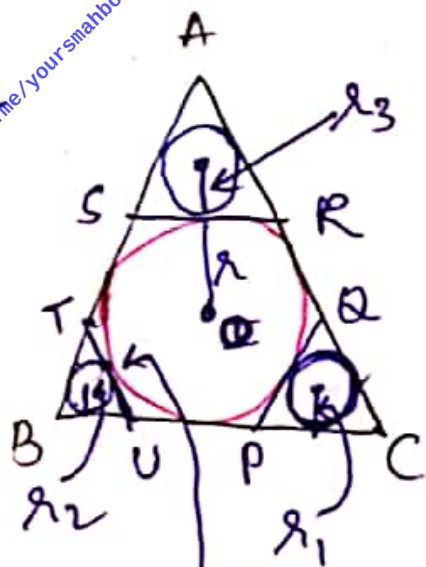
$$s = x+y+z$$

$$\Delta = \sqrt{(x+y+z)xyz}$$

$$r = \frac{\Delta}{s} = \frac{\sqrt{(x+y+z)xyz}}{x+y+z}$$

$$r = \sqrt{\frac{xyz}{x+y+z}}$$

😊 Satyam Padhyay



$PQ \parallel AB$

$RS \parallel BC$

$TU \parallel AC$

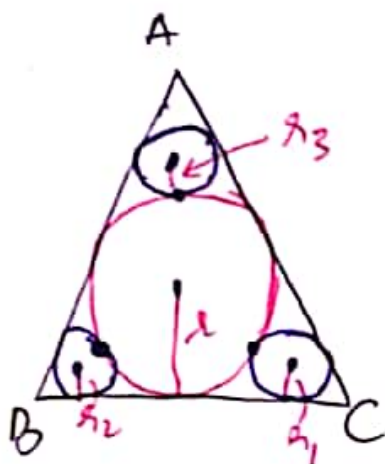
Gudhyang

Result:-

$$r = r_1 + r_2 + r_3$$



(यहाँ Touch नहीं कर रहे हैं वृत्त)



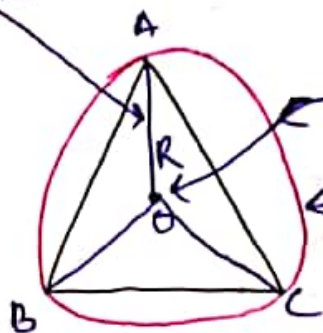
$$r = \sqrt{r_1 r_2} + \sqrt{r_2 r_3} + \sqrt{r_3 r_1}$$

②

Circum radius of a triangle → (परित्रिज्या)

⑨

Circumradius (R)
(परित्रिज्या)



Circum centre
परिकेन्द्र

Circum circle
परिघट्ट

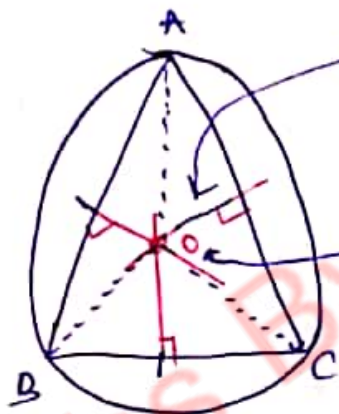
$$OA = OB = OC = R$$

Circumradius.

1) Circum centre may lie outside the circle.
(बाहर भी हो सकता है घट्ट)

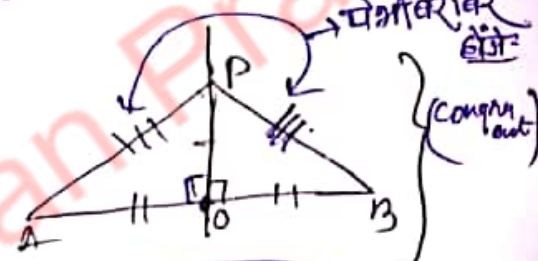
2) Circum centre is intersection of all perpendicular bisector of all 3 sides.

eg.



Perpendicular bisector

Circum centre



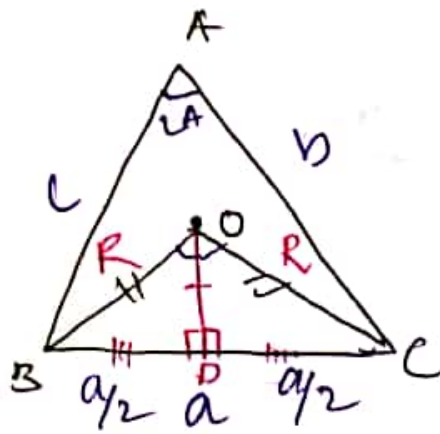
$$AP = BP$$

#



Result:-

$$\left. \begin{aligned} \angle BOC &= 2\angle BAC \\ \angle AOC &= 2\angle ABC \\ \angle AOB &= 2\angle ACB \end{aligned} \right\}$$



O - Circum radius - (R)

$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$$

$$R = \frac{abc}{4\Delta}$$

By RHS congruency
Both $\Delta \{ \triangle BOC \cong \triangle COO \}$

PROOF -

$$\sin A = \frac{a/2}{R}$$

$$R = \frac{a}{2\sin A}$$

$$\Delta = \frac{1}{2} bc \sin A$$

$$\sin A = \frac{2\Delta}{bc}$$

$$R = \frac{a}{2 \times \frac{2\Delta}{bc}}$$

$$\frac{abc}{4\Delta}$$

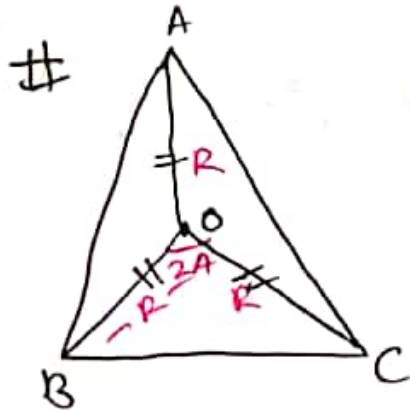
$$\frac{abc}{4\Delta}$$

Similarly -

$$\left\{ R = \frac{b}{2\sin B} = \frac{c}{2\sin C} \right\}$$

Relation b/w r and R and angles of Triangle (17)

$$r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$



[Ar - Area]

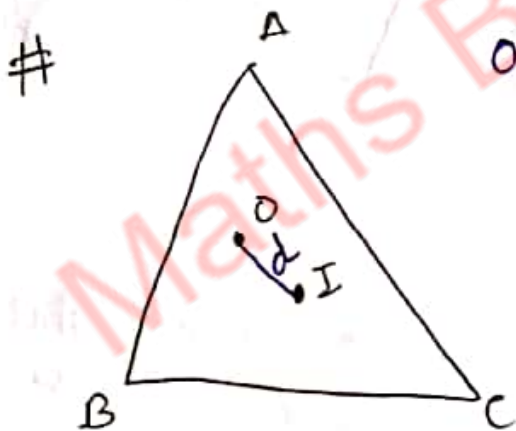
$$\text{Ar. } \triangle ABC = \text{Ar. } \triangle BOC + \text{Ar. } \triangle AOC + \text{Ar. } \triangle AOB$$

$$\Delta = \frac{1}{2} R^2 \sin 2A + \frac{1}{2} R^2 \sin 2B + \frac{1}{2} R^2 \sin 2C$$

$$\Delta = \frac{1}{2} R^2 (\sin 2A + \sin 2B + \sin 2C)$$

$$\Delta = \frac{1}{2} R^2 \times 4 \sin A \sin B \sin C$$

$$\Delta = 2R^2 \sin A \sin B \sin C$$



O - Circumradius
I - Incentre

$$\frac{1}{R+d} + \frac{1}{R-d} = \frac{1}{r}$$

R - Circumradius

r - Inradius

d - distance b/w R and r.

On Solving-

$$d = \sqrt{R^2 - 2Rr}$$

(17)

$$d^2 = R^2 - 2Rr$$

$$d^2 \geq 0 \Rightarrow R^2 - 2Rr \geq 0$$

$$R \cdot (R - 2r) \geq 0$$

$$\Rightarrow R - 2r \geq 0$$

$$R \geq 2r$$

$$\boxed{\frac{R}{r} \geq 2}$$

for any Triangle

$$\left\{ \frac{R}{r} \text{ min. value} \rightarrow 2 \right\}$$

In Equilateral Δ

*#

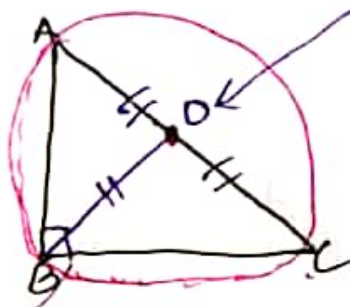
Position of Circum Centre -

① Acute angle triangle (अमकोण)



Circumcentre always अन्दर होगा
Acute angle triangle में

② Right angle Δ — (समकोण)

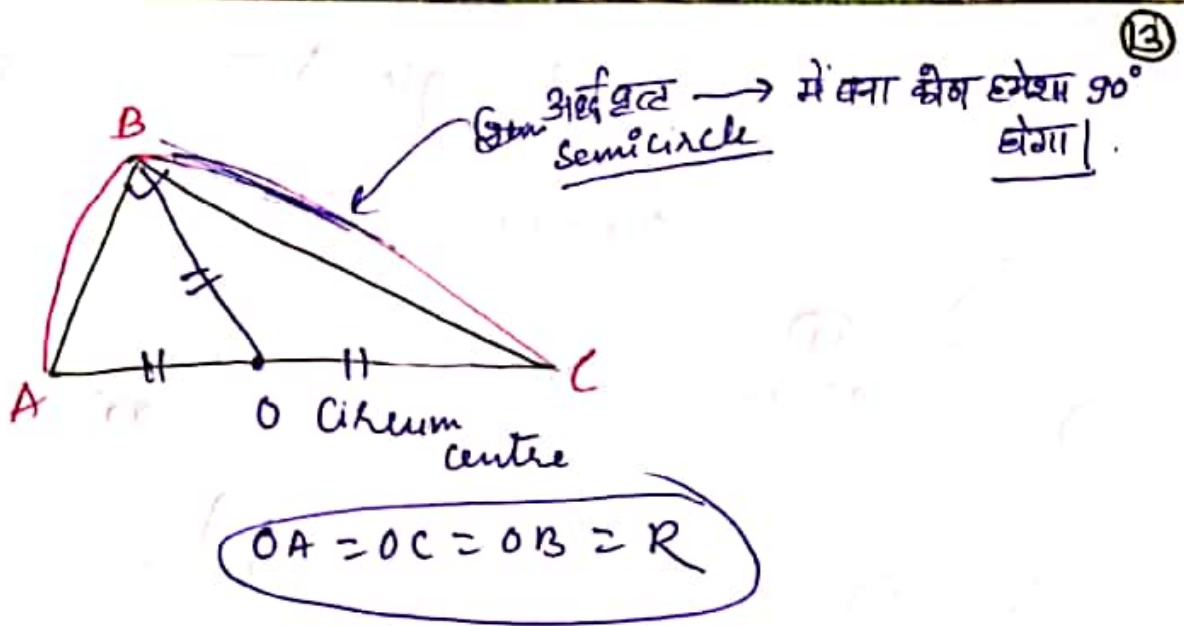


Circumcentre is mid point of hypotenuse.

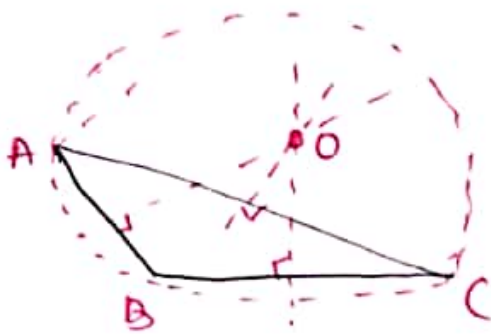
$$\boxed{OA = OB = OC = R}$$

$$OB = R = \frac{H}{2}$$

#* OB is median of Hypotenuse }
SHORTEST median

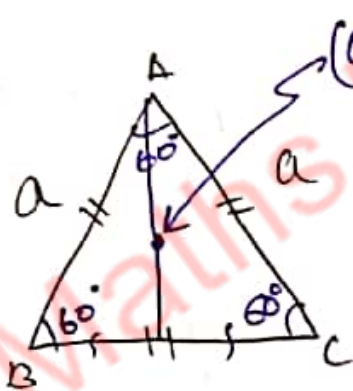


③ Obtuse angle Δ (अधिक कोण Δ)



Circum centre is always outside of the Δ {largest side के समीप होगा}

Circum radius of Equilateral Δ —



(O, I, H)

$$R = \frac{2H}{3} = \frac{a}{\sqrt{3}}$$

$$R = \frac{a}{2 \times \sin 60} = \frac{a}{2 \times \frac{\sqrt{3}}{2}} = \left(\frac{a}{\sqrt{3}} \right)$$

$$r = \frac{a}{2\sqrt{3}}$$

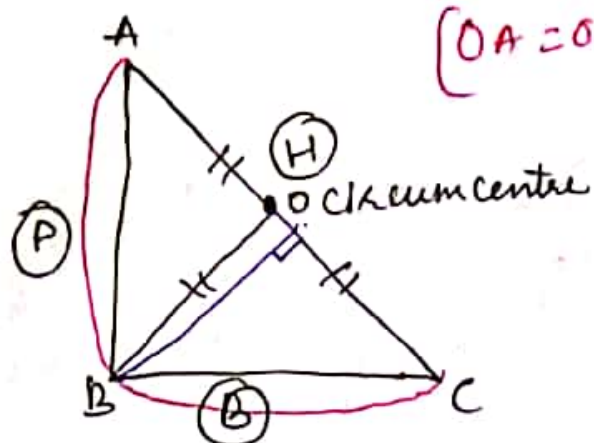
$$AO : DO = 2 : 1$$

$$R : r$$

$$R : r = 2 : 1$$

Circum Radius of Right angle Δ -

(14)



$$[OA = OB = OC = R]$$

$$R = \frac{H}{2 \sin 90^\circ}$$

$$R = \frac{H}{2}$$

$$s = \frac{P+B-H}{2}$$

$$H = 2R$$

$$2s = P+B-H$$

$$2s + H = P+B$$

$$2s + 2R = P+B$$

*

$$2(s+R) = P+B$$

#

$$(s+R) = \frac{P+B}{2}$$

$$\Delta = s \cdot S$$

$$\Delta = \left(\frac{P+B-H}{2} \right) \cdot S$$

$$\Delta = \left(\frac{P+B+H-2H}{2} \right) \cdot S$$

$$\Delta = (S-H) S$$

$$H = 2R$$

*

$$\Delta = S \cdot (S - 2R)$$

Relation b/w r, R and $\Delta \rightarrow$

Right angle Δ
area...

$$\text{Result } \Delta = r^2 + 2rR$$

PROOF

$$r = \frac{P+B-H}{2}$$

$$(2r+H)^2 = (P+B)^2$$

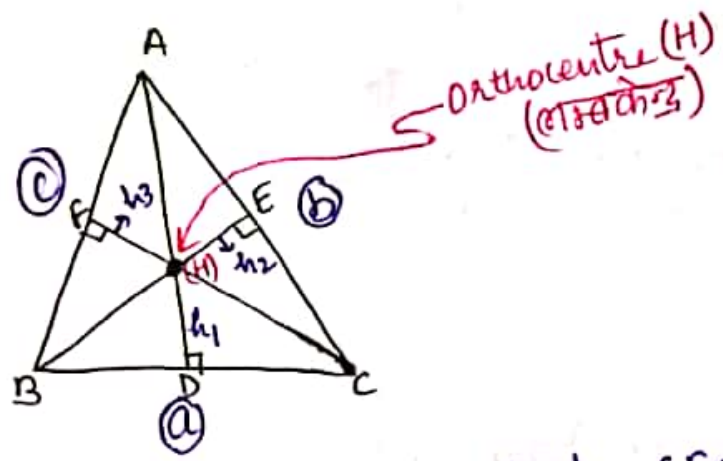
$$4r^2 + H^2 + 4rH = P^2 + B^2 + 2PB$$

$$4r^2 + 4rH = 2PB$$

$$r^2 + rH = \frac{1}{2} PB$$

$$\Delta = r^2 + 2rR$$

Orthocentre of triangle (त्रिभुज का लम्ब केन्द्र)



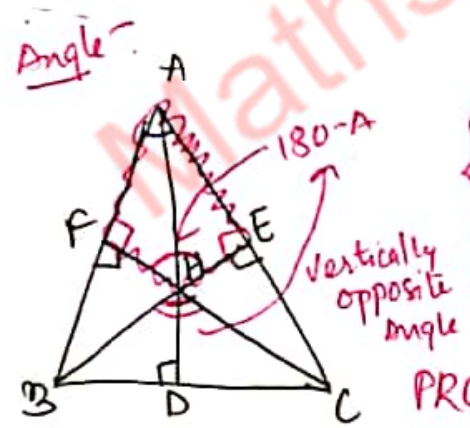
$AD = h_1, BE = h_2, CF = h_3$

→ Orthocentre is Intersection of all 3 altitude of triangle.
(लम्ब केन्द्र)

→ Orthocentre may lie outside the triangle.
(बाहर)

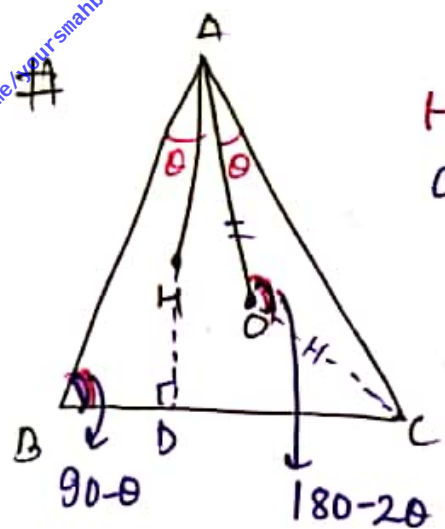
$\Delta = \frac{1}{2} ah_1 = \frac{1}{2} bh_2 = \frac{1}{2} ch_3$

$\Rightarrow ah_1 = bh_2 = ch_3$
 $h_1 : h_2 : h_3 = \frac{1}{a} : \frac{1}{b} : \frac{1}{c}$



$\left\{ \begin{array}{l} \angle BHC = 180^\circ - A \\ \angle AHC = 180^\circ - B \\ \angle AHB = 180^\circ - C \end{array} \right.$

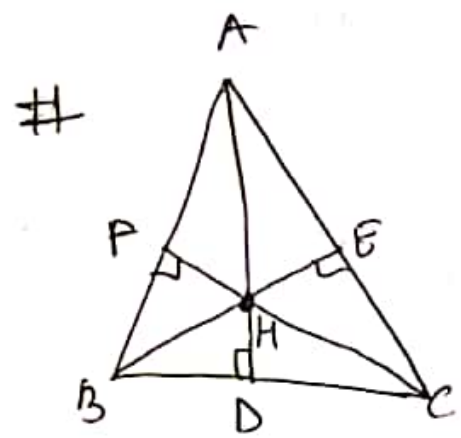
PROOF — $\square AFHE$ में —
 $\angle F + \angle E = 180^\circ$
Then $\angle A = 180^\circ - \angle A$
Then — $\angle BHC = 180^\circ - \angle A$ [Vertically opposite angle]



H - Orthocentre
O - Circumcentre

$$\angle BAH = \angle OAC$$

[Other side ka sum 2θ dena chahiye]
इसलिए $\angle A = 2\theta$ and $\angle C = \theta$.



$$(AB + BC + CA) > (AD + BE + CF)$$

$$\frac{AB + BC + CA}{AD + BE + CF} > 1$$

$$\begin{aligned} BC &> FC \\ AB &> BE \\ AC &> AD \end{aligned}$$

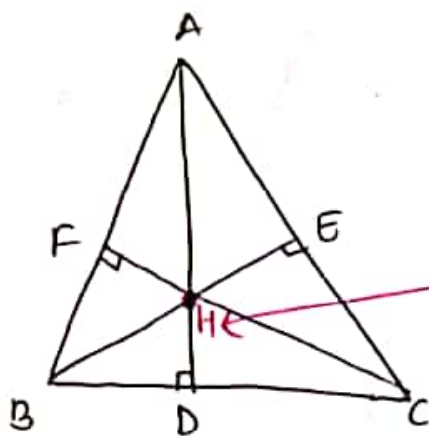
$$\left\{ (AB + BC + CA) > (AD + BE + CF) \right\}$$

$$\Rightarrow \frac{AB + BC + CA}{AD + BE + CF} > 1$$

Position of orthocentre -

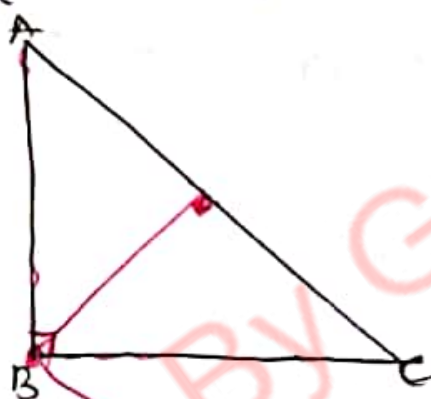
(17)

① Acute angle triangle - (भुजकोण Δ)

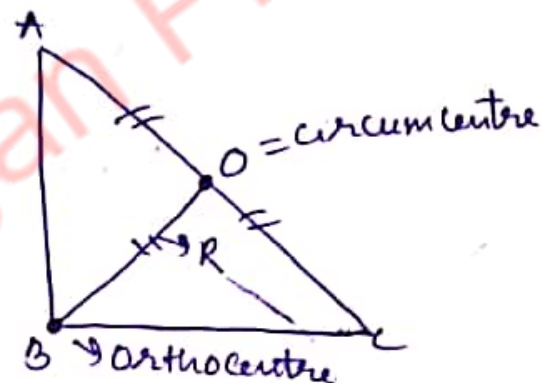


Orthocentre (लम्ब केन्द्र)
(हमेशा Inside होगा भुजकोण Δ में)

② Right angle triangle - (समकोण Δ)



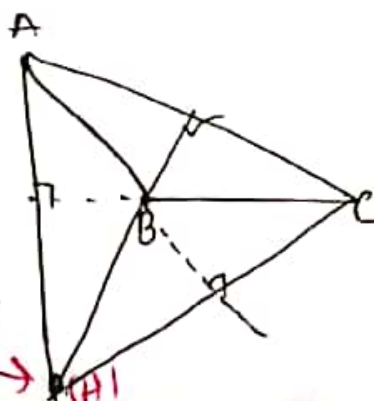
Orthocentre at 90° vertex.



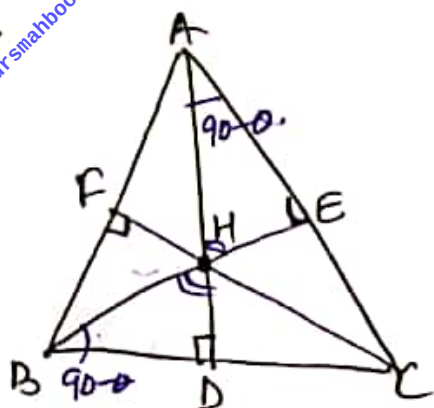
OB $\rightarrow$ distance
b/w Orthocentre
and Circumcentre

$$R = \frac{H}{2}$$

③ Obtuse angle triangle (अधिक कोण Δ)



(हमेशा Δ के बाहर होगा
अधिक कोण Δ में)
Orthocentre $\rightarrow$



$$AH \times HD = BH \times HE = CH \times HF$$

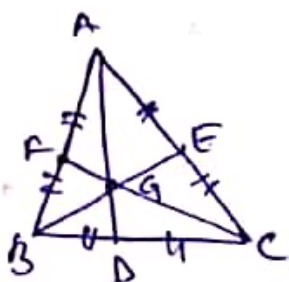
$\triangle BHD$ & $\triangle EHA$ में

$$\Rightarrow \triangle BHD \sim \triangle EHA$$

$$\frac{AH}{BH} = \frac{HE}{HD}$$

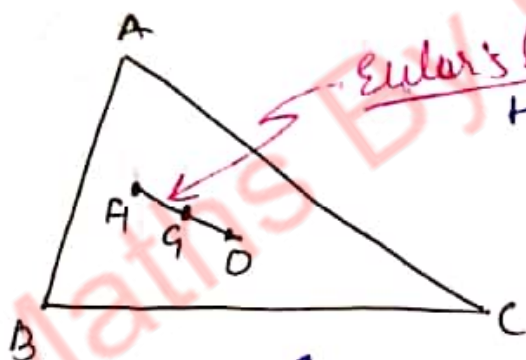
$$AH \times HD = BH \times HE$$

Centroid -



$$AG:GD = BG:GE = CG:CF = 2:1$$

#



Euler's line

H = Orthocentre

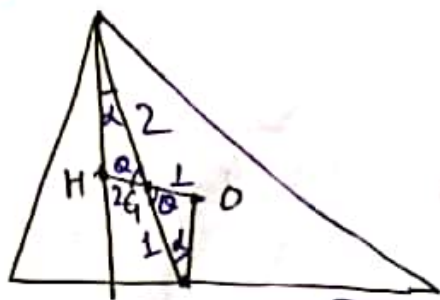
G = Centroid

O = Circum Centre

H, G & O एक सीधी रेखा में होते हैं।

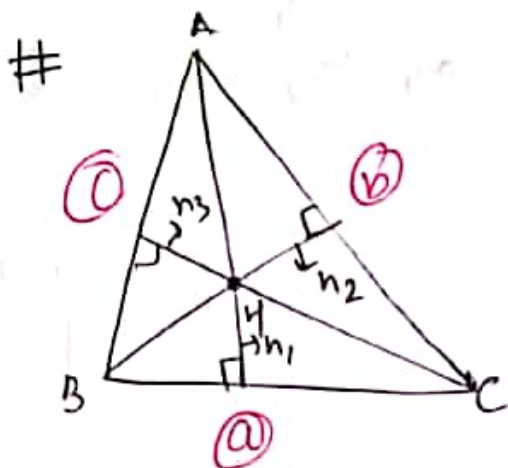
$$[HG:GO = 2:1]$$

any triangle



$BE = EC$ — $OE \perp BC$

By Similarity
Proof



$$\frac{h_1 h_2}{h_1 + h_2} < h_3 < \frac{h_1 h_2}{h_1 - h_2}$$

difference

$$\text{Area } \Delta = \frac{1}{2} a h_1 = \frac{1}{2} b h_2 = \frac{1}{2} c h_3$$

$$a = \frac{2\Delta}{h_1}, \quad b = \frac{2\Delta}{h_2}, \quad c = \frac{2\Delta}{h_3}$$

$$\Rightarrow |a - b| < c < |a + b|$$

$$\Rightarrow \left| \frac{2\Delta}{h_1} - \frac{2\Delta}{h_2} \right| < \frac{2\Delta}{h_3} < \left| \frac{2\Delta}{h_1} + \frac{2\Delta}{h_2} \right|$$

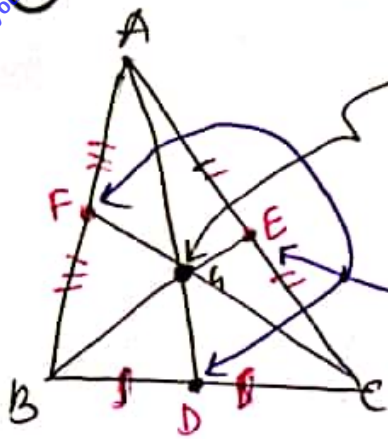
$$= \frac{h_2 - h_1}{h_1 h_2} < \frac{1}{h_3} < \frac{h_1 + h_2}{h_1 h_2}$$

$$\Rightarrow \frac{h_1 h_2}{h_1 - h_2} > h_3 > \frac{h_1 h_2}{h_1 + h_2}$$



Centroid of triangle →

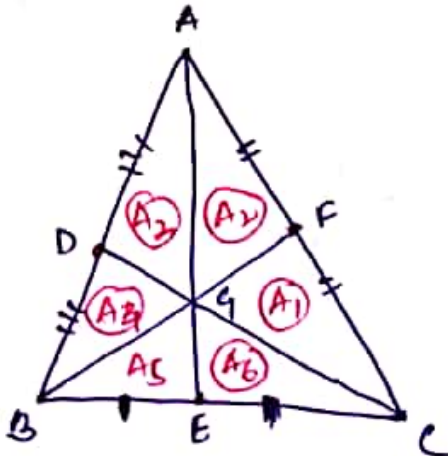
(20)



Centroid
(G)

$$AG:GH = BG:GE = CG:GF = 2:1$$

median
(माध्यिका)

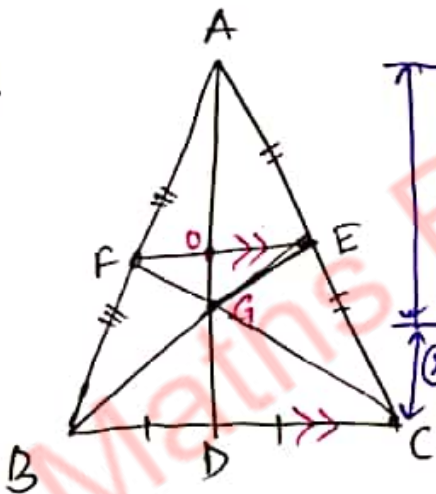


$$A_1 = A_2 = A_3 = A_4 = A_5 = A_6$$

A - Area

6 equal Area में divide करता है।

#



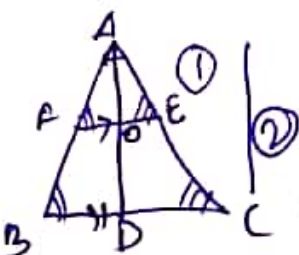
$$AG = 4\text{ cm}$$

$$AO:OG:GD = 3:1:2$$

$$\left\{ \begin{array}{l} EF \parallel BC \\ EF = \frac{BC}{2} \end{array} \right\}$$

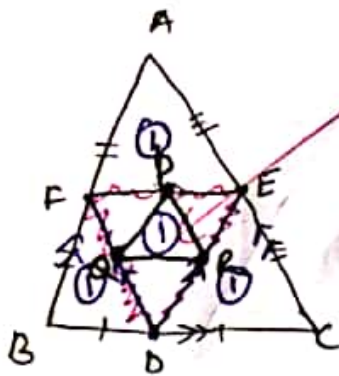
$$\therefore \left\{ \begin{array}{l} AO = 3\text{ cm} \\ OG = 1\text{ cm} \\ GD = 2\text{ cm} \end{array} \right\}$$

$$\therefore \text{total} = 6\text{ cm}$$

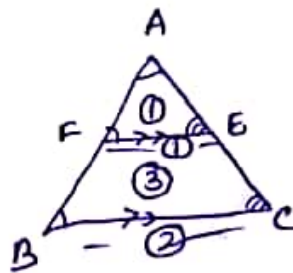


Medial Triangle:-

(21)



$FE \parallel BC$ by midpoint point theorem.



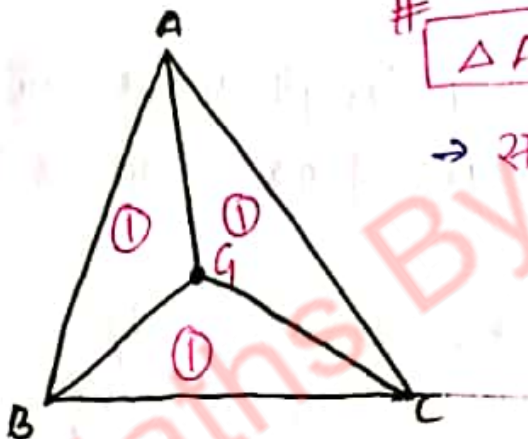
$$\frac{\Delta AFE}{\Delta ABC} = \frac{1}{4}, \quad \frac{\Delta BDF}{\Delta ABC} = \frac{1}{4}, \quad \frac{\Delta CED}{\Delta ABC} = \frac{1}{4}$$

$\Delta DEF = \frac{1}{4} \times \text{Area of } \Delta ABC$

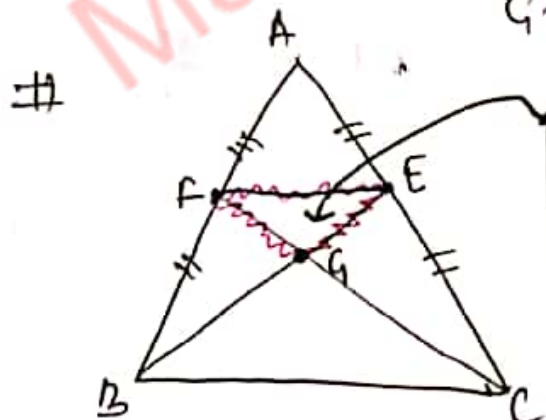
$\Delta PQR = \frac{1}{4} \times \Delta DEF$

$\Delta AGC = \Delta BGC = \Delta AGB = 1$

→ सभी का क्षेत्रफल बराबर होता है।
(केन्द्रक) (Centroid)

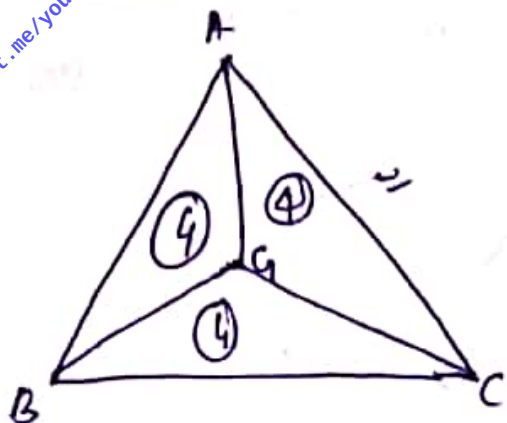


G - Centroid of ΔABC

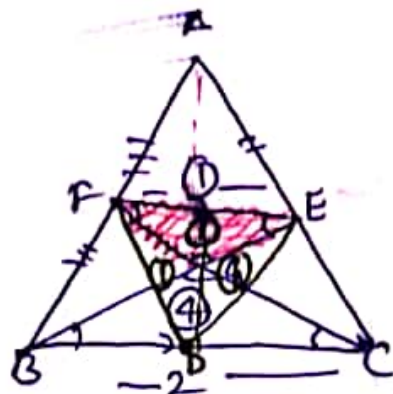


$\text{Area of } \Delta GEF = \frac{1}{12} \text{ of Ar. of } \Delta ABC$





Assume



$$FE : BC = 1 : 2$$

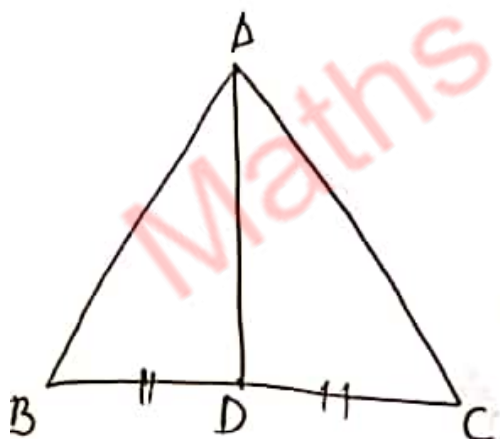
$$\Delta FGE : \Delta BGC = 1 : 4$$

$$\frac{\Delta FGE}{\Delta ABC} = \frac{1}{12}$$

$$Ar ABC = 4 + 4 + 4 = 12 \text{ cm}^2$$

$$\left[\frac{\Delta DEF (\text{medial})}{Ar ABC} = \frac{3}{12} = \frac{1}{4} \right]$$

Apollonius theorem $\rightarrow$ [To find length of median]
(भारमिका की लंब कर्षत करने कोलेए)



$$BD = DC$$

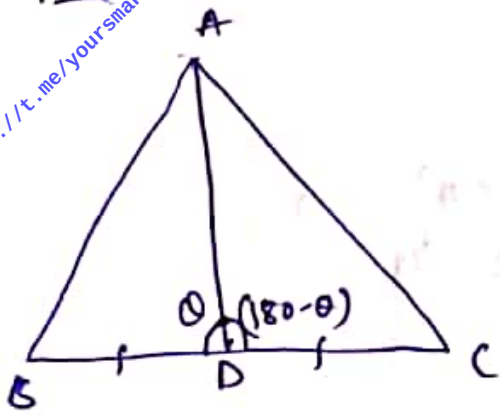
AD = median of side BC

$$AB^2 + AC^2 = 2(AD^2 + BD^2)$$

Apollonius theorem -

OR:

$$AB^2 + AC^2 = 2\left(AD^2 + \frac{BC^2}{4}\right)$$



By using cosine rule—

In $\triangle ADB$ $AB^2 = AD^2 + BD^2 - 2AD \cdot BD \cos \theta$ — (1)

In $\triangle ADC$ $AC^2 = AD^2 + DC^2 - 2AD \cdot DC \cdot \cos(180 - \theta)$
 $AC^2 = AD^2 + DC^2 + 2AD \cdot DC \cdot \cos \theta$ — (2) $\left[\because \cos(180 - \theta) = -\cos \theta \right]$

eqⁿ (1) + eqⁿ (2)

$$AB^2 = AD^2 + BD^2 - 2AD \cdot BD \cos \theta$$

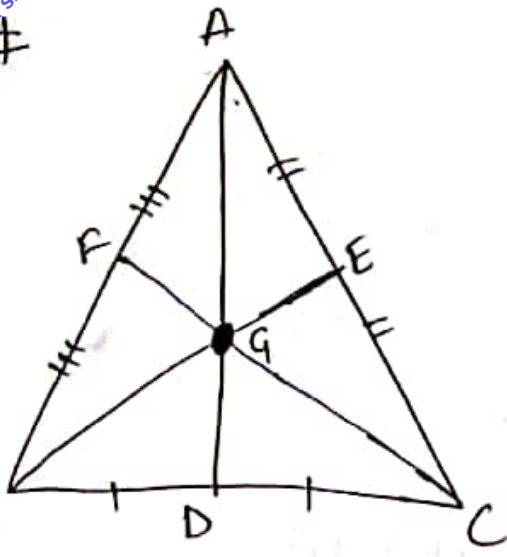
$$AC^2 = AD^2 + DC^2 + 2AD \cdot DC \cos \theta$$

$\therefore BD = DC$

$AB^2 + AC^2 = 2(AD^2 + BD^2)$

Apollonius theorem

→ To find length of median —



$$\frac{AB^2 + BC^2 + CA^2}{AD^2 + BE^2 + CF^2} = \frac{4}{3}$$

PROOF

By using Apollonius theorem -

$$AB^2 + AC^2 = 2\left(AD^2 + \frac{BC^2}{4}\right) \quad \text{--- (i)}$$

$$AB^2 + BC^2 = 2\left(BE^2 + \frac{AC^2}{4}\right) \quad \text{--- (ii)}$$

$$AC^2 + BC^2 = 2\left(CF^2 + \frac{AB^2}{4}\right) \quad \text{--- (iii)}$$

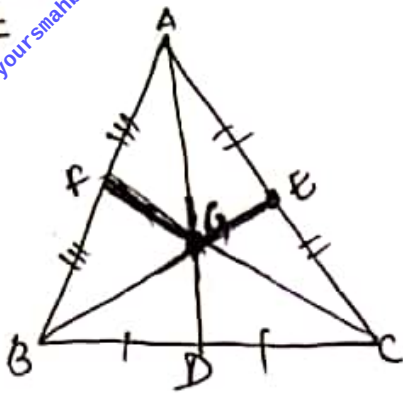
Add $\rightarrow 2(AB^2 + BC^2 + AC^2) = 2(AD^2 + BE^2 + CF^2) + \left(\frac{BC^2 + AC^2 + AB^2}{2}\right)$

$$\# \left(2 - \frac{1}{2} = \frac{3}{2}\right)$$

$$\frac{3}{2}(AB^2 + BC^2 + CA^2) = 2(AD^2 + BE^2 + CF^2)$$

$$\frac{AB^2 + BC^2 + CA^2}{AD^2 + BE^2 + CF^2} = \frac{4}{3}$$

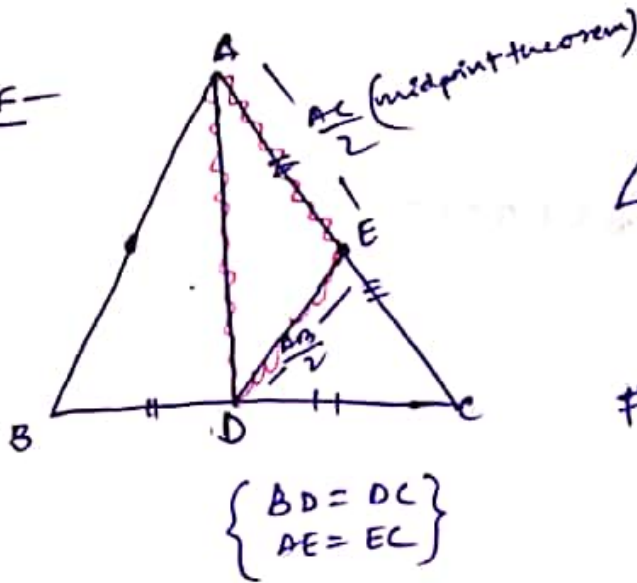
☺ इति सिद्धम्



$$1 < \frac{AB+BC+CA}{AD+BE+CF} < \frac{4}{3}$$

[AD, BE, CF → median]
G → centroid

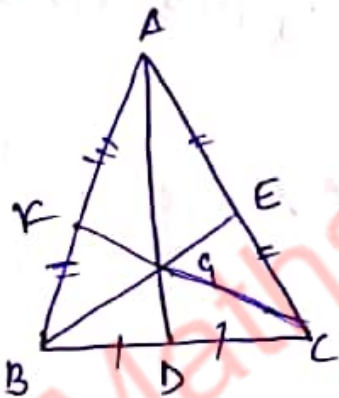
PROOF-



$\triangle ADE \sim \triangle ABC$

$$\frac{AB}{2} + \frac{AC}{2} > AD$$

$$\# \boxed{AB+AC > 2AD}$$



$$AB+AC > 2AD$$

$$AB+BC > 2BE$$

$$AC+BC > 2CF$$

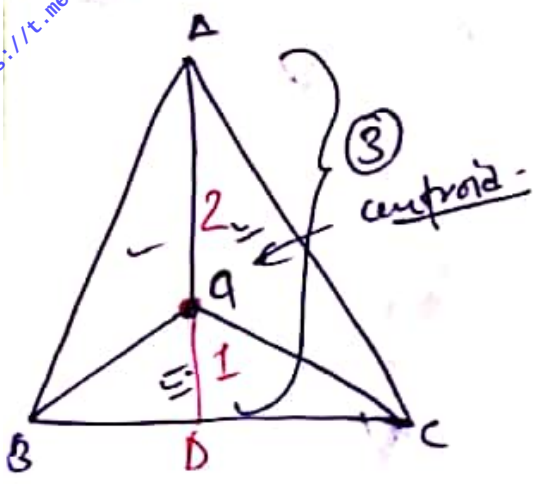
$$\text{Add - } \cancel{2}(AB+BC+CA) > \cancel{2}(AD+BE+CF) \times \cancel{2}$$

$$\& \circledast (AB+BC+CA) > (AD+BE+CF)$$

$$\left\{ \frac{AB+BC+CA}{AD+BE+CF} > 1 \right\}$$

[1st condition proved]

2nd condition.



$$\begin{aligned} BG + GC &> BC \\ AG + GC &> AC \\ AG + BG &> AB \end{aligned}$$

Add - $2(AG + BG + GC) > (AB + BC + CA)$

$$\left\{ \begin{aligned} AG &= \frac{2}{3} AD \\ BG &= \frac{2}{3} BE \\ CG &= \frac{2}{3} CF \end{aligned} \right\}$$

$$2 \times \left(\frac{2}{3} AD + \frac{2}{3} BE + \frac{2}{3} CF \right) > (AB + BC + CA)$$

$$4(AD + BE + CF) > 3(AB + BC + CA)$$

OR

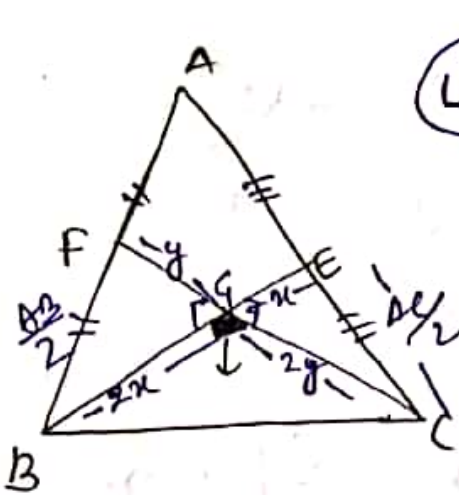
$$\frac{AB + BC + CA}{AD + BE + CF} < \frac{4}{3}$$

2nd condition proved

इति सिद्धम्



When two median Intersects each other at $90^\circ \rightarrow$ (27)



$$\angle BGC = 90^\circ$$

$$AB^2 + AC^2 = BC^2$$

PROOF

$$4x^2 + y^2 = \frac{AB^2}{4} \quad \text{--- (i)}$$

$$x^2 + 4y^2 = \frac{AC^2}{4} \quad \text{--- (ii)}$$

$$\underline{\text{Add}} - 5(x^2 + y^2) = \frac{AB^2 + AC^2}{4}$$

$$5(4x^2 + 4y^2) = AB^2 + AC^2$$

$$5((2x)^2 + (2y)^2) = AB^2 + AC^2$$

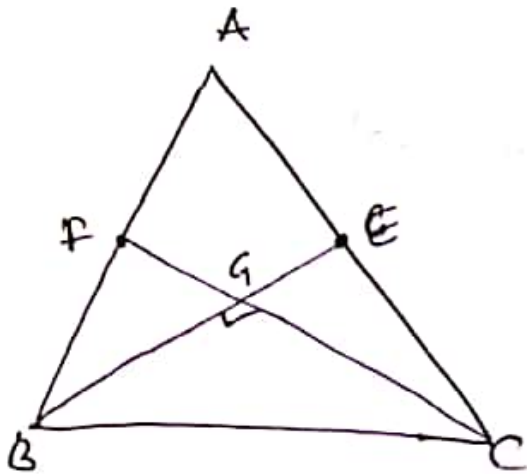
$$\boxed{BC^2 = AB^2 + AC^2}$$

$$\left[\text{In } \Delta BGC \right] \\ \left[(2x)^2 + (2y)^2 = BC^2 \right]$$

☺ [इति सिद्धम्]

Special Case $\rightarrow$ [Isosceles Δ]

(28)



$$AB^2 + AC^2 = 4BC^2$$

$$2AB^2 = 4BC^2$$

$$\frac{AB^2}{BC^2} = \frac{4}{2}$$

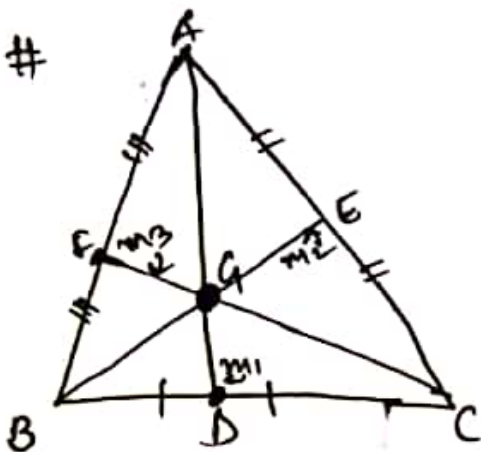
(Tier-II) ✓

OR.

$$\frac{AB}{BC} = \sqrt{\frac{4}{2}}$$

(CPD) ✓

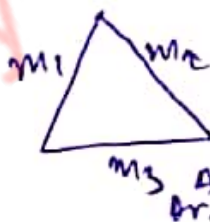
$AB = BC$ (median BE and CF are perpendicular to each other)



To Find:

Area of Δ when length of all 3 medians given.

median $\left\{ \begin{array}{l} AD = m_1 \\ BE = m_2 \\ CF = m_3 \end{array} \right\}$



$$S_m = \frac{m_1 + m_2 + m_3}{2}$$

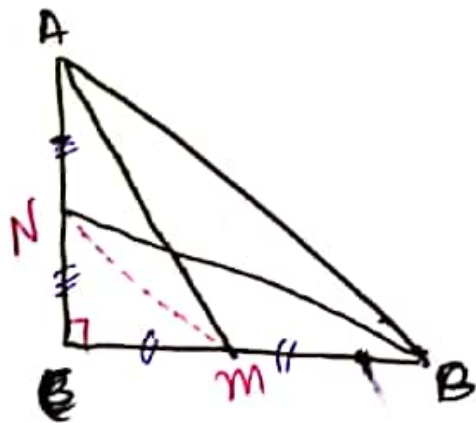
$$Ar. \Delta ABC = \frac{4}{3} \times \sqrt{S_m (S_m - m_1) (S_m - m_2) (S_m - m_3)}$$

#

$$Ar. \Delta ABC = \frac{4}{3} \times \sqrt{S_m (S_m - m_1) (S_m - m_2) (S_m - m_3)}$$

$$Ar. \Delta ABC = \frac{4}{3} \times Ar. \Delta \text{ made by using length of median}$$

median in right angle triangle - (29)



$$AM^2 + BN^2 = AB^2 + MN^2 \quad \text{--- (1)}$$

$$[MN = \frac{AB}{2}] \text{ (midpoint theorem)}$$

AM & NB are the median of $\triangle ABC$ -
 $\angle C = 90^\circ$

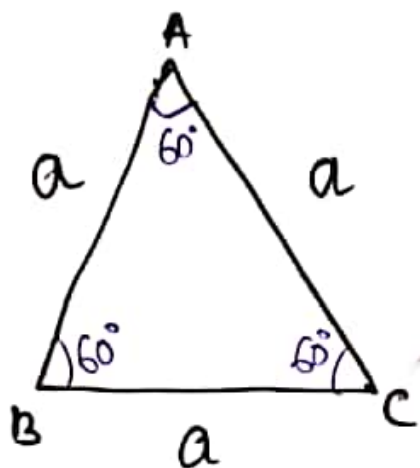
put the value of MN in eqⁿ --- (1)

$$AM^2 + BN^2 = \frac{AB^2}{4}$$

$$* \quad 4(AM^2 + BN^2) = AB^2$$

OR $\left[\frac{AM^2 + BN^2}{AB^2} = \frac{1}{4} \right] \quad \text{--- (2)}$

1) Equilateral Triangle (समबाहु त्रिभुज)



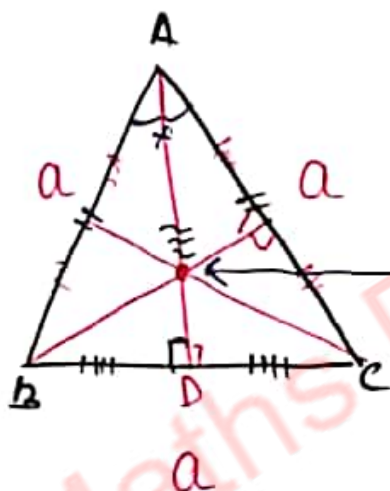
$$AB = BC = CA = a$$
$$\angle A = \angle B = \angle C = 60^\circ$$

$$\text{Area} = \frac{1}{2} \times a \times a \times \frac{\sqrt{3}}{2}$$

$$[\sin 60^\circ = \frac{\sqrt{3}}{2}]$$

$$\boxed{\text{Area} = \frac{\sqrt{3}}{4} \times a^2}$$

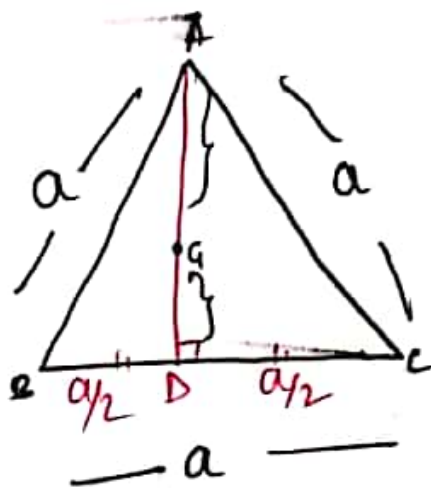
⇒



{ AD → Altitude, perpendicular bisector
AD → median ; Angle bisector

(H, I, G, O)

⇒



$$AD = \sqrt{a^2 - \frac{a^2}{4}}$$

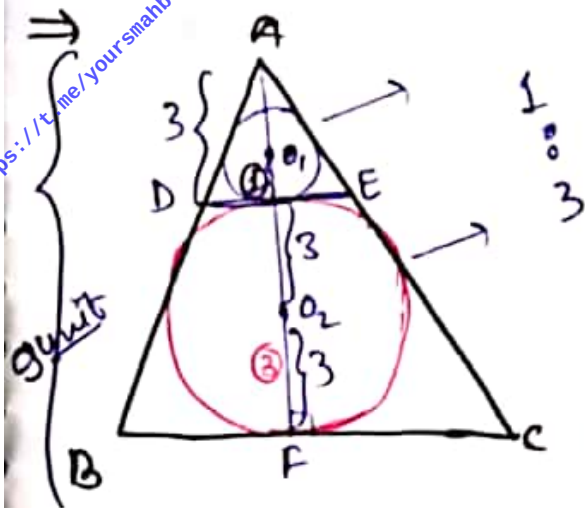
$$\boxed{AD = \frac{\sqrt{3}}{2} a = \text{height}}$$

inradius

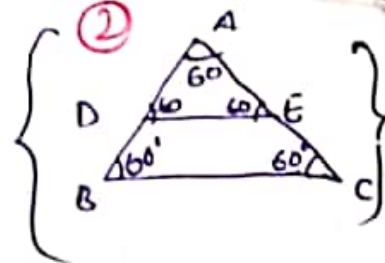
$$r = \frac{\text{height}}{3} = \frac{a}{2\sqrt{3}}$$

circumradius

$$R = \frac{\text{height} \times 2}{3} = \frac{a}{\sqrt{3}}$$



[$\triangle ABC$ - समबाहु $\triangle$]

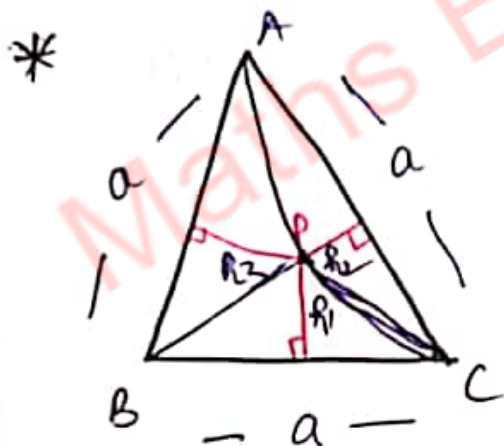


$$\left\{ \begin{array}{l} \text{Radius} = 1.83 \\ \text{Area} = 1.89 \end{array} \right\}$$

$$\begin{aligned} \Rightarrow \text{Side} &= a \\ \text{height} &= \frac{\sqrt{3}}{2} a \\ \text{Inradius} &= \frac{a}{2\sqrt{3}} \\ \text{Circumradius} &= \frac{a}{\sqrt{3}} \\ \text{area} &= \frac{\sqrt{3}}{4} a^2 \end{aligned}$$

Let

$$\begin{aligned} \text{Side} &= 2 \quad (XK) \\ \text{height} &= \sqrt{3} \quad (XK) \\ \text{Inradius} &= \frac{1}{\sqrt{3}} \quad (XK) \\ \text{Circumradius} &= \frac{2}{\sqrt{3}} \quad (XK) \\ \text{area} &= \sqrt{3} \quad (XK^2) \end{aligned}$$



$$h = h_1 + h_2 + h_3$$

$$\frac{\sqrt{3}}{4} a^2 = \frac{1}{2} \times a \times h_1 + \frac{1}{2} \times a \times h_2 + \frac{1}{2} \times a \times h_3$$

$$\frac{\sqrt{3}}{4} a \times a = \frac{1}{2} a [h_1 + h_2 + h_3]$$

$$\frac{\sqrt{3} a}{2} = h_1 + h_2 + h_3$$

$$\therefore h = \frac{\sqrt{3} a}{2}$$

$$h = h_1 + h_2 + h_3$$

$$a = \frac{2}{\sqrt{3}} (h_1 + h_2 + h_3)$$

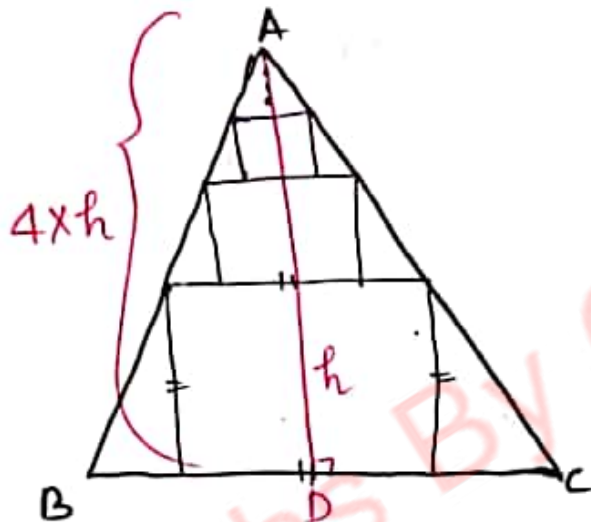
(3)

$$\text{Area} = \frac{\sqrt{3}}{4} \times \frac{1}{3\sqrt{3}} (h_1 + h_2 + h_3)^2$$

$$\text{Area} = \frac{1}{\sqrt{3}} (h_1 + h_2 + h_3)^2$$

(5)

Sum of perimeter of Infinite Square in a Δ



AD = h (corresponding to base BC)

$\Rightarrow 4h \rightarrow$ Sum of perimeter of all the square = 4h

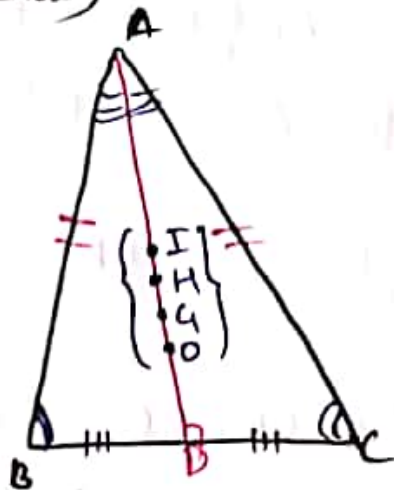
(6)

Isosceles Δ :- (समद्विबाहु Δ)

(4)

$AB = AC$ (lateral side)

$BC = \text{Base}$

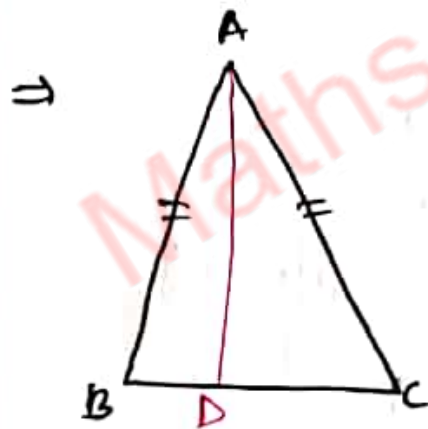


ΔAOB & ΔAOC
(congruent)

$$\left\{ \begin{array}{l} AB = AC \\ \Rightarrow LB = LC \\ BD = DC \end{array} \right\}$$

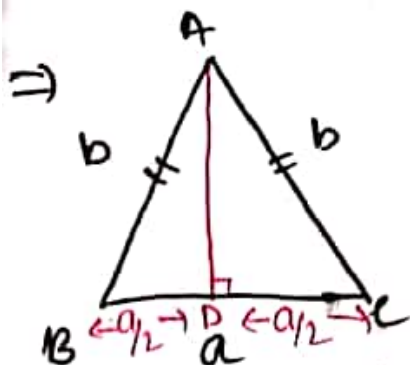
(I, H, G, O) एक ही लाइन में होते हैं,
समद्विबाहु Δ में

$AD \rightarrow$ { Altitude, median,
perpendicular bisector
of side BC
- Angle bisector.
[By congruency] }



$AB = AC$
 $D \rightarrow$ any point on base BC

$$AD^2 = AB^2 - BD \times DC$$



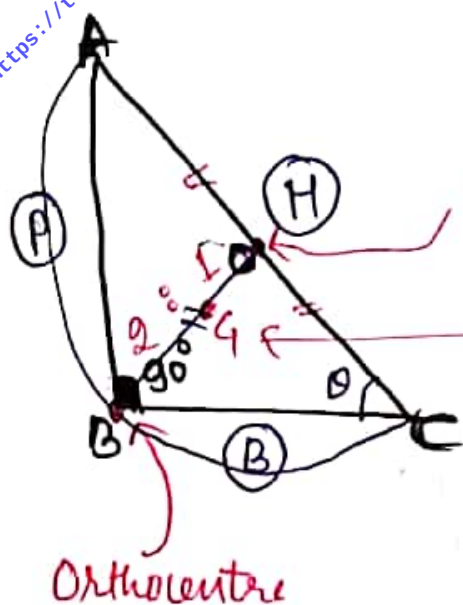
$$AD = \sqrt{b^2 - \frac{a^2}{4}}$$

$$AD = \frac{\sqrt{4b^2 - a^2}}{2}$$

$$* \text{ Area} = \frac{1}{2} \times a \times \frac{\sqrt{4b^2 - a^2}}{2} = \frac{a}{4} \sqrt{4b^2 - a^2}$$

Right angle Triangle - (समकोण Δ)

(S)



AB & BC $\rightarrow$ legs
AC = hypotenuse

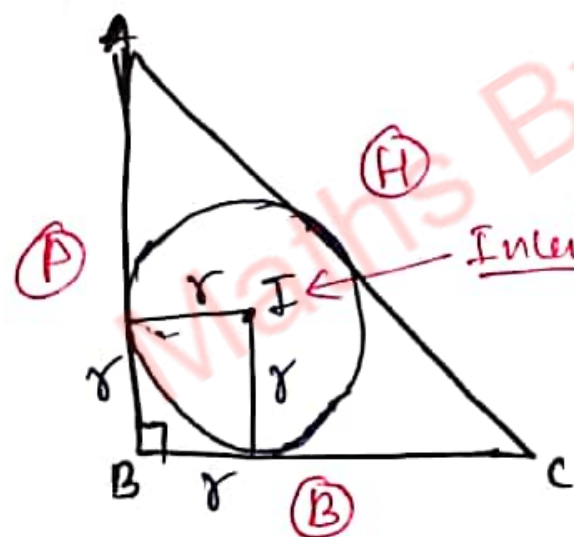
Circumcentre

Centroid

$$H^2 = P^2 + B^2$$

$$OA = OB = OC = R \text{ (Circum radius)} \\ = \frac{H}{2}$$

$$[OB \rightarrow \text{Shortest median} = R = \frac{H}{2}]$$



Incentre

$$r = \frac{P+B-H}{2}$$

$$2r = P+B-H$$

$$2r+H = P+B$$

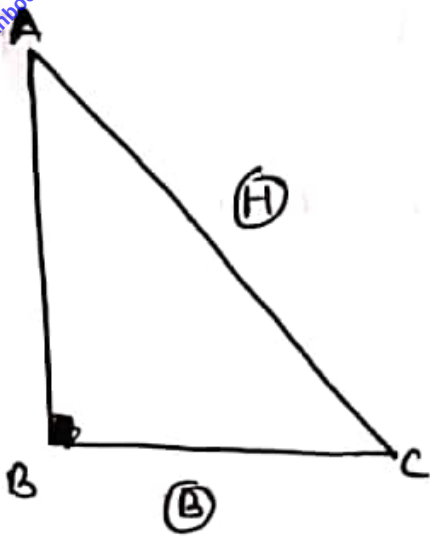
$$[H=2R]$$

$$2r+2R = P+B$$

$$2(r+R) = P+B$$

(6)

(P)



Area:

$$\Delta = \frac{1}{2} PB \sin 90^\circ = \frac{1}{2} PB$$

$$s = \frac{P+B+H}{2}$$

$$\begin{aligned} \Delta &= s \times s \\ &= \left(\frac{P+B+H}{2} \right) \times s \\ &= \left(\frac{P+B+H-2H}{2} \right) \times s \\ &= \left(\frac{P+B+H}{2} - H \right) \times s \end{aligned}$$

(Area)

$$\Delta = (s - H) \times s$$

OR.

$$\Delta = (s - 2R) s$$

(Area)

😊

⇒

Inradius = r

Circumradius = R

(Area)

$$\Delta = r^2 + 2rR$$

$$r = \frac{P+B-H}{2}$$

$$2r + H = P + B \quad \text{--- (Square) Both Side}$$

$$(2r + H)^2 = (P + B)^2$$

$$4r^2 + \cancel{H^2} + 4rH = \cancel{P^2 + B^2} + 2PB$$

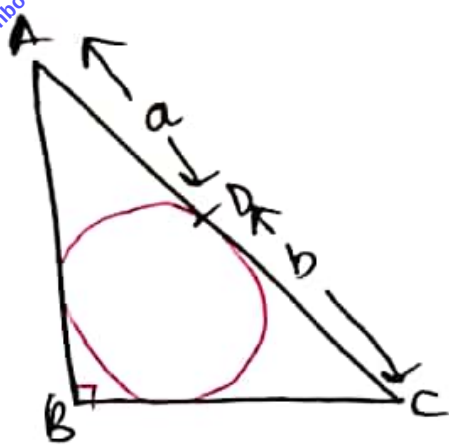
$$r^2 + rH = \frac{1}{2} PB$$

$$r^2 + 2rR = \frac{1}{2} PB = \Delta$$

area
(right angle Δ)

$$\text{Area} = a \times b$$

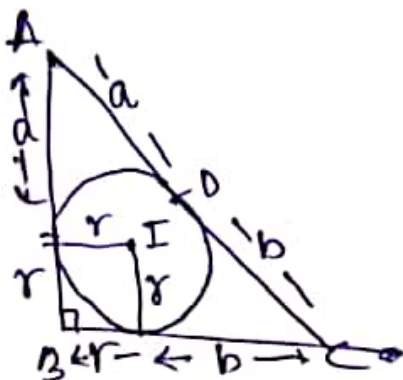
⑦



$$AD = a$$

$$CD = b$$

PROOF—



$$S = a + b + r$$

$$\Delta = \frac{1}{2} \times S$$

$$\Delta = \frac{1}{2} \times (a + b + r)$$

$$(a+r)^2 + (b+r)^2 = (a+b)^2$$

$$a^2 + r^2 + 2ar + b^2 + 2br + r^2 = a^2 + b^2 + 2ab$$

$$2r^2 + 2ar + 2br = 2ab$$

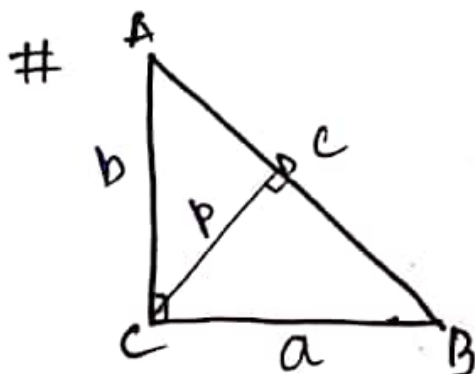
$$r(a+b+r) = ab$$

OR

$$\Delta = ab$$

⑧

इति सिद्धम्



$$\text{Area} = \text{Area}$$

$$\frac{1}{2} \times a \times b = \frac{1}{2} \times c \times p$$

$$p = \frac{ab}{c}$$

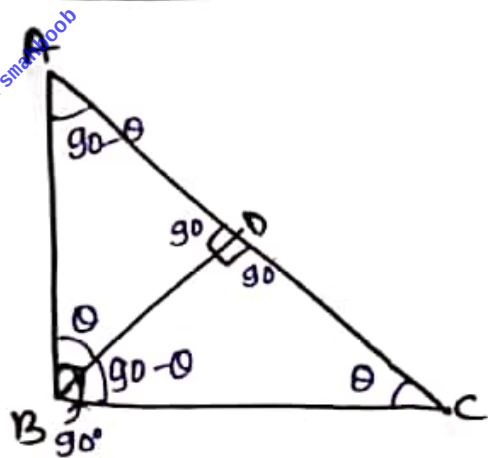
Reciprocal of

$$\text{OR } \left[\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} \right]$$

$$\frac{1}{p^2} = \frac{c^2}{a^2 b^2} = \frac{a^2 + b^2}{a^2 b^2}$$

$$\left[\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} \right]$$

#



$$\Rightarrow BD \perp AC$$

⑧

[similarity]

$$\triangle ABC \sim \triangle BDC$$

$$\frac{AB}{BD} = \frac{BC}{DC} = \frac{AC}{BC}$$

OR

$$BC^2 = AC \times CD \quad * \text{---} ①$$

$$BD = \frac{AB \times BC}{AC}$$

$$\triangle ADB \sim \triangle BDC$$

$$\frac{AD}{BD} = \frac{DB}{DC} = \frac{AB}{BC}$$

$$BD^2 = AD \times DC \quad * \text{---} ③$$

$$\triangle ABC \sim \triangle ADB$$

$$\frac{AB}{AD} = \frac{BC}{DB} = \frac{CA}{BA}$$

$$AB^2 = AD \times AC \quad * \text{---} ②$$

$$[eq^{\circ} ② \div eq^{\circ} ①]$$

$$\frac{AB^2}{BC^2} = \frac{AC \times AD}{AC \times CD}$$

$$\frac{AB^2}{BC^2} = \frac{AD}{CD}$$

Result - ###

$$1) BC^2 = AC \times CD$$

$$2) BD = \frac{AB \times BC}{AC}$$

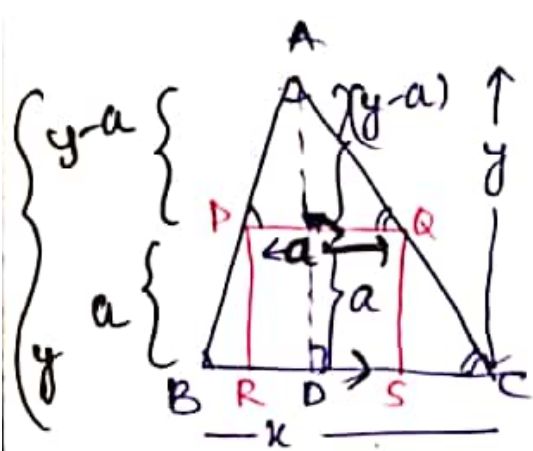
$$3) AB^2 = AD \times AC$$

$$4) BD^2 = AD \times DC$$

$$5) \frac{AB^2}{BC^2} = \frac{AD}{CD}$$

Square inside a right angle triangle

⇒ Square in ^{side} General Δ



side $\rightarrow$ (a) $\rightarrow$ PQRS is a square of maximum possible size on base BC.

$$\Delta ASR \sim \Delta ABC$$

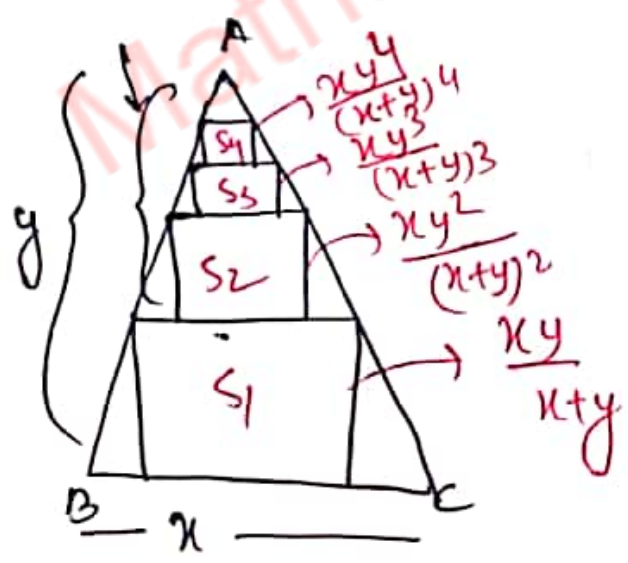
$$\frac{a}{x} = \frac{y-a}{y}$$

$$ay = xy - ax$$

$$a(x+y) = xy \quad \text{###}$$

$$a = \frac{xy}{x+y}$$

$$y - \frac{xy}{x+y} = \frac{y^2}{x+y}$$



Sum of perimeter of all Infinite square = $4y$

S_2 of side.

$$\text{Side} = \frac{\frac{xy}{x+y} \times \frac{y^2}{(x+y)}}{\frac{xy + y^2}{(x+y)}}$$

{ ऐसे सभी square की side, perimeter निकाल सकते हैं }

$$\left\{ \text{Side} = \frac{xy^2}{(x+y)xy(x+y)} \Rightarrow \frac{xy^2}{(x+y)^2} \right\}$$

$$\Rightarrow \text{Area} (S_1 + S_2 + S_3 + \dots \infty) \quad (10)$$

$$\left(\frac{xy}{x+y}\right)^2 + \left\{\frac{xy^2}{(x+y)^2}\right\}^2 + \left\{\frac{xy^3}{(x+y)^3}\right\}^2 + \dots \infty$$

$$= \left(\frac{xy}{x+y}\right)^2 \left[1 + \frac{y^2}{(x+y)^2} + \frac{y^4}{(x+y)^4} + \dots \infty \right]$$

$$= \frac{x^2 y^2}{(x+y)^2} \times \left[\frac{1}{1 - \frac{y^2}{(x+y)^2}} \right]$$

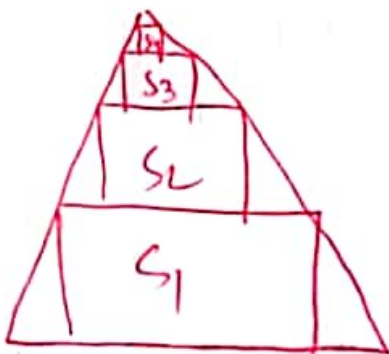
$$= \frac{x^2 y^2}{(x+y)^2} \times \frac{(x+y)^2}{x(x+2y)}$$

$$\text{Area} = \frac{xy^2}{x+2y}$$

[Infinite square & Area के
सम का formula]
(Sum)

$$\left\{ \frac{xy^2}{x+2y} \right\}$$

(For Any triangle)



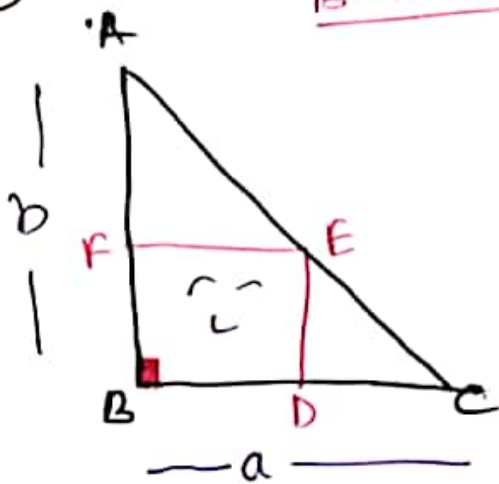
Square inside a right angle Triangle? - (1)

Base - BC

BDEF $\rightarrow$ Square

$$\left[\text{Side of Square BDEF} = \frac{ab}{a+b} \right]$$

Language (Maximum Area ki square बनाइ जाई)



(2) When one side of a square is on hypotenuse.

$$\text{Side of Square PQRS} = \frac{cx \cdot p}{c+p}$$

$$\left[p = \frac{ab}{c} \right]$$

$$= \frac{cx \cdot \frac{ab}{c}}{c + \frac{ab}{c}}$$

$$\boxed{\text{Side} = \frac{abc}{c^2 + ab}}$$

$\square$ PQRS

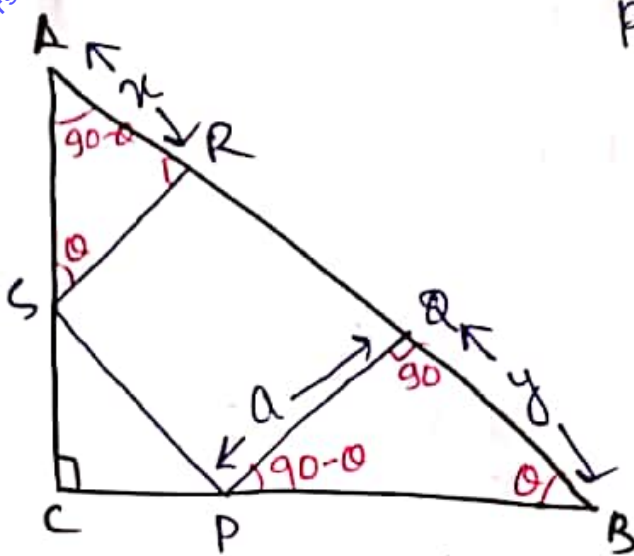
$$c^2 = a^2 + b^2$$

OR $\Rightarrow \frac{abc}{a^2 + b^2 + ab}$

2nd maximum Area of square ~~TEE~~

1st wala ~~TEE~~
max side of square

③



PQRS $\rightarrow$ Square

⑫

$$\begin{cases} AR = x \\ QB = y \end{cases}$$

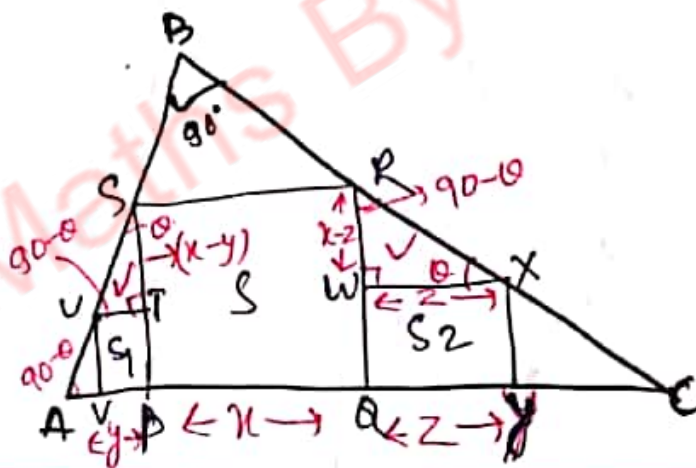
$$\triangle ARS \sim \triangle PQB$$

$$\frac{a}{x} = \frac{y}{a}$$

$$a^2 = xy$$

$$a = \sqrt{xy}$$

④



ALL THREE ARE $\rightarrow$ Square

$$\triangle SUT \sim \triangle KRW$$

$$\frac{y}{x-z} = \frac{x-y}{z}$$

$$yz = x^2 - xy - zx + yz$$

$$x(y+z) = x^2$$

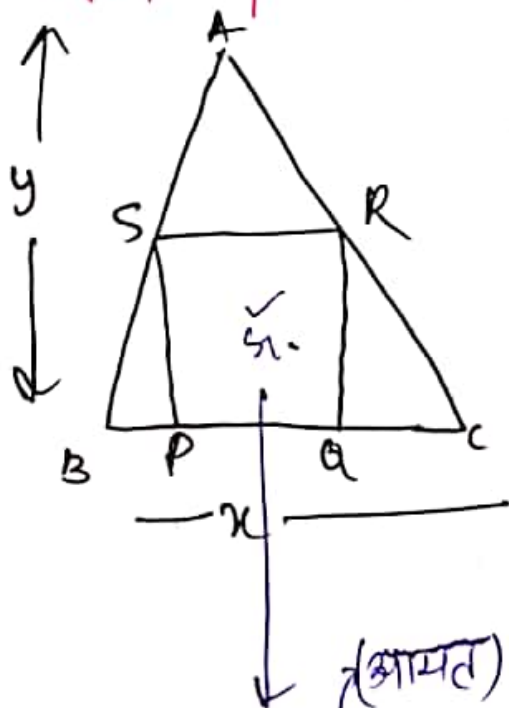
$$x = y+z$$

$\sqrt{x^2} = \sqrt{y^2} + \sqrt{z^2}$
Area $\rightarrow \sqrt{S} = \sqrt{S_1} + \sqrt{S_2}$

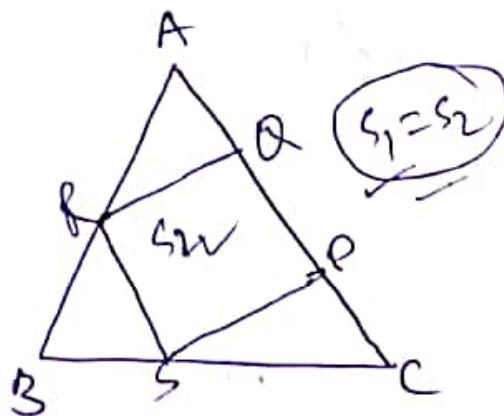
(x, y, z $\rightarrow$ Side of square)

Rectangle inside in triangle →

(13)



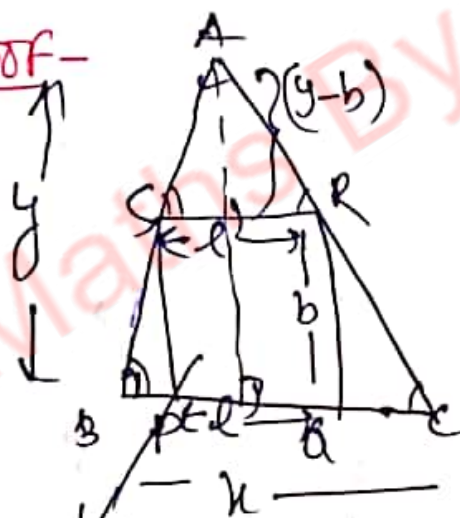
PQRS → Rectangle, max area on base.
BC of $\triangle ABC$



Area of Rectangle (PQRS)
 $= \frac{1}{2}$ of Area of $\triangle ABC$

{ किसी भी Base पर ^{max} आयत Area }
Original $\triangle$ का आधा होगा

PROOF-



$$\frac{l}{x} = \frac{y-b}{y}$$

$$\frac{l}{x} = 1 - \frac{b}{y}$$

$$\frac{l}{x} + \frac{b}{y} = 1$$

By using differential calculation

$$l = \frac{x}{2}$$

$$b = \frac{y}{2}$$

Maximize the Area of Rectangle $\frac{1}{2}(\triangle ABC)$ $Am \geq Gm$

$$\frac{\frac{l}{x} + \frac{b}{y}}{2} \geq \sqrt{\frac{l}{x} \times \frac{b}{y}}$$

$$\Rightarrow \frac{1}{2} \geq \sqrt{\frac{l}{x} \times \frac{b}{y}}$$

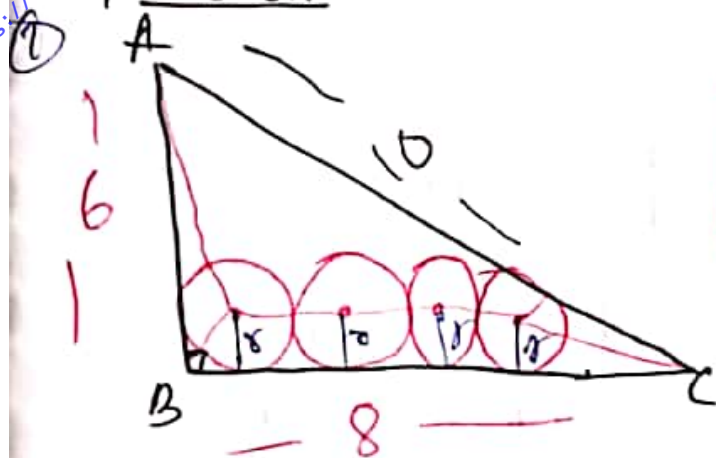
$$\Rightarrow \frac{1}{4} \geq \left(\frac{lb}{xy} \right)$$

$$\Rightarrow \frac{1}{2} \times \left(\frac{1}{2} \times xy \right) \geq lb$$

Area of $\triangle ABC$ $\geq$ Area of Rectangle

Circles Inside Right angle Triangle - By Question -

(14)

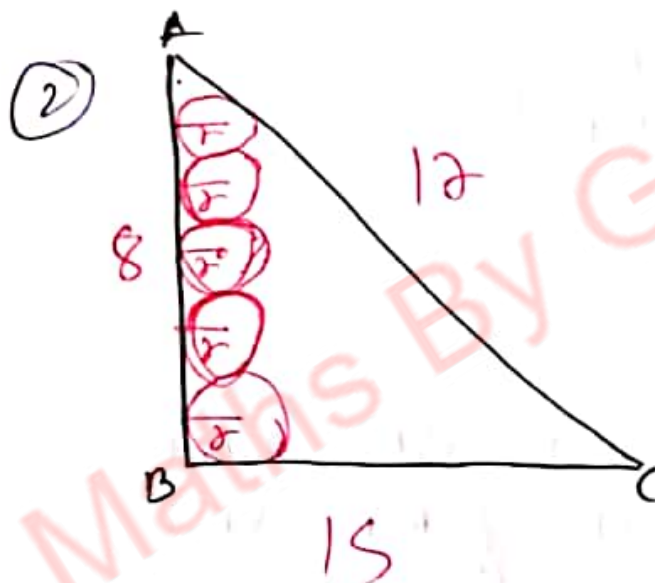


$$r = \frac{\Delta}{S + (n+1) \times h}$$

$$r = \frac{24}{12 + 3 \times 6}$$

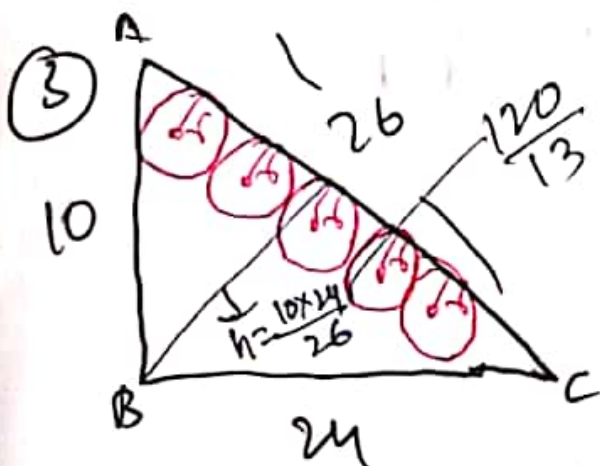
$$r = \frac{24}{30} = \frac{4}{5}$$

$$r = \frac{4}{5} = 0.8 \text{ cm}$$



$$r = \frac{60}{20 + 4 \times 15}$$

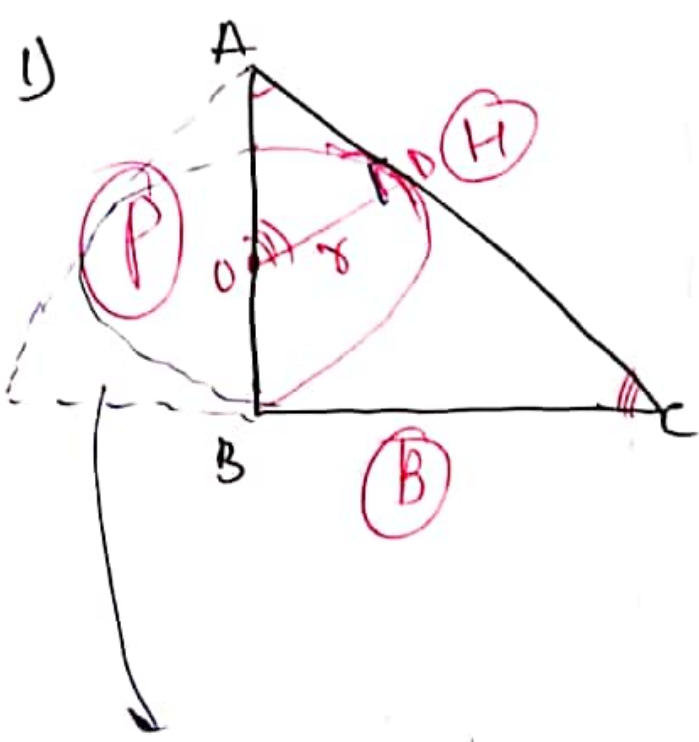
$$r = \frac{60}{80} = \frac{3}{4} = 0.75 \text{ cm}$$



$$r = \frac{120}{30 + 4 \times \frac{120}{13}} \Rightarrow \frac{4}{1 + \frac{16}{13}}$$

$$r = \frac{52 \text{ cm}}{29}$$

Semi Circle inside a right angle $\Delta \rightarrow$ (15)



$$\frac{r}{B} = \frac{P-r}{H}$$

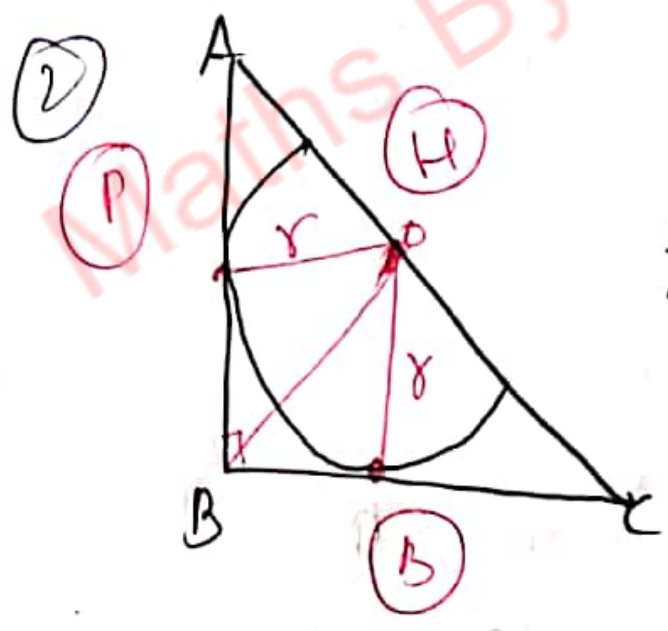
$$rH = PB - Br$$

$$r(B+H) = PB$$

$$r = \frac{PB}{B+H}$$

$$r = \frac{\Delta}{S}$$

$$= \frac{\frac{1}{2} \times \angle B \times P}{(H+B)} = \frac{BP}{H+B}$$



$$Ar. \Delta ABC = \Delta BOC + \Delta AOB$$

$$\frac{1}{2} \times PB = \frac{1}{2} \times B \times r + \frac{1}{2} \times P \times r$$

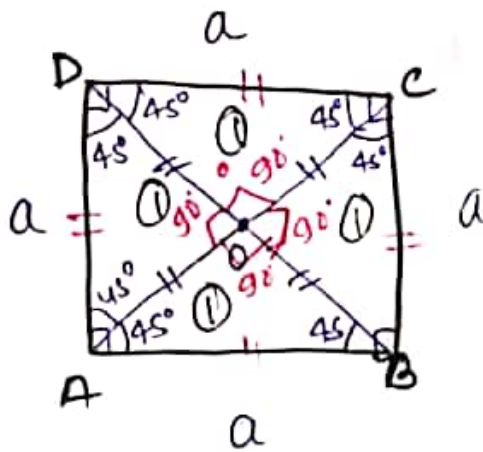
$$PB = r(P+B)$$

$$r = \frac{PB}{P+B}$$

Syedhyang

Square का →

①



- # $AB = BC = CD = AD = a$
- # $\angle A = \angle B = \angle C = \angle D = 90^\circ$
- # Perimeter (परिमाप) $= a + a + a + a = 4a$

Area (क्षेत्रफल) $= a^2$

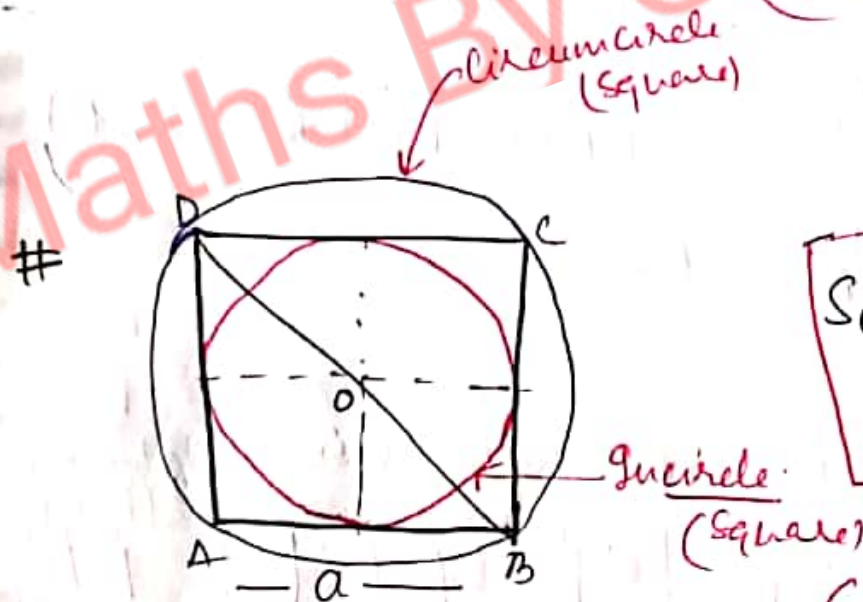
diagonals $= AC = BD = a\sqrt{2}$
(विकर्ण)

diagonals bisect vertex angle

diagonals bisect each other at 90°

$AO = BO = CO = DO$

$Ar[\Delta AOB = \Delta COD = \Delta BOC = \Delta AOD]$

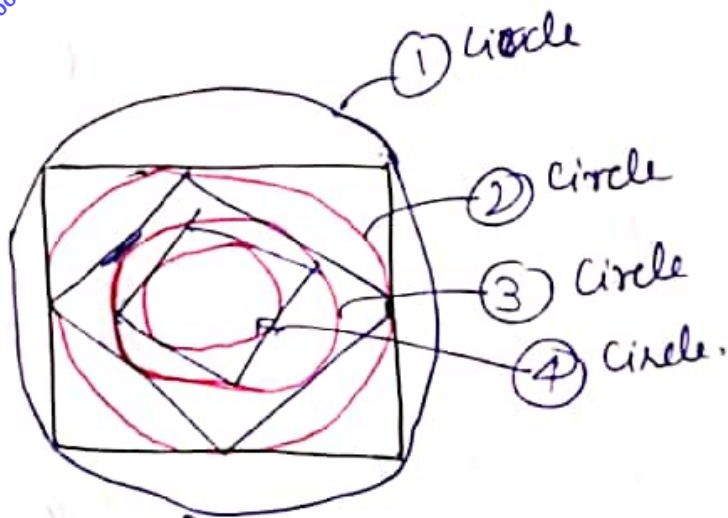


(Incircle)
Square के Circle के
radius $= r = \frac{a}{2}$

Circumcircle Area : Area
Incircle = 2:1

Circumcircle
Square के Circumcircle
के Radius $= R = \frac{a}{\sqrt{2}}$

$R:r = \frac{a}{\sqrt{2}} : \frac{a}{2} = \sqrt{2} : 1$

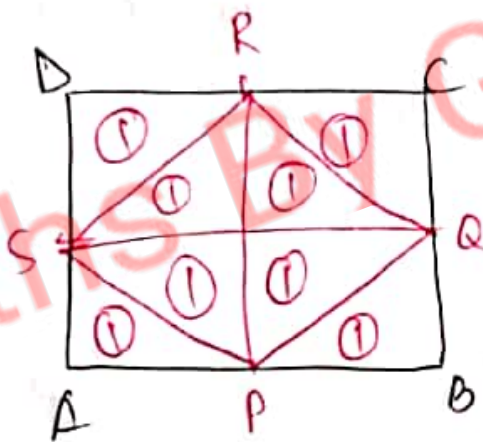


[Area का Ratio] इस प्रकार होगा ↓



$$16 : 8 : 4 : 2 : 1$$

#

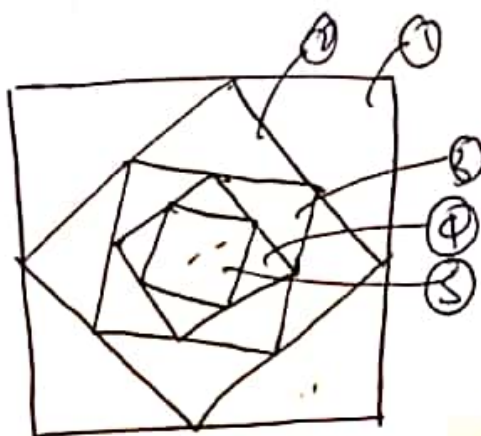


P, Q, R, S → midpoint

$$\square ABCD : \square PQRS = 2 : 1$$

Area

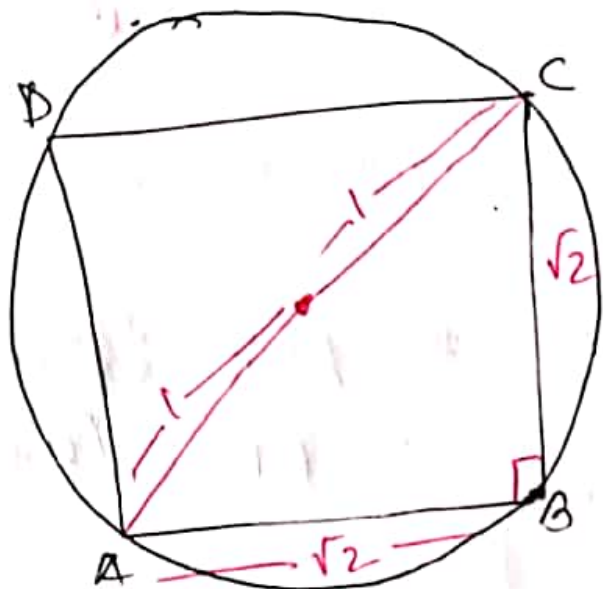
#



⇒ Area का अनुपात आया होता जाता है —

$$\square : \square : \square : \square : \square$$

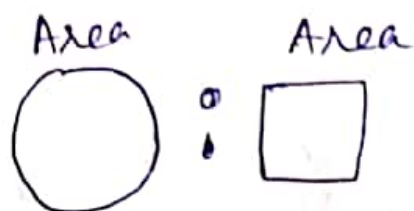
$$16 : 8 : 4 : 2 : 1$$



ABCD $\rightarrow$ Square.

③

अनुपात (Ratio)

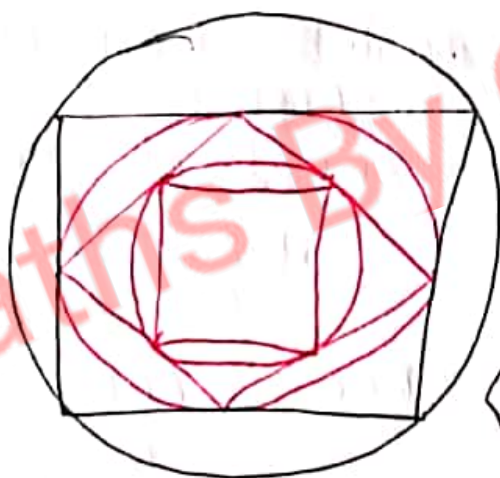


$$\pi(1)^2 : (\sqrt{2})^2$$

$$\boxed{\pi : 2}$$

OR. $\frac{22}{7} : 2$

$$\Rightarrow 11 : 7$$



$\square \rightarrow$ Square.

$$\bigcirc : \square : \bigcirc : \square : \bigcirc : \square$$

Area $\rightarrow 16\pi : 32 : 8\pi : 16 : 4\pi : 8$

$$\pi : 2$$

Half

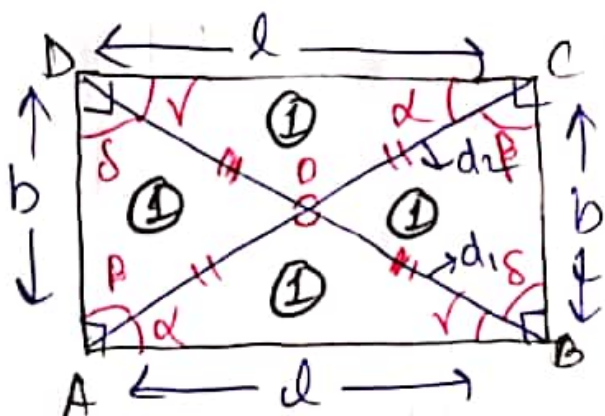
OR. $\Rightarrow 44 : 28 : 22 : 14 : 11 : 7$

$$11 : 7$$



Rectangle - आयत

(4)



$$\# \begin{cases} AC = \sqrt{l^2 + b^2} \\ BD = \sqrt{l^2 + b^2} \end{cases}$$

$$\# \begin{cases} d_1 = d_2 = \sqrt{l^2 + b^2} \end{cases}$$

$$\# AO = CO = BO = DO$$

$$\# \begin{cases} AO = CO = \frac{AC}{2} \\ DO = BO = \frac{BD}{2} \end{cases}$$

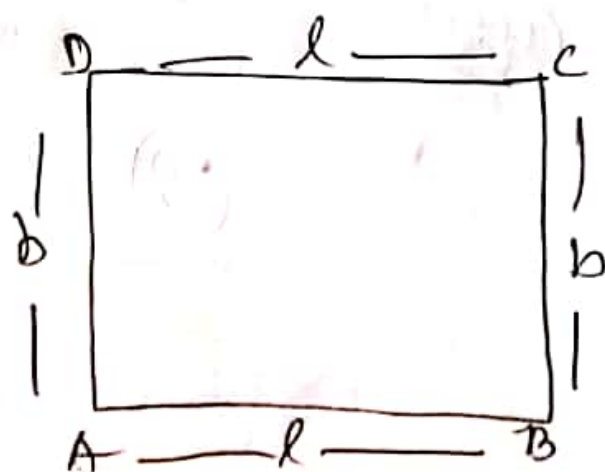
diagonals bisect each other but not at 90°

diagonals do not bisect Vertex angle..

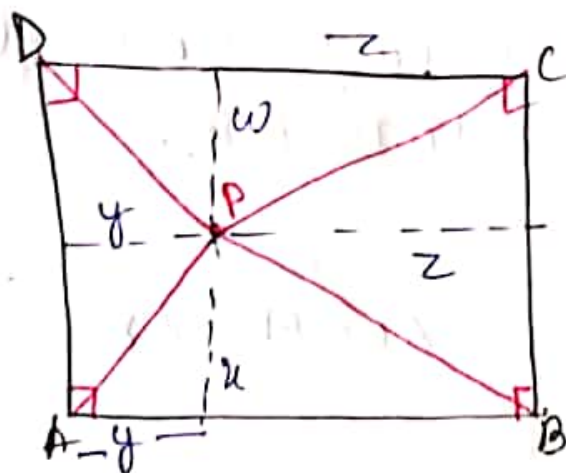
Area:-

$$\Delta AOB = \Delta COD = \Delta BOC = \Delta AOD$$

#



$$\begin{cases} \text{Area (क्षेत्रफल)} = l \times b \\ \text{Perimeter (परिमाप)} = 2(l + b) \end{cases}$$



(5)

① # $\Delta PAB + \Delta PCD = \Delta PBC + \Delta PAD = \frac{1}{2} \text{ area of rectangle } (ABCD)$

$PA^2 + PC^2 = PD^2 + PB^2$

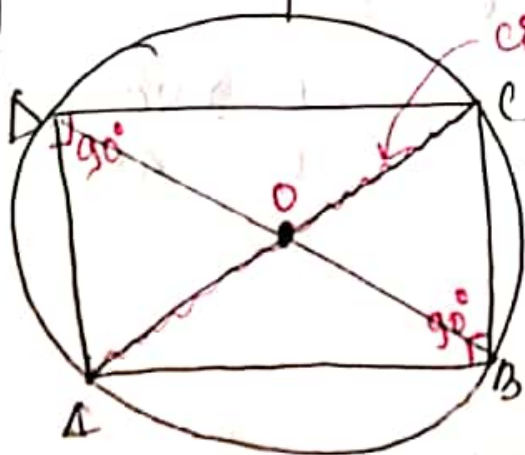
$x^2 + y^2 + w^2 + z^2 = x^2 + z^2 + w^2 + y^2$

BOTH ARE EQUAL

Proved. इति सिद्धम्



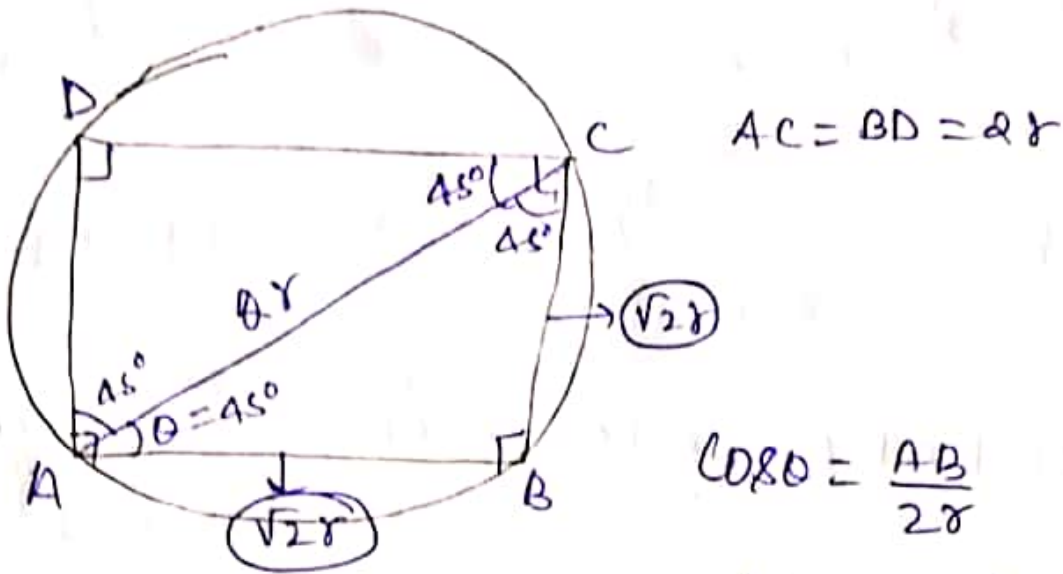
$\Rightarrow$ Every Rectangle is a cyclic quadrilateral.



circle as diameters होगा ?

(6)

Circle के अन्दर maximum area का rectangle (आयत) हमेशा एक Square (वर्ग) होगा.



$$\cos \theta = \frac{AB}{2r}$$

$$AB = 2r \cos \theta$$

$$\sin \theta = \frac{BC}{2r}$$

$$BC = 2r \sin \theta$$

Rectangle.

$$\text{Area of } \square ABCD = 4r^2 \sin \theta \cdot \cos \theta$$

$$= (2r)^2 (2 \cdot \sin \theta \cdot \cos \theta)$$

$$= 2r^2 \sin 2\theta$$

$$\text{For max. Area} \rightarrow \theta \rightarrow 1$$

$$2\theta \rightarrow 90^\circ$$

$$\theta \rightarrow 45^\circ$$

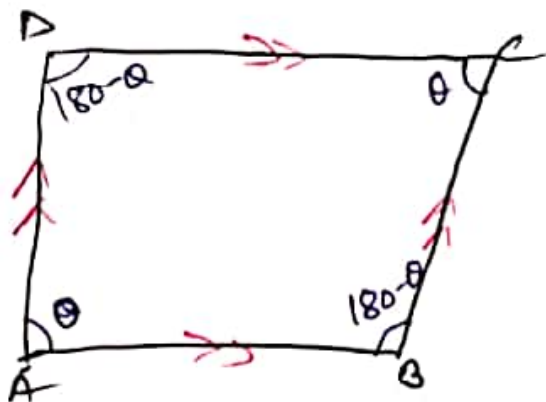
$$\left\{ \begin{array}{l} \text{max. Area} = 2r^2 \\ \text{Perimeter} = 4\sqrt{2}r \end{array} \right\}$$



#

Parallelogram (समांतर चतुर्भुज) $\square \rightarrow$ (11gm)

7

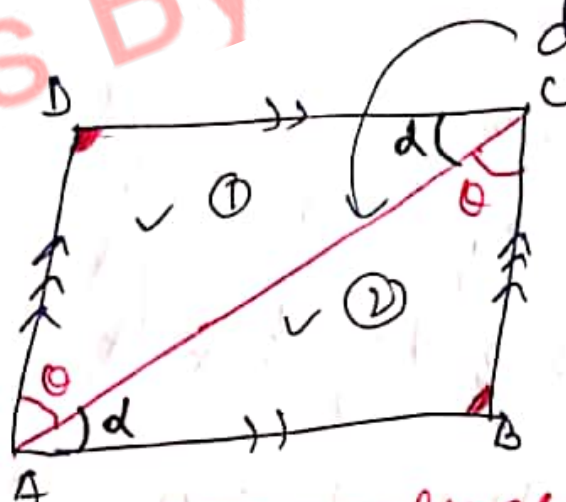


$$\left. \begin{aligned} &AB \parallel CD \text{ \& } AD \parallel BC \\ \Rightarrow &ABCD \text{ is a } \parallel^{\text{gm}}. \\ &LA = LC \\ &LB = LD \end{aligned} \right\}$$

Sum of two consecutive angles in a $\parallel^{\text{gm}}$ is 180° .

Opposite angles ~~are~~ equal

{ Angle } $\Rightarrow$ ratio $\Rightarrow x : y : x : y \Rightarrow \parallel^{\text{gm}}$ होगा



$\Rightarrow$ divide ($\parallel^{\text{gm}}$ area)
 $\downarrow$
Two Equal part
 $\text{Area 1} = \text{Area 2}$

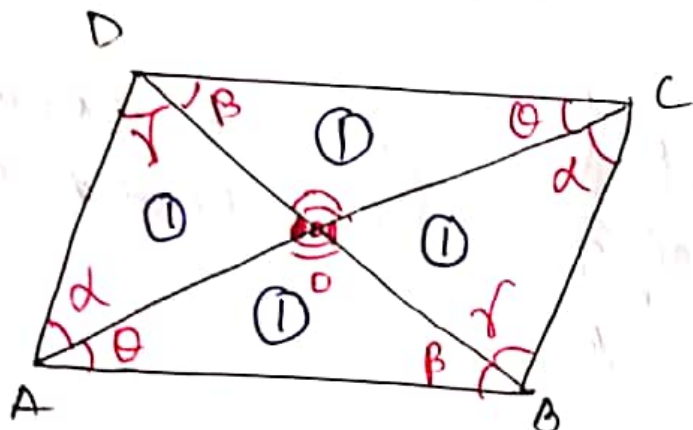
$\triangle ADC \cong \triangle CBA$ (congruent)

$$\left\{ \begin{aligned} &AD = CB \\ &DC = BA \end{aligned} \right\} \quad \left\{ \begin{aligned} &BC = AD \\ &AB = CD \end{aligned} \right\}$$

$\parallel^{\text{gm}}$ area = ① + ②
 $\Rightarrow$ diagonals do not bisect vertex angle
 $\boxed{\angle \neq 0}$

⑧

#



$\triangle AOB \cong \triangle COD$

$AO = CO$

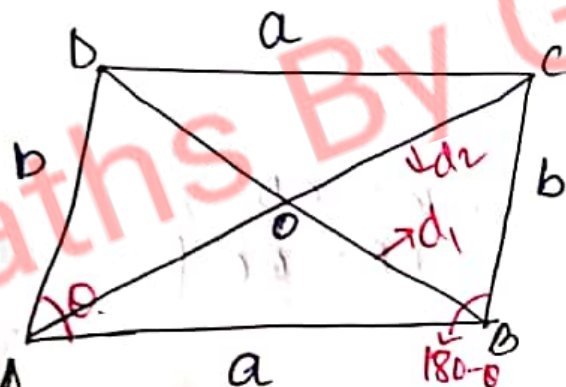
$BO = DO$

(निष्कर्ष)
Diagonals bisect each other?
but not at 90°

$\Rightarrow$ #

$$\triangle AOB = \triangle BOC = \triangle AOD = \triangle DOC = \frac{1}{4} \times \text{area of } \text{parallelogram } ABCD$$

#



$BD \rightarrow d_1$

$AC \rightarrow d_2$

$d_1 \neq d_2$

PROOF A

By Cosine rule-

$$d_1^2 = a^2 + b^2 - 2ab \cos \theta \quad \text{--- (1) } \checkmark$$

$$d_2^2 = a^2 + b^2 - 2ab \cos (180 - \theta) \quad \text{--- (2) } \checkmark$$

$$d_2^2 = a^2 + b^2 + 2ab \cos \theta \quad \text{--- (2) } \checkmark$$

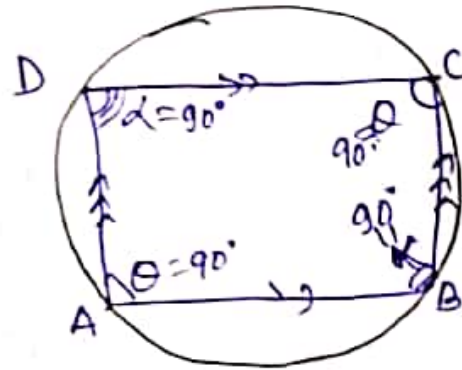
next $\rightarrow$

eqⁿ ① + eqⁿ ②

⑨

$$d_1^2 + d_2^2 = 2(a^2 + b^2)$$

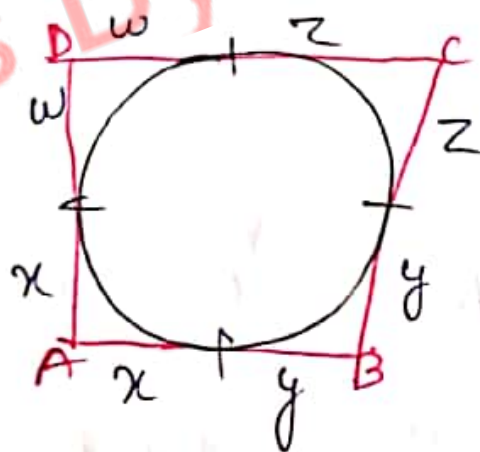
1) Circle के अन्दर बना ॥^{gm} हमेशा एक आयत होता है।



$$2\theta \rightarrow 180^\circ \Rightarrow \theta = 90^\circ$$

$$2\alpha \rightarrow 180^\circ \Rightarrow \alpha = 90^\circ$$

2) parallelogram Circumscribed about a circle is always a rhombus.



$$AB + CD = BC + AD$$

$$\begin{cases} AB = CD \\ AD = BC \end{cases}$$

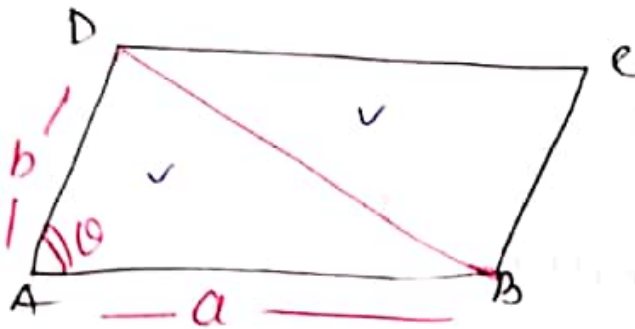
$$\Rightarrow 2AB = 2BC$$

$$AB = BC$$



Area of $\triangle ABD \rightarrow$

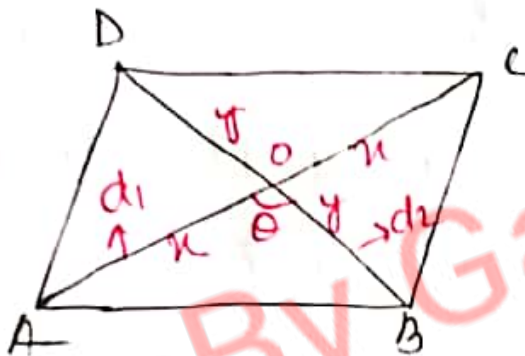
(10)



$$\text{Area}(\triangle ABD) = \frac{1}{2} ab \sin \theta$$

Area of $\triangle ABCD = ab \sin \theta$

(2)



11 gm chl Area = $\frac{1}{2} \times d_1 \times d_2 \sin \theta$

$$\begin{cases} AO = OC \\ OB = OD \end{cases}$$

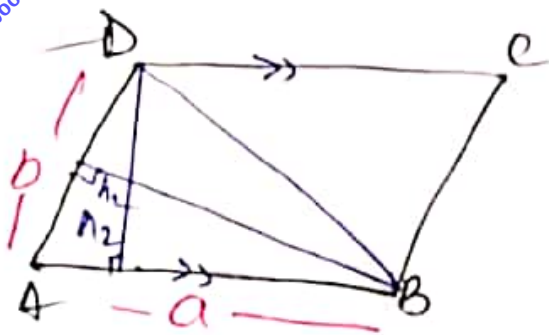
$$\triangle AOB = \frac{1}{2} xy \sin \theta$$

$$11 \text{ gm } ABCD = \frac{1}{2} \times (2x) \times (2y) \sin \theta$$

$\downarrow \quad \downarrow$
 $d_1 \quad d_2$

Area 11 gm $ABCD = \frac{1}{2} \times d_1 \times d_2 \sin \theta$

③



⑪

$$\text{Area of } \triangle ABD = \frac{1}{2} \times a \times h_1$$

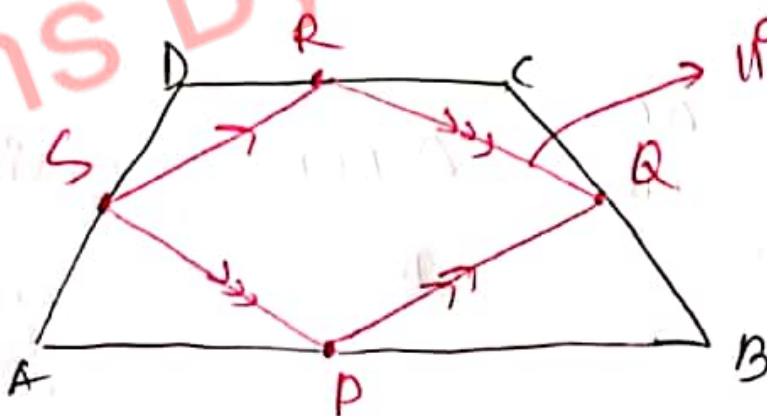
[Double होता है (11^{gm} का area) diagonal]

$$11^{\text{gm}} \text{ का area} = (a h_1)$$

$$11^{\text{gm}} \text{ का area} = (b h_2)$$

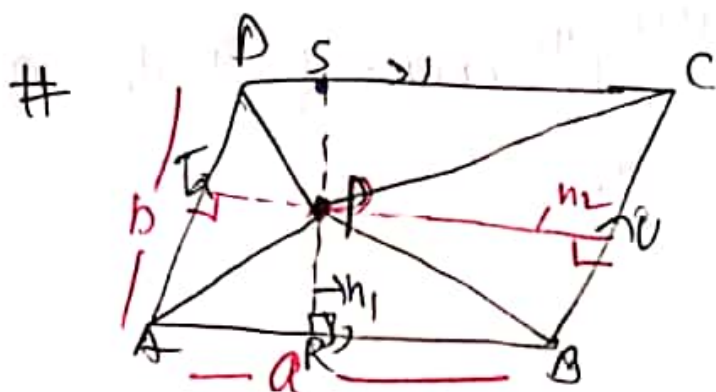
$$a h_1 = b h_2$$

किसी भी quadrilateral के mid point को मिलाने पर हमेशा एक 11^{gm} बनता है। -



11^{gm} होगा (हमेशा)

PQRS → midpoint



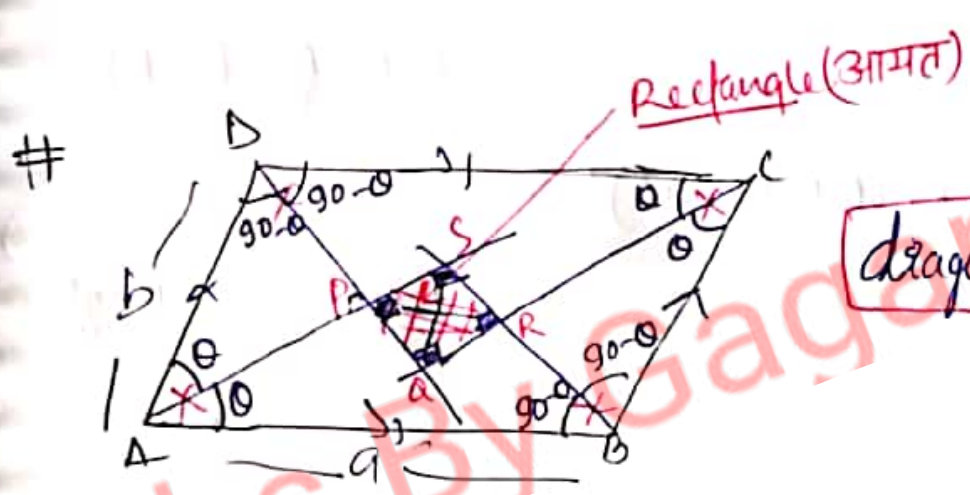
Area:

$$\triangle PAB + \triangle PCD = \triangle PBC + \triangle PAD$$

$$\begin{aligned} & \triangle PAB + \triangle PBD \\ & \Rightarrow \frac{1}{2} a \times PR + \frac{1}{2} a \times PS \\ & = \left(\frac{1}{2} a \times h_1 \right) \\ & = \frac{1}{2} \text{area of } \parallel^{\text{gm}} \end{aligned}$$

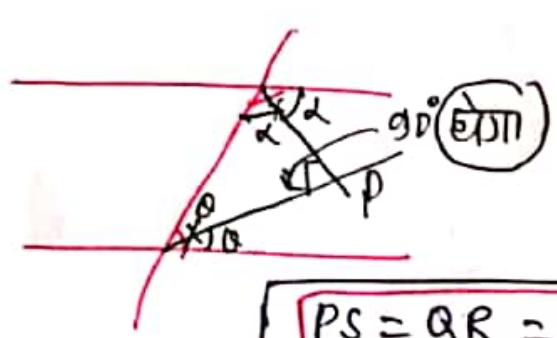
$$\begin{aligned} & \triangle PAD + \triangle PCD \\ & \Rightarrow \frac{1}{2} b \times PR + \frac{1}{2} b \times PS \\ & = \frac{1}{2} b \times h_2 \\ & = \frac{1}{2} \text{area of } \parallel^{\text{gm}} \end{aligned}$$

Both are equal. इति सिद्धम्



diagonals $\rightarrow PR = QS = a - b$
 difference

$\Rightarrow$ किसी भी $\parallel^{\text{gm}}$ के angle bisector से हमेशा एक Rectangle बनता है।



$$\begin{aligned} \triangle ASB \quad \cos \theta &= \frac{AS}{a} \Rightarrow AS = a \cos \theta \\ \triangle APD \quad \cos \theta &= \frac{AP}{b} \Rightarrow AP = b \cos \theta \end{aligned}$$

$PS = QR = AS - AP = (a - b) \cos \theta$

$\triangle DQC$ में —

$$\cos(90^\circ - \theta) = \frac{DQ}{a} \Rightarrow DQ = a \sin \theta$$

$\triangle DPA$

$$\cos(90^\circ - \theta) = \frac{DP}{b} \Rightarrow DP = b \sin \theta$$

$$PQ = RS = DQ - DP = (a - b) \sin \theta$$

$$PR = QS = \sqrt{PQ^2 + PS^2} = a - b$$

let $x = (a - b)$

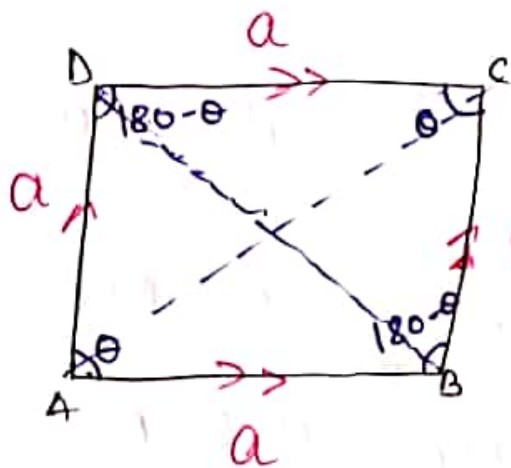
#

$$\text{Perimeter} = 2x(\cos \theta + \sin \theta)$$

@ $\rightarrow$ half vertex angle

Rhombus (समचतुर्भुज)

14



$$AB = BC = CD = DA = a$$

$$\angle A = \angle C = \theta$$

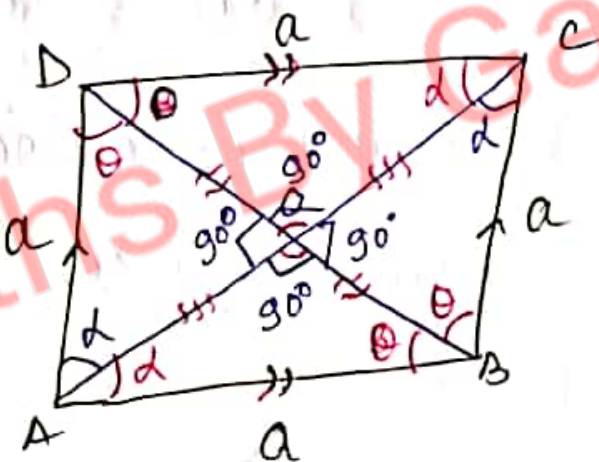
$$\angle B = \angle D = 180 - \theta$$

$$AC \neq BD$$

diagonals not equal

$\theta \rightarrow$ BD (acute) $\rightarrow$ छोटा

$180 - \theta \rightarrow$ AC (obtuse) $\rightarrow$ बड़ा



$$AC \neq BD$$

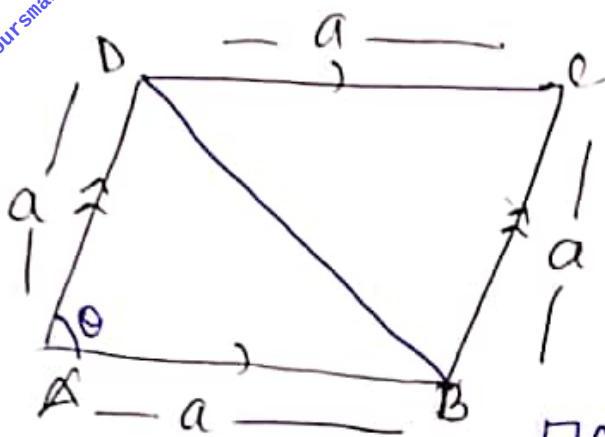
diagonal bisect vertex angle

$$DO = BO = \frac{BD}{2}$$

$$CO = AO = \frac{AC}{2}$$

diagonal bisect each other at 90°

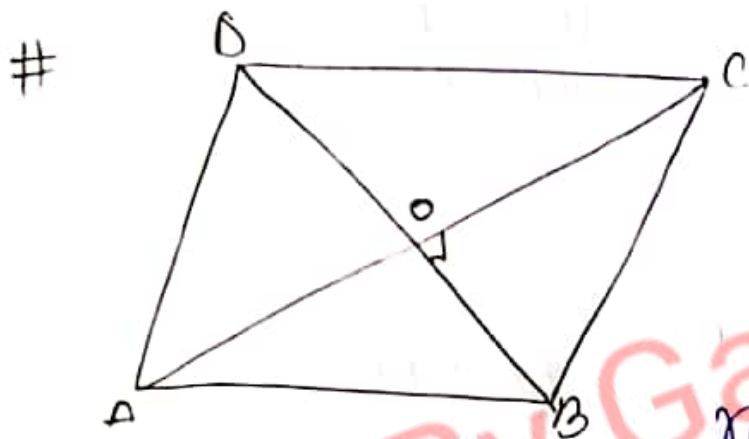
$$\begin{cases} 2\alpha + 2\theta = 180 \\ \alpha + \theta = 90 \end{cases}$$



$$\text{Perimeter} = 4a$$

$$\square ABCD \text{ Area} = \frac{1}{2} a \times a \sin \theta \times 2$$

$$\text{Area} = a^2 \sin \theta$$



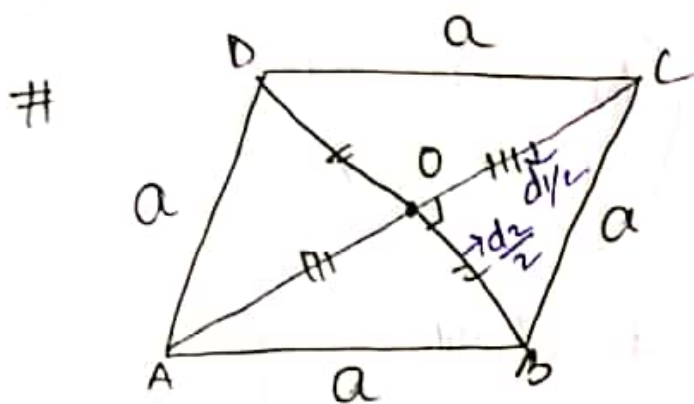
$$\begin{cases} AC = d_1 \\ BD = d_2 \end{cases}$$

$$\text{Rhombus area} = \frac{1}{2} d_1 d_2 \sin \theta$$

$$\text{Rhombus area} \Rightarrow \theta = 90^\circ$$

$$(\sin 90^\circ = 1)$$

$$\text{Area} = \frac{1}{2} d_1 \times d_2$$



$$\begin{aligned} AC &= d_1 \\ BD &= d_2 \end{aligned}$$

$$OC = \frac{AC}{2} = \frac{d_1}{2}$$

$$OB = \frac{BD}{2} = \frac{d_2}{2}$$

$$\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2 = a^2$$

$$d_1^2 + d_2^2 = 4a^2$$



Any 2
1) Quadrilateral

midpoints

parallelogram

2) Square

midpoint

Square

3) Rectangle

midpoint

Rhombus

4) Rhombus

midpoint

Rectangle

5) 11 cm

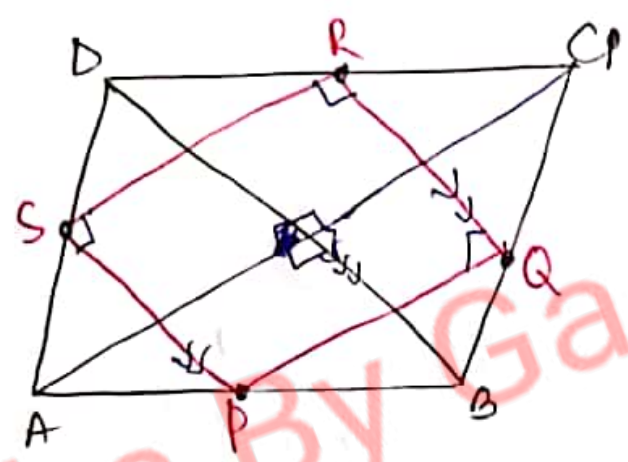
midpoint

11 cm

11 cm & angle bisector → rectangle

[सब के mid point को मिलावे पर बने चतुर्भुज]

#



PQRS → rectangle

$$PQ = \frac{AC}{2} \quad RS \text{ (midpoint)}$$

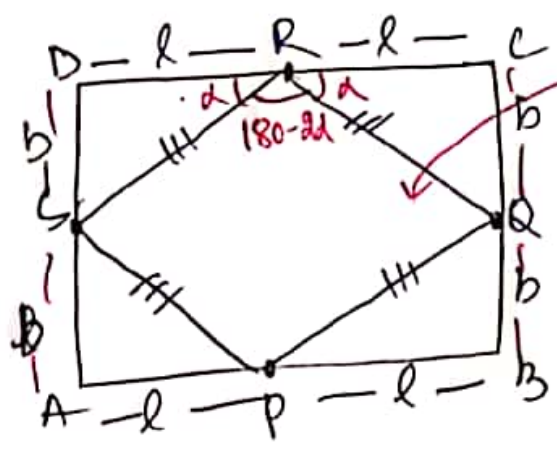
$$RQ = \frac{BD}{2} = SP \text{ (midpoint)}$$

$$AC \neq BD$$

चारों length आपस में बराबर नहीं हैं



#



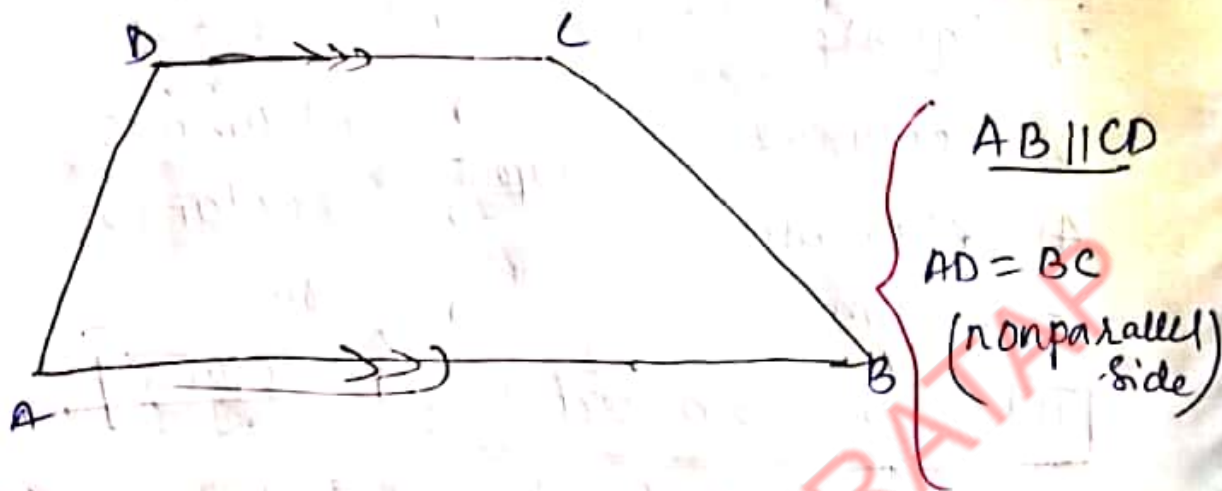
$$d \neq 45^\circ \quad (180 - 2A \neq 90)$$

PQRS → Rhombus

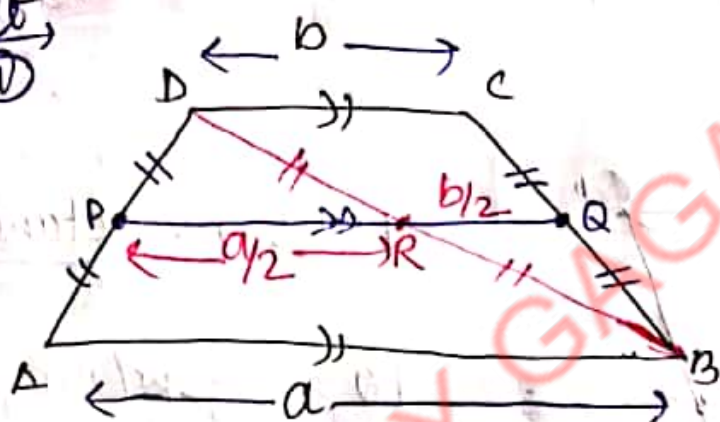
Gagan Prakashan

Trapezium:- (समलम्ब)

①



Result,
①



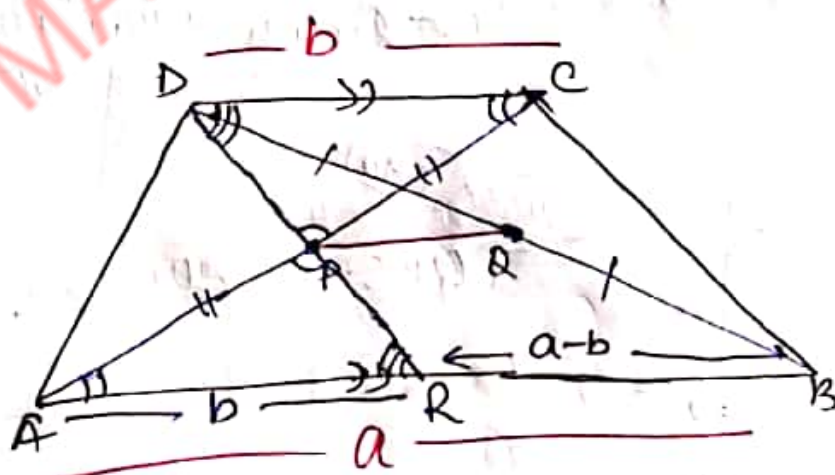
$\{PQ \parallel AB \parallel CD\}$

$$PQ = \frac{a+b}{2}$$

PROOF

$$\left\{ \begin{aligned} PQ &= PR + RQ \\ RQ &= \frac{a}{2} + \frac{b}{2} = \left(\frac{a+b}{2} \right) \end{aligned} \right\}$$

②
#



$$PQ = \frac{a-b}{2}$$

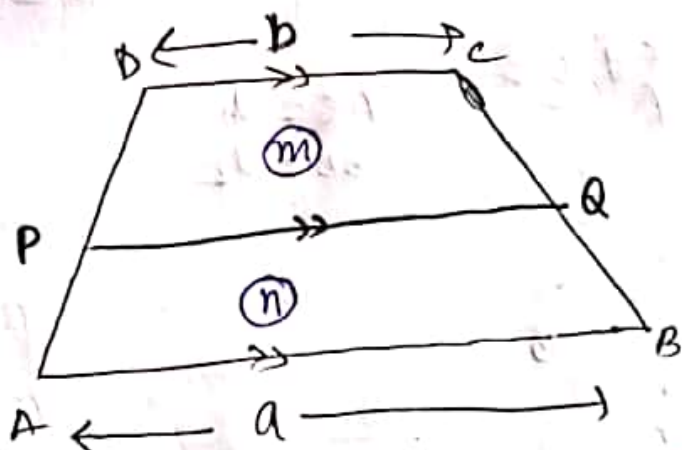
$\triangle DRB$ में

$$PQ = \frac{RB}{2} = \frac{a-b}{2}$$

By: midpoint
Theorem

③

②



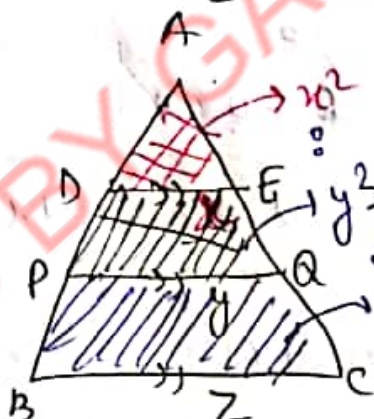
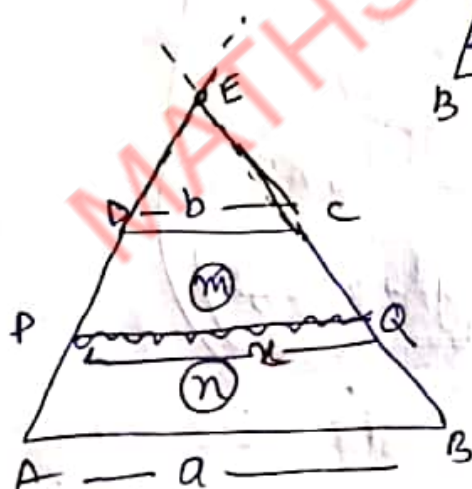
$$\left. \begin{aligned} PQ \parallel AB \parallel DC \\ \square PQCD : \square APQB = m : n \\ (\text{Area}) \end{aligned} \right\} PQ = ?$$

$$PQ = \sqrt{\frac{ma^2 + nb^2}{m+n}}$$

if $m = n$ or $m:n = 1:1$

$$PQ = \sqrt{\frac{a^2 + b^2}{2}}$$

PROOF: concept:



Area Ratio

$$\left. \begin{aligned} \triangle ADE : \triangle APQ : \triangle ABC \\ x^2 : y^2 : z^2 \end{aligned} \right\}$$

EPQR में (OR PQ के Area)

$$x^2 - b^2 = m \quad \text{--- (1)}$$

AEB में

$$\square APQB \text{ का Area}$$

$$a^2 - x^2 = n \quad \text{--- (2)}$$

$$(1) \div (2)$$

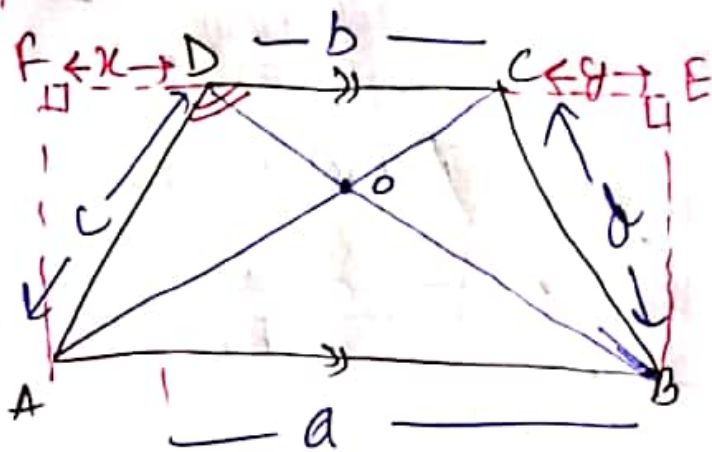
$$\frac{x^2 - b^2}{a^2 - x^2} = \frac{m}{n}$$

$$= nx^2 - b^2n = a^2m - mx^2$$

$$\Rightarrow \sqrt{x^2 = \frac{ma^2 + nb^2}{m+n}}$$

④

③



$$AC = d_1$$

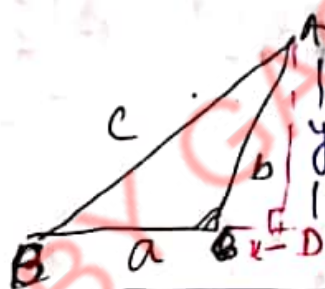
$$BD = d_2$$

Result →

Imp.

$$d_1^2 + d_2^2 = c^2 + d^2 + 2ab$$

Concept:-
Concept:-



$$c^2 = a^2 + b^2 + 2ax$$

ΔADB

$$c^2 = y^2 + (a+x)^2$$

$$c^2 = y^2 + a^2 + x^2 + 2ax$$

$$c^2 = a^2 + (y^2 + x^2) + 2ax$$

$$c^2 = a^2 + b^2 + 2ax$$

ΔADC

$$y^2 + x^2 = b^2$$

ΔADC में-

$$d_1^2 = c^2 + b^2 + 2bx \quad \text{--- (1)}$$

ΔDCB में-

$$d_2^2 = b^2 + d^2 + 2by \quad \text{--- (2)}$$

ADD →

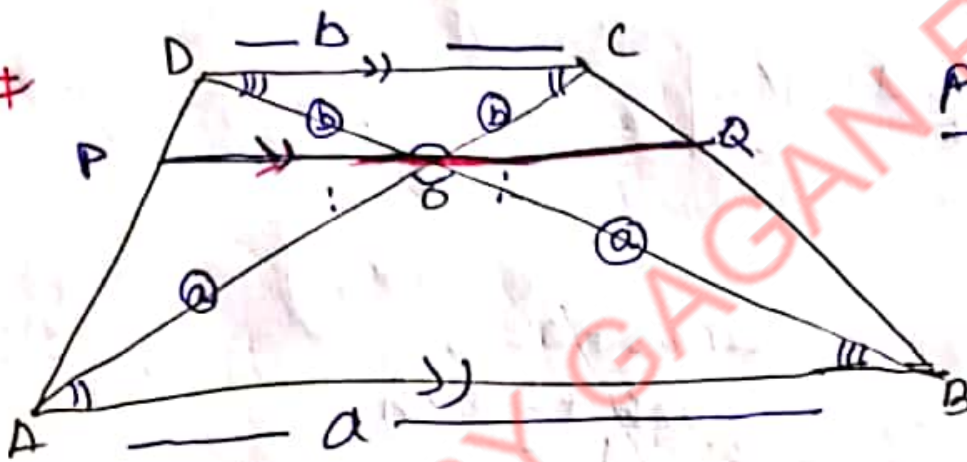
$$d_1^2 + d_2^2 = c^2 + b^2 + b^2 + d^2 + 2b(x+y)$$

↓

$$\Rightarrow d_1^2 + d_2^2 = e^2 + d^2 + 2b(b+x+y) \quad [b+x+y=a]$$

$$\Rightarrow \boxed{d_1^2 + d_2^2 = e^2 + d^2 + 2ab} \quad \text{इति सिद्धम्} \quad (3)$$

(5) #



PQ || AB || CD

#

$$\boxed{PQ = \frac{2ab}{a+b}}$$

PROOF -

$\triangle AOB \sim \triangle COD$

$$\left\{ \begin{array}{l} DO : OB = b : a \\ CO : OA = b : a \\ DC : AB = b : a \end{array} \right.$$

By Similarity

$\triangle DBA$

$$\frac{PO}{a} = \frac{b}{a+b}$$

$$PO = \frac{ab}{a+b}$$

Similarly $\triangle DCB \quad QO = \frac{ab}{a+b}$

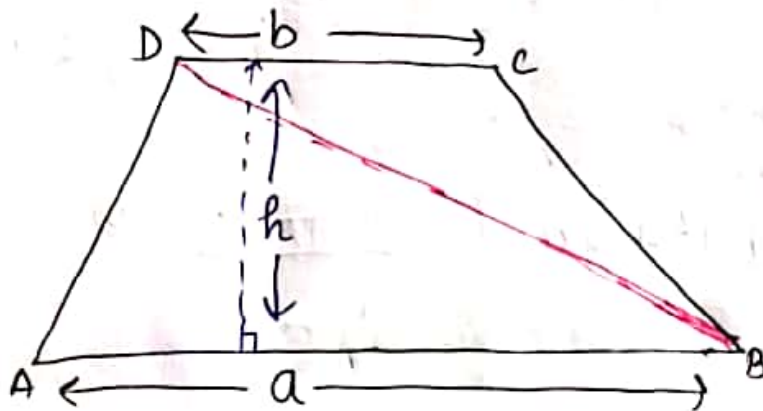
$$PQ = PO + QO$$

$$\boxed{PQ = \frac{2ab}{a+b}}$$



Satyam

Area of Trapezium —



$$\text{Area} = \frac{1}{2}(a+b) \times h$$

PROOF.

Area.

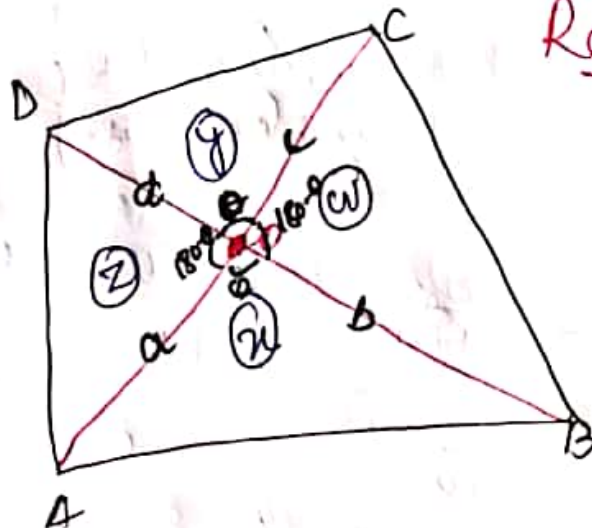
$$\square ABCD = \triangle ADB + \triangle DCB$$

$$\text{Area} = \frac{1}{2} a \times h + \frac{1}{2} b \times h$$

$$\text{Area} = \frac{1}{2}(a+b) \times h =$$

For any general quadrilateral —

$$\left. \begin{array}{l} \text{Area} \\ \triangle AOB = x \\ \triangle BOC = w \\ \triangle COD = y \\ \triangle AOD = z \end{array} \right\}$$



Result —

$$x \times y = z \times w$$

PROOF

⑥

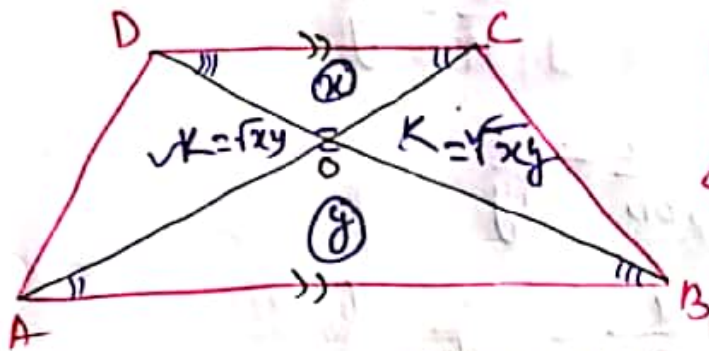
$$X \times Y = \frac{1}{2} xab \sin \theta \times \frac{1}{2} cd \sin \theta$$

$$Z \times W = \frac{1}{2} xbc \sin \theta \times \frac{1}{2} xad \sin \theta$$

→ BOTH ARE EQUAL

$$X \times Y = Z \times W$$

#



$$\begin{cases} \Delta AOB = y \\ \Delta COD = x \end{cases}$$

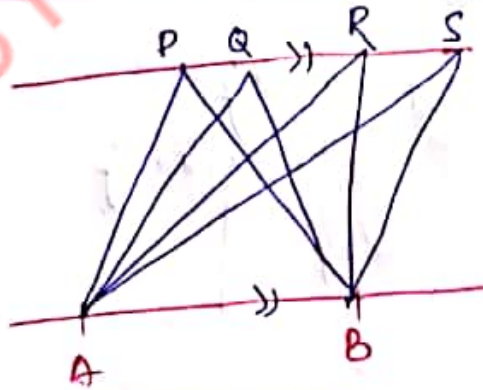
$$X \times Y = K \times K$$

$$K = \sqrt{xy}$$

* ΔAOB & ΔCOD will always be similar triangle

* $\text{Area of } \Delta AOD = \text{Area of } \Delta BOC$

Theorem -



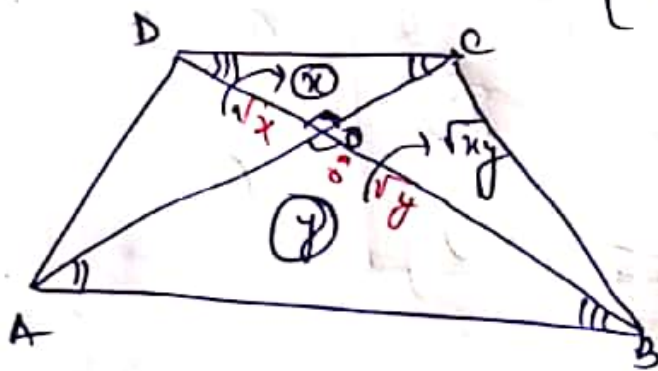
$$\text{Area } [\Delta APB = \Delta AQB = \Delta ARB = \Delta ASB]$$

→ दो समान्तर रेखाओं पर एक ही आधार पर बने Δ , $\square$, त्रिभुज, चतुर्भुज (Triangle) (quadrilateral) के क्षेत्रफल हमेशा बराबर होंगे।
(Area) (Equal)

Explanation -

$$\left\{ \begin{array}{l} \sqrt{x} = x \\ \sqrt{y} = y \end{array} \right\}$$

(7)



$$\left\{ \begin{array}{l} \text{area } \triangle COD = x \text{ cm}^2 \\ \text{area } \triangle AOB = y \text{ cm}^2 \end{array} \right\}$$

$$\frac{DO}{OB} = \frac{\sqrt{x}}{\sqrt{y}}$$

$$\frac{OC}{OA} = \frac{\sqrt{x}}{\sqrt{y}}$$

Details explanation,

$$\frac{\triangle DOC}{\triangle AOD} = \frac{\sqrt{x}}{\sqrt{y}}$$

$$\frac{x}{\triangle AOD} = \frac{\sqrt{x}}{\sqrt{y}}$$

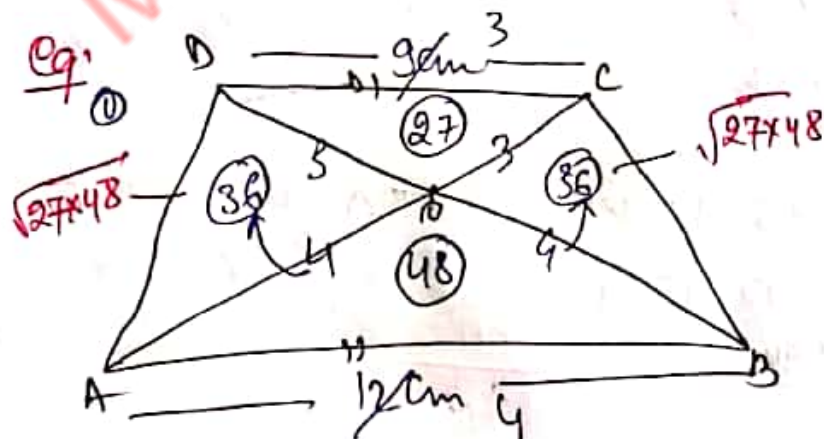
$$\triangle AOD = \sqrt{xy}$$

$$\square ABCD = x + y + 2\sqrt{xy}$$

$$= (\sqrt{x})^2 + (\sqrt{y})^2 + 2\sqrt{x} \times \sqrt{y}$$

#

$$\text{Area } \square ABCD = (\sqrt{x} + \sqrt{y})^2$$



$$\triangle COD = 27 \text{ cm}^2$$

$$\triangle AOB = 48 \text{ cm}^2$$

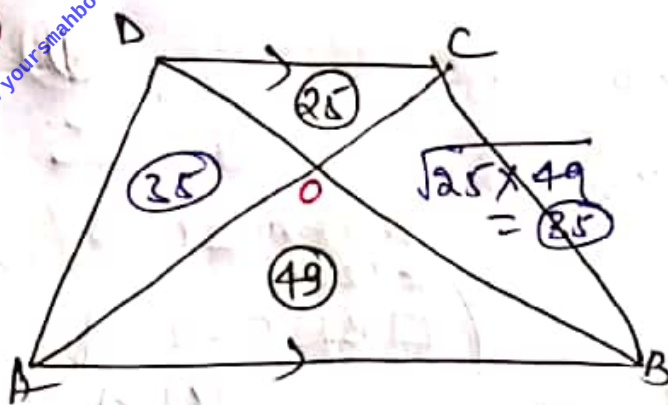
$$\triangle BOC = 36 \text{ cm}^2$$

$$\triangle AOD = 36 \text{ cm}^2$$

$$\square ABCD = 147 \text{ cm}^2$$

$$\begin{array}{l} 9:16 \\ \times 3 \quad \times 3 \\ 27 \quad 48 \end{array}$$

②



Area

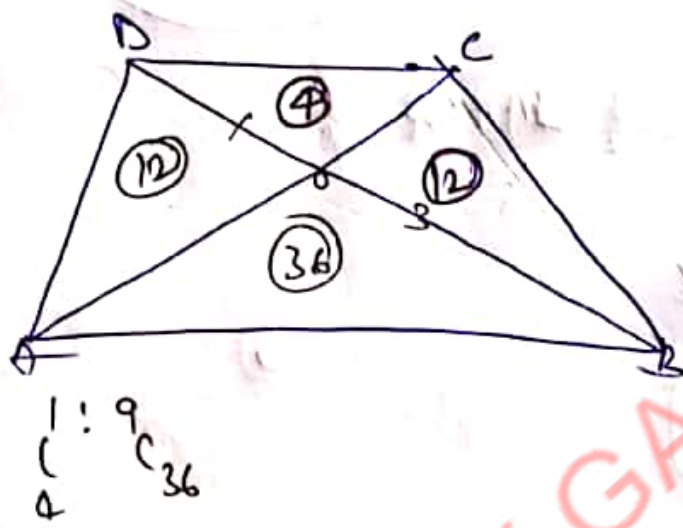
$\square ABCD = ?$

$= 144 \text{ unit}^2$
Ans

$\sqrt{xy}$

⑧

③

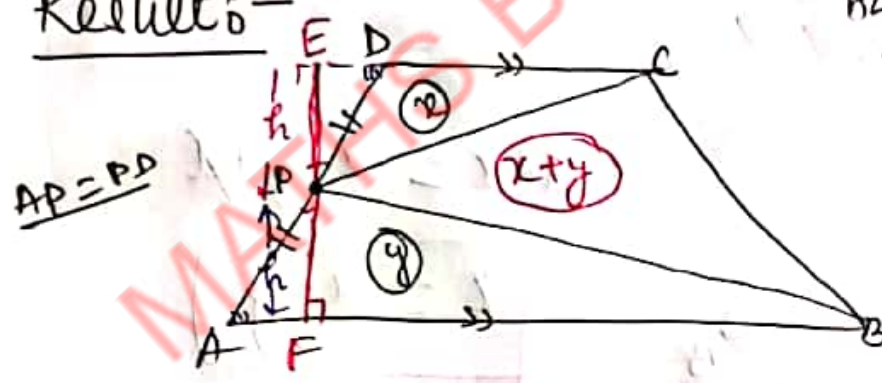


Area

$\square ABCD = ?$

64 Ans

Result:-



$AP = PD$

$\triangle BPC = x+y$

$= \frac{1}{2} \square ABCD$

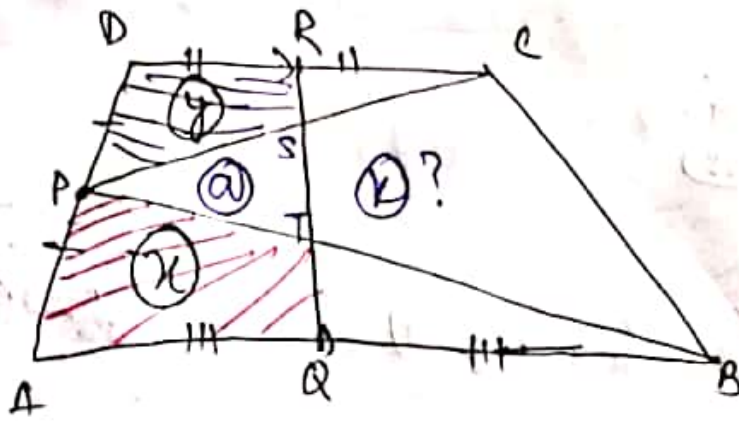
$\triangle APF \cong \triangle DPE$

$x+y = \frac{1}{2} b \times h + \frac{1}{2} a \times h$
 $= \frac{1}{2} (a+b) \times (2h) \times \frac{1}{2}$

Trapezium ki Area

$x+y = \frac{\square ABCD}{2}$

😊

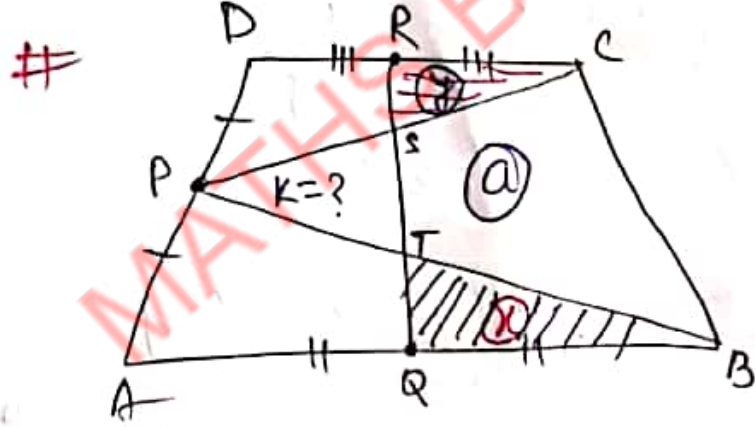


Area $\rightarrow$
 $\square BTSC = ?$
 $= (x+y)$

$$\triangle BPC = \square AQRD = \frac{1}{2} \square ABCD$$

$$x+k = x+y+x$$

$$k = x+y$$

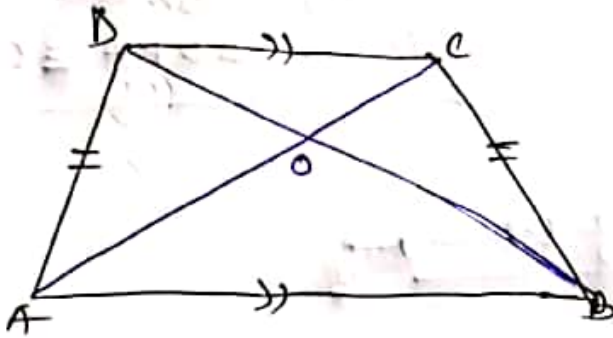


$$x+k = x+y+x$$

$$k = x+y$$

$$\text{Area } \triangle PTS = (x+y)$$

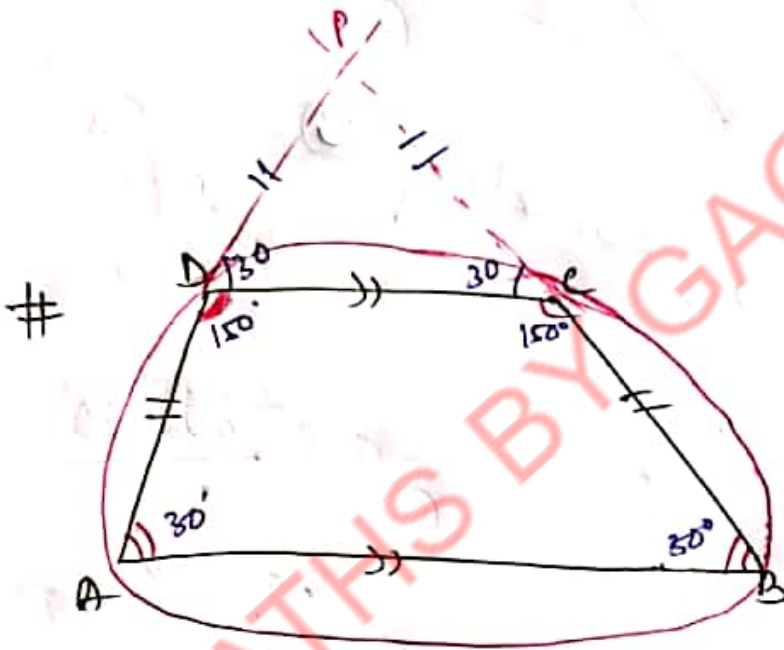
Isosceles trapezium (समद्विबाहु समलम्ब चतुर्भुज) (10)



$AB \parallel CD$ & $AD = BC$
 Then: $\triangle ABC$ Isosceles होगा

Result

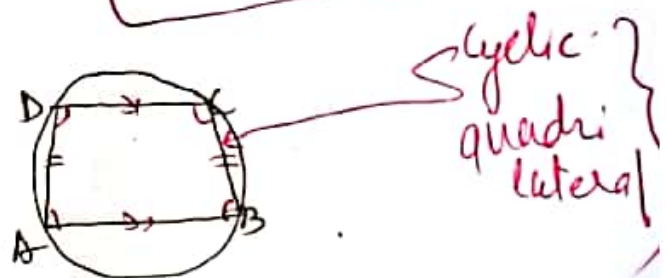
$AC = BD$
 diagonals are equal.
 But diagonal do not bisect each other



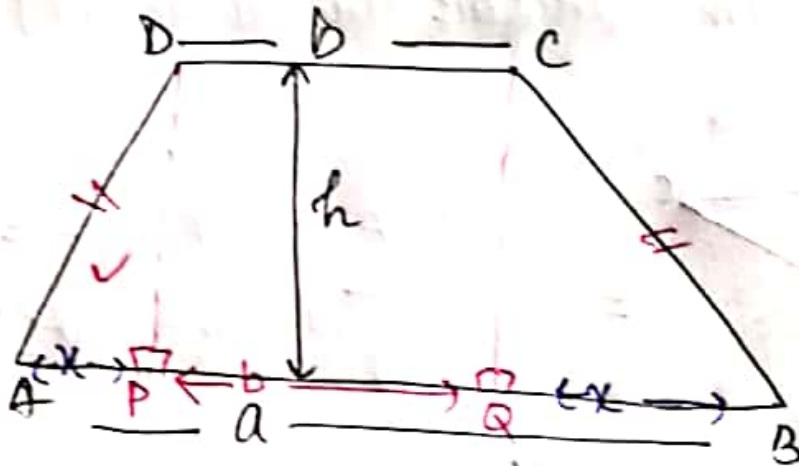
$\angle A = \angle B$
 $\angle D = \angle C$

$\angle A + \angle D = 180^\circ$
 $\angle B + \angle C = 180^\circ$
 $\angle A + \angle C = 180^\circ$
 $\angle B + \angle D = 180^\circ$

{ Isosceles trapezium is always
 a cyclic quadrilateral.
 (चक्रीय) }



#



congruent $\triangle$ s.

$$\triangle DPA \cong \triangle CQB$$

$$\square ABCD = (b+x) \times h$$

Satyam Padhyay

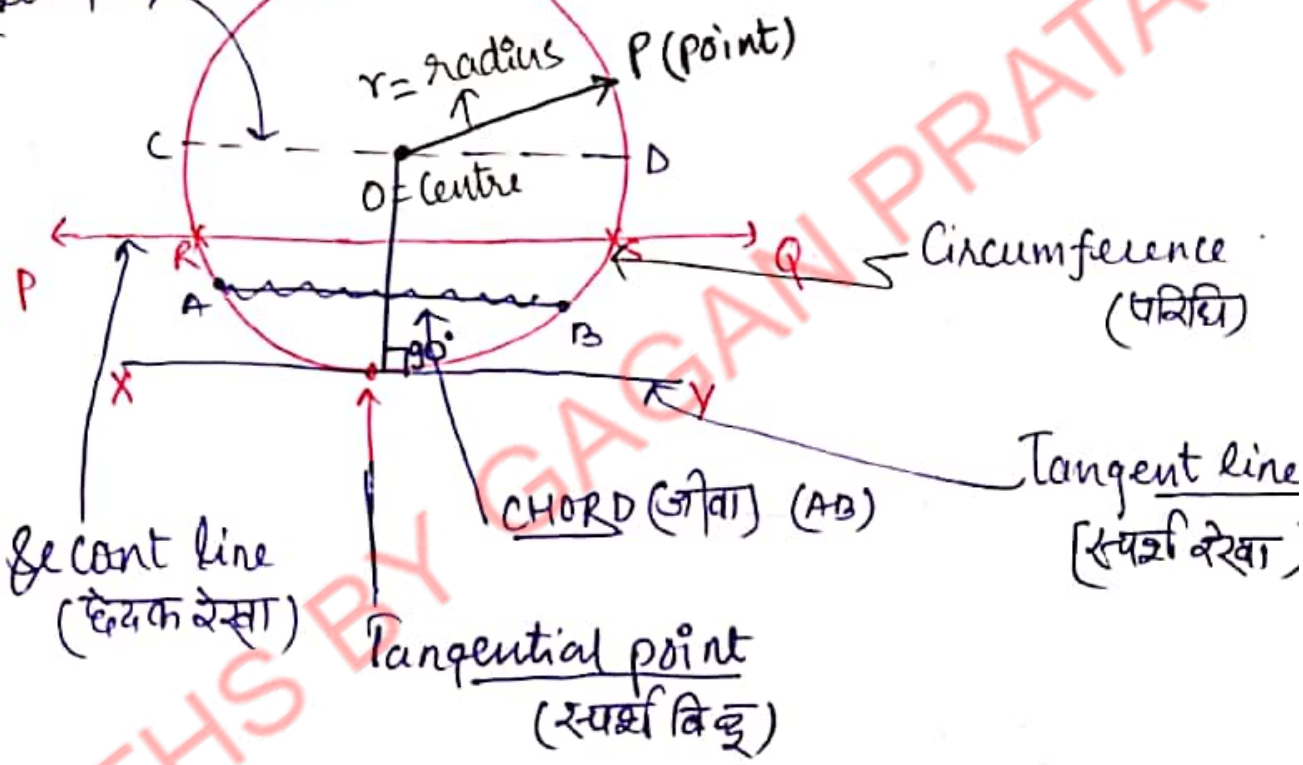
END Geometry 😊

[Circle] वृत्त

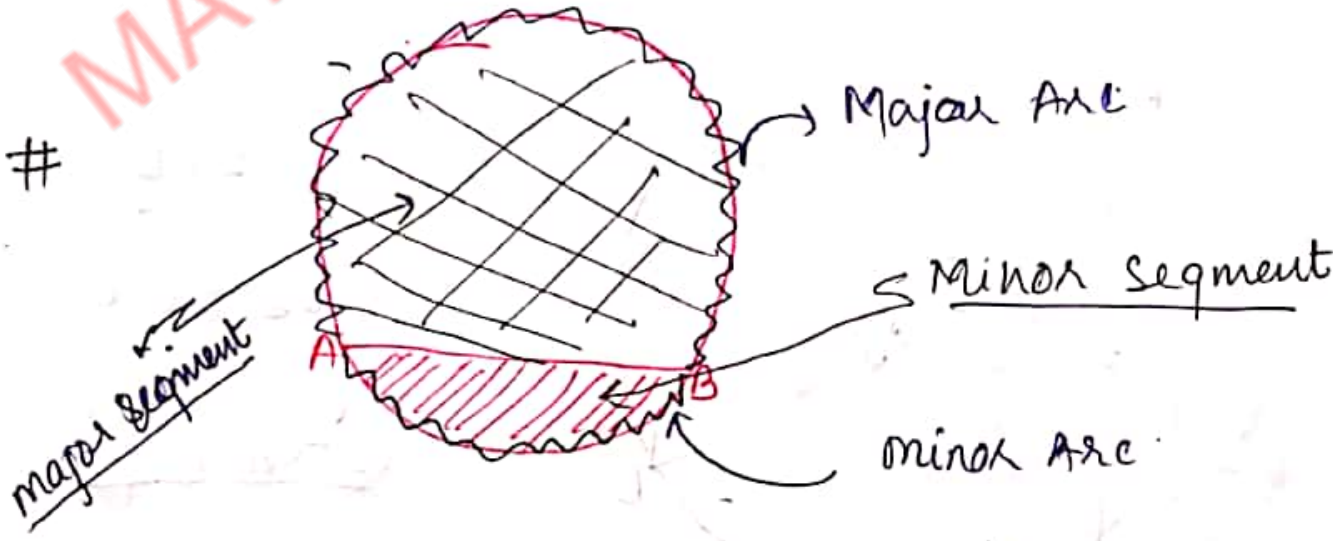
①

#

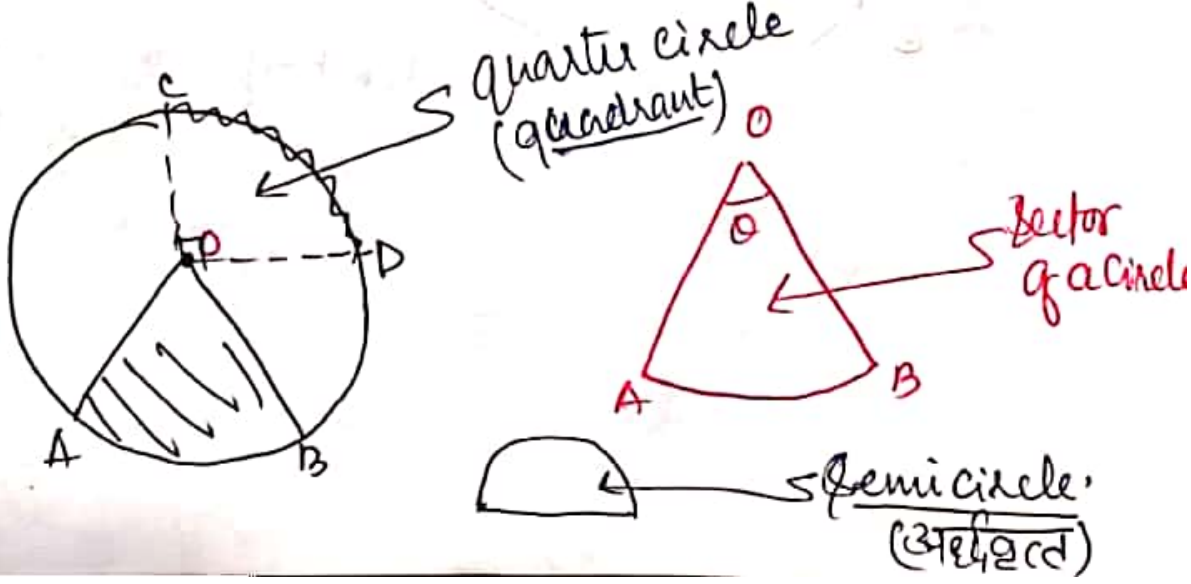
(Longest CHORD) or diameter (व्यास)

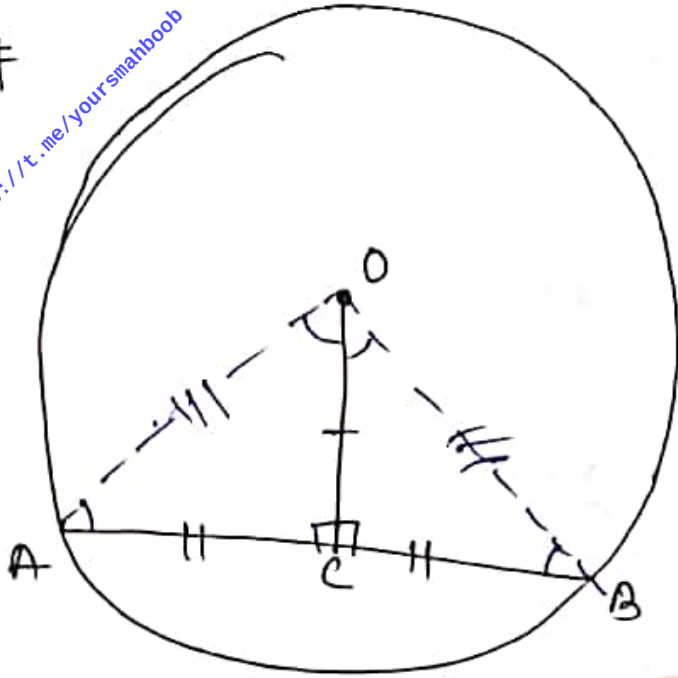


#



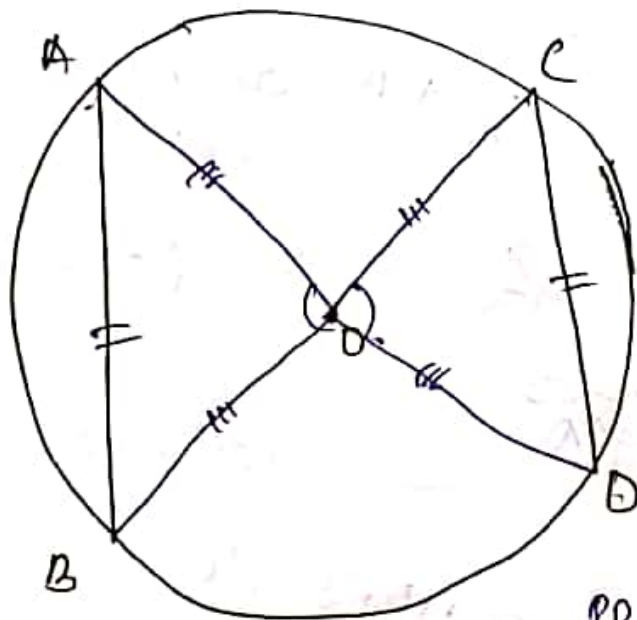
#





②
* Circle के Centre से
Chord पर डाला गया
perpendicular
Chord को bisect करता है

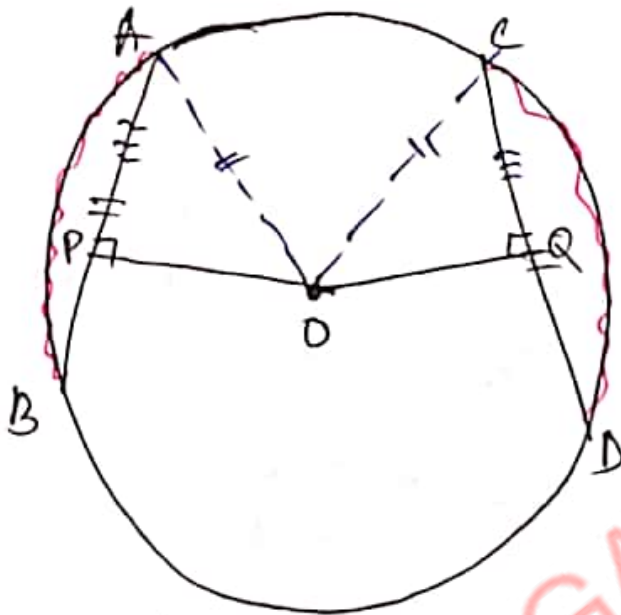
PROOF $\rightarrow \triangle OCA \cong \triangle OCB$
 $\downarrow$
 [Congruent हो गे]
 [OA = OB, CA = CB]



98
CHORD $[AB = CD]$
 $\Rightarrow \angle AOB = \angle COD$
 Central angle

PROOF $\rightarrow$ { Both Triangle
are congruent }

(3)

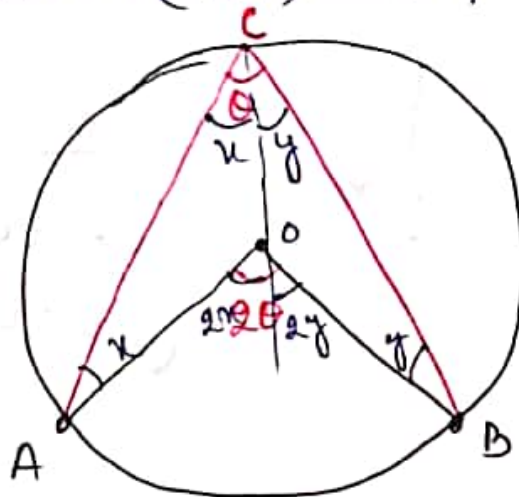


95 $\left\{ \begin{array}{l} AB = CD \\ \widehat{AB} = \widehat{CD} \end{array} \right\}$ Arc

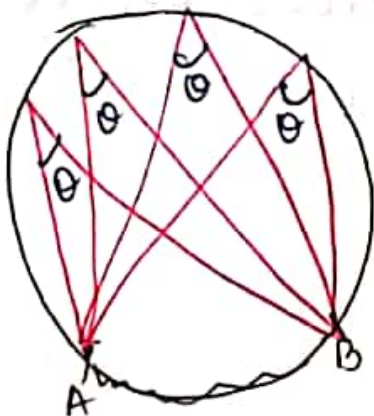
$\Rightarrow OP = OQ$

Because (both Δ are congruent)

Circle के किसी Arc द्वारा Centre पर बना angle वही Arc द्वारा Circumference (परिधि) पर बने angle का Double (दोगुना) होता है -

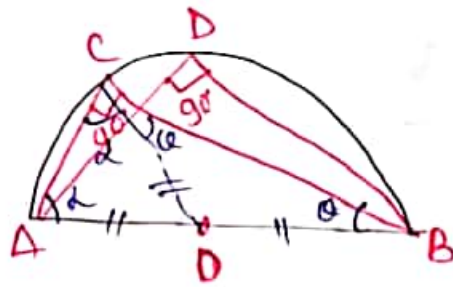


$\left\{ \begin{array}{l} \angle AOB = 2 \angle ACB \\ \downarrow \\ 2(x+y) = 2 \times (x+y) \end{array} \right\}$



{ Circle के किसी Arc द्वारा परिधि पर बने angle वही angle बराबर होते हैं (यही भी बने एक ही Arc द्वारा) }

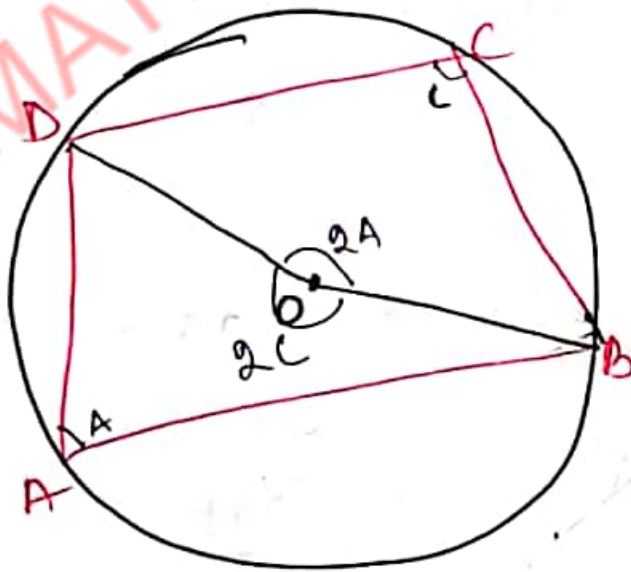
Semicircle में बना angle हमेशा 90° होता है - (4)
(अष्टवृत्त)



$$\begin{cases} \angle ACB = 90^\circ \\ \angle ADB = 90^\circ \end{cases}$$

PROOF $\begin{cases} 2(\alpha + \theta) = 180^\circ \\ \alpha + \theta = 90^\circ \end{cases}$

Cyclic quadrilateral :-
(चक्रीय चतुर्भुज)



$\square ABCD \rightarrow$ Cyclic quadrilateral.

Condition -

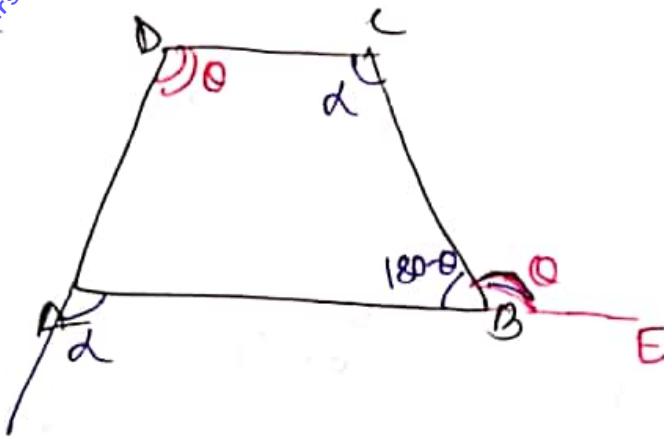
$$\begin{cases} \angle A + \angle C = 180^\circ \\ \angle B + \angle D = 180^\circ \end{cases}$$

PROOF - $\begin{cases} 2A + 2C = 360^\circ \\ \angle A + \angle C = 180^\circ \end{cases}$

Similarly $\angle B + \angle D = 180^\circ$

Satyam Padhyay





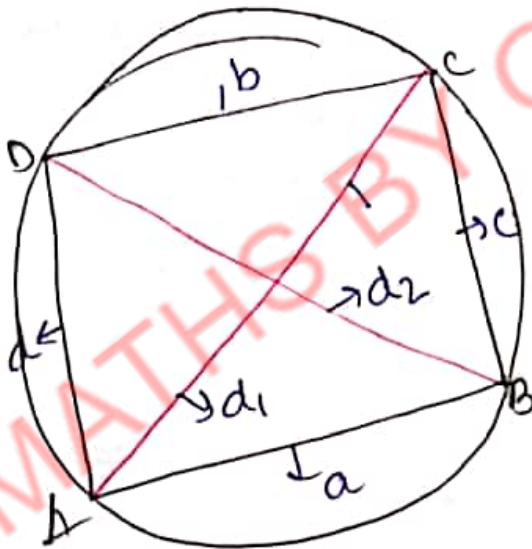
⑤
□ ABCD → cyclic quadrilateral

$$\angle CBE = \angle ADC$$

□ ABCD → cyclic qua —

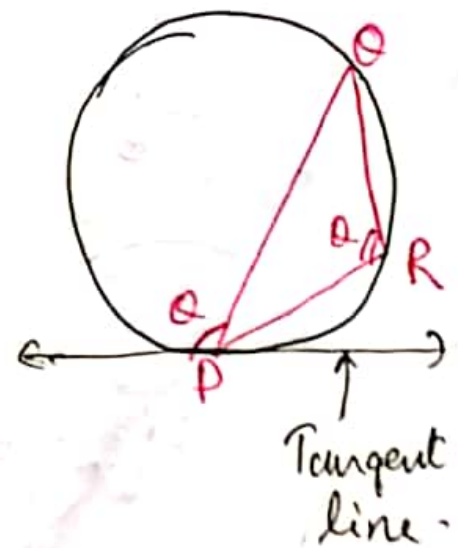
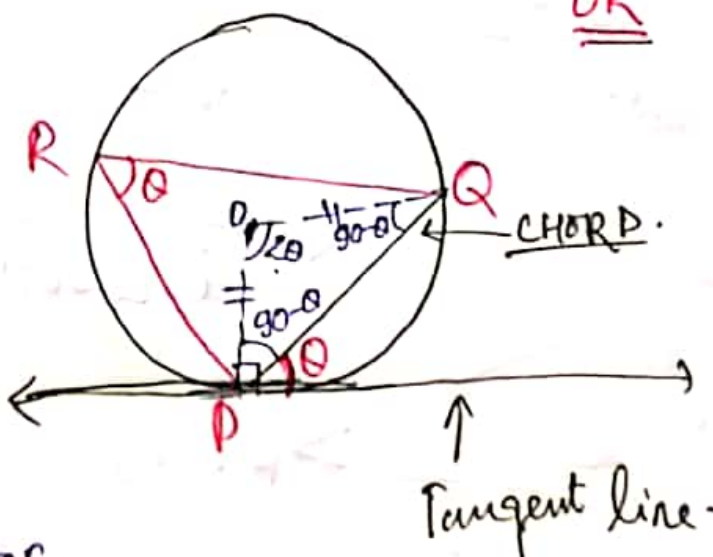
Condition :-

$$* \boxed{ab + cd = d_1 d_2}$$



Alternate Segment Theorem :-

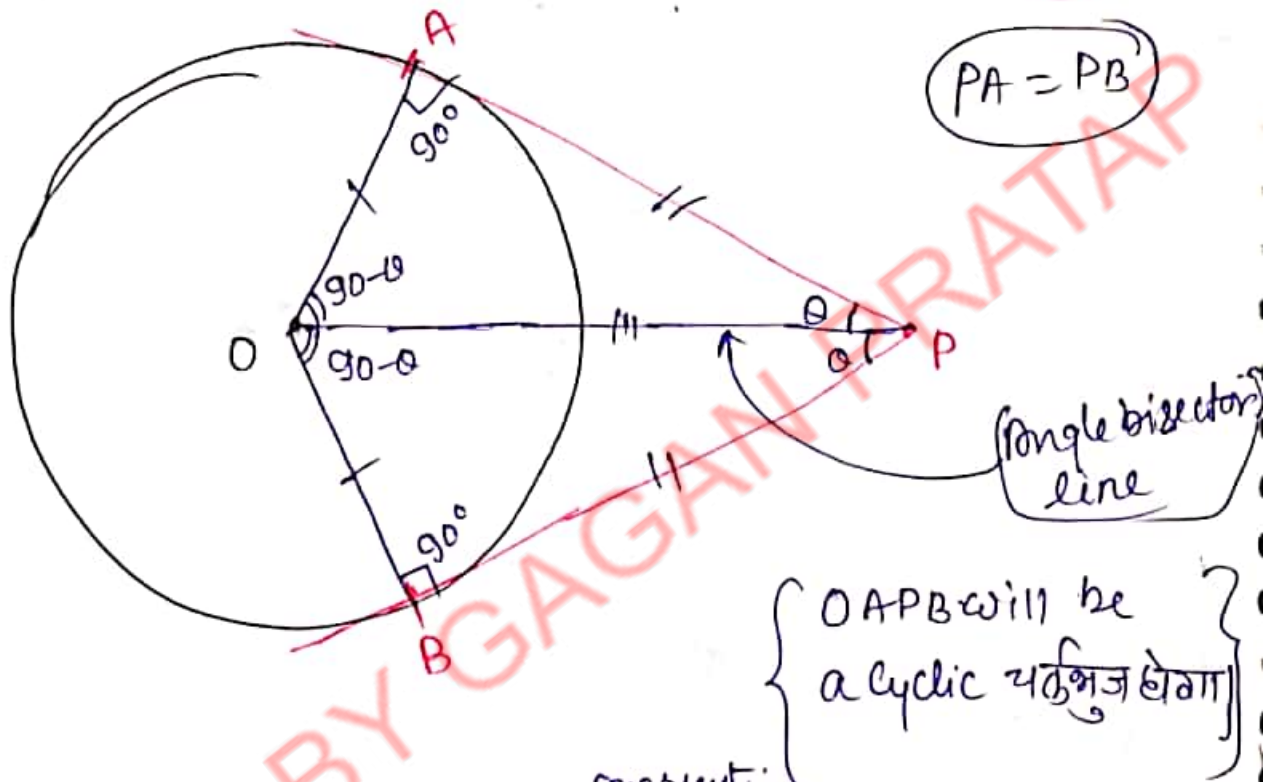
OR



PROOF:

{ CHORD PQ → Centre → 2a }

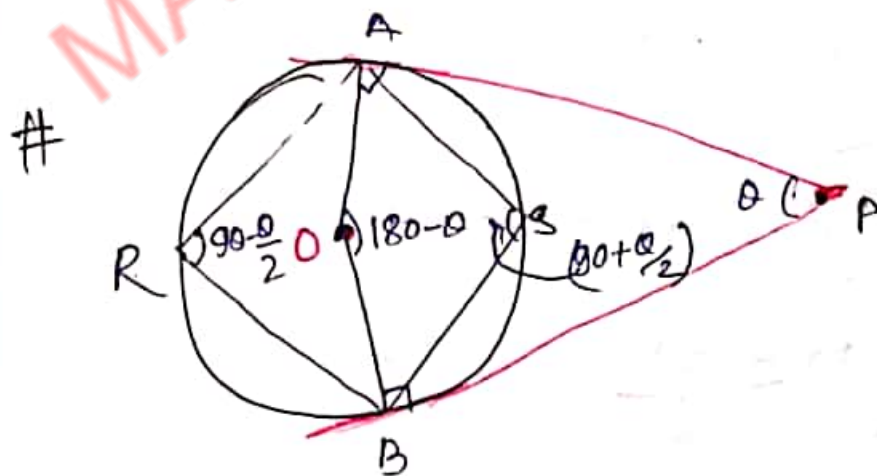
{ PQ-काट परी मी पर कोण बननेका }



Proof

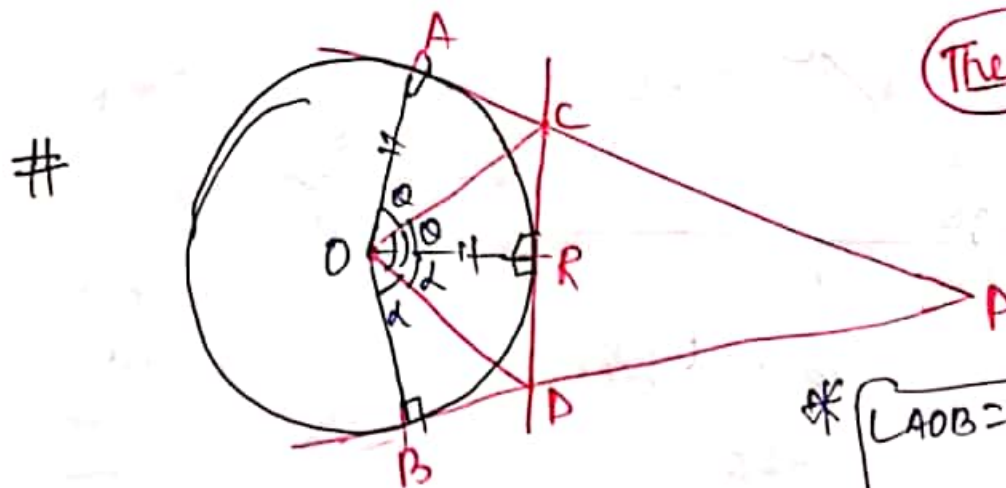
$\triangle OAP \cong \triangle OBP$ (congruent)

$PA = PB$

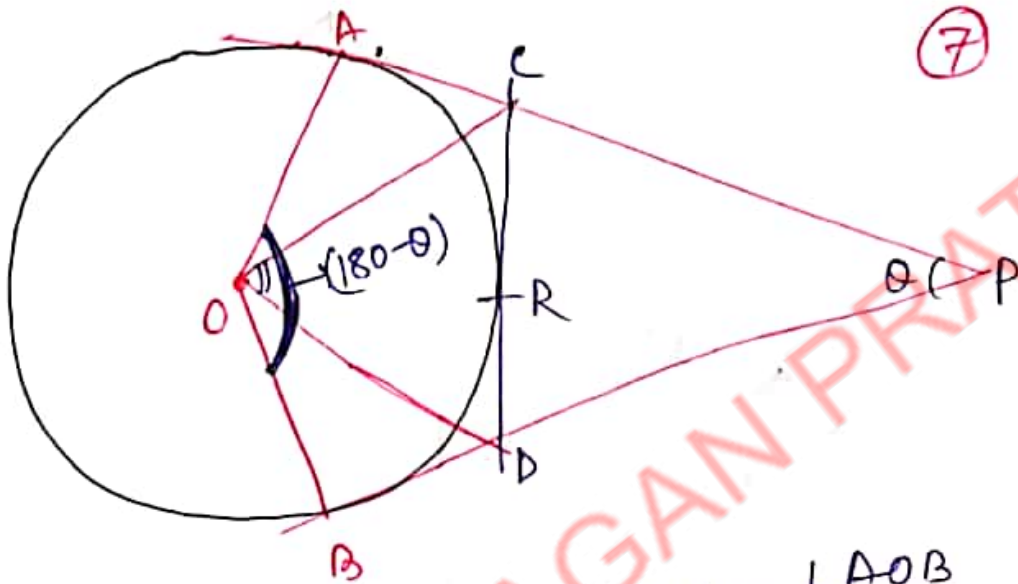


Theorem

$\angle COD = \frac{\angle AOB}{2}$



$\angle AOB = 2(\theta + \alpha)$
 $\angle COD = \theta + \alpha$



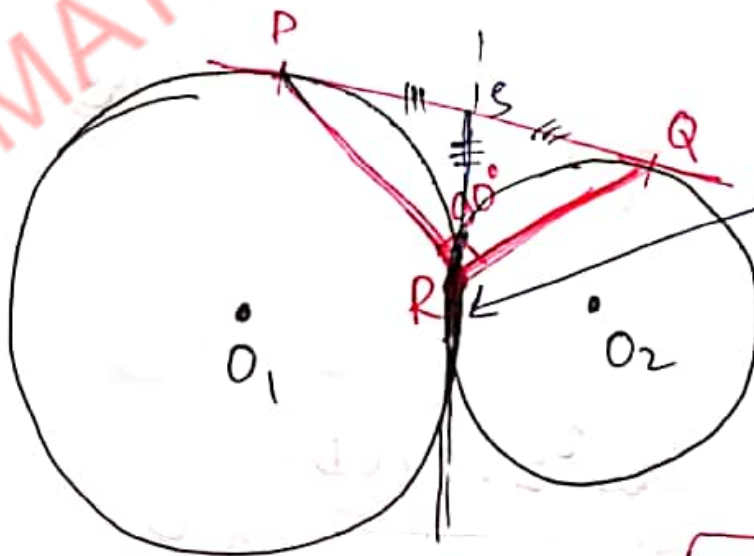
$$\angle COD = \frac{\angle AOB}{2}$$

$$\angle COD = \frac{180 - \theta}{2}$$

#

$$\angle COD = 90 - \frac{\theta}{2}$$

#

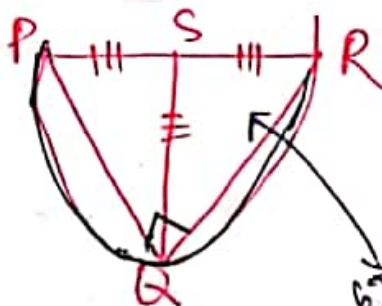


Touch on R hai.

$$\angle PRQ = 90^\circ$$

$$SP = SR = SQ$$

$$SR = \frac{PQ}{2}$$



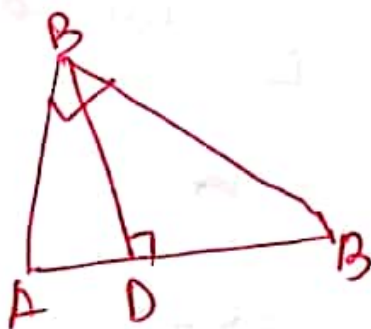
Right angle triangle hai.



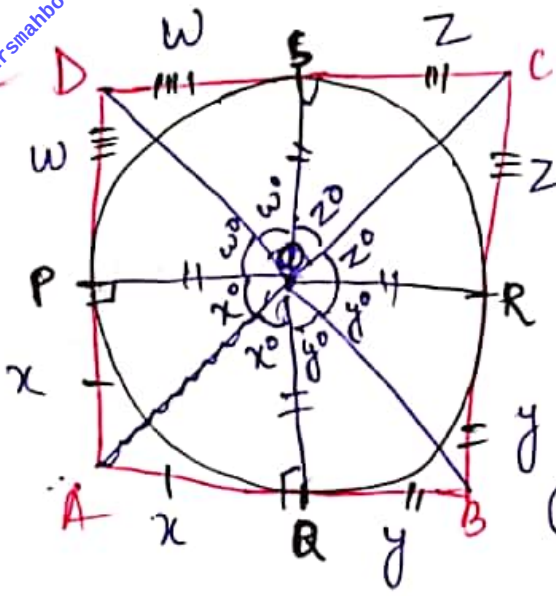
PROOF
 $\angle(x + 0) = 180^\circ$

$$Q + d = 90^\circ$$

By similarity,



$$BD^2 = AD \times CD$$



When a circle is inscribed in a quadrilateral.

Result -

①

$$AB + CD = BC + AD$$

②

$$\left. \begin{aligned} \angle AOB + \angle COD &= 180^\circ \\ \angle BOC + \angle AOD &= 180^\circ \end{aligned} \right\}$$

PROOF ①

$$\left. \begin{aligned} AB + CD &= x + y + z + w \\ AD + BC &= x + w + y + z \end{aligned} \right\} \text{ BOTH ARE EQUAL}$$

$$AB + CD = BC + AD \quad \text{इति सिद्धम् ①}$$

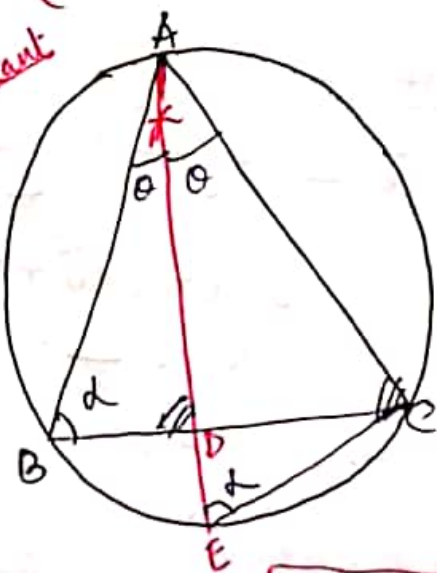
PROOF

②

$$x^\circ + y^\circ + z^\circ + w^\circ = 180^\circ$$

$$\left\{ \begin{aligned} \angle AOB + \angle COD &= 180^\circ \\ \angle BOC + \angle AOD &= 180^\circ \end{aligned} \right\} \text{ proved इति सिद्धम् ②}$$

most important



$$AB \times AC + AE \times DE = AE^2$$

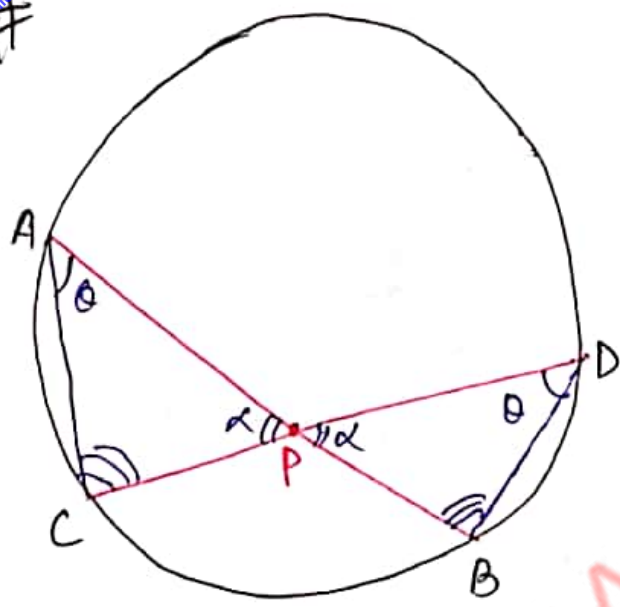
Similar Triangle

$$\frac{AD}{AC} = \frac{AB}{AE}$$

$$AD \times AE = AB \times AC \quad \text{--- ①}$$

$$AD \times AE + AE \times DE = AB \times AC + AE \times DE \Rightarrow AE^2 \quad \text{इति सिद्धम्}$$

$$AE = AD + DE$$



Result-

$$PA \times PB = PC \times PD$$

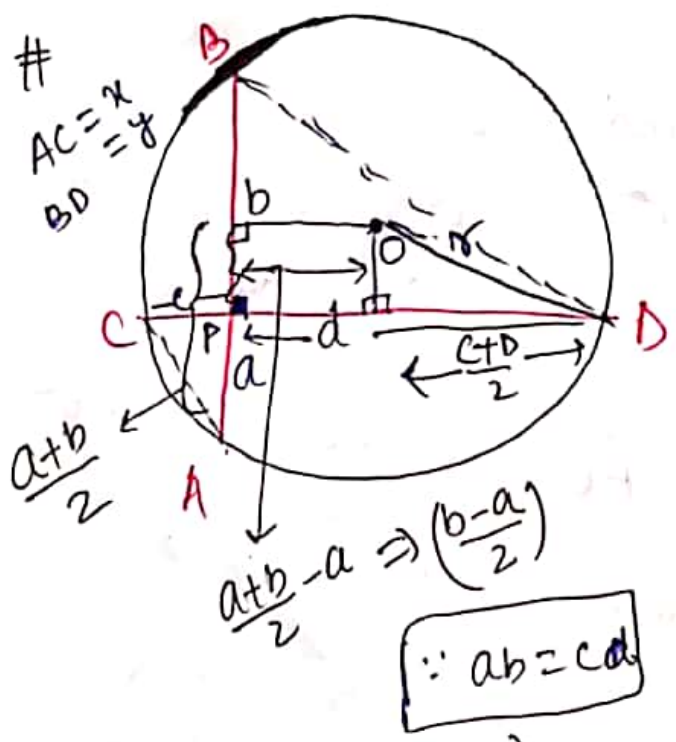
Similar Δ

PROOF -

$$\frac{PA}{PD} = \frac{PC}{PB}$$

$$PA \times PB = PC \times PD$$

सिद्धि (सिद्ध) ☺



$$r = \frac{\sqrt{x^2 + y^2}}{2}$$

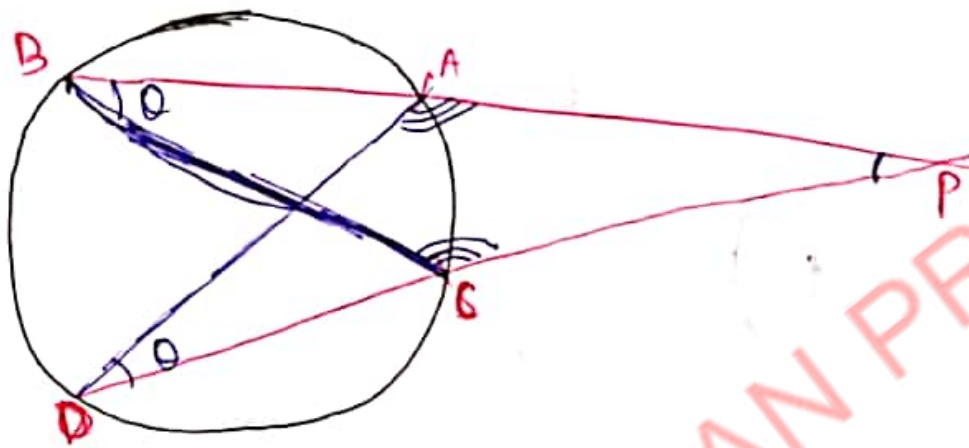
PROOF.

$$\begin{aligned} r^2 &= \frac{(b-a)^2}{4} + \frac{(c+d)^2}{4} \\ &= \frac{a^2 + b^2 - 2ab + c^2 + d^2 + 2cd}{4} \end{aligned}$$

$$r^2 = \frac{x^2 + y^2}{4}$$

$$r = \frac{\sqrt{x^2 + y^2}}{2}$$

सिद्धि (सिद्ध) ☺



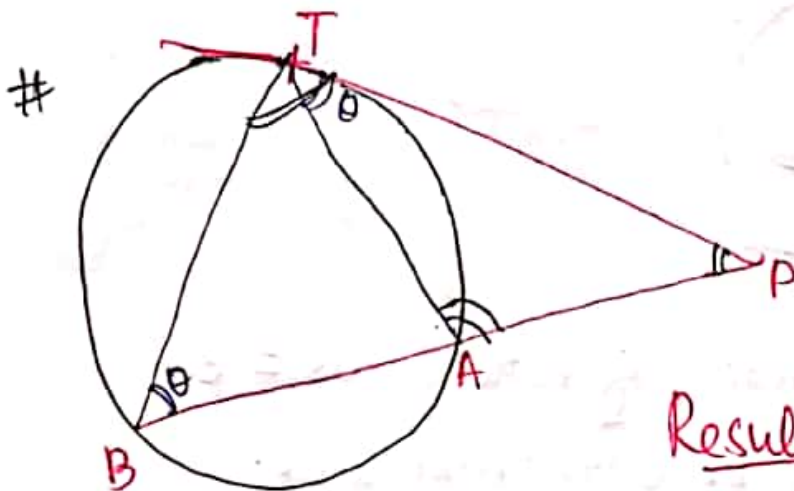
Result

$$PA \times PB = PC \times PD$$

PROOF - Similarity

$$\frac{PC}{PA} = \frac{PB}{PD}$$

$$\Rightarrow PA \times PB = PC \times PD$$



Result -

$$PT^2 = PA \times PB$$

PROOF - Similar Δ

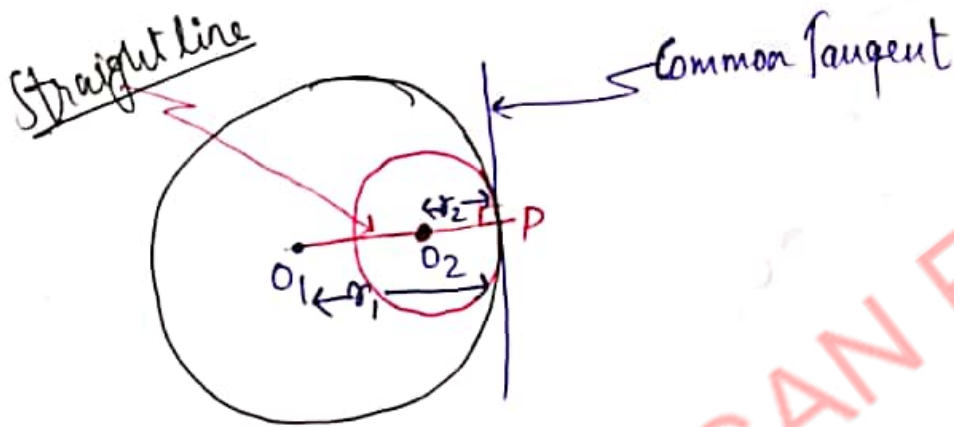
$$\frac{PT}{PB} = \frac{PA}{PT}$$

$$PT^2 = PA \times PB$$

☺ (कवि सिद्धेश)

Internally Touch Circle: -

(12)



* maximum common tangent of both circle = 1

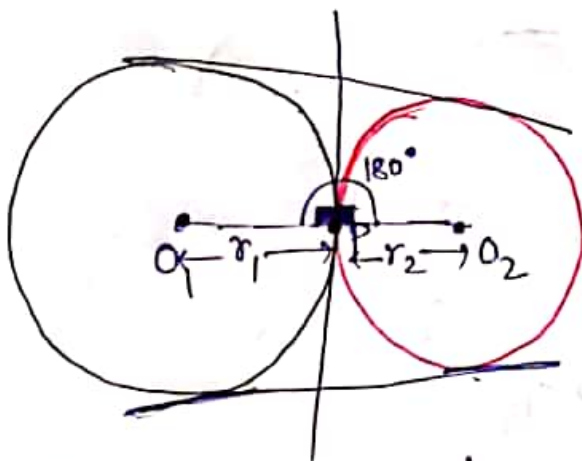
* O_1O_2P will always be a straight line.

Distance

$$O_1O_2 = O_1P - O_2P$$

$$O_1O_2 = r_1 - r_2$$

(2) Externally Touch

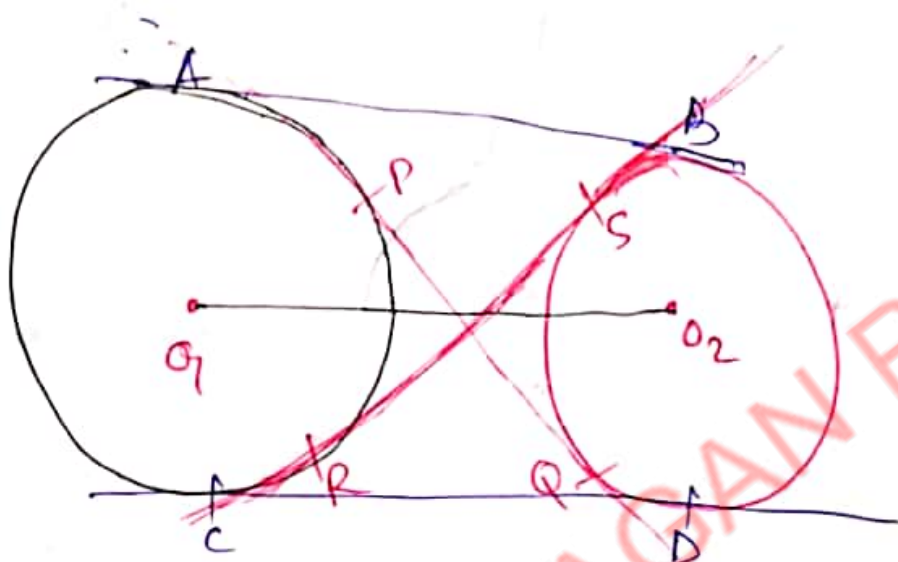


* maximum Common tangent of both Circles = 3

* O_1PO_2 will always be a straight line.

$$O_1O_2 = O_1P + O_2P$$

$$O_1O_2 = r_1 + r_2 \quad [\text{Distance}]$$

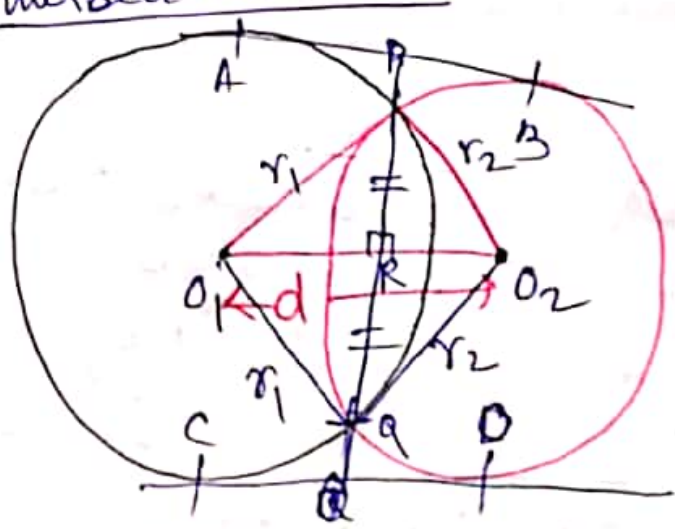


* Maximum common tangent of both circle = 4

$AB = CD = \text{Direct Common tangent (DCT)}$

$PQ = RS = \text{Transverse common tangent (TCT)}$

④ Intersect each other —



$AB = CD = \text{DCT}$

$PQ = \text{Common Chord}$
(उभयनिष्ठ जीवा)

$PR = QR = \frac{PQ}{2}$

* maximum common tangent of both circle = 2.

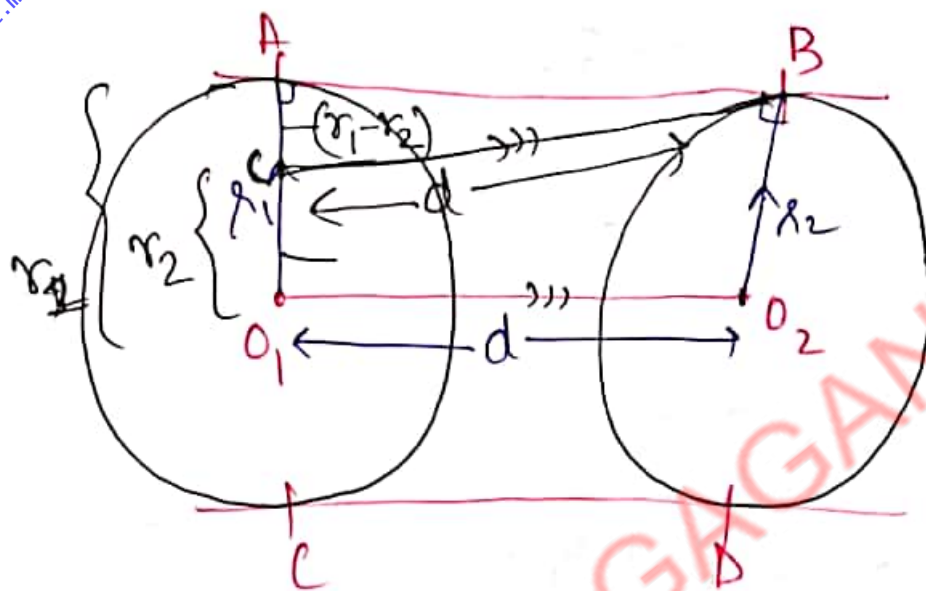
$O_1 P O_2 \cong O_1 Q O_2$

$PR \perp O_1 O_2$
 $QR \perp O_1 O_2$

#

Direct Common tangent (D.C.T) →

(14)



given:

$$AB = CD$$

$$O_1O_2 = d$$

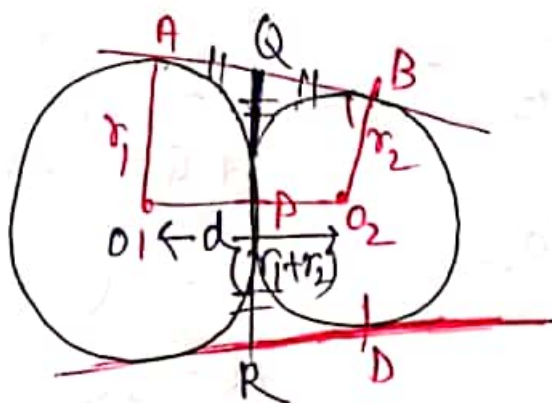
$$\begin{cases} O_1A = r_1 \\ O_2B = r_2 \end{cases}$$

$\triangle ACB$ में $(\angle A = 90^\circ)$

$$DCT = AB^2 = d^2 - (r_1 - r_2)^2$$

$$* DCT = AB = \sqrt{d^2 - (r_1 - r_2)^2}$$

When Circle touch each other externally —



Result

$$\begin{cases} QP = AQ \\ QP = QB \\ \hline QP = AB \end{cases}$$

Similarly

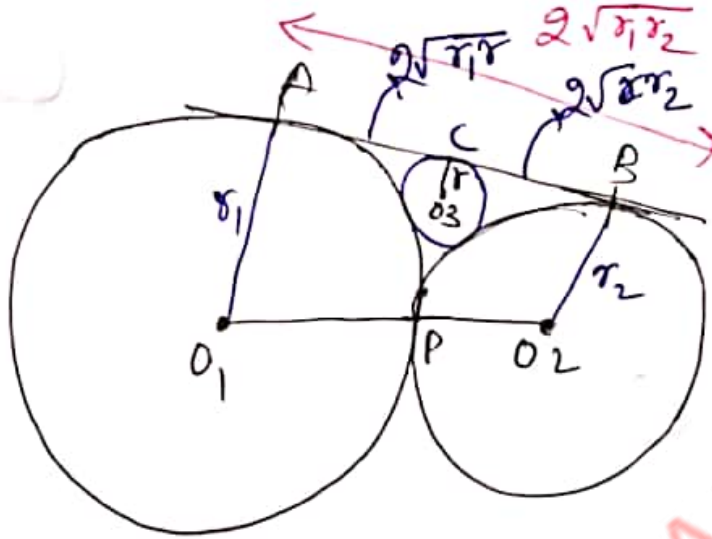
$$2PR = CD$$

$$\left\{ AB = \sqrt{(r_1 + r_2)^2 - (r_1 - r_2)^2} = 2\sqrt{r_1 r_2} \right\}$$

① + ②

$$2QR = 2AB$$

$$QR = AB = CD$$



PROOF

$$\begin{cases} AC = 2\sqrt{r_1 r_3} \\ BC = 2\sqrt{r_2 r_3} \end{cases}$$

$$AB = 2\sqrt{r_1 r_2}$$

Result-

$$\frac{1}{\sqrt{r}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}}$$

$$AB = AC + BC$$

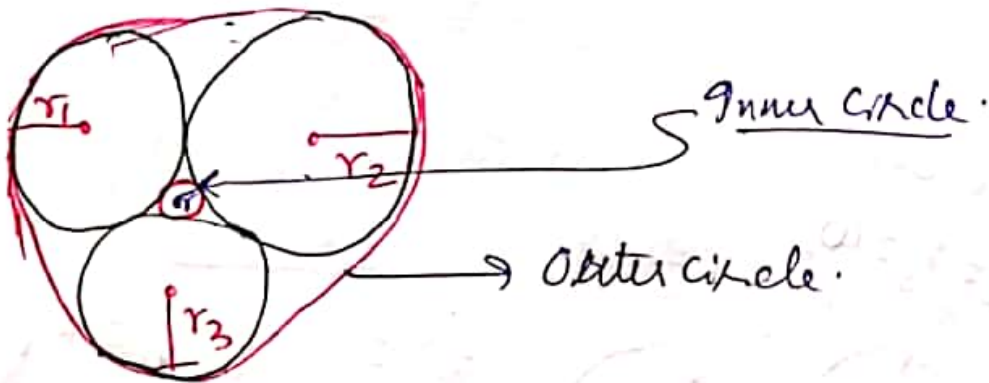
$$2\sqrt{r_1 r_2} = 2\sqrt{r_1 r_3} + 2\sqrt{r_2 r_3}$$

divide $\rightarrow \frac{2\sqrt{r_1 r_2}}{2\sqrt{r_1 r_1 r_2}} = \frac{2\sqrt{r_1 r_3}}{2\sqrt{r_1 r_1 r_2}} + \frac{2\sqrt{r_2 r_3}}{2\sqrt{r_1 r_1 r_2}}$

$$\frac{1}{\sqrt{r}} = \frac{1}{\sqrt{r_2}} + \frac{1}{\sqrt{r_1}}$$

Proved. @

Circle discaerts theorem $\rightarrow$



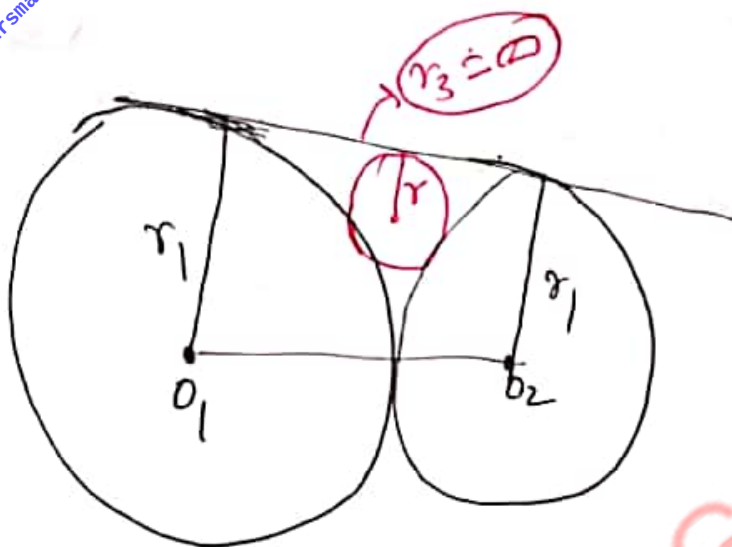
Result

$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \pm 2\sqrt{\frac{1}{r_1 r_2} + \frac{1}{r_2 r_3} + \frac{1}{r_3 r_1}}$$

$\frac{1}{r} = \text{curvature}$

-ive — Outer circle.
+ive — inner circle.

(16)



$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \pm 2 \sqrt{\frac{1}{r_1 r_2} + \frac{1}{r_2 r_3} + \frac{1}{r_3 r_1}}$$

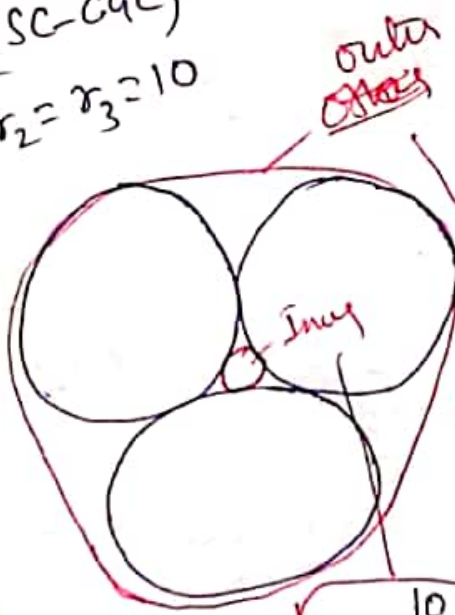
$r_3 = 0$

$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + 2 \sqrt{\frac{1}{r_1 r_2}}$$

$$\left(\frac{1}{\sqrt{r}}\right)^2 = \left(\frac{1}{\sqrt{r_1}}\right)^2 + \left(\frac{1}{\sqrt{r_2}}\right)^2 + 2 \sqrt{\frac{1}{r_1}} \times \frac{1}{\sqrt{r_2}}$$

$$\frac{1}{\sqrt{r}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}}$$

eg. (SSC-CGL)
 $r_1 = r_2 = r_3 = 10$



$$\frac{1}{r} = \frac{1}{10} + \frac{1}{10} + \frac{1}{10} \pm 2 \sqrt{\frac{1}{10 \times 10} + \frac{1}{10 \times 10} + \frac{1}{10 \times 10}}$$

$$\frac{1}{r} = \frac{3}{10} \pm 2 \frac{\sqrt{3}}{10}$$

$$\frac{1}{r} = \frac{3 \pm 2\sqrt{3}}{10}$$

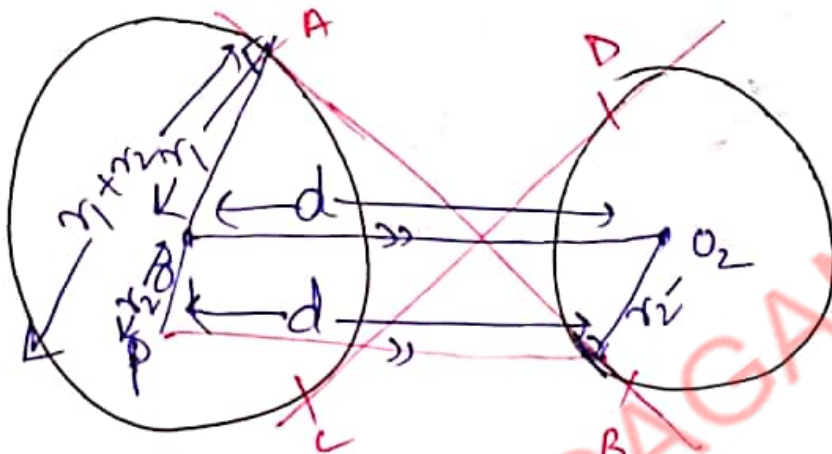
$$r = \frac{10}{3 \pm 2\sqrt{3}}$$

$$r = \frac{10}{3 + 2\sqrt{3}}$$

$$r = \frac{10}{2\sqrt{3} - 3}$$

Transverse Common tangent -

(17)



$$AB = CD$$

ΔPAB में :-

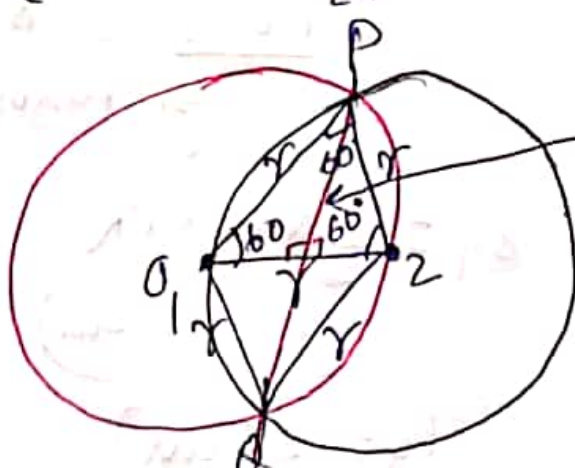
$$* AB = TCT = \sqrt{d^2 - (r_1 + r_2)^2}$$

$$* DCT = \sqrt{d^2 - (r_1 - r_2)^2}$$

$$DCT > TCT$$

Common Chord (उभयनिष्ठ जीवा) -

① When two circles pass through each other centre (एक दूसरे के केंद्रों से गुजरें)

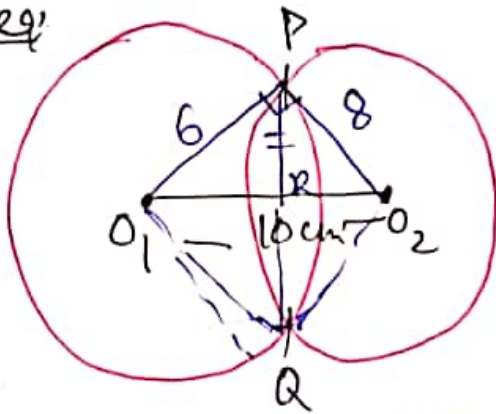


$$\text{Common Chord} = 2 \times \frac{\sqrt{3}}{2} r$$

$$C.C = \sqrt{3} r$$

2

eg



given

$$\begin{cases} r_1 = 6 \text{ cm} \\ r_2 = 8 \text{ cm} \\ PQ = ? \end{cases}$$

$$PR = RQ$$

$$PR = \frac{6 \times 8}{10}$$

$$PR = 4.8 \text{ cm}$$

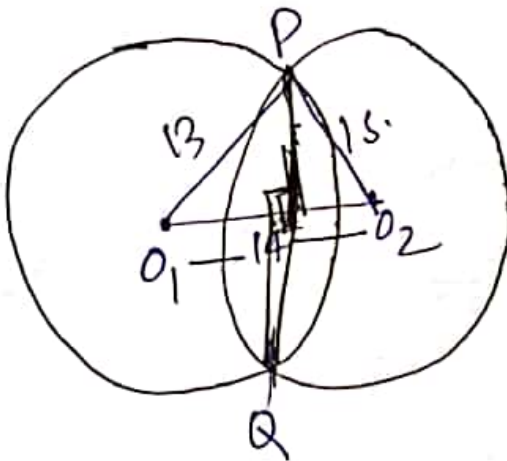
$$PQ = 9.6 \text{ cm}$$

$$PR = \frac{PQ}{2}$$

Result

$$PQ = \frac{2r_1 r_2}{d}$$

3



given

$$r_1 = 13 \text{ cm}$$

$$r_2 = 15 \text{ cm}$$

$$PR = RQ$$

$$A_{\Delta PO_1O_2} = 84$$

[By Heron's formula]

$$84 = \frac{1}{2} \times 14 \times PR$$

$$PR = 12 \text{ cm}$$

$$PQ = 24 \text{ cm}$$

Satyam Upadhyay

$$(2XR)^2 = (2AX)^2 + (2PR)^2$$

(20)

$$XY^2 = AB^2 + PQ^2$$

proved.

इति सिद्धम्

(21)

- * line and angle.
- * Inequality of Δ
- * Angle in Δ
- * Sine rule
- * Cosine rule
- * area related problem of Δ
- * MPG (Mass point Geometry)
- * Similarity
- * Congruency
- * Centres of Δ 's
 - Incentre
 - Circumcentre
 - Orthocentre
 - Centroid
 - Exocentre

- * Equilateral Δ
- * Right angle Δ
- * Isosceles Δ
- * acute / obtuse angle Δ
- * Square.
- * Rectangle
- * Parallelogram (11)
- * Rhombus
- * Trapezium.
- * Circle.
 - * Angles
 - * Chords
 - * Secants
 - * Tangents.

Line and Angle (रेखा और कोण)

- * Point (बिन्दु)
 - (लम्बाई) No length, (चौड़ाई) No breadth, (ऊँचाई) no height
 - Points is a zero dimensional figure.
 - Each point has unique location. (प्रत्येक बिन्दु)

(•) Point (A)

[Always Capital letter]

* Line (रेखा)

- A line is a set of points having only length.
- Line is one (1) dimensional.

eg.



denotes $\overleftrightarrow{AB}$ OR line $\overleftrightarrow{AB}$

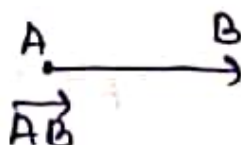


* Line Segment (रेखाखण्ड) —

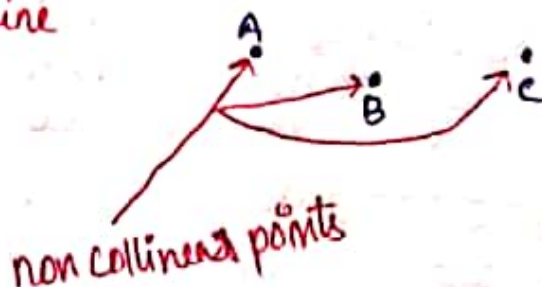
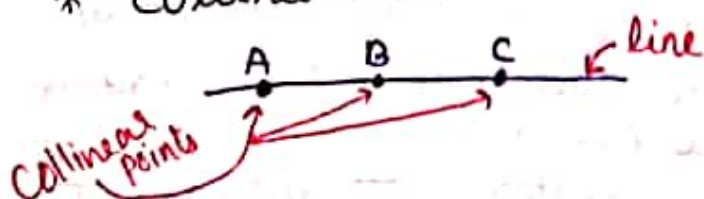


Denotes $\overline{AB}$

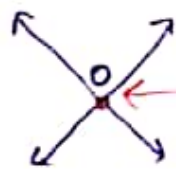
* Rays (किरण) —



* Collinear and non-collinear points (संरेखीय और असंरेखीय बिन्दु)

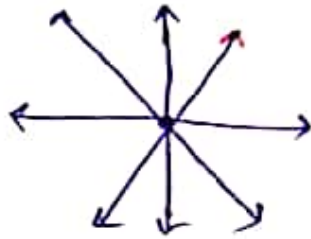


Intersecting lines (प्रतिच्छेदी रेखाएँ)

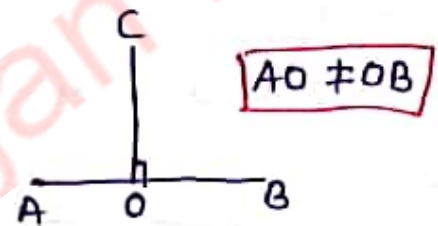


Intersecting points
प्रतिच्छेदी रेखाएँ

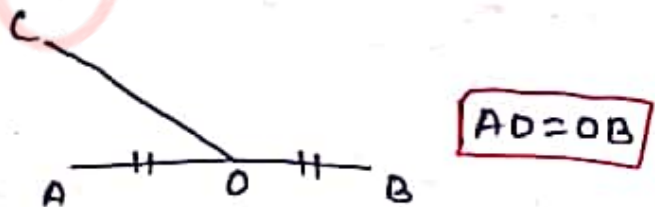
* Con-current lines: - [बिन्दुगामी रेखाएँ] → तीन या तीन से अधिक रेखाएँ एक ही बिन्दु से होकर जाती हैं।



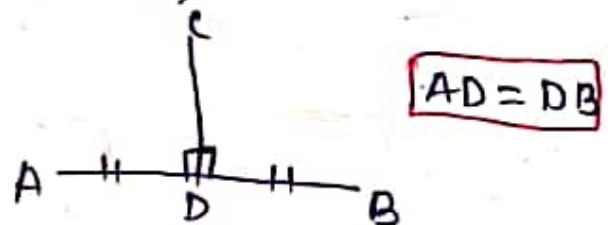
* Perpendicular line (लम्ब रेखा)



* Bisector अर्धक -



* Perpendicular Bisector (लम्ब अर्धक रेखा)



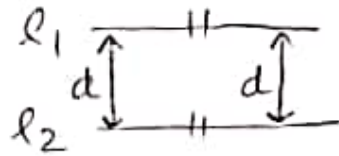
* Parallel lines :- ("समान्तर रेखाएँ")

→ Parallel lines do not have any common point or they do not intersect each other but parallel lines must be co-planar.

→ समान्तर रेखाएँ के पास न तो कोई अभ्यनिष्ठ बिन्दु होता है और न ही वे एक दूसरे को स्पर्श करते हैं (अनन्त तक) लेकिन

लेकिन समान्तर रेखाओं के बीच की दूरी सर्वत्र बराबर होती है। -

eg. <https://t.me/yourmathsnotes>



$$l_1 \parallel l_2$$

#



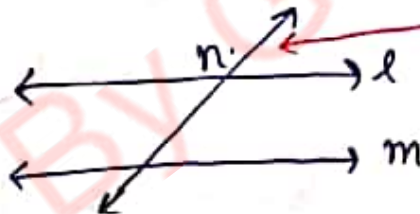
$$l_1 \parallel l_2$$

$$l_2 \parallel l_3$$

THEN $l_1 \parallel l_3$ [एक ही समतल में]

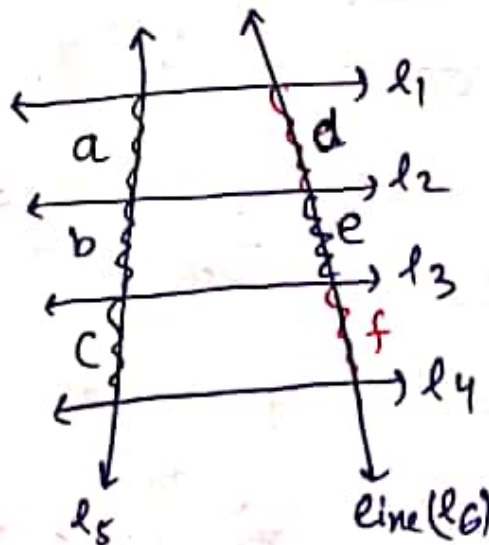
Transversal line (तिर्यकच्छेदी रेखाएँ)

Transversal line



THEOREM (प्रमेय) → (I)

Corresponding ratios are always same.



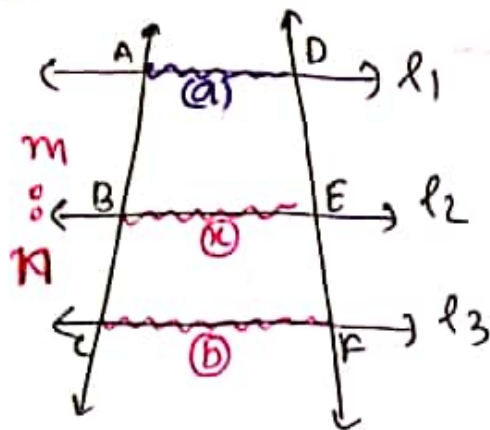
$l_1 \parallel l_2 \parallel l_3 \parallel l_4$
ratios are equal

$$\left\{ \begin{array}{l} \frac{a}{b} = \frac{d}{e} \\ \frac{b}{c} = \frac{e}{f} \end{array} \right\}$$

$$\text{OR } \left[\frac{a+b}{b+c} = \frac{d+e}{e+f} \right]$$

$$\text{OR } \left[\frac{b}{a+b+c} = \frac{e}{d+e+f} \right]$$

THEOREM प्रमेय - (II)



Given:

$$AD = a$$

$$CF = b$$

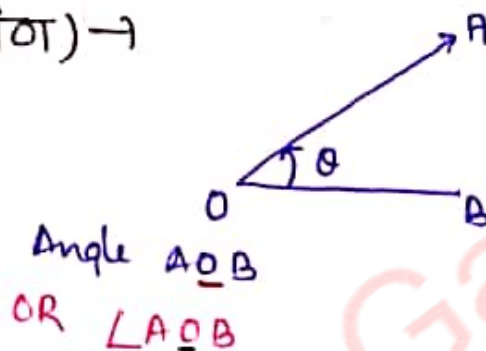
$$AB : BC = m : n$$

$$DE : EF = m : n$$

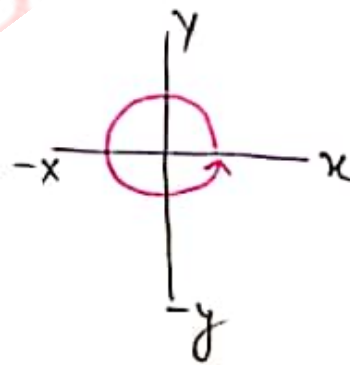
Find $x = ?$

$$x = \frac{axn + bxm}{m+n}$$

Angles (कोण) $\rightarrow$

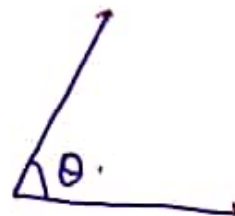


[+ive angle]



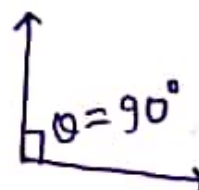
TYPES OF ANGLE (कोण के प्रकार)

1. Acute Angle (न्यून कोण)



$$0^\circ < \theta < 90^\circ$$

2. Right angle (समकोण)



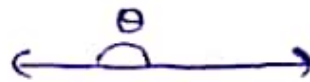
$$\theta = 90^\circ$$

3. Obtuse Angle (अधिक कोण)



$$90^\circ < \theta < 180^\circ$$

4. Straight angle (सुत कोण)



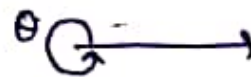
$$\theta = 180^\circ$$

5. Reflex Angle (वृद्ध कोण)



$$180^\circ < \theta < 360^\circ$$

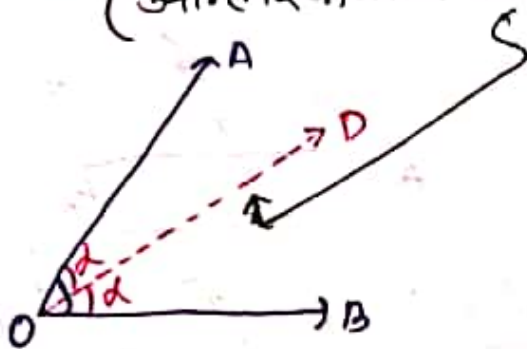
6. Complete angle (सम्पूर्ण कोण)



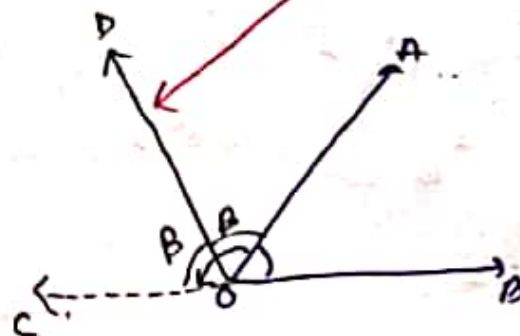
$$\theta = 360^\circ$$

Angle bisector :- (कोण समद्विभाजक)

1) Interior angle bisector (आन्तरिक कोण द्विभाजक)



2) Exterior angle bisector (बाह्य कोण समद्विभाजक)



$\angle AOC$
↓
[Exterior angle]

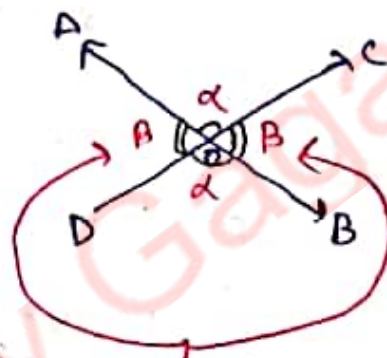


$$\alpha + \alpha + B + B = 180$$

$$\alpha + B = 90^\circ$$

Exterior Angle (आन्तरिक कोण) Exterior Angle (अन्तर कोण)

Vertically opposite angle (शीर्षा विमुख कोण)

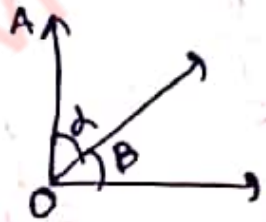


* Vertically opposite angle are same.

शीर्षा विमुखकोण आपस में बराबर होते हैं।

Vertically opposite angle

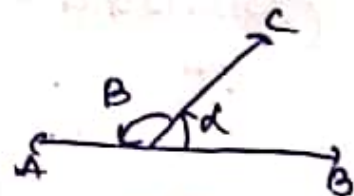
* Complementary angle (पूरक कोण)



$$\alpha + B = 90^\circ$$

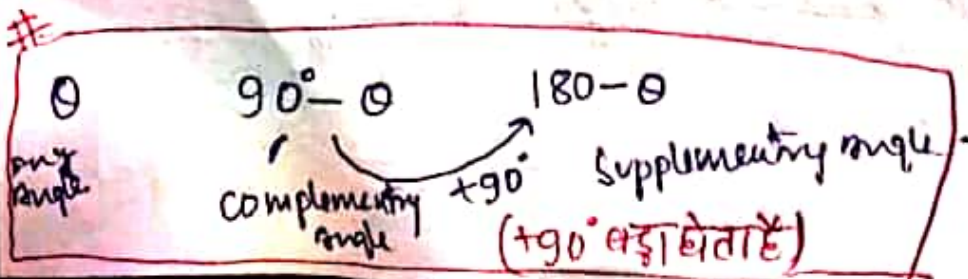
complementary to each other

* Supplementary angle (सम्पूरक कोण)



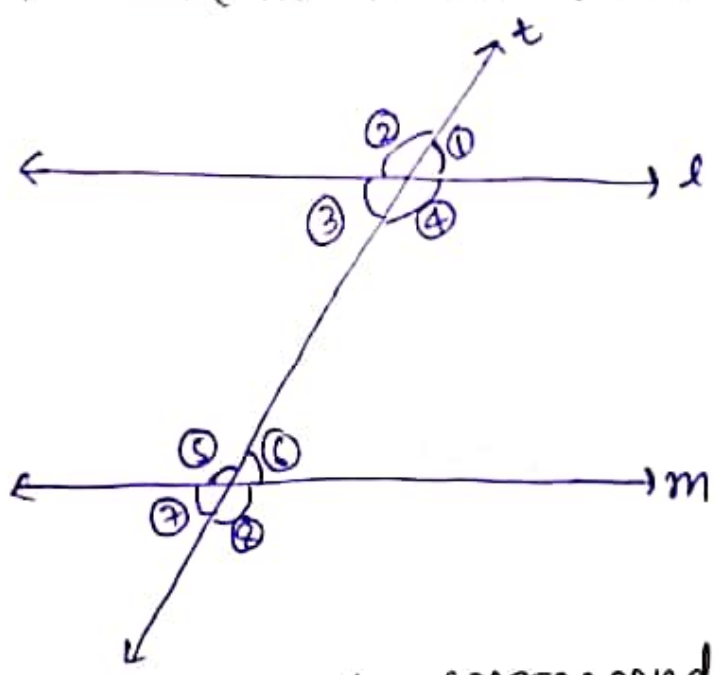
$$\alpha + B = 180^\circ$$

Supplementary to each other

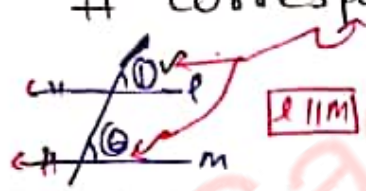


If "l" and "m" are two parallel lines and "t" is a transversal line then..

(यदि "l" और "m" दो समांतर रेखाओं और "t" तिर्यक रेखा हो तब) —

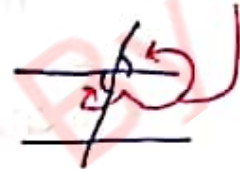


corresponding angle (संगत कोण)



$\angle 1 = \angle 5, \angle 4 = \angle 8$
 $\angle 2 = \angle 6, \angle 3 = \angle 7$

Vertically opposite angle (शीर्षाभिमुख कोण)



$\angle 1 = \angle 3, \angle 2 = \angle 4$
 $\angle 5 = \angle 7, \angle 6 = \angle 8$

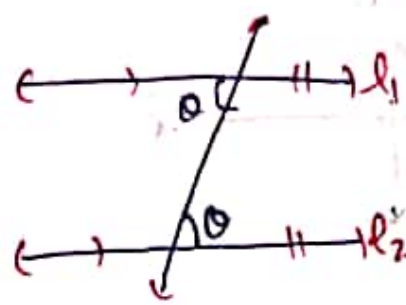
Alternate angle (रैकान्तर कोण)



$\angle 4 = \angle 5, \angle 3 = \angle 6$

→ Its vice versa are true →

eg.



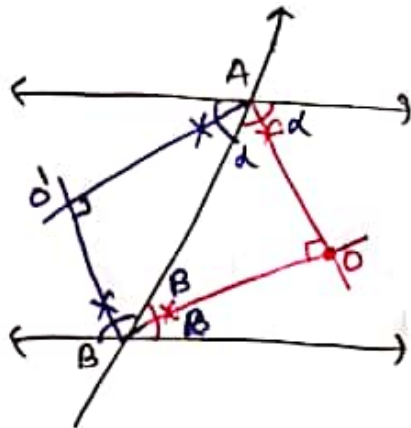
then $\boxed{l_1 \parallel l_2}$

(Alternate angle)

Interior Angle (आन्तरिक कोण)

$$[\angle 4 + \angle 6 = \angle 3 + \angle 5 = 180^\circ]$$

#



$$\angle AOB = 90^\circ$$

$$\angle A'O'B' = 90^\circ$$

PROOF $\Rightarrow \alpha + \alpha + \beta + \beta = 180$

$$\boxed{\alpha + \beta = 90^\circ}$$

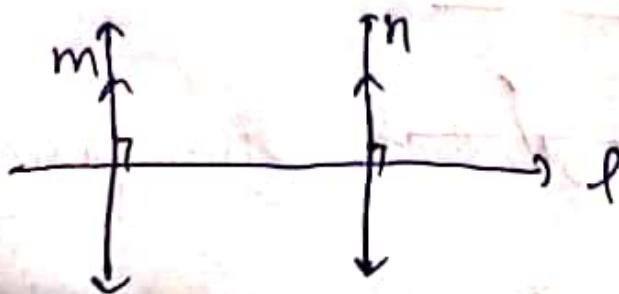
then $\angle O = 90^\circ$

Because

$\triangle AOB$ में दो Angle का सम 90° है तो विसर्ग कोण 90° होगा क्योंकि $\triangle$ के तीनों भुजाओं का योग 180° होता है।

Point * ~~if two lines are perpendicular to each other then they are parallel to each other~~

if $l \perp m$ and $l \perp n$ then $m \parallel n$

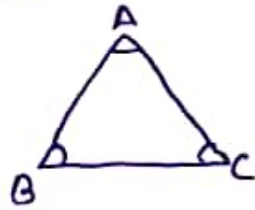


TYPES OF TRIANGLE

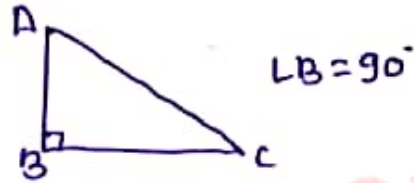
Δ - त्रिभुज

① Based upon Angles (कोणों के आधार पर)

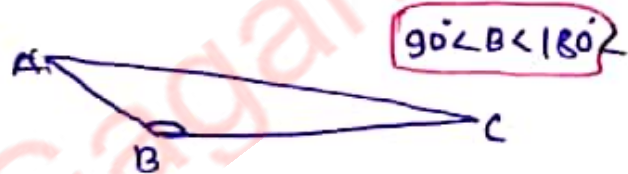
a) Acute angle triangle (न्यून कोण त्रिभुज)



b) Right angle Δ
(समकोण Δ)

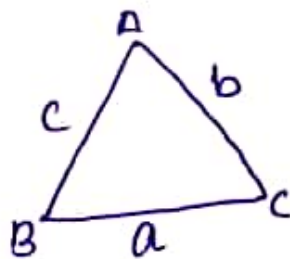


③ obtuse angle Δ
(अधिक कोण Δ)



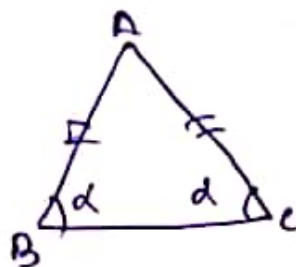
② Based upon Sides (भुजाओं के आधार पर)

① Scalene Δ
(विषमबाहु Δ)



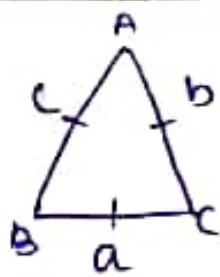
$$a \neq b \neq c$$
$$\angle A \neq \angle B \neq \angle C$$

② Isosceles Δ
(समबाहु Δ)

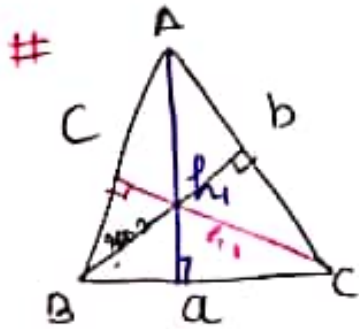


$$AB = AC$$
$$\angle B = \angle C$$

③ Equilateral Δ (समबाहु Δ)



$$\left[a = b = c \right. \\ \left. \angle A = \angle B = \angle C = 60^\circ \right]$$

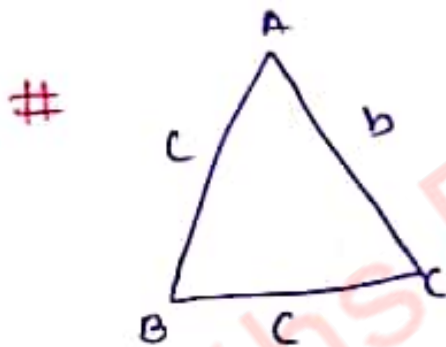


$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\left[\text{Area} = \frac{1}{2} a h_1 = \frac{1}{2} b h_2 = \frac{1}{2} c h_3 \right]$$

$$a h_1 = b h_2 = c h_3$$

$$h_1 : h_2 : h_3 = \frac{1}{a} : \frac{1}{b} : \frac{1}{c}$$



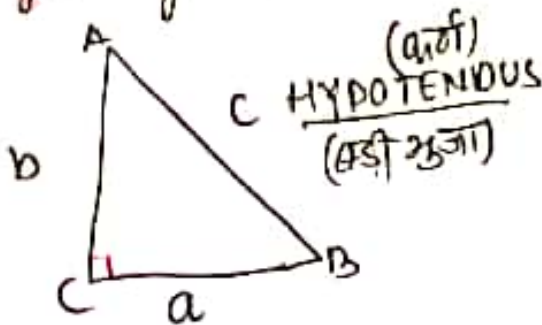
$$\text{Perimeter (परिमाप)} = a + b + c$$

$$\text{Semiperimeter (अर्धपरिमाप)} = \frac{a + b + c}{2}$$

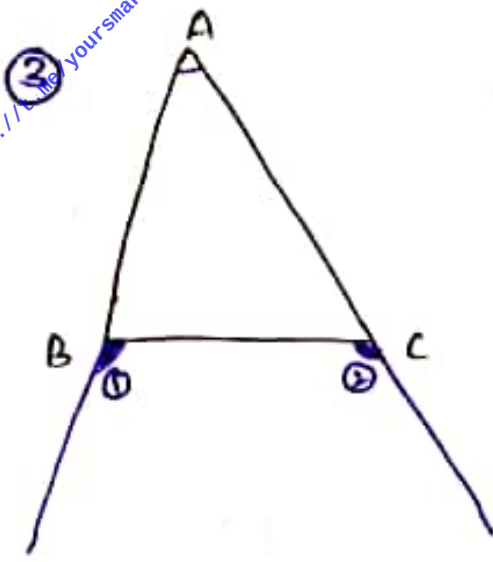
$$\text{Area of } \Delta = \sqrt{S(S-a)(S-b)(S-c)}$$

Heron's FORMULA

Right angle Δ -



$$c^2 = a^2 + b^2$$



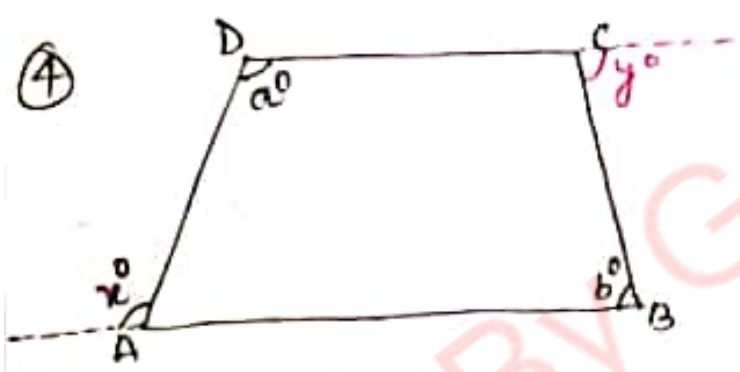
$$* \quad \textcircled{1} + \textcircled{2} = 180^\circ + A^\circ$$

PROOF -

$$\begin{aligned} \textcircled{1} + \textcircled{2} &= 180^\circ - LB + 180^\circ - LC \\ &= 360^\circ - (LB + LC) \end{aligned}$$

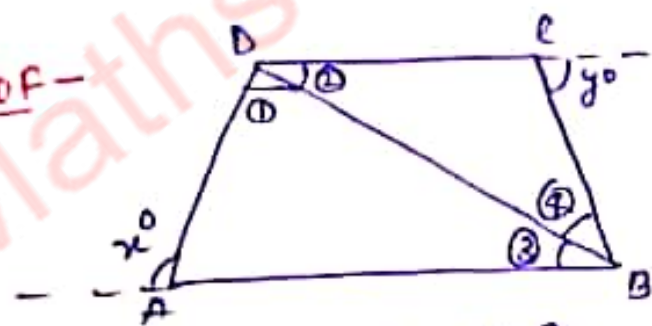
$$\therefore LB + LC = 180^\circ - LA$$

$$\textcircled{1} + \textcircled{2} = 180^\circ + LA^\circ$$



$$* \quad a^\circ + b^\circ = x^\circ + y^\circ$$

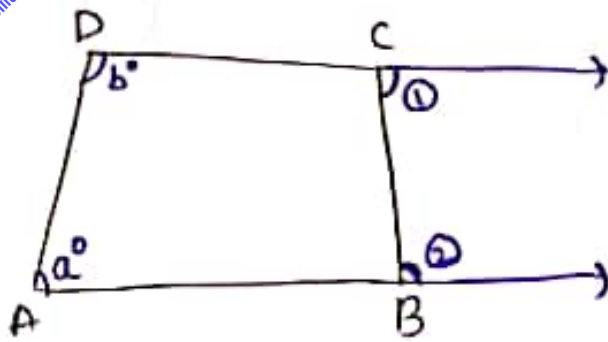
PROOF -



$$\begin{aligned} x^\circ &= \textcircled{1} + \textcircled{3} \\ y^\circ &= \textcircled{2} + \textcircled{4} \end{aligned}$$

$$* \quad x + y = a + b$$

5



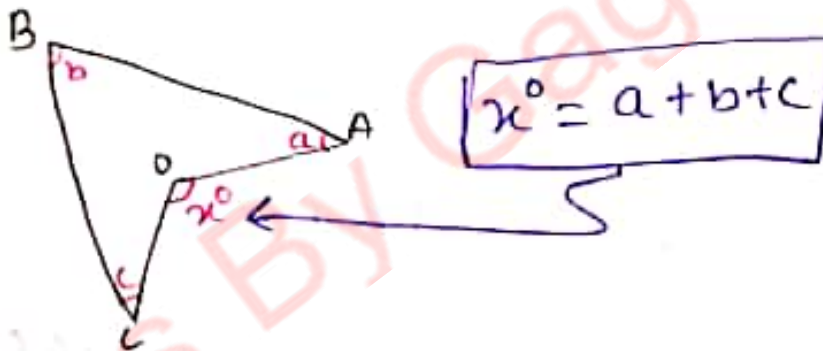
Result

$$\textcircled{1} + \textcircled{2} = a^\circ + b^\circ$$

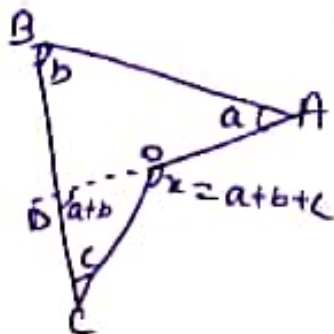
PROOF :-

$$\begin{aligned} \textcircled{1} + \textcircled{2} &= a^\circ + b^\circ \\ &\downarrow \quad \quad \downarrow \\ 180^\circ - \angle C + 180^\circ - \angle B \\ &= 360^\circ - (\angle B + \angle C) \\ &\downarrow \\ \angle A + \angle B + \angle C + \angle D - \angle B - \angle C \\ &= \boxed{a^\circ + b^\circ} \quad \# \end{aligned}$$

6

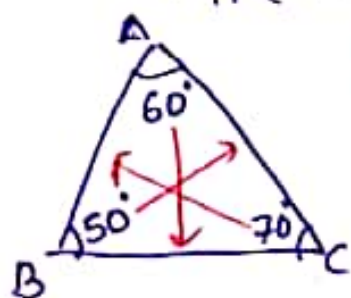


PROOF



[By sum of oppo interior angle is equal to its opposite exterior angle]

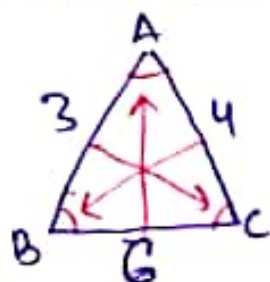
① किसी Δ में बड़े कोण के सामने की भुजा बड़ी तथा छोटे कोण के सामने की भुजा छोटी तथा बराबर कोण के सामने की भुजा बराबर होती है। -



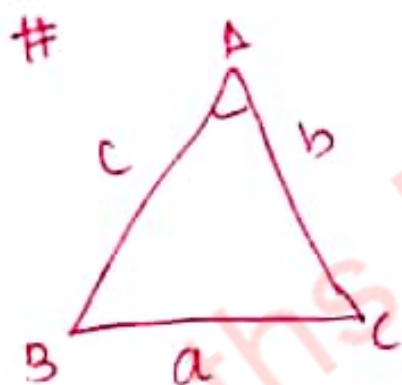
$$AB > BC > AC$$

(Sides)

② किसी Δ में बड़ी भुजा के सामने का कोण बड़ा छोटी भुजा के सामने का कोण छोटा व बराबर भुजा के सामने का कोण बराबर होता है।



$$LA > LB > LC$$



Angle $\downarrow$
 $LA = \text{largest} \rightarrow a = \text{largest}$
 $LB = \text{smallest} \Rightarrow b = \text{smallest}$

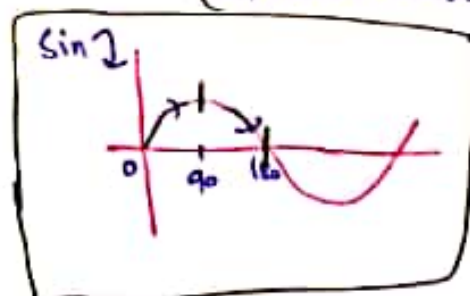
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k = (2R)$$

(Circumradius)

$$a = k \sin A$$

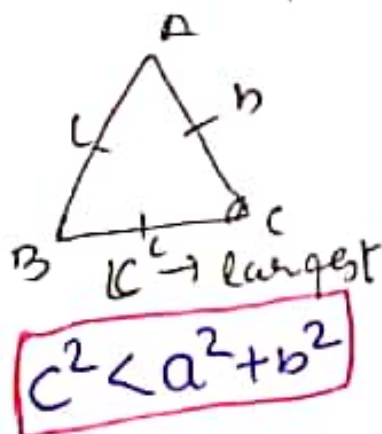
$$b = k \sin B$$

$$c = k \sin C$$

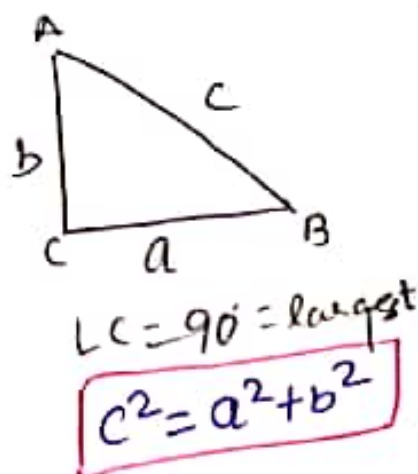


9th Δ $\rightarrow$

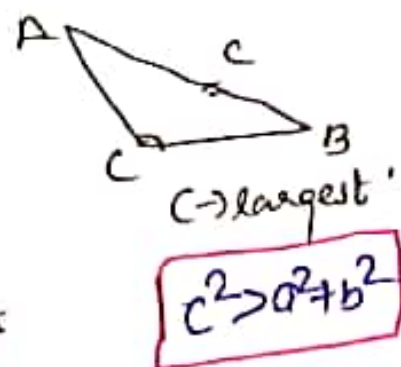
① Acute angle Δ



② Right angle Δ

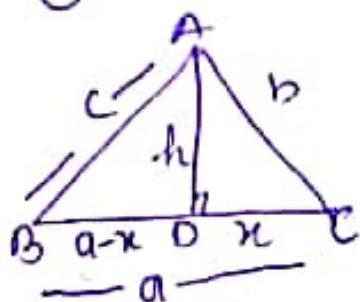


③ Obtuse angle Δ



PROOF :-

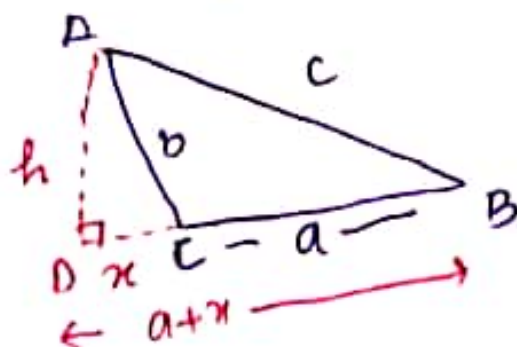
① Acute Angle (न्यूनकोण) $\Delta \Rightarrow$



$$\begin{aligned} \Rightarrow c^2 &= (a-x)^2 + h^2 \\ c^2 &= a^2 + x^2 - 2ax + h^2 \\ c^2 &= a^2 + b^2 - 2ax \\ \boxed{c^2 + 2ax} &= a^2 + b^2 \end{aligned}$$

$\Rightarrow c^2 < a^2 + b^2$
(always)

② Obtuse angle Δ
(अधिक कोण Δ)



$$\begin{aligned} c^2 &= h^2 + (a+x)^2 \\ c^2 &= h^2 + a^2 + x^2 + 2ax \\ c^2 &= a^2 + b^2 + 2ax \end{aligned}$$

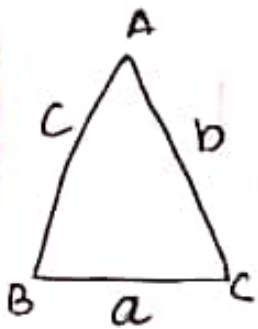
$$c^2 = a^2 + b^2 + 2ax$$

$\boxed{c^2 > a^2 + b^2}$

(add sth add कर रहे हैं)

* Sum of any two side of a Δ is always greater than Semiperimeter of Δ .

(Δ में किसी दो भुजाओं का योग ~~हिसी~~ अर्धपरिमाप से सदैव बड़ा होता है)



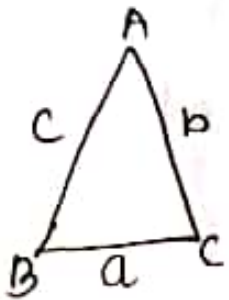
PROOF

$$\begin{aligned} a+b &> c \rightarrow a+b+\overset{\text{add}}{a+b} > c+\overset{\text{add}}{a+b} \quad \text{Perimeter} \\ b+c &> a \quad 2(a+b) > P \quad \therefore (P=a+b+c) \\ a+c &> b \quad a+b > \frac{P}{2} \quad (\because \frac{P}{2}=S) \end{aligned}$$

(Conclusion) $\rightarrow$ $\boxed{a+b > S}$ (Semiperimeter)

* Length of any side of a Δ is always less than its Semiperimeter.

(Δ की किसी भी भुजा की लम्बाई उसे अर्धपरिमाप से छोटी होती है)



$$\begin{aligned} a+b &> c \Rightarrow a+b+\overset{\text{Add}}{c} > c+\overset{\text{add}}{c} \\ P &> 2c \quad \therefore P=a+b+c \\ \frac{P}{2} &> c \quad (\frac{P}{2}=S) \end{aligned}$$

Conclusion $\rightarrow$ $\boxed{S > c}$ Semiperimeter

Similarly - $\begin{cases} S > a \\ S > b \end{cases}$

OR Δ की $\boxed{\text{largest side} < \frac{P}{2}}$

* Smallest side of a Δ is always less than or equal to one-third of Perimeter.

(Δ की छोटी भुजा $\frac{P}{3}$ से छोटी या उसके बराबर होगी)

* $\boxed{\text{Smallest side} \leq \frac{P}{3}}$

PROOF:-

$$a+b+c = P$$

$$\downarrow \downarrow \downarrow$$

$$c+c+c \leq P$$

$$c \leq \frac{P}{3}$$

$$a \geq b \geq c$$

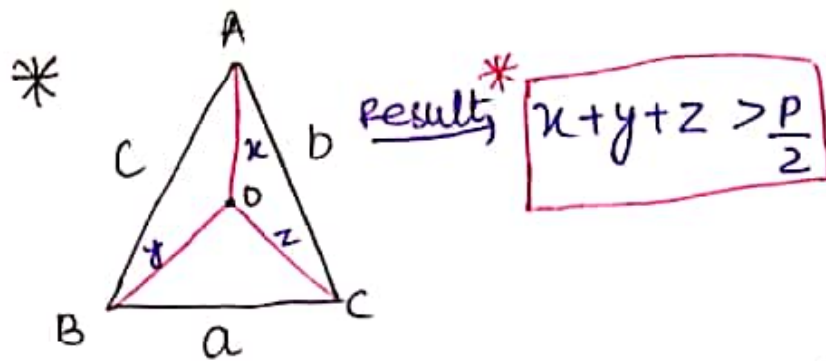
$$\downarrow \downarrow$$

$$a \leq \frac{P}{2} \quad c \leq \frac{P}{3}$$

eg. $5+4+3 = 12$

$$\downarrow \downarrow \downarrow$$

$$3 \ 3 \ 3 \leq 9$$



PROOF,

$$\begin{aligned} \text{In } \triangle OBA & \rightarrow x+y > c \quad \text{--- (1)} \\ \triangle OCB & \rightarrow y+z > a \quad \text{--- (2)} \\ \triangle OAC & \rightarrow x+z > b \quad \text{--- (3)} \end{aligned}$$

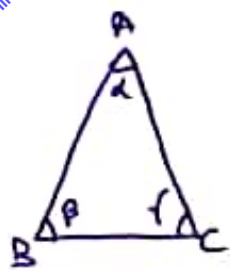
$$\text{Add} \rightarrow 2(x+y+z) > a+b+c \quad \therefore (P=a+b+c)$$

$$\#* \quad (x+y+z) > \frac{P}{2}$$

Number of possible Triangle with perimeter of Δ (Integer Side)

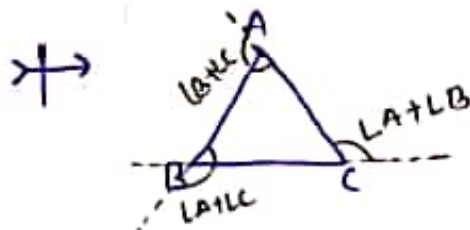
no. of possible Δ (Integer Side)	Perimeter	no. of possible Δ Integer side	Perimeter
X	1	2 ✓	← 7
X	2	1	← 8
(equilateral) ✓ (1)	3		
X	← 4		
1 ✓	← 5		
equilateral 1 ✓	← 6		
		Result →	

Triangle (त्रिभुज)



Triangle:— (त्रिभुज में)

- 3- vertices (शीर्ष)
 - 3- Interior angle / exterior angle (आन्तरिक व बाहरी कोण)
 - 3- Sides (भुजा), Perpendicular, height etc.
- $\angle A + \angle B + \angle C = 180$ * Triangle is smallest polygon.
(Δ सबसे छोटा बहुभुज है)



⇒ Exterior angle is equal to sum of its opposite angle.

[Δ बाहरी कोण आन्तरिक दो कोणों के सम के बराबर होता है]

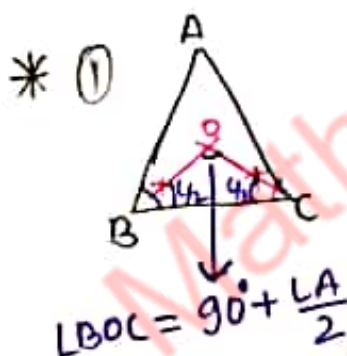
* किसी Δ के बाहरी कोण का योग 360° होता है।

Sum of → All Exterior Angle in a triangle is 360° .

Conclusion:- किसी बहुभुज में n भुजा हो तो -

* Sum of interior Angle = $(n-2) \times 180^\circ$

* Sum of exterior angle = 360°



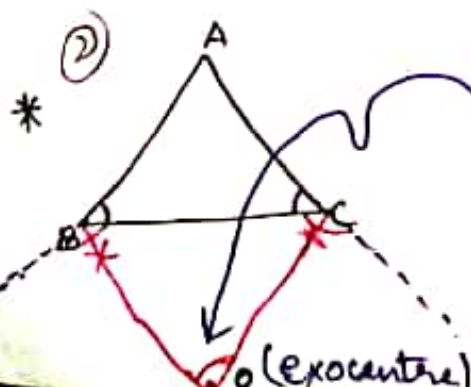
PROOF → $\angle BOC = 180^\circ - \left(\frac{B}{2} + \frac{C}{2}\right)$

$= 180^\circ - \left(\frac{B+C}{2}\right)$

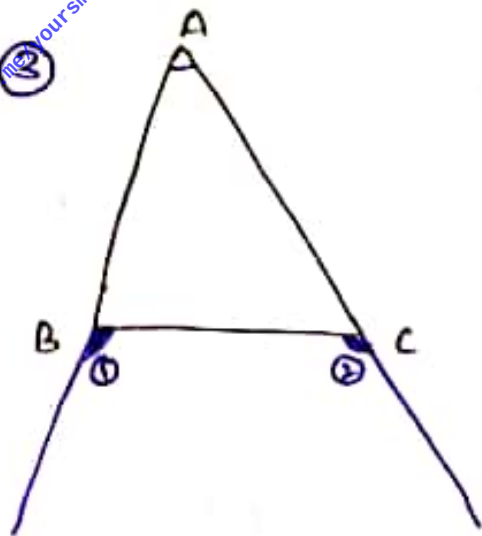
$= 180 - 90 + \frac{\angle A}{2}$

$\angle BOC = 90 + \frac{\angle A}{2}$

$\therefore B+C = (180-A)$



$\angle BOC = 90 - \frac{\angle A}{2}$



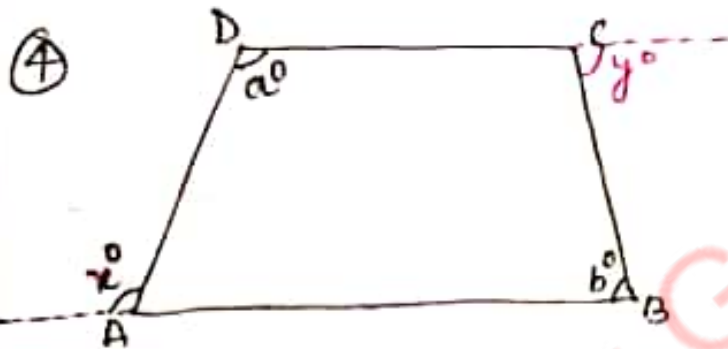
$$* \quad \textcircled{1} + \textcircled{2} = 180^\circ + A^\circ$$

PROOF -

$$\begin{aligned} \textcircled{1} + \textcircled{2} &= 180^\circ - LB + 180^\circ - LC \\ &= 360^\circ - (LB + LC) \end{aligned}$$

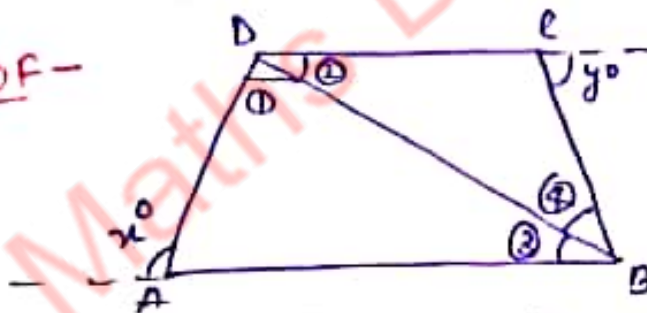
$$\therefore LB + LC = 180^\circ - LA$$

$$\textcircled{1} + \textcircled{2} = 180^\circ + LA^\circ$$



$$* \quad a^\circ + b^\circ = x^\circ + y^\circ$$

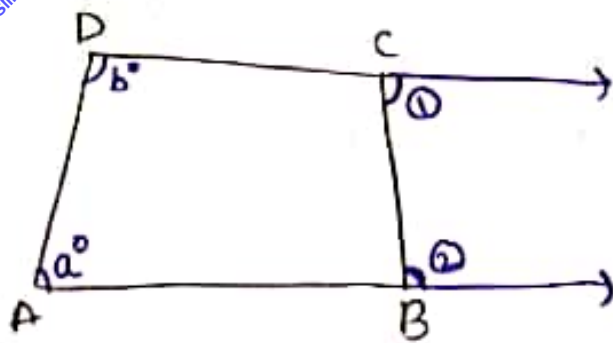
PROOF -



$$\begin{aligned} x^\circ &= \textcircled{1} + \textcircled{3} \\ y^\circ &= \textcircled{2} + \textcircled{4} \end{aligned}$$

$$* \quad x + y = a + b$$

⑤



Result

$$\textcircled{1} + \textcircled{2} = a^\circ + b^\circ$$

PROOF :-

$$\textcircled{1} + \textcircled{2} = a^\circ + b^\circ$$

$$180^\circ - \angle C + 180^\circ - \angle B$$

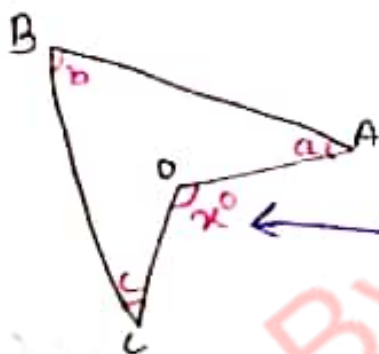
$$360^\circ - (\angle B + \angle C)$$

$$\downarrow$$

$$\angle A + \angle B + \angle C + \angle D - \angle B - \angle C$$

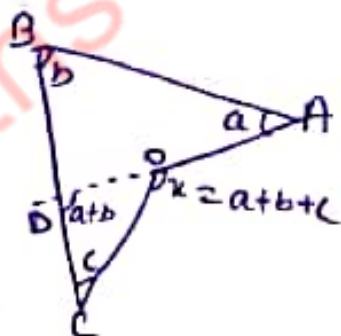
$$= \boxed{a^\circ + b^\circ} \quad \#$$

⑥



$$x^\circ = a + b + c$$

PROOF



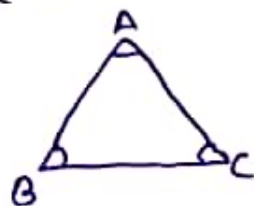
[By sum of oppo interior angle is equal to its opposite exterior angle]

TYPES OF TRIANGLE $\Rightarrow$

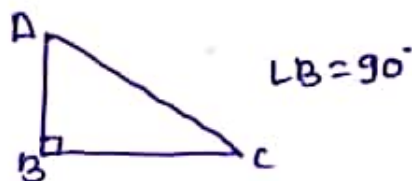
Δ - त्रिभुज

① Based upon Angles (कोणों के आधार पर)

a) Acute angle triangle (न्यून कोण त्रिभुज)



b) Right angle Δ
(समकोण Δ)

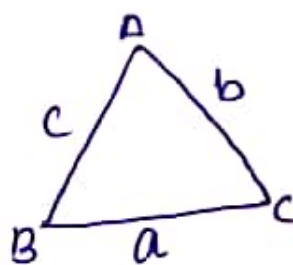


③ obtuse angle Δ
(अधिक कोण Δ)



② Based upon Sides (भुजाओं के आधार पर)

① Scalene Δ
(विक्रमसङ्कु Δ)



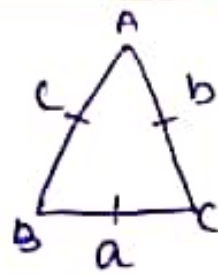
$$a \neq b \neq c$$
$$\angle A \neq \angle B \neq \angle C$$

② Isosceles Δ
(समबाहु Δ)

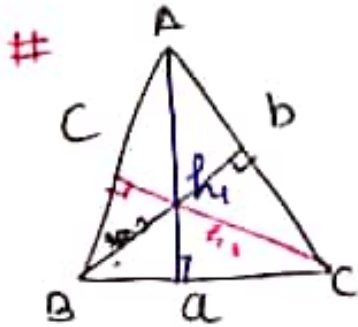


$$AB = AC$$
$$\angle B = \angle C$$

③ Equilateral Δ (समबाहु Δ)



$$\left[\begin{aligned} a &= b = c \\ \angle A &= \angle B = \angle C = 60^\circ \end{aligned} \right]$$



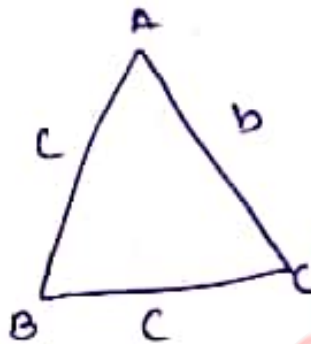
$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\left[\text{Area} = \frac{1}{2} a h_1 = \frac{1}{2} b h_2 = \frac{1}{2} c h_3 \right]$$

$$a h_1 = b h_2 = c h_3$$

$$\left[h_1 : h_2 : h_3 = \frac{1}{a} : \frac{1}{b} : \frac{1}{c} \right]$$

#

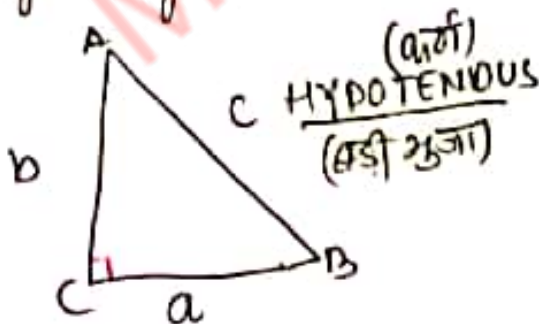


$$\text{Perimeter (परिमाप)} = a + b + c$$

$$\text{Semiperimeter (अर्धपरिमाप)} = \frac{a+b+c}{2} \quad (S)$$

$$\left[\begin{aligned} \text{Area of } \Delta &= \sqrt{S(S-a)(S-b)(S-c)} \\ \text{Heron's FORMULA} \end{aligned} \right]$$

Right angle Δ -



(कर्ण)
c HYPOTENOUS
(अड़ी भुजा)

$$\left[c^2 = a^2 + b^2 \right]$$

① किसी Δ में बड़े कोण के सामने की भुजा बड़ी तथा छोटे कोण के सामने की भुजा छोटी तथा बराबर कोण के सामने की भुजा बराबर होती है। -



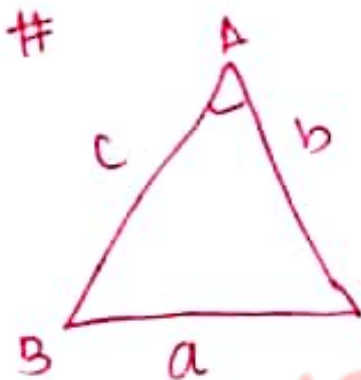
$$AB > BC > AC$$

(Sides)

② किसी Δ में बड़ी भुजा के सामने का कोण बड़ा छोटी भुजा के सामने का कोण छोटा व बराबर भुजा के सामने का कोण बराबर होता है।



$$\angle A > \angle B > \angle C$$



Angle $\downarrow$
 $\angle A = \text{largest} \rightarrow a = \text{largest}$
 $\angle B = \text{smallest} \Rightarrow b = \text{smallest}$

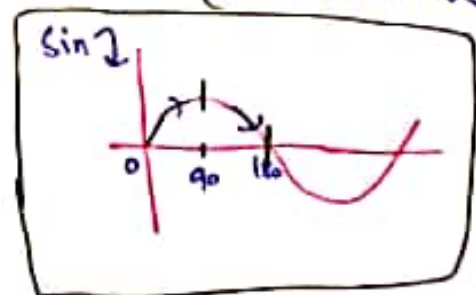
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k = (2R)$$

(Circumradius)

$$a = k \sin A$$

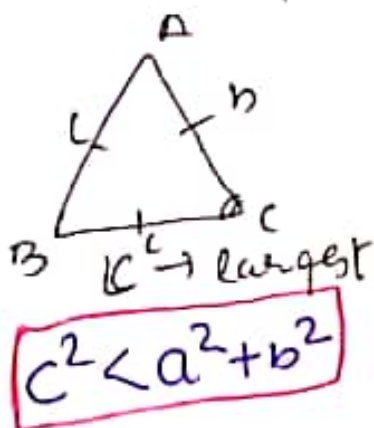
$$b = k \sin B$$

$$c = k \sin C$$

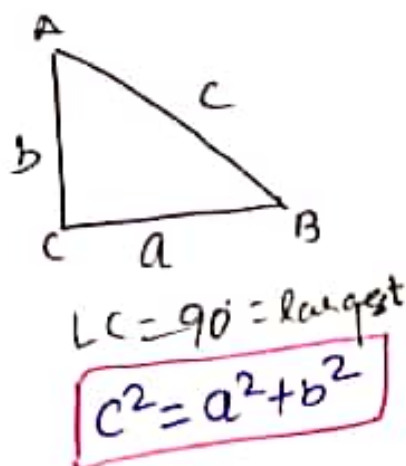


9th → Δ

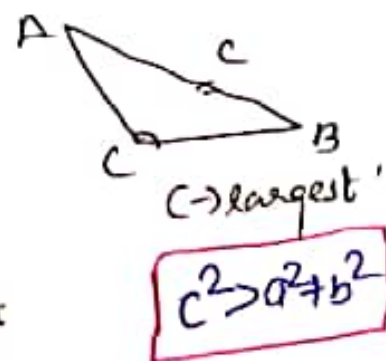
① Acute angle Δ



② Right angle Δ

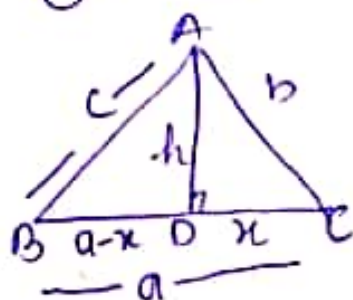


③ Obtuse angle Δ



PROOF :-

① Acute Angle (मनकोण) Δ ⇒



$$\Rightarrow c^2 = (a-x)^2 + h^2$$

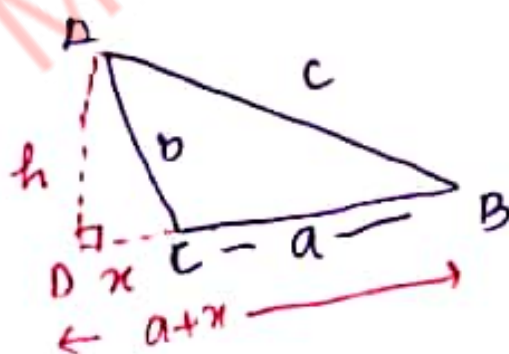
$$c^2 = a^2 + x^2 - 2ax + h^2$$

$$c^2 = a^2 + b^2 - 2ax$$

$$c^2 + 2ax = a^2 + b^2$$

↑
more add करने पर
 $\Rightarrow c^2 < a^2 + b^2$
(always)

② Obtuse angle Δ (अधिक कोण Δ)



$$c^2 = h^2 + (a+x)^2$$

$$c^2 = h^2 + a^2 + x^2 + 2ax$$

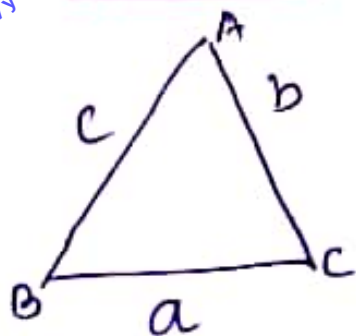
$$c^2 = a^2 + b^2 + 2ax$$

$$c^2 + a^2 = a^2 + b^2 + 2ax$$

$$c^2 > a^2 + b^2$$

(add
sth add
कराएँ)

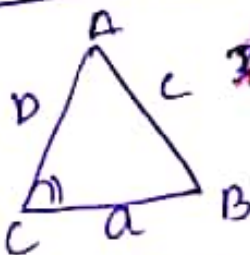
Cosine rule -



$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

By Cosine rule:-

Acute



$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

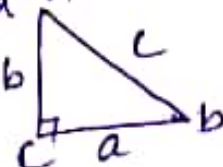
↑
+ve

$$+ve = a^2 + b^2 - c^2$$

$$c^2 + ve = a^2 + b^2$$

$$c^2 < a^2 + b^2$$

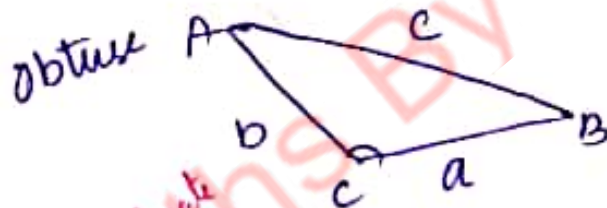
Right angle A



$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

↑
0

$$a^2 + b^2 = c^2$$



$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

↑
-ve

$$c^2 - k = a^2 + b^2$$

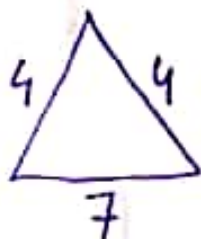
$$c^2 > a^2 + b^2$$

Properties of Δ -

① Sum of any sides of a triangle is always greater than third side.

(Δ की दो भुजाओं का योग तिसरी भुजा से सदैव बड़ा होता है)

eg:



$$4 + 4 > 7$$

✓ possible Δ



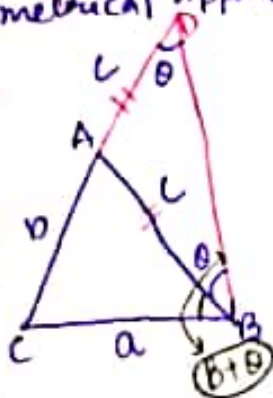
$$4 + 3 = 7$$

$$7 = 7$$

X Δ नहीं बनेगा
not possible.

Geometrical Approach -

②



ΔBCD में

$$LB + LD > LB$$

$$CD > BC$$

$$b + c > a \Rightarrow b > a - c$$

Similarly $\begin{cases} c + a > b \\ a + b > c \end{cases} \Rightarrow \begin{cases} c > b - a \\ a > b - c \end{cases}$

$b > |a - c|$ Always +ve value

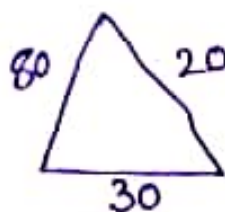
② difference of any side of a triangle is always less than the third side.

(Δ की दो भुजाओं का अन्तर तीसरी भुजा से छोटा होता है)



$$4 - 4 < 7$$

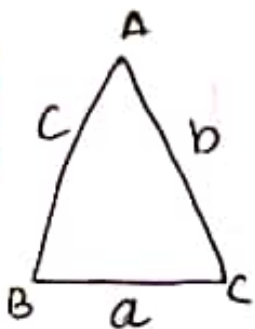
✓ possible.



X not possible.

* Sum of any two sides of a Δ is always greater than Semiperimeter of Δ .

(Δ में किसी दो भुजाओं का योग किसी अर्धपरिभाप से सदैव बड़ा होता है)



PROOF

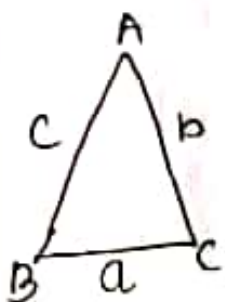
$$\begin{aligned} a+b &> c \rightarrow a+b + \overset{\text{add}}{a+b} > c + \overset{\text{add}}{a+b} \quad \text{Perimeter} \\ b+c &> a \quad 2(a+b) > P \quad \therefore (P = a+b+c) \\ a+c &> b \quad a+b > \frac{P}{2} \quad (\because \frac{P}{2} = S) \end{aligned}$$

(Conclusion) $\rightarrow a+b > S$

(Semiperimeter)

* Length of any side of a Δ is always less than its Semiperimeter.

(Δ का किसी भी भुजा की लम्बाई उसे अर्धपरिभाप से छोटी होती है)



$$\begin{aligned} a+b &> c \Rightarrow a+b + \overset{\text{add}}{c} > c + \overset{\text{add}}{c} \\ P &> 2c \quad \therefore P = a+b+c \\ \frac{P}{2} &> c \quad (\frac{P}{2} = S) \end{aligned}$$

Conclusion $\rightarrow S > c$

Semiperimeter

Similarly - $\begin{cases} S > a \\ S > b \end{cases}$

OR Δ की $\boxed{\text{largest side} < \frac{P}{2}}$

* Smallest side of a Δ is always less than or equal to one-third of Perimeter.

(Δ की छोटी भुजा $\frac{P}{3}$ से छोटी या उसके बराबर होती है)

* $\boxed{\text{Smallest side} \leq \frac{P}{3}}$

PROOF:-

$$a+b+c = P$$

Small $\downarrow \downarrow \downarrow$

$$c+c+c \leq P$$

$$c \leq \frac{P}{3}$$

$$a \geq b \geq c$$

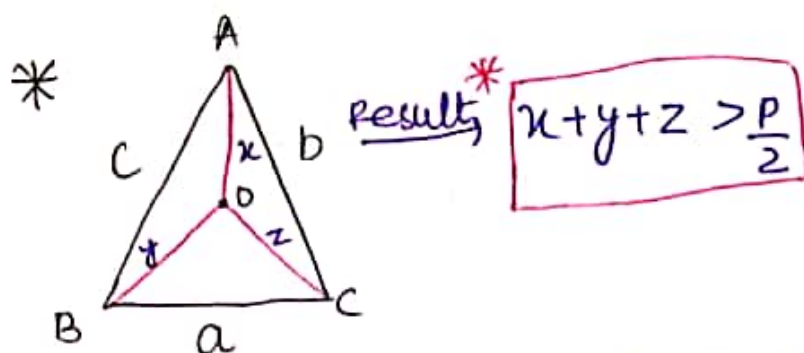
$$\downarrow \quad \downarrow$$

$$a \leq \frac{P}{2} \quad c \leq \frac{P}{3}$$

eg. $5+4+3 = 12$

Small $\downarrow \downarrow \downarrow$

$$3 \quad 3 \quad 3 \leq 9$$



PROOF,

$$\begin{aligned} \text{In } \triangle OBA & \rightarrow x+y > c \quad \text{--- (1)} \\ \triangle OCB & \rightarrow y+z > a \quad \text{--- (2)} \\ \triangle OAC & \rightarrow x+z > b \quad \text{--- (3)} \end{aligned}$$

Add $\rightarrow$

$$2(x+y+z) > a+b+c$$

$$\therefore (P=a+b+c)$$

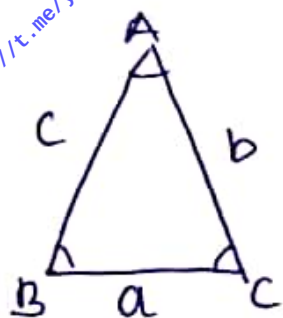
$$x+y+z > \frac{P}{2}$$

Number of possible Triangle with perimeter of Δ (Integer Side)

no. of possible Δ (Integer Side)	Perimeter	no. of possible Δ Integer side	Perimeter
X	1	2 $\checkmark$	7
X	2	1	8
(equilateral) Δ (1)	3		
X	4		
1 $\checkmark$	5		
equilateral Δ 1 $\checkmark$	6		

Result $\rightarrow$

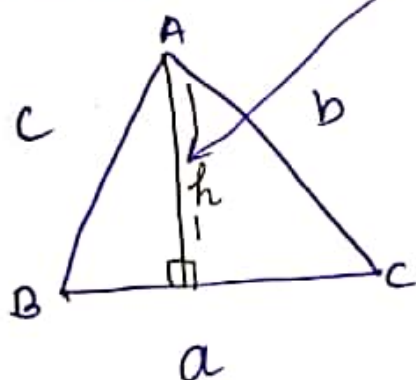
Sine rule:-



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k \text{ (Constant)}$$

$2R$
Circumradius
(परित्रिज्या)

PROOF →



(Perpendicular)

In $\triangle ABD$

$$\sin B = \frac{h}{c}$$

$$h = c \sin B$$

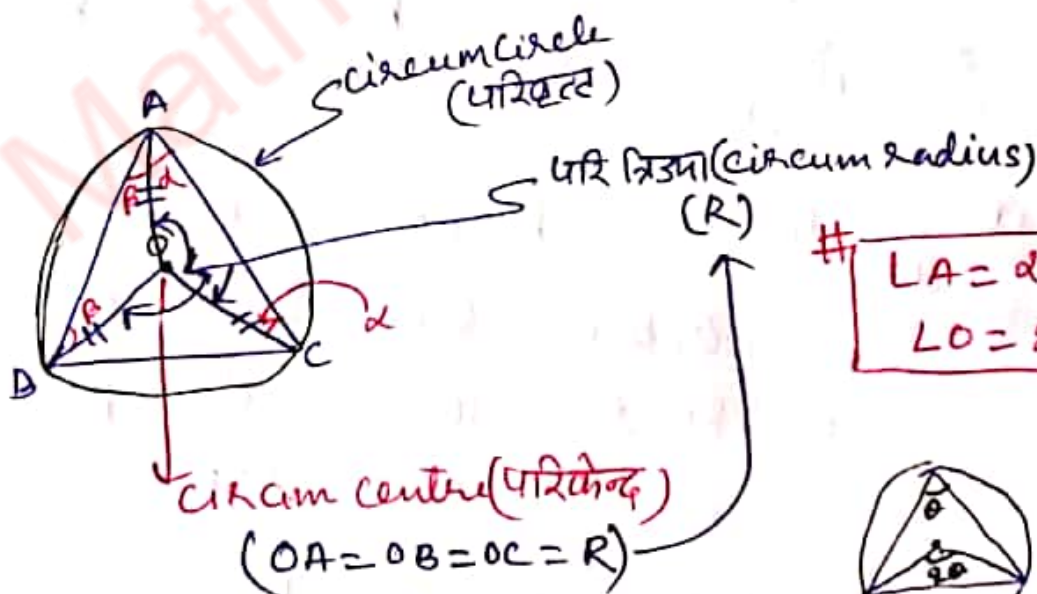
In $\triangle ADC$

$$\sin C = \frac{h}{b}$$

$$h = b \sin C$$

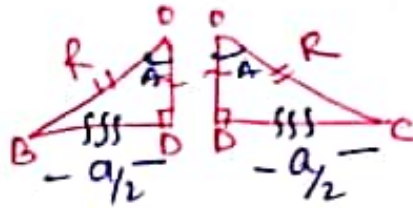
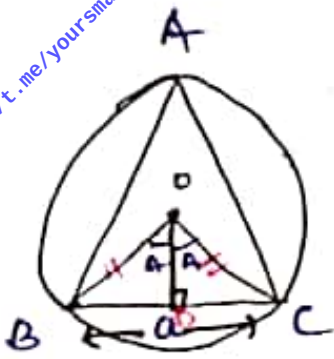
$$c \sin B = b \sin C$$

$$\frac{b}{\sin B} = \frac{c}{\sin C} = \frac{a}{\sin A} \checkmark$$



$$\begin{aligned} LA &= \alpha + B \\ LO &= 2(\alpha + B) \end{aligned}$$





We know that $\angle BOC = 2\angle A$

$$\sin A = \frac{a/2}{R}$$

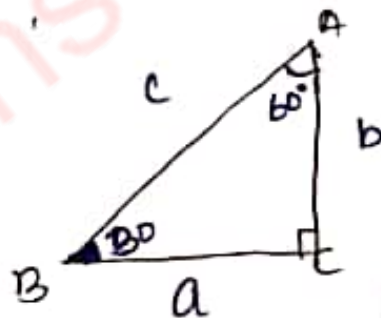
$$2R = \frac{a}{\sin A}$$

$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$$

↓
Circumradius
(परित्रिज्या)

* Standard Result:-

①-



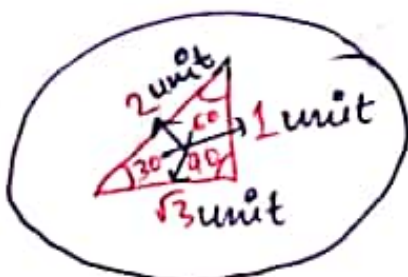
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$$

$$a = k \sin A, b = k \sin B, c = k \sin C$$

$$a : b : c = \sin A : \sin B : \sin C$$

$$a : b : c = \sin 30^\circ : \sin 60^\circ : \sin 90^\circ$$

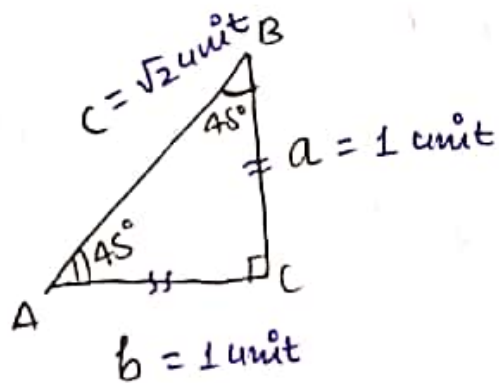
$$= \frac{1}{2} : \frac{\sqrt{3}}{2} : 1 \times 2$$



$$a : b : c = 1 : \sqrt{3} : 2$$

↓ 30° ↓ 60° ↓ 90°

2-



By Sine rule:-

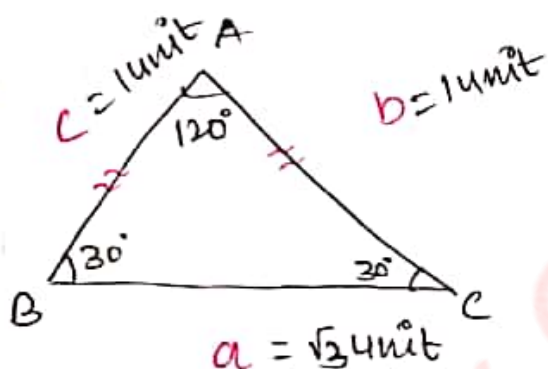
$$a:b:c = \sin 45^\circ : \sin 45^\circ : \sin 90^\circ$$

$$= \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{2}} : 1 \times \sqrt{2}$$

$$a:b:c = 1:1:\sqrt{2}$$

Arrows indicate the mapping: $1 \rightarrow 45^\circ$, $1 \rightarrow 45^\circ$, $\sqrt{2} \rightarrow 90^\circ$.

3-



By Sine rule,

$$a:b:c = \sin 120^\circ : \sin 30^\circ : \sin 30^\circ$$

$$= \frac{\sqrt{3}}{2} \times 2 : \frac{1}{2} \times 2 : \frac{1}{2} \times 2$$

$$a:b:c = \sqrt{3}:1:1$$

Arrows indicate the mapping: $\sqrt{3} \rightarrow 120^\circ$, $1 \rightarrow 30^\circ$, $1 \rightarrow 30^\circ$.

Imp:

[Result values]-

$$\# \sin 15^\circ = \cos 75^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$\# \sin 75^\circ = \cos 15^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$$

$$\# \tan 15^\circ = [\cot 75^\circ] = \frac{\sin 15^\circ}{\cos 15^\circ} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2-\sqrt{3}$$

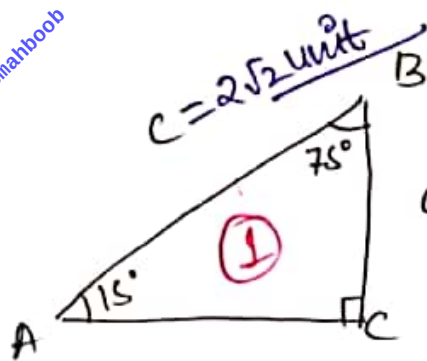
$$\# \tan 75^\circ = [\cot 15^\circ] = \frac{\sqrt{3}+1}{\sqrt{3}-1} = 2+\sqrt{3}$$

$$\# \textcircled{1} \sin 15^\circ = \sin(45^\circ - 30^\circ)$$

$$= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ$$

$$= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} = \frac{\sqrt{3}-1}{2\sqrt{2}} \checkmark$$

14-



$$a = (\sqrt{3} - 1) \text{ unit}$$

$$b = (\sqrt{3} + 1) \text{ unit}$$

Sine rule:-

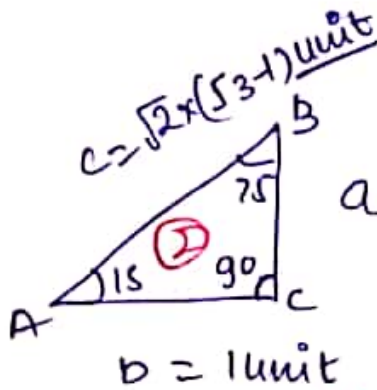
$$a : b : c = \sin 15^\circ : \sin 75^\circ : \sin 90^\circ$$

$\downarrow \quad \downarrow \quad \downarrow$
 $15 \quad 75 \quad 90$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}} : \frac{\sqrt{3} + 1}{2\sqrt{2}} : 1$$

$$a : b : c = \sqrt{3} - 1 : \sqrt{3} + 1 : 2\sqrt{2}$$

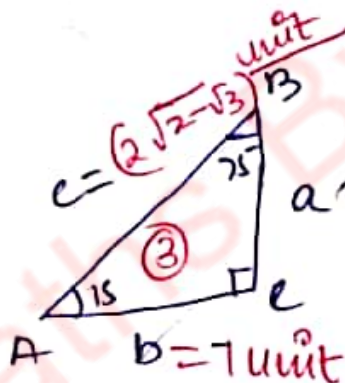
OR.



$$a = \frac{\sqrt{3} - 1}{\sqrt{3} + 1} \text{ unit}$$

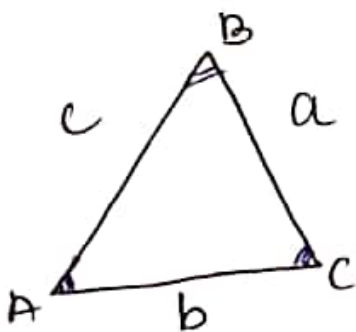
$$\tan 15^\circ = \frac{\sqrt{3} - 1}{\sqrt{3} + 1} \Leftarrow \frac{a}{b}$$

OR.



$$a = (2 - \sqrt{3}) \text{ unit} \quad \tan 15^\circ = \frac{2 - \sqrt{3}}{1} \Leftarrow \frac{a}{b}$$

Cosine rule :-



$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

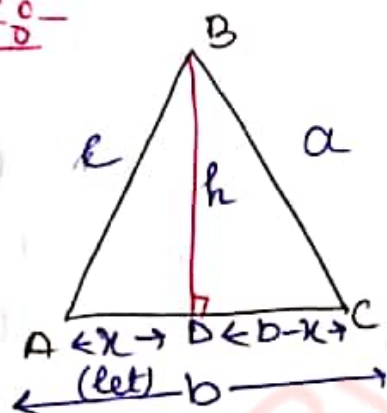
$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\left\{ \begin{array}{l} a^2 = b^2 + c^2 - 2bc \cos A \\ b^2 = a^2 + c^2 - 2ac \cos B \\ c^2 = a^2 + b^2 - 2ab \cos C \end{array} \right.$$

When one angle given ~~one~~ two side given.
Then by using cosine rule we
~~can~~ easily find Side Third.

PROOFS-



$$[h = h]$$

$$c^2 - x^2 = a^2 - (b-x)^2$$

$$(b-x)^2 - x^2 = a^2 - c^2$$

$$b^2 + \cancel{x^2} - 2bx - \cancel{x^2} = a^2 - c^2$$

$$\# \quad x = \frac{b^2 + c^2 - a^2}{2b}$$

In $\triangle ADB$

$$\Rightarrow \boxed{\cos A = \frac{x}{c}}$$

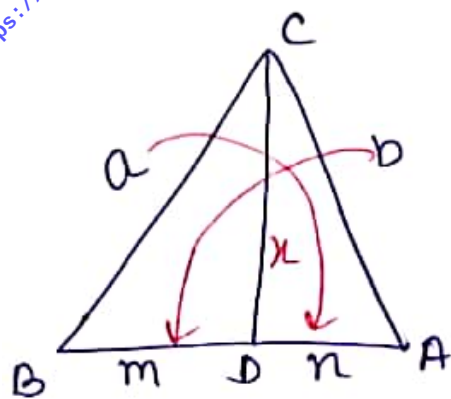
$$\frac{x}{c} = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\# \quad \boxed{\cos A = \frac{b^2 + c^2 - a^2}{2bc}}$$

Similarly: - we find:-

$$\boxed{\cos B, \cos C}$$

(Stewart) Stewart Theorem: $\rightarrow$

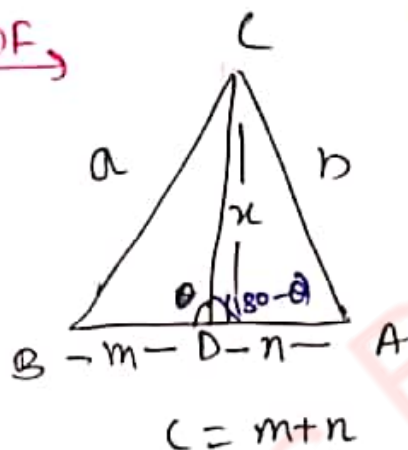


$$AB = c = m + n$$

$$a^2n + b^2m = x^2c + mnc$$

Stewart Theorem

PROOF $\rightarrow$



Using cosine rule $\rightarrow$

$$\cos(180 - \theta) = -\cos \theta$$

$$a^2 = x^2 + m^2 - 2mx \cos \theta \quad \text{--- (1)}$$

$$b^2 = x^2 + n^2 - 2nx \cos(180 - \theta)$$

$$b^2 = x^2 + n^2 + 2nx \cos \theta \quad \text{--- (2)}$$

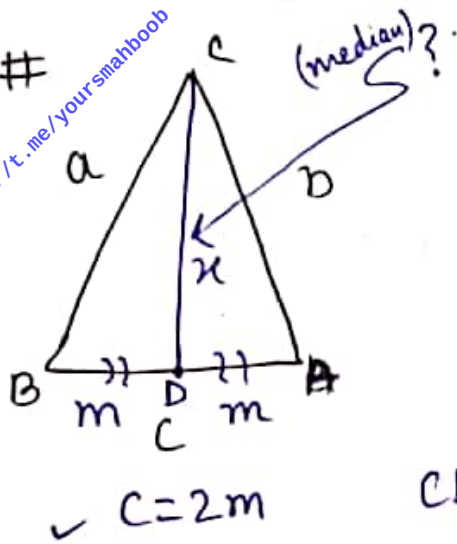
multiply (1) $\rightarrow n$
(2) $\rightarrow m$ And Add both.

$$a^2n + b^2m = x^2(n+m) + m^2n + n^2m$$

$$a^2n + b^2m = x^2(n+m) + mn(m+n)$$

$$m+n = c$$

$$a^2n + b^2m = x^2c + mnc$$



Ex

Apolloneus theorem:-

$$a^2 + b^2 = 2(x^2 + m^2)$$

$$m = \frac{c}{2}$$

OR $\left\{ a^2 + b^2 = 2\left(x^2 + \frac{c^2}{4}\right) \right\}$

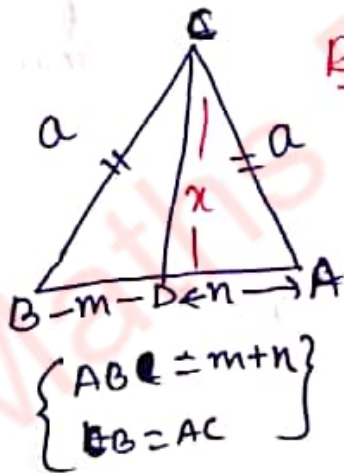
CD $\rightarrow$ median

Stewart theorem

$$a^2m + b^2m = x^2 \times 2m + m \times m \times 2m$$

$$a^2 + b^2 = 2(x^2 + m^2)$$

* Special Case in Isosceles $\Delta \rightarrow$



* Result $\rightarrow x^2 = a^2 - mn$

Stewart theorem

$$a^2n + a^2m = x^2c + mnc$$

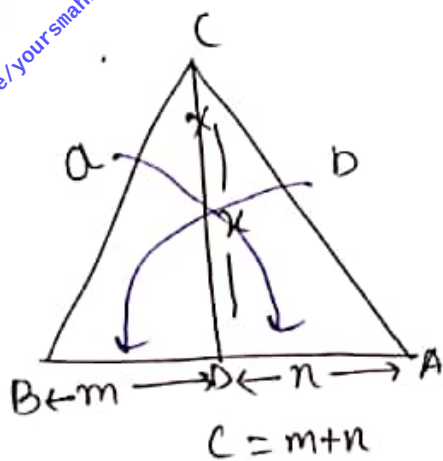
$$a^2x = x(x^2 + mn)$$

$$a^2 = x^2 + mn$$

$$x^2 = a^2 - mn$$

[length of angle bisector] $\rightarrow$ [Same result]

#



CD $\rightarrow$ angle bisector.

#

$$x^2 = axb - m \times n$$

$$\boxed{x^2 = ab - mn}$$

By using Stewart's theorem,

$$a^2n + b^2m = x^2c + mnc$$

$$\frac{a}{b} = \frac{m}{n} \left[\text{By interior angle bisector theorem} \right]$$

$$\boxed{an = bm}$$

$$\Rightarrow \underline{an} \times a + \underline{bm} \times b = c(x^2 + mn)$$

$$\Rightarrow bma + anb = c(x^2 + mn)$$

$$\Rightarrow ab(\cancel{m+n}) = \cancel{c}(x^2 + mn) \quad (\because m+n=c)$$

$$\boxed{x^2 = ab - mn}$$

SPECIAL TRIANGLE $\rightarrow$

(बड़ा कोण)

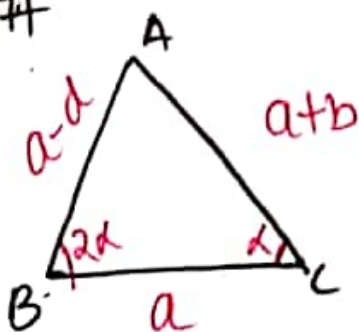
$\Rightarrow$ All side of a Δ are in A.P. largest angle is twice the smallest angle (दोय कोण)
(दो गुना)

Side are

(Always) Side ratio

$$\rightarrow \boxed{4 : 5 : 6}$$

#



(a-d)
 $\downarrow$
Smallest

(a) (a+d)
 $\downarrow$
largest

$\rightarrow$ Using sine rule $\rightarrow$

By using Sine rule:-

$$\frac{a+d}{a-d} = \frac{\sin 2\alpha}{\sin \alpha} = \frac{2\cancel{\sin \alpha} \cdot \cos \alpha}{\cancel{\sin \alpha}}$$

$$\boxed{\cos \alpha = \frac{a+d}{2(a-d)}} \quad \text{--- (1)}$$

By using Cosine rule-

$$\cos \alpha = \frac{a^2 + (a+d)^2 - (a-d)^2}{2a(a+d)}$$

OR.

$$\cos \alpha = \frac{a+4d}{2(a+d)} \quad \text{--- (2)}$$

$$\textcircled{1} = \textcircled{2}$$

$$\frac{a+b}{2(a-d)} = \frac{a+4d}{2(a+d)}$$

$$a^2 + d^2 + 2ad = a^2 + 3ad - 4d^2$$

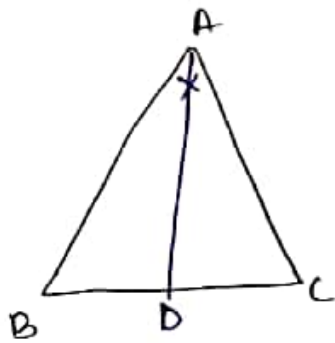
$$5d^2 = ad$$

$$\frac{a}{d} = \frac{5}{1}$$

$$(a-d) : a : (a+d) \quad \textcircled{\sim}$$

$$* \boxed{4 : 5 : 6}$$

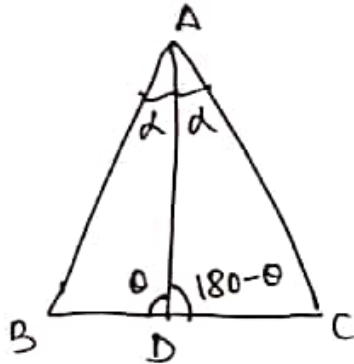
Interior angle bisector theorem:- (आन्तरिक)



AD → angle bisector of angle A

$$\boxed{\frac{AB}{AC} = \frac{BD}{DC}} \quad \#$$

PROOF



Sine rule:- ①

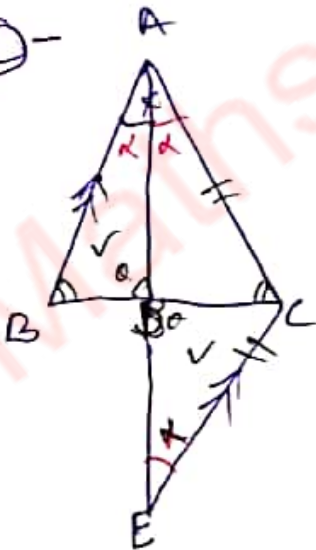
$$\frac{AB}{BD} = \frac{\sin \theta}{\sin \alpha}$$

$$\frac{AC}{DC} = \frac{\sin(180-\theta)}{\sin \alpha} = \frac{\sin \theta}{\sin \alpha}$$

$$\frac{AB}{BD} = \frac{AC}{DC}$$

OR. $\boxed{\frac{AB}{AC} = \frac{BD}{DC}}$ (Interior angle bisector theorem)

OR. ②-

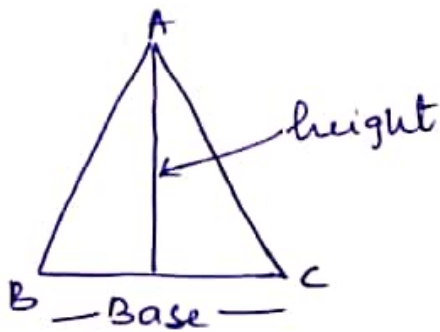


$$\frac{AB}{EC} = \frac{BD}{DC}$$

$$\# \boxed{EC = AC}$$

$$* \boxed{\frac{AB}{AC} = \frac{BD}{DC}}$$

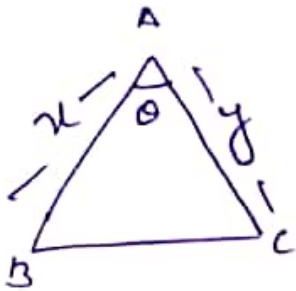
Area of Triangle!—



#①

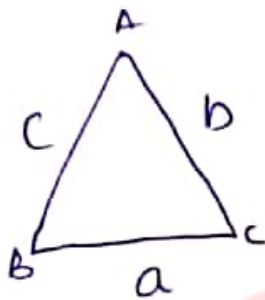
$$\Delta = \frac{1}{2} \times \text{Base} \times \text{height}$$

Δ - Area of triangle



#②

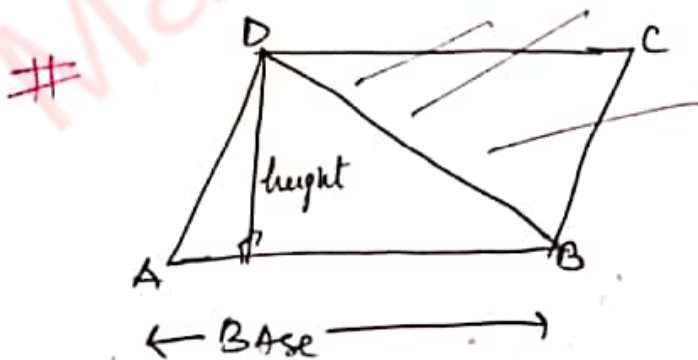
$$\Delta = \frac{1}{2} xy \sin \theta$$



#③

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

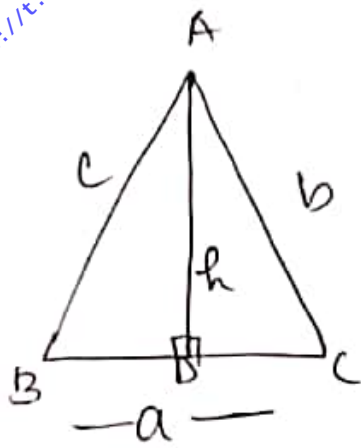
$s \rightarrow$ semiperimeter (अर्धपरिमाप)
(Heron's formula) $\left(\frac{a+b+c}{2}\right)$



Area of parallelogram Area = Base $\times$ height

Area of Δ Area = $\frac{1}{2} \times \text{Base} \times \text{height}$

आप



$$\sin B = \frac{h}{c} \Rightarrow h = c \sin B$$

$$\sin C = \frac{h}{b} \Rightarrow h = b \sin C$$

$$\left\{ \begin{aligned} \Delta &= \frac{1}{2} \times a \times h \\ \Delta &= \frac{1}{2} ac \sin B \\ \text{Similarly } \Delta &= \frac{1}{2} ab \sin C \\ \Delta &= \frac{1}{2} bc \sin A \end{aligned} \right.$$



$$\boxed{\text{area} = \frac{1}{2} xy \sin \theta}$$

PROOF HERON'S FORMULA



$$\Delta = \frac{1}{2} bc \sin A$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} \Rightarrow \sin A = \sqrt{1 - \left(\frac{b^2 + c^2 - a^2}{2bc} \right)^2}$$

$$\Delta = \frac{1}{2} \times bc \times \sqrt{1 - \left(\frac{b^2 + c^2 - a^2}{2bc} \right)^2}$$

$$= \frac{1}{2} \times \cancel{bc} \times \frac{1}{\cancel{2bc}} \times \sqrt{(2bc)^2 - (b^2 + c^2 - a^2)^2}$$

$$= \frac{1}{4} \sqrt{(2bc + b^2 + c^2 - a^2) \times (2bc - b^2 - c^2 + a^2)}$$

$$\Delta = \frac{1}{4} \sqrt{\{(b+c)^2 - a^2\} \times \{a^2 - (b+c)^2\}}$$

$$\Delta = \frac{1}{4} \sqrt{(b+c+a)(b+c-a) \times (a+b-c)(a-b+c)}$$

$$\Delta = \sqrt{\frac{a+b+c}{2} \times \frac{b+c-a}{2} \times \frac{a+b-c}{2} \times \frac{a-b+c}{2}}$$

we know that $\left[s = \frac{a+b+c}{2} \right]$

*

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

(इति सिद्धम्)
(Proved)

☺

* # Similarity (समरूपता)

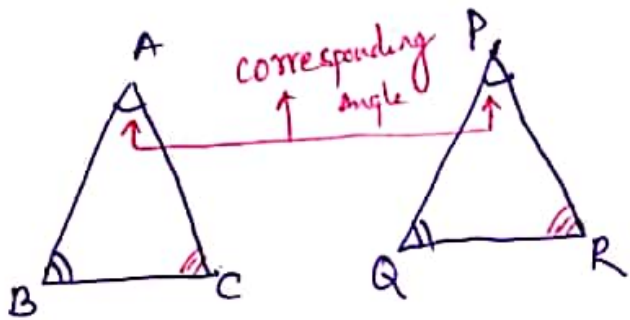
* Similarity of triangles $\rightarrow$ gbs. ~~Size~~ Size may be different or not but their shape is same.
[आकार $\rightarrow$ समान होता है।]

How to Find: —

In Triangle: —

A - Angle

(I) A - A - A Similarity! —
OR A - A Similarity! —

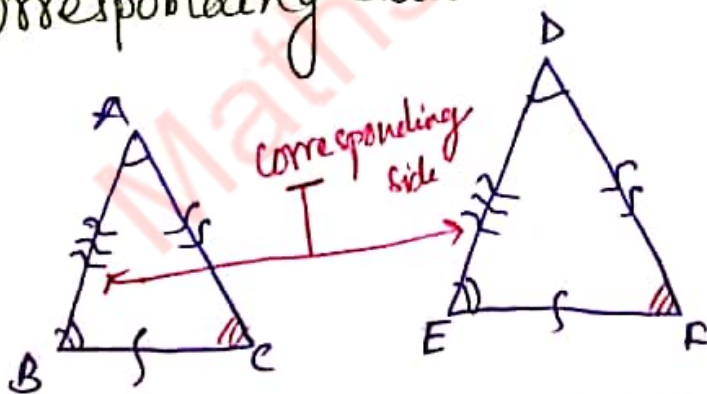


gbs $LA = LP$
 $LB = LQ$
then we know —
 $LC = LR$

$$\triangle ABC \sim \triangle PQR$$

(Similarity Indication)
(Identification)

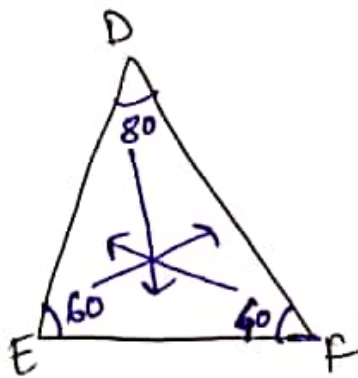
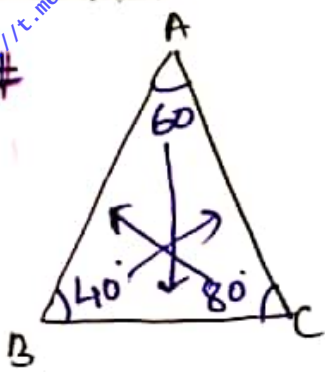
Corresponding Side: —



* Equal Angles के सामने की Side. Corresponding Side होती है।

$$\left\{ \frac{BC}{EF} = \frac{AC}{DF} = \frac{AB}{DE} \right\} = K \text{ (Scaling Factor)} \\ \text{OR (multiplying factor)}$$

Example:-

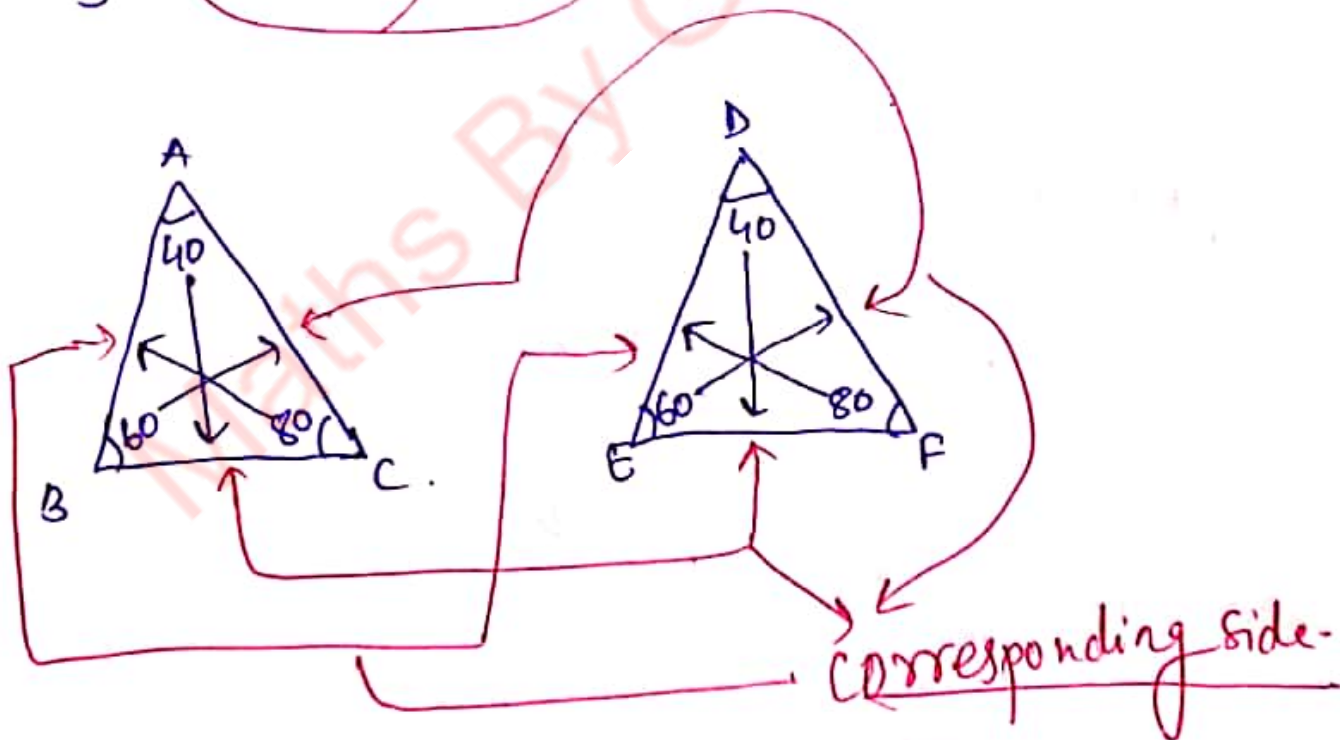
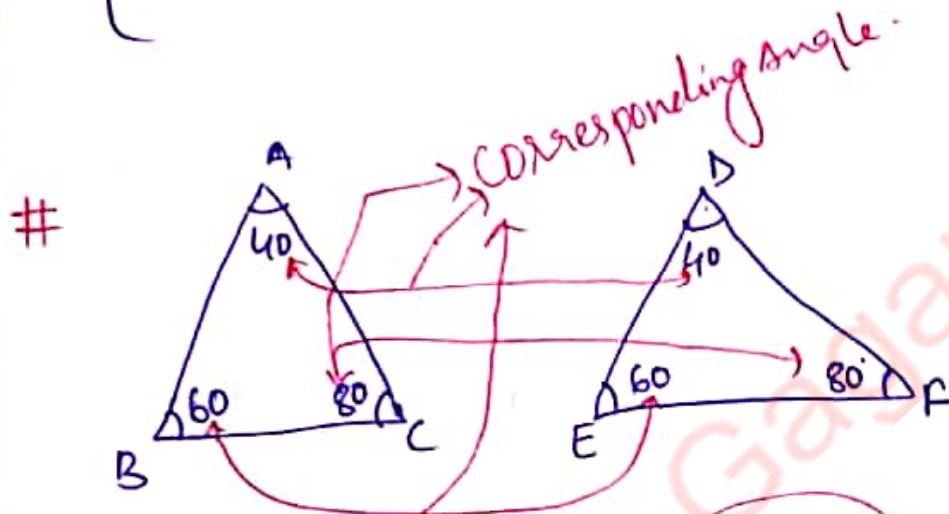


[Angle के सामने वाली भुजा संगत भुजा होगी]

$$\triangle ABC \sim \triangle FED \quad [LA=LE, LB=LF, LC=LD]$$

Its ratio is same [In Similarity]

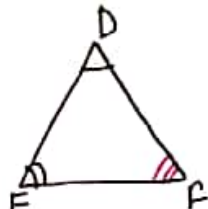
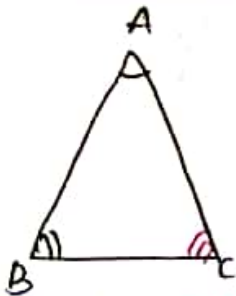
$$\left\{ \frac{AB}{FE} = \frac{BC}{FD} = \frac{AC}{ED} = k \right\} \text{[Scaling factor]}$$



Then - $\triangle ABC \sim \triangle DEF$

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = k \text{ (scaling factor)}$$

Result:-



of $\triangle ABC \sim \triangle DEF$

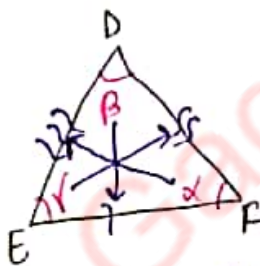
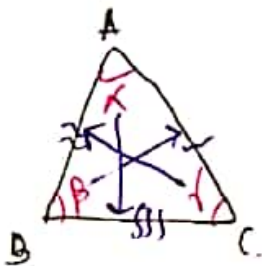
Where $\rightarrow$ $\left\{ \begin{array}{l} r \rightarrow \text{Inradius of } \triangle \\ R \rightarrow \text{Circumradius of } \triangle \\ m \rightarrow \text{medians of } \triangle \\ h \rightarrow \text{Altitude of } \triangle \\ a \rightarrow \text{Angle bisector of } \triangle \\ A \rightarrow \text{Area of } \triangle \end{array} \right.$

then:

$$\frac{BC}{EF} = \frac{AC}{DF} = \frac{AB}{DE} = \frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle DEF} = \frac{r_1}{r_2} = \frac{R_1}{R_2} = \frac{h_1}{h_2} = \frac{m_1}{m_2} = \frac{a_1}{a_2} = k \text{ (Scaling factor)}$$

$$= \left(\frac{A_1}{A_2} \right)^{\frac{1}{2}}$$

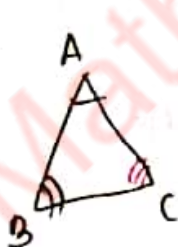
#



$\triangle ABC \sim \triangle FDE$

$$\frac{BC}{DE} = \frac{AC}{EF} = \frac{AB}{DF}$$

#



$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle DEF} = \left(\frac{BC}{EF} \right)^2 = \left(\frac{AC}{DF} \right)^2 = \left(\frac{AB}{DE} \right)^2 = k^2$$



$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle DEF} = \frac{\frac{1}{2} \times BC \times h_1}{\frac{1}{2} \times EF \times h_2}$$

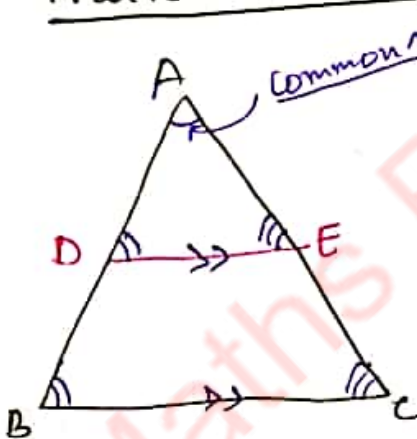
$$= \frac{BC}{EF} \times \frac{BC}{EF} = \left(\frac{BC}{EF}\right)^2$$

We know that in similar

$$\text{where } \frac{h_1}{h_2} = \frac{BC}{EF}$$

Corresponding length — $\times K$
 Area — $\times K^2$
 Volume — $\times K^3$

Thales theorem:-



$$\frac{AD}{BD} = \frac{AE}{EC}$$

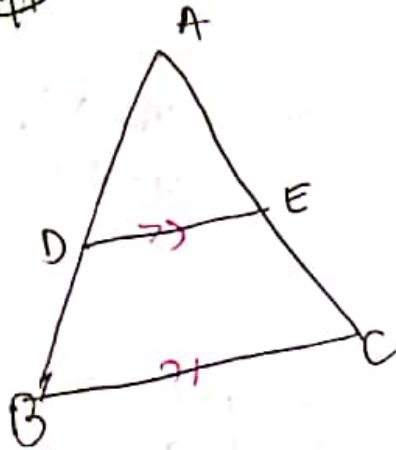
$$\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$$

By similarity

$$\frac{AB}{AD} = \frac{AC}{AE} - 1$$

$$\frac{AB - AD}{AD} = \frac{AC - AE}{AE} \Rightarrow \frac{BD}{AD} = \frac{EC}{AE} \Rightarrow \frac{AD}{BD} = \frac{AE}{EC}$$

#



$$\left\{ \begin{array}{l} \frac{AD}{BD} = \frac{AE}{EC} \text{ (given)} \\ \text{OR } \frac{AD}{AB} = \frac{AE}{AC} \text{ (given)} \end{array} \right\}$$

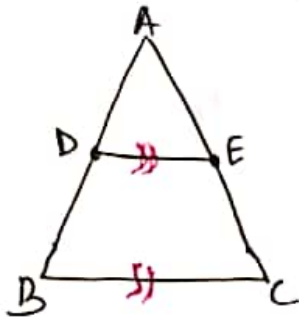


Then,

$$\boxed{DE \parallel BC}$$

(Conversion is also true)

Mid point theorem :-



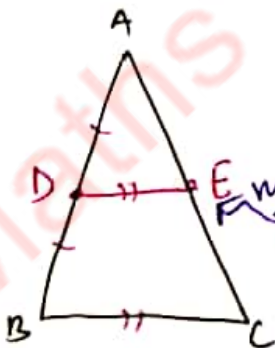
D - mid point of AB (given)
E - mid point of AC

then

$$\boxed{\begin{array}{l} DE \parallel BC \\ DE = \frac{BC}{2} \end{array}}$$

Result #

Convergence of mid point theorem: →

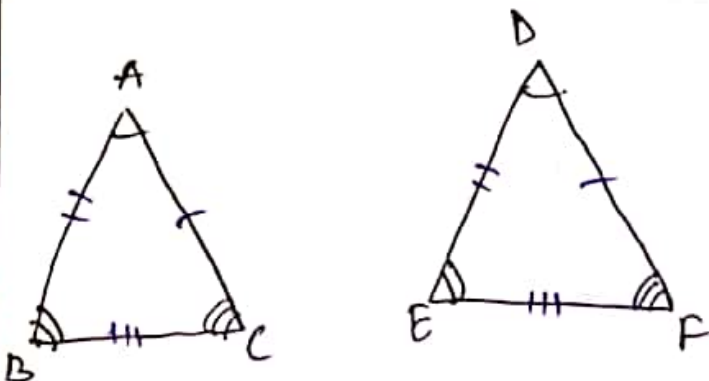


D is mid point of AB D से BC के
parallel line खिंचते हैं।

∴ E will be the mid point of AC

Congruency of triangle:-

[Size and Shape $\rightarrow$ Same] then the Δ is congruent.



Given-

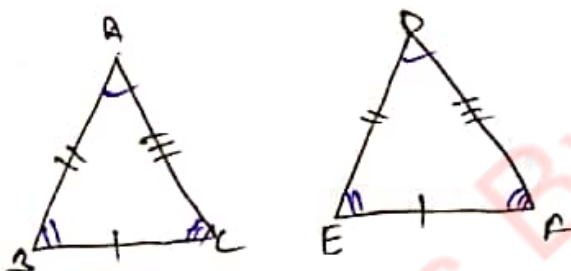
$$\begin{cases} \angle A = \angle D \\ \angle B = \angle E \\ \angle C = \angle F \end{cases}$$

Then $AB = DE$
 $BC = EF$
 $AC = DF$

$$\triangle ABC \cong \triangle DEF$$

(Congruency Identification)

① SSS - (Side Side Side)
 भुजा - भुजा - भुजा



$$\begin{aligned} AB &= DE \\ BC &= EF \\ AC &= DF \end{aligned}$$

Then $\triangle ABC \cong \triangle DEF$

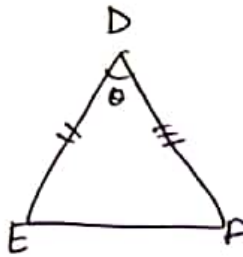
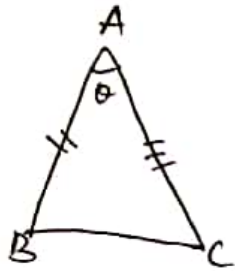
$$\text{Area of } \triangle ABC = \text{Area of } \triangle DEF$$

$$\frac{1}{2} \times AB \times AC \sin A = \frac{1}{2} \times DE \times DF \sin D$$

$$\sin A = \sin D$$

$$\begin{cases} \angle A = \angle D \\ \angle B = \angle E \\ \angle C = \angle F \end{cases}$$

② SAS (Side-Angle-Side) Congruency.
(शुजा-कोण-शुजा)

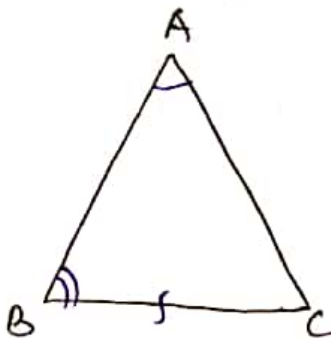


given:
 $\angle A = \angle D$
 $AB = DE$
 $AC = DF$

then $BC = EF$

$$\triangle ABC \cong \triangle DEF$$

③ AAS (Angle-Angle-Side) Congruency.
(कोण-कोण-शुजा)



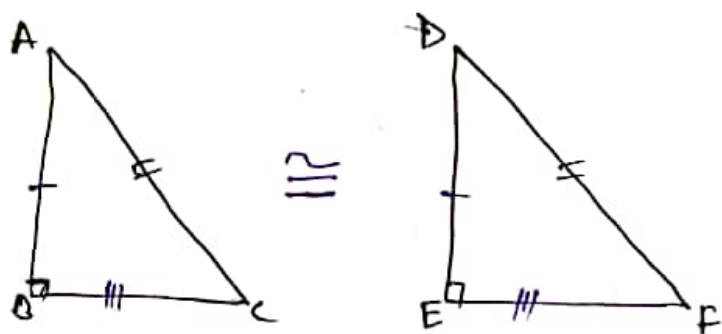
given:
 $\angle A = \angle D$
 $\angle B = \angle E$
 $BC = EF$

then

$$\begin{aligned} \angle C &= \angle F \\ AB &= DE \\ AC &= DF \end{aligned}$$

$$\triangle ABC \cong \triangle DEF$$

④ RHS = (Right angle-hypotenuse-side)
(Congruency)

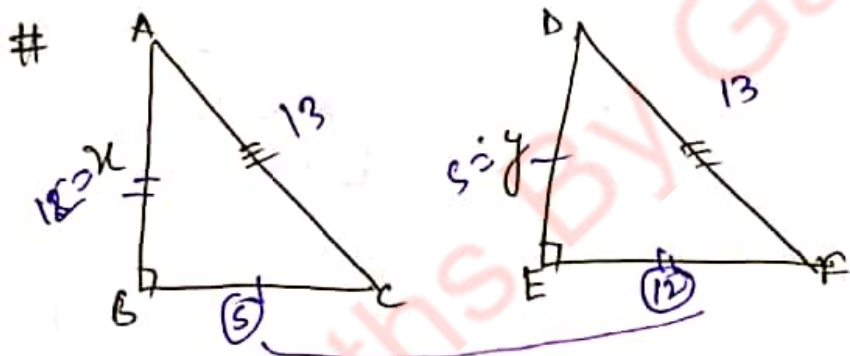


Given $\angle B = \angle E = 90^\circ$

$$AC = DF$$

$$AB = DE$$

then $\left\{ \begin{array}{l} \triangle ABC \cong \triangle DEF \\ BC = EF \end{array} \right\}$ Result



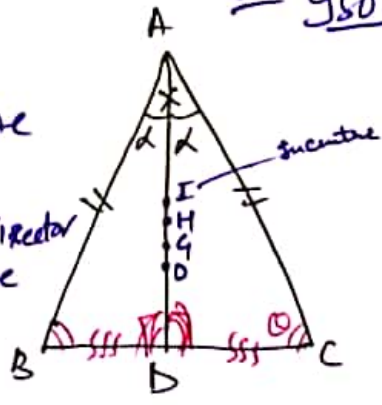
Given $\rightarrow \left\{ \begin{array}{l} BC + EF = x + y \\ BC \sim EF = x \sim y \end{array} \right\}$

then $\triangle ABC \cong \triangle DEF$
congruent

#

Ex: Isosceles Δ

$AB = AC$
AD is
angle bisector
of $\angle BAC$



$\left\{ \begin{array}{l} AD - \text{angle bisector} \\ \text{Altitude/median} \end{array} \right\}$

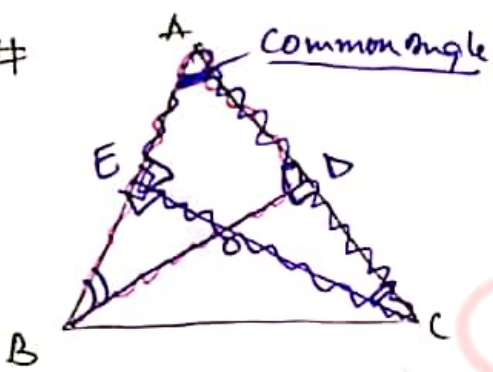
$AB = AC$
 $\angle BAD = \angle CAD$
 $AD = AD \text{ (common)}$

$BD = DC$

Then AD is median.

Isosceles Δ में चारो centre एक ही लाइन में होते हैं (AD)

#



Common angle

Altitude $(BD = CE)$ दो Altitude
 $\Downarrow$
then Δ is isosceles Δ

$(BD = CE = AF)$ (तीनो altitude)
equilateral Δ

$BD = CE$

$\Rightarrow \boxed{AB = AC}$

2nd method.

$$\frac{1}{2} \times AC \times BD = \frac{1}{2} \times AB \times CE$$

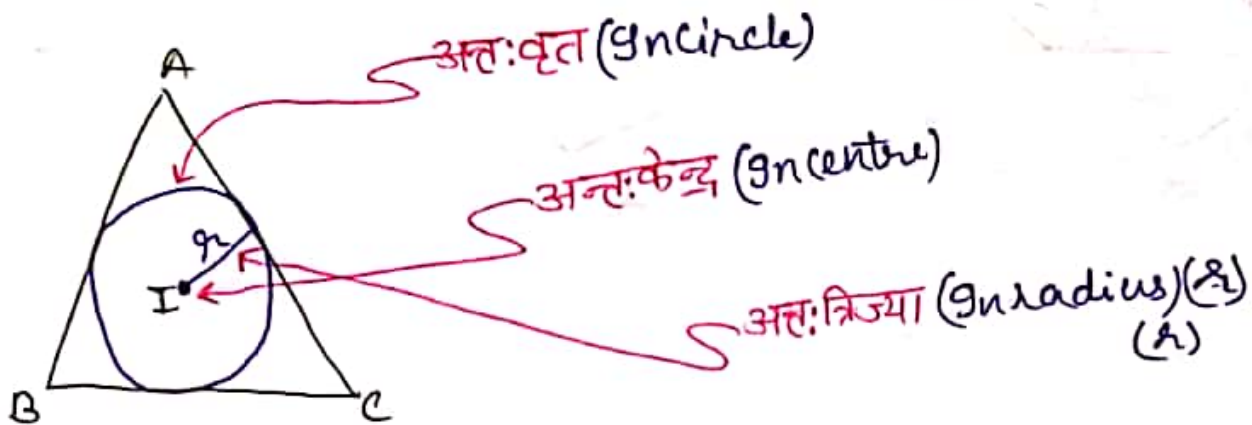
$\boxed{AC = AB}$



Centre of Triangle.

①

(I) - Incentre (अंतःकेन्द्र)

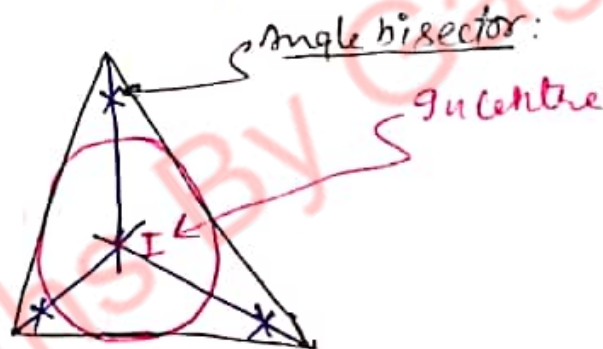


⇒ Incentre is equidistant from all the side.
(बराबर दूरी)

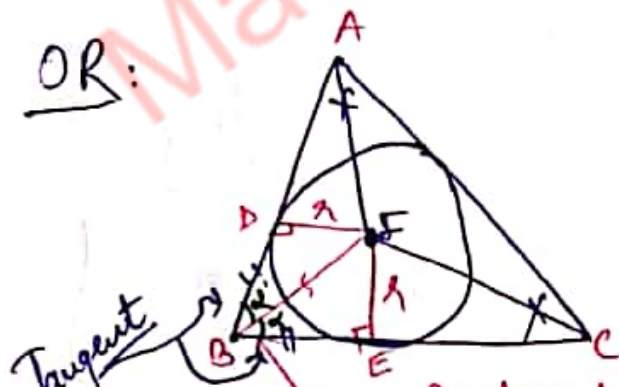
⇒ Incentre always lies inside the Δ .
(अन्दर ही होगा Incentre)
(हमेशा)

⇒ Incentre is Intersection of all the internal angle bisector of Δ triangle.

eg.

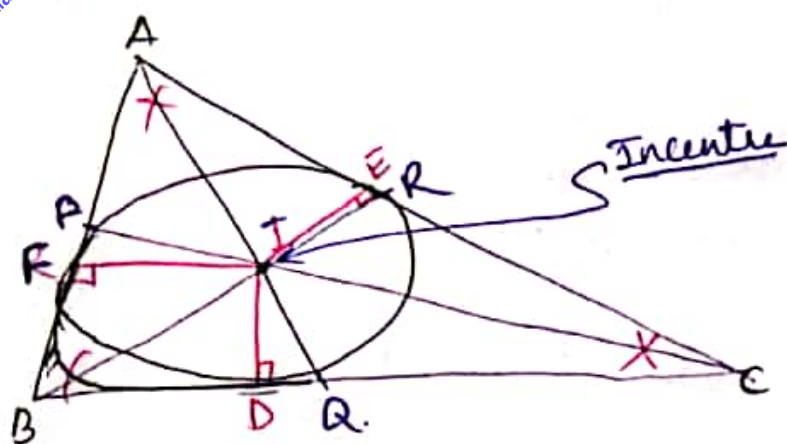


OR:

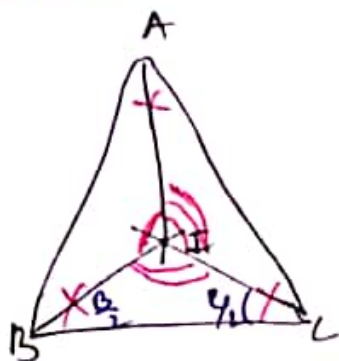


$\triangle BID \cong \triangle CIE$
[are congruent]

मेरी interior angle bisector होगा।



Result -



$$\begin{aligned} \angle BIC &= 90^\circ + \frac{\angle A}{2} \\ \angle AIC &= 90^\circ + \frac{\angle B}{2} \\ \angle AIB &= 90^\circ + \frac{\angle C}{2} \end{aligned}$$

PROOF

$$\begin{aligned} \angle BIC &= 180^\circ - \left(\frac{\angle B}{2} + \frac{\angle C}{2} \right) \\ &= 180^\circ - \left(\frac{\angle B + \angle C}{2} \right) \end{aligned}$$

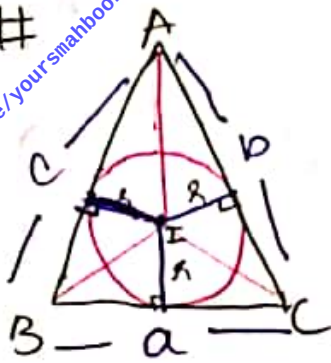
$$\because \angle B + \angle C = 180^\circ - \angle A$$

$$= 180^\circ - 90^\circ + \frac{\angle A}{2}$$

$$\angle BIC = 90^\circ + \frac{\angle A}{2}$$

Similarly,

$$\begin{cases} \angle AIC = 90^\circ + \frac{\angle B}{2} \\ \angle AIB = 90^\circ + \frac{\angle C}{2} \end{cases}$$



$$r = \frac{\Delta}{S}$$

(Ar - Area)

$$Ar \Delta ABC = Ar \Delta BIC + Ar \Delta AIC + Ar \Delta AIB$$

$$\Delta = \frac{1}{2} \times a \times r + \frac{1}{2} b \times r + \frac{1}{2} c \times r$$

$$\Delta = r \left(\frac{a+b+c}{2} \right)$$

$$\therefore \frac{a+b+c}{2} = S$$

$$\Delta = r \cdot S$$

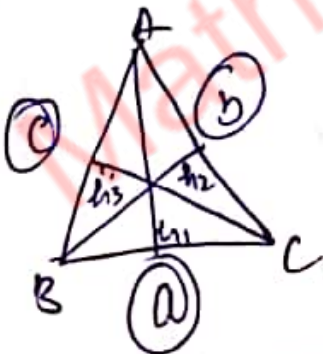
$$r = \frac{\Delta}{S} \text{ (General formula)}$$

h_1, h_2 & h_3 are the height of a Δ . the find
Inradius of Δ .

$\Rightarrow$ Result $\rightarrow$

$$\frac{1}{r} = \frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3}$$

PROOF



$$\Delta = \frac{1}{2} a h_1 = \frac{1}{2} b h_2 = \frac{1}{2} c h_3$$

$$a = \frac{2\Delta}{h_1}, b = \frac{2\Delta}{h_2}, c = \frac{2\Delta}{h_3}$$

$$a+b+c = 2\Delta \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3} \right)$$

$$\frac{a+b+c}{2\Delta} = \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3} \right)$$

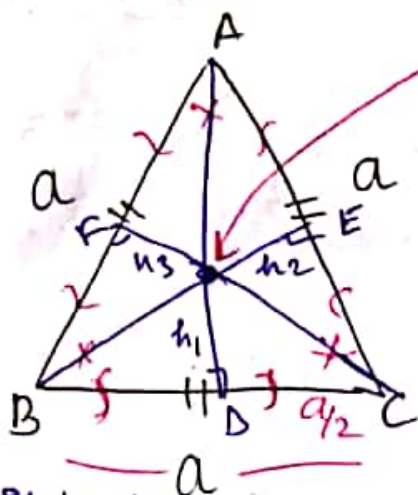
$$\frac{1}{r} = \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3} \right)$$



$$\therefore \frac{a+b+c}{2} = S$$

* Inradius of Equilateral Triangle -

(4)



I, G, H, O

equ! - Area of Δ

$$\Delta = \frac{\sqrt{3}a^2}{4}$$

$$r = \frac{\Delta}{s}$$

$$r = \frac{\frac{\sqrt{3}a^2}{4}}{\frac{3a}{2}}$$

$$r = \frac{a}{2\sqrt{3}}$$

😊

In equilateral Δ

$$h = h_1 = h_2 = h_3 = m$$

median

$$AD = \sqrt{a^2 - \frac{a^2}{4}}$$

$$AD = \frac{\sqrt{3}a}{2} = h$$

OR

$$h = \frac{\sqrt{3}a}{2} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$h = 3 \times \frac{a}{2\sqrt{3}}$$

$\therefore \frac{a}{2\sqrt{3}} = r$ Inradius. [Only equilateral Δ]

$$r = \frac{h}{3}$$

median = angle bisector

$$R = \frac{2H}{3}$$

$$AG : GD = 2 : 1$$

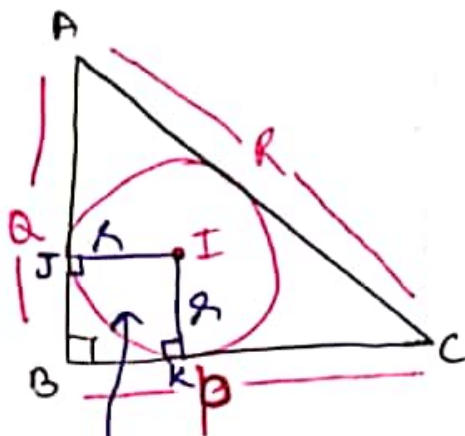
$$R : r = 2 : 1$$

R $\rightarrow$ Circumradius

r $\rightarrow$ Inradius

Radius of Right angle Triangle →

⑤



$$* \quad r = \frac{Q + P - R}{2}$$

$$\# \quad r = \frac{\text{Base} + \text{Perpendicular} - \text{Hypotenuse}}{2}$$

(Square) (अनेम) (JIKB is square)
at.

PROOF

$$r = \frac{\Delta}{S}$$

$$r = \frac{\frac{1}{2} PQ}{\frac{Q+P+R}{2}} \times \frac{P+Q-R}{P+Q-R}$$

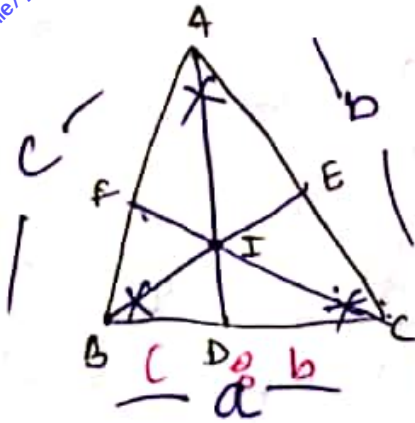
$$r = \frac{PQ(P+Q-R)}{(Q+P)^2 - R^2}$$

$$r = \frac{PQ(P+Q-R)}{Q^2 + P^2 + 2PQ - R^2}$$

$$r = \frac{P+Q-R}{2}$$



Result:-



$$\left\{ \begin{array}{l} \frac{AI}{ID} = \frac{b+c}{a} \\ \frac{BI}{IE} = \frac{a+c}{b} \\ \frac{CI}{IF} = \frac{a+b}{c} \end{array} \right.$$

I - Incentre
AD, BE & CF are angle bisector.

PROOF -

$$BD : DC = c : b$$

In $\triangle ABD$

By angle bisector theorem

$$\rightarrow BD = \left[a \times \frac{c}{c+b} \right]$$

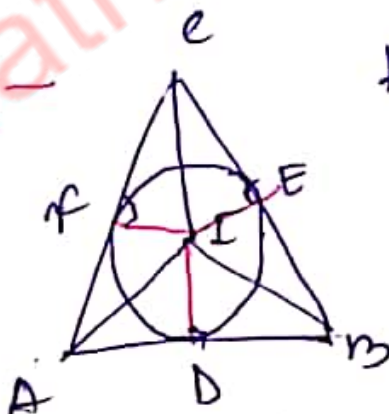
$$\frac{AI}{ID} = \frac{AB}{BD}$$

$$= \frac{a}{a \times \frac{c}{c+b}} = \frac{b+c}{a} \quad \text{☺}$$

Proved

Similarly $\left\{ \begin{array}{l} \frac{BI}{IE} = \frac{a+c}{b} \\ \frac{CI}{IF} = \frac{a+b}{c} \end{array} \right.$

Result:-



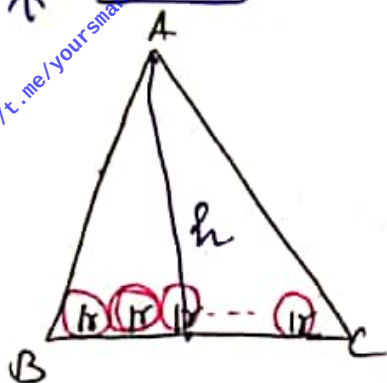
Result

$$\Delta = r^2 \left(\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} \right)$$

☺

* Result -

7

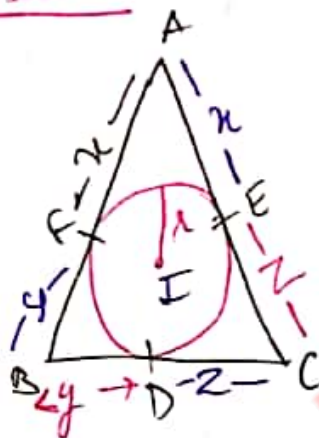


n Circle of equal radius.

$$r = \frac{\Delta}{s + (n-1)h}$$

- Δ - area of $\triangle ABC$
- s - Semiperimeter (अर्धपरिमाप)
- n - no. of circles
- * h - corresponding height (संगत अर्धवृत्त)

Result -



$$r = \sqrt{\frac{xyz}{x+y+z}}$$

Proof: Sides $\rightarrow$ $(x+y), (y+z), (z+x)$

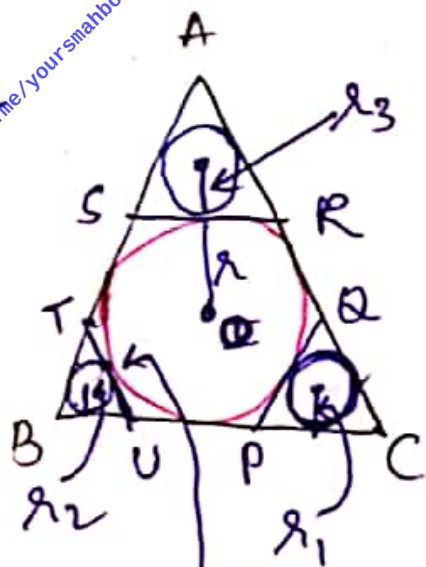
$$s = x+y+z$$

$$\Delta = \sqrt{(x+y+z)xyz}$$

$$r = \frac{\Delta}{s} = \frac{\sqrt{(x+y+z)xyz}}{x+y+z}$$

$$r = \sqrt{\frac{xyz}{x+y+z}}$$

😊 Satyam Padhyay



$PQ \parallel AB$

$RS \parallel BC$

$TU \parallel AC$

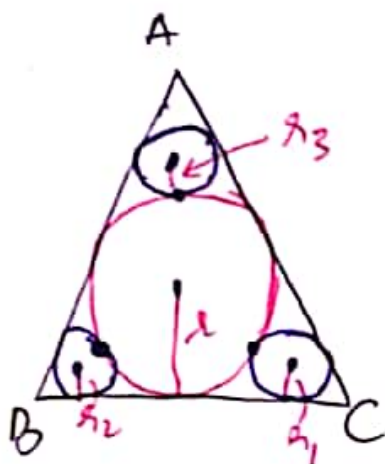
Gudhyang

Result:-

$$r = r_1 + r_2 + r_3$$



(यहाँ Touch नहीं कर रहे हैं वृत्त)



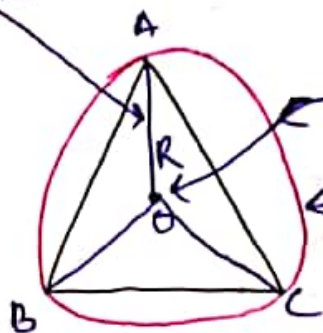
$$r = \sqrt{r_1 r_2} + \sqrt{r_2 r_3} + \sqrt{r_3 r_1}$$

②

Circum radius of a triangle → (परित्रिज्या)

⑨

Circumradius (R)
(परित्रिज्या)



Circum centre
परिकेन्द्र

Circum circle
परिघट्ट

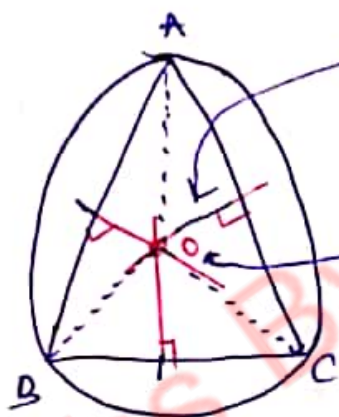
$$OA = OB = OC = R$$

Circumradius.

1) Circum centre may lie outside the circle.
(बाहर भी हो सकता है घट्ट)

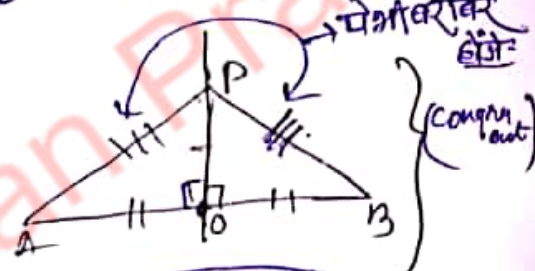
2) Circum centre is intersection of all perpendicular bisector of all 3 sides.

eg.



Perpendicular bisector

Circum centre



$$PA = PB$$

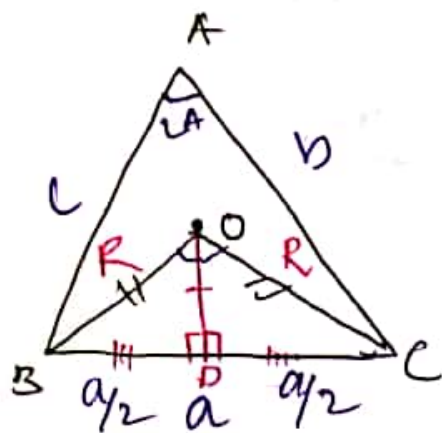
पेक्षा कर लेंगे
(Congruent)

#



Result:-

$$\left. \begin{aligned} \angle BOC &= 2\angle BAC \\ \angle AOC &= 2\angle ABC \\ \angle AOB &= 2\angle ACB \end{aligned} \right\}$$



O - Circum radius - (R)

$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$$

$$R = \frac{abc}{4\Delta}$$

By RHS congruency

Both $\Delta \{ \triangle BOC \cong \triangle COO \}$

PROOF -

$$\sin A = \frac{a/2}{R}$$

$$R = \frac{a}{2\sin A}$$

$$\Delta = \frac{1}{2} bc \sin A$$

$$\sin A = \frac{2\Delta}{bc}$$

$$R = \frac{a}{2 \times \frac{2\Delta}{bc}}$$

$$\frac{abc}{4\Delta}$$

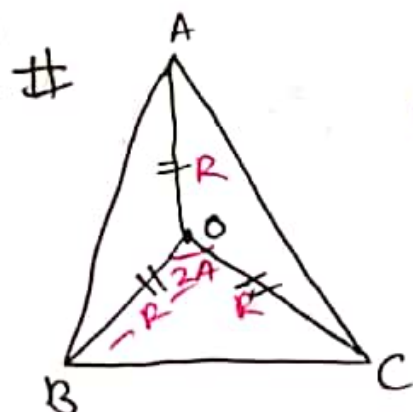
$$\frac{abc}{4\Delta}$$

Similarly -

$$\left\{ R = \frac{b}{2\sin B} = \frac{c}{2\sin C} \right\}$$

Relation b/w r and R and angles of Triangle (17)

$$r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$



[Ar - Area]

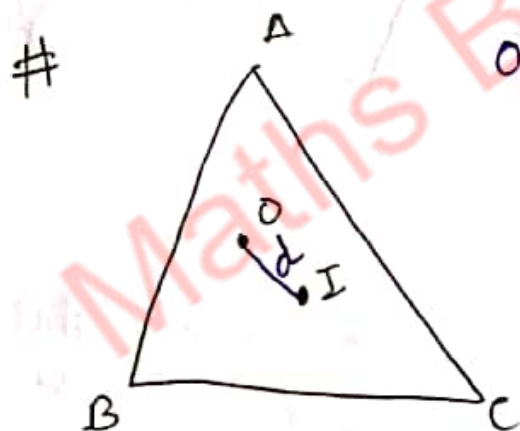
$$\text{Ar. } \triangle ABC = \text{Ar. } \triangle BOC + \text{Ar. } \triangle AOC + \text{Ar. } \triangle AOB$$

$$\Delta = \frac{1}{2} R^2 \sin 2A + \frac{1}{2} R^2 \sin 2B + \frac{1}{2} R^2 \sin 2C$$

$$\Delta = \frac{1}{2} R^2 (\sin 2A + \sin 2B + \sin 2C)$$

$$\Delta = \frac{1}{2} R^2 \times 4 \sin A \sin B \sin C$$

$$\Delta = 2R^2 \sin A \sin B \sin C$$



O - Circumradius
I - Incentre

$$\frac{1}{R+d} + \frac{1}{R-d} = \frac{1}{r}$$

R - Circumradius

r - Inradius

d - distance b/w R and r.

On Solving-

$$d = \sqrt{R^2 - 2Rr}$$

(17)

$$d^2 = R^2 - 2Rr$$

$$d^2 \geq 0 \Rightarrow R^2 - 2Rr \geq 0$$

$$R \cdot (R - 2r) \geq 0$$

$$\Rightarrow R - 2r \geq 0$$

$$R \geq 2r$$

$$\boxed{\frac{R}{r} \geq 2}$$

for any Triangle

$$\left\{ \frac{R}{r} \text{ min. value} \rightarrow 2 \right\}$$

In Equilateral Δ

*#

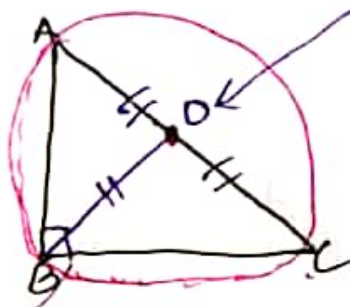
Position of Circum Centre -

① Acute angle triangle (अमकोण)



Circumcentre always अन्दर होगा
Acute angle triangle में

② Right angle Δ — (समकोण)

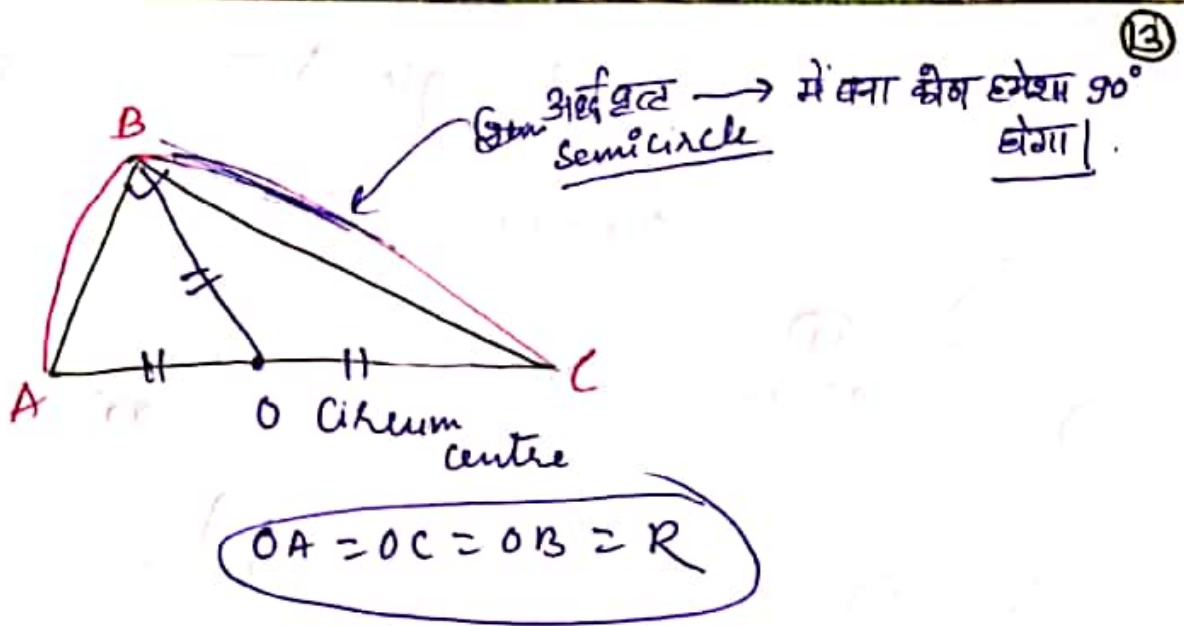


Circumcentre is mid point of hypotenuse.

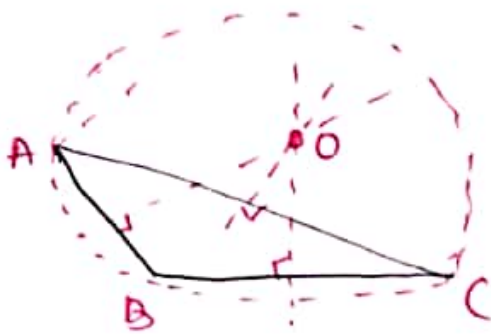
$$\boxed{OA = OB = OC = R}$$

$$OB = R = \frac{H}{2}$$

#* OB is median of Hypotenuse }
SHORTEST median

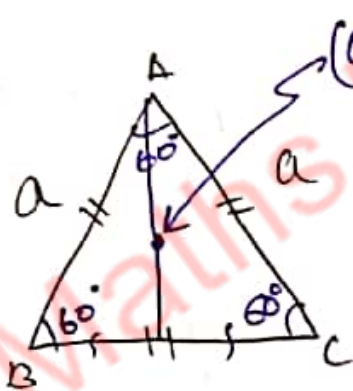


③ Obtuse angle Δ (अधिक कोण Δ)



Circum centre is always outside of the Δ {largest side के समीप होगा}

Circum radius of Equilateral Δ —



(O, E, H) $R = \frac{2H}{3} = \frac{a}{\sqrt{3}}$

$$R = \frac{a}{2 \times \sin 60} = \frac{a}{2 \times \frac{\sqrt{3}}{2}} = \left(\frac{a}{\sqrt{3}} \right)$$

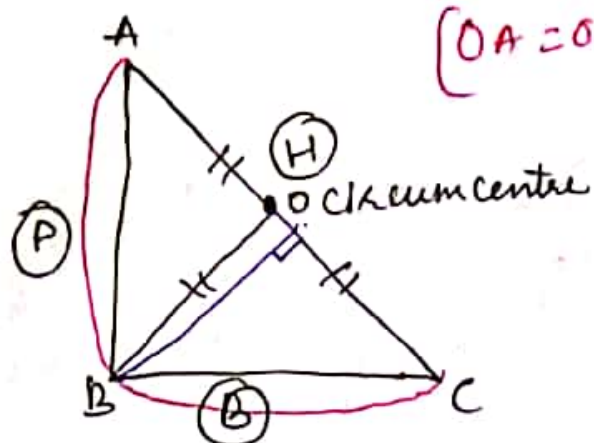
$$r = \frac{a}{2\sqrt{3}}$$

$AO : DO = 2 : 1$
 $R : r$

$R : r = 2 : 1$

Circum Radius of Right angle Δ -

(14)



$$[OA = OB = OC = R]$$

$$R = \frac{H}{2 \sin 90^\circ}$$

$$R = \frac{H}{2}$$

$$s = \frac{P+B-H}{2}$$

$$H = 2R$$

$$2s = P+B-H$$

$$2s + H = P+B$$

$$2s + 2R = P+B$$

*

$$2(s+R) = P+B$$

#

$$(s+R) = \frac{P+B}{2}$$

$$\Delta = s \cdot S$$

$$\Delta = \left(\frac{P+B-H}{2} \right) \cdot S$$

$$\Delta = \left(\frac{P+B+H-2H}{2} \right) \cdot S$$

$$\Delta = (S-H) \cdot S$$

$$H = 2R$$

*

$$\Delta = S \cdot (S - 2R)$$

Relation b/w r, R and $\Delta \rightarrow$

Right angle Δ
area...

$$\text{Result } \Delta = r^2 + 2rR$$

PROOF

$$r = \frac{P+B-H}{2}$$

$$(2r+H)^2 = (P+B)^2$$

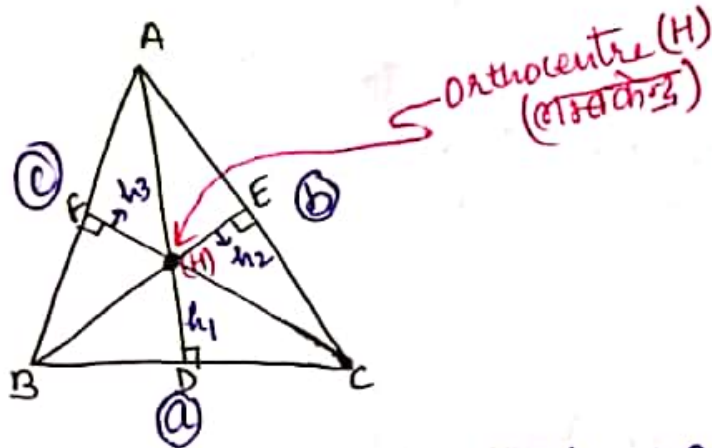
$$4r^2 + H^2 + 4rH = P^2 + B^2 + 2PB$$

$$4r^2 + 4rH = 2PB$$

$$r^2 + rH = \frac{1}{2} PB$$

$$\Delta = r^2 + 2rR$$

Orthocentre of triangle (त्रिभुज का लम्ब केन्द्र)



$$AD = h_1, BE = h_2, CF = h_3$$

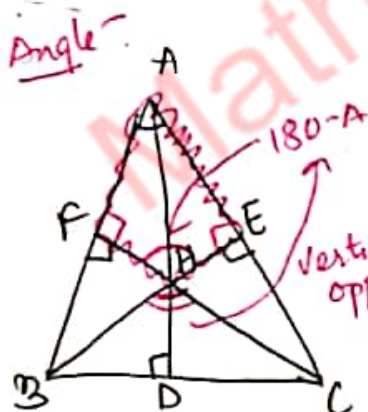
→ Orthocentre is Intersection of all 3 altitude of triangle.
(लम्ब केन्द्र)

→ Orthocentre may lie outside the triangle.
(बाहर)

$$\Delta = \frac{1}{2} ah_1 = \frac{1}{2} bh_2 = \frac{1}{2} ch_3$$

$$\Rightarrow ah_1 = bh_2 = ch_3$$

$$h_1 : h_2 : h_3 = \frac{1}{a} : \frac{1}{b} : \frac{1}{c}$$

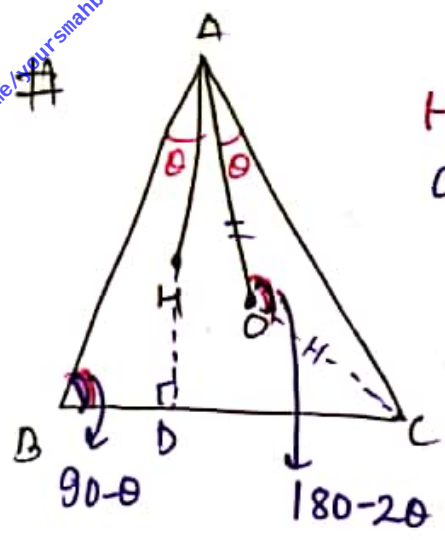


$$\left\{ \begin{array}{l} \angle BHC = 180^\circ - A \\ \angle AHC = 180^\circ - B \\ \angle AHB = 180^\circ - C \end{array} \right.$$

PROOF — $\square AFHE$ में —
 $\angle F + \angle E = 180^\circ$

Then $\angle A = 180^\circ - \angle A$

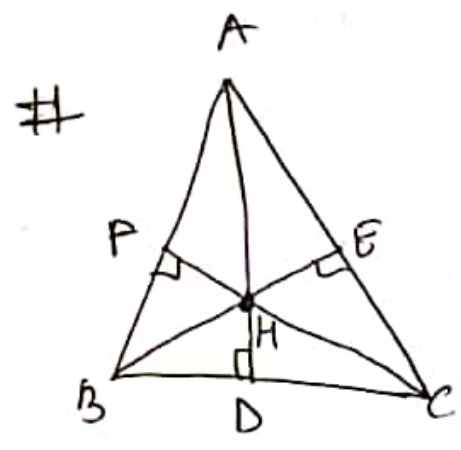
Then — $\angle BHC = 180^\circ - \angle A$ [Vertically opposite angle]



H - Orthocentre
O - Circumcentre

$$\angle BAH = \angle OAC$$

[Other side ka sum 2θ dena chahiye]
इसलिए $\angle A = 2\theta$ and $\angle C = \theta$.



$$(AB + BC + CA) > (AD + BE + CF)$$

$$\frac{AB + BC + CA}{AD + BE + CF} > 1$$

$$\begin{aligned} BC &> FC \\ AB &> BE \\ AC &> AD \end{aligned}$$

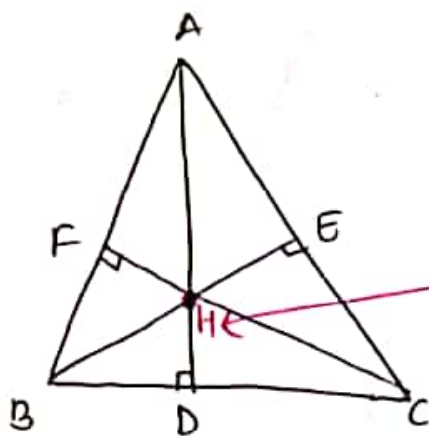
$$\left\{ (AB + BC + CA) > (AD + BE + CF) \right\}$$

$$\Rightarrow \frac{AB + BC + CA}{AD + BE + CF} > 1$$

Position of orthocentre -

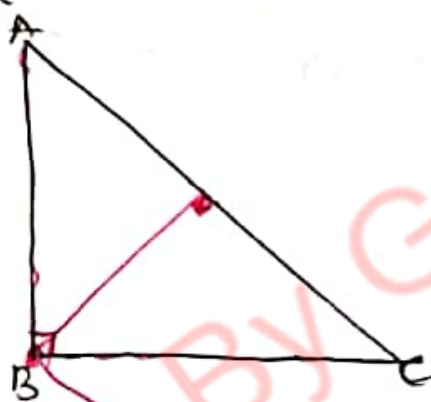
(17)

① Acute angle triangle - (भुजकोण Δ)

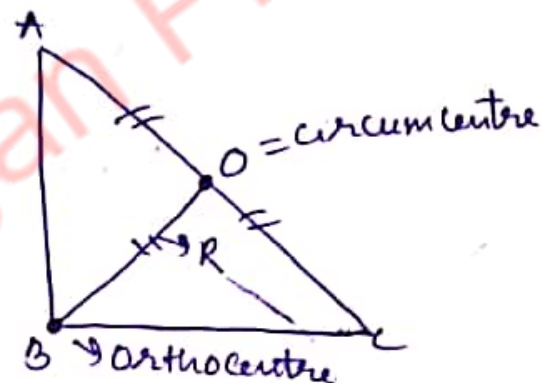


Orthocentre (लम्ब केन्द्र)
(हमेशा Inside होगा भुजकोण Δ में)

② Right angle triangle - (समकोण Δ)



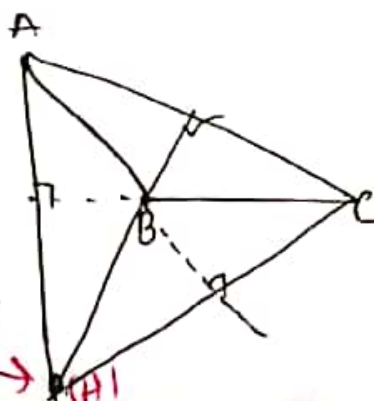
Orthocentre at 90° vertex.



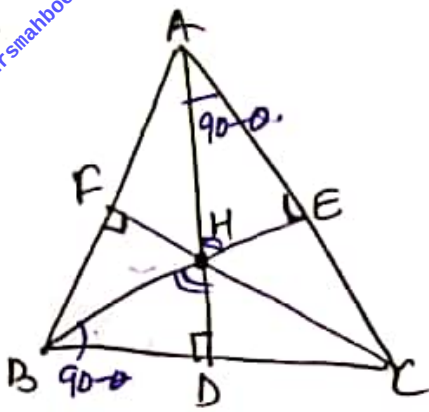
OB $\rightarrow$ distance
b/w Orthocentre
and Circumcentre

$$R = \frac{H}{2}$$

③ Obtuse angle triangle (अधिक कोण Δ)



(हमेशा Δ के बाहर होगा
अधिक कोण Δ में)
Orthocentre $\rightarrow$



$$AH \times HD = BH \times HE = CH \times HF$$

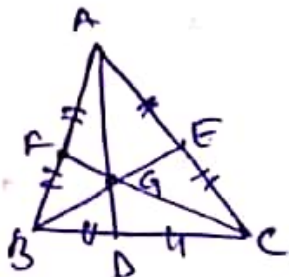
$\triangle BHD$ & $\triangle EHA$ में

$$\Rightarrow \triangle BHD \sim \triangle EHA$$

$$\frac{AH}{BH} = \frac{HE}{HD}$$

$$AH \times HD = BH \times HE$$

Centroid -



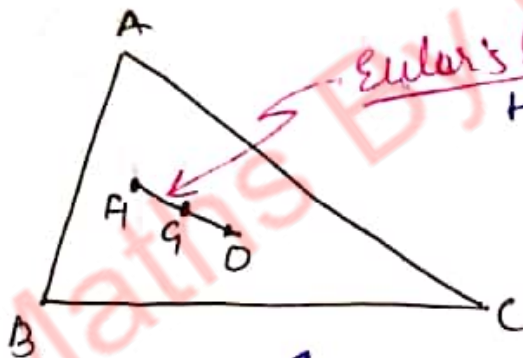
$$AG:GD = BG:GE = CG:CF = 2:1$$

Euler's line

H = Orthocentre

G = Centroid

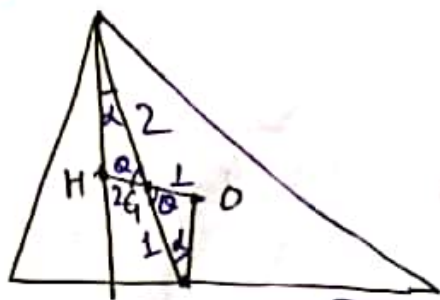
O = Circum Centre



H, G & O एक सीधी रेखा में होते हैं।

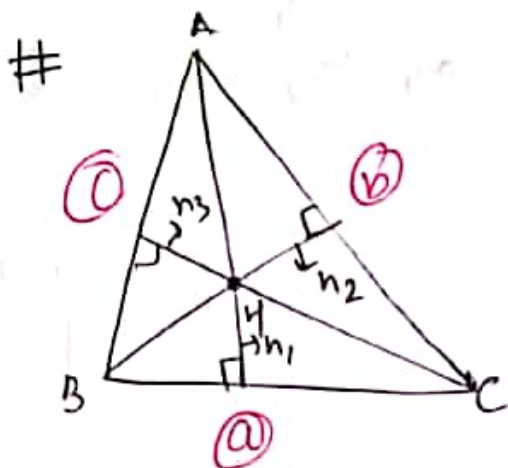
$$[HG:GO = 2:1]$$

any triangle



$BF = EC$ — $OE \perp BC$

By Similarity
Proof



$$\frac{h_1 h_2}{h_1 + h_2} < h_3 < \frac{h_1 h_2}{h_1 - h_2}$$

difference

$$\text{Area } \Delta = \frac{1}{2} a h_1 = \frac{1}{2} b h_2 = \frac{1}{2} c h_3$$

$$a = \frac{2\Delta}{h_1}, \quad b = \frac{2\Delta}{h_2}, \quad c = \frac{2\Delta}{h_3}$$

$$\Rightarrow |a - b| < c < |a + b|$$

$$\Rightarrow \left| \frac{2\Delta}{h_1} - \frac{2\Delta}{h_2} \right| < \frac{2\Delta}{h_3} < \left| \frac{2\Delta}{h_1} + \frac{2\Delta}{h_2} \right|$$

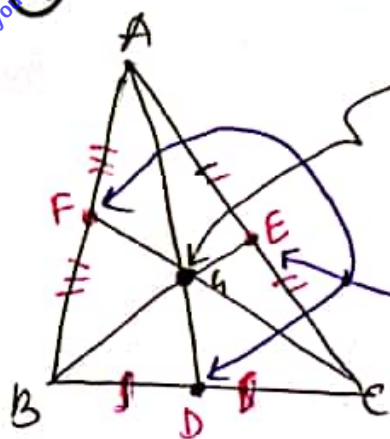
$$= \frac{h_2 - h_1}{h_1 h_2} < \frac{1}{h_3} < \frac{h_1 + h_2}{h_1 h_2}$$

$$\Rightarrow \frac{h_1 h_2}{h_1 - h_2} > h_3 > \frac{h_1 h_2}{h_1 + h_2}$$



Centroid of triangle →

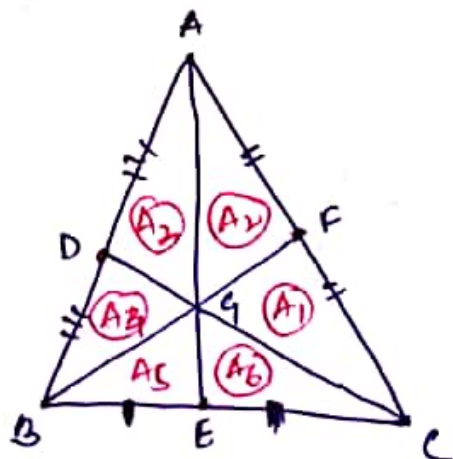
(20)



Centroid
(G)

$$AG:GH = BG:GE = CG:GF = 2:1$$

median
(माध्यिका)

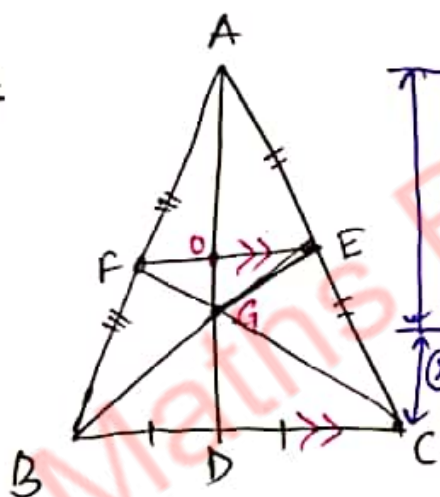


$$A_1 = A_2 = A_3 = A_4 = A_5 = A_6$$

A - Area

6 equal Area में divide करता है।

#



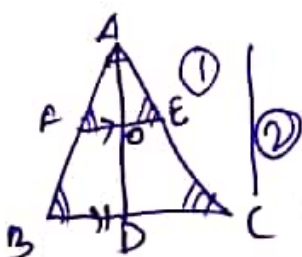
$$AG = 4 \text{ cm}$$

$$AO:OG:GD = 3:1:2$$

$$\left\{ \begin{array}{l} EF \parallel BC \\ EF = \frac{BC}{2} \end{array} \right\}$$

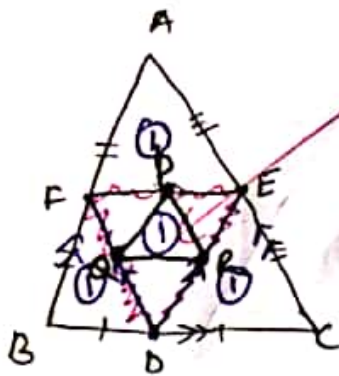
$$\therefore \left\{ \begin{array}{l} AO = 3 \text{ cm} \\ OG = 1 \text{ cm} \\ GD = 2 \text{ cm} \end{array} \right\}$$

$$\therefore \text{total} = 6 \text{ cm}$$

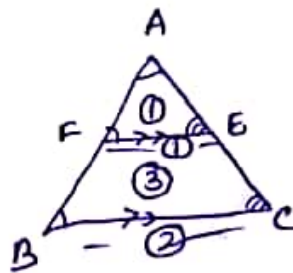


Medial Triangle:-

(21)



$FE \parallel BC$ by midpoint point theorem.



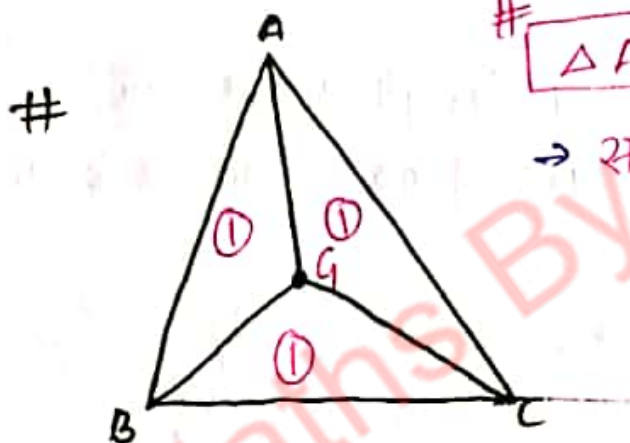
$$\frac{\Delta AFE}{\Delta ABC} = \frac{1}{4}, \quad \frac{\Delta BDF}{\Delta ABC} = \frac{1}{4}, \quad \frac{\Delta CED}{\Delta ABC} = \frac{1}{4}$$

$\Delta DEF = \frac{1}{4} \times \text{Area of } \Delta ABC$

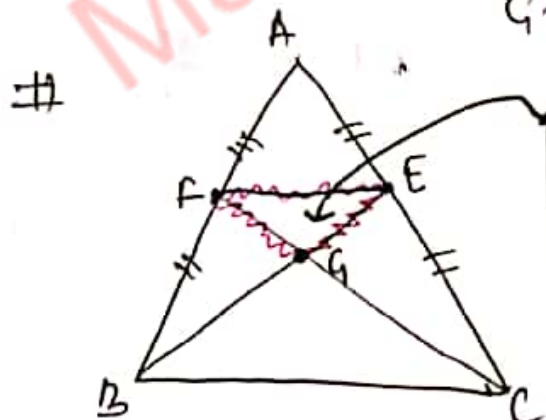
$\Delta PQR = \frac{1}{4} \times \Delta DEF$

$\Delta AGC = \Delta BGC = \Delta AGB = 1$

→ सभी का क्षेत्रफल बराबर होता है।
(केन्द्रक) (Centroid)

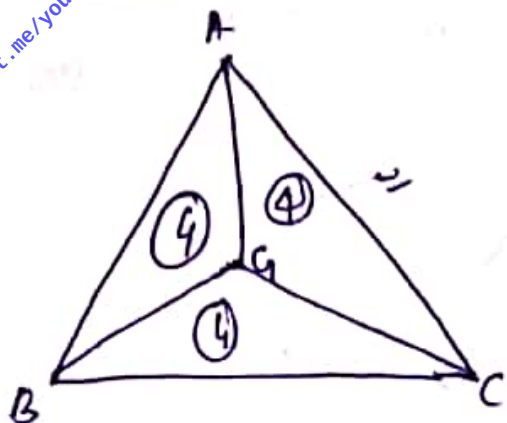


G - Centroid of ΔABC

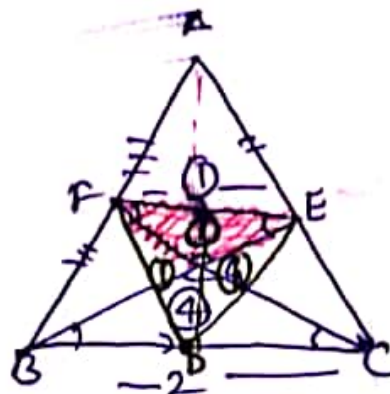


$\text{Area of } \Delta GEF = \frac{1}{12} \text{ of Ar. of } \Delta ABC$





Assume



$$FE : BC = 1 : 2$$

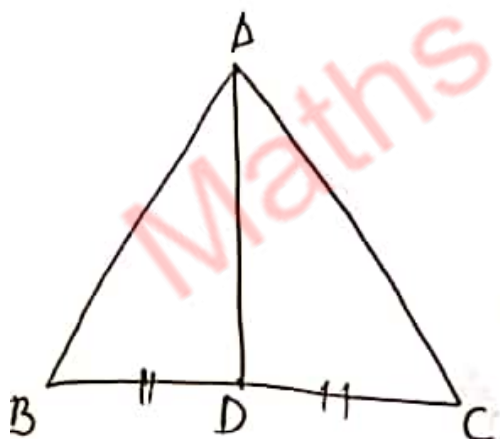
$$\Delta FGE : \Delta BGC = 1 : 4$$

$$\frac{\Delta FGE}{\Delta ABC} = \frac{1}{12}$$

$$Ar ABC = 4 + 4 + 4 = 12 \text{ cm}^2$$

$$\left[\frac{\Delta DEF (\text{medial})}{Ar ABC} = \frac{3}{12} = \frac{1}{4} \right]$$

Apollonius theorem $\rightarrow$ [To find length of median]
(भारमिका की लंब कर्षण करने कोलेए)



$$BD = DC$$

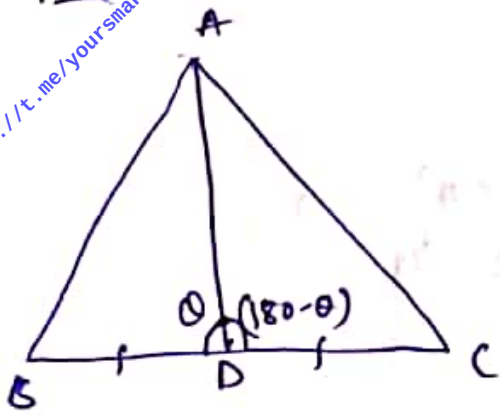
AD = median of side BC

$$AB^2 + AC^2 = 2(AD^2 + BD^2)$$

Apollonius theorem -

OR:

$$AB^2 + AC^2 = 2\left(AD^2 + \frac{BC^2}{4}\right)$$



By using cosine rule—

In $\triangle ADB$ $AB^2 = AD^2 + BD^2 - 2AD \cdot BD \cos \theta$ — (1)

In $\triangle ADC$ $AC^2 = AD^2 + DC^2 - 2AD \cdot DC \cdot \cos(180-\theta)$
 $AC^2 = AD^2 + DC^2 + 2AD \cdot DC \cdot \cos \theta$ — (2) $\left[\because \cos(180-\theta) = -\cos \theta \right]$

eqⁿ (1) + eqⁿ (2)

$$AB^2 = AD^2 + BD^2 - 2AD \cdot BD \cos \theta$$

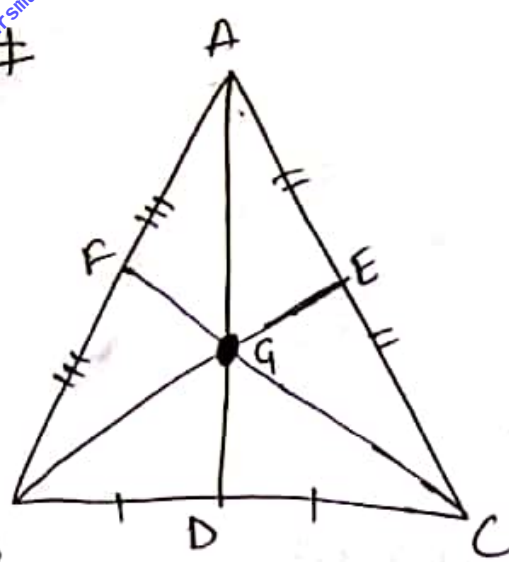
$$AC^2 = AD^2 + DC^2 + 2AD \cdot DC \cos \theta$$

$\therefore BD = DC$

$AB^2 + AC^2 = 2(AD^2 + BD^2)$

Apollonius theorem

→ To find length of median —



$$\frac{AB^2 + BC^2 + CA^2}{AD^2 + BE^2 + CF^2} = \frac{4}{3}$$

PROOF

By using Apollonius theorem-

$$AB^2 + AC^2 = 2\left(AD^2 + \frac{BC^2}{4}\right) \quad \text{--- (i)}$$

$$AB^2 + BC^2 = 2\left(BE^2 + \frac{AC^2}{4}\right) \quad \text{--- (ii)}$$

$$AC^2 + BC^2 = 2\left(CF^2 + \frac{AB^2}{4}\right) \quad \text{--- (iii)}$$

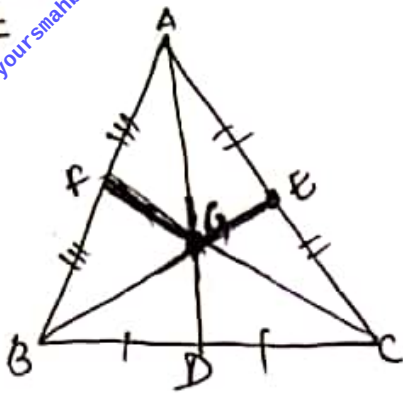
Add $\rightarrow 2(AB^2 + BC^2 + AC^2) = 2(AD^2 + BE^2 + CF^2) + \left(\frac{BC^2 + AC^2 + AB^2}{2}\right)$

$$\# \left(2 - \frac{1}{2} = \frac{3}{2}\right)$$

$$\frac{3}{2}(AB^2 + BC^2 + CA^2) = 2(AD^2 + BE^2 + CF^2)$$

$$\frac{AB^2 + BC^2 + CA^2}{AD^2 + BE^2 + CF^2} = \frac{4}{3}$$

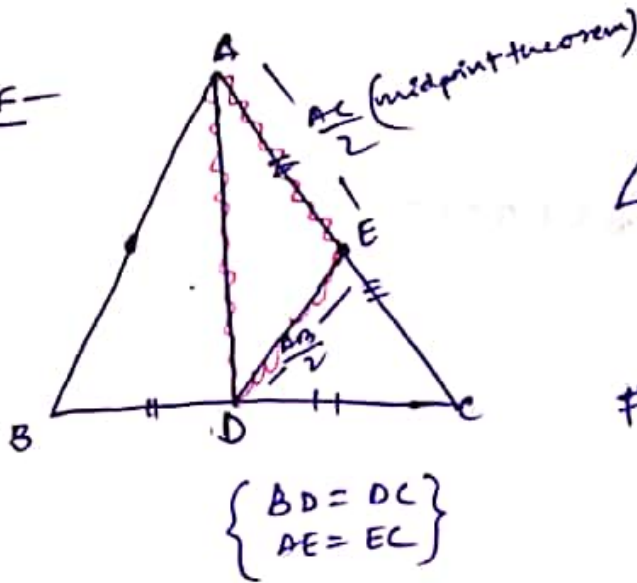
☺ इति सिद्धम्



$$1 < \frac{AB+BC+CA}{AD+BE+CF} < \frac{4}{3}$$

[AD, BE, CF → median]
G → centroid

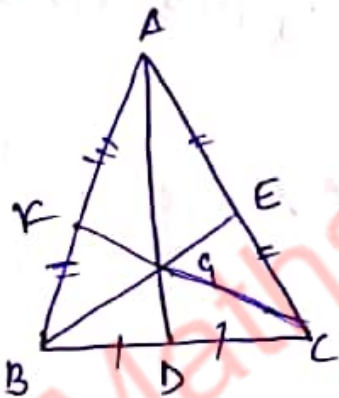
PROOF-



$\triangle ADE \sim \triangle ABC$

$$\frac{AB}{2} + \frac{AC}{2} > AD$$

$$\# \quad AB + AC > 2AD$$



$$AB + AC > 2AD$$

$$AB + BC > 2BE$$

$$AC + BC > 2CF$$

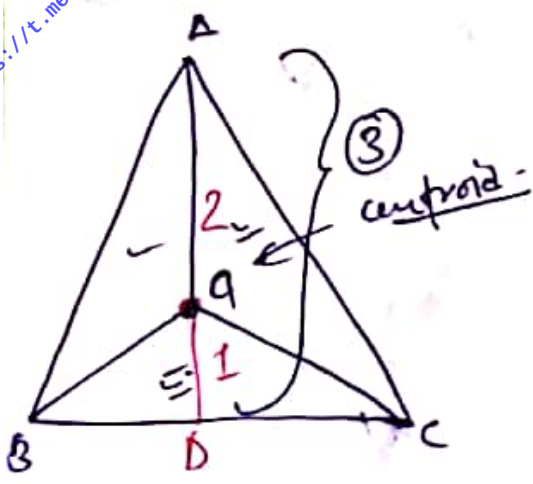
$$\text{Add - } (AB + BC + CA) > (AD + BE + CF) \times 2$$

$$\therefore (AB + BC + CA) > (AD + BE + CF)$$

$$\left\{ \frac{AB+BC+CA}{AD+BE+CF} > 1 \right\}$$

[1st condition proved]

2nd condition.



$$\begin{aligned} BG + GC &> BC \\ AG + GC &> AC \\ AG + BG &> AB \end{aligned}$$

Add - $2(AG + BG + GC) > (AB + BC + CA)$

$$\left\{ \begin{aligned} AG &= \frac{2}{3} AD \\ BG &= \frac{2}{3} BE \\ CG &= \frac{2}{3} CF \end{aligned} \right\}$$

$$2 \times \left(\frac{2}{3} AD + \frac{2}{3} BE + \frac{2}{3} CF \right) > (AB + BC + CA)$$

$$4(AD + BE + CF) > 3(AB + BC + CA)$$

OR

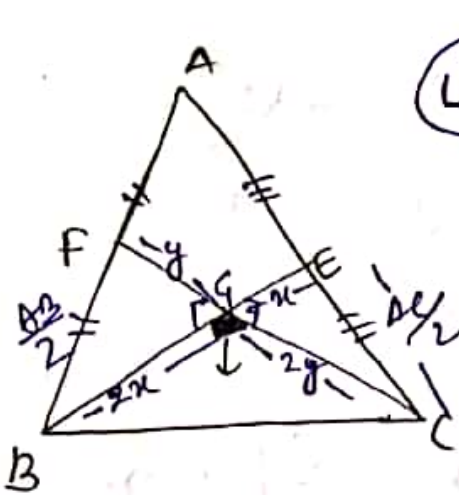
$$\frac{AB + BC + CA}{AD + BE + CF} < \frac{4}{3}$$

2nd condition
Proved

इति सिद्धम्



When two median Intersects each other at $90^\circ \rightarrow$ (27)



$$\angle BGC = 90^\circ$$

$$AB^2 + AC^2 = BC^2$$

PROOF

$$4x^2 + y^2 = \frac{AB^2}{4} \quad \text{--- (i)}$$

$$x^2 + 4y^2 = \frac{AC^2}{4} \quad \text{--- (ii)}$$

$$\underline{\text{Add}} - 5(x^2 + y^2) = \frac{AB^2 + AC^2}{4}$$

$$5(4x^2 + 4y^2) = AB^2 + AC^2$$

$$5((2x)^2 + (2y)^2) = AB^2 + AC^2$$

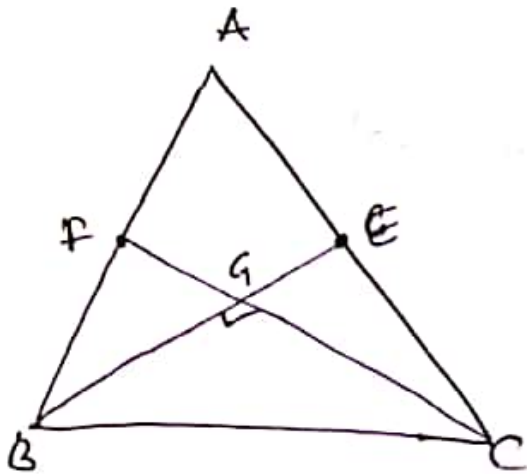
$$\boxed{BC^2 = AB^2 + AC^2}$$

$$\begin{aligned} &[\text{In } \Delta BGC] \\ &[(2x)^2 + (2y)^2 = BC^2] \end{aligned}$$

☺ [इति सिद्धम्]

Special Case $\rightarrow$ [Isosceles Δ]

(28)



$$AB^2 + AC^2 = 4BC^2$$

$$2AB^2 = 4BC^2$$

$$\frac{AB^2}{BC^2} = \frac{2}{1}$$

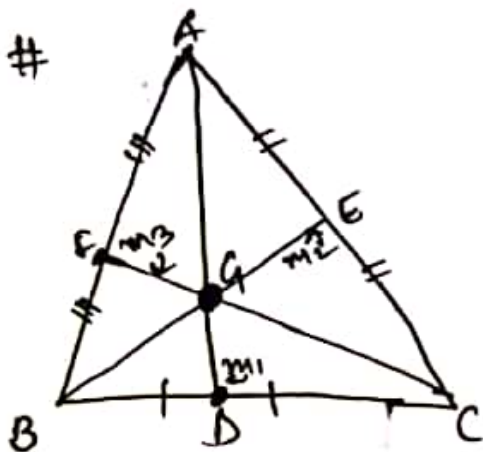
(Tier-II) ✓

OR.

$$\frac{AB}{BC} = \sqrt{\frac{2}{1}}$$

(CPD) ✓

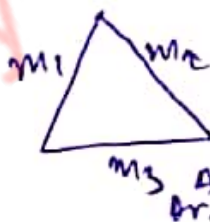
$AB = BC$ (median BE and CF are perpendicular to each other)



To Find:

Area of Δ when length of all 3 medians given.

median $\left\{ \begin{array}{l} AD = m_1 \\ BE = m_2 \\ CF = m_3 \end{array} \right\}$



$$s_m = \frac{m_1 + m_2 + m_3}{2}$$

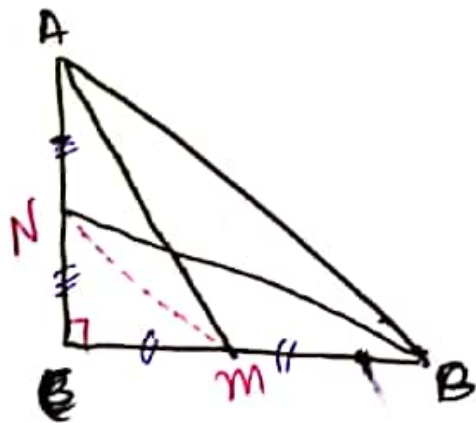
$$Ar. \Delta ABC = \frac{4}{3} \times \sqrt{s_m (s_m - m_1) (s_m - m_2) (s_m - m_3)}$$

#

$$Ar. \Delta ABC = \frac{4}{3} \times \sqrt{s_m (s_m - m_1) (s_m - m_2) (s_m - m_3)}$$

$$Ar. \Delta ABC = \frac{4}{3} \times Ar. \Delta \text{ made by using length of median}$$

median in right angle triangle - (29)



$$AM^2 + BN^2 = AB^2 + MN^2 \quad \text{--- (1)}$$

$$[MN = \frac{AB}{2}] \text{ (midpoint theorem)}$$

AM & NB are the median of $\triangle ABC$ -
 $\angle C = 90^\circ$

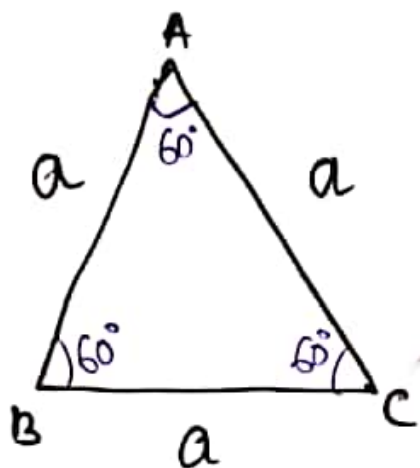
put the value of MN in eqⁿ --- (1)

$$AM^2 + BN^2 = \frac{AB^2}{4}$$

$$* \quad 4(AM^2 + BN^2) = AB^2$$

OR.. $\left[\frac{AM^2 + BN^2}{AB^2} = \frac{1}{4} \right] \quad \text{--- (2)}$

1) Equilateral Triangle (समबाहु त्रिभुज)



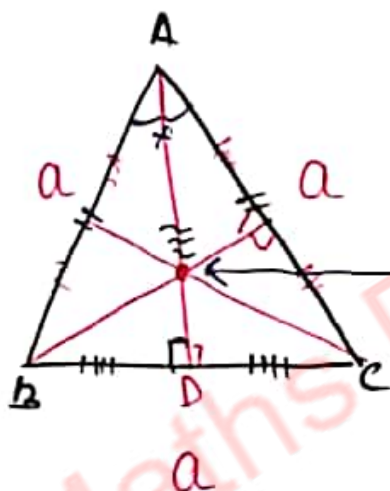
$$AB = BC = CA = a$$
$$\angle A = \angle B = \angle C = 60^\circ$$

$$\text{Area} = \frac{1}{2} \times a \times a \times \frac{\sqrt{3}}{2}$$

$$[\sin 60^\circ = \frac{\sqrt{3}}{2}]$$

$$\boxed{\text{Area} = \frac{\sqrt{3}}{4} \times a^2}$$

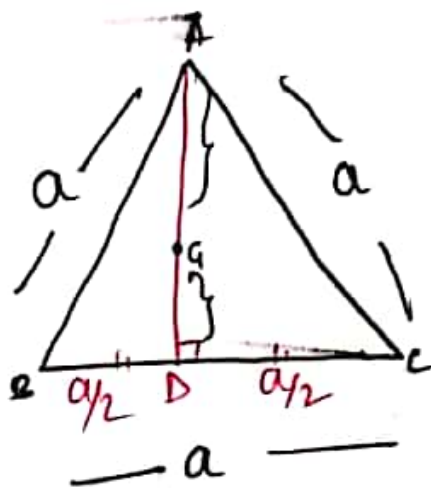
⇒



{ AD → Altitude, perpendicular bisector
AD → median ; Angle bisector

(H, I, G, O)

⇒



$$AD = \sqrt{a^2 - \frac{a^2}{4}}$$

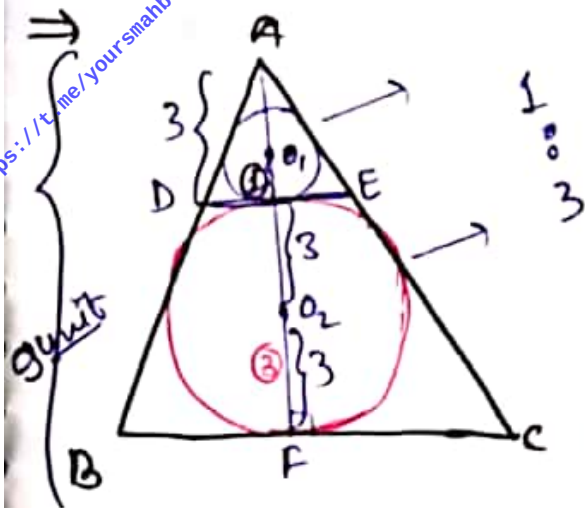
$$\boxed{AD = \frac{\sqrt{3}}{2} a = \text{height}}$$

inradius

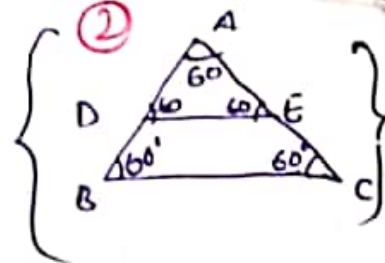
$$r = \frac{\text{height}}{3} = \frac{a}{2\sqrt{3}}$$

circumradius

$$R = \frac{\text{height} \times 2}{3} = \frac{a}{\sqrt{3}}$$



[$\triangle ABC$ - समबाहु $\triangle$]

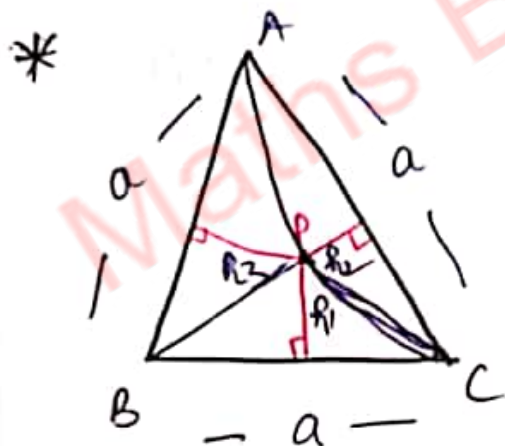


$$\left\{ \begin{array}{l} \text{Radius} = 1.83 \\ \text{Area} = 1.89 \end{array} \right\}$$

$$\Rightarrow \begin{aligned} \text{Side} &= a \\ \text{height} &= \frac{\sqrt{3}}{2} a \\ \text{Inradius} &= \frac{a}{2\sqrt{3}} \\ \text{Circumradius} &= \frac{a}{\sqrt{3}} \\ \text{area} &= \frac{\sqrt{3}}{4} a^2 \end{aligned}$$

Let

$$\begin{aligned} \text{Side} &= 2 \quad (XK) \\ \text{height} &= \sqrt{3} \quad (XK) \\ \text{Inradius} &= \frac{1}{\sqrt{3}} \quad (XK) \\ \text{Circumradius} &= \frac{2}{\sqrt{3}} \quad (XK) \\ \text{area} &= \sqrt{3} \quad (XK^2) \end{aligned}$$



$$h = h_1 + h_2 + h_3$$

$$\frac{\sqrt{3}}{4} a^2 = \frac{1}{2} \times a \times h_1 + \frac{1}{2} \times a \times h_2 + \frac{1}{2} \times a \times h_3$$

$$\frac{\sqrt{3}}{4} a \times a = \frac{1}{2} a [h_1 + h_2 + h_3]$$

$$\frac{\sqrt{3} a}{2} = h_1 + h_2 + h_3$$

$$\therefore h = \frac{\sqrt{3} a}{2}$$

$$h = h_1 + h_2 + h_3$$

$$a = \frac{2}{\sqrt{3}} (h_1 + h_2 + h_3)$$

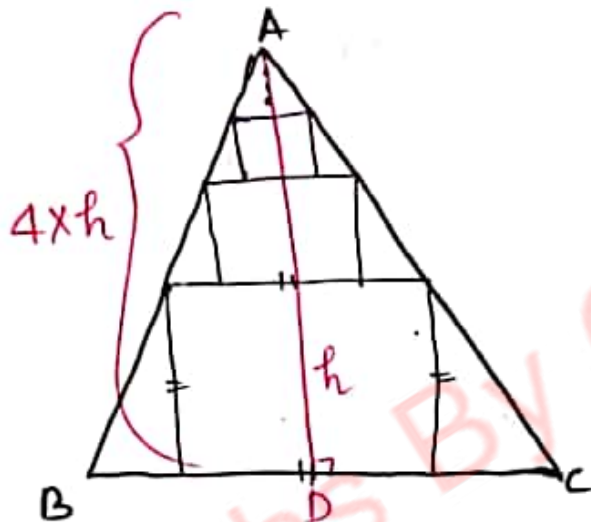
(3)

$$\text{Area} = \frac{\sqrt{3}}{4} \times \frac{1}{3\sqrt{3}} (h_1 + h_2 + h_3)^2$$

$$\text{Area} = \frac{1}{\sqrt{3}} (h_1 + h_2 + h_3)^2$$

(5)

Sum of perimeter of Infinite Square in a Δ



AD = h (corresponding to base BC)

$\Rightarrow 4h \rightarrow$ Sum of perimeter of all the square = 4h

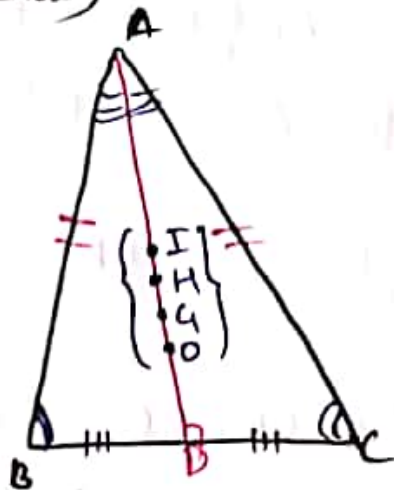
(6)

Isosceles Δ :- (समद्विबाहु Δ)

(4)

$AB = AC$ (lateral side)

$BC = \text{Base}$

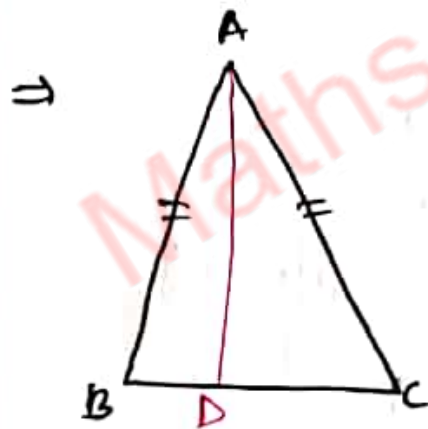


ΔAOB & ΔAOC
(congruent)

$$\left\{ \begin{array}{l} AB = AC \\ \Rightarrow LB = LC \\ BD = DC \end{array} \right\}$$

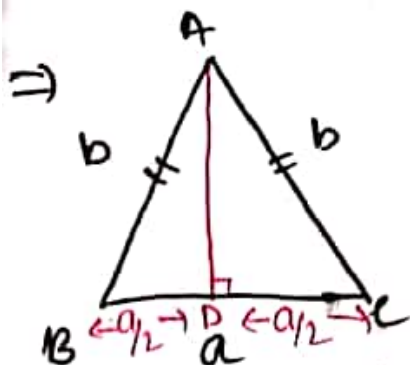
(I, H, G, O) एक ही लाइन में होते हैं,
समद्विबाहु Δ में

$AD \rightarrow$ { Altitude, median,
perpendicular bisector
of side BC
- Angle bisector.
[By congruency] }



$AB = AC$
 $D \rightarrow$ Any point on Base BC

$$AD^2 = AB^2 - BD \times DC$$



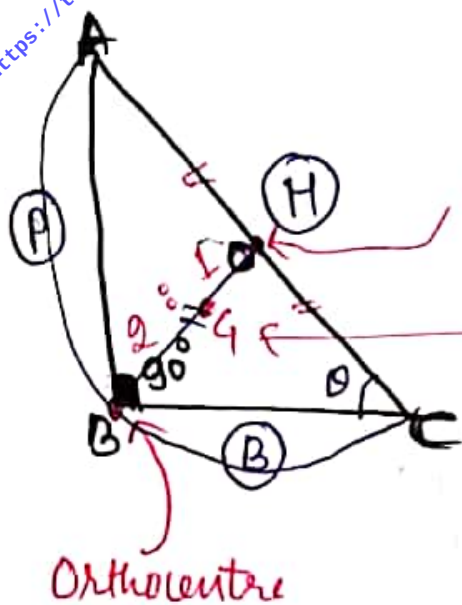
$$AD = \sqrt{b^2 - \frac{a^2}{4}}$$

$$AD = \frac{\sqrt{4b^2 - a^2}}{2}$$

$$* \text{ Area} = \frac{1}{2} \times a \times \frac{\sqrt{4b^2 - a^2}}{2} = \frac{a}{4} \sqrt{4b^2 - a^2}$$

Right angle Triangle - (समकोण Δ)

(S)



AB & BC $\rightarrow$ legs

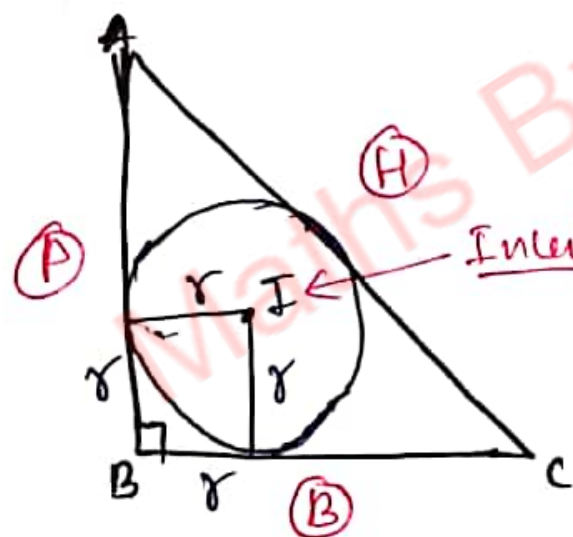
Circumcentre AC = hypotenuse

Centroid

$$H^2 = P^2 + B^2$$

$$OA = OB = OC = R \text{ (Circum radius)} \\ = \frac{H}{2}$$

$$[OB \rightarrow \text{Shortest median} = R = \frac{H}{2}]$$



Incentre

$$r = \frac{P+B-H}{2}$$

$$2r = P+B-H$$

$$2r+H = P+B$$

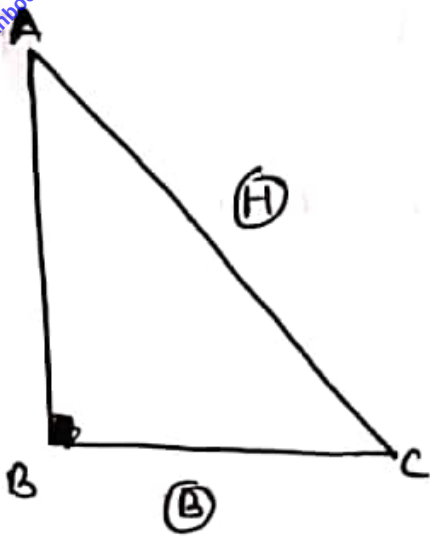
$$[H=2R]$$

$$2r+2R = P+B$$

$$2(r+R) = P+B$$

(6)

(P)



Area:

$$\Delta = \frac{1}{2} PB \sin 90^\circ = \frac{1}{2} PB$$

$$s = \frac{P+B+H}{2}$$

$$\begin{aligned} \Delta &= s \times s \\ &= \left(\frac{P+B+H}{2} \right) \times s \\ &= \left(\frac{P+B+H-2H}{2} \right) \times s \\ &= \left(\frac{P+B+H}{2} - H \right) \times s \end{aligned}$$

(Area)

$$\Delta = (s - H) \times s$$

OR.

$$\Delta = (s - 2R) s$$

(Area)

😊

⇒

Inradius = r

Circumradius = R

(Area)

$$\Delta = r^2 + 2rR$$

$$r = \frac{P+B-H}{2}$$

$$2r + H = P + B \quad \text{--- (Square) Both Side}$$

$$(2r + H)^2 = (P + B)^2$$

$$4r^2 + \cancel{H^2} + 4rH = \cancel{P^2 + B^2} + 2PB$$

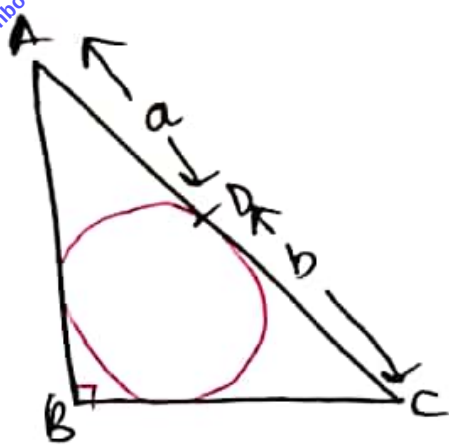
$$r^2 + rH = \frac{1}{2} PB$$

$$r^2 + 2rR = \frac{1}{2} PB = \Delta$$

area
(right angle Δ)

$$\boxed{\text{Area} = a \times b}$$

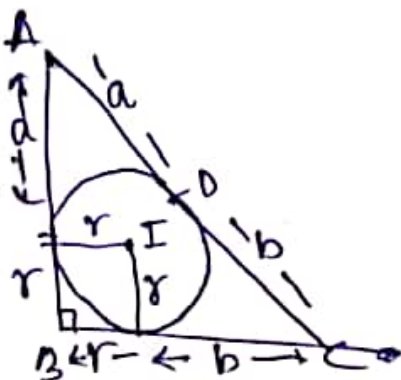
⑦



$$AD = a$$

$$CD = b$$

PROOF—



$$S = a + b + r$$

$$\Delta = \frac{1}{2} \times S$$

$$\Delta = \frac{1}{2} \times (a + b + r)$$

$$(a+r)^2 + (b+r)^2 = (a+b)^2$$

$$a^2 + r^2 + 2ar + b^2 + 2br + r^2 = a^2 + b^2 + 2ab$$

$$2r^2 + 2ar + 2br = 2ab$$

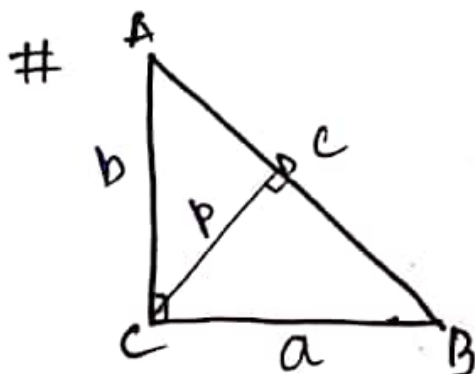
$$r(a+b+r) = ab$$

OR

$$\boxed{\Delta = ab}$$

⑧

इति सिद्धम्



$$\text{Area} = \text{Area}$$

$$\frac{1}{2} \times a \times b = \frac{1}{2} \times c \times p$$

$$\boxed{p = \frac{ab}{c}}$$

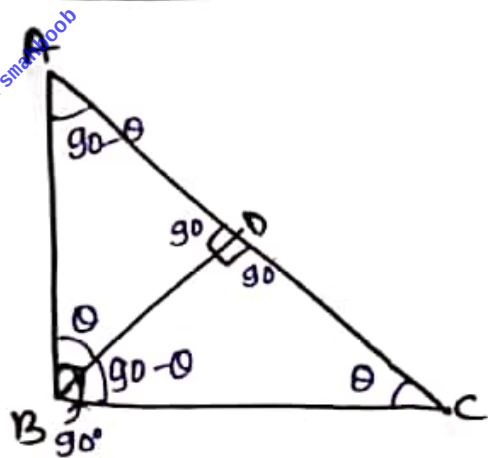
Reciprocal of

$$\text{OR } \left[\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} \right]$$

$$\frac{1}{p^2} = \frac{c^2}{a^2 b^2} = \frac{a^2 + b^2}{a^2 b^2}$$

$$\left[\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} \right]$$

#



$$\Rightarrow BD \perp AC$$

⑧

[similarity]

$$\triangle ABC \sim \triangle BDC$$

$$\frac{AB}{BD} = \frac{BC}{DC} = \frac{AC}{BC}$$

OR

$$BC^2 = AC \times CD \quad \text{--- ①}$$

$$BD = \frac{AB \times BC}{AC}$$

$$\triangle ADB \sim \triangle BDC$$

$$\frac{AD}{BD} = \frac{DB}{DC} = \frac{AB}{BC}$$

$$BD^2 = AD \times DC \quad \text{--- ③}$$

$$\triangle ABC \sim \triangle ADB$$

$$\frac{AB}{AD} = \frac{BC}{DB} = \frac{CA}{BA}$$

$$AB^2 = AD \times AC \quad \text{--- ②}$$

$$[eq^{\circ} 2 \div eq^{\circ} 1]$$

$$\frac{AB^2}{BC^2} = \frac{AC \times AD}{AC \times CD}$$

$$\frac{AB^2}{BC^2} = \frac{AD}{CD}$$

Result - ###

$$1) BC^2 = AC \times CD$$

$$2) BD = \frac{AB \times BC}{AC}$$

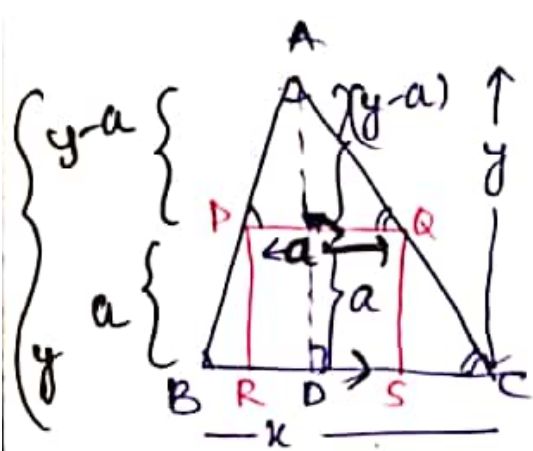
$$3) AB^2 = AD \times AC$$

$$4) BD^2 = AD \times DC$$

$$5) \frac{AB^2}{BC^2} = \frac{AD}{CD}$$

Square inside a right angle triangle

⇒ Square in ^{side} General Δ



→ PQRS is a square of maximum possible size on base BC.

$$\Delta ASR \sim \Delta ABC$$

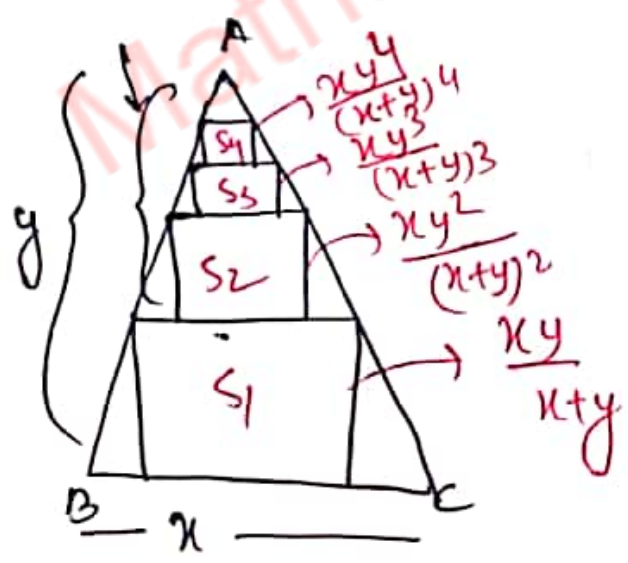
$$\frac{a}{x} = \frac{y-a}{y}$$

$$ay = xy - ax$$

$$a(x+y) = xy \quad \text{###}$$

$$a = \frac{xy}{x+y}$$

$$y - \frac{xy}{x+y} = \frac{y^2}{x+y}$$



Sum of perimeter of all Infinite square = $4y$

S_2 of side.

$$= \frac{xy}{x+y} \times \frac{y^2}{(x+y)}$$

$$\text{Side} = \frac{xy + y^2}{(x+y)}$$

{ ऐसे सभी square की side, perimeter निकाल सकते हैं }

$$\left\{ \text{Side} = \frac{xy^2}{(x+y)xy(x+y)} \Rightarrow \frac{xy^2}{(x+y)^2} \right\}$$

$$\Rightarrow \text{Area} (S_1 + S_2 + S_3 + \dots \infty) \quad (10)$$

$$\left(\frac{xy}{x+y}\right)^2 + \left\{\frac{xy^2}{(x+y)^2}\right\}^2 + \left\{\frac{xy^3}{(x+y)^3}\right\}^2 + \dots \infty$$

$$= \left(\frac{xy}{x+y}\right)^2 \left[1 + \frac{y^2}{(x+y)^2} + \frac{y^4}{(x+y)^4} + \dots \infty \right]$$

$$= \frac{x^2 y^2}{(x+y)^2} \times \left[\frac{1}{1 - \frac{y^2}{(x+y)^2}} \right]$$

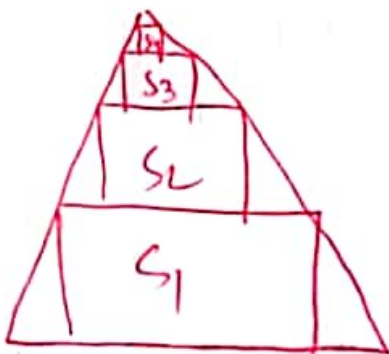
$$= \frac{x^2 y^2}{(x+y)^2} \times \frac{(x+y)^2}{x(x+2y)}$$

$$\text{Area} = \frac{xy^2}{x+2y}$$

[Infinite square & Area के
सम का formula]
(Sum)

$$\left\{ \frac{xy^2}{x+2y} \right\}$$

(For Any triangle)



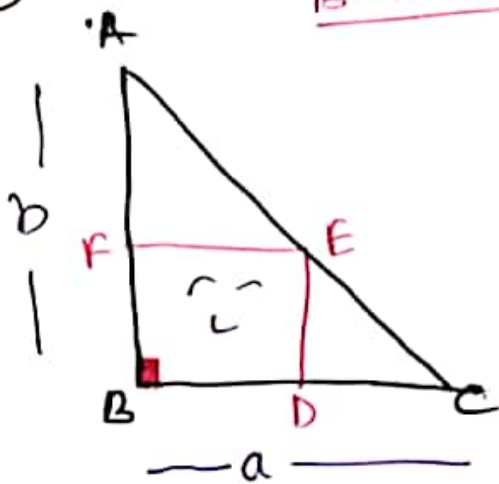
Square inside a right angle Triangle? - (1)

Base - BC

BDEF $\rightarrow$ Square

$$\left[\text{Side of Square BDEF} = \frac{ab}{a+b} \right]$$

~~max~~ (Maximum Area of square बनाना है)



(2) When one side of a square is on hypotenuse.

$$\text{Side of Square PQRS} = \frac{cx \cdot p}{c+p}$$

$$\left[p = \frac{ab}{c} \right]$$

$$= \frac{cx \cdot \frac{ab}{c}}{c + \frac{ab}{c}}$$

$$\boxed{\text{Side} = \frac{abc}{c^2 + ab}}$$

$\square$ PQRS

$$c^2 = a^2 + b^2$$

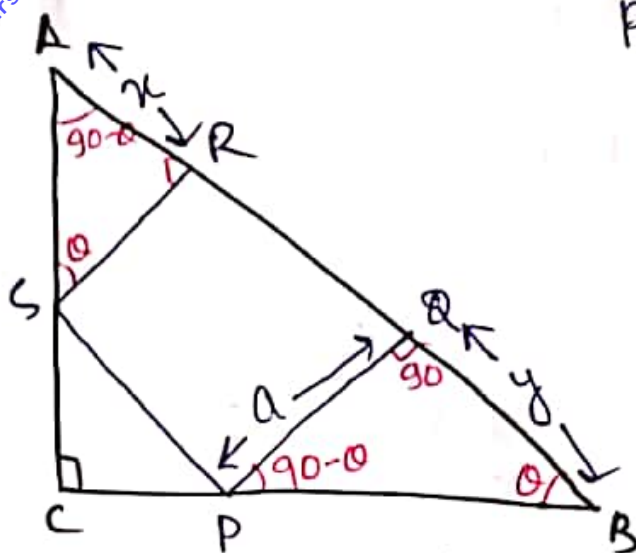
OR $\Rightarrow \frac{abc}{a^2 + b^2 + ab}$

2nd maximum Area of square है

1st wala है
max side of square

~~1st wala~~

③



PQRS $\rightarrow$ Square

⑫

$$\begin{cases} AR = x \\ QB = y \end{cases}$$

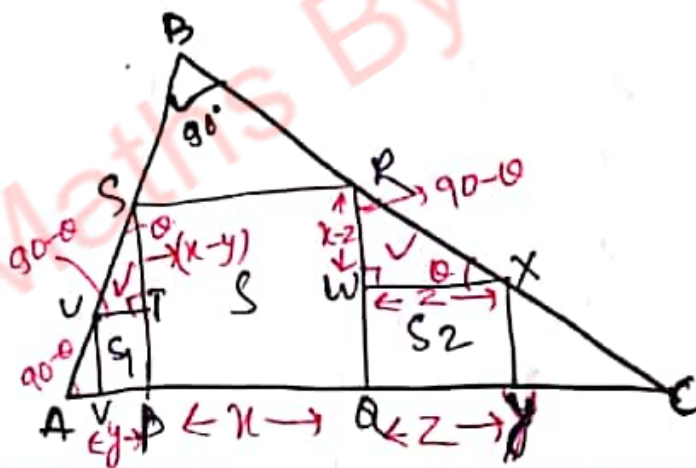
$$\triangle ARS \sim \triangle PQB$$

$$\frac{a}{x} = \frac{y}{a}$$

$$a^2 = xy$$

$$a = \sqrt{xy}$$

④



ALL THREE ARE $\rightarrow$ Square

$$\triangle SUT \sim \triangle KRW$$

$$\frac{y}{x-z} = \frac{x-y}{z}$$

$$yz = x^2 - xy - zx + yz$$

$$x(y+z) = x^2$$

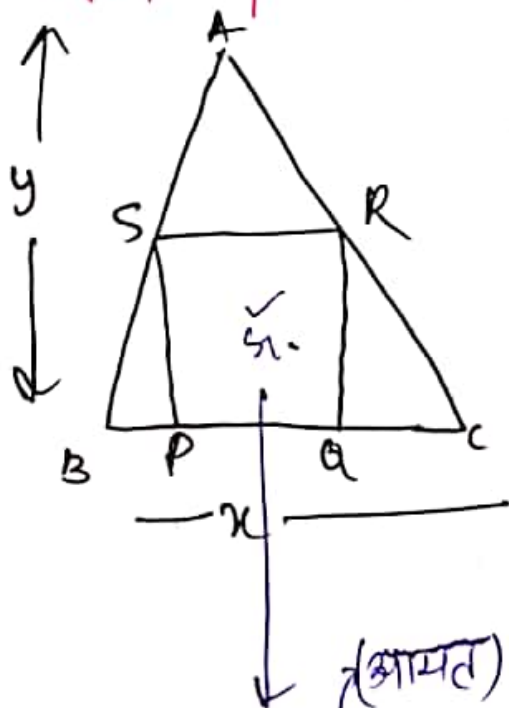
$$x = y+z$$

$\sqrt{x^2} = \sqrt{y^2} + \sqrt{z^2}$
Area $\rightarrow \sqrt{S} = \sqrt{S_1} + \sqrt{S_2}$

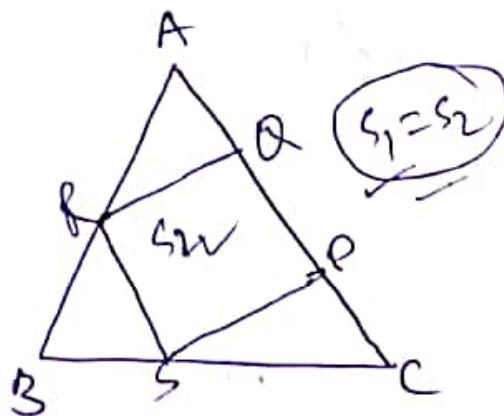
(x, y, z $\rightarrow$ Side of square)

Rectangle inside in triangle →

(13)



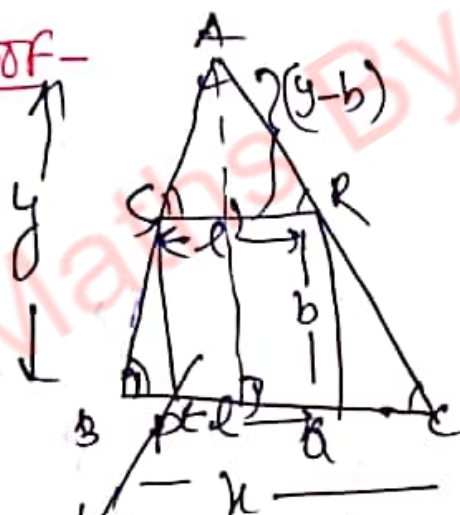
PQRS → Rectangle, max area on base.
BC of $\triangle ABC$



Area of Rectangle (PQRS)
= $\frac{1}{2}$ of Area of $\triangle ABC$

किसी भी Base पर Δ के Max Area का आधा होगा

PROOF-



$$\frac{l}{x} = \frac{y-b}{y}$$

$$\frac{l}{x} = 1 - \frac{b}{y}$$

$$\frac{l}{x} + \frac{b}{y} = 1$$

By using differential calculation

$$l = \frac{x}{2}$$

$$b = \frac{y}{2}$$

Maximize the Area of Rectangle $\frac{1}{2}(\triangle ABC)$
 $\frac{l+b}{2} \geq \sqrt{\frac{l \times b}{y}}$
 $\Rightarrow \frac{1}{2} \geq \sqrt{\frac{l \times b}{y}}$

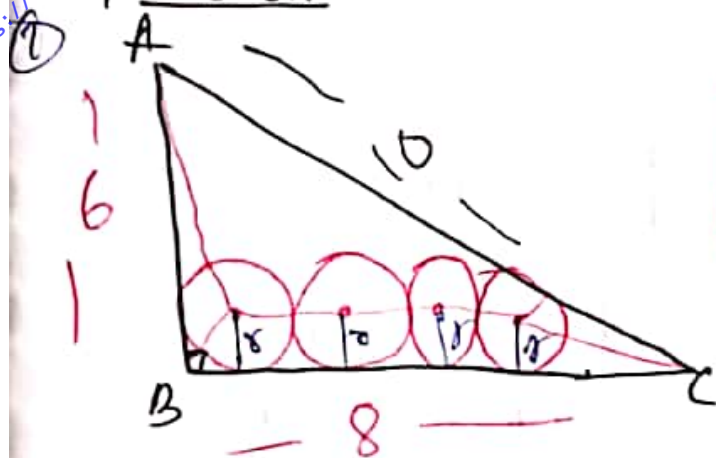
$$\Rightarrow \frac{1}{4} \geq \left(\frac{lb}{xy} \right)$$

$$\Rightarrow \frac{1}{2} \times \left(\frac{1}{2} \times xy \right) \geq lb$$

Area of Δ Area of Rectangle

Circles Inside Right angle Triangle - By Question -

(14)

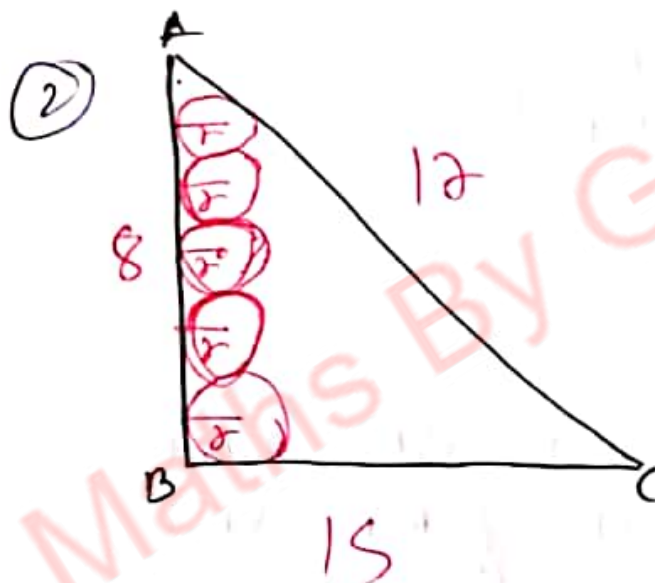


$$r = \frac{\Delta}{S + (n+1) \times h}$$

$$r = \frac{24}{12 + 3 \times 6}$$

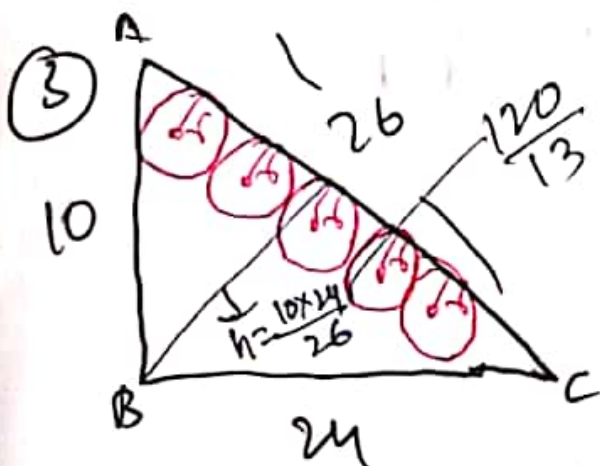
$$r = \frac{24}{30} = \frac{4}{5}$$

$$r = \frac{4}{5} = 0.8 \text{ cm}$$



$$r = \frac{60}{20 + 4 \times 15}$$

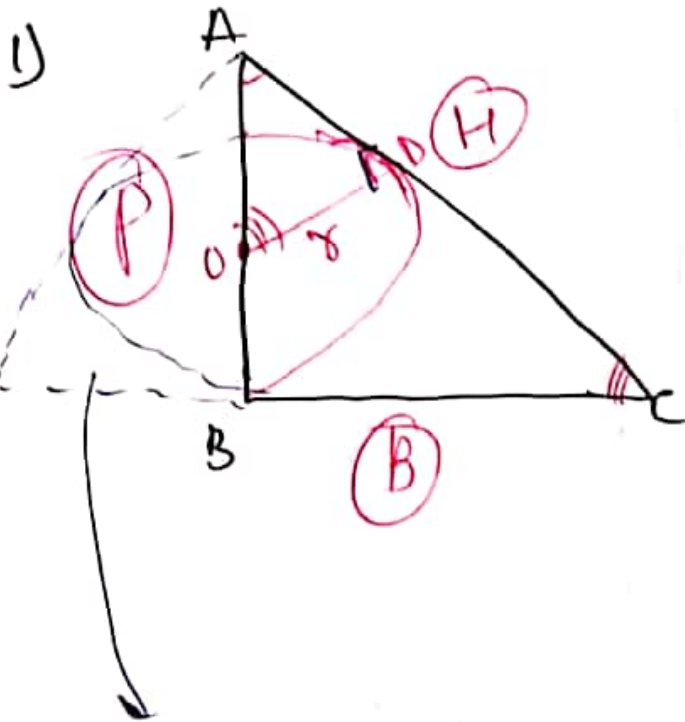
$$r = \frac{60}{80} = \frac{3}{4} = 0.75 \text{ cm}$$



$$r = \frac{120}{30 + 4 \times \frac{120}{13}} \Rightarrow \frac{4}{1 + \frac{16}{13}}$$

$$r = \frac{52 \text{ cm}}{29}$$

Semi Circle inside a right angle $\Delta \rightarrow$ (15)



$$\frac{r}{B} = \frac{P-r}{H}$$

$$rH = PB - Br$$

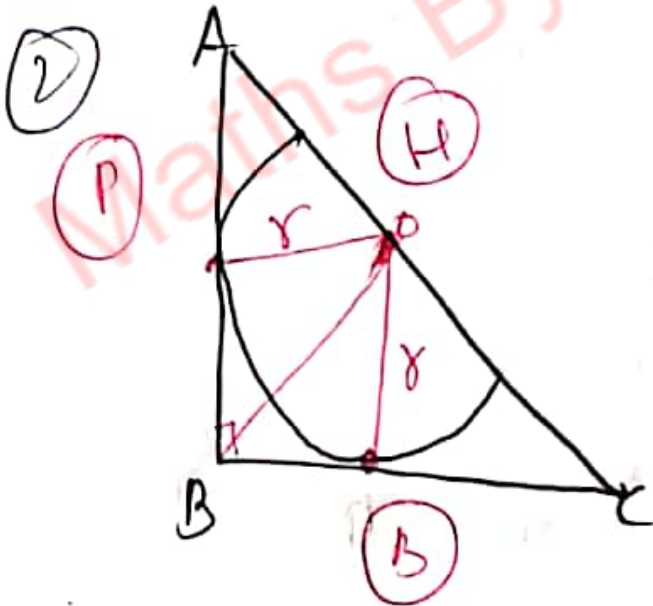
$$r(B+H) = PB$$

$$r = \frac{PB}{B+H}$$

$$r = \frac{\Delta}{S}$$

$$= \frac{\frac{1}{2} \times \angle B \times P}{(H+B)}$$

$$= \frac{BP}{H+B}$$



$$\text{Ar. } \Delta ABC = \Delta BOC + \Delta AOB$$

$$\frac{1}{2} \times PB = \frac{1}{2} \times B \times r + \frac{1}{2} \times P \times r$$

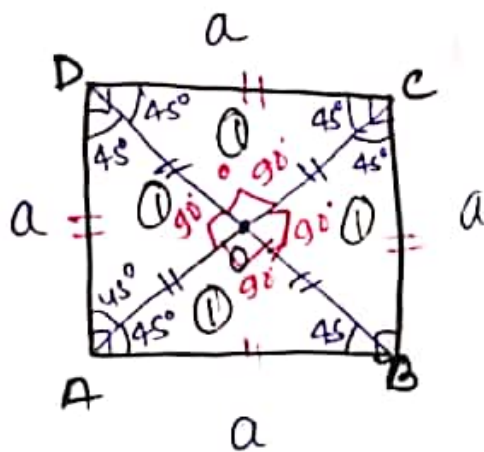
$$PB = r(P+B)$$

$$r = \frac{PB}{P+B}$$

Syedhyang

Square का →

①



- # $AB = BC = CD = AD = a$
- # $\angle A = \angle B = \angle C = \angle D = 90^\circ$
- # Perimeter (परिमाप) $= a + a + a + a = 4a$

Area (क्षेत्रफल) $= a^2$

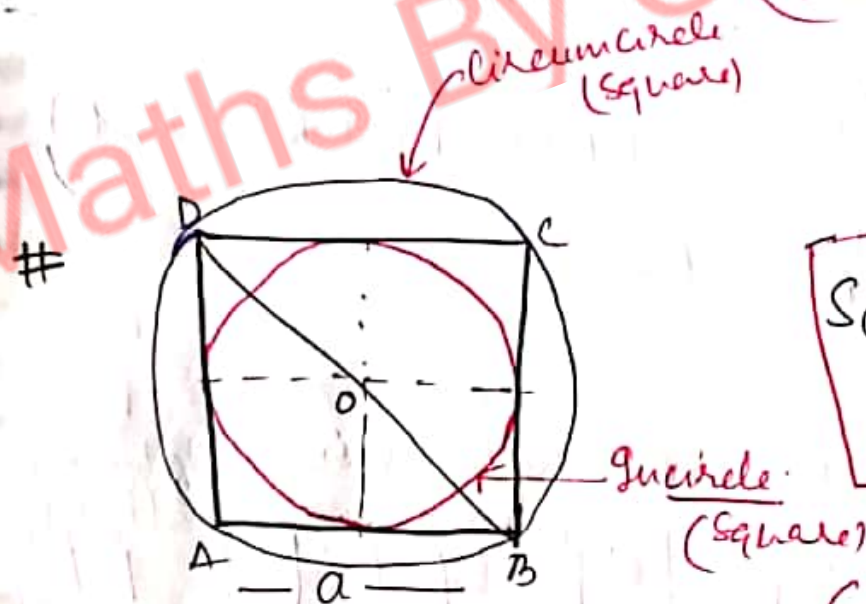
diagonals $= AC = BD = a\sqrt{2}$
(विकर्ष)

diagonals bisect vertex angle

diagonals bisect each other at 90°

$AO = BO = CO = DO$

$Ar[\Delta AOB = \Delta COD = \Delta BOC = \Delta AOD]$

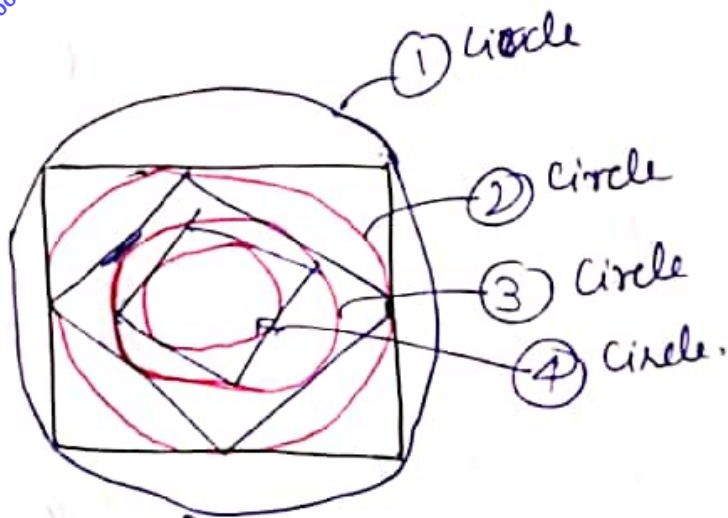


(Incircle)
Square के Circle के
radius $= r = \frac{a}{2}$

Circumcircle $\frac{\text{Area}}{\text{Incircle}} = 2:1$

Circumcircle
Square के Circumcircle
के Radius $= R = \frac{a}{\sqrt{2}}$

$R:r = \frac{a}{\sqrt{2}} : \frac{a}{2} = \sqrt{2}:1$

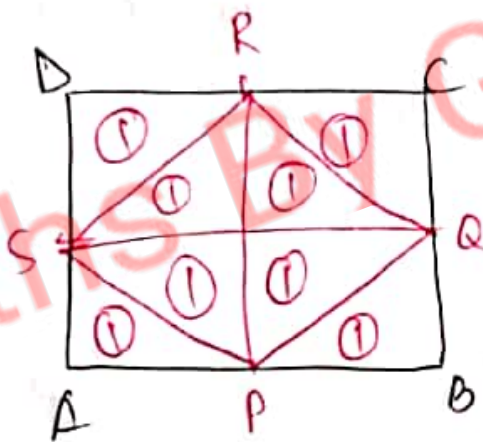


[Area का Ratio] इस प्रकार होगा ↓



$$16 : 8 : 4 : 2 : 1$$

#

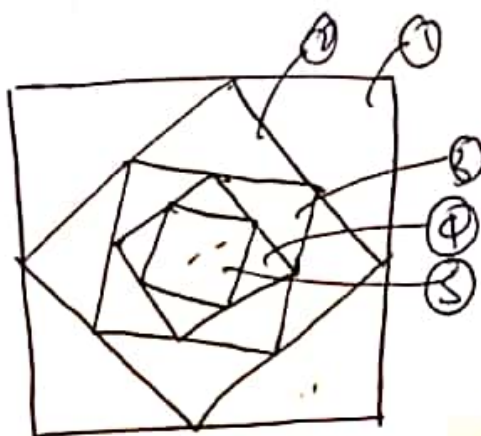


P, Q, R, S → midpoint

$$\square ABCD : \square PQRS = 2 : 1$$

Area ↓

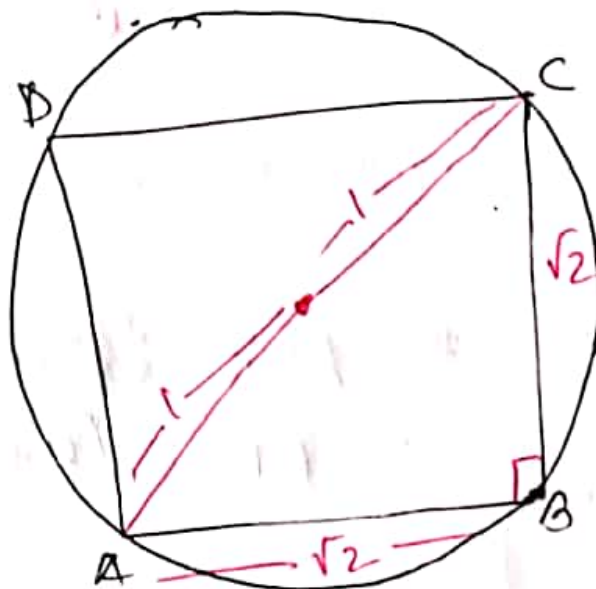
#



⇒ Area का अनुपात आया होता जाता है —

$$\square : \square : \square : \square : \square$$

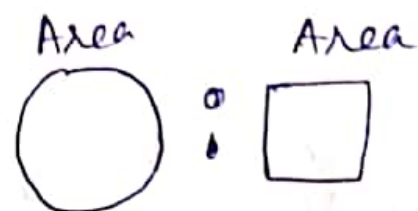
$$16 : 8 : 4 : 2 : 1$$



ABCD $\rightarrow$ Square.

③

अनुपात (Ratio)

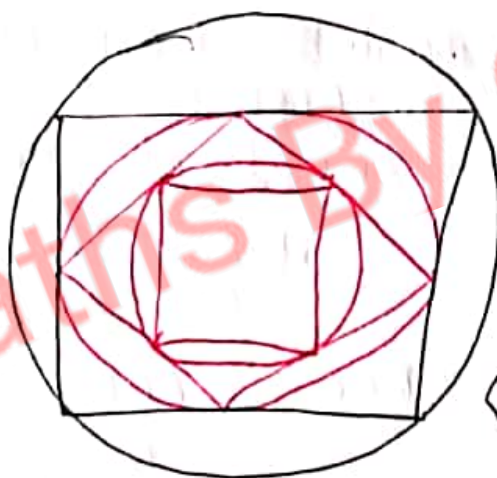


$$\pi(1)^2 : (\sqrt{2})^2$$

$$\boxed{\pi : 2}$$

OR. $\frac{22}{7} : 2$

$$\Rightarrow 11 : 7$$



$\square \rightarrow$ Square.

$$\bigcirc : \square : \bigcirc : \square : \bigcirc : \square$$

Area $\rightarrow 16\pi : 32 : 8\pi : 16 : 4\pi : 8$

$$\pi : 2$$

Half

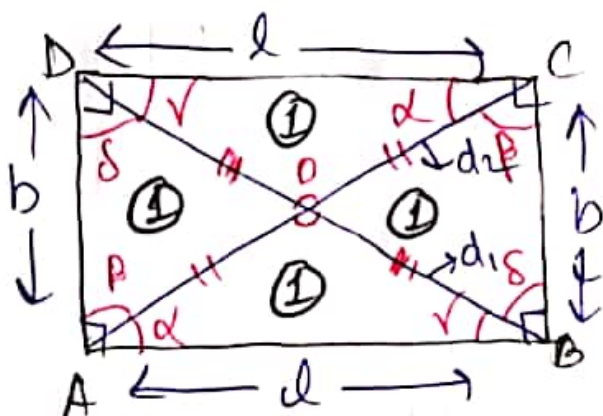
OR. $\Rightarrow 44 : 28 : 22 : 14 : 11 : 7$

$$11 : 7$$



Rectangle - आयत

(4)



$$\# \begin{cases} AC = \sqrt{l^2 + b^2} \\ BD = \sqrt{l^2 + b^2} \end{cases}$$

$$\# \{d_1 = d_2 = \sqrt{l^2 + b^2}\}$$

$$\# AO = CO = BO = DO$$

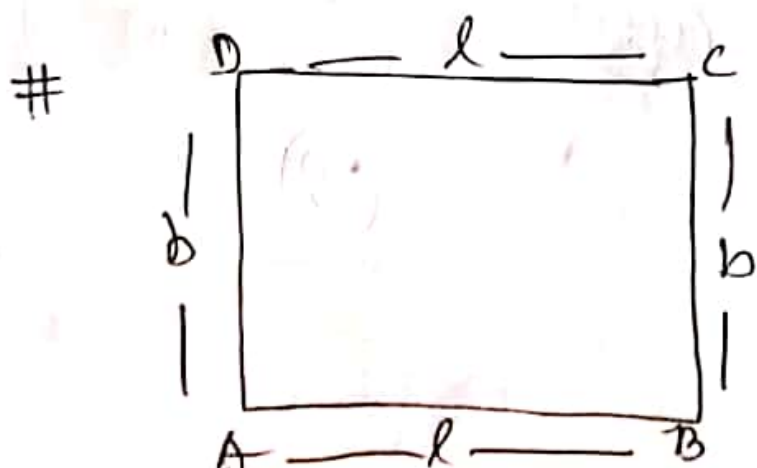
$$\# \begin{cases} AO = CO = \frac{AC}{2} \\ DO = BO = \frac{BD}{2} \end{cases}$$

diagonals bisect each other but not at 90°

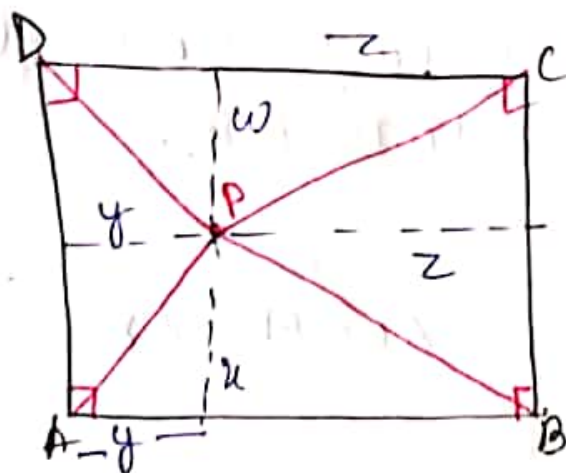
diagonals do not bisect Vertex angle..

Area:-

$$\Delta AOB = \Delta COD = \Delta BOC = \Delta AOD$$



$$\begin{cases} \text{Area (क्षेत्रफल)} = l \times b \\ \text{Perimeter (परिमाप)} = 2(l + b) \end{cases}$$



⑤

① # $\Delta PAB + \Delta PCD = \Delta PBC + \Delta PAD = \frac{1}{2} \text{ area of rectangle } (ABCD)$

$PA^2 + PC^2 = PD^2 + PB^2$

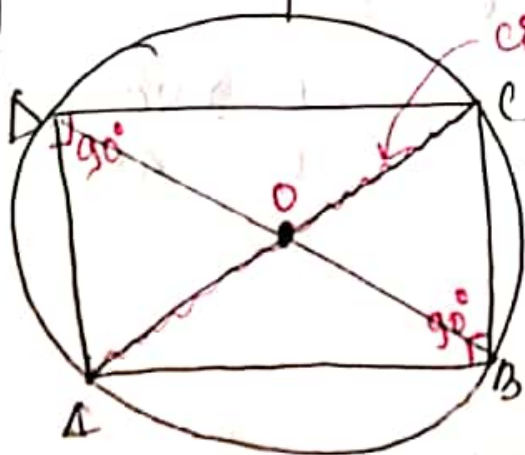
$x^2 + y^2 + w^2 + z^2 = x^2 + z^2 + w^2 + y^2$

BOTH ARE EQUAL

Proved. इति सिद्धम्



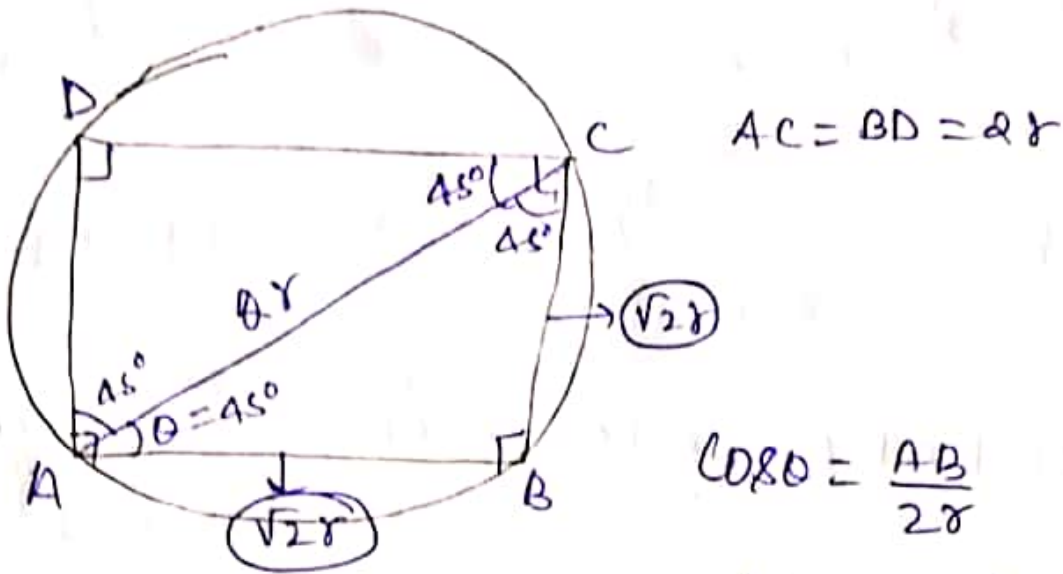
$\Rightarrow$ Every Rectangle is a cyclic quadrilateral.



circle as diameters होगा ?

(6)

Circle के अन्दर maximum area का rectangle (आयत) हमेशा एक Square (वर्ग) होगा.



$$\cos \theta = \frac{AB}{2r}$$

$$AB = 2r \cos \theta$$

$$\sin \theta = \frac{BC}{2r}$$

$$BC = 2r \sin \theta$$

Rectangle.

$$\text{Area of } \square ABCD = 4r^2 \sin \theta \cdot \cos \theta$$

$$= (2r)^2 (2 \cdot \sin \theta \cdot \cos \theta)$$

$$= 2r^2 \sin 2\theta$$

$$\text{For max. Area} \rightarrow \theta \rightarrow 1$$

$$2\theta \rightarrow 90^\circ$$

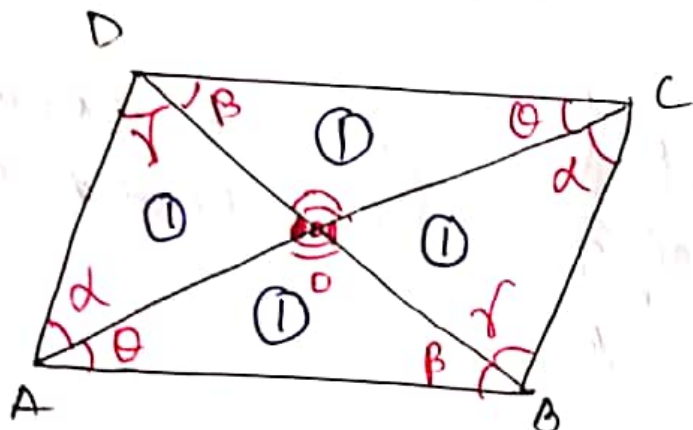
$$\theta \rightarrow 45^\circ$$

$$\left\{ \begin{array}{l} \text{max. Area} = 2r^2 \\ \text{Perimeter} = 4\sqrt{2}r \end{array} \right\}$$



⑧

#



$\triangle AOB \cong \triangle COD$

$AO = CO$

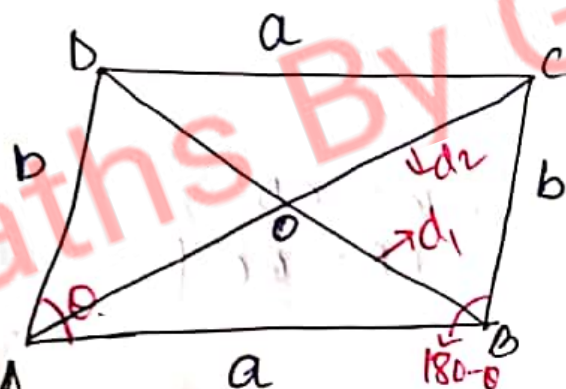
$BO = DO$

(निष्कर्ष)
Diagonals bisect each other?
but not at 90°

$\Rightarrow$ #

$$\triangle AOB = \triangle BOC = \triangle AOD = \triangle DOC = \frac{1}{4} \times \text{area of } \text{parallelogram } ABCD$$

#



$BD \rightarrow d_1$

$AC \rightarrow d_2$

$$d_1 \neq d_2$$

PROOF A

By Cosine rule-

$$d_1^2 = a^2 + b^2 - 2ab \cos \theta \quad \text{--- (1) } \checkmark$$

$$d_2^2 = a^2 + b^2 - 2ab \cos (180 - \theta) \quad \text{--- (2) } \checkmark$$

$$d_2^2 = a^2 + b^2 + 2ab \cos \theta \quad \text{--- (2) } \checkmark$$

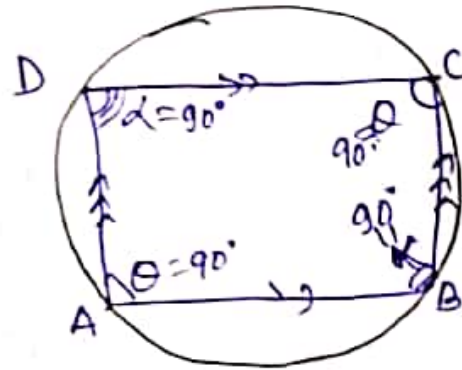
next $\rightarrow$

eqⁿ ① + eqⁿ ②

⑨

$$d_1^2 + d_2^2 = 2(a^2 + b^2)$$

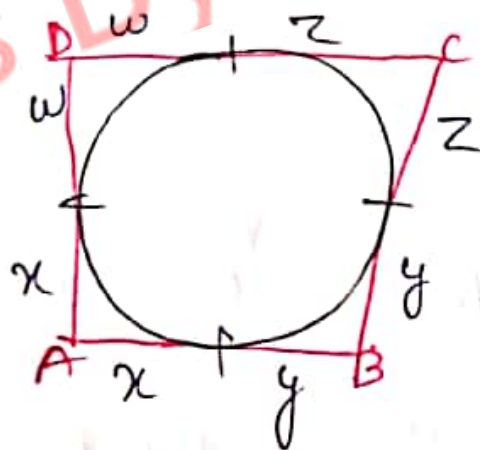
1) Circle के अन्दर बना ॥^{gm} हमेशा एक आयत होता है।



$$2\theta \rightarrow 180^\circ \Rightarrow \theta = 90^\circ$$

$$2\alpha \rightarrow 180^\circ \Rightarrow \alpha = 90^\circ$$

2) parallelogram Circumscribed about a circle is always a rhombus.



$$AB + CD = BC + AD$$

$$\begin{cases} AB = CD \\ AD = BC \end{cases}$$

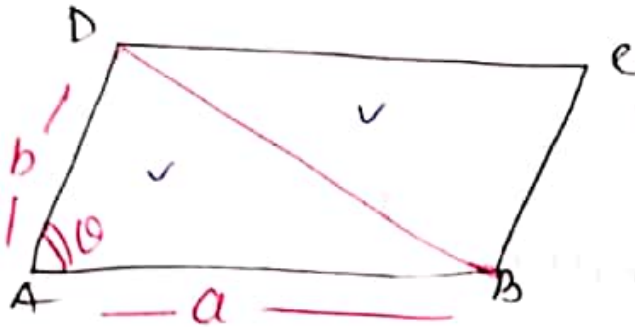
$$\Rightarrow 2AB = 2BC$$

$$AB = BC$$



Area of $\triangle ABD \rightarrow$

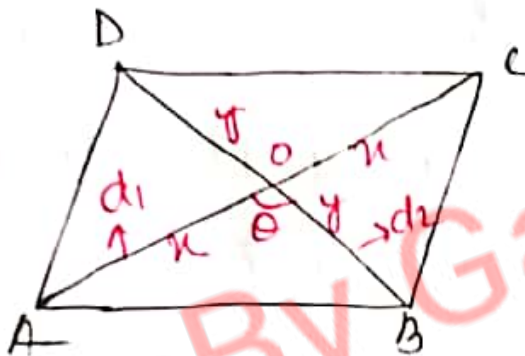
(10)



$$\text{Area}(\triangle ABD) = \frac{1}{2} ab \sin \theta$$

Area of $\triangle ABCD = ab \sin \theta$

(2)



Area = $\frac{1}{2} \times d_1 \times d_2 \sin \theta$

$$\begin{cases} AO = OC \\ OB = OD \end{cases}$$

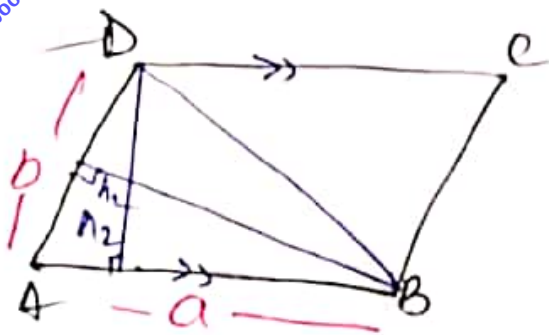
$$\triangle AOB = \frac{1}{2} xy \sin \theta$$

$$\text{Area } ABCD = \frac{1}{2} \times (2x) \times (2y) \sin \theta$$

$\downarrow \quad \downarrow$
 $d_1 \quad d_2$

Area of $\triangle ABCD = \frac{1}{2} \times d_1 \times d_2 \sin \theta$

③



⑪

$$\text{Area of } \triangle ABD = \frac{1}{2} \times a \times h_1$$

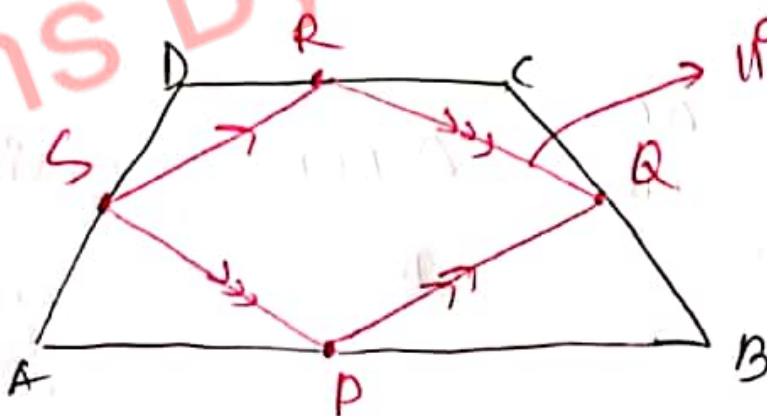
[Double होता है (11^{gm} का area) diagonal]

$$11^{\text{gm}} \text{ का area} = (a h_1)$$

$$11^{\text{gm}} \text{ का area} = (b h_2)$$

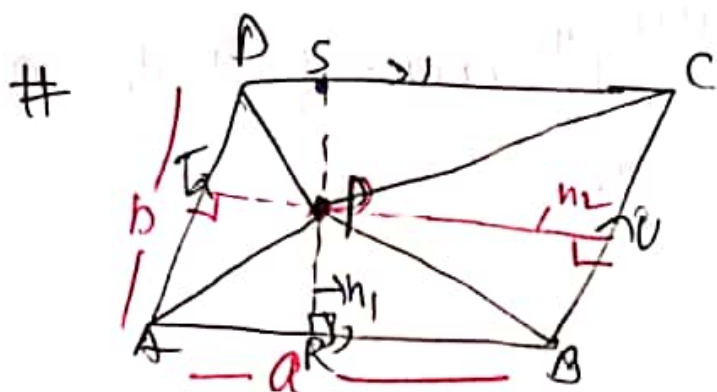
$$a h_1 = b h_2$$

किसी भी quadrilateral के mid point को मिलाने पर हमेशा एक 11^{gm} बनता है। -



11^{gm} होगा (हमेशा)

PQRS → midpoint



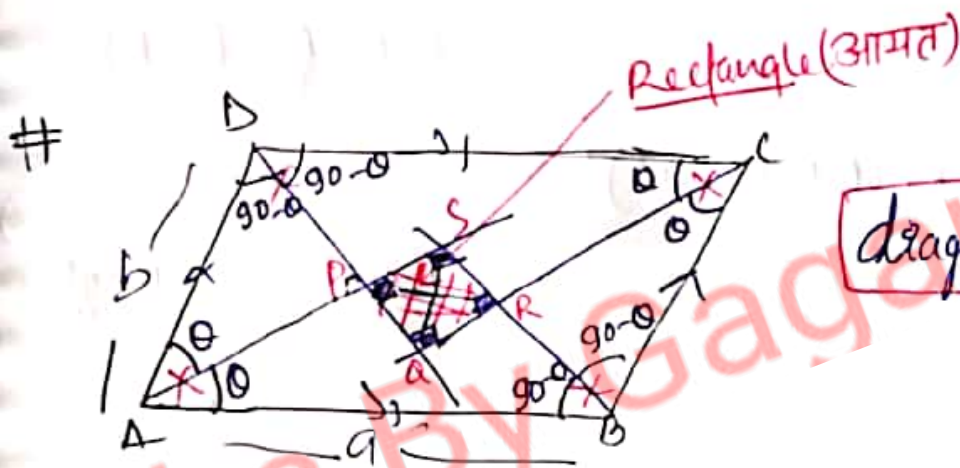
Area:

$$\triangle PAB + \triangle PCD = \triangle PBC + \triangle PAD$$

$$\begin{aligned} & \triangle PAB + \triangle PBD \\ & \Rightarrow \frac{1}{2} a \times PR + \frac{1}{2} a \times PS \\ & = \left(\frac{1}{2} a \times h_1 \right) \\ & = \frac{1}{2} \text{ area of } \parallel^{\text{gm}} \end{aligned}$$

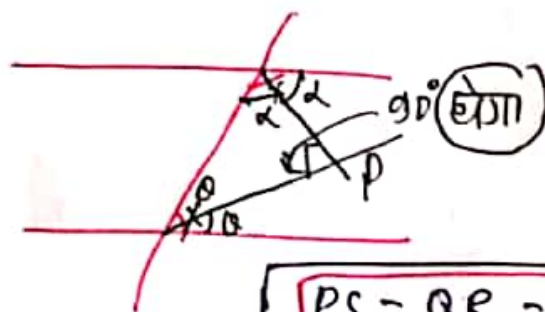
$$\begin{aligned} & \triangle PAD + \triangle PCD \\ & \Rightarrow \frac{1}{2} b \times PR + \frac{1}{2} b \times PS \\ & = \frac{1}{2} b \times h_2 \\ & = \frac{1}{2} \text{ area of } \parallel^{\text{gm}} \end{aligned}$$

Both are equal. इति सिद्धम्



diagonals $\rightarrow PR = QS = a - b$
 difference

$\Rightarrow$ किसी भी $\parallel^{\text{gm}}$ के angle bisector से हमेशा एक Rectangle (आमत) बनता है।



$$\frac{\triangle ASB}{\cos \theta = \frac{AS}{a} \Rightarrow AS = a \cos \theta}$$

$$\frac{\triangle APD}{\cos \theta = \frac{AP}{b} \Rightarrow AP = b \cos \theta}$$

$PS = QR = AS - AP = (a - b) \cos \theta$

(13)

$\triangle DQC$ में —

$$\cos(90^\circ - \theta) = \frac{DQ}{a} \Rightarrow DQ = a \sin \theta$$

$\triangle DPA$

$$\cos(90^\circ - \theta) = \frac{DP}{b} \Rightarrow DP = b \sin \theta$$

$$PQ = RS = DQ - DP = (a - b) \sin \theta$$

$$PR = QS = \sqrt{PQ^2 + PS^2} = a - b$$

let $x = (a - b)$

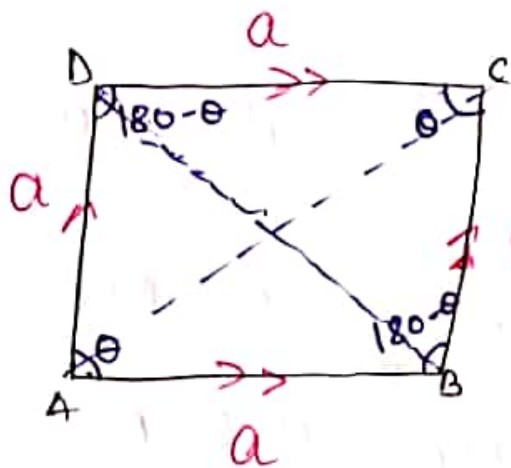
#

$$\text{Perimeter} = 2x(\cos \theta + \sin \theta)$$

@ $\rightarrow$ half vertex angle

Rhombus (समचतुर्भुज)

14



$$AB = BC = CD = DA = a$$

$$\angle A = \angle C = \theta$$

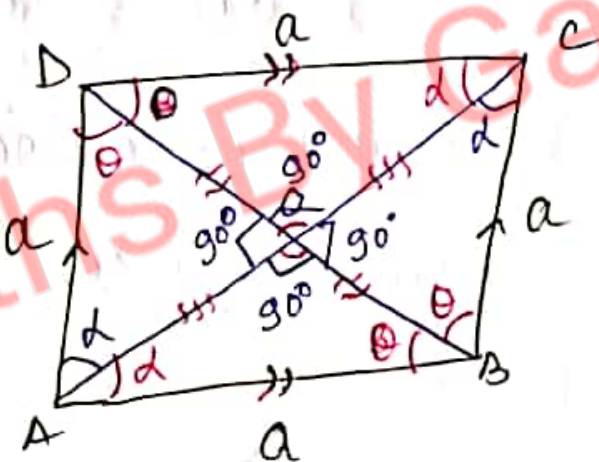
$$\angle B = \angle D = 180 - \theta$$

$$AC \neq BD$$

diagonals not equal

$\theta \rightarrow$ BD (acute) $\rightarrow$ छोटा

$180 - \theta \rightarrow$ AC (obtuse) $\rightarrow$ बड़ा



$$AC \neq BD$$

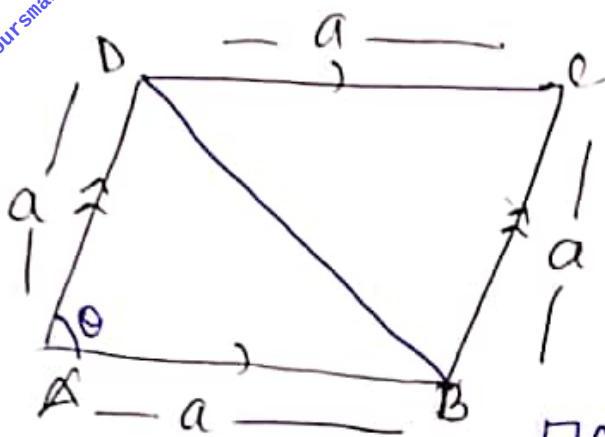
diagonal bisect vertex angle

$$DO = BO = \frac{BD}{2}$$

$$CO = AO = \frac{AC}{2}$$

diagonal bisect each other at 90°

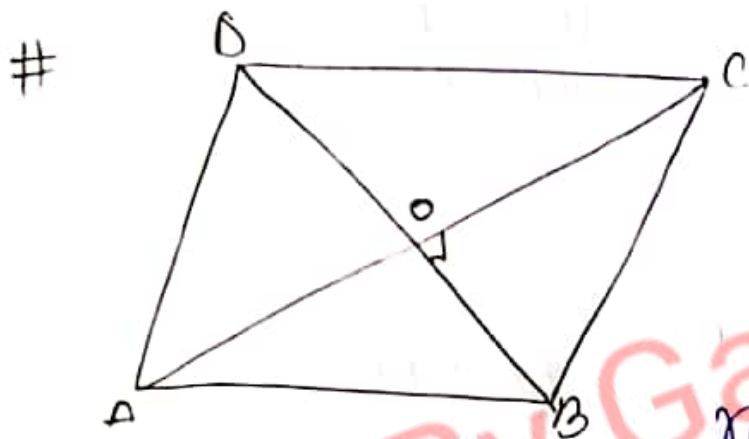
$$\begin{cases} 2\alpha + 2\theta = 180 \\ \alpha + \theta = 90 \end{cases}$$



$$\text{Perimeter} = 4a$$

$$\square ABCD \text{ Area} = \frac{1}{2} a \times a \sin \theta \times 2$$

$$\text{Area} = a^2 \sin \theta$$



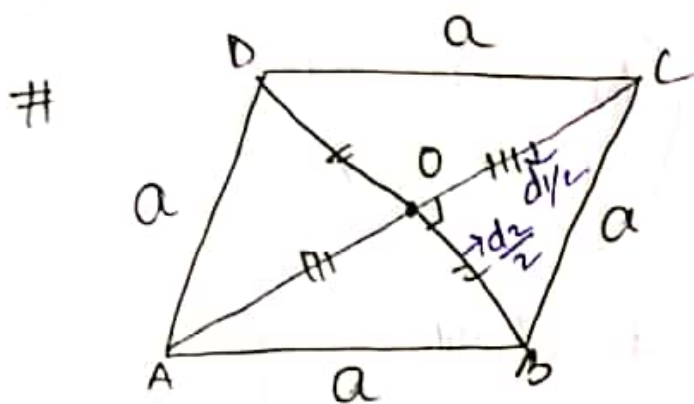
$$\begin{cases} AC = d_1 \\ BD = d_2 \end{cases}$$

$$\text{Rhombus area} = \frac{1}{2} d_1 d_2 \sin \theta$$

$$\text{Rhombus area} \Rightarrow \theta = 90^\circ$$

$$(\sin 90^\circ = 1)$$

$$\text{Area} = \frac{1}{2} d_1 \times d_2$$



$$\begin{aligned} AC &= d_1 \\ BD &= d_2 \end{aligned}$$

$$OC = \frac{AC}{2} = \frac{d_1}{2}$$

$$OB = \frac{BD}{2} = \frac{d_2}{2}$$

$$\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2 = a^2$$

$$d_1^2 + d_2^2 = 4a^2$$



Any 2
1) Quadrilateral

midpoints

parallelogram

2) Square

midpoint

Square

3) Rectangle

midpoint

Rhombus

4) Rhombus

midpoint

Rectangle

5) 11 cm

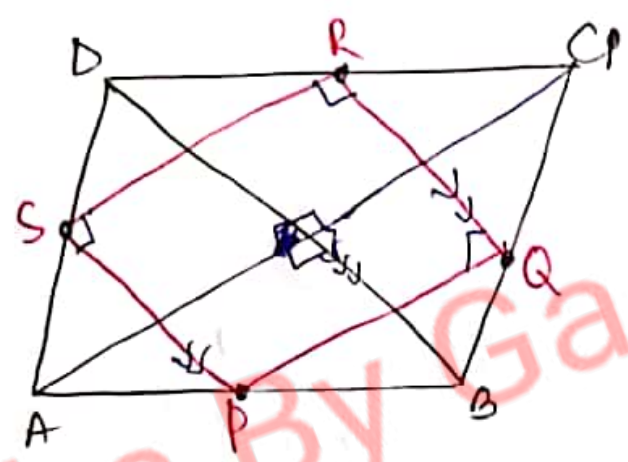
midpoint

11 cm

11 cm & angle bisector → rectangle

[सब के mid point को मिलावे पर बने चतुर्भुज]

#



PQRS → rectangle

$$PQ = \frac{AC}{2} \quad RS \text{ (midpoint)}$$

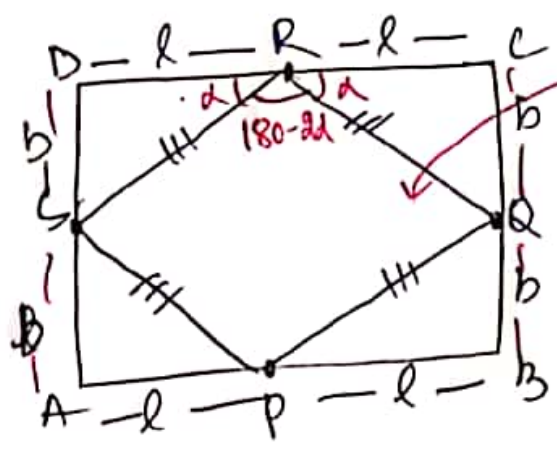
$$RQ = \frac{BD}{2} = SP \text{ (midpoint)}$$

$$AC \neq BD$$

चारों length आपस में बराबर नहीं हैं



#



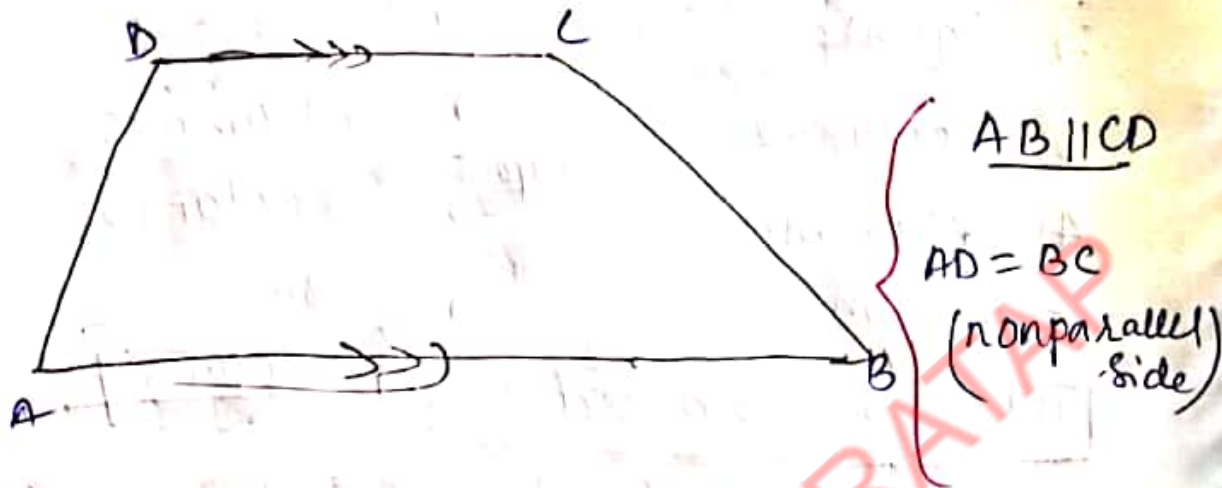
$$d \neq 45^\circ \quad (180 - 2A \neq 90)$$

PQRS → rhombus

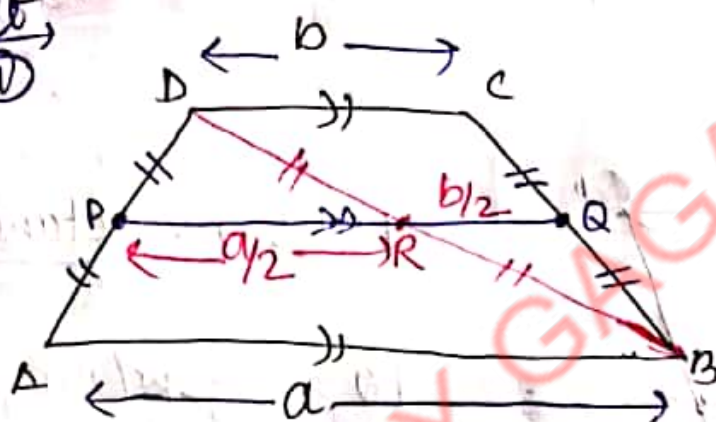
Gagan Prakashan

Trapezium: - (समलम्ब)

①



Result,
①



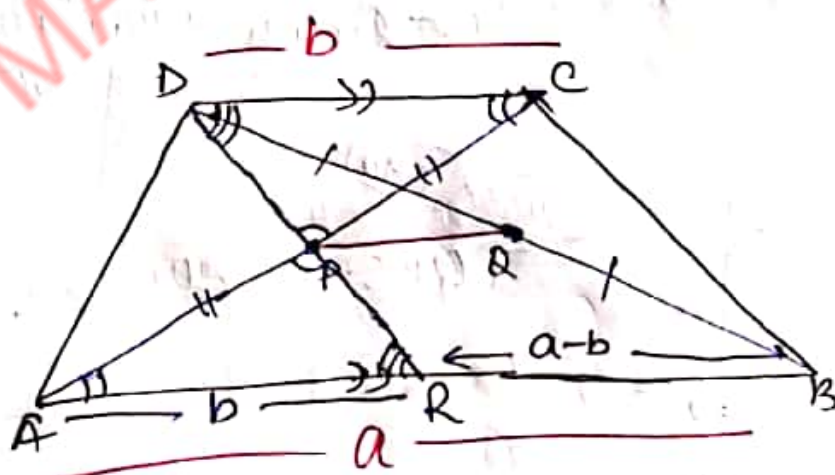
$\{PQ \parallel AB \parallel CD\}$

$$PQ = \frac{a+b}{2}$$

PROOF

$$\left\{ \begin{aligned} PQ &= PR + RQ \\ RQ &= \frac{a}{2} + \frac{b}{2} = \left(\frac{a+b}{2} \right) \end{aligned} \right\}$$

②
#



$$PQ = \frac{a-b}{2}$$

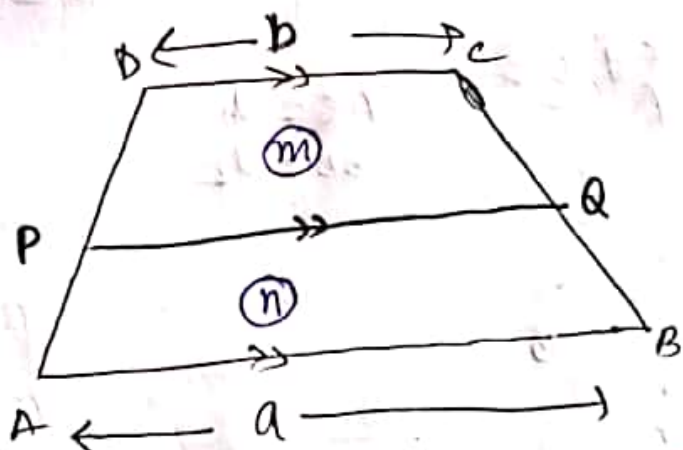
$\triangle DRB$ में

$$PQ = \frac{RB}{2} = \frac{a-b}{2}$$

By: midpoint
Theorem

③

②



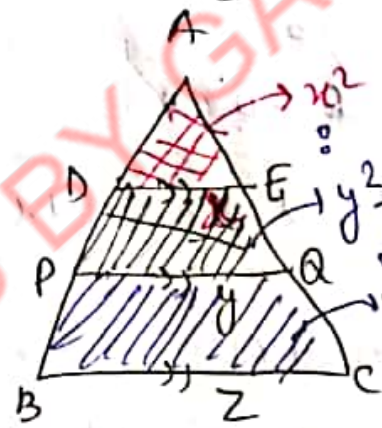
$PQ \parallel AB \parallel DC$
 $\square PQCD : \square APQB = m:n$
 (Area)
 $PQ = ?$

$$PQ = \sqrt{\frac{ma^2 + nb^2}{m+n}}$$

if $m = n$ or $m:n = 1:1$

$$PQ = \sqrt{\frac{a^2 + b^2}{2}}$$

PROOF: Concept



Area Ratio
 $\triangle ADE : \triangle APQ : \triangle ABC$
 $x^2 : y^2 : z^2$

EPQR में (OPC PQ का Area)

$$x^2 - b^2 = m \quad \text{--- (1)}$$

AEB में

$$\square APQB \text{ का Area}$$

$$a^2 - x^2 = n \quad \text{--- (2)}$$

(1) $\div$ (2)

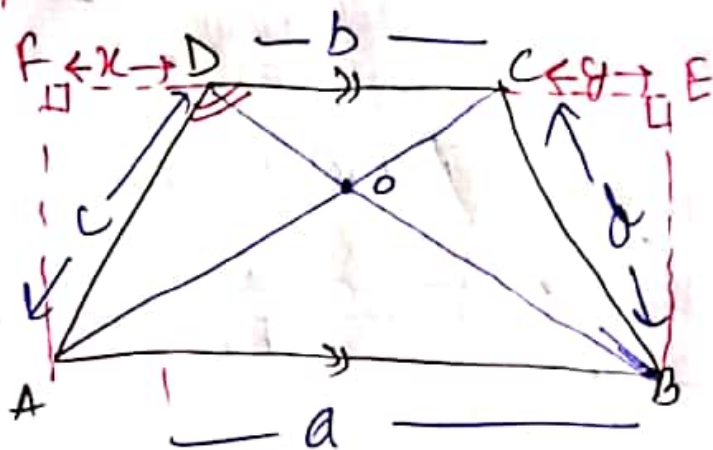
$$\frac{x^2 - b^2}{a^2 - x^2} = \frac{m}{n}$$

$$= nx^2 - b^2n = a^2m - mx^2$$

$$\Rightarrow \sqrt{x^2 = \frac{ma^2 + nb^2}{m+n}} \quad \text{[शिव सिद्धम]} \quad \text{Suryam}$$

④

③



$$AC = d_1$$

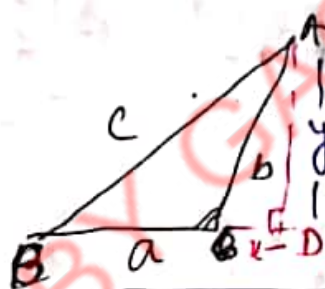
$$BD = d_2$$

Result →

Imp.

$$d_1^2 + d_2^2 = c^2 + d^2 + 2ab$$

Concept:-
Concept:-



$$c^2 = a^2 + b^2 + 2ax$$

$\triangle ADB$

$$c^2 = y^2 + (a+x)^2$$

$$c^2 = y^2 + a^2 + x^2 + 2ax$$

$$c^2 = a^2 + (y^2 + x^2) + 2ax$$

$$c^2 = a^2 + b^2 + 2ax$$

$\triangle ADC$

$$y^2 + x^2 = b^2$$

$\triangle ADC$ में -

$$d_1^2 = c^2 + b^2 + 2bx \quad \text{--- (1)}$$

$\triangle DCB$ में -

$$d_2^2 = b^2 + c^2 + 2by \quad \text{--- (2)}$$

ADD →

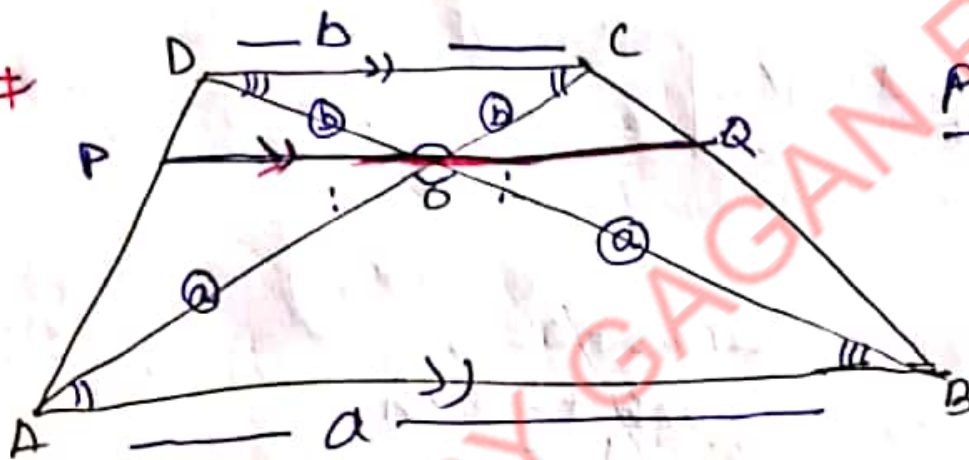
$$d_1^2 + d_2^2 = c^2 + b^2 + b^2 + c^2 + 2b(x+y)$$

↓

$$\Rightarrow d_1^2 + d_2^2 = e^2 + d^2 + 2b(b+x+y) \quad [b+x+y=a]$$

$$\Rightarrow \boxed{d_1^2 + d_2^2 = e^2 + d^2 + 2ab} \quad \text{इति सिद्धम्} \quad (3)$$

(5) #



PQ || AB || CD

#

$$\boxed{PQ = \frac{2ab}{a+b}}$$

PROOF -

$$\Delta AOB \sim \Delta COD$$

$$\left\{ \begin{array}{l} DO : OB = b : a \\ CO : OA = b : a \\ DC : AB = b : a \end{array} \right.$$

By Similarity

ΔDBA

$$\frac{PO}{a} = \frac{b}{a+b}$$

$$PO = \frac{ab}{a+b}$$

Similarly $\Delta DCB \quad QO = \frac{ab}{a+b}$

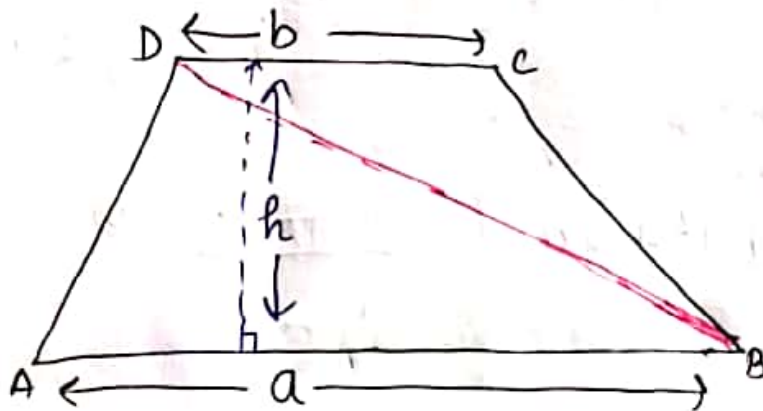
$$PQ = PO + QO$$

$$\boxed{PQ = \frac{2ab}{a+b}}$$



Satyam

Area of Trapezium —



$$\text{Area} = \frac{1}{2}(a+b) \times h$$

PROOF.

Area.

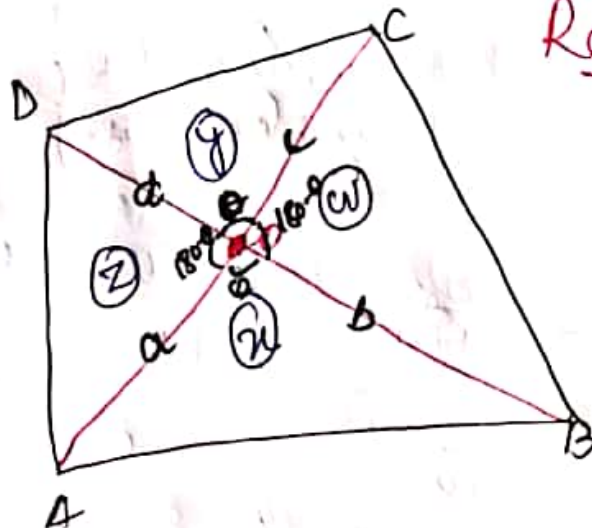
$$\square ABCD = \triangle ADB + \triangle DCB$$

$$\text{Area} = \frac{1}{2} a \times h + \frac{1}{2} b \times h$$

$$\text{Area} = \frac{1}{2}(a+b) \times h =$$

For any general quadrilateral —

$$\left. \begin{array}{l} \text{Area} \\ \triangle AOB = x \\ \triangle BOC = w \\ \triangle COD = y \\ \triangle AOD = z \end{array} \right\}$$



Result —

$$x \times y = z \times w$$

PROOF

⑥

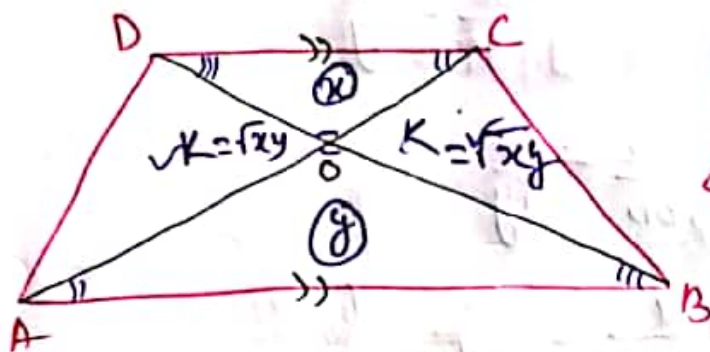
$$X \times Y = \frac{1}{2} xab \sin \theta \times \frac{1}{2} cd \sin \theta$$

$$Z \times W = \frac{1}{2} xbc \sin \theta \times \frac{1}{2} xad \sin \theta$$

→ BOTH ARE EQUAL

$$X \times Y = Z \times W$$

#



$$\begin{cases} \Delta AOB = y \\ \Delta COD = x \end{cases}$$

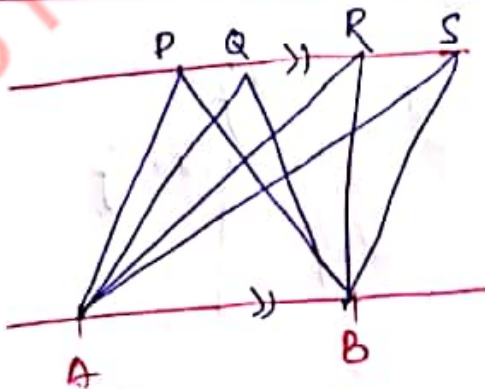
$$X \times Y = k \times k$$

$$k = \sqrt{xy}$$

* ΔAOB & ΔCOD will always be similar triangle

* Area of $\Delta AOD = \text{Area of } \Delta BOC$

Theorem -



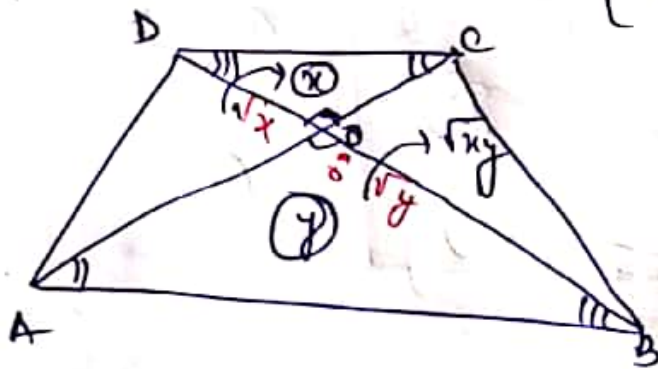
$$\text{Area} [\Delta APB = \Delta AQB = \Delta ARB = \Delta ASB]$$

→ दो समान्तर रेखाओं पर एक ही आधार पर बने Δ , $\square$, त्रिभुज, चतुर्भुज (Triangle) (quadrilateral) के क्षेत्रफल हमेशा बराबर होंगे। (Area) (Equal)

Explanation -

$$\left\{ \begin{array}{l} \sqrt{x} = x \\ \sqrt{y} = y \end{array} \right.$$

(7)



$$\left\{ \begin{array}{l} \text{area } \triangle COD = x \text{ cm}^2 \\ \text{area } \triangle AOB = y \text{ cm}^2 \end{array} \right.$$

$$\frac{DO}{OB} = \frac{\sqrt{x}}{\sqrt{y}}$$

$$\frac{OC}{OA} = \frac{\sqrt{x}}{\sqrt{y}}$$

Details explanation,

$$\frac{\triangle DOC}{\triangle AOD} = \frac{\sqrt{x}}{\sqrt{y}}$$

$$\frac{x}{\triangle AOD} = \frac{\sqrt{x}}{\sqrt{y}}$$

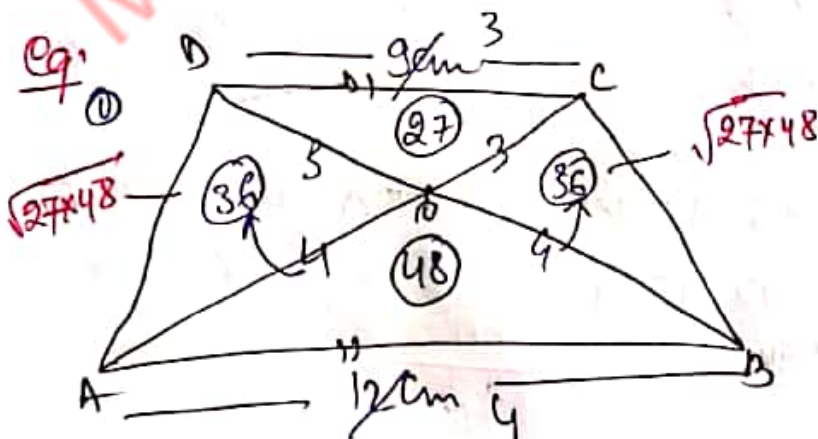
$$\triangle AOD = \sqrt{xy}$$

$$\square ABCD = x + y + 2\sqrt{xy}$$

$$= (\sqrt{x})^2 + (\sqrt{y})^2 + 2 \times \sqrt{x} \times \sqrt{y}$$

#

$$\text{Area } \square ABCD = (\sqrt{x} + \sqrt{y})^2$$



$$\triangle COD = 27 \text{ cm}^2$$

$$\triangle AOB = 48 \text{ cm}^2$$

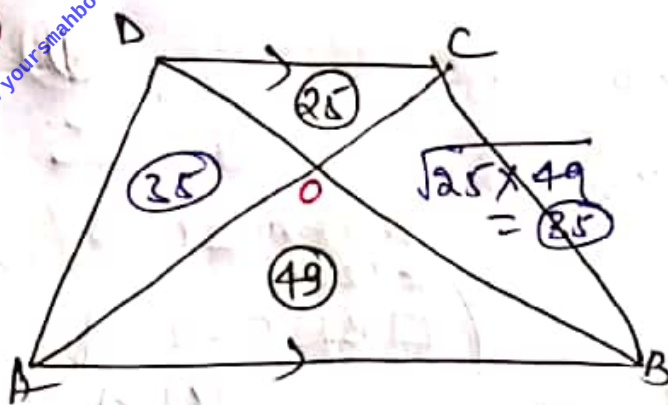
$$\triangle BOC = 36 \text{ cm}^2$$

$$\triangle AOD = 36 \text{ cm}^2$$

$$\square ABCD = 147 \text{ cm}^2$$

$$\begin{array}{l} 9:16 \\ \times 3 \quad \times 3 \\ 27 \quad 48 \end{array}$$

②



Area

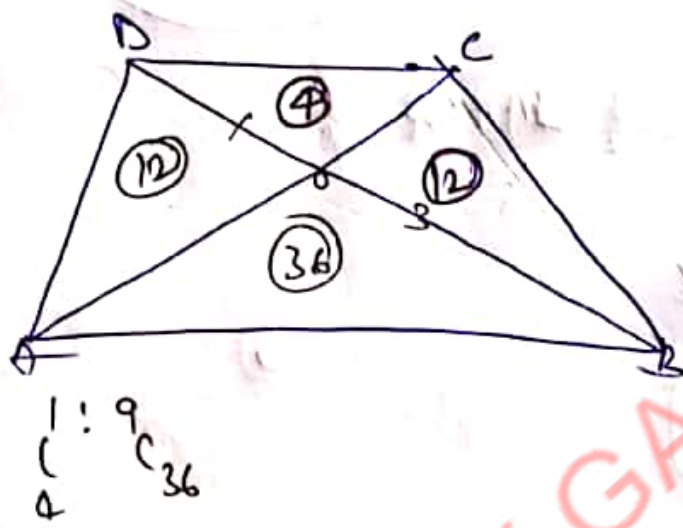
$\square ABCD = ?$

$= 144 \text{ unit}^2$
Ans

$\sqrt{xy}$

⑧

③

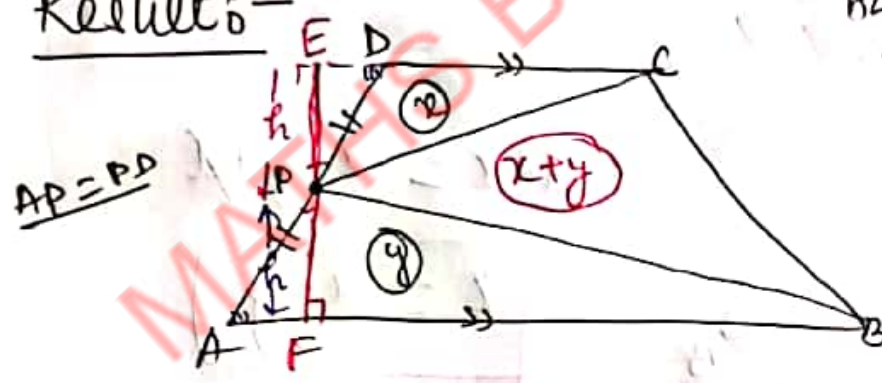


Area

$\square ABCD = ?$

64 Ans.

Result:-



Area

$\triangle DPC = x$
 $\triangle APB = y$

$\triangle BPC = ?$

$\triangle BPC = x+y$

$= \frac{1}{2} \square ABCD$

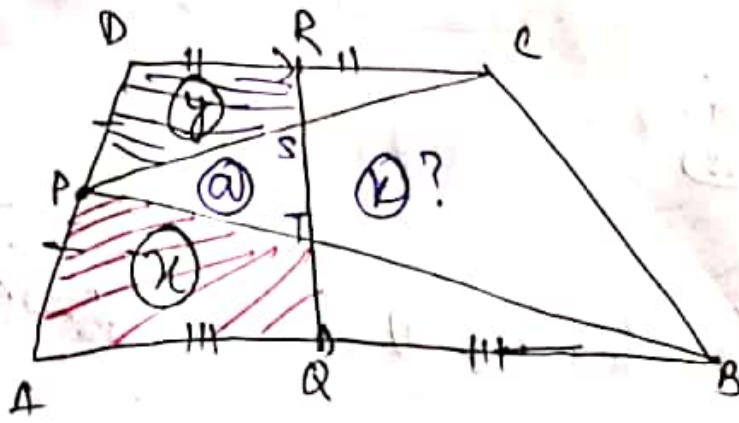
$\triangle APF \cong \triangle DPE$

$x+y = \frac{1}{2} b \times h + \frac{1}{2} a \times h$
 $= \frac{1}{2} (a+b) \times (2h) \times \frac{1}{2}$

Trapezium ki Area

$x+y = \frac{\square ABCD}{2}$

😊

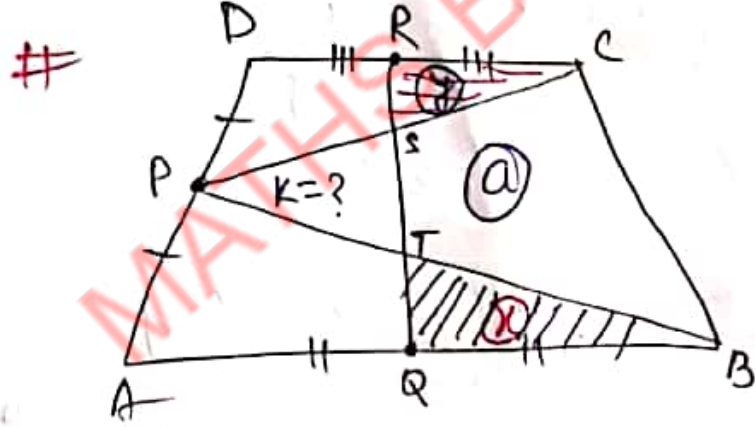


Area $\rightarrow$
 $\square BTSC = ?$
 $= (x+y)$

$$\triangle BPC = \square AQRD = \frac{1}{2} \square ABCD$$

$$x+k = x+y+x$$

$$k = x+y$$

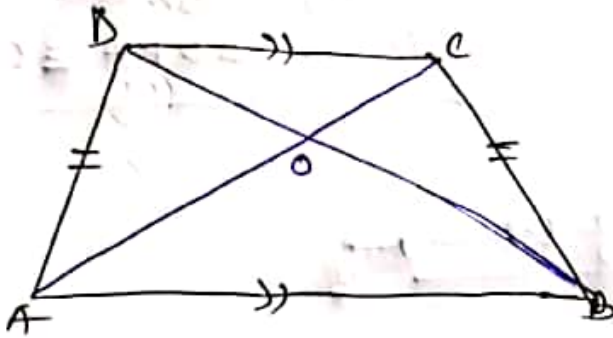


$$x+k = x+y+x$$

$$k = x+y$$

$$\text{Area } \triangle PTS = (x+y)$$

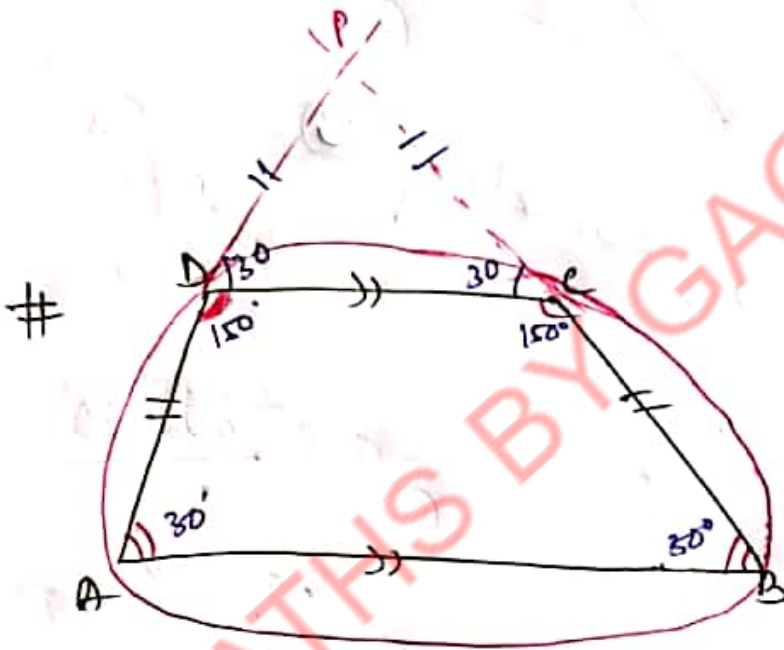
Isosceles trapezium (समद्विबाहु समलम्ब चतुर्भुज) (10)



$AB \parallel CD$ & $AD = BC$
Then: $\triangle ABC$ Isosceles होगा

Result

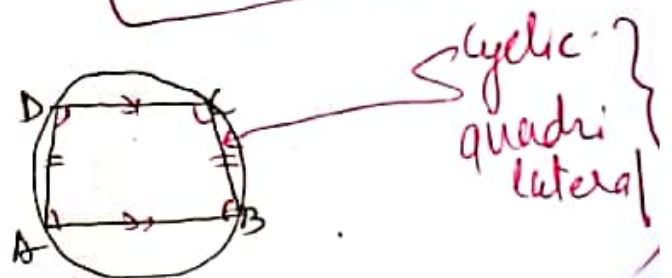
$AC = BD$
diagonals are equal.
But diagonal do not bisect each other



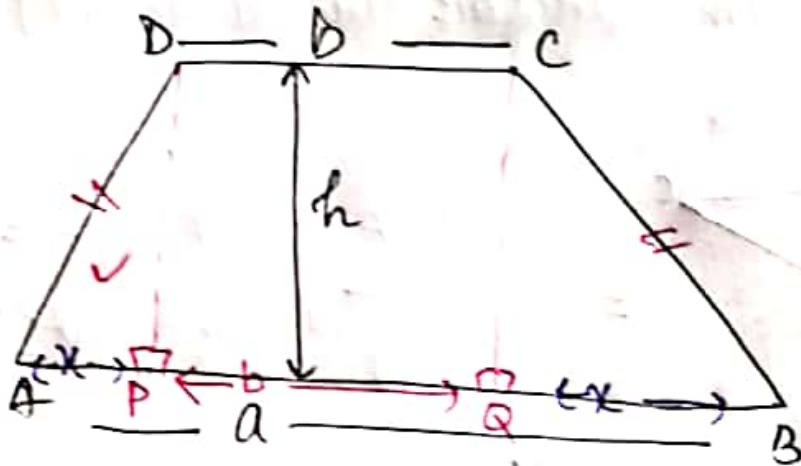
$$\begin{aligned} \angle A &= \angle B \\ \angle D &= \angle C \end{aligned}$$

$$\begin{aligned} \angle A + \angle D &= 180^\circ \\ \angle B + \angle C &= 180^\circ \\ \angle A + \angle C &= 180^\circ \\ \angle B + \angle D &= 180^\circ \end{aligned}$$

{ Isosceles trapezium is always
a cyclic quadrilateral.
(चक्रीय) }



#



congruent $\triangle$ s.

$$\triangle DPA \cong \triangle CQB$$

(11)

$$\square ABCD = (b+x) \times h$$

Satyam Padhyay

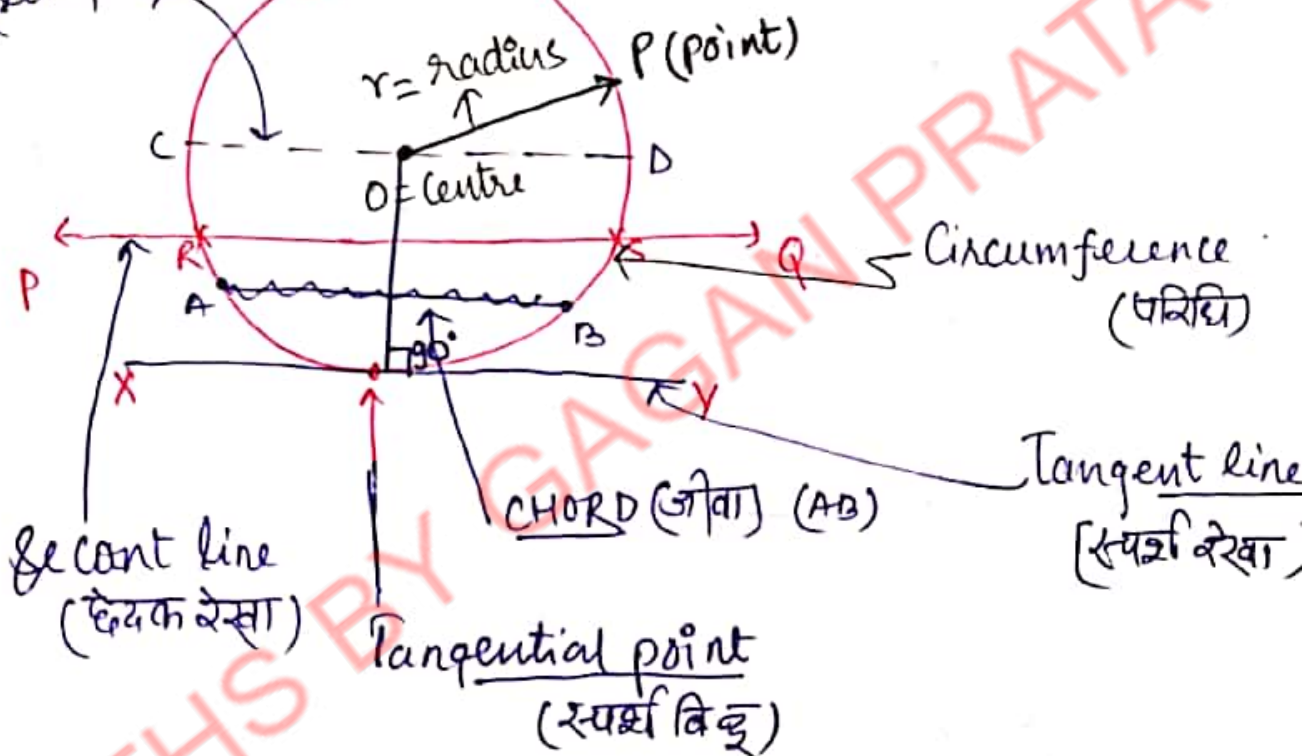
END Geometry 😊

[Circle] वृत्त

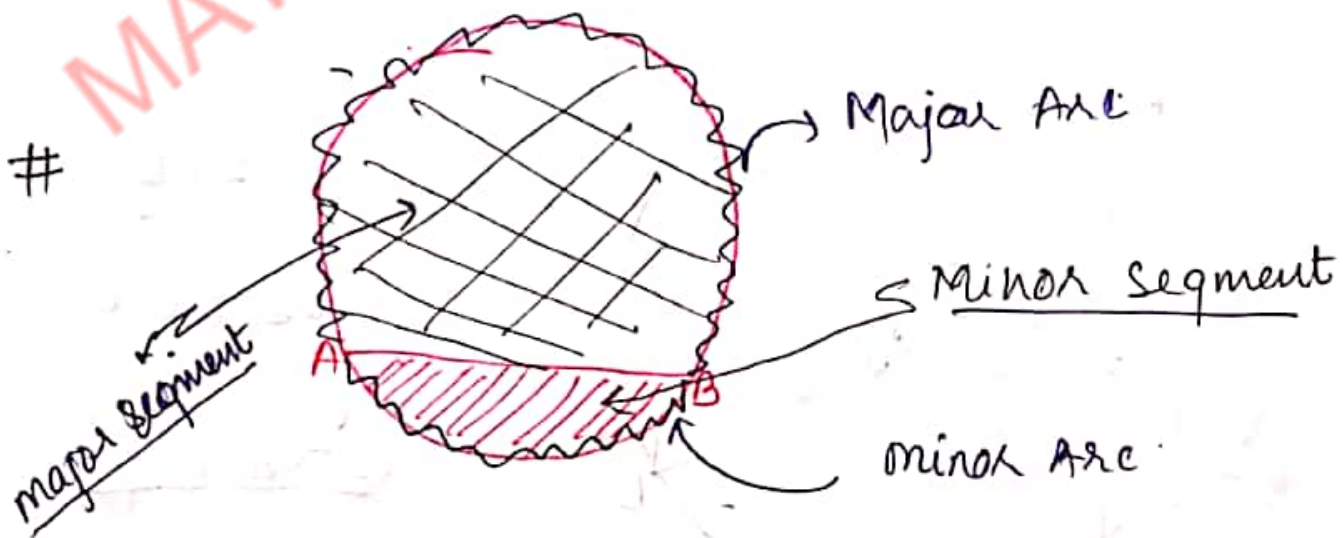
①

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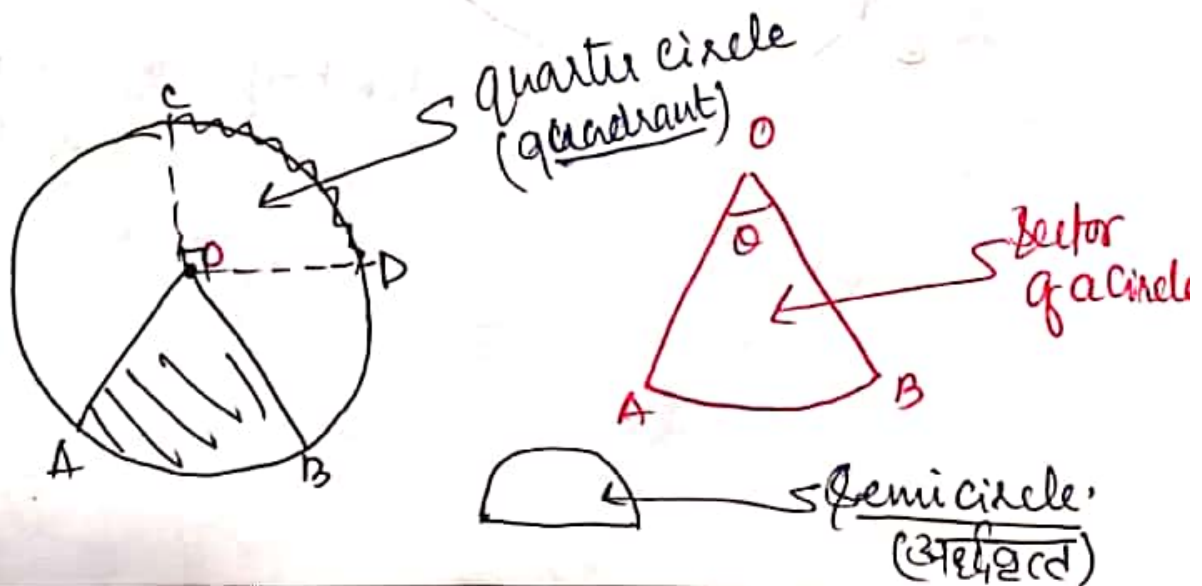
(longest CHORD) or diameter (व्यास)

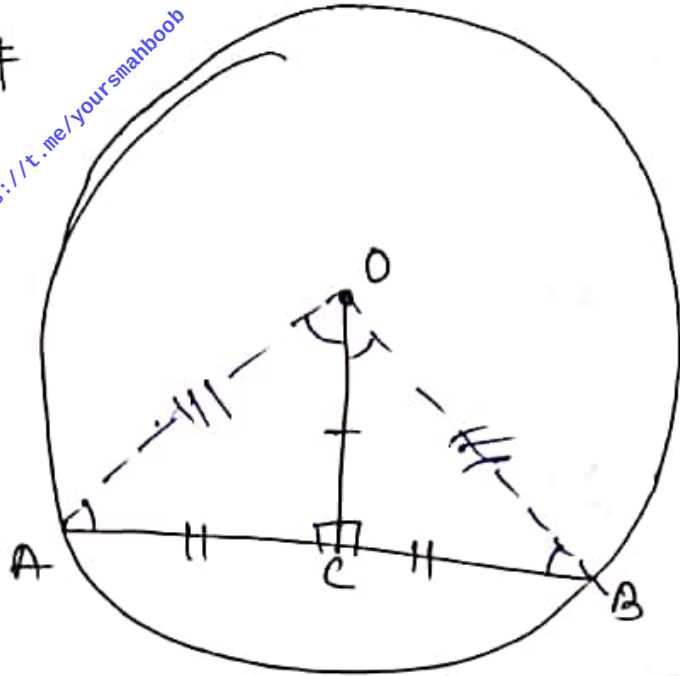


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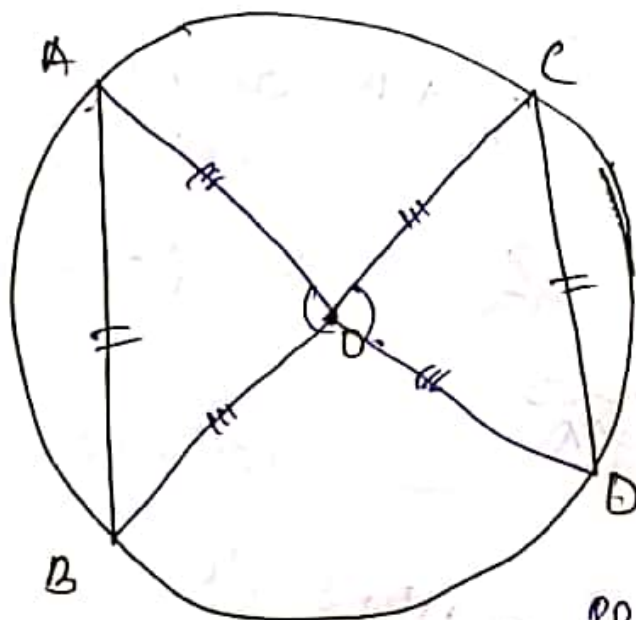




②
* Circle के Centre से
Chord पर डाला गया
perpendicular
Chord को bisect करता है

PROOF $\rightarrow \triangle OCA \cong \triangle OCB$
 $\downarrow$
 [Congruent हो गे]
 [OA = OB, CA = CB]

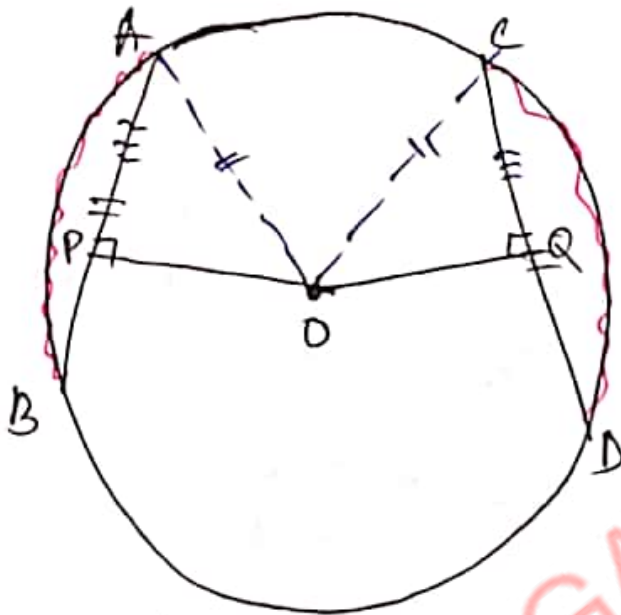
#



98
CHORD [AB = CD]
 $\Rightarrow \angle AOB = \angle COD$
 Central angle

PROOF $\rightarrow$ { Both Triangle
are congruent }

(3)

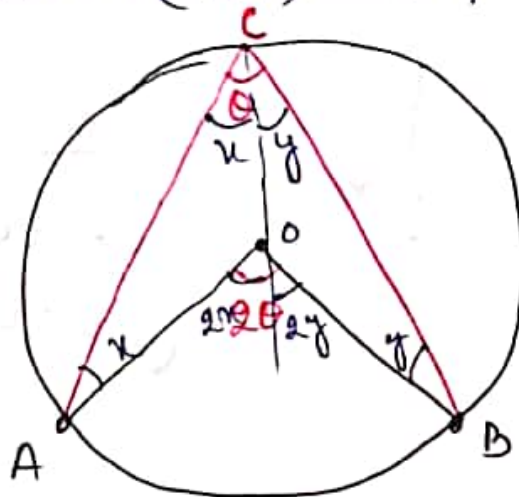


95 $\left\{ \begin{array}{l} AB = CD \\ \widehat{AB} = \widehat{CD} \end{array} \right.$ Arc

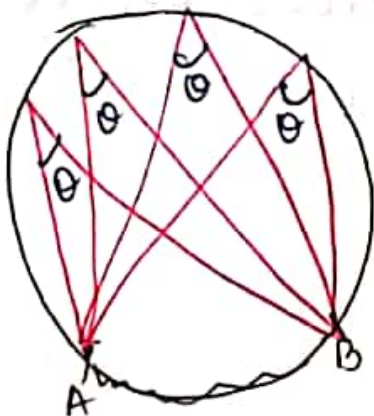
$\Rightarrow OP = OQ$

Because (both Δ are congruent)

Circle के किसी Arc द्वारा Centre पर बना angle वही Arc द्वारा Circumference (परिधि) पर बने angle का Double (दोगुना) होता है -

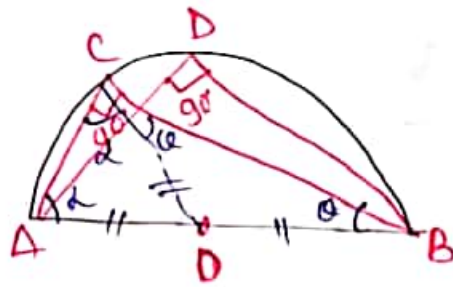


$\left\{ \begin{array}{l} \angle AOB = 2 \angle ACB \\ \downarrow \\ 2(x+y) = 2 \times (x+y) \end{array} \right.$



{ Circle के किसी Arc द्वारा परिधि पर बने angle वही angle बराबर होते हैं (यही भी बने एक ही Arc द्वारा) }

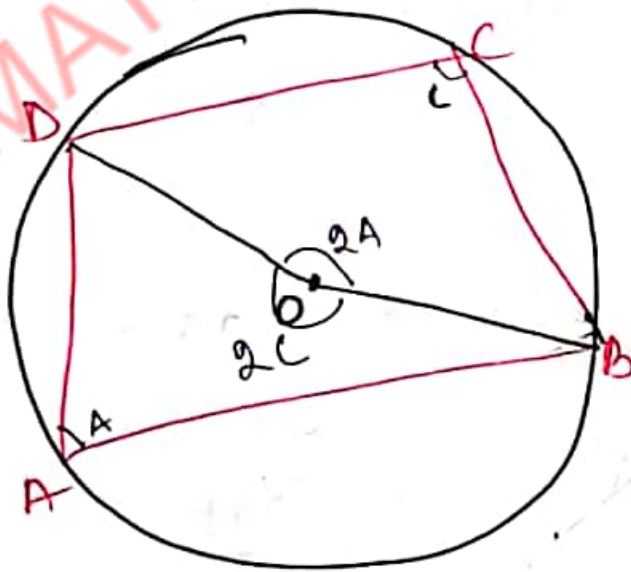
Semicircle में बना angle हमेशा 90° होता है - (4)
(अविवृत)



$$\begin{cases} \angle ACB = 90^\circ \\ \angle ADB = 90^\circ \end{cases}$$

PROOF - $\begin{cases} 2(\alpha + \theta) = 180^\circ \\ \alpha + \theta = 90^\circ \end{cases}$

Cyclic quadrilateral :-
(चक्रीय चतुर्भुज)



$\square ABCD \rightarrow$ Cyclic quadrilateral.

Condition -

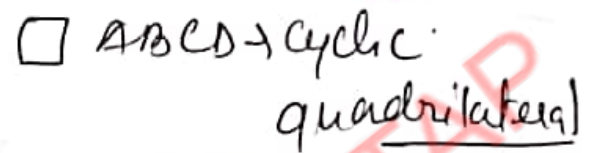
$$\begin{cases} \angle A + \angle C = 180^\circ \\ \angle B + \angle D = 180^\circ \end{cases}$$

PROOF - $\begin{cases} 2A + 2C = 360^\circ \\ \angle A + \angle C = 180^\circ \end{cases}$

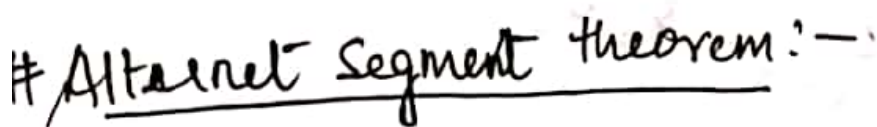
Similarly - $\angle B + \angle D = 180^\circ$

Satyam Padhyay

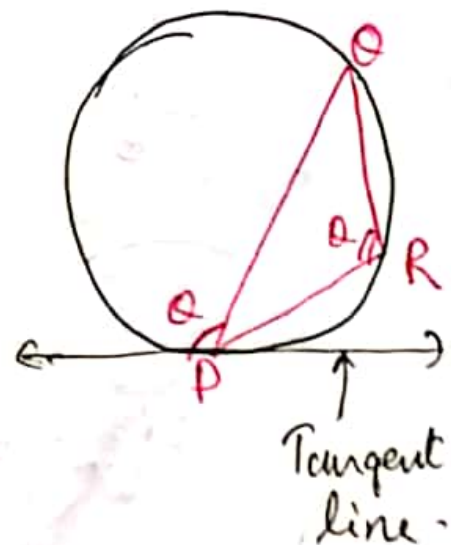




□ ABCD → cyclic quad —

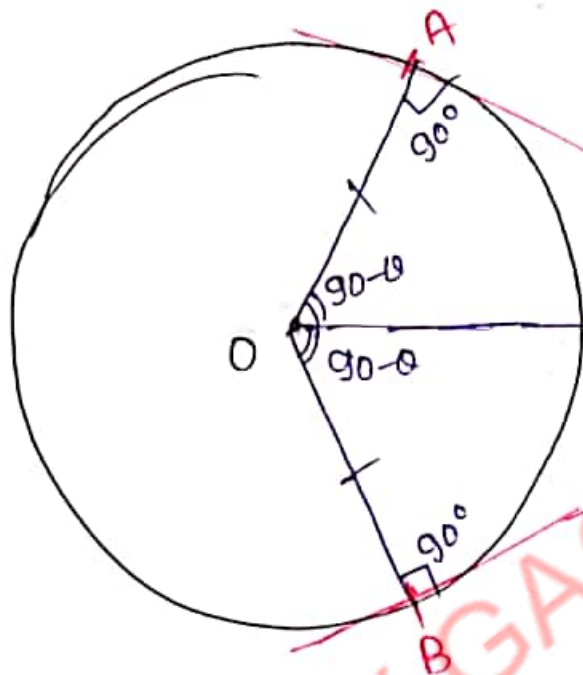
$$ab + cd = d_1 d_2$$


OR



PROOF: (CHORD $PQ \rightarrow$ Centre $\rightarrow 2O$)

પ્રા.કારા પરીક્ષા પર લગાડેલ ઓફીસ કનેક્ટા



(B)
 $PA = PB$

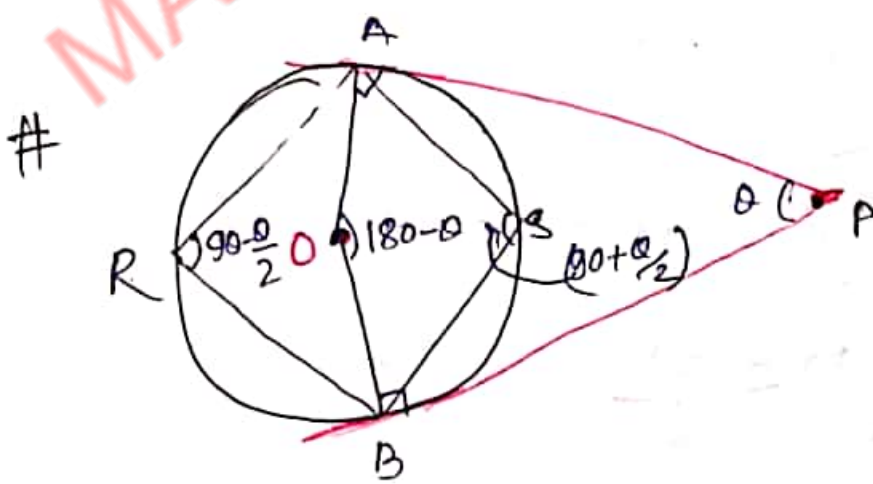
(angle bisector line)

{ OAPB will be a cyclic चतुर्भुज होगा }

Proof

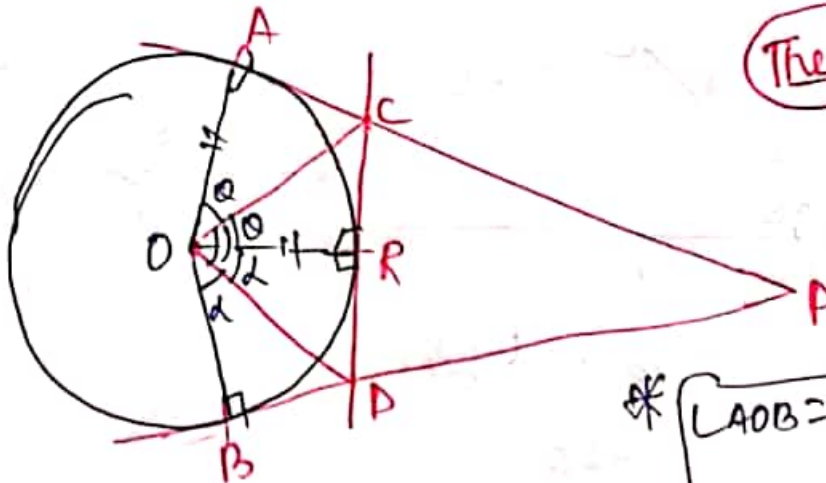
$\triangle OAP \cong \triangle OBP$ (congruent)

$PA = PB$



#

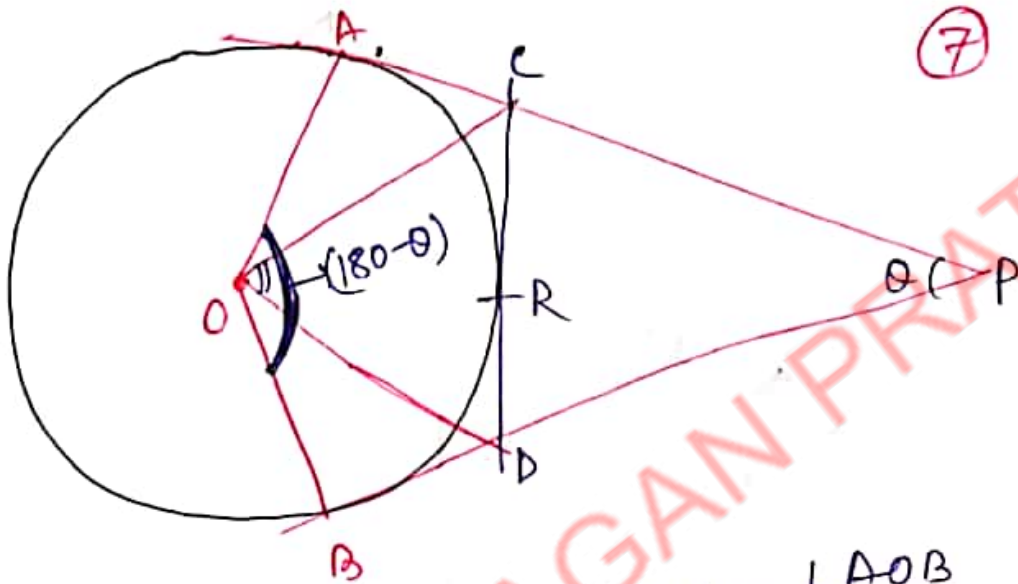
#



Theorem

* $\angle COB = \frac{\angle AOB}{2}$

* $\angle AOB = 2(\theta + \alpha)$
↓
 $\angle COB = (\theta + \alpha)$

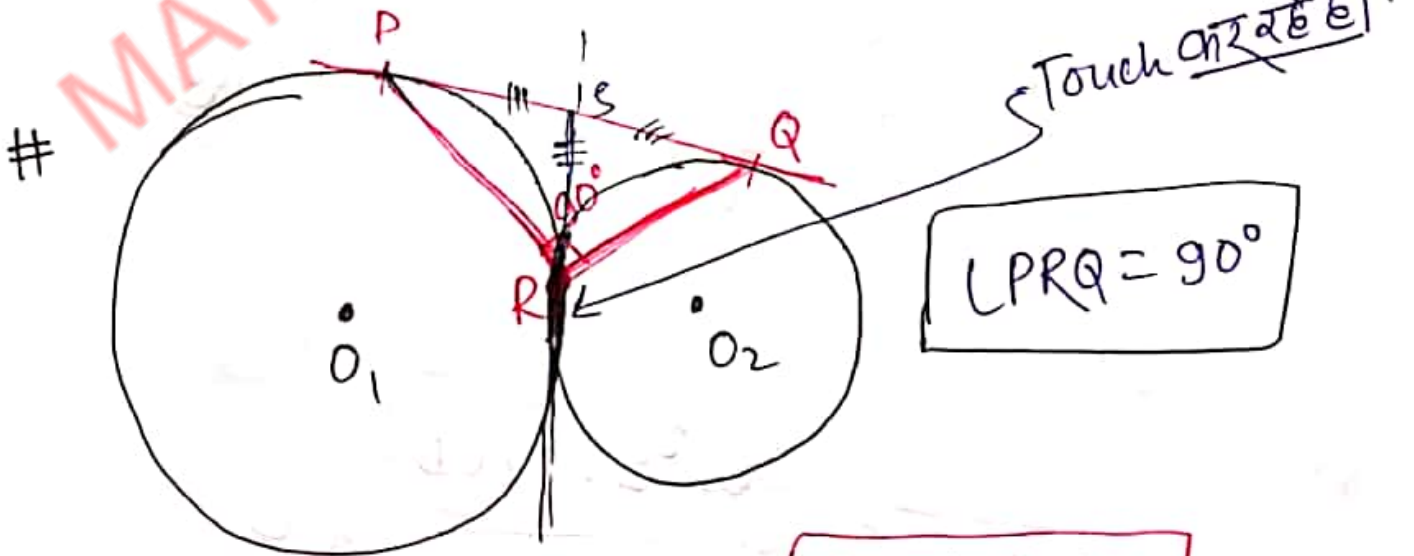


$$\angle COD = \frac{\angle AOB}{2}$$

$$\angle COD = \frac{180 - \theta}{2}$$

#

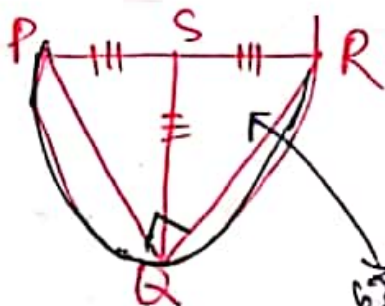
$$\angle COD = 90 - \frac{\theta}{2}$$



$$\angle PRQ = 90^\circ$$

$$SP = SR = SQ$$

$$SR = \frac{PQ}{2}$$



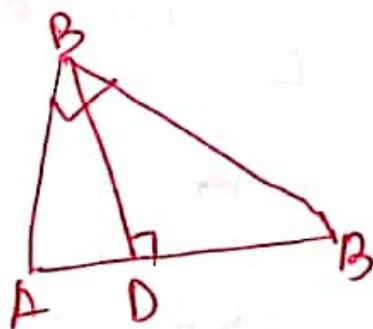
Right angle triangle hai



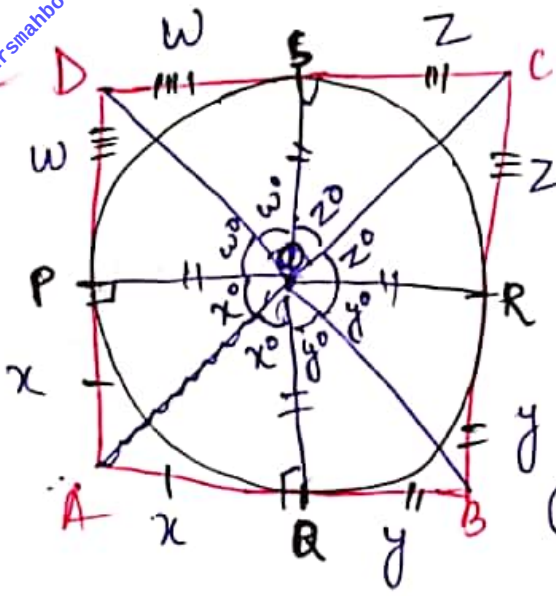
PROOF
 $\angle(x + \theta) = 180^\circ$

$$Q + d = 90^\circ$$

By similarity,



$$BD^2 = AD \times CD$$



When a circle is inscribed in a quadrilateral.

Result -

①

$$AB + CD = BC + AD$$

②

$$\left. \begin{aligned} \angle AOB + \angle COD &= 180^\circ \\ \angle BOC + \angle AOD &= 180^\circ \end{aligned} \right\}$$

PROOF ①

$$\left. \begin{aligned} AB + CD &= x + y + z + w \\ AD + BC &= x + w + y + z \end{aligned} \right\} \text{ BOTH ARE EQUAL}$$

$$AB + CD = BC + AD \quad \text{इति सिद्धम् ①}$$

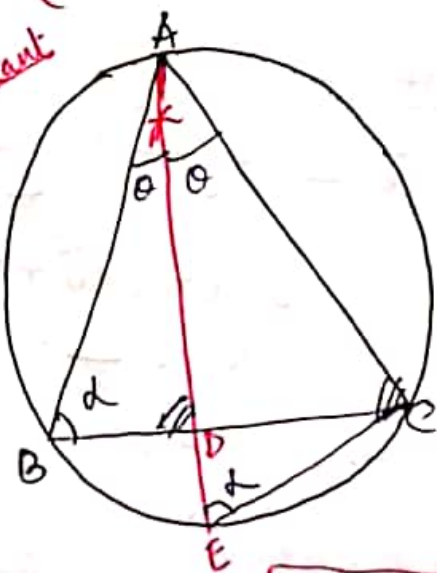
PROOF

②

$$x^\circ + y^\circ + z^\circ + w^\circ = 180^\circ$$

$$\left\{ \begin{aligned} \angle AOB + \angle COD &= 180^\circ \\ \angle BOC + \angle AOD &= 180^\circ \end{aligned} \right\} \text{ proved इति सिद्धम् ②}$$

most important



$$AB \times AC + AE \times DE = AE^2$$

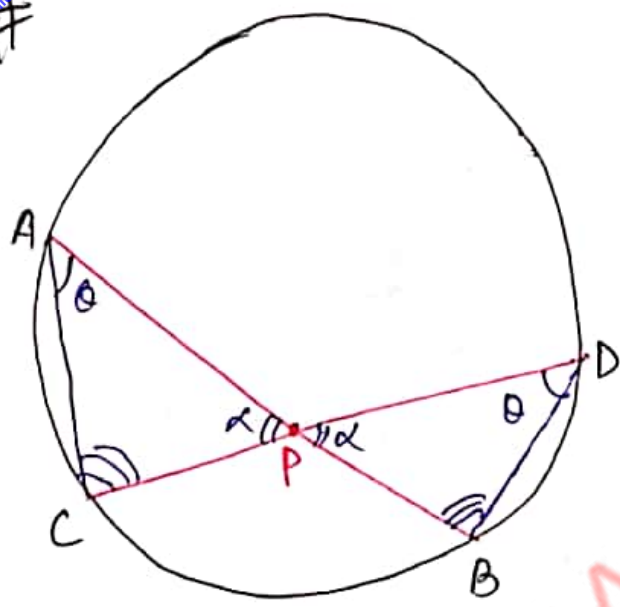
Similar Triangle

$$\frac{AD}{AC} = \frac{AB}{AE}$$

$$AD \times AE = AB \times AC \quad \text{--- ①}$$

$$AD \times AE + AE \times DE = AB \times AC + AE \times DE \Rightarrow AE^2 \quad \text{इति सिद्धम्}$$

$$AE = AD + DE$$



Result-

$$PA \times PB = PC \times PD$$

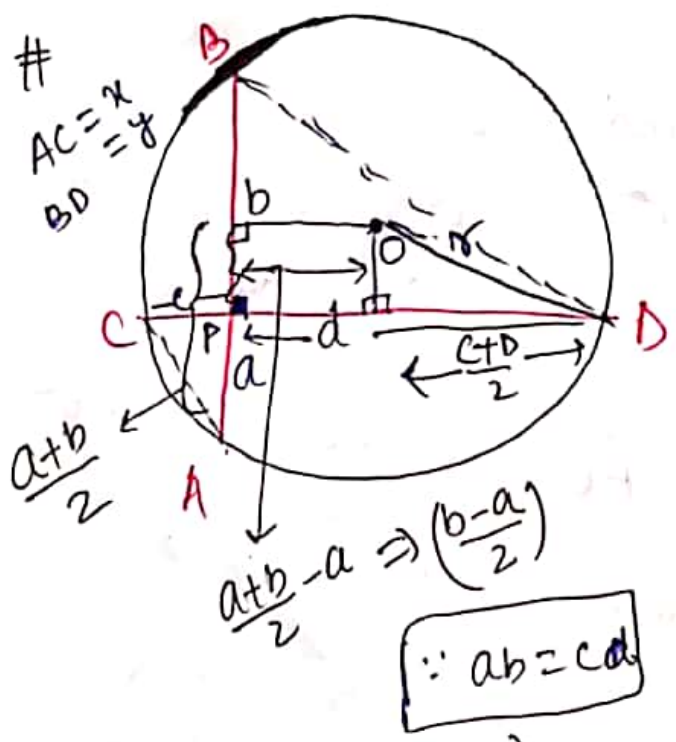
Similar Δ

PROOF -

$$\frac{PA}{PD} = \frac{PC}{PB}$$

$$PA \times PB = PC \times PD$$

सिद्धि (सिद्ध) ☺



$$r = \frac{\sqrt{x^2 + y^2}}{2}$$

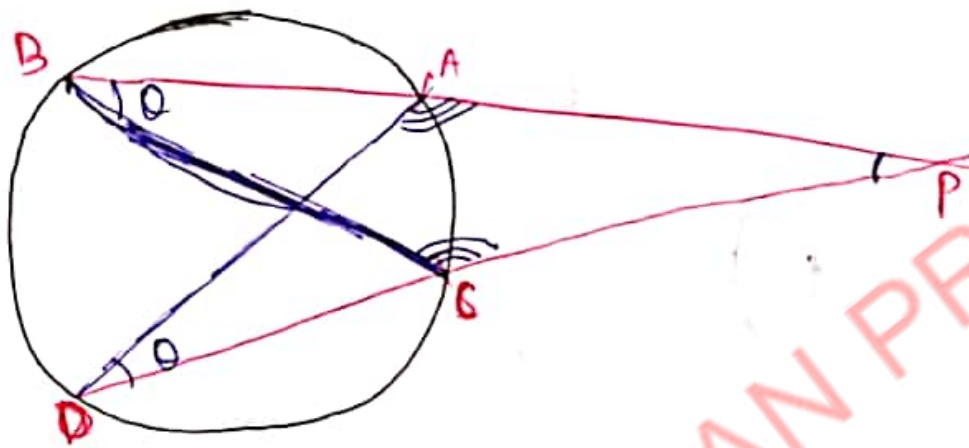
PROOF.

$$\begin{aligned} r^2 &= \frac{(b-a)^2}{4} + \frac{(c+d)^2}{4} \\ &= \frac{a^2 + b^2 - 2ab + c^2 + d^2 + 2cd}{4} \end{aligned}$$

$$r^2 = \frac{x^2 + y^2}{4}$$

$$r = \frac{\sqrt{x^2 + y^2}}{2}$$

सिद्धि (सिद्ध) ☺



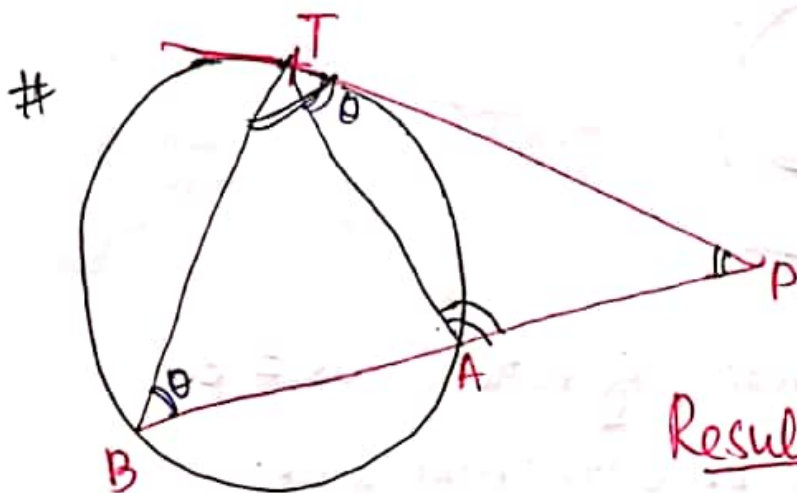
Result

$$PA \times PB = PC \times PD$$

PROOF - Similarity

$$\frac{PC}{PA} = \frac{PB}{PD}$$

$$\Rightarrow PA \times PB = PC \times PD$$



Result -

$$PT^2 = PA \times PB$$

PROOF - Similar Δ

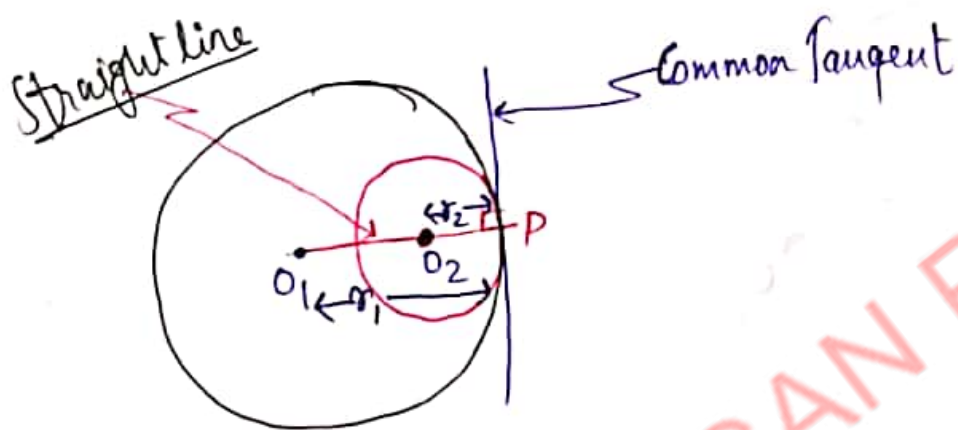
$$\frac{PT}{PB} = \frac{PA}{PT}$$

$$PT^2 = PA \times PB$$

☺ (कवि सिद्धेश्वर)

Internally Touch Circle: -

(12)



* maximum common tangent of both circle = 1

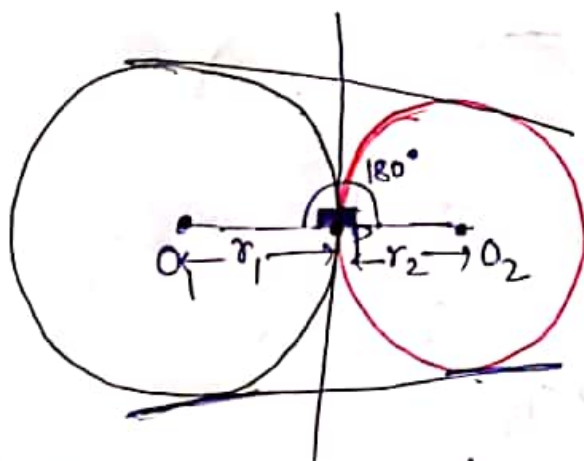
* O_1O_2P will always be a straight line.

Distance

$$O_1O_2 = O_1P - O_2P$$

$$O_1O_2 = r_1 - r_2$$

(2) Externally Touch

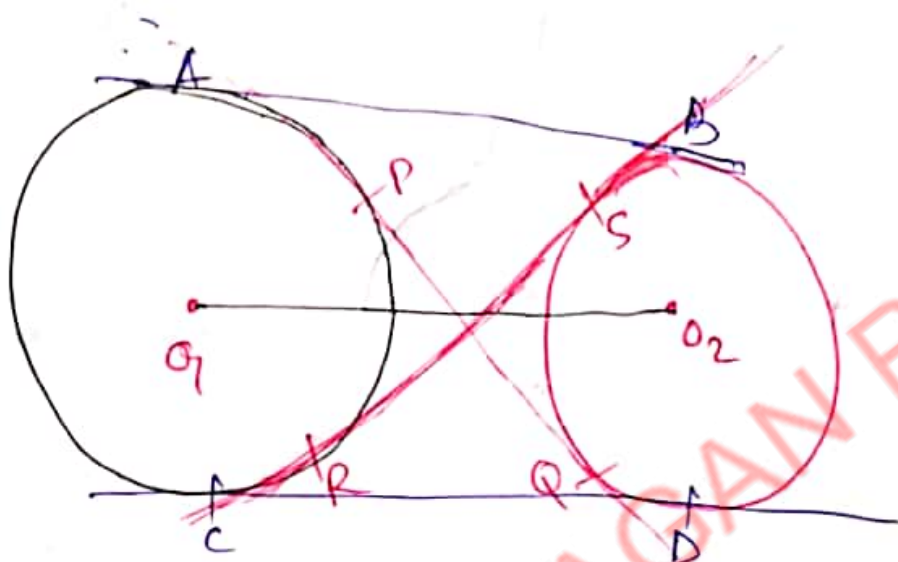


* maximum Common tangent of both Circles = 3

* O_1PO_2 will always be a straight line.

$$O_1O_2 = O_1P + O_2P$$

$$O_1O_2 = r_1 + r_2 \quad [\text{Distance}]$$

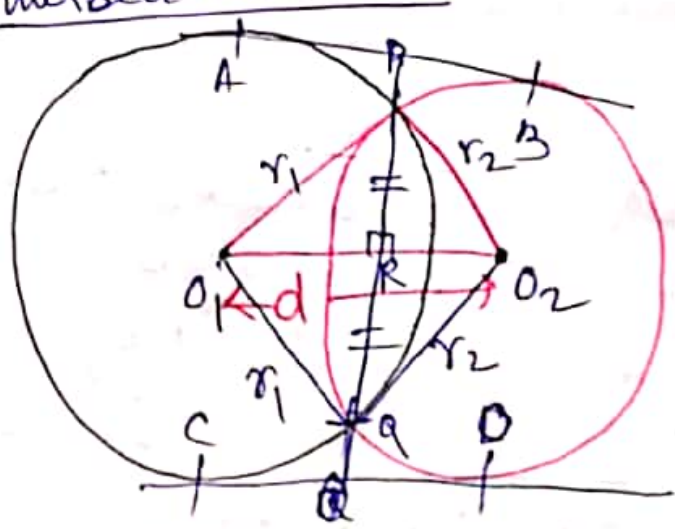


* Maximum common tangent of both circle = 4

$AB = CD = \text{Direct Common tangent (DCT)}$

$PQ = RS = \text{Transverse common tangent (TCT)}$

④ Intersect each other -



$AB = CD = \text{DCT}$

$PQ = \text{Common Chord}$
(उभयनिष्ठ जीवा)

$PR = QR = \frac{PQ}{2}$

* maximum common tangent of both circle = 2.

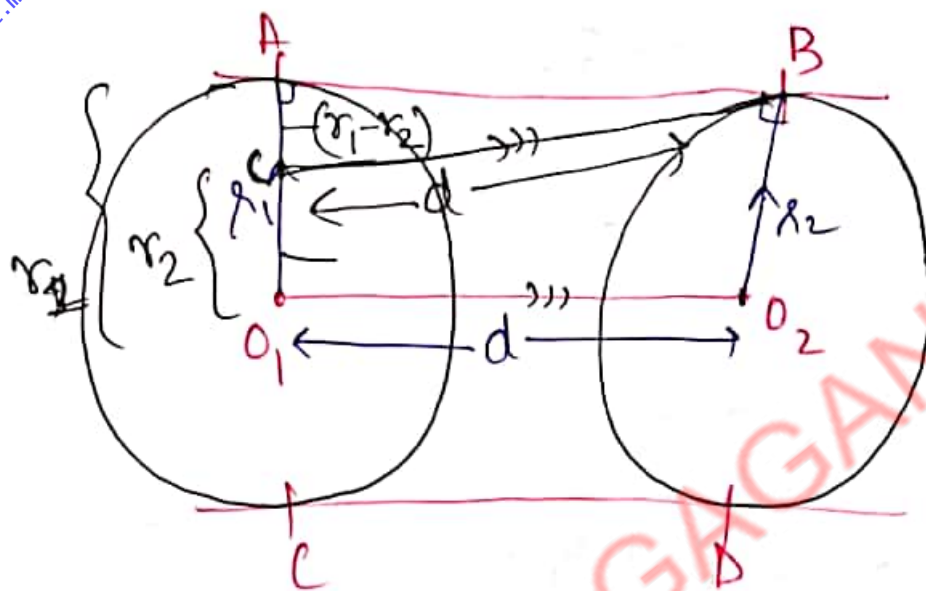
$O_1PO_2 \cong O_1QO_2$

$PR \perp O_1O_2$
 $QR \perp O_1O_2$

#

Direct Common tangent (D.C.T) →

(14)



given:

$$AB = CD$$

$$O_1O_2 = d$$

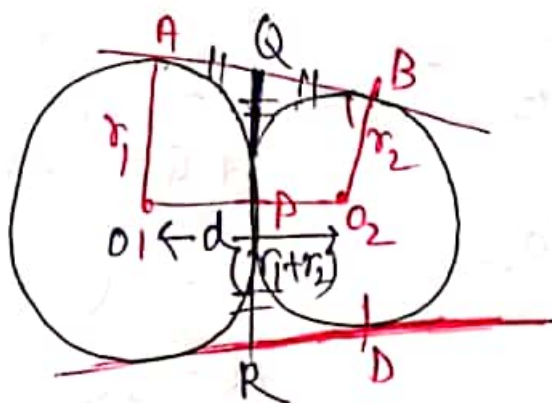
$$\begin{cases} O_1A = r_1 \\ O_2B = r_2 \end{cases}$$

$\triangle ACB$ में $(\angle A = 90^\circ)$

$$DCT = AB^2 = d^2 - (r_1 - r_2)^2$$

$$* DCT = AB = \sqrt{d^2 - (r_1 - r_2)^2}$$

When Circle touch each other externally —



Result

$$\begin{cases} QP = AQ \\ QP = QB \\ \hline QP = AB \end{cases}$$

Similarly

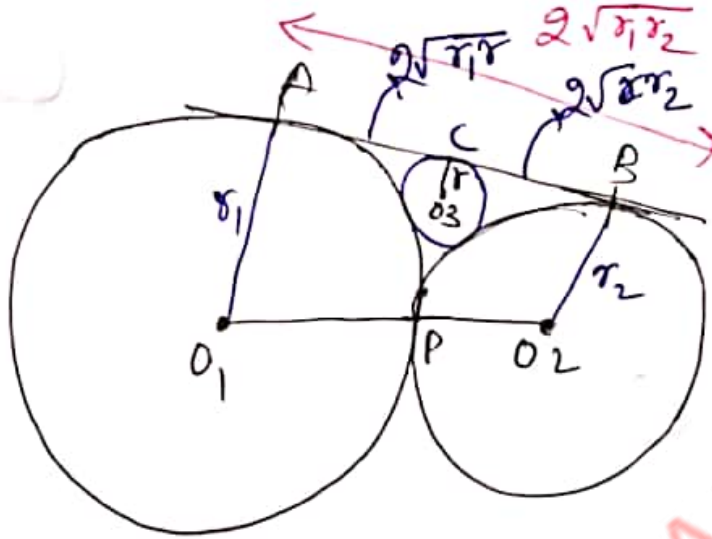
$$2PR = CD$$

$$\left\{ AB = \sqrt{(r_1 + r_2)^2 - (r_1 - r_2)^2} = 2\sqrt{r_1 r_2} \right\}$$

① + ②

$$2QR = 2AB$$

$$QR = AB = CD$$



PROOF

$$\begin{cases} AC = 2\sqrt{r_1 r_3} \\ BC = 2\sqrt{r_2 r_3} \end{cases}$$

$$AB = 2\sqrt{r_1 r_2}$$

Result-

$$\frac{1}{\sqrt{r}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}}$$

$$AB = AC + BC$$

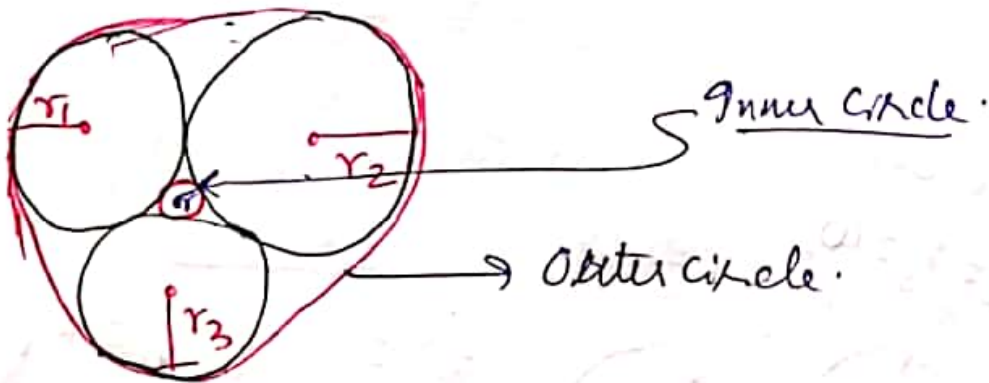
$$2\sqrt{r_1 r_2} = 2\sqrt{r_1 r_3} + 2\sqrt{r_2 r_3}$$

divide $\rightarrow \frac{2\sqrt{r_1 r_2}}{2\sqrt{r_1 r_1 r_2}} = \frac{2\sqrt{r_1 r_3}}{2\sqrt{r_1 r_1 r_2}} + \frac{2\sqrt{r_2 r_3}}{2\sqrt{r_2 r_1 r_2}}$

$$\frac{1}{\sqrt{r}} = \frac{1}{\sqrt{r_2}} + \frac{1}{\sqrt{r_1}}$$

Proved. @

Circle discaerts theorem $\rightarrow$

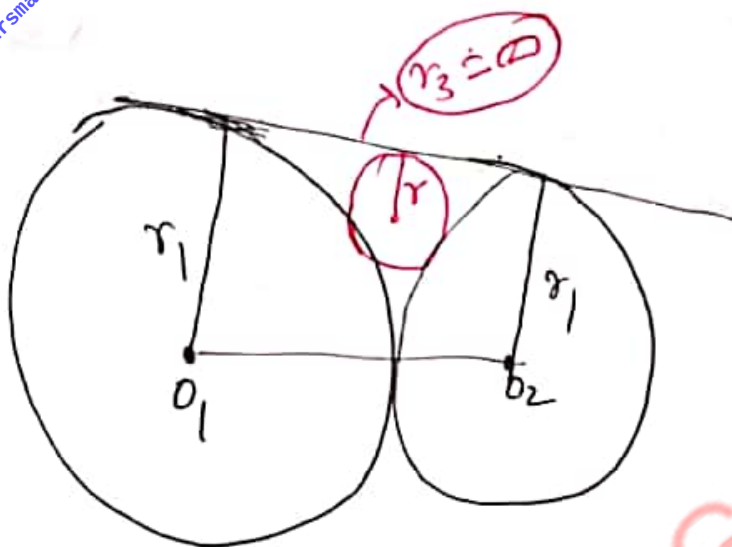


Result

$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \pm 2\sqrt{\frac{1}{r_1 r_2} + \frac{1}{r_2 r_3} + \frac{1}{r_3 r_1}}$$

$\frac{1}{r} = \text{curvature}$

-ive — Outer circle.
+ive — Inner circle.



$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \pm 2 \sqrt{\frac{1}{r_1 r_2} + \frac{1}{r_2 r_3} + \frac{1}{r_3 r_1}}$$

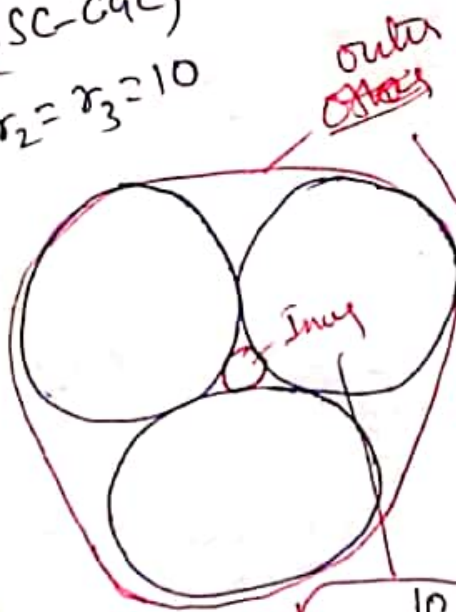
$r_3 = 0$

$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + 2 \sqrt{\frac{1}{r_1 r_2}}$$

$$\left(\frac{1}{\sqrt{r}}\right)^2 = \left(\frac{1}{\sqrt{r_1}}\right)^2 + \left(\frac{1}{\sqrt{r_2}}\right)^2 + 2 \sqrt{\frac{1}{r_1}} \times \frac{1}{\sqrt{r_2}}$$

$$\frac{1}{\sqrt{r}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}}$$

eg. (SSC-CGL)
 $r_1 = r_2 = r_3 = 10$



$$\frac{1}{r} = \frac{1}{10} + \frac{1}{10} + \frac{1}{10} \pm 2 \sqrt{\frac{1}{10 \times 10} + \frac{1}{10 \times 10} + \frac{1}{10 \times 10}}$$

$$\frac{1}{r} = \frac{3}{10} \pm 2 \frac{\sqrt{3}}{10}$$

$$\frac{1}{r} = \frac{3 \pm 2\sqrt{3}}{10}$$

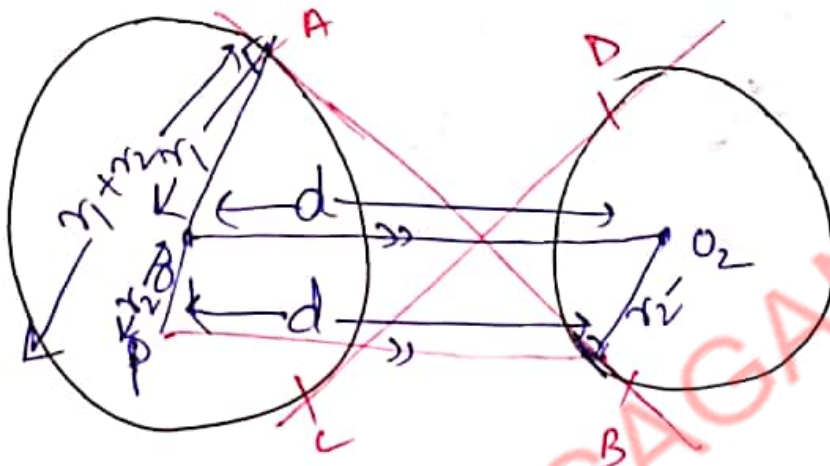
$$r = \frac{10}{3 \pm 2\sqrt{3}}$$

$$r = \frac{10}{3 + 2\sqrt{3}}$$

$$r = \frac{10}{2\sqrt{3} - 3}$$

Transverse Common tangent -

(17)



$$AB \neq CD$$

ΔPAB में :-

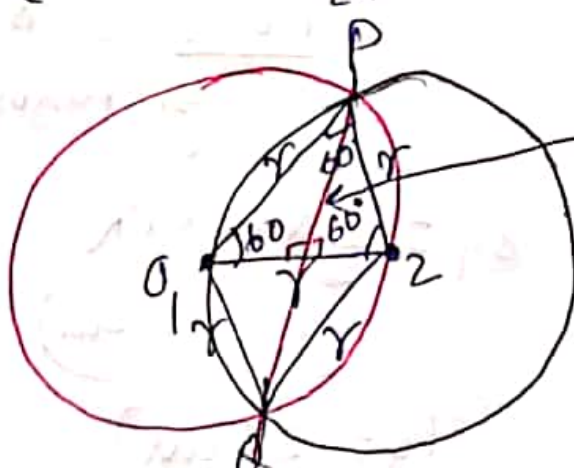
$$* AB = TCT = \sqrt{d^2 - (r_1 + r_2)^2}$$

$$* DCT = \sqrt{d^2 - (r_1 - r_2)^2}$$

$$DCT > TCT$$

Common Chord (उभयनिष्ठ जीवा) -

① When two circles pass through each other centre (एक दूसरे के केंद्रों से गुजरें)

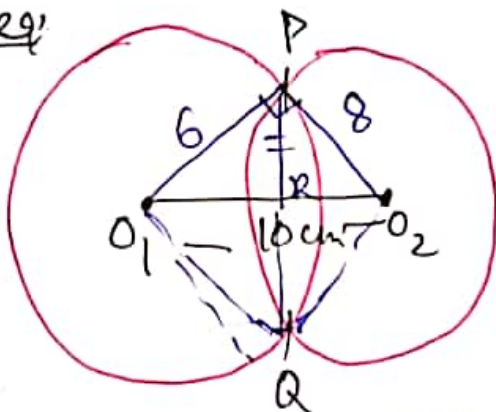


Common Chord
 $= 2 \times \frac{\sqrt{3}}{2} r$

$$C.C = \sqrt{3}r$$

2

eg



given

$$\begin{cases} r_1 = 6 \text{ cm} \\ r_2 = 8 \text{ cm} \\ PQ = ? \end{cases}$$

$$PR = RQ$$

$$PR = \frac{6 \times 8}{10}$$

$$PR = 4.8 \text{ cm}$$

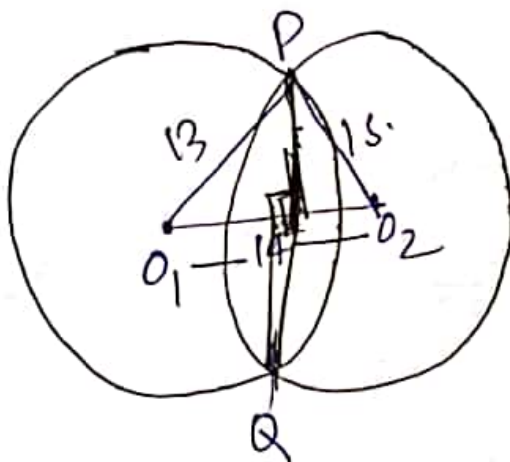
$$PQ = 9.6 \text{ cm}$$

$$PR = \frac{PQ}{2}$$

Result

$$PQ = \frac{2r_1 r_2}{d}$$

3



given

$$r_1 = 13 \text{ cm}$$

$$r_2 = 15 \text{ cm}$$

$$PR = RQ$$

$$A_{\Delta PO_1O_2} = 84$$

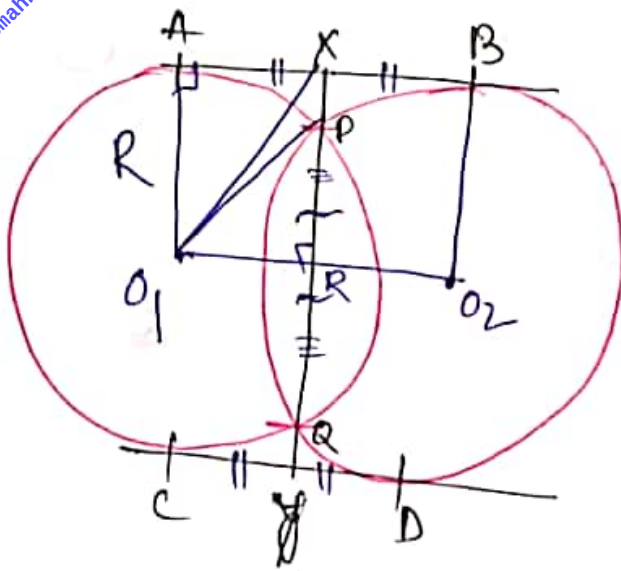
[By Heron's formula]

$$84 = \frac{1}{2} \times 14 \times PR$$

$$PR = 12 \text{ cm}$$

$$PQ = 24 \text{ cm}$$

Satyam Upadhyay



Result -

$$XY^2 = AB^2 + PQ^2$$

PROOF

$$AX = BX =$$

↓

$$\text{Similarly } CY = DY = \frac{AB}{2} = \frac{CD}{2}$$

$$\begin{cases} AX^2 = XP \times XQ \\ BX^2 = XP \times XQ \end{cases}$$

$$PR = QR$$

$$PX = QY$$

$$RX = RY$$

$$\Delta O_1AX \text{ में } O_1X^2 = R^2 + AX^2 \quad \checkmark$$

$$\Delta O_1RX \text{ में } O_1X^2 = O_1R^2 + XR^2 \quad \checkmark$$

$$\Delta O_1RP \text{ में } O_1P^2 = O_1R^2 + PR^2$$

$$R^2 = O_1R^2 + PR^2$$

$$R^2 + AX^2 = R^2 - PR^2 + XR^2$$

$$4AX^2 = 4AX^2 + 4PR^2$$

↓

$$(2XR)^2 = (2AX)^2 + (2PR)^2$$

(20)

$$XY^2 = AB^2 + PQ^2$$

proved.

इति सिद्धम्

(21)